

Owner comparison matrix - beam-v03 (EC3) vs MasterSeries

63 cases, 118 runs, generated 2026-09-20. Basis: S275, 6.0 m simply supported fork ends, EC3 + UK NA, ULS 1.35G + 1.5Q, SLS 1.0Q; base loading G UDL 10 kN/m + Q UDL 8 kN/m + Q 30 kN at 2.0 m. One brief per run follows in this order. Verdicts: PASS 21, FAIL 90, NOT VERIFIED 3, REFUSED 4, ERROR 0.

Method: eigen = FE eigenvalue M_cr; standard = SN003a / P385 closed forms; full = fully restrained; refused = not run. Annex A = EN 1993-6 LTB + torsion (≥ 99 (capacity exhausted) = the amplifier is unbounded, $M_{y,Ed} \geq M_{cr}$). A channel's standard-route M_cr is the kappa-chain value (SN003a value in brackets where offered). N_Ed and M_z,Ed of L5..L7 are ULS design values entered directly. Comparison column: MasterSeries figures.

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b,Rd	Annex A	defl.	Comparison
UB-L1	UB 457 x 191 x 82	L1: no torsion	eigen	PASS	LTB M_Ed/M_b,Rd (0.581)	393.1	290.4	-	0.193	
UB-L1	UB 457 x 191 x 82	L1: no torsion	standard	PASS	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L2	UB 457 x 191 x 82	L2: horizontal ecc. e = +40	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.903)	393.1	290.4	0.903	0.193	
UB-L2	UB 457 x 191 x 82	L2: horizontal ecc. e = +40	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-L3t	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	eigen	PASS	LTB M_Ed/M_b,Rd (0.719)	283.9	234.8	-	0.193	
UB-L3t	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	standard	NOT VERIFIED	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L3b	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	eigen	PASS	LTB M_Ed/M_b,Rd (0.491)	542.7	343.8	-	0.193	
UB-L3b	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	standard	PASS	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L4	UB 457 x 191 x 82	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.113)	283.9	234.8	1.113	0.193	
UB-L4	UB 457 x 191 x 82	L4: e + z_g	standard	NOT VERIFIED	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-L5	UB 457 x 191 x 82	L5: e + z_g + N_Ed	eigen	FAIL	Member buckling z-z (Eq 6.62) (1.193)	283.9	234.8	1.113	0.193	
UB-L5	UB 457 x 191 x 82	L5: e + z_g + N_Ed	standard	FAIL	Member buckling z-z (Eq 6.62) (1.032)	398.4	293.4	0.897	0.193	
UB-L6	UB 457 x 191 x 82	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (2.671)	292.8	241.1	1.787	0.277	
UB-L6	UB 457 x 191 x 82	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (2.671)	416.4	303.6	1.407	0.277	
UB-L7	UB 457 x 191 x 82	L7: L6 + couple 40 kN.m	eigen	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (2.908)	297.7	243.5	1.995	0.316	
UB-L7	UB 457 x 191 x 82	L7: L6 + couple 40 kN.m	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (2.908)	415.5	303.1	1.513	0.316	
UC-L1	UC 203 x 203 x 60	L1: no torsion	eigen	FAIL	Deflection (1.169)	277.9	151.8	-	1.169	
UC-L1	UC 203 x 203 x 60	L1: no torsion	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L2	UC 203 x 203 x 60	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.650)	277.9	151.8	1.650	1.169	
UC-L2	UC 203 x 203 x 60	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.640)	281.9	153.0	1.640	1.169	
UC-L3t	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB M_Ed/M_b,Rd (1.189)	219.6	141.0	-	1.169	
UC-L3t	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L3b	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (1.169)	350.7	160.5	-	1.169	
UC-L3b	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L4	UC 203 x 203 x 60	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.927)	219.6	141.0	1.927	1.169	
UC-L4	UC 203 x 203 x 60	L4: e + z_g	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.640)	281.9	153.0	1.640	1.169	
UC-L5	UC 203 x 203 x 60	L5: e + z_g + N_Ed	eigen	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (3.308)	219.6	141.0	1.927	1.169	
UC-L5	UC 203 x 203 x 60	L5: e + z_g + N_Ed	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (3.308)	281.9	153.0	1.640	1.169	
UC-L6	UC 203 x 203 x 60	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (8.823)	226.9	144.0	5.285	1.678	
UC-L6	UC 203 x 203 x 60	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (8.823)	294.8	156.4	2.435	1.678	
UC-L7	UC 203 x 203 x 60	L7: L6 + couple 40 kN.m	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	228.6	143.7	≥ 99 (capacity exhausted)	1.913	
UC-L7	UC 203 x 203 x 60	L7: L6 + couple 40 kN.m	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (9.931)	294.1	156.3	2.682	1.913	
RHS-L1	RHS 200 x 100 x 8.0 (HF)	L1: no torsion	eigen	FAIL	Deflection (3.210)	899.1	77.5	-	3.210	
RHS-L1	RHS 200 x 100 x 8.0 (HF)	L1: no torsion	standard	FAIL	Deflection (3.210)	916.2	77.5	-	3.210	
RHS-L2	RHS 200 x 100 x 8.0 (HF)	L2: horizontal ecc. e = +40	eigen	FAIL	Deflection (3.210)	899.1	77.5	-	3.210	
RHS-L2	RHS 200 x 100 x 8.0 (HF)	L2: horizontal ecc. e = +40	standard	FAIL	Deflection (3.210)	916.2	77.5	-	3.210	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b,Rd	Annex A	defl.	Comparison
RHS-L3t	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (3.210)	876.7	77.5		- 3.210	
RHS-L3t	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L3b	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (3.210)	921.9	77.5		- 3.210	
RHS-L3b	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L4	RHS 200 x 100 x 8.0 (HF)	L4: e + z_g	eigen	FAIL	Deflection (3.210)	876.7	77.5		- 3.210	
RHS-L4	RHS 200 x 100 x 8.0 (HF)	L4: e + z_g	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L5	RHS 200 x 100 x 8.0 (HF)	L5: e + z_g + N_Ed	eigen	FAIL	Bending+axial cross-section (6.2.9) (3.621)	876.7	77.5		- 3.210	
RHS-L5	RHS 200 x 100 x 8.0 (HF)	L5: e + z_g + N_Ed	standard	FAIL	Bending+axial cross-section (6.2.9) (3.621)	916.2	77.5		- 3.210	
RHS-L6	RHS 200 x 100 x 8.0 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (6.070)	910.8	77.5		- 4.605	
RHS-L6	RHS 200 x 100 x 8.0 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Biaxial bending + axial (6.2.9.1) (6.070)	958.8	77.5		- 4.605	
RHS-L7	RHS 200 x 100 x 8.0 (HF)	L7: L6 + couple 40 kN.m	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (7.033)	895.8	77.5		- 5.249	
RHS-L7	RHS 200 x 100 x 8.0 (HF)	L7: L6 + couple 40 kN.m	standard	FAIL	Biaxial bending + axial (6.2.9.1) (7.033)	956.0	77.5		- 5.249	
SHS-L1	SHS 150 x 150 x 6.3 (HF)	L1: no torsion	eigen	FAIL	Deflection (5.867)	1190.1	52.8		- 5.867	
SHS-L1	SHS 150 x 150 x 6.3 (HF)	L1: no torsion	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L2	SHS 150 x 150 x 6.3 (HF)	L2: horizontal ecc. e = +40	eigen	FAIL	Deflection (5.867)	1190.1	52.8		- 5.867	
SHS-L2	SHS 150 x 150 x 6.3 (HF)	L2: horizontal ecc. e = +40	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L3t	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (5.867)	1162.3	52.8		- 5.867	
SHS-L3t	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L3b	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (5.867)	1218.4	52.8		- 5.867	
SHS-L3b	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L4	SHS 150 x 150 x 6.3 (HF)	L4: e + z_g	eigen	FAIL	Deflection (5.867)	1162.3	52.8		- 5.867	
SHS-L4	SHS 150 x 150 x 6.3 (HF)	L4: e + z_g	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L5	SHS 150 x 150 x 6.3 (HF)	L5: e + z_g + N_Ed	eigen	FAIL	Bending+axial cross-section (6.2.9) (7.031)	1162.3	52.8		- 5.867	
SHS-L5	SHS 150 x 150 x 6.3 (HF)	L5: e + z_g + N_Ed	standard	FAIL	Bending+axial cross-section (6.2.9) (7.031)	1212.8	52.8		- 5.867	
SHS-L6	SHS 150 x 150 x 6.3 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (11.579)	1207.9	52.8		- 8.417	
SHS-L6	SHS 150 x 150 x 6.3 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Biaxial bending + axial (6.2.9.1) (11.579)	1269.4	52.8		- 8.417	
SHS-L7	SHS 150 x 150 x 6.3 (HF)	L7: L6 + couple 40 kN.m	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (13.483)	1187.5	52.8		- 9.595	
SHS-L7	SHS 150 x 150 x 6.3 (HF)	L7: L6 + couple 40 kN.m	standard	FAIL	Biaxial bending + axial (6.2.9.1) (13.483)	1265.5	52.8		- 9.595	
PFC-L1	PFC 200 x 75 x 23	L1: no torsion	eigen	FAIL	LTB M_Ed/M_b,Rd (6.475)	35.8	25.6		- 3.652	
PFC-L1	PFC 200 x 75 x 23	L1: no torsion	standard	FAIL	LTB M_Ed/M_b,Rd (18.794)	9.1 (35.7)	8.8		- 3.652	
PFC-L2	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-L2	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b.Rd	Annex A	defl.	Comparison
PFC-L2w	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-L2w	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L3t	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L3t	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L3b	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	41.4	28.2	>= 99 (capacity exhausted)	3.652	
PFC-L3b	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L4	PFC 200 x 75 x 23	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L4	PFC 200 x 75 x 23	L4: e + z_g	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L5	PFC 200 x 75 x 23	L5: e + z_g + N_Ed	eigen	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L5	PFC 200 x 75 x 23	L5: e + z_g + N_Ed	standard	FAIL	Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L6	PFC 200 x 75 x 23	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	32.0	23.7	>= 99 (capacity exhausted)	5.239	
PFC-L6	PFC 200 x 75 x 23	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	5.239	
PFC-L7	PFC 200 x 75 x 23	L7: L6 + couple 40 kN.m	eigen	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	31.9	23.6	>= 99 (capacity exhausted)	5.972	
PFC-L7	PFC 200 x 75 x 23	L7: L6 + couple 40 kN.m	standard	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	5.972	
UB-R1	UB 457 x 191 x 82	R1: fully restrained	full	PASS	Bending+torsion cross-section (P385 3.1.2) (0.415)	-	503.3	-	0.193	
UB-R2	UB 457 x 191 x 82	R2: unrestrained (reference)	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.903)	393.1	290.4	0.903	0.193	
UB-R2	UB 457 x 191 x 82	R2: unrestrained (reference)	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-R3	UB 457 x 191 x 82	R3: restrained at 2, 4 m	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.568)	2355.7	485.7	0.568	0.193	
UB-R3	UB 457 x 191 x 82	R3: restrained at 2, 4 m	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-R4	UB 457 x 191 x 82	R4: warping restrained	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.590)	669.5	372.0	0.590	0.193	
UB-R4	UB 457 x 191 x 82	R4: warping restrained	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.743)	398.4	293.4	0.743	0.193	
UB-R5	UB 457 x 191 x 82	R5: warping free	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.903)	393.1	290.4	0.903	0.193	
UB-R5	UB 457 x 191 x 82	R5: warping free	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-R6	UB 457 x 191 x 82	R6: destabilising switch, z_g = 0	eigen	NOT VERIFIED	LTB+torsion (EN 1993-6 Annex A) (0.903)	393.1	290.4	0.903	0.193	
UB-R6	UB 457 x 191 x 82	R6: destabilising switch, z_g = 0	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.052)	306.3	248.1	1.052	0.193	
UB-R7	UB 457 x 191 x 82	R7: stabilising z_g = -D/2	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.781)	542.7	343.8	0.781	0.193	
UB-R7	UB 457 x 191 x 82	R7: stabilising z_g = -D/2	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	398.4	293.4	0.897	0.193	
UB-R8	UB 457 x 191 x 82	R8: L_E override 0.85 L	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.903)	393.1	290.4	0.903	0.193	
UB-R8	UB 457 x 191 x 82	R8: L_E override 0.85 L	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.800)	510.6	335.1	0.800	0.193	
UB-R9	UB 457 x 191 x 82	R9: top flange restrained (= full)	full	PASS	Bending+torsion cross-section (P385 3.1.2) (0.415)	-	503.3	-	0.193	
UB-R10	UB 457 x 191 x 82	R10: bottom flange restrained	refused	REFUSED	-	-	-	-	-	
UB-R11	UB 457 x 191 x 82	R11: partially restrained	refused	REFUSED	-	-	-	-	-	
PFC-R1	PFC 200 x 75 x 23	R1: fully restrained	full	FAIL	Bending+torsion cross-section (P385 3.1.2) (14.062)	-	62.4	-	3.652	
PFC-R2	PFC 200 x 75 x 23	R2: unrestrained (reference)	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R2	PFC 200 x 75 x 23	R2: unrestrained (reference)	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R3	PFC 200 x 75 x 23	R3: restrained at 2, 4 m	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	119.0	46.3	>= 99 (capacity exhausted)	3.652	
PFC-R3	PFC 200 x 75 x 23	R3: restrained at 2, 4 m	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R4	PFC 200 x 75 x 23	R4: warping restrained	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	41.6	28.2	>= 99 (capacity exhausted)	3.652	
PFC-R4	PFC 200 x 75 x 23	R4: warping restrained	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R5	PFC 200 x 75 x 23	R5: warping free	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R5	PFC 200 x 75 x 23	R5: warping free	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R6	PFC 200 x 75 x 23	R6: destabilising switch, z_g = 0	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b.Rd	Annex A	defl.	Comparison
PFC-R6	PFC 200 x 75 x 23	R6: destabilising switch, z_g = 0	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	6.3	6.3	≥ 99 (capacity exhausted)	3.652	
PFC-R7	PFC 200 x 75 x 23	R7: stabilising z_g = -D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	41.4	28.2	≥ 99 (capacity exhausted)	3.652	
PFC-R7	PFC 200 x 75 x 23	R7: stabilising z_g = -D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	9.1	8.8	≥ 99 (capacity exhausted)	3.652	
PFC-R8	PFC 200 x 75 x 23	R8: L_E override 0.85 L	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	35.8	25.6	≥ 99 (capacity exhausted)	3.652	
PFC-R8	PFC 200 x 75 x 23	R8: L_E override 0.85 L	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (≥ 99 (capacity exhausted))	12.6	11.5	≥ 99 (capacity exhausted)	3.652	
PFC-R9	PFC 200 x 75 x 23	R9: top flange restrained (= full)	full	FAIL	Bending+torsion cross-section (P385 3.1.2) (14.062)	-	62.4	-	3.652	
PFC-R10	PFC 200 x 75 x 23	R10: bottom flange restrained	refused	REFUSED	-	-	-	-	-	
PFC-R11	PFC 200 x 75 x 23	R11: partially restrained	refused	REFUSED	-	-	-	-	-	

UB 457 x 191 x 82, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

Beam & Beam-Portion (Member)

Member UB-L1 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m (kN/m)

Q UDL 8.000 0-6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.8044 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G, \text{inf}} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.50Q [$y_{G, inf} = 0.9$, EQU set A]

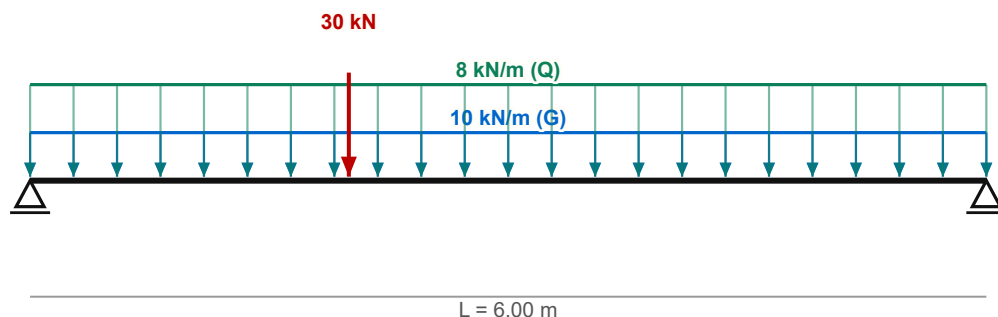
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

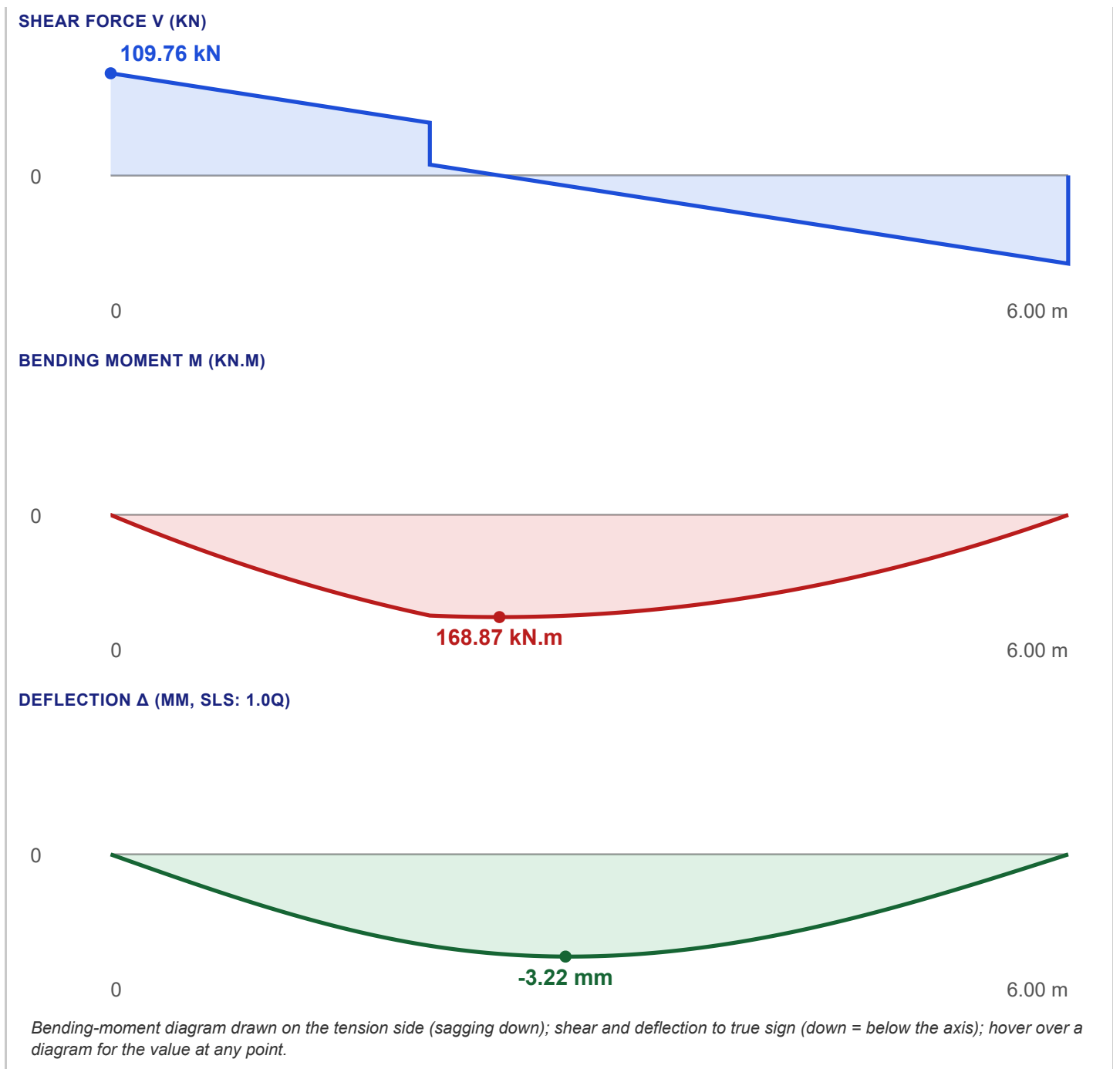
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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LOADING





Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.179 \quad \text{OK 7.2} \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

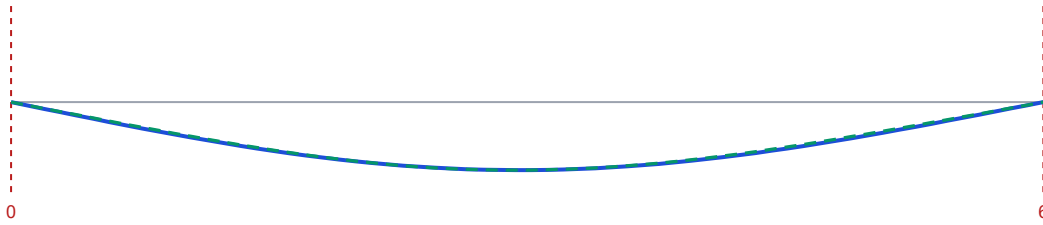
Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = M_{cr}/M_{cr,uniform} & \quad 393.107 / 342.775 \text{ kN.m (shape only: } z_g = z_j = 0) \quad 1.147 \quad k_c \text{ only} \\ C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) \text{ (standard, comparison)} & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \text{ — general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 \text{ kN.m} \quad 1.162 \quad \text{Serna} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = \text{portion between restraints} & \quad \text{End 1 v, } \varphi = 0; \text{ End 2 v, } \varphi = 0 \quad 6 \text{ m} \quad \text{FE} \\ M_{cr} = \text{FE eigenvalue (n}_{Elem}, \text{ mesh error)} & \quad 64 \text{ elements, } 0.000 \% ; 1 \text{ combination(s), } 2 \text{ solve(s) + 1 cached} \quad 393.107 \text{ kN.m} \quad \text{converged} \\ \lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}} & \quad \sqrt{1830.0 \times 275 / 393.107} \quad 1.131 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 1.131, 1.159, 0.49 \quad 0.562 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.562, 1.131, 0.934, 0.974 \quad 0.577 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.577 \times 1830.0 \times 275 \leq 503.250 = 290.421 \text{ kN.m} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 290.421 \quad 0.581 \quad \text{OK} \end{aligned}$$

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m 3.22 mm OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.581

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.581$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

Beam & Beam-Portion (Member)

Member UB-L1 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m (kN/m)

0 UDL 8.000 0-6 m (kN/m)

O PY 30.000 @ 2 m (kN)

G SW 0.8044 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G, \text{inf}} = 1.0$, STR set B]

Uls C2: Uls: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

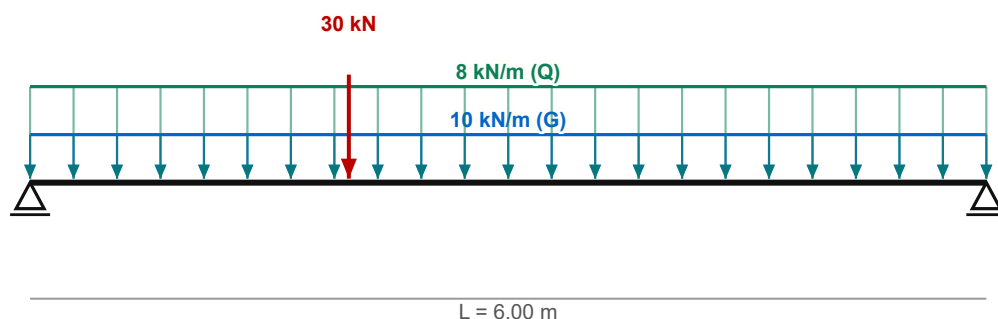
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

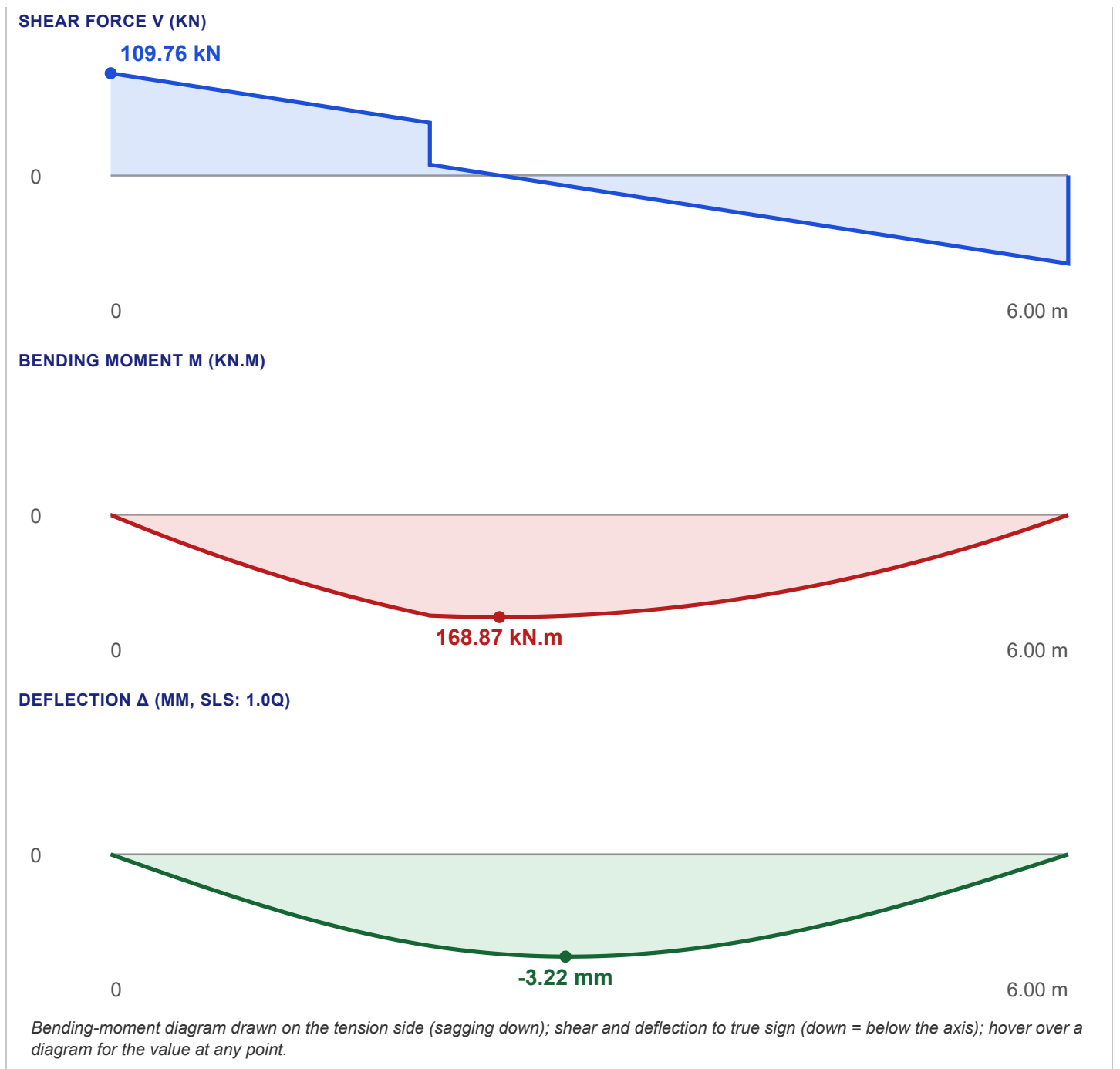
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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LOADING





Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / 1$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.179 \quad \text{OK } 7.2 \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \quad 1.162 \quad \text{Serna} \\ C_1 \text{ basis} & \quad \text{general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{\max}=168.9 \text{ kN}\cdot\text{m} \quad \text{SN003a} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = 1 L & \quad 1 \times 6 = 6 \text{ m} \\ M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E) & \quad 1.162, 6.000, 1870, 69.2, 0.922, 210000; z_g = 0 \quad 398.376 \text{ kN}\cdot\text{m} \quad \text{SN003a} \\ \lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}} & \quad \sqrt{1830.0 \times 275 / 398.376} \quad 1.124 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 1.124, 1.151, 0.49 \quad 0.566 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.566, 1.124, 0.928, 0.971 \quad 0.583 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.583 \times 1830.0 \times 275 \leq 503.250 = 293.432 \text{ kN}\cdot\text{m} \\ \lambda_{LT} \text{ (P362 6.55 simplified) ; } M_{b,Rd} & \quad (1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364; \chi_{LT,mod} = 0.451; \quad 226.988 \text{ kN}\cdot\text{m} \quad \text{P362 6.55} \\ \lambda_{LT} \text{ (comparison only)} & \quad M_{y,Ed}/M_{b,Rd,simp} = 0.744 \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 293.432 \quad 0.575 \quad \text{OK} \end{aligned}$$

Deflection Check - Load Case 1 (SLS: 1.0Q)

$$\text{In-span } \delta \leq \text{Span}/360 \quad 3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m} \quad 3.22 \text{ mm} \quad \text{OK}$$

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_{Rd}	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.575$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at $e = +40$ mm (torque), $z_g = 0$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

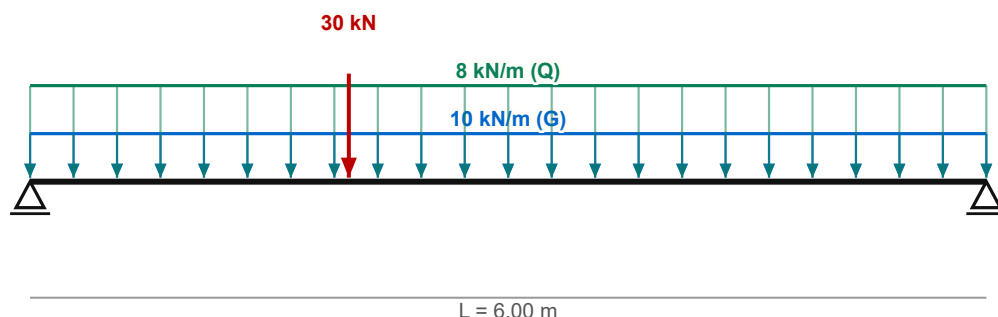
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.26	109.76	0.00	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	-3.66	-94.76	0.00	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

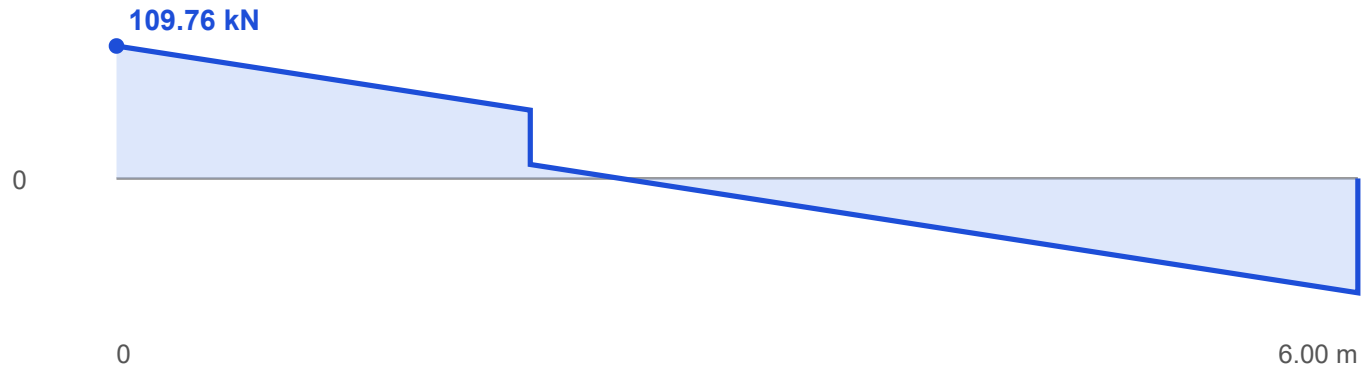
Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model: M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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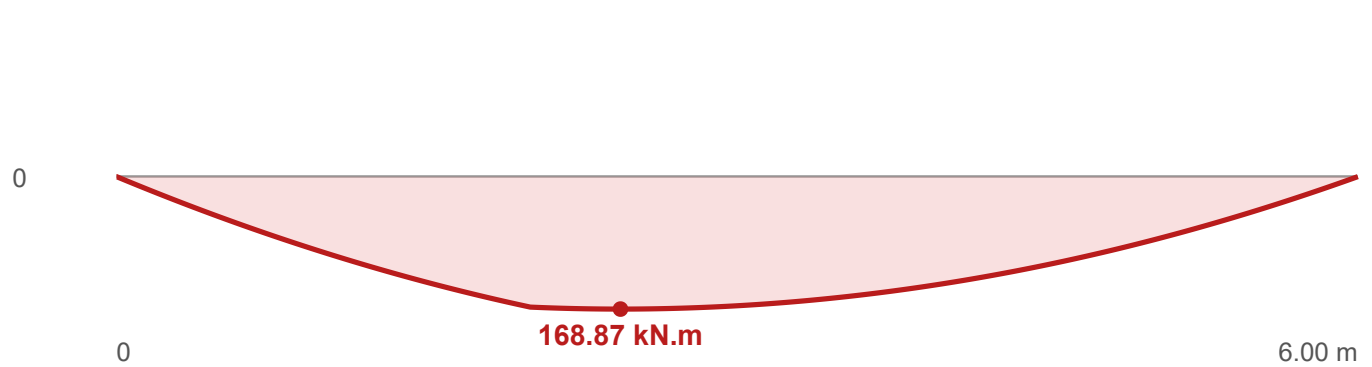
LOADING



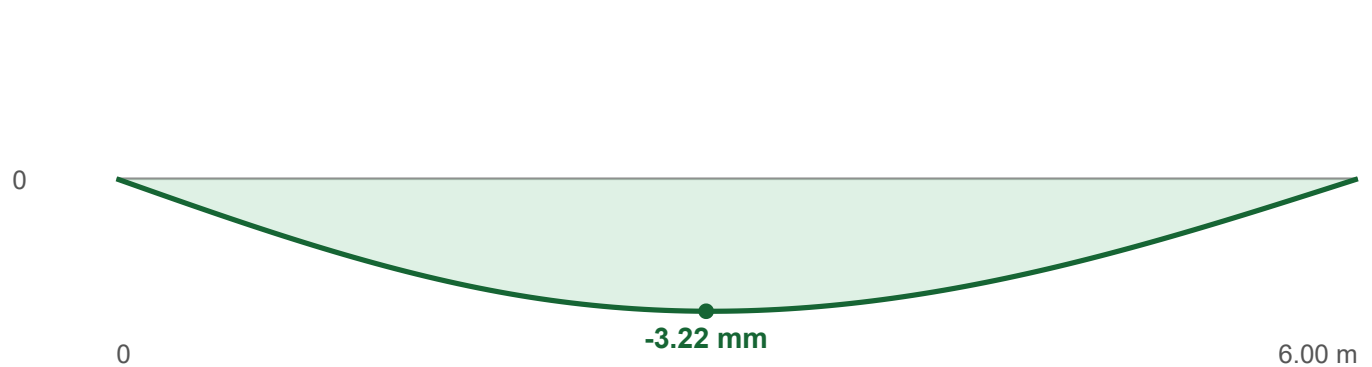
SHEAR FORCE V (kN)



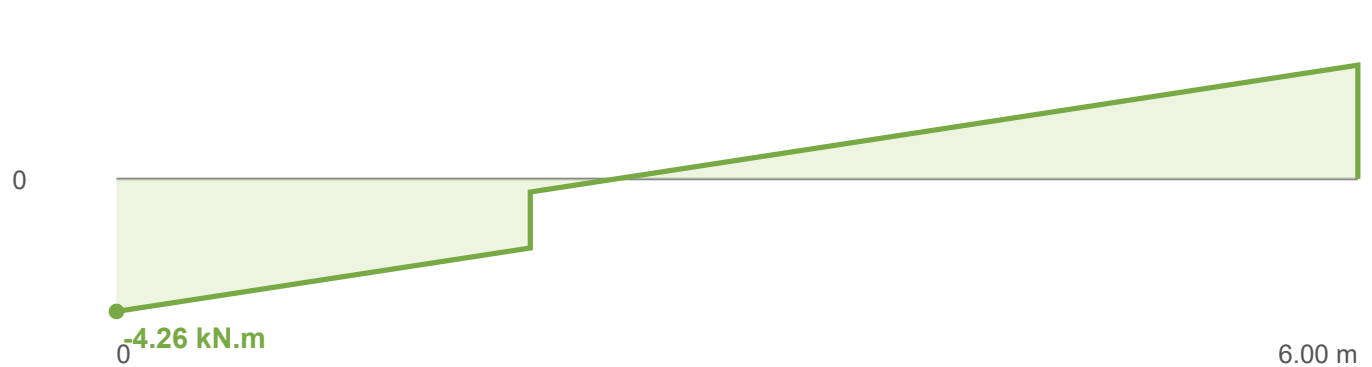
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038\text{e-}5 \text{ rad/mm}$, $\phi'' = 1.724\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.103\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$ P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
σ_x, τ , (6.1) at the 4 section points	$f_y / \gamma_{M0} = 275$		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

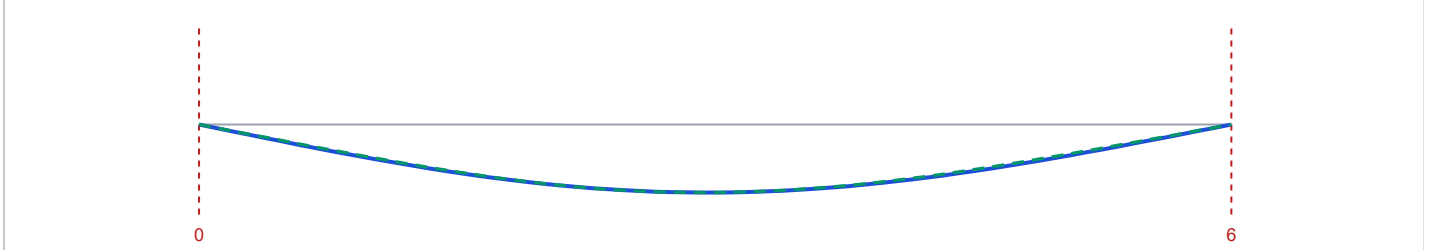
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.93 m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w ϕ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.753	1.025	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.365 m: 168.802/282.919 + 1.000x9.873/83.600 + 0.663x0.882x1.753x7.403/40.255	0.903	OK EN 1993-6 A
End torques T _t	St Venant part G I _T ϕ' at x = 0 / x = L; total T = G I _T ϕ' - E I _w $\phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902 \text{ kN.m}$, $V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.903	0.903

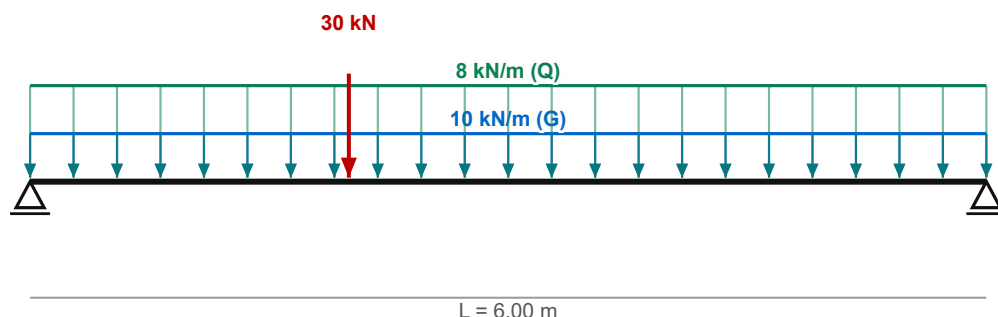
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.903

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

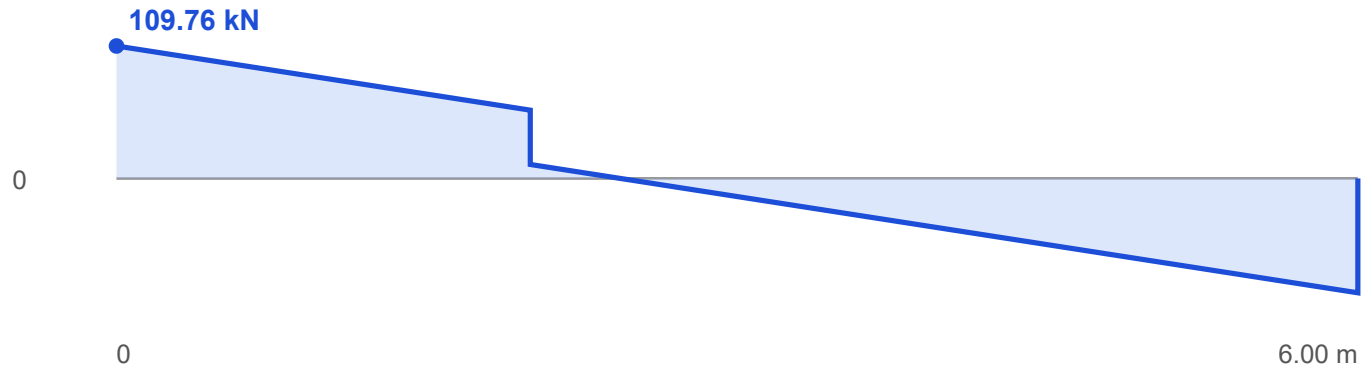
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at $e = +40$ mm (torque), $z_g = 0$

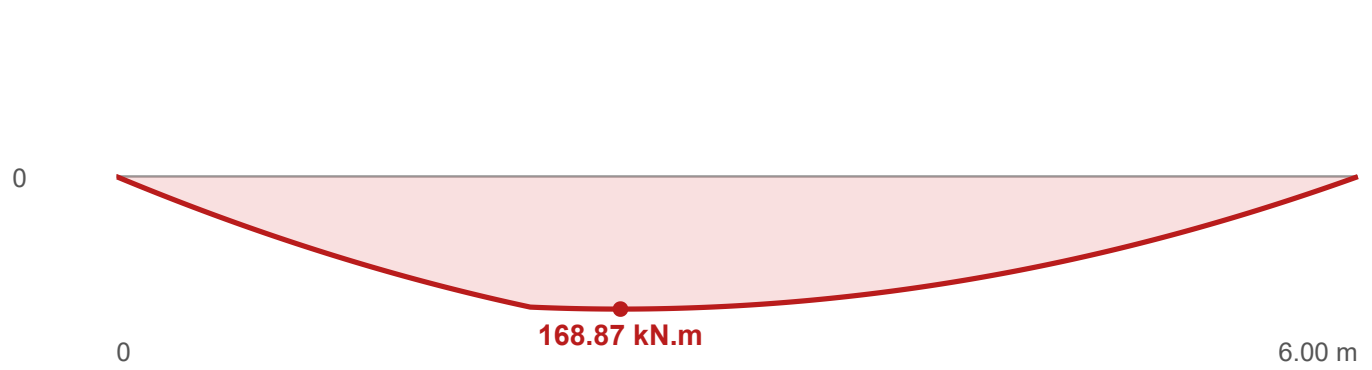
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)



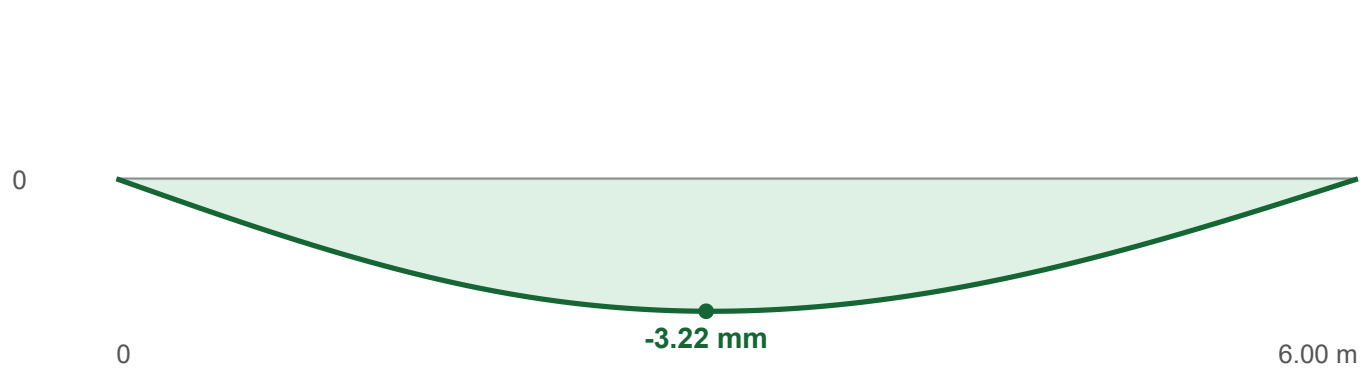
SHEAR FORCE V (kN)



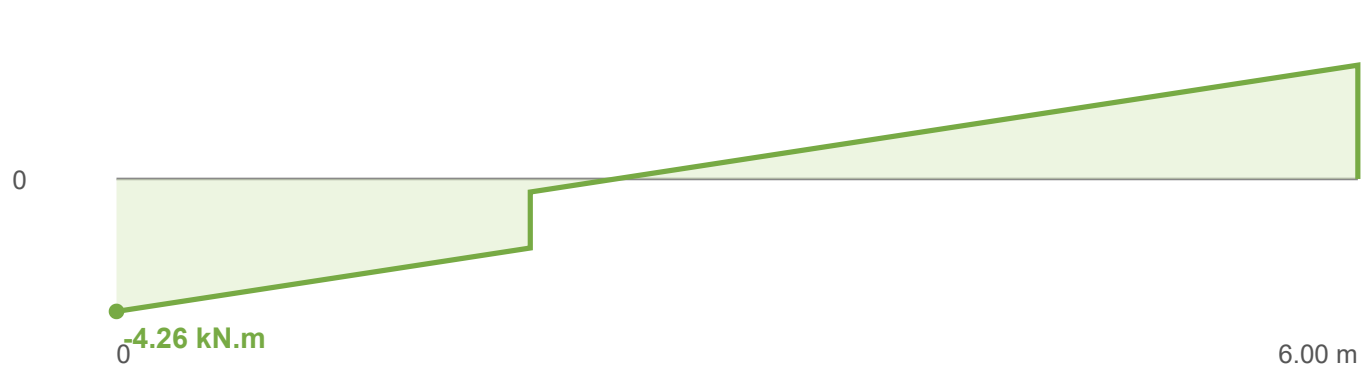
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038\text{e-}5 \text{ rad/mm}$, $\phi'' = 1.724\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.103\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$ P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
σ_x, τ , (6.1) at the 4 section points	$f_y / \gamma_{M0} = 275$		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	10.129 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 40.255, M _{pl,z} = 83.600	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.736	1.015	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.365 m: 168.802/285.039 + 1.000x9.873/83.600 + 0.663x0.882x1.736x7.403/40.255	0.897	OK EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' – EI _w φ''' = 4.260 / 3.660 kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 1.902 kN.m, V = 109.758 kN	27.22 N/mm ²	P385
V _{pl,T,Rd} = √(1 – T _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 27.22 N/mm ² ; V _{pl,Rd} = 756.260; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.929	702.492 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +230$ mm (D/2, destabilising). $e = 0$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

G SW 0.8044 kN/m 0-6 m (automatic)

ULS C2: ULS: 1.35G + 1.50Q [$y_{G, inf} = 0.9$, EQU set A]

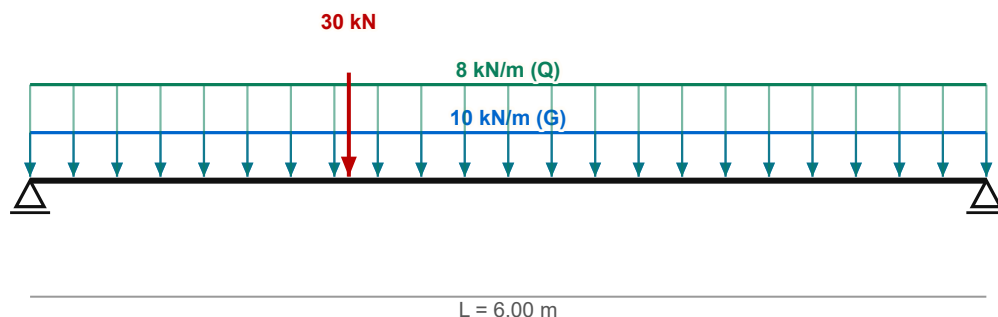
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

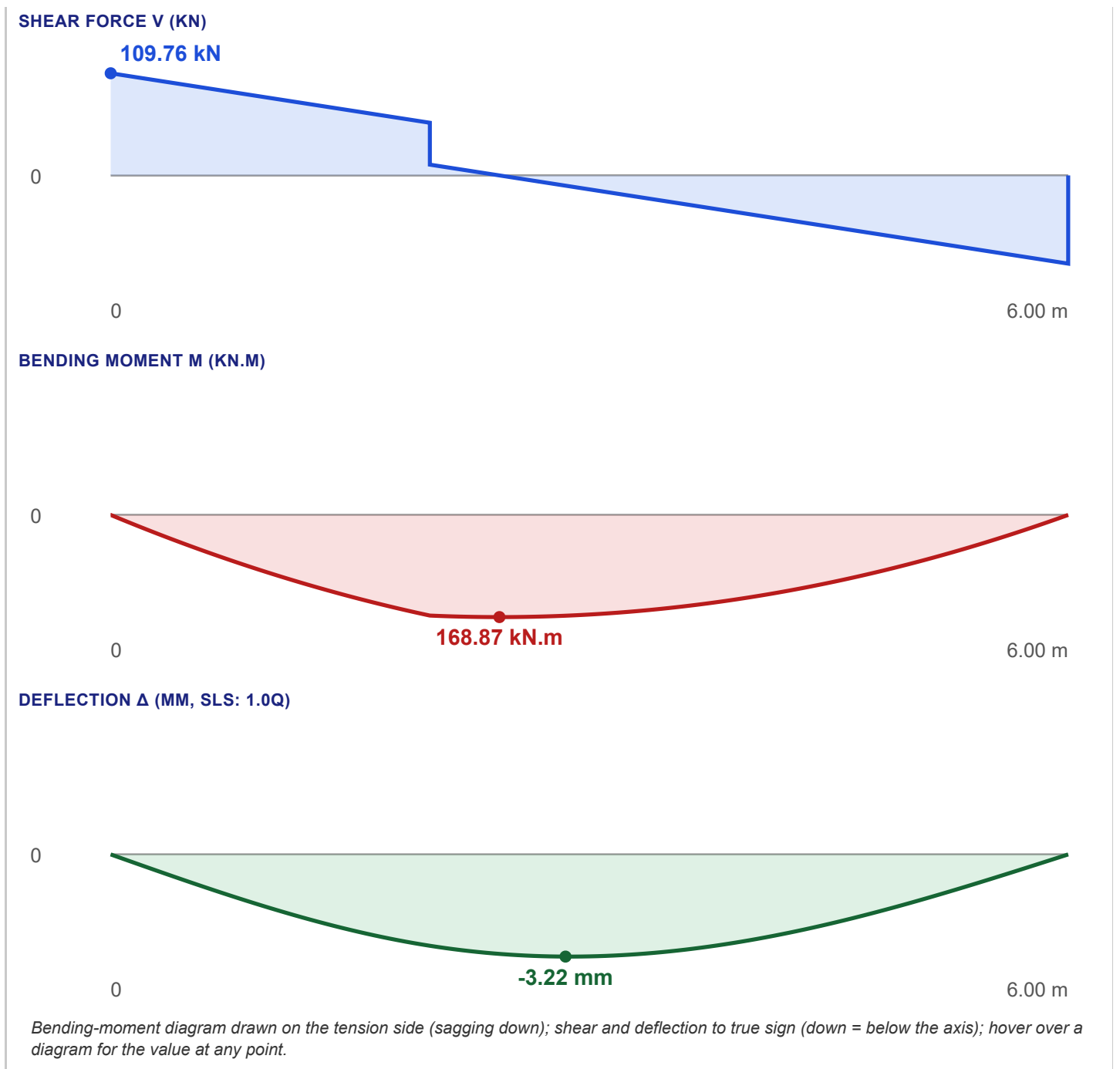
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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LOADING





Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / 1$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.179 \quad \text{OK 7.2} \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

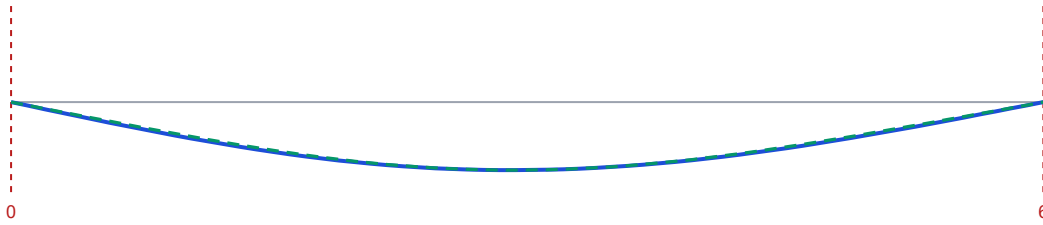
Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = M_{cr}/M_{cr,uniform} & \quad 393.107 / 342.775 \text{ kN.m (shape only: } z_g = z_j = 0) \quad 1.147 \quad k_c \text{ only} \\ C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) \text{ (standard, comparison)} & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \text{ — general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 \text{ kN.m} \quad 1.162 \quad \text{Serna} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = \text{portion between restraints} & \quad \text{End 1 v, } \varphi = 0; \text{ End 2 v, } \varphi = 0 \quad 6 \text{ m} \quad \text{FE} \\ M_{cr} = \text{FE eigenvalue (n}_{Elem}, \text{ mesh error)} & \quad 64 \text{ elements, } 0.000 \% ; z_g = 230, 0 \text{ mm; } 1 \text{ combination(s), } 1 \text{ solve(s) + } 2 \text{ cached} \quad 283.858 \text{ kN.m} \quad \text{converged} \\ \lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}} & \quad \sqrt{1830.0 \times 275 / 283.858} \quad 1.332 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 1.332, 1.393, 0.49 \quad 0.460 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.460, 1.332, 0.934, 0.986 \quad 0.467 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.467 \times 1830.0 \times 275 \leq 503.250 = 234.796 \text{ kN.m} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 234.796 \quad 0.719 \quad \text{OK} \end{aligned}$$

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 3.22 mm OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.336	0.719	0.193	0.145	0.251	0.257	0.719

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.719$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L3t · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **NOT VERIFIED** · governing LTB $M_{Ed}/M_{b,Rd} = 0.575$
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +230$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (NOT VERIFIED)
Member UB-L3t UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

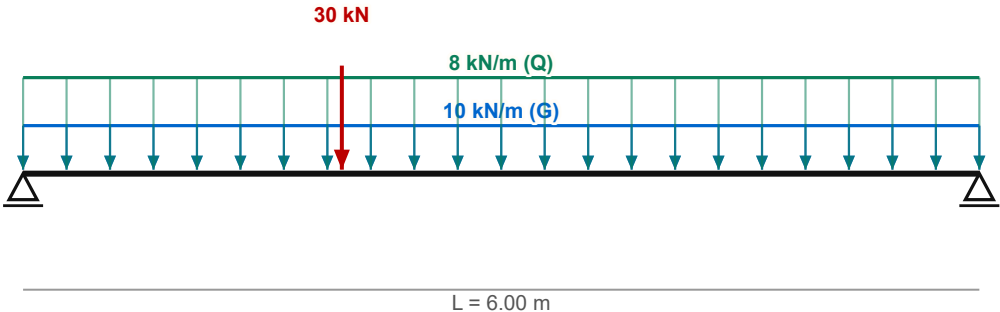
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 0$ mm $z_g = 230$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 0$ mm $z_g = 230$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 230$ mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

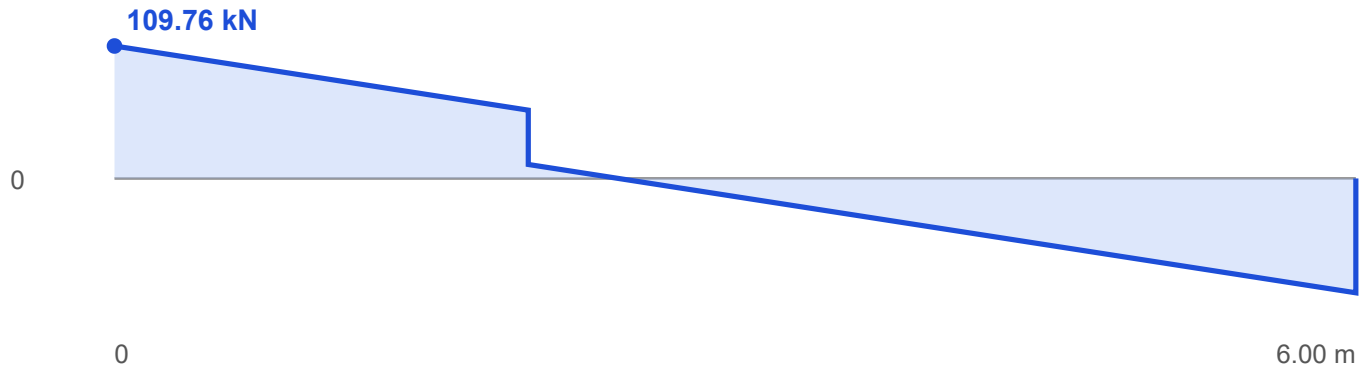
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

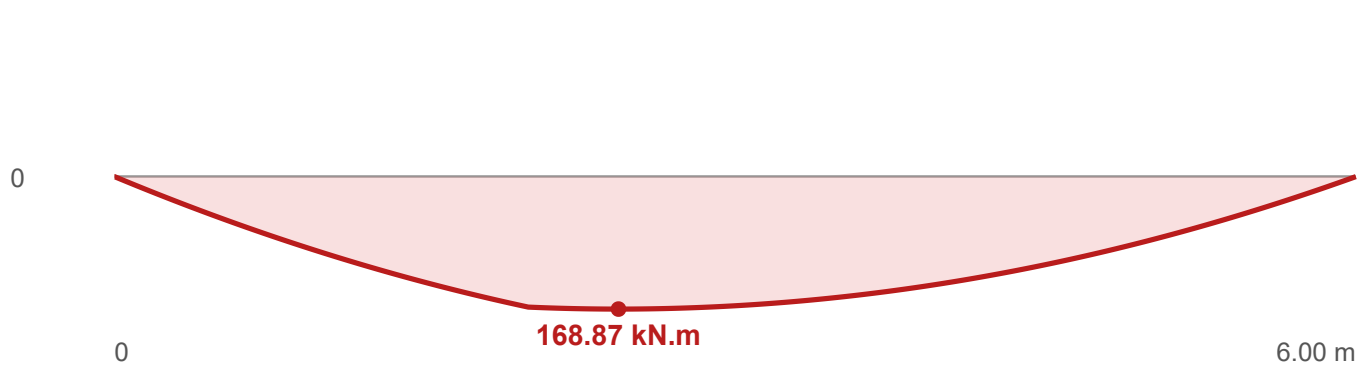
LOADING



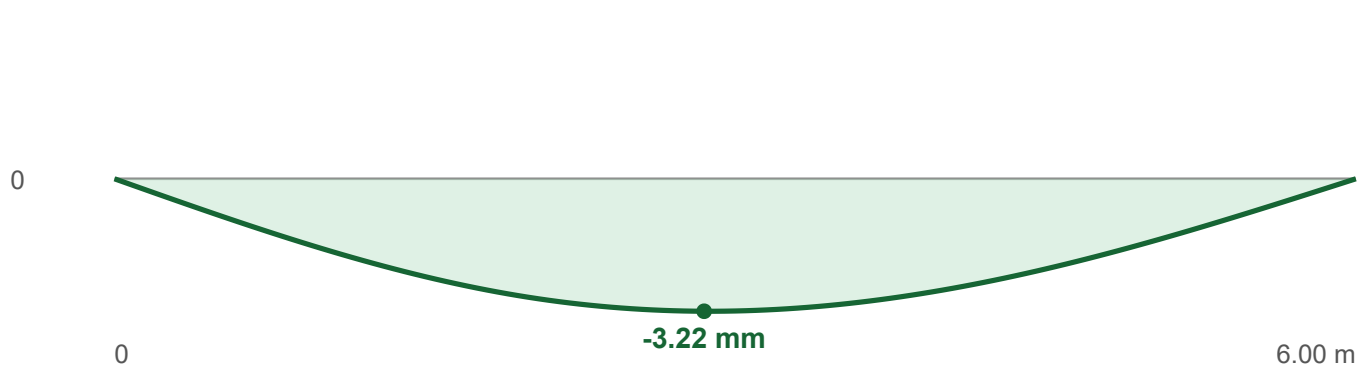
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
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$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	398.376 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_y f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
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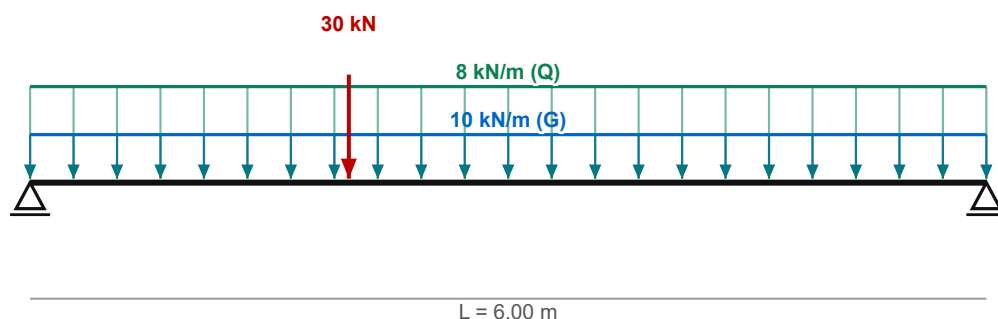
MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

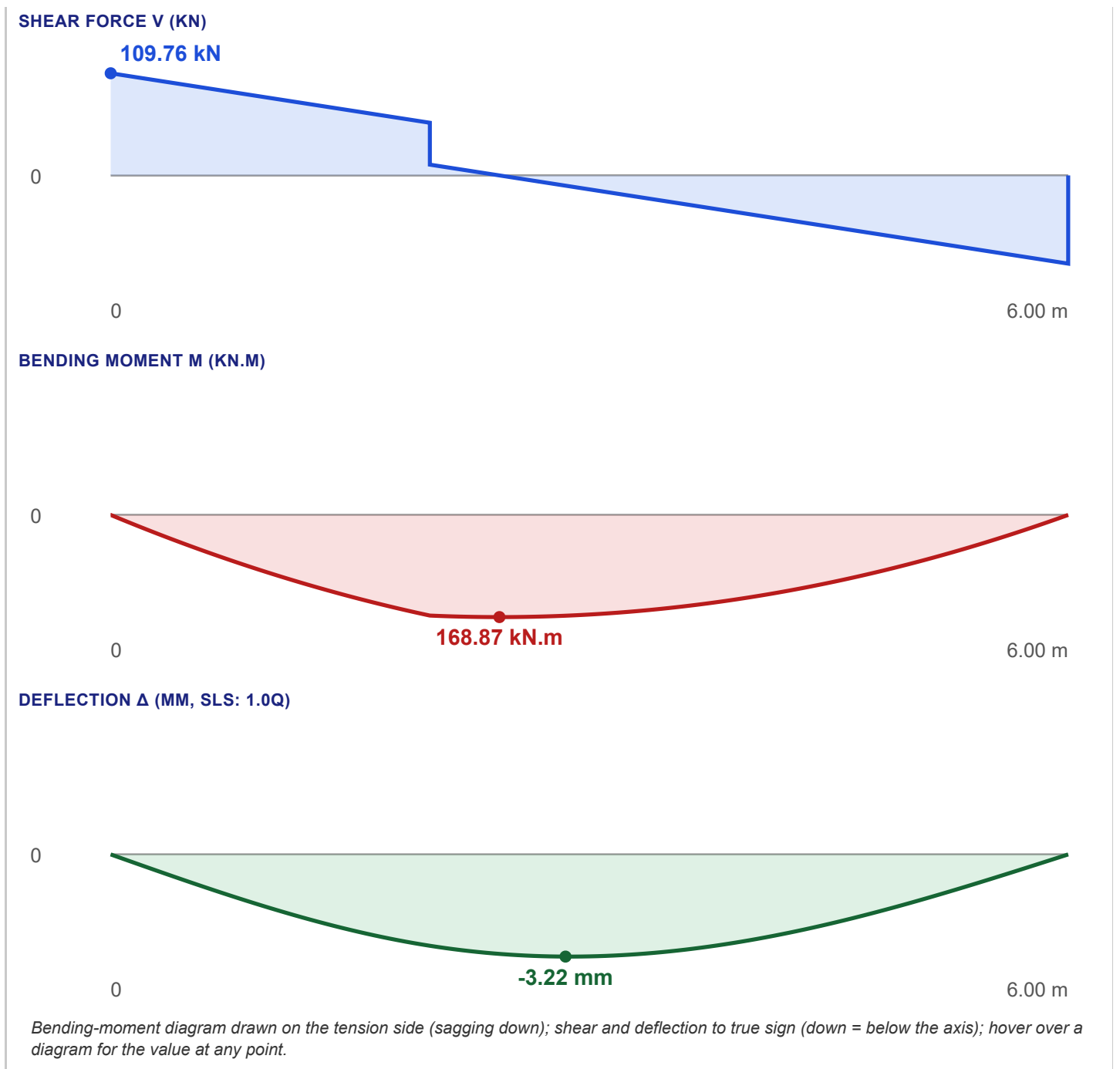
NOT VERIFIED — governing LTB $M_{Ed}/M_{b,Rd} = 0.575$

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.





Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / 1$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.179 \quad \text{OK 7.2} \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

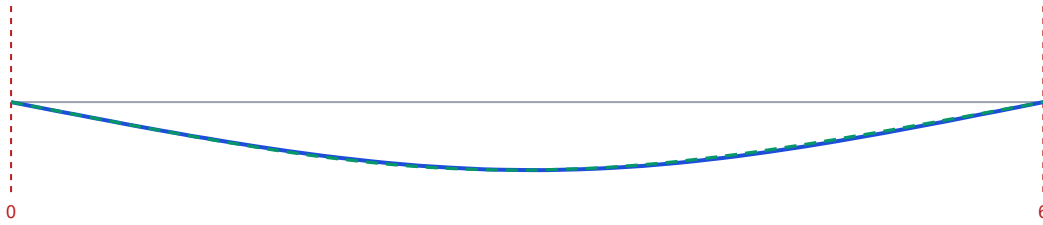
Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = M_{cr}/M_{cr,uniform} & \quad 393.107 / 342.775 \text{ kN.m (shape only: } z_g = z_j = 0) \quad 1.147 \quad k_c \text{ only} \\ C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) \text{ (standard, comparison)} & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \text{ — general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 \text{ kN.m} \quad 1.162 \quad \text{Serna} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = \text{portion between restraints} & \quad \text{End 1 v, } \varphi = 0; \text{ End 2 v, } \varphi = 0 \quad 6 \text{ m} \quad \text{FE} \\ M_{cr} = \text{FE eigenvalue (n}_{Elem}, \text{ mesh error)} & \quad 64 \text{ elements, } 0.000 \% ; z_g = -230, 0 \text{ mm; } 1 \text{ combination(s), } 1 \text{ solve(s) + } 2 \text{ cached} \quad 542.681 \text{ kN.m} \quad \text{converged} \\ \lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}} & \quad \sqrt{1830.0 \times 275 / 542.681} \quad 0.963 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 0.963, 0.986, 0.49 \quad 0.662 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.662, 0.963, 0.934, 0.969 \quad 0.683 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.683 \times 1830.0 \times 275 \leq 503.250 = \quad 343.812 \text{ kN.m} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 343.812 \quad 0.491 \quad \text{OK} \end{aligned}$$

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 3.02$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

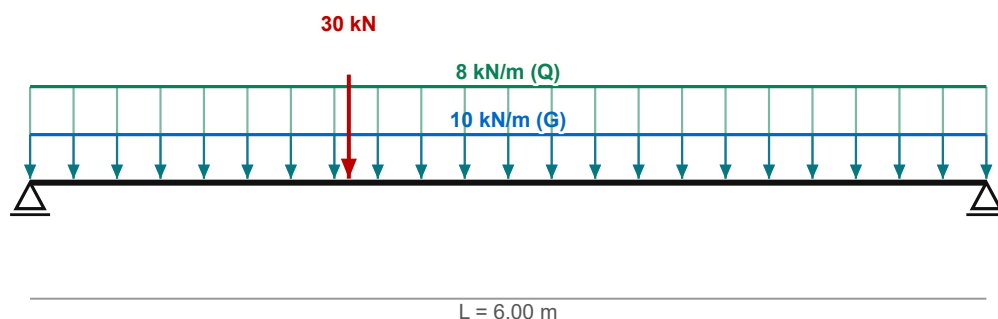
In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m 3.22 mm OK

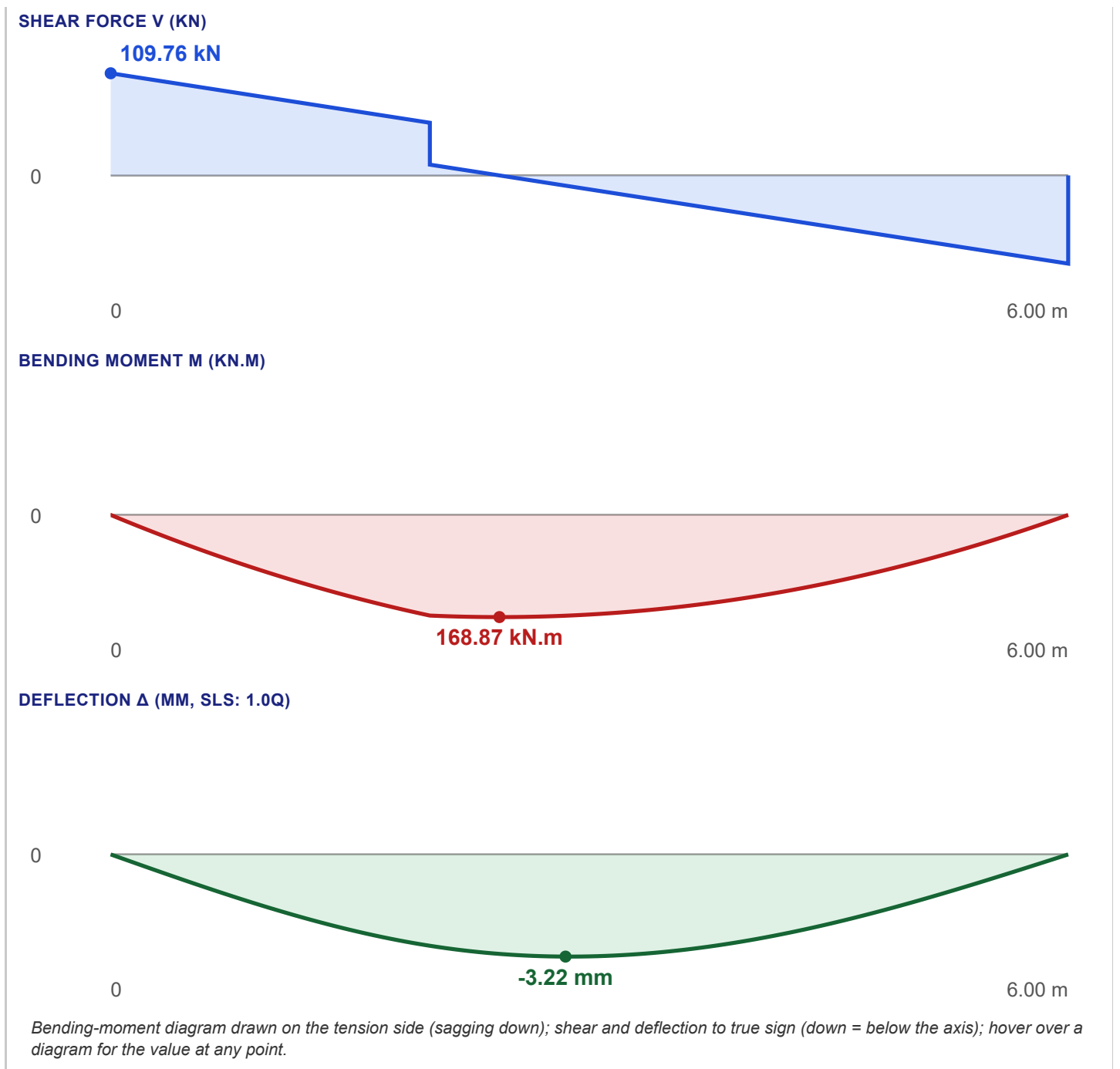
MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.336	0.491	0.193	0.145	0.251	0.257	0.491

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.491$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.





Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / } 437.340 = 0.251 \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: 1.35G + 1.5Q)} \quad 0.179 \quad \text{OK 7.2} \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \quad 1.162 \quad \text{Serna} \\ C_1 \text{ basis} & \quad \text{general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{\max}=168.9 \text{ kN}\cdot\text{m} \quad \text{SN003a} \end{aligned}$$

Lateral Buckling Check M.b.Rd

$$\begin{aligned} L_e = 1 L & \quad 1 \times 6 = 6 \text{ m} \\ M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E) & \quad 1.162, 6.000, 1870, 69.2, 0.922, 210000; z_g = -230 \text{ mm (below the shear centre, stabilising) — not applied: } C_2 \text{ not published for this moment diagram; } M_{cr} \text{ is taken with the load at the shear centre (conservative)} \quad 398.376 \text{ kN}\cdot\text{m} \quad \text{SN003a} \\ \lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}} & \quad \sqrt{1830.0 \times 275 / 398.376} \quad 1.124 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 1.124, 1.151, 0.49 \quad 0.566 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.566, 1.124, 0.928, 0.971 \quad 0.583 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.583 \times 1830.0 \times 275 \leq 503.250 = 293.432 \text{ kN}\cdot\text{m} \\ \lambda_{LT} \text{ (P362 6.55 simplified) ; } M_{b,Rd} & \quad (1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364; \chi_{LT,mod} = 0.451; M_{y,Ed}/M_{b,Rd,simp} = 0.744 \quad 226.988 \text{ kN}\cdot\text{m} \quad \text{P362 6.55 comparison} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 293.432 \quad 0.575 \quad \text{OK} \end{aligned}$$

Deflection Check - Load Case 1 (SLS: 1.0Q)

$$\text{In-span } \delta \leq \text{Span}/360 \quad 3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m} \quad 3.22 \text{ mm} \quad \text{OK}$$

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

PASS — governing LTB M_Ed/M_b,Rd = 0.575

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at $e = +40$ mm and $z_g = +230$ mm (top flange)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

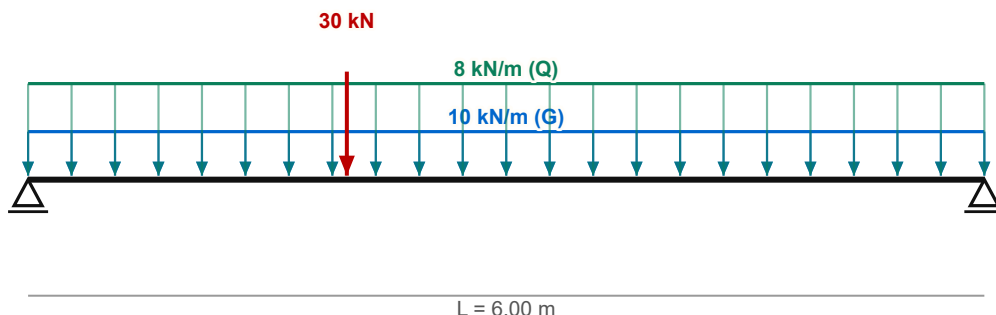
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

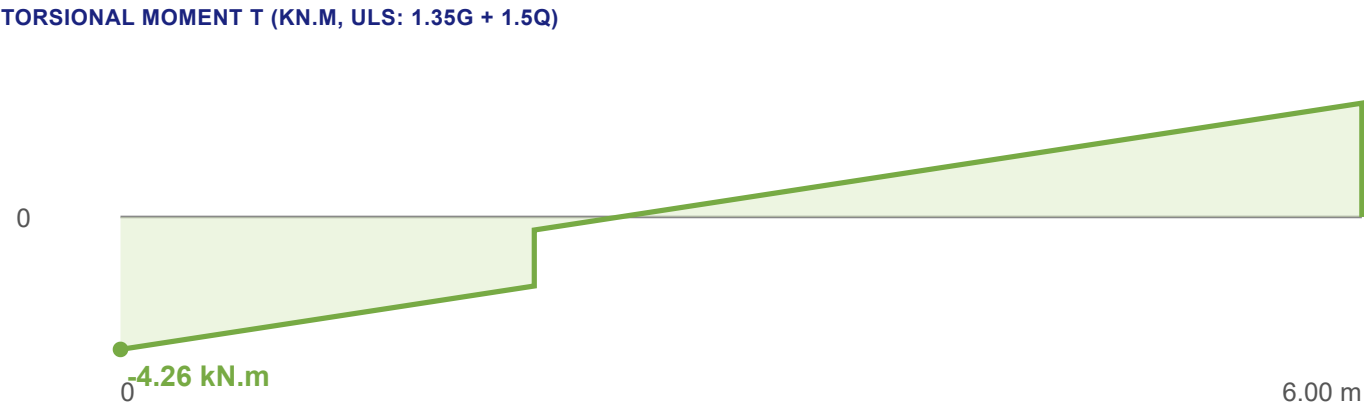
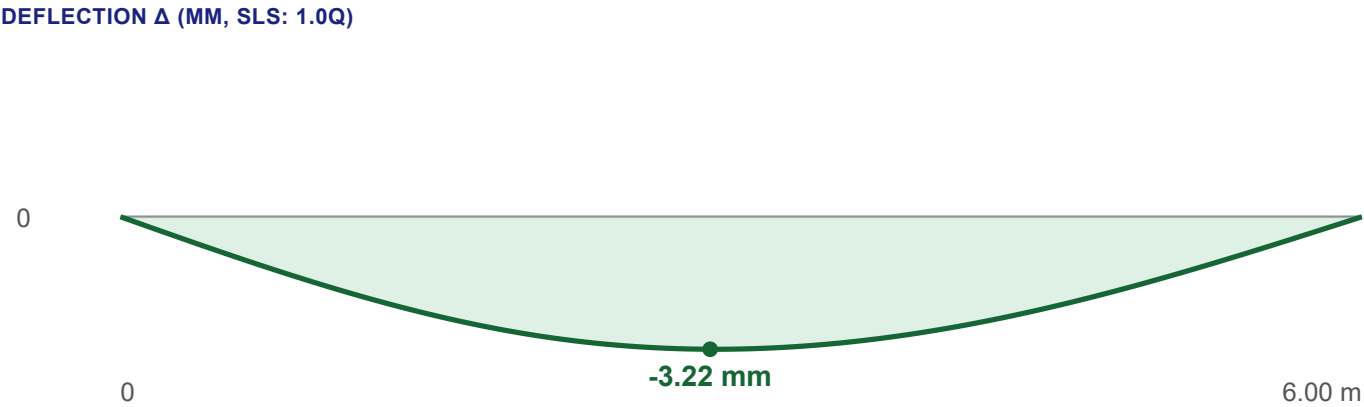
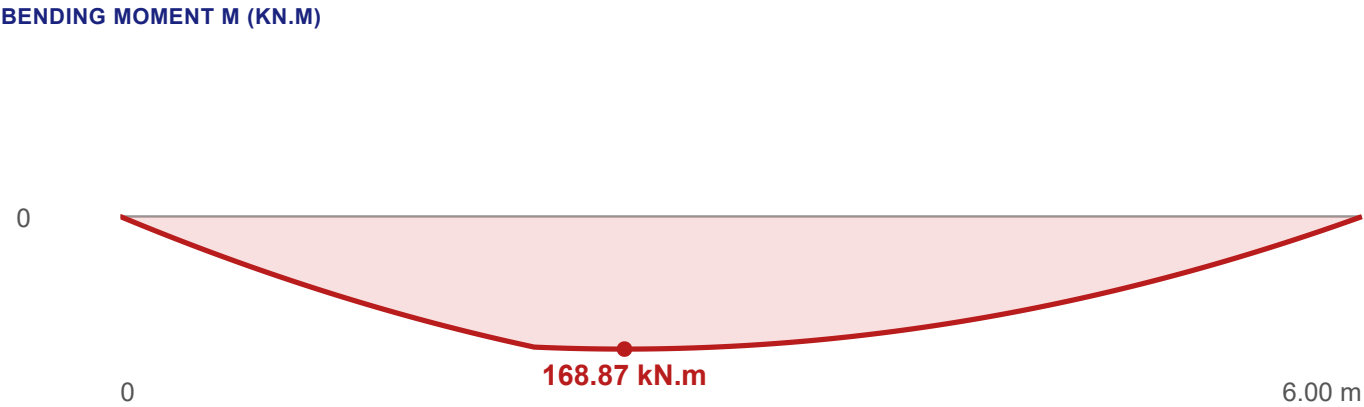
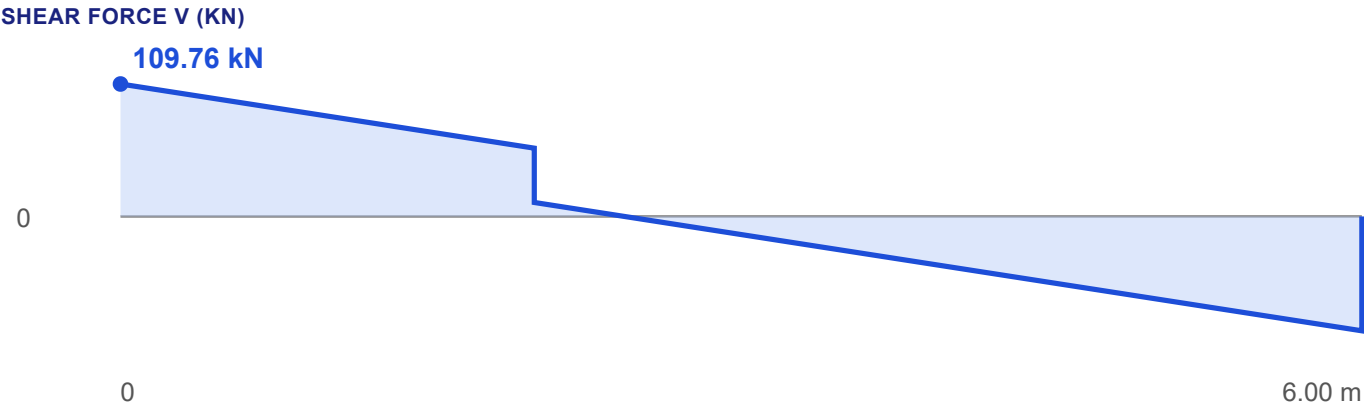
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.26	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-3.66	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

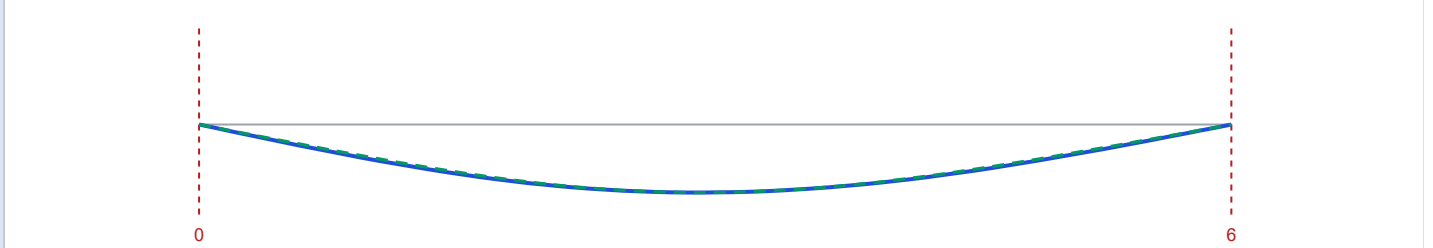
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 230$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	283.858 kN.m	converged
$\lambda_{LT} = \sqrt{W_{fy}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 283.858}$	1.332	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.332, 1.393, 0.49	0.460	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.460, 1.332, 0.934, 0.986	0.467	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.467 \times 1830.0 \times 275 \leq 503.250 =$	234.796 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 234.796	0.719	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = $E I_w \phi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
M _{w,Ed} = $E I_w \phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.883 \times 2.469$	1.446	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.29$ m: $168.587/231.414 + 1.000 \times 9.749/83.600 + 0.663 \times 0.883 \times 2.469 \times 7.472/40.255$	1.113	Warning EN 1993-6 A
End torques T _t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total T = $GI_T \phi' - E I_w \phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902 \text{ kN.m}$, $V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))}.V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm}$ @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

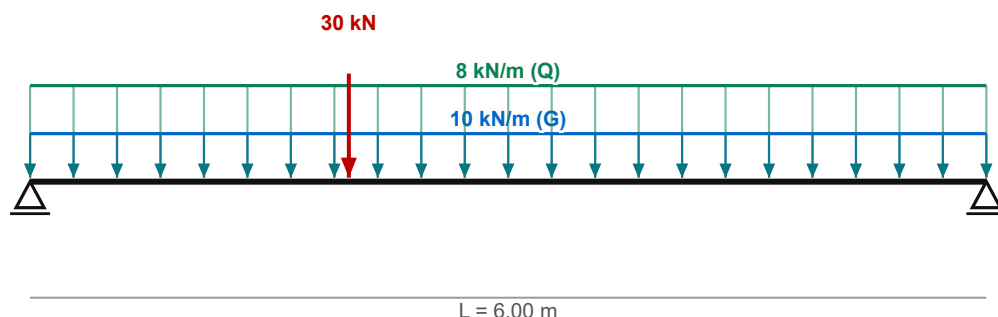
MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.719	0.193	0.145	0.251	0.257	0.415	0.710	0.156	1.113	1.113

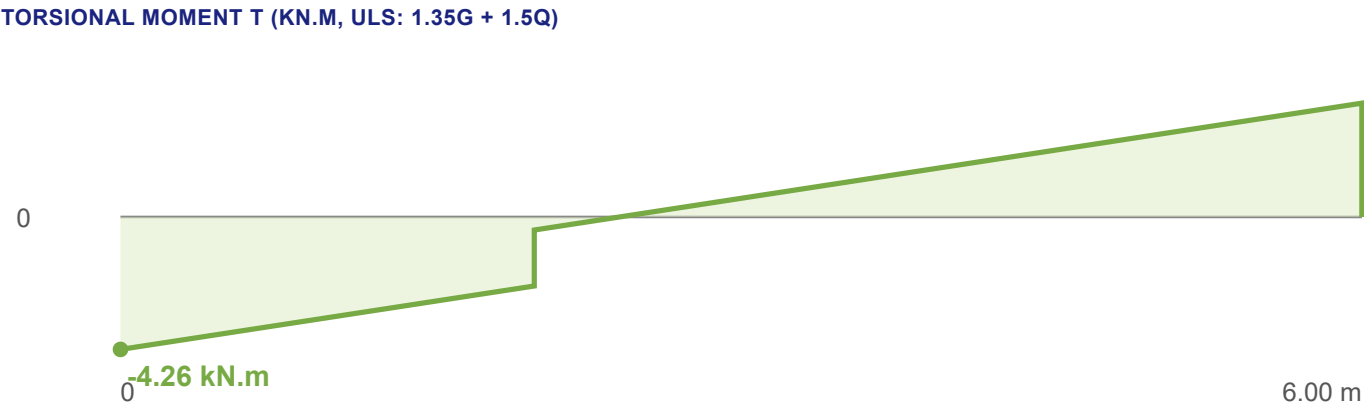
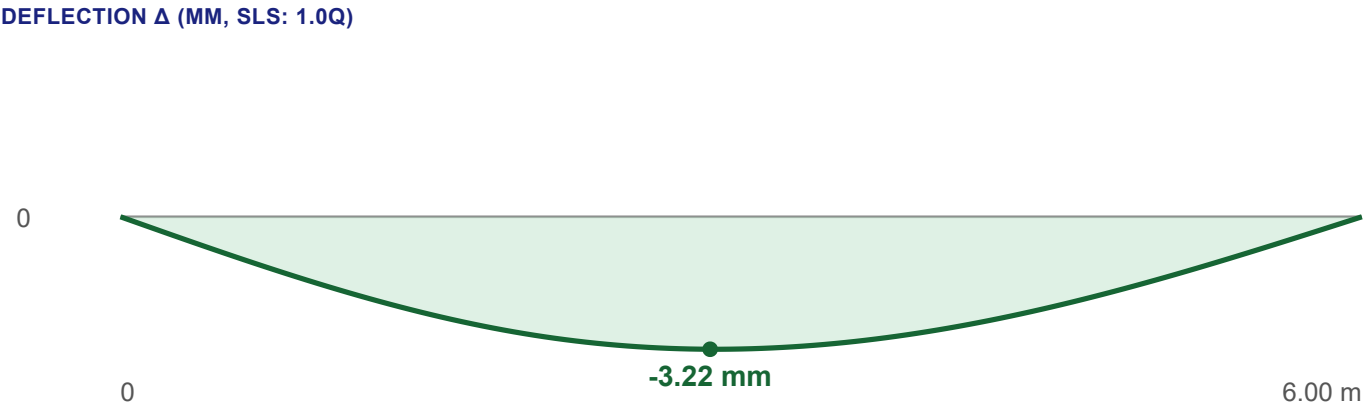
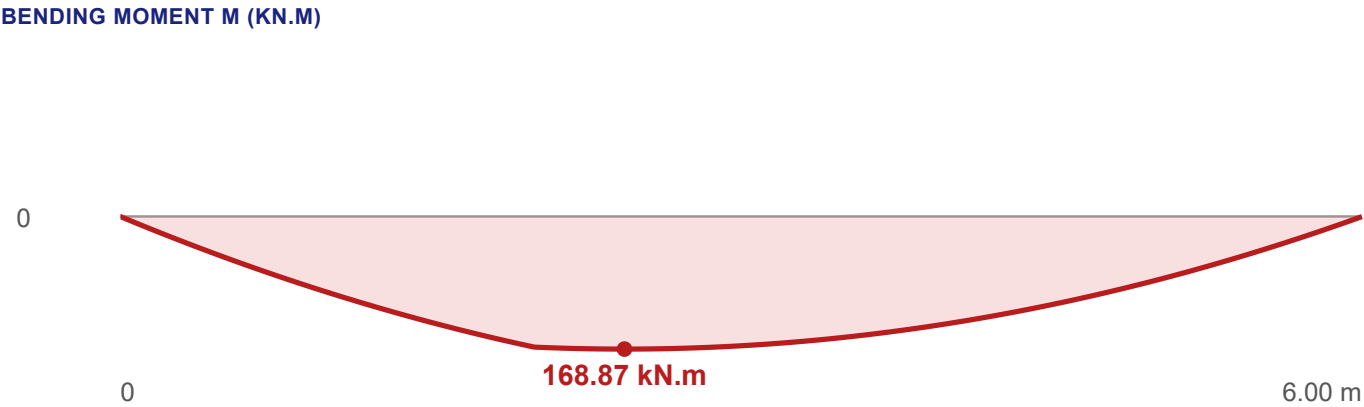
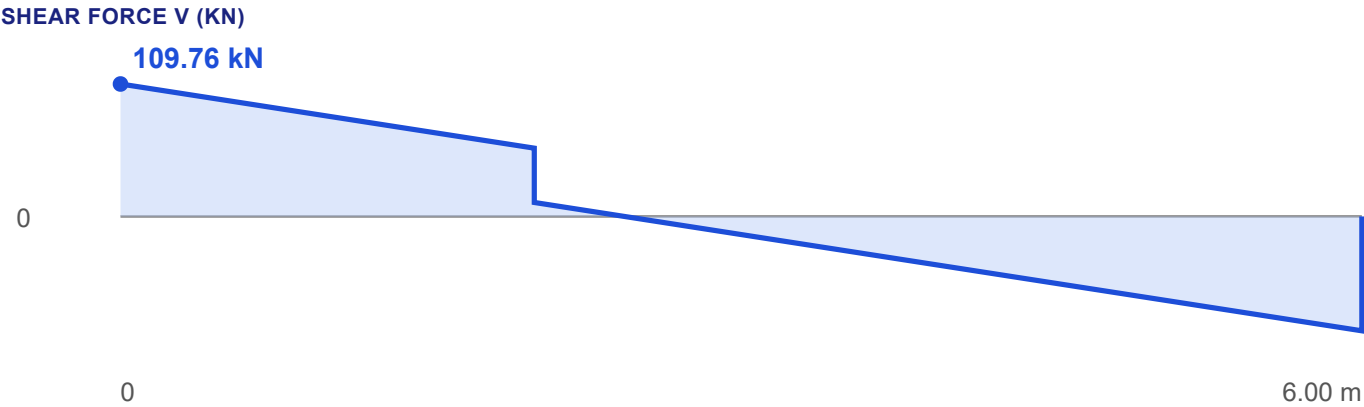
FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.113

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at $e = +40$ mm and $z_g = +230$ mm (top flange)





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833 ; 0.833 \times 192.79$	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)

F_{Ed}/F_{Rd}	$F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000)$ $= 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	X_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9 \text{ kN}\cdot\text{m}$		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	398.376 kN.m	BLOCKED
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
$\bar{\lambda}_{LT}$ (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\bar{\lambda}_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
$\varphi_{\max} \text{ (ULS)}$	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi'' / (h - t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34 \text{ m}$: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.882 \times 1.736$	1.015	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.365 \text{ m}$: 168.802/285.039 + $1.000 \times 9.873/83.600 +$ $0.663 \times 0.882 \times 1.736 \times 7.403/40.255$	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660 \text{ kN}\cdot\text{m}$	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $\tau_t = 1.902 \text{ kN.m, } V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

NOT VERIFIED — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L5 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **FAIL** · governing Member buckling z-z (Eq 6.62) = 1.193
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UB 457 x 191 x 82, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at $e = +40$ mm, $z_g = +230$ mm

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member UB-L5 UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

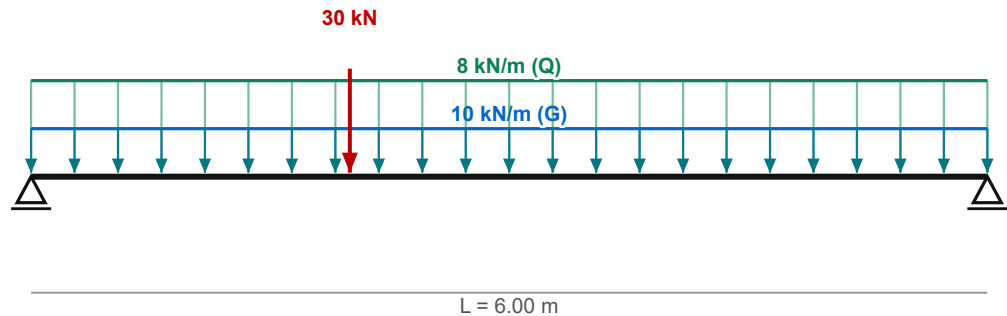
Member Loading and Member Forces

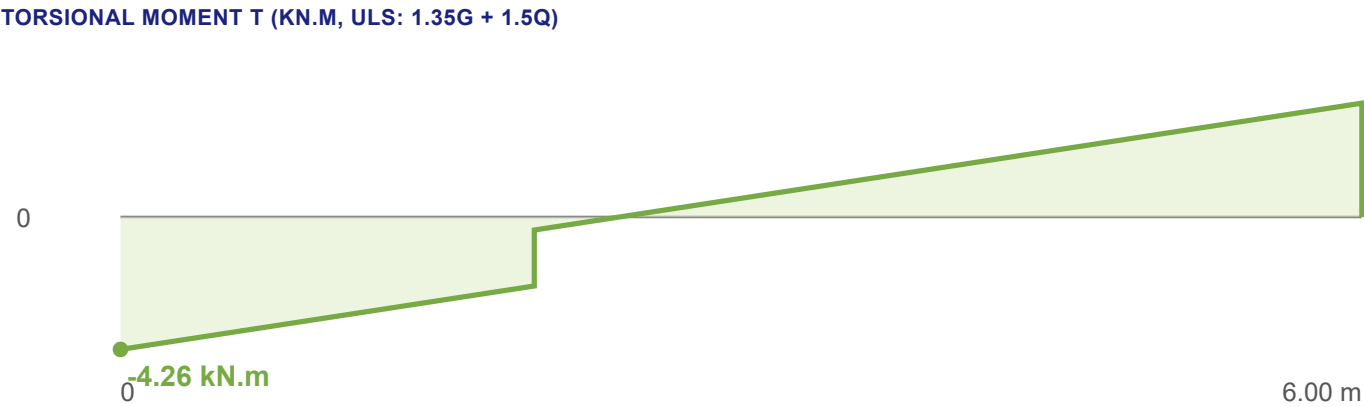
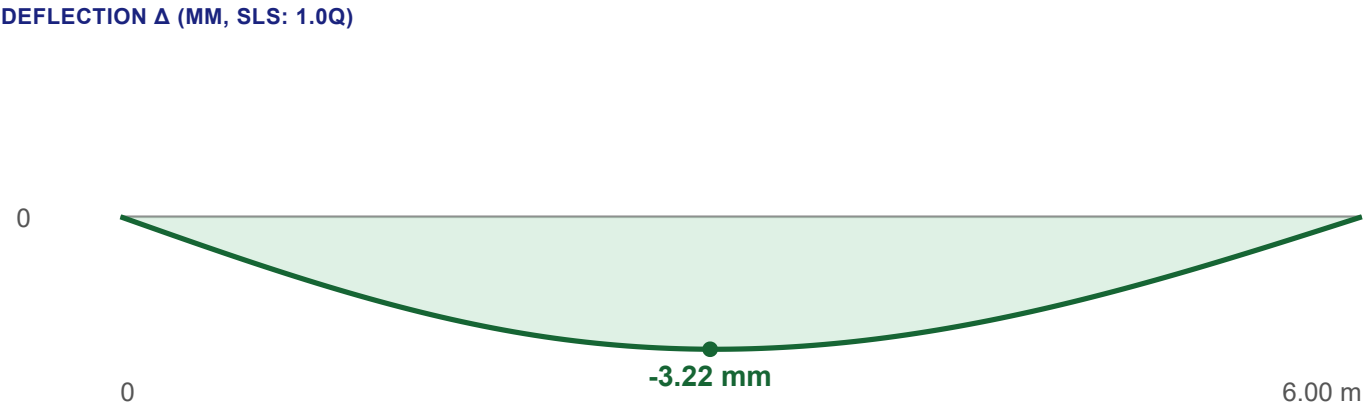
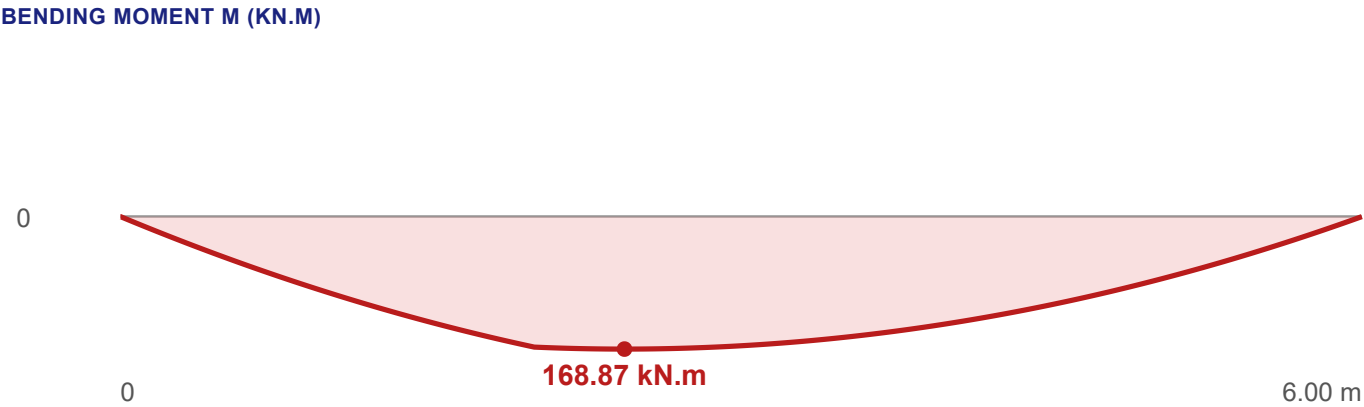
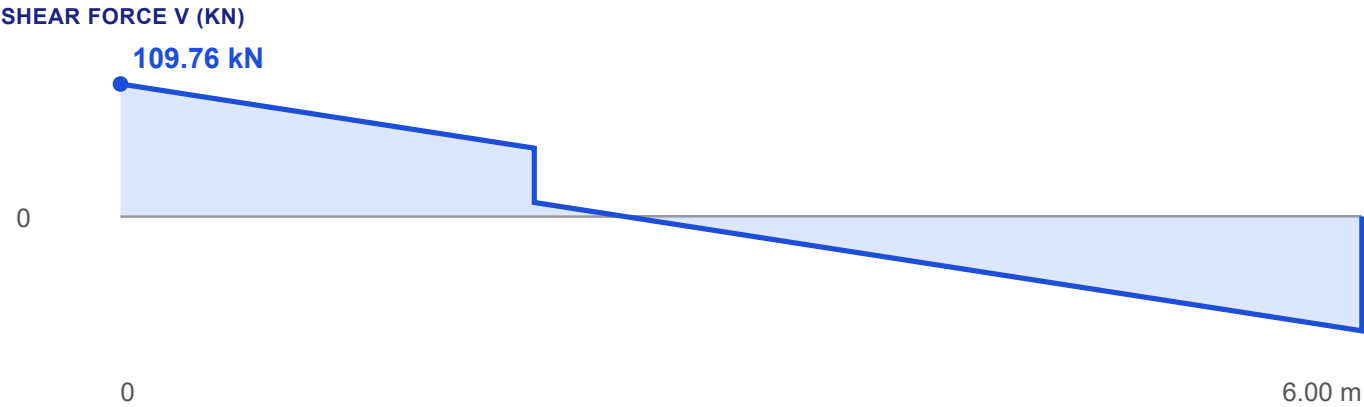
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 40$ mm $z_g = 230$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 40$ mm $z_g = 230$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 230$ mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.26	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	150.00C	-3.66	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]			
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 168.87, 0.00			Class 1
	(Axial: Slender web)			
Web classified for	bending + compression: $\alpha = 0.568, \psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε		Class 1	Table 5.2

$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl.y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl.y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72\epsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl.y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c.y,Rd} = f_y W_{pl.y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c.y,Rd}$	$168.868 / 503.250 =$	0.336	OK
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$104.00 \times 275 / 1$ (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} f_y / \gamma_{M0}$	$100.66 \times 275 / 1$ (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2860.000 =$ (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c,Rd}$	$150.000 / 2768.024 =$	0.054	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	$1830.0, 47.632, 0.052$	1830.0 cm ³	
$M_{N.y,Rd} = W_{pl.N.y} f_y / \gamma_{M0}$	$1830.0 \times 275 / 1$	503.250 kN.m	6.2.9.1(4)
$(M_{y,Ed} / M_{N.y,Rd})^\alpha + (M_{z,Ed} / M_{N.z,Rd})^\beta$	$(168.868 / 503.250)^2 + (0)^1 =$	0.113	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.266 m, P1 flange tip	@ $x = 2.266 \text{ m}$ (ULS: $1.35G + 1.5Q$): $M_y = 168.483 \text{ kN.m}$, $V = 4.526 \text{ kN}$, $\phi = 0.05759 \text{ rad}$, $\phi' = 9.950\text{e-}6 \text{ rad/mm}$, $\phi'' = 1.719\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.110\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.702 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el.y} + M_z / W_{el.z} + E W_{n0} \phi'' = 14.42 + 104.65 + 49.50 + 76.52 = 245.09 \text{ N/mm}^2$; $\tau = G t_f \phi' = 12.90 \text{ N/mm}^2$; $(245.09 / 275)^2 + 3(12.90 / 275)^2 =$	0.801	OK 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 104.65 + 49.50 + 76.52 = 245.09$, $\tau = 12.90 \rightarrow 0.801$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 100.82 + 2.57 = 117.81$, $\tau = 0.26 + 12.90 + 0.45 = 13.60 \rightarrow 0.191$; P3 web-flange junction, web: $\sigma_x = 14.42 + 97.18 + 2.57 = 114.18$, $\tau = 0.84 + 7.98 = 8.82 \rightarrow 0.175$; P4 web mid-depth: $\sigma_x = 14.42 + 2.57 = 16.99$, $\tau = 1.12 + 7.98 = 9.10 \rightarrow 0.007$ ($y_w = t_w / 2 = 4.95 \text{ mm}$; $S_f / 2 =$ half flange one side of the web)	$f_y / \gamma_{M0} = 275$	
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00 / 16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0		6.5(1)
$a =$ distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)
Governing station $x = 0 \text{ m}$: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]			
s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)

$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833; 0.833 \times 192.79$	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) } / 437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 + N_{Ed} / N_{pl,Rd} = 150.000 / 2860.000 = 0.052) = 0.293 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	0.209	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.209	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.287	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.185	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} \cdot f_y; \lambda = \sqrt{(A_{eff} \cdot f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot \chi \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10 / 1 =$	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot \chi \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z; \lambda_T; N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

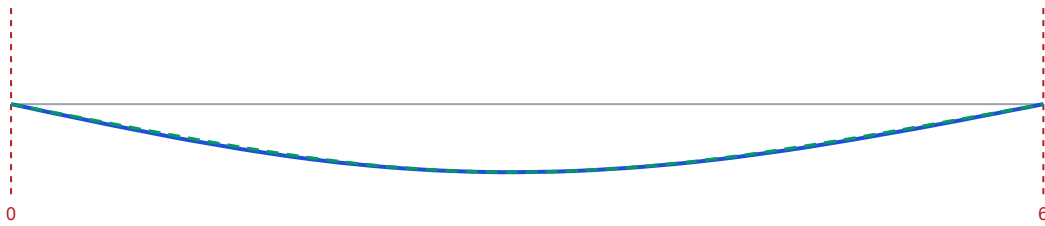
$C_1 = M_{cr} / M_{cr,uniform}$	$393.107 / 342.775 \text{ kN.m (shape only: } z_g = z_j = 0)$	1.147	k _c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	$0.0, 0.0, 164.6, 1.000, 300.000$ — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 \text{ kN.m}$	1.162	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 164.64, \psi = 0.859, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 10.129 twist $\phi \cdot M_y = 10.129$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 164.64, \psi = 0.859, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 230, 0 \text{ mm}$; 1 combination(s), 0 solve(s) + 3 cached	283.858 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 283.858}$	1.332	

$\chi_{LT} = F_n(\chi_{LT}, \Phi_{LT}, \alpha_{LT})$	1.332, 1.393, 0.49	0.460	Curve c
$\chi_{LT,mod} = F_n(\chi_{LT}, \chi_{LT}, k_c, f)$	0.460, 1.332, 0.934, 0.986	0.467	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.467 \times 1830.0 \times 275 \leq 503.250 =$	234.796 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 234.796	0.719	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed}/(\chi_y N_{Rk}/\gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed}/(\chi_z N_{Rk}/\gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} M_{y,Rk}/\gamma_{M1})$	$168.868 / 206.569$ ($M_{y,Rk} = 442.750$; $W_{el,y}/W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.817	OK
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(0.000 \text{ imposed} + 10.129 \text{ twist } \phi \cdot M_y = 10.129) / 53.900$	0.188	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+0.6\chi_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.362 \times 0.056\} \leq 1.000(1+0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\chi_z U_{N,z}\}$	$1.000\{1+0.6 \times 1.607 \times 0.177\} \leq 1.000(1+0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\chi_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\chi_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.056+1.012 \times 0.817+1.106 \times 0.188$	1.092	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.177+0.988 \times 0.817+1.106 \times 0.188$	1.193	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.883 \times 2.469$	1.446	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2.29 m: $168.587/231.414 + 1.000 \times 9.749/83.600 + 0.663 \times 0.883 \times 2.469 \times 7.472/40.255$	1.113	Warning EN 1993-6 A
Basis with N _{Ed} / M _{z,Ed}	No expression in EN 1993-1-1 or EN 1993-6 combines N _{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to M _{z,Ed} , and EN 1993-6 (A.1) with M _{z,Ed} = M _z + $\phi \cdot M_y$		EN 1993-1-1, EN 1993-6

superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only		$\frac{N_{Ed}}{(X_Z N_{Rk}/Y_{M1})} + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_{\alpha} M_w/M_{f,Rd} = 1.193$ (Eq 6.62 with $M_{z,tot} = 10.129$) + 0.280 (warping term of (A.1) @ $x = 2$ m: $k_w = 0.662$, $k_{zw} = 0.892$, $k_{\alpha} = 2.469$, $M_w = 7.736$ kN.m) =		1.473	information only																																	
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0$ / $x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260$ / 3.660 kN.m	1.902	1.724 kN.m																																			
Torsion Shear Design @ 0 m																																						
T_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, $V = 109.758$ kN	27.22 N/mm ²		P385																																		
$V_{pl.T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))}.V_{pl,Rd}$	@ $x = 0$ m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl.T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)																																			
$V_{Ed}/V_{pl.T,Rd}$	109.758 / 702.492	0.156		OK																																		
Deflection Check - Load Case 1 (SLS: 1.0Q)																																						
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm		OK																																		
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	0.0252 rad = $1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°		OK																																		
<table><tr><td>N_{Ed}/N_{c,Rd}</td><td>U_{Ny,z}</td><td>U_{My,z}</td><td>A_x+M_{6.6.1}</td><td>A_x+M_{6.6.2}</td><td>Deflection</td><td>V/V_{pl}</td><td>MA/M_c</td><td>L_{TB}</td><td>F/F_{Rd}</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>L_{TB}+T</td><td>Max</td></tr><tr><td>0.054</td><td>0.113</td><td>0.177</td><td>0.817</td><td>1.092</td><td>1.193</td><td>0.193</td><td>0.145</td><td>0.336</td><td>0.719</td><td>0.251</td><td>0.287</td><td>0.415</td><td>0.801</td><td>0.156</td><td>1.113</td><td>1.193</td></tr></table>						N _{Ed} /N _{c,Rd}	U _{Ny,z}	U _{My,z}	A _x +M _{6.6.1}	A _x +M _{6.6.2}	Deflection	V/V _{pl}	MA/M _c	L _{TB}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	L _{TB} +T	Max	0.054	0.113	0.177	0.817	1.092	1.193	0.193	0.145	0.336	0.719	0.251	0.287	0.415	0.801	0.156	1.113	1.193
N _{Ed} /N _{c,Rd}	U _{Ny,z}	U _{My,z}	A _x +M _{6.6.1}	A _x +M _{6.6.2}	Deflection	V/V _{pl}	MA/M _c	L _{TB}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	L _{TB} +T	Max																							
0.054	0.113	0.177	0.817	1.092	1.193	0.193	0.145	0.336	0.719	0.251	0.287	0.415	0.801	0.156	1.113	1.193																						

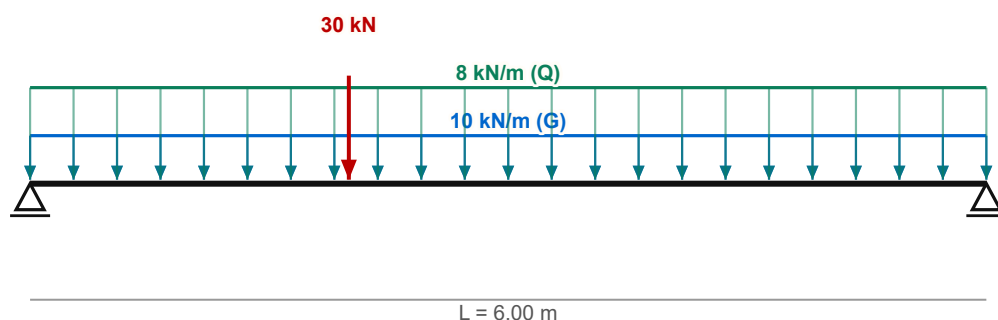
FAIL — governing Member buckling z-z (Eq 6.62) = 1.193

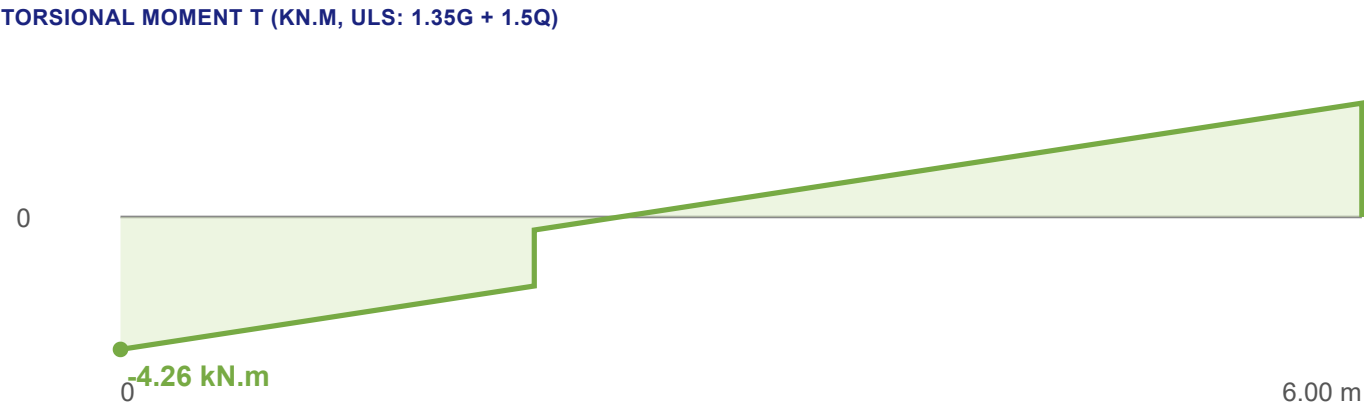
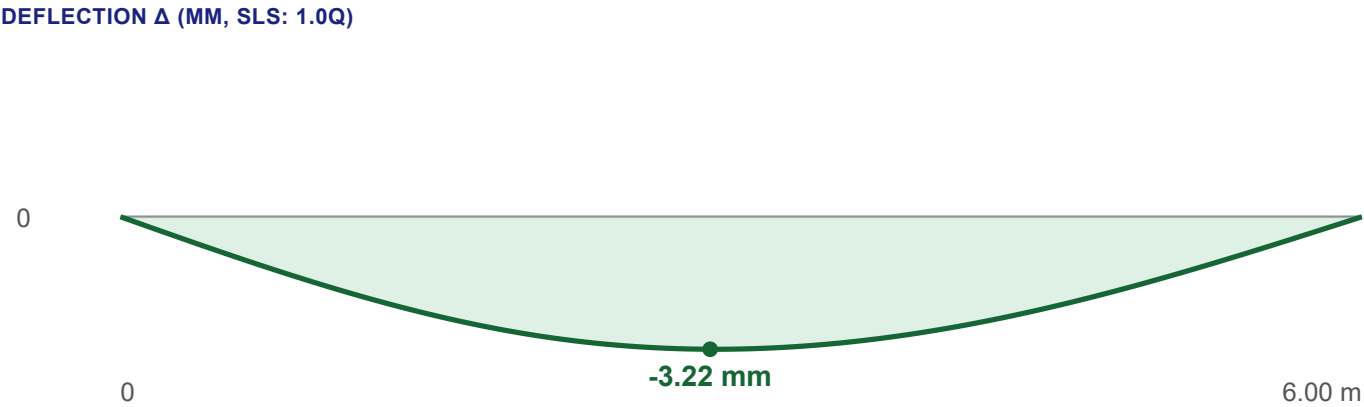
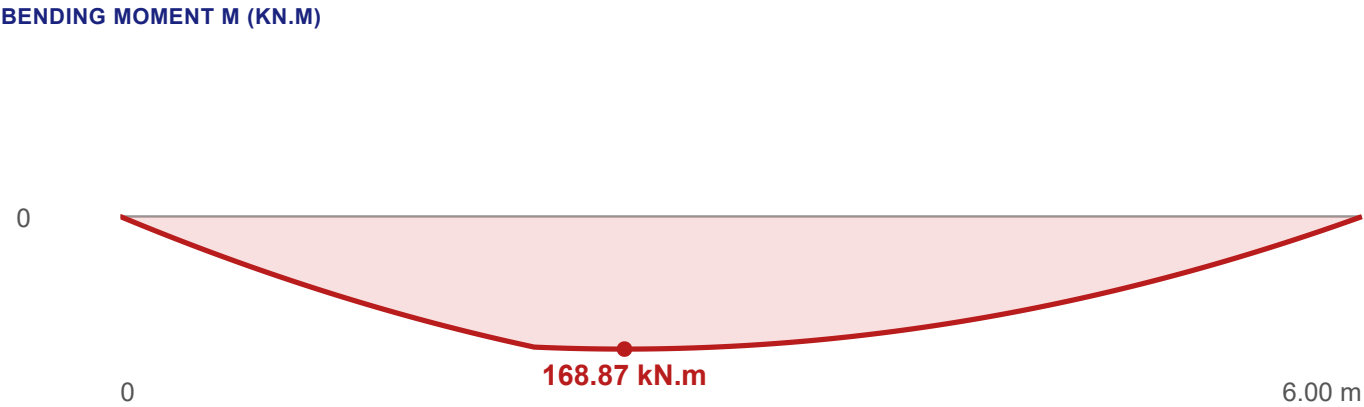
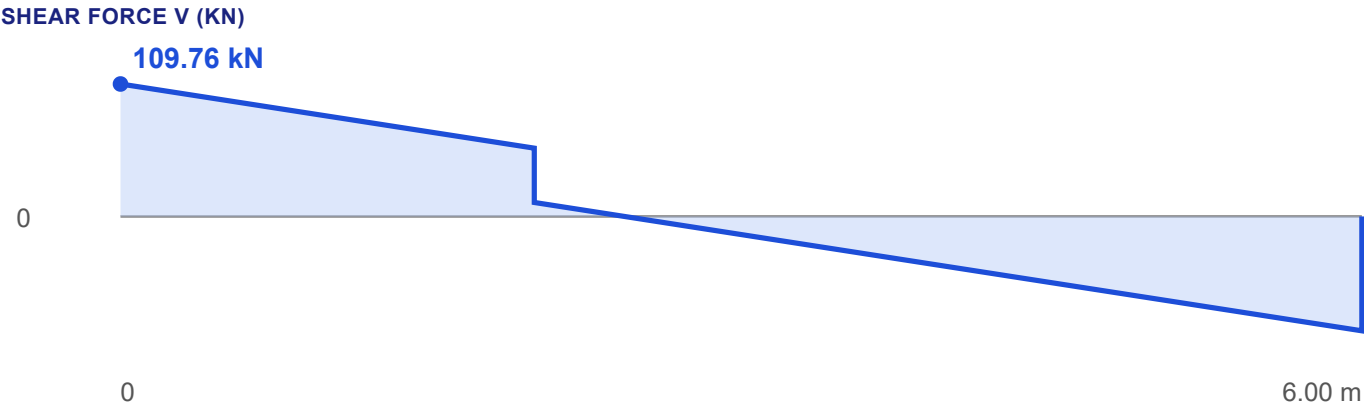
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_Z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed} / (X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot} / (M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_{\alpha} M_{w,Ed} / M_{f,Rd} = 1.193$ (Eq 6.62 with $M_{z,tot} = 10.13$ kN.m) + 0.280 (warping term of (A.1) at $x = 2.00$ m, $k_w = 0.662$, $k_{zw} = 0.892$, $k_{\alpha} = 2.469$) = 1.473. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UB 457 x 191 x 82, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z g = +230 mm





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]			
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 168.87, 0.00			Class 1
	(Axial: Slender web)			
Web classified for	bending + compression: $\alpha = 0.568, \psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε		Class 1	Table 5.2

$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\varepsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\varepsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72\varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$104.00 \times 275 / 1$ (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} f_y / \gamma_{M0}$	$100.66 \times 275 / 1$ (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2860.000 =$ (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c,Rd}$	$150.000 / 2768.024 =$	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$1830.0, 47.632, 0.052$	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$1830.0 \times 275 / 1$	503.250 kN.m	6.2.9.1(4)
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(168.868 / 503.250)^2 + (0)^1 =$	0.113	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.266 m, P1 flange tip	@ $x = 2.266 \text{ m}$ (ULS: $1.35G + 1.5Q$): $M_y = 168.483 \text{ kN.m}$, $V = 4.526 \text{ kN}$, $\phi = 0.05759 \text{ rad}$, $\phi' = 9.950\text{e-}6 \text{ rad/mm}$, $\phi'' = 1.719\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.110\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.702 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 14.42 + 104.65 + 49.50 + 76.52 = 245.09 \text{ N/mm}^2$; $\tau = G t_f \phi' = 12.90 \text{ N/mm}^2$; $(245.09 / 275)^2 + 3(12.90 / 275)^2 =$	0.801	OK 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 104.65 + 49.50 + 76.52 = 245.09$, $\tau = 12.90 \rightarrow 0.801$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 100.82 + 2.57 = 117.81$, $\tau = 0.26 + 12.90 + 0.45 = 13.60 \rightarrow 0.191$; P3 web-flange junction, web: $\sigma_x = 14.42 + 97.18 + 2.57 = 114.18$, $\tau = 0.84 + 7.98 = 8.82 \rightarrow 0.175$; P4 web mid-depth: $\sigma_x = 14.42 + 2.57 = 16.99$, $\tau = 1.12 + 7.98 = 9.10 \rightarrow 0.007$ ($y_w = t_w / 2 = 4.95 \text{ mm}$; $S_f / 2 =$ half flange one side of the web)	$f_y / \gamma_{M0} = 275$	
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.	NOT VERIFIED	
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30\varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00 / 16.00)^2 = 14.311$ if $\lambda_{\bar{t}} > 0.5$, else 0		6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1; L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/Y_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 + N_{Ed}/N_{pl,Rd} = 150.000/2860.000 = 0.052) = 0.293 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.209	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.209	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.287	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.185	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); $c =$ distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} \cdot f_y$; $\lambda = \sqrt{(A_{eff} \cdot f_y/N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y/N_{cr}}$	$\sqrt{100.66 \times 275/21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot X \cdot f_y / Y_{M1}$	$100.66 \times 0.963 \times 275/10/1 =$	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y/N_{crz}}$	$\sqrt{100.66 \times 275/1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot X \cdot f_y / Y_{M1}$	$100.66 \times 0.306 \times 275/10/1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 164.64$, $\psi = 0.859$, $\alpha_h = 0.000$	1.000	Table B.3

$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 0.000 \text{ imposed} + 10.129 \text{ twist } \varphi.M_y = 10.129; \psi = 1 \text{ for the imposed constant } M_z, \text{ the twist part taken at } C_{mz} = 1.0 \text{ (Table B.3 upper bound, conservative)}$	1.000	Table B.3
$C_{my} = 1.0 \text{ (diagram outside Table B.3)}$	$M_h = 0.00, M_s = 164.64, \psi = 0.859, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	$1.162, 6.000, 1870, 69.2, 0.922, 210000; z_g = +230 \text{ mm entered — NOT applied: } C_2 \text{ not published for this moment diagram; } M_{cr} \text{ is at the shear centre (PASS blocked)}$	398.376 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_{fy}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	$1.124, 1.151, 0.49$	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	$0.566, 1.124, 0.928, 0.971$	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
$\lambda_{LT} \text{ (P362 6.55 simplified) ; } M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364; \chi_{LT,mod} = 0.451;$ $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff,y} = 2768.024 \text{ kN}; M_{y,Rk} = W_{eff,y}f_y \text{ with } W_{eff,y} = W_{el,y} = 1610.0 \text{ cm}^3 \text{ (flanges Class } \leq 3, \text{ web Class 4 only in uniform compression, } \Delta M = 0) \text{ [assumption]; } k_{ij} \text{ from the Class 3/4 rows}$		6.3.3
$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 2664.536 \text{ (} N_{Rk} = 2768.024 \text{)}$	0.056	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	$168.868 / 258.156 \text{ (} M_{y,Rk} = 442.750; W_{el,y}/W_{pl,y} = 0.880 \text{ applied to } M_{b,Rd} \text{)}$	0.654	OK
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(0.000 \text{ imposed} + 10.129 \text{ twist } \varphi.M_y = 10.129) / 53.900$	0.188	OK
$k_{zy} \text{ method}$	$M_{b,Rd} < M_{c,y,Rd} \text{ therefore susceptible to LTB, using Table B.2}$		Table B.2
$k_{yy} = C_{my}\{1+0.6\chi_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.362 \times 0.056\} \leq 1.000(1+0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\chi_z U_{N,z}\}$	$1.000\{1+0.6 \times 1.607 \times 0.177\} \leq 1.000(1+0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25); \lambda_z = 1.607, U_{N,z} = 0.177, C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.056+1.012 \times 0.654+1.106 \times 0.188$	0.926	OK
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.177+0.988 \times 0.654+1.106 \times 0.188$	1.032	Warning

Torsion Design

J, H, a, Q _f , Q _w	$69.2 \text{ cm}^4, 0.922 \text{ dm}^6, 1859 \text{ mm}, \text{—}, \text{—}$		P385 App A
W_{n0}, S_{w1}	$212 \text{ cm}^2, 1620 \text{ cm}^4$		P385 App A

Torsion Bending Design @ 2.34 m

$T_{Ed} \text{ (max)}$	combination ULS: $1.35G + 1.5Q$	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	$0.0605 \text{ rad} = 3.47^\circ$	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi.M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34 \text{ m: } 168.747, 7.426, 9.835 \text{ kN.m; } M_{pl,y} = 503.250, M_{pl,f} = 40.255, M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.882 \times 1.736$	1.015	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.365 \text{ m: } 168.802/285.039 + 1.000 \times 9.873/83.600 + 0.663 \times 0.882 \times 1.736 \times 7.403/40.255$	0.897	OK EN 1993-6 A

Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi_i M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi_i M_y$			EN 1993-1-1, EN 1993-6
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 1.032$ (Eq 6.62 with $M_{z,tot} = 10.129$) + 0.197 (warping term of (A.1) @ $x = 2$ m: $k_w = 0.662$, $k_{zw} = 0.892$, $k_\alpha = 1.736$, $M_w = 7.736$ kN.m) =			1.228 information only
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260 / 3.660$ kN.m			1.902 / 1.724 kN.m
Torsion Shear Design @ 0 m				
T_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, $V = 109.758$ kN			27.22 N/mm ² P385
$V_{pl.T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl.T,Rd}/V_{pl,Rd} = 0.929$			702.492 kN 6.2.7(9)
$V_{Ed}/V_{pl.T,Rd}$	109.758 / 702.492			0.156 OK
Deflection Check - Load Case 1 (SLS: 1.0Q)				
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m			3.22 mm OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	0.0252 rad = $1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)			1.44° OK
$N_{Ed}/N_{c,Rd}$	UNyz	UMyz	Ax+M_6.62	Ax+M_6.62
0.054	0.113	0.177	0.654	0.926
			1.032	0.193
			0.145	0.336
			0.575	0.251
			0.287	0.415
			0.801	0.156
			0.897	1.032

FAIL — governing Member buckling z-z (Eq 6.62) = 1.032

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.032$ (Eq 6.62 with $M_{z,tot} = 10.13$ kN.m) + 0.197 (warping term of (A.1) at $x = 2.00$ m, $k_w = 0.662$, $k_{zw} = 0.892$, $k_\alpha = 1.736$) = 1.228. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

UB-L6 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **FAIL** · governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.671

Owner case: ... and axial load on beam, load applied at angle on beam

UB 457 x 191 x 82, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +230, factored by the combination) plus an entered CONSTANT $M_{z,Ed}$ = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment $H a b / L = 10.261 \times 2 \times 4 / 6 = 13.681$ kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member ($C_{mz} = 1.0$), not as the true triangular diagram, and the torque of the horizontal component acting at z_g ($H \times z_g = 10.261 \times 0.230 = 2.36$ kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member UB-L6 UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm $z_g = 230$ mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm $z_g = 230$ mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm $z_g = 230$ mm (kN)

Q PY 28.191 @ 2 m e = 40 mm $z_g = 230$ mm (kN)

G SW 0.8044 kN/m 0-6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

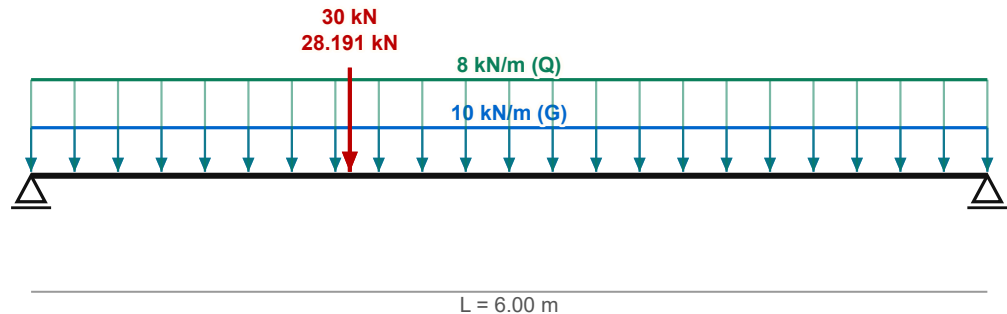
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.39	137.95	0.00	0.00	20.52	222.73	20.52	-4.61	137.95	—
	End 2, x = 6	150.00C	-4.22	-108.85	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	108.85	—

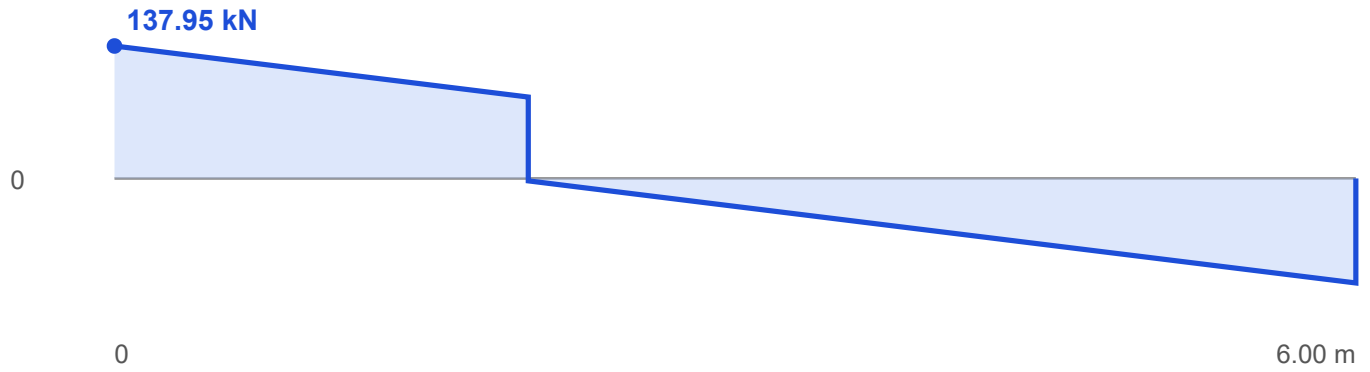
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

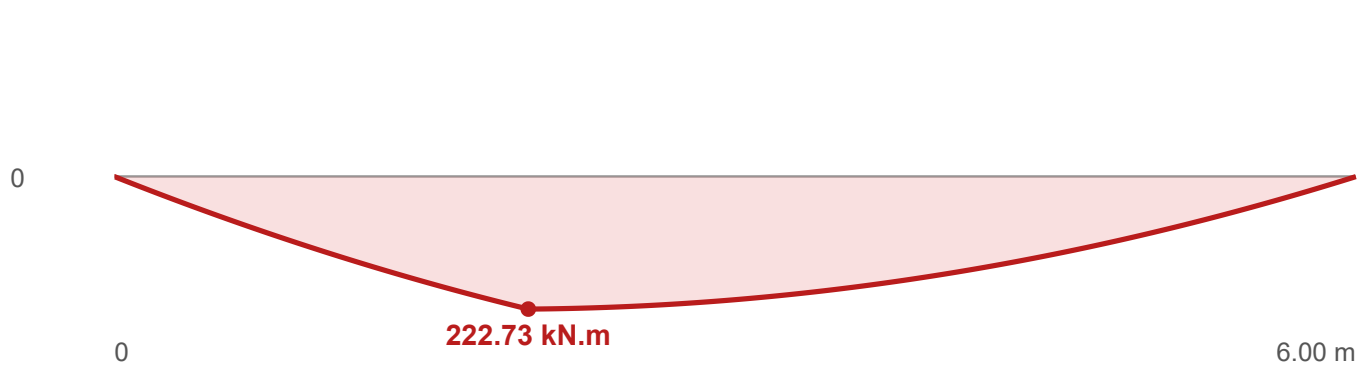
LOADING



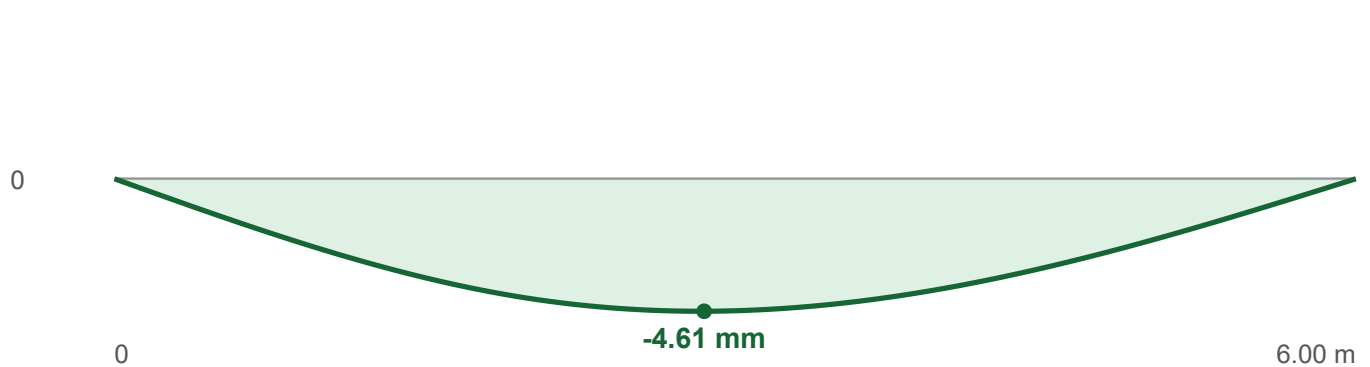
SHEAR FORCE V (kN)



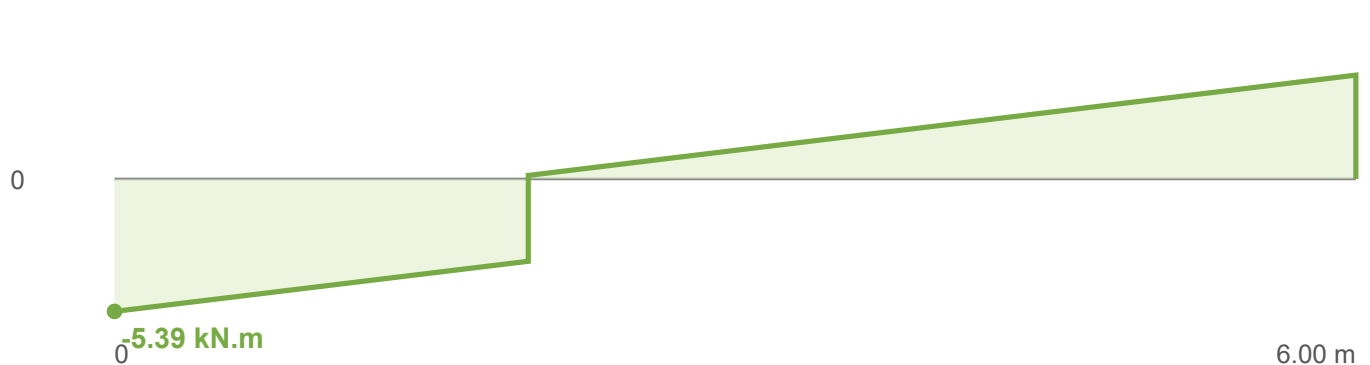
BENDING MOMENT M (kN.M)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 222.73, 20.52	Class 1
	(Axial: Slender web)	
Web classified for	bending + compression: $\alpha = 0.568$, $\psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1 Table 5.2

Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 169.3, tip 257.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 136.2, tip 48.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² $\times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed}/V_{pl,y,Rd}$	137.949 / 756.260 =	0.182	OK
$h_w/t_w \leq 72\epsilon/\eta$	43.23 $\leq 72 \times 0.924/1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	2.510 / 756.260 =	0.003	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	275 $\times 1830.0/1$	503.250 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	222.726 / 503.250 =	0.443	OK
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 $\times 304.0/1$	83.600 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	104.00 $\times 275/1$ (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} \cdot f_y/\gamma_{M0}$	100.66 $\times 275/1$ (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed}/N_{c,Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	1830.0 $\times 275/1$	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	304.0 $\times 275/1$	83.600 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(222.726/503.250)^2 + (20.521/83.600)^1 =$	0.441	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ $x = 2 \text{ m}$ (ULS: 1.35G + 1.5Q): $M_y = 222.726 \text{ kN.m}$, $V = 41.134 \text{ kN}$, $\phi = 0.07008 \text{ rad}$, $\phi' = 1.806\text{e-}5 \text{ rad/mm}$, $\phi'' = 2.483\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 1.206\text{e-}11 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 15.608 = 36.129 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + EW_{n0}\phi'' = 14.42 + 138.34 + 184.33 + 110.53 = 447.63 \text{ N/mm}^2$; $\tau = Gt\phi' = 23.41 \text{ N/mm}^2$; $(447.63/275)^2 + 3(23.41/275)^2 =$	2.671	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 138.34 + 184.33 + 110.53 = 447.63$, $\tau = 23.41 \rightarrow 2.671$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 133.28 + 9.56 = 157.26$, $\tau = 2.35 + 23.41 + 2.56 = 28.33 \rightarrow 0.359$; P3 web-flange junction, web: $\sigma_x = 14.42 + 128.47 + 9.56 = 152.46$, $\tau = 7.61 + 14.49 = 22.10 \rightarrow 0.327$; P4 web mid-depth: $\sigma_x = 14.42 + 9.56 = 23.99$, $\tau = 10.15 + 14.49 = 24.63 \rightarrow 0.032$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); min[227.03, 192.79]	192.79 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 3)}$	0.600 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.949$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.315 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.315 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 + N_{Ed} / N_{pl,Rd} = 150.000 / 2860.000 = 0.052) = 0.357 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.255 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	137.949	ULS: 1.35G + 1.5Q	0.315	0.255	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.416	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	108.853	ULS: 1.35G + 1.5Q	0.249	0.208	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} f_y$; $\lambda = \sqrt{(A_{eff} f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10 / 1 =$	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	408.966 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.193	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 206.9, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=177.0$, $M(L/2)=206.9$, $M(3L/4)=133.4$, $M_{max}=222.7$ kN·m	1.215	Serna

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 206.92, \psi = 0.785, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 16.609 twist $\phi \cdot M_y = 37.130$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 206.92, \psi = 0.785, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 230, 0$ mm; 1 combination(s), 2 solve(s) + 1 cached	292.804 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 292.804}$	1.311	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.311, 1.368, 0.49	0.469	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.469, 1.311, 0.916, 0.980	0.479	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.479 \times 1830.0 \times 275 \leq 503.250 =$	241.095 kN.m	
$M_{y,Ed} / M_{b,Rd}$	$222.726 / 241.095$	0.924	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$222.726 / 212.111$ ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	1.050	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$(20.521 \text{ imposed} + 16.609 \text{ twist } \phi \cdot M_y = 37.130) / 53.900$	0.689	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 \{1 + 0.6 \times 0.056\}$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 \{1 + 0.6 \times 0.177\}$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 1.607, U_{N,z} = 0.177, C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 1.050 + 1.106 \times 0.689$	1.881	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 1.050 + 1.106 \times 0.689$	1.977	Warning

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 3.23$	P385 App C
ϕ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	$0.0771 \text{ rad} = 4.42^\circ$	
$B_{Ed} = E I_w \phi''$ (max)	@ $x = 2$ m	4.807 kN.m ²	
$M_{w,Ed} = E I_w \phi'' / (h - t_f)$	max over span	10.827 kN.m	

$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 217.915 x 76.22e-3 (@ x = 2.515 m) = 20.521 + 16.609 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	37.130 kN.m	P385 3.1.2																																		
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ x = 2 m: 222.726, 10.827, 36.129 kN.m ($M_{z,tot} = 20.521 + 15.608$); $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.897	OK P385 3.1.2																																		
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3																																		
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.646 \times 0.568 \times 4.178$ ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 36.129/83.600$)	1.533	EN 1993-6 A																																		
$M_y/M_{b,Rd} + C_{mz}M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2 m: 222.726/236.229 + 1.000x36.129/83.600 + 0.646x0.568x4.178x10.827/40.255 ($M_{z,tot} = 20.521$ imposed + 15.608 twist $\phi \cdot M_y$)	1.787	Warning EN 1993-6 A																																		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6																																		
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 1.977$ (Eq 6.62 with $M_{z,tot} = 37.130$) + 0.412 (warping term of (A.1) @ x = 2 m: $k_w = 0.646$, $k_{zw} = 0.568$, $k_\alpha = 4.178$, $M_w = 10.827$ kN.m) =	2.390	information only																																		
End torques T_t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total $T = GI_T \phi' - EI_w \phi''' = 5.388 / 4.224$ kN.m	2.460 / 2.113 kN.m																																			
Torsion Shear Design @ 0 m																																					
T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 2.460$ kN.m, $V = 137.949$ kN	35.20 N/mm²	P385																																		
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 35.20$ N/mm²; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.907$	685.927 kN	6.2.7(9)																																		
$V_{Ed}/V_{pl,T,Rd}$	137.949 / 685.927	0.201	OK																																		
Deflection Check - Load Case 1 (SLS: 1.0Q)																																					
In-span $\delta \leq \text{Span}/360$	$4.61 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	4.61 mm	OK																																		
Torq in Case 1 @ 2.714 m: $\theta_{max} \leq 2.00^\circ$	0.0363 rad = $2.08^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	2.08°	> 2° advisory																																		
<table><tr><td>$N_{Ed}/N_{c,Rd}$</td><td>α_{Ed}</td><td>U_{Ny}</td><td>U_{My}</td><td>$Ax+M_{6.6}$</td><td>$Ax+M_{6.6}$</td><td>Deflection</td><td>V/Vpl</td><td>MA/Mc</td><td>LTB</td><td>F/F_Rd</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>LTB+T</td><td>Max</td></tr><tr><td>0.054</td><td>0.441</td><td>0.177</td><td>1.050</td><td>1.881</td><td>1.977</td><td>0.277</td><td>0.182</td><td>0.443</td><td>0.924</td><td>0.315</td><td>0.416</td><td>0.897</td><td>2.671</td><td>0.201</td><td>1.787</td><td>2.671</td></tr></table>				$N_{Ed}/N_{c,Rd}$	α_{Ed}	U_{Ny}	U_{My}	$Ax+M_{6.6}$	$Ax+M_{6.6}$	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	0.054	0.441	0.177	1.050	1.881	1.977	0.277	0.182	0.443	0.924	0.315	0.416	0.897	2.671	0.201	1.787	2.671
$N_{Ed}/N_{c,Rd}$	α_{Ed}	U_{Ny}	U_{My}	$Ax+M_{6.6}$	$Ax+M_{6.6}$	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max																					
0.054	0.441	0.177	1.050	1.881	1.977	0.277	0.182	0.443	0.924	0.315	0.416	0.897	2.671	0.201	1.787	2.671																					

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.671

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.977$ (Eq 6.62 with $M_{z,tot} = 37.13$ kN.m) + 0.412 (warping term of (A.1) at x = 2.00 m, $k_w = 0.646$, $k_{zw} = 0.568$, $k_\alpha = 4.178$) = 2.390. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

UB-L6 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **FAIL** · governing Elastic yield criterion (6.1) with torsion, $cl\ 6.2.7(5) = 2.671$

Owner case: ... and axial load on beam, load applied at angle on beam

UB 457 x 191 x 82, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, $e = +40$, $z_g = +230$, factored by the combination) plus an entered CONSTANT $M_{z,Ed} = 20.521$ kN.m (= 1.5×13.681 , the ULS design value of the Q-case moment $H\ a\ b / L = 10.261 \times 2 \times 4 / 6 = 13.681$ kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member ($C_{mz} = 1.0$), not as the true triangular diagram, and the torque of the horizontal component acting at z_g ($H \times z_g = 10.261 \times 0.230 = 2.36$ kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member UB-L6 UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x\ U_y\ U_z\ R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y\ U_z\ R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m $e = 40\ mm\ z_g = 230\ mm$ (kN/m)

Q UDL 8.000 0-6 m $e = 40\ mm\ z_g = 230\ mm$ (kN/m)

Q PY 30.000 @ 2 m $e = 40\ mm\ z_g = 230\ mm$ (kN)

Q PY 28.191 @ 2 m $e = 40\ mm\ z_g = 230\ mm$ (kN)

G SW 0.8044 kN/m 0-6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

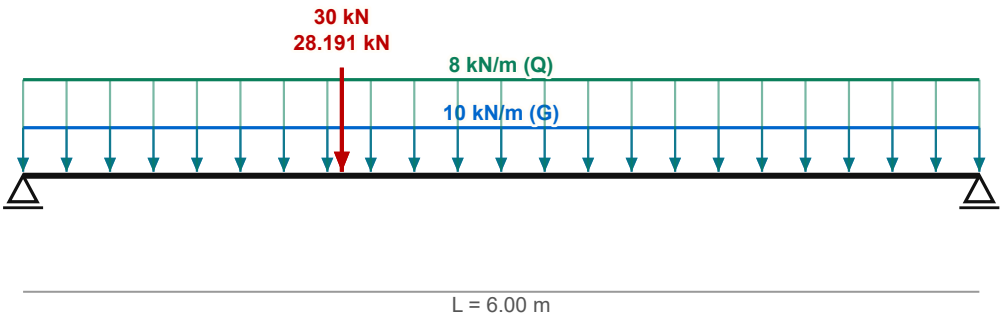
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.39	137.95	0.00	0.00	20.52	222.73	20.52	-4.61	137.95	—
	End 2, x = 6	150.00C	-4.22	-108.85	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	108.85	—

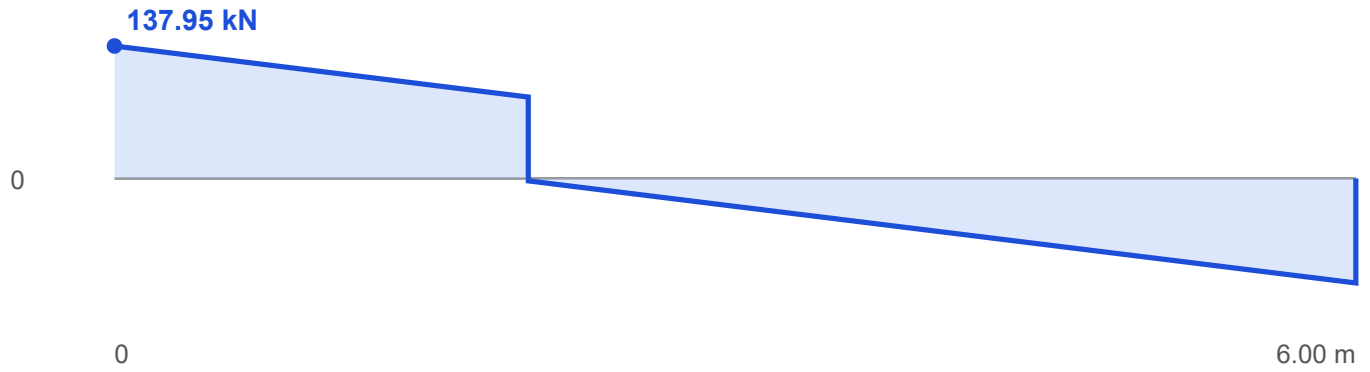
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

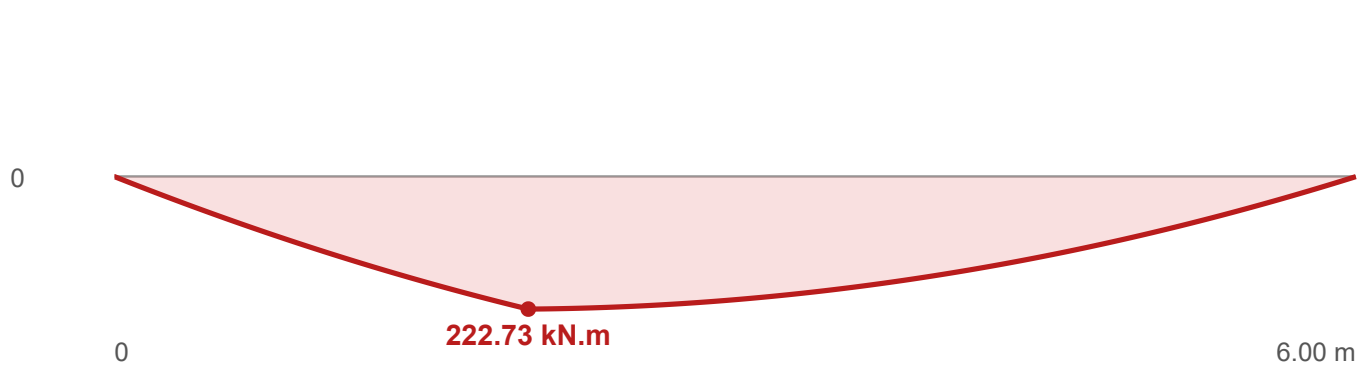
LOADING



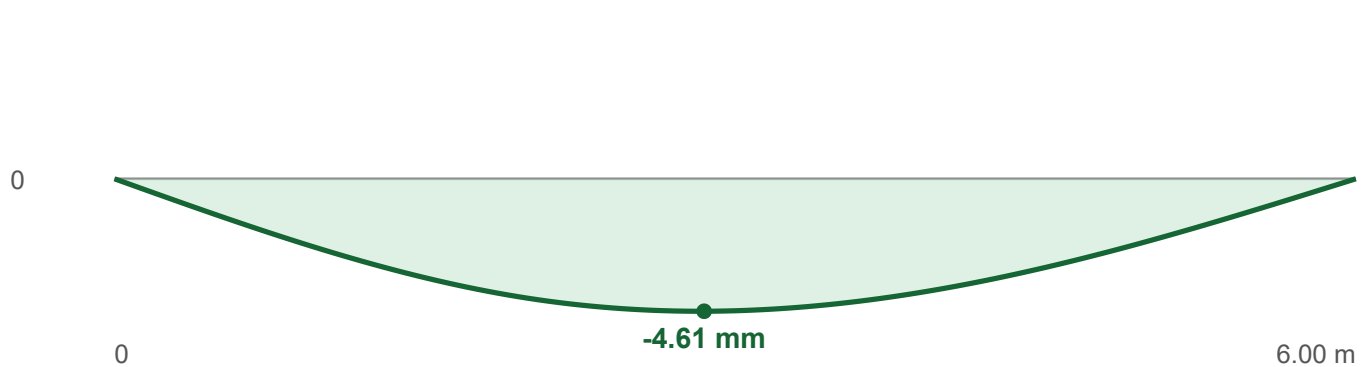
SHEAR FORCE V (kN)



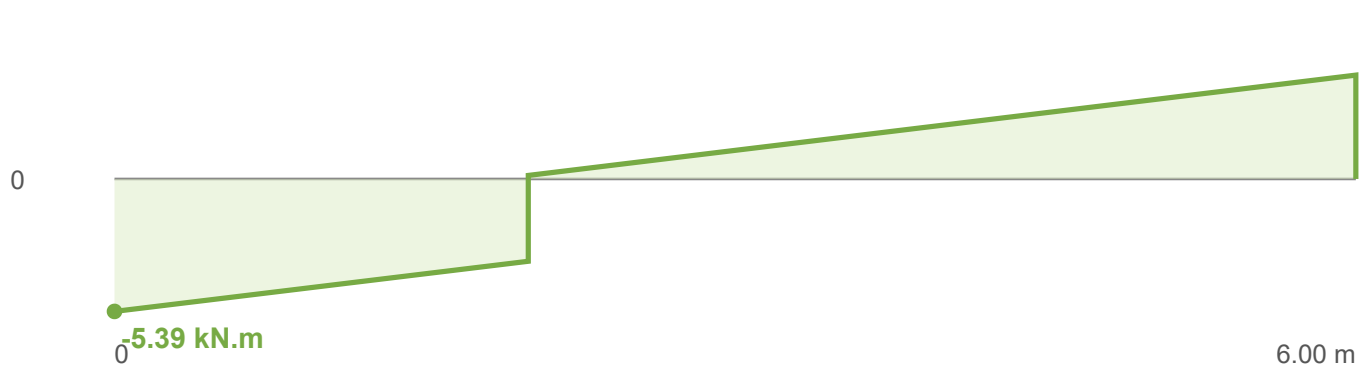
BENDING MOMENT M (kN.M)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 222.73, 20.52	Class 1
	(Axial: Slender web)	
Web classified for	bending + compression: $\alpha = 0.568$, $\psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1 Table 5.2

Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 169.3, tip 257.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 136.2, tip 48.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² x 275/($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed}/V_{pl,y,Rd}$	137.949 / 756.260 =	0.182	OK
$h_w/t_w \leq 72\epsilon/\eta$	43.23 $\leq 72 \times 0.924/1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	2.510 / 756.260 =	0.003	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	222.726 / 503.250 =	0.443	OK
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 x 304.0/1	83.600 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	104.00 x 275/1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} \cdot f_y/\gamma_{M0}$	100.66 x 275/1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed}/N_{c,Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	1830.0 x 275/1	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	304.0 x 275/1	83.600 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(222.726/503.250)^2 + (20.521/83.600)^1 =$	0.441	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ $x = 2 \text{ m}$ (ULS: 1.35G + 1.5Q): $M_y = 222.726 \text{ kN.m}$, $V = 41.134 \text{ kN}$, $\phi = 0.07008 \text{ rad}$, $\phi' = 1.806\text{e-}5 \text{ rad/mm}$, $\phi'' = 2.483\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 1.206\text{e-}11 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 15.608 = 36.129 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + EW_{n0}\phi'' = 14.42 + 138.34 + 184.33 + 110.53 = 447.63 \text{ N/mm}^2$; $\tau = Gt\phi' = 23.41 \text{ N/mm}^2$; $(447.63/275)^2 + 3(23.41/275)^2 =$	2.671	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 138.34 + 184.33 + 110.53 = 447.63$, $\tau = 23.41 \rightarrow 2.671$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 133.28 + 9.56 = 157.26$, $\tau = 2.35 + 23.41 + 2.56 = 28.33 \rightarrow 0.359$; P3 web-flange junction, web: $\sigma_x = 14.42 + 128.47 + 9.56 = 152.46$, $\tau = 7.61 + 14.49 = 22.10 \rightarrow 0.327$; P4 web mid-depth: $\sigma_x = 14.42 + 9.56 = 23.99$, $\tau = 10.15 + 14.49 = 24.63 \rightarrow 0.032$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000$ mm, the full member length (conservative)	6.4(1)

Governing station $x = 0$ m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 137.949$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.315	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.315 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 + N_{Ed}/N_{pl,Rd} = 150.000/2860.000 = 0.052) = 0.357 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.255	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	137.949	ULS: 1.35G + 1.5Q	0.315	0.255	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.416	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	108.853	ULS: 1.35G + 1.5Q	0.249	0.208	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y)$: held in position at both ends, in direction at neither: 1.0 L	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} f_y$; $\lambda = \sqrt{(A_{eff} f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10/1 =$	2664.536 kN	Curve a

$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 206.9, 1.000, 300.000	1.215	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=177.0$, $M(L/2)=206.9$, $M(3L/4)=133.4$, $M_{max}=222.7$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 206.92$, $\psi = 0.785$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 16.609 twist ϕ . $M_y = 37.130$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 206.92$, $\psi = 0.785$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	1 x 6 =	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.215, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	416.357 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 416.357}$	1.099	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.099, 1.125, 0.49	0.580	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.580, 1.099, 0.907, 0.962	0.603	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.603 \times 1830.0 \times 275 \leq 503.250 =$	303.596 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\chi_{C1}) \cdot 0.9 \lambda_z \sqrt{\beta_w} = 1.334$; $\chi_{LT,mod} = 0.468$; $M_{y,Ed} / M_{b,Rd,simp} = 0.946$	235.463 kN.m	P362 6.55 comparison
$M_{y,Ed} / M_{b,Rd}$	$222.726 / 303.596$	0.734	OK

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$222.726 / 267.098$ ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.834	OK
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$(20.521 \text{ imposed} + 16.609 \text{ twist } \phi \cdot M_y = 37.130) / 53.900$	0.689	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 (1 + 0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 (1 + 0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 0.834 + 1.106 \times 0.689$	1.663	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 0.834 + 1.106 \times 0.689$	1.764	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ³ , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
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Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C																																		
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0771 rad = 4.42°																																			
B _{Ed} = EI _w φ" (max)	@ x = 2 m	4.807 kN.m ²																																			
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	10.827 kN.m																																			
M _{z,tot} = M _{z,Ed} + φ.M _{y,Ed}	20.521 + 217.915 x 76.22e-3 (@ x = 2.515 m) = 20.521 + 16.609 (imposed constant M _{z,Ed} + max coincident twist moment)	37.130 kN.m	P385 3.1.2																																		
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _{z,tot} /M _{pl,z}	@ x = 2 m: 222.726, 10.827, 36.129 kN.m (M _{z,tot} = 20.521 + 15.608); M _{pl,y} = 503.250, M _{pl,f} = 40.255, M _{pl,z} = 83.600	0.897	OK P385 3.1.2																																		
C _{mz} (EN 1993-6 A.1)	C _{mz} = 1.0: the imposed M _z is a constant diagram (psi = 1, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3																																		
k = k _w .k _{zw} .k _α	0.646 x 0.568 x 2.150 (k _{zw} = 1 - M _{z,tot} /M _{z,Rk} = 1 - 36.129/83.600)	0.789	EN 1993-6 A																																		
M _y /M _{b,Rd} + C _{mz} M _{z,tot} /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 222.726/292.053 + 1.000x36.129/83.600 + 0.646x0.568x2.150x10.827/40.255 (M _{z,tot} = 20.521 imposed + 15.608 twist φ.M _y)	1.407	Warning EN 1993-6 A																																		
Basis with N _{Ed} / M _{z,Ed}	No expression in EN 1993-1-1 or EN 1993-6 combines N _{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} , and EN 1993-6 (A.1) with M _{z,Ed} = M _z + phi.M _y		EN 1993-1-1, EN 1993-6																																		
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	N _{Ed} /(X _z N _{Rk} /Y _{M1}) + k _{zy} M _{y,Ed} /(X _{LT} M _{y,Rk} /Y _{M1}) + k _{zz} M _{z,tot} /(M _{z,Rk} /Y _{M1}) + k _w k _{zw} k _α M _w /M _{f,Rd} = 1.764 (Eq 6.62 with M _{z,tot} = 37.130) + 0.212 (warping term of (A.1)) @ x = 2 m: k _w = 0.646, k _{zw} = 0.568, k _α = 2.150, M _w = 10.827 kN.m) =	1.976	information only																																		
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	2.460 / 2.113 kN.m																																			
Torsion Shear Design @ 0 m																																					
T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 2.460 kN.m, V = 137.949 kN	35.20 N/mm ²	P385																																		
V _{pl,T,Rd} = √(1 - τ _t /(1.25f _y /√3)).V _{pl,Rd}	@ x = 0 m: τ _t = 35.20 N/mm ² ; V _{pl,Rd} = 756.260; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.907	685.927 kN	6.2.7(9)																																		
V _{Ed} /V _{pl,T,Rd}	137.949 / 685.927	0.201	OK																																		
Deflection Check - Load Case 1 (SLS: 1.0Q)																																					
In-span δ ≤ Span/360	4.61 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	4.61 mm	OK																																		
Torq in Case 1 @ 2.714 m: θ _{max} ≤ 2.00°	0.0363 rad = 2.08° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	2.08°	> 2° advisory																																		
<table><tr><td>N_{Ed}/N_{c,Rd}</td><td>ρ_{ed}</td><td>UNyz</td><td>UMyz</td><td>Ax+M_6.6</td><td>Ax+M_6.6</td><td>Deflection</td><td>V/Vpl</td><td>MA/Mc</td><td>LTB</td><td>F/F_Rd</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>LTB+T</td><td>Max</td></tr><tr><td>0.054</td><td>0.441</td><td>0.177</td><td>0.834</td><td>1.663</td><td>1.764</td><td>0.277</td><td>0.182</td><td>0.443</td><td>0.734</td><td>0.315</td><td>0.416</td><td>0.897</td><td>2.671</td><td>0.201</td><td>1.407</td><td>2.671</td></tr></table>				N _{Ed} /N _{c,Rd}	ρ _{ed}	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	0.054	0.441	0.177	0.834	1.663	1.764	0.277	0.182	0.443	0.734	0.315	0.416	0.897	2.671	0.201	1.407	2.671
N _{Ed} /N _{c,Rd}	ρ _{ed}	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max																					
0.054	0.441	0.177	0.834	1.663	1.764	0.277	0.182	0.443	0.734	0.315	0.416	0.897	2.671	0.201	1.407	2.671																					

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.671

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(\chi_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(\chi_{LT} M_{y,Rk}/\gamma_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.764$ (Eq 6.62 with $M_{z,tot} = 37.13 \text{ kN}\cdot\text{m}$) + 0.212 (warping term of (A.1) at $x = 2.00 \text{ m}$, $k_w = 0.646$, $k_{zw} = 0.568$, $k_\alpha = 2.150$) = 1.976. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

UB-L7 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **FAIL** · governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.908

Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER

UB 457 x 191 x 82, 6 m SS, unrestrained; L7 = L6 + "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member UB-L7 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40 \text{ mm}$ $z_g = 230 \text{ mm}$ (kN/m)

Q UDL 8.000 0–6 m $e = 40 \text{ mm}$ $z_g = 230 \text{ mm}$ (kN/m)

Q PY 30.000 @ 2 m $e = 40 \text{ mm}$ $z_g = 230 \text{ mm}$ (kN)

Q PY 28.191 @ 2 m $e = 40 \text{ mm}$ $z_g = 230 \text{ mm}$ (kN)

Q M 40.000 @ 4 m (kN.m)

G SW 0.8044 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

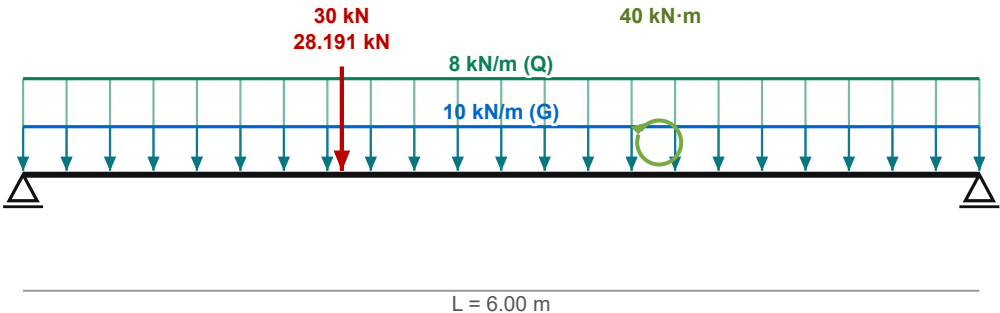
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.39	147.95	0.00	0.00	20.52	243.78	20.52	-5.26	147.95	—
	End 2, x = 6	150.00C	-4.22	-98.85	0.00	0.00	20.52	@ 2.280	@ —	@ 2.850	98.85	—

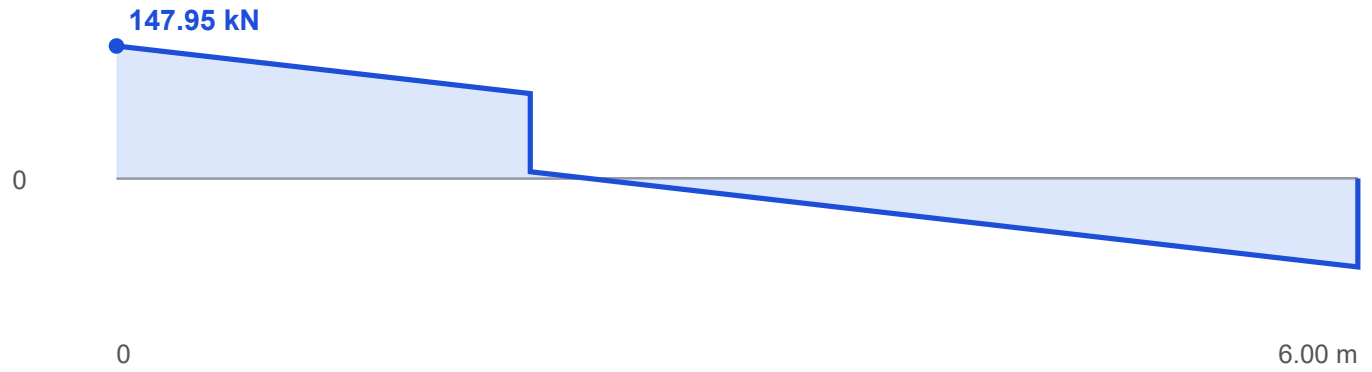
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

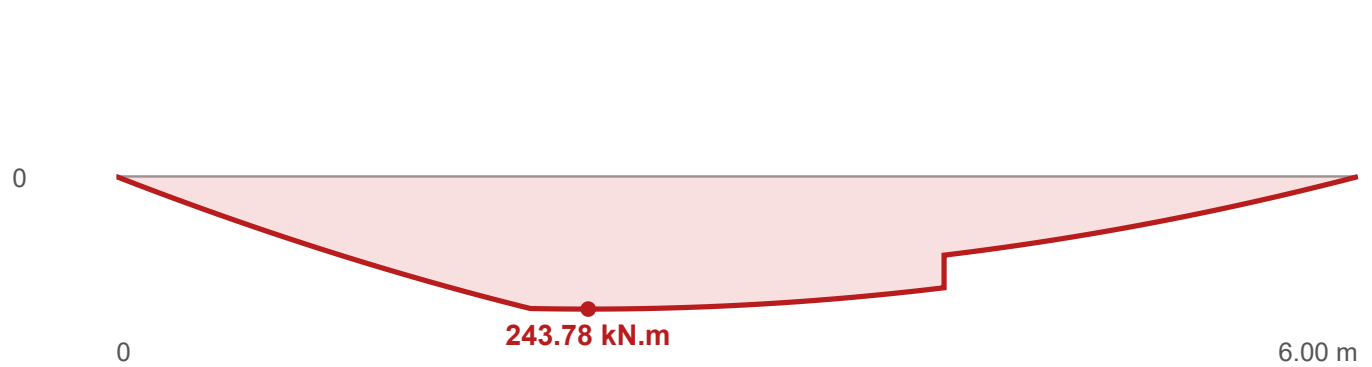
LOADING



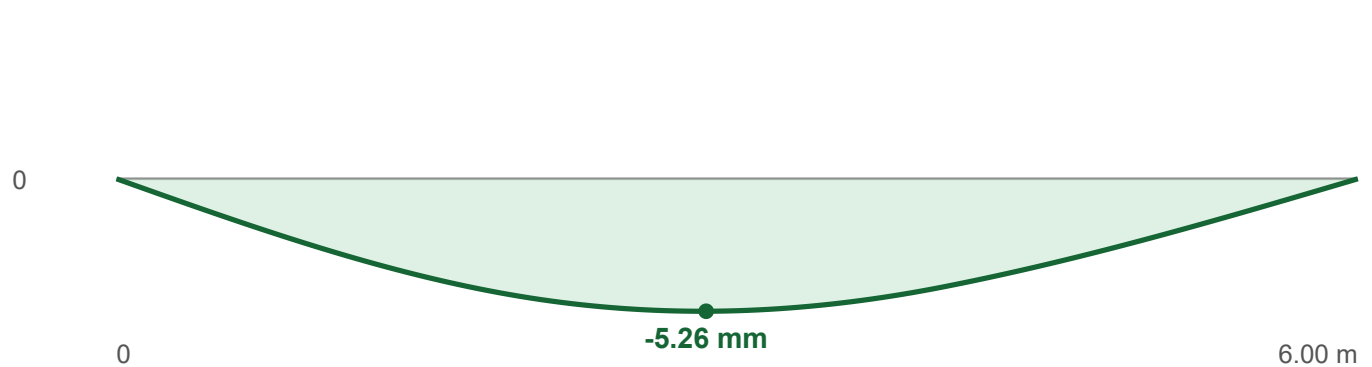
SHEAR FORCE V (kN)



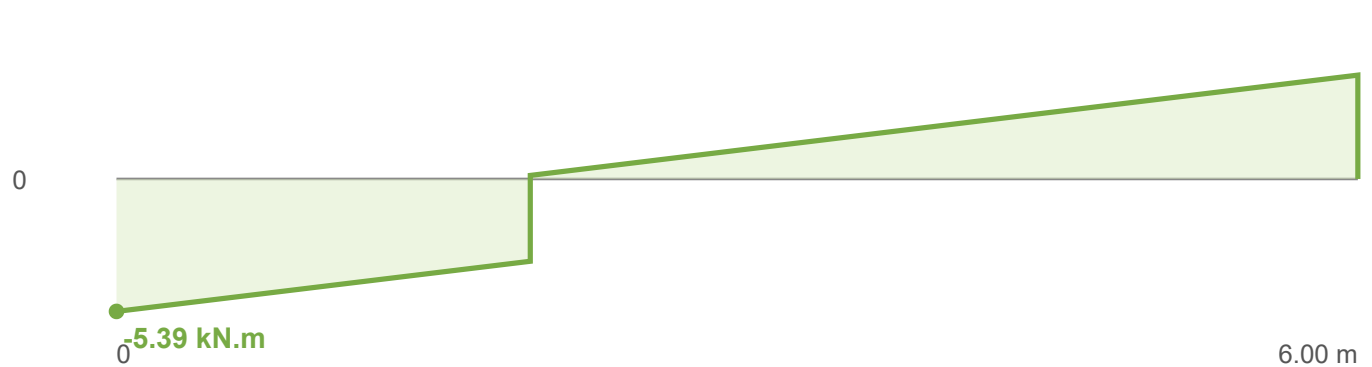
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 243.78, 20.52		Class 1
	(Axial: Slender web)		
Web classified for	bending + compression: $\alpha = 0.568, \psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2

Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 182.4, tip 270.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 149.3, tip 61.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² $\times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed}/V_{pl,y,Rd}$	147.949 / 756.260 =	0.196	OK
$h_w/t_w \leq 72\epsilon/\eta$	43.23 $\leq 72 \times 0.924/1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	0.046 / 756.260 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	275 $\times 1830.0/1$	503.250 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	243.781 / 503.250 =	0.484	OK
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 $\times 304.0/1$	83.600 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	104.00 $\times 275/1$ (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} \cdot f_y/\gamma_{M0}$	100.66 $\times 275/1$ (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed}/N_{c,Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	1830.0 $\times 275/1$	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	304.0 $\times 275/1$	83.600 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(243.781/503.250)^2 + (20.521/83.600)^1 =$	0.480	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ $x = 2 \text{ m}$ (ULS: 1.35G + 1.5Q): $M_y = 242.726 \text{ kN.m}$, $V = 51.134 \text{ kN}$, $\phi = 0.07008 \text{ rad}$, $\phi' = 1.806\text{e-}5 \text{ rad/mm}$, $\phi'' = 2.483\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 1.206\text{e-}11 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 17.010 = 37.531 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + EW_{n0}\phi'' = 14.42 + 150.76 + 191.48 + 110.53 = 467.20 \text{ N/mm}^2$; $\tau = Gt\phi' = 23.41 \text{ N/mm}^2$; $(467.20/275)^2 + 3(23.41/275)^2 =$	2.908	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 150.76 + 191.48 + 110.53 = 467.20$, $\tau = 23.41 \rightarrow 2.908$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 145.24 + 9.93 = 169.60$, $\tau = 2.93 + 23.41 + 2.56 = 28.90 \rightarrow 0.413$; P3 web-flange junction, web: $\sigma_x = 14.42 + 140.01 + 9.93 = 164.37$, $\tau = 9.46 + 14.49 = 23.95 \rightarrow 0.380$; P4 web mid-depth: $\sigma_x = 14.42 + 9.93 = 24.36$, $\tau = 12.62 + 14.49 = 27.10 \rightarrow 0.037$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 147.949$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.338 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.338 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 + N_{Ed} / N_{pl,Rd} = 150.000 / 2860.000 = 0.052) = 0.380 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.272 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	147.949	ULS: 1.35G + 1.5Q	0.338	0.272	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.438	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	98.853	ULS: 1.35G + 1.5Q	0.226	0.191	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} f_y$; $\lambda = \sqrt{(A_{eff} f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10 / 1 =$	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

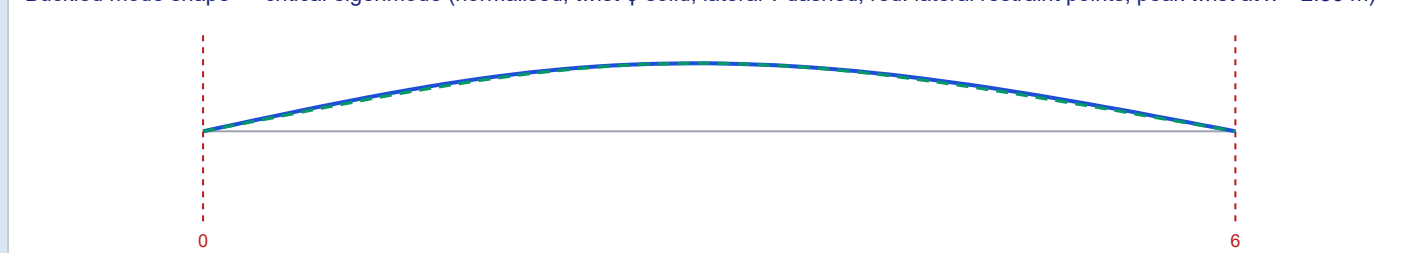
$C_1 = M_{cr} / M_{cr,uniform}$	402.212 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.173	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 236.9, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=192.0$, $M(L/2)=236.9$, $M(3L/4)=118.4$, $M_{max}=243.8$ kN.m	1.212	Serna

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 236.92, \psi = 0.664, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 18.592 twist $\phi \cdot M_y = 39.113$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 236.92, \psi = 0.664, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 230, 0$ mm; 1 combination(s), 3 solve(s)	297.660 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 297.660}$	1.300	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.300, 1.355, 0.49	0.475	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.475, 1.300, 0.923, 0.981	0.484	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.484 \times 1830.0 \times 275 \leq 503.250 =$	243.466 kN.m	
$M_{y,Ed} / M_{b,Rd}$	243.781 / 243.466	1.001	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.86$ m)



Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$243.781 / 214.197$ ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	1.138	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$(20.521 \text{ imposed} + 18.592 \text{ twist } \phi \cdot M_y = 39.113) / 53.900$	0.726	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 \{1 + 0.6 \times 0.056\}$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 \{1 + 0.6 \times 0.177\}$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 1.607, U_{N,z} = 0.177, C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 1.138 + 1.106 \times 0.726$	2.011	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 1.138 + 1.106 \times 0.726$	2.105	Warning

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.066 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 3.23$	P385 App C
ϕ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0771 rad = 4.42°	
$B_{Ed} = E I_w \phi''$ (max)	@ $x = 2$ m	4.807 kN.m ²	
$M_{w,Ed} = E I_w \phi'' / (h - t_f)$	max over span	10.827 kN.m	

$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 242.084 x 76.8e-3 (@ x = 2.639 m) = 20.521 + 18.592 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	39.113 kN.m	P385 3.1.2																																		
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ x = 2.066 m: 243.164, 10.656, 37.840 kN.m ($M_{z,tot} = 20.521 + 17.319$); $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.951	OK P385 3.1.2																																		
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3																																		
$k = k_w \cdot k_{zw} \cdot k_\alpha$	0.646 x 0.551 x 5.524 ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 37.531/83.600$)	1.967	EN 1993-6 A																																		
$M_y/M_{b,Rd} + C_{mz}M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2 m: 242.726/238.794 + 1.000x37.531/83.600 + 0.646x0.551x5.524x10.827/40.255 ($M_{z,tot} = 20.521$ imposed + 17.010 twist $\phi \cdot M_y$)	1.995	Warning EN 1993-6 A																																		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6																																		
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 2.105$ (Eq 6.62 with $M_{z,tot} = 39.113$) + 0.529 (warping term of (A.1)) @ x = 2 m: $k_w = 0.646$, $k_{zw} = 0.551$, $k_\alpha = 5.524$, $M_w = 10.827$ kN.m) =	2.634	information only																																		
End torques T_t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total $T = GI_T \phi' - EI_w \phi''' = 5.388 / 4.224$ kN.m	2.460 / 2.113 kN.m																																			
Torsion Shear Design @ 0 m																																					
T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 2.460$ kN.m, $V = 147.949$ kN	35.20 N/mm²	P385																																		
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 35.20$ N/mm²; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.907$	685.927 kN	6.2.7(9)																																		
$V_{Ed}/V_{pl,T,Rd}$	147.949 / 685.927	0.216	OK																																		
Deflection Check - Load Case 1 (SLS: 1.0Q)																																					
In-span $\delta \leq \text{Span}/360$	$5.26 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	5.26 mm	OK																																		
Torq in Case 1 @ 2.714 m: $\theta_{max} \leq 2.00^\circ$	0.0363 rad = $2.08^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	2.08°	> 2° advisory																																		
<table><tr><td>$N_{Ed}/N_{c,Rd}$</td><td>α_{Ed}</td><td>U_{Ny}</td><td>U_{My}</td><td>$Ax+M_{6.6}$</td><td>$Ax+M_{6.6}$</td><td>Deflection</td><td>V/Vpl</td><td>MA/Mc</td><td>LTB</td><td>F/F_{Rd}</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>LTB+T</td><td>Max</td></tr><tr><td>0.054</td><td>0.480</td><td>0.177</td><td>1.138</td><td>2.011</td><td>2.105</td><td>0.316</td><td>0.196</td><td>0.484</td><td>1.001</td><td>0.338</td><td>0.438</td><td>0.951</td><td>2.908</td><td>0.216</td><td>1.995</td><td>2.908</td></tr></table>				$N_{Ed}/N_{c,Rd}$	α_{Ed}	U_{Ny}	U_{My}	$Ax+M_{6.6}$	$Ax+M_{6.6}$	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	0.054	0.480	0.177	1.138	2.011	2.105	0.316	0.196	0.484	1.001	0.338	0.438	0.951	2.908	0.216	1.995	2.908
$N_{Ed}/N_{c,Rd}$	α_{Ed}	U_{Ny}	U_{My}	$Ax+M_{6.6}$	$Ax+M_{6.6}$	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max																					
0.054	0.480	0.177	1.138	2.011	2.105	0.316	0.196	0.484	1.001	0.338	0.438	0.951	2.908	0.216	1.995	2.908																					

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.908

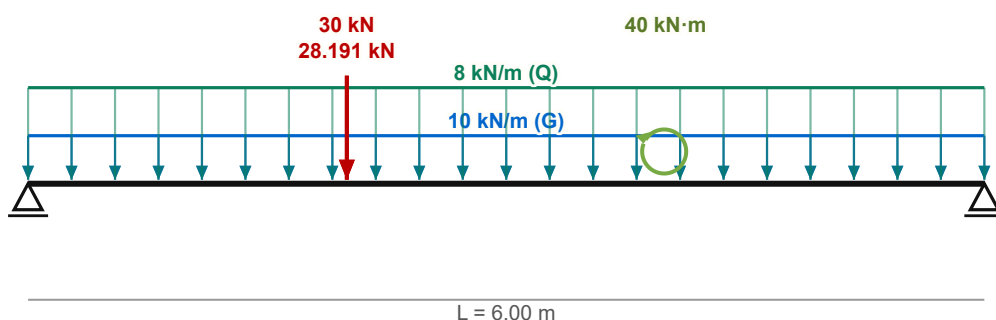
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

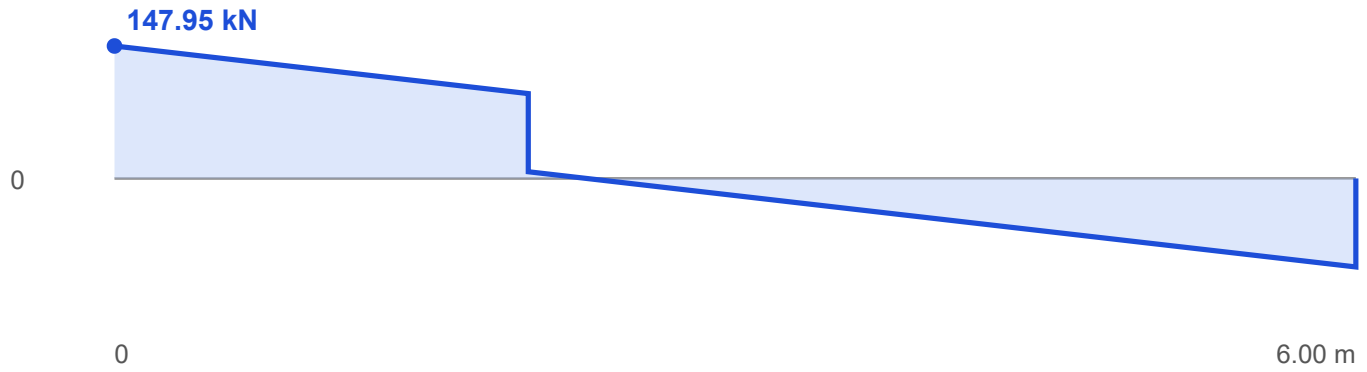
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 2.105$ (Eq 6.62 with $M_{z,tot} = 39.11$ kN.m) + 0.529 (warping term of (A.1) at x = 2.00 m, $k_w = 0.646$, $k_{zw} = 0.551$, $k_\alpha = 5.524$) = 2.634. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

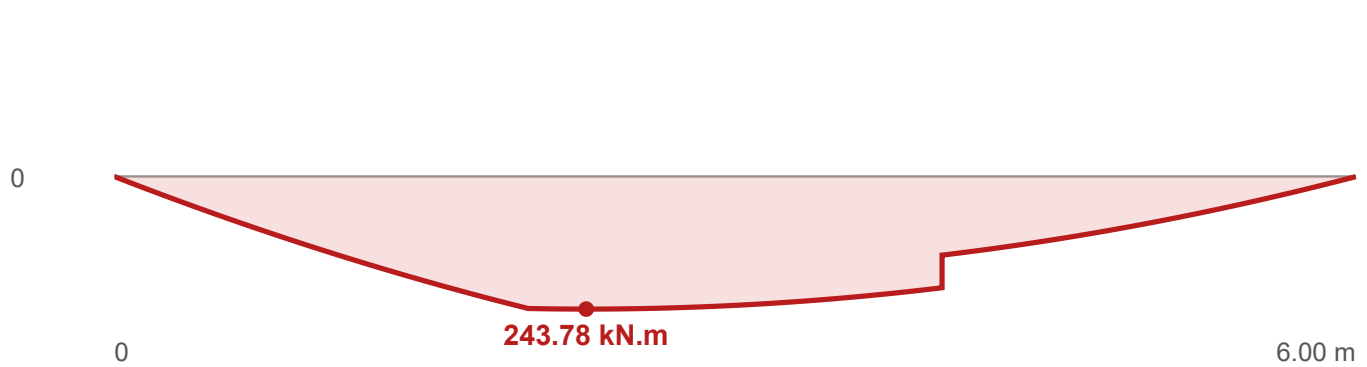
UB 457 x 191 x 82, 6 m SS, unrestrained; $L7 = L6 + \text{"EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM"}: \text{couple } 40 \text{ kN.m (Q, factored by the combination) at } 4 \text{ m, applied as a pure } M_y \text{ in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z g); only forces with a lever arm do}$



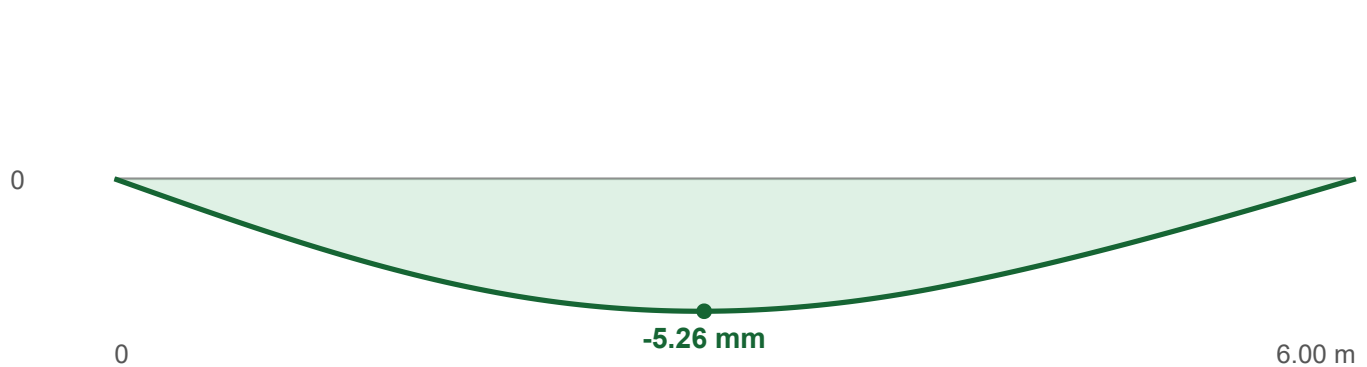
SHEAR FORCE V (kN)



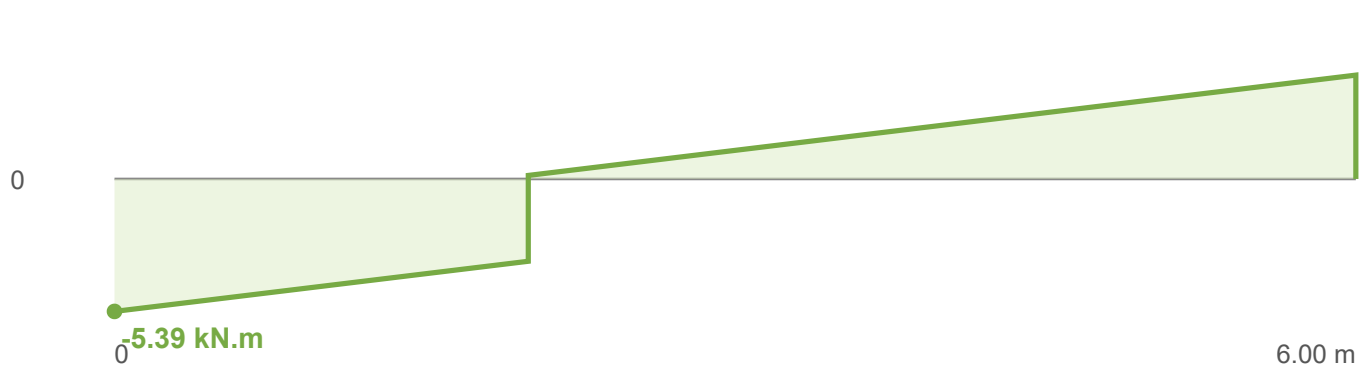
BENDING MOMENT M (kN.M)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 243.78, 20.52	Class 1
	(Axial: Slender web)	
Web classified for	bending + compression: $\alpha = 0.568$, $\psi = -0.895$; limits 62.1ε / 71.5ε / 112.1ε (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1 Table 5.2

Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 182.4, tip 270.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 149.3, tip 61.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (\delta/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² x 275/($\sqrt{3} \times 1$); $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed}/V_{pl,y,Rd}$	147.949 / 756.260 =	0.196	OK
$h_w/t_w \leq 72\epsilon/\eta$	43.23 $\leq 72 \times 0.924/1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	0.046 / 756.260 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	243.781 / 503.250 =	0.484	OK
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 x 304.0/1	83.600 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	104.00 x 275/1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} \cdot f_y/\gamma_{M0}$	100.66 x 275/1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed}/N_{c,Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	1830.0 x 275/1	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	304.0 x 275/1	83.600 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(243.781/503.250)^2 + (20.521/83.600)^1 =$	0.480	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ $x = 2 \text{ m}$ (ULS: 1.35G + 1.5Q): $M_y = 242.726 \text{ kN.m}$, $V = 51.134 \text{ kN}$, $\phi = 0.07008 \text{ rad}$, $\phi' = 1.806\text{e-}5 \text{ rad/mm}$, $\phi'' = 2.483\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 1.206\text{e-}11 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 17.010 = 37.531 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + E W_{n0} \phi'' = 14.42 + 150.76 + 191.48 + 110.53 = 467.20 \text{ N/mm}^2$; $\tau = G t \phi' = 23.41 \text{ N/mm}^2$; $(467.20/275)^2 + 3(23.41/275)^2 =$	2.908	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 14.42 + 150.76 + 191.48 + 110.53 = 467.20$, $\tau = 23.41 \rightarrow 2.908$ (governs); P2 web-flange junction, flange: $\sigma_x = 14.42 + 145.24 + 9.93 = 169.60$, $\tau = 2.93 + 23.41 + 2.56 = 28.90 \rightarrow 0.413$; P3 web-flange junction, web: $\sigma_x = 14.42 + 140.01 + 9.93 = 164.37$, $\tau = 9.46 + 14.49 = 23.95 \rightarrow 0.380$; P4 web mid-depth: $\sigma_x = 14.42 + 9.93 = 24.36$, $\tau = 12.62 + 14.49 = 27.10 \rightarrow 0.037$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000$ mm, the full member length (conservative)	6.4(1)

Governing station $x = 0$ m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 147.949$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 = 0.338 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 + N_{Ed}/N_{pl,Rd} = 150.000/2860.000 = 0.052) = 0.380 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.338	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.272	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	147.949	ULS: 1.35G + 1.5Q	0.338	0.272	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.438	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	98.853	ULS: 1.35G + 1.5Q	0.226	0.191	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y)$: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} f_y$; $\lambda = \sqrt{(A_{eff} f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10/1 =$	2664.536 kN	Curve a

$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot \lambda_z \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 845.783 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.177	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 236.9, 1.000, 300.000	1.212	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=192.0$, $M(L/2)=236.9$, $M(3L/4)=118.4$, $M_{max}=243.8$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 236.92$, $\psi = 0.664$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 18.592 twist ϕ . $M_y = 39.113$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 236.92$, $\psi = 0.664$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	1 x 6 =	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.212, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	415.519 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 415.519}$	1.101	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.101, 1.126, 0.49	0.580	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.580, 1.101, 0.908, 0.962	0.602	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.602 \times 1830.0 \times 275 \leq 503.250 =$	303.127 kN.m	
λ_{LT} (P362 6.55 simplified); $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1}) \cdot 0.9 \lambda_z \sqrt{\beta_w} = 1.336$; $\chi_{LT,mod} = 0.467$; $M_{y,Ed} / M_{b,Rd,simp} = 1.037$	235.066 kN.m	P362 6.55 comparison
$M_{y,Ed} / M_{b,Rd}$	243.781 / 303.127	0.804	OK

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$243.781 / 266.685$ ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.914	OK
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$(20.521 \text{ imposed} + 18.592 \text{ twist } \phi \cdot M_y = 39.113) / 53.900$	0.726	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \lambda_z U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 (1 + 0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 (1 + 0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 0.914 + 1.106 \times 0.726$	1.784	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 0.914 + 1.106 \times 0.726$	1.884	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ³ , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.066 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
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Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
$\varphi_{\max}$ (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	$0.0771 \text{ rad} = 4.42^\circ$	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2 \text{ m}$	4.807 kN.m^2	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	10.827 kN.m	
$M_{z,tot} = M_{z,Ed} + \varphi \cdot M_{y,Ed}$	$20.521 + 242.084 \times 76.8\text{e-}3$ (@ $x = 2.639 \text{ m}$) = $20.521 + 18.592$ (imposed constant $M_{z,Ed}$ + max coincident twist moment)	39.113 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2.066 \text{ m}$: $243.164, 10.656, 37.840 \text{ kN.m}$ ($M_{z,tot} = 20.521 + 17.319$); $M_{pl,y} = 503.250, M_{pl,f} = 40.255, M_{pl,z} = 83.600$	0.951	OK P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.647 \times 0.547 \times 2.419$ ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 37.840/83.600$)	0.857	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.066 \text{ m}$: $243.164/291.733 + 1.000 \times 37.840/83.600 + 0.647 \times 0.547 \times 2.419 \times 10.656/40.255$ ($M_{z,tot} = 20.521$ imposed + 17.319 twist $\varphi \cdot M_y$)	1.513	Warning EN 1993-6 A
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi_i M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi_i M_y$		EN 1993-1-1, EN 1993-6
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 1.884$ (Eq 6.62 with $M_{z,tot} = 39.113$) + 0.232 (warping term of (A.1)) @ $x = 2 \text{ m}$: $k_w = 0.646, k_{zw} = 0.551, k_\alpha = 2.419, M_w = 10.827 \text{ kN.m}$ =	2.115	information only
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 5.388 / 4.224 \text{ kN.m}$	$2.460 / 2.113 \text{ kN.m}$	
Torsion Shear Design @ 0 m			
T_t at the V-T station	@ $x = 0 \text{ m}$ (ULS: $1.35G + 1.5Q$): $T_t = 2.460 \text{ kN.m}$, $V = 147.949 \text{ kN}$	35.20 N/mm^2	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $\tau_t = 35.20 \text{ N/mm}^2$; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.907$	685.927 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	$147.949 / 685.927$	0.216	OK
Deflection Check - Load Case 1 (SLS: 1.0Q)			
In-span $\delta \leq \text{Span}/360$	$5.26 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	5.26 mm	OK
Torq in Case 1 @ 2.714 m : $\theta_{\max} \leq 2.00^\circ$	$0.0363 \text{ rad} = 2.08^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	2.08°	> 2° advisory
$N_{Ed}/N_{c,Rd}$	UN_yz	UM_yz	$AX+M_{6.6}$
0.054	0.480	0.177	0.914
			1.784 1.884
			0.316
			0.196
			0.484
			0.804
			0.338
			0.438
			0.951
			2.908
			0.216
			1.513 2.908

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 2.908

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(\chi_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(\chi_{LT} M_{y,Rk}/\gamma_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.884$ (Eq 6.62 with $M_{z,tot} = 39.11 \text{ kN}\cdot\text{m}$) + 0.232 (warping term of (A.1) at $x = 2.00 \text{ m}$, $k_w = 0.646$, $k_{zw} = 0.551$, $k_\alpha = 2.419$) = 2.115. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

UC-L1 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 1.169

Owner case: vertical loads resulting in no torsion

UC 203 x 203 x 60, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre (e = 0, $z_g = 0$)

Beam & Beam-Portion (Member) (FAIL)
Member UC-L1 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

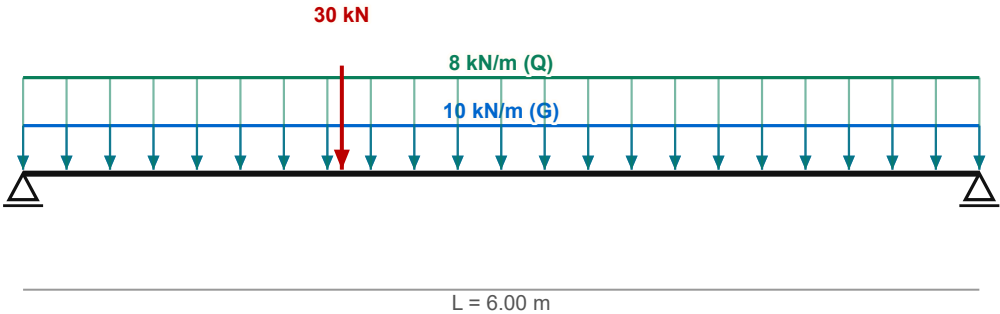
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

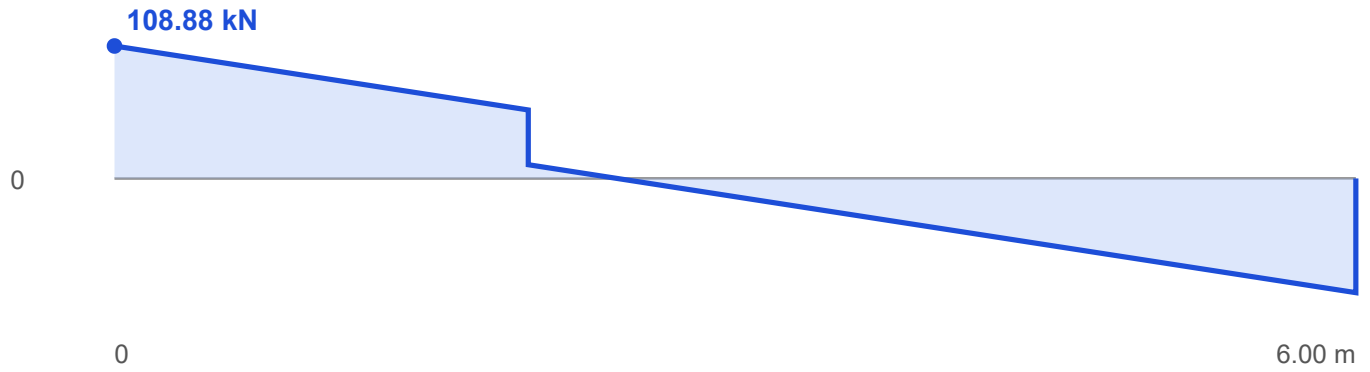
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

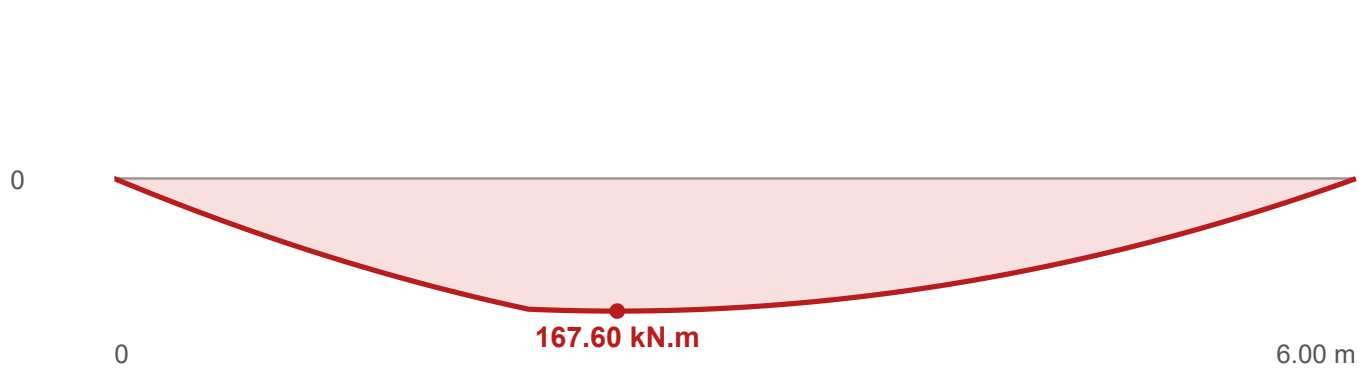
LOADING



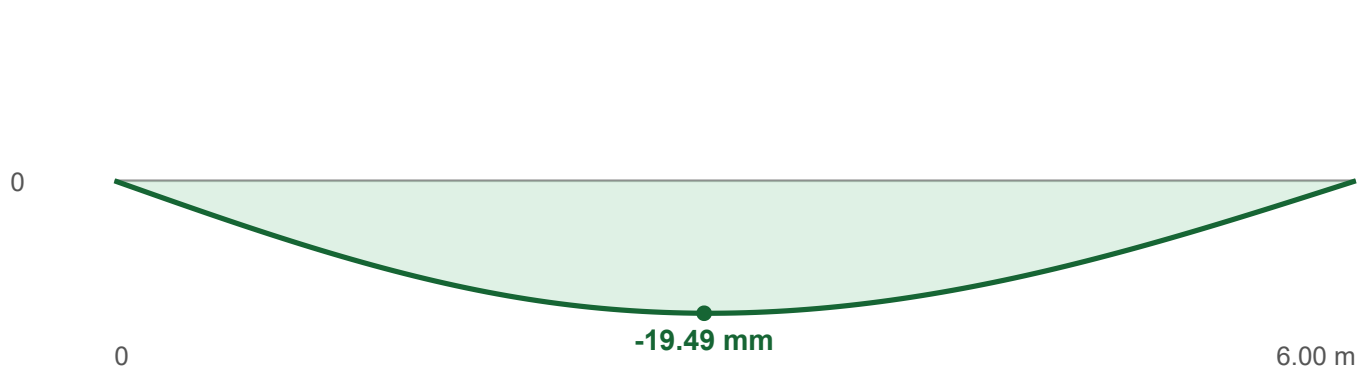
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

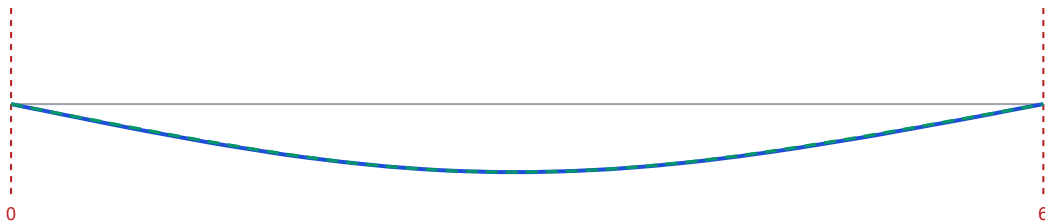
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	277.879 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.146	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN.m	1.163	Serna

Lateral Buckling Check $M_{b,Rd}$

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	277.879 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 277.879}$	0.806	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.806, 0.812, 0.34	0.814	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.814, 0.806, 0.934, 0.967	0.842	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.842 \times 656.0 \times 275 \leq 180.400 =$	151.844 kN.m	
$M_{y,Ed}/M_{b,Rd}$	167.604 / 151.844	1.104	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.104	1.169	0.309	0.253	0.600	1.104

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L1 · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 1.169

Owner case: vertical loads resulting in no torsion

UC 203 x 203 x 60, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre (e = 0, z_g = 0)

Beam & Beam-Portion (Member) (FAIL)
Member UC-L1 UC 203 x 203 x 60
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

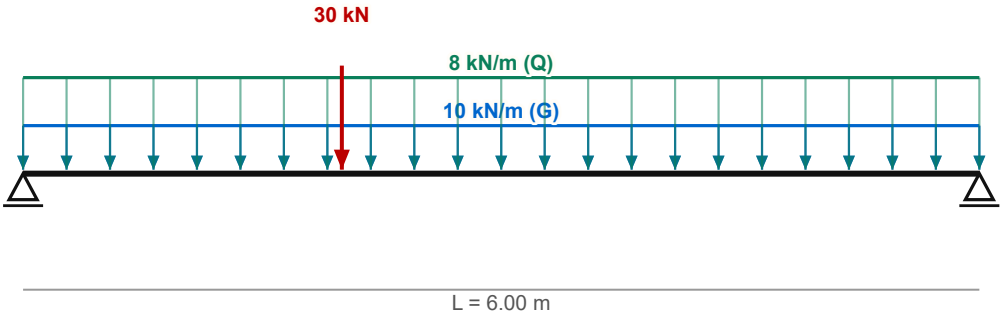
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)

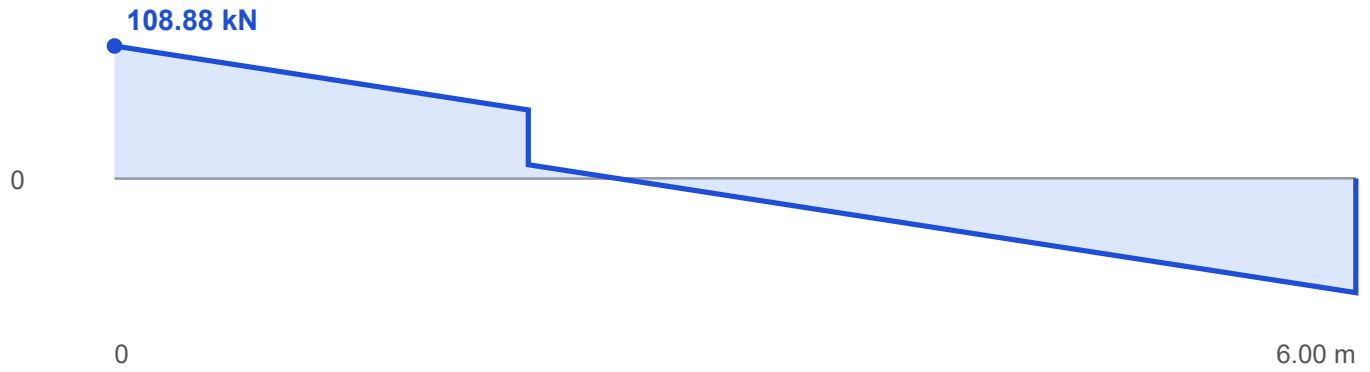
$R_{min} \geq 0$

OK

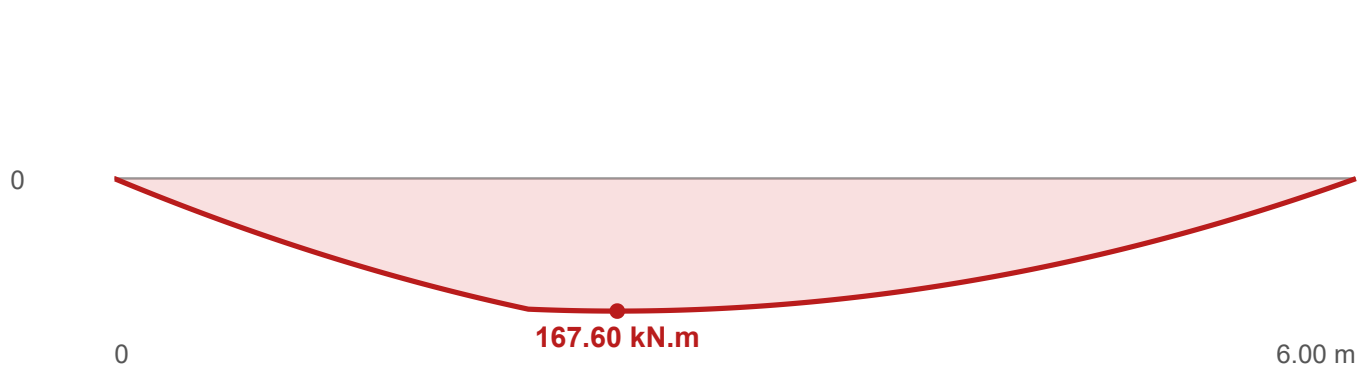
LOADING



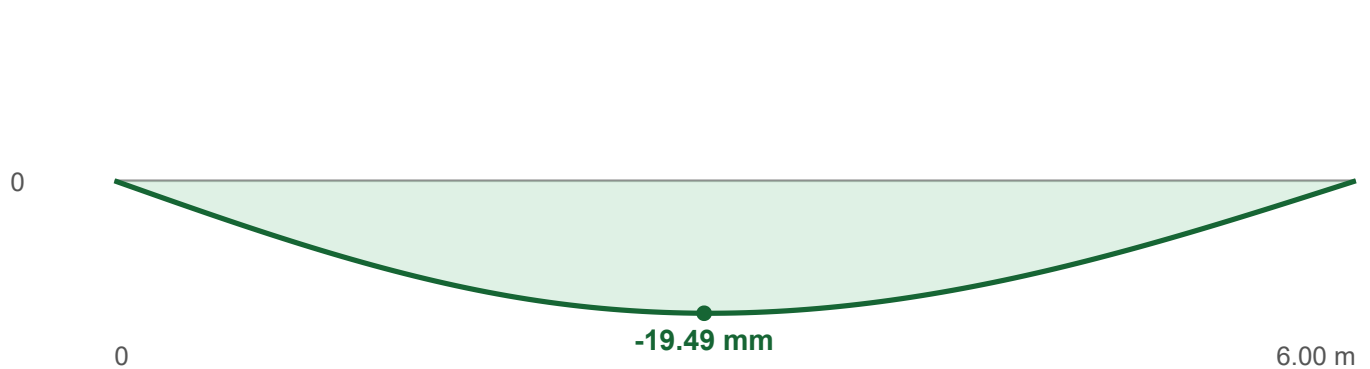
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{\max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	$1.163, 6.000, 2060, 47.2, 0.197, 210000$; $z_g = 0$ (load through the shear centre)	281.930 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$167.604 / 152.975$	1.096	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
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MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L2 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.650
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member UC-L2 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.5886 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.26	4.26	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6.00C	-3.66	-3.66	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

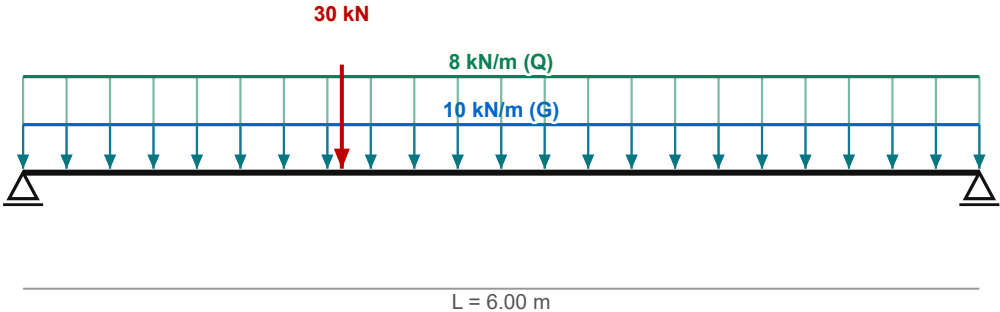
Uplift

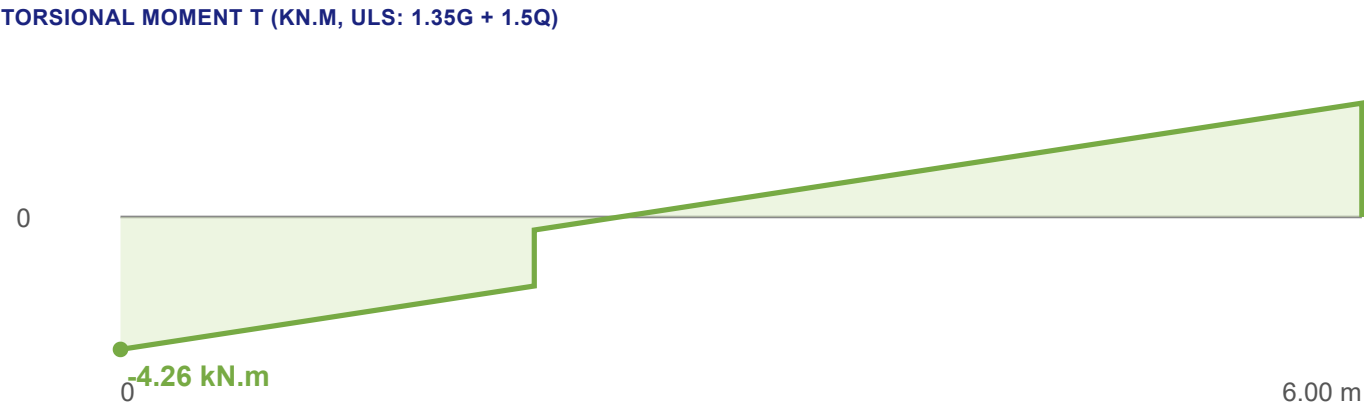
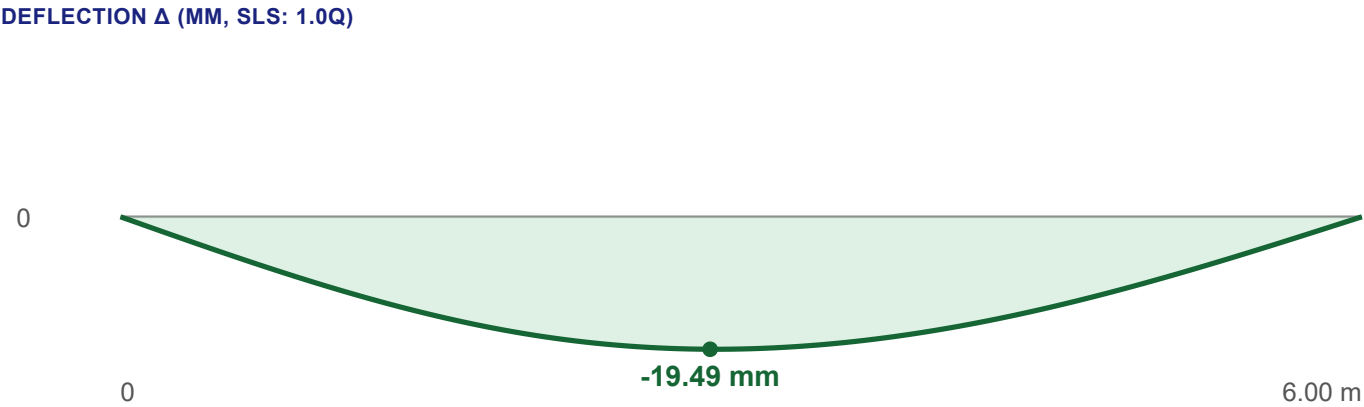
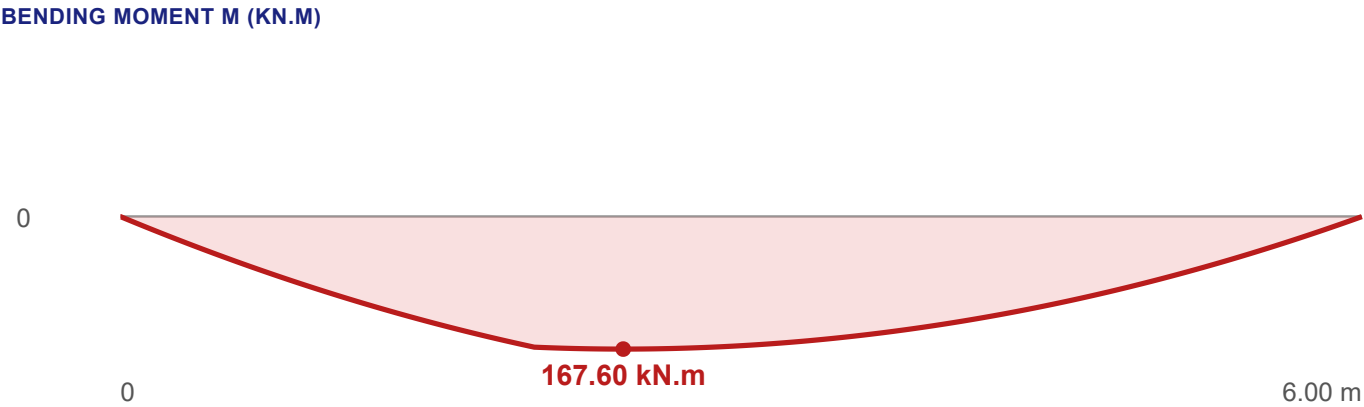
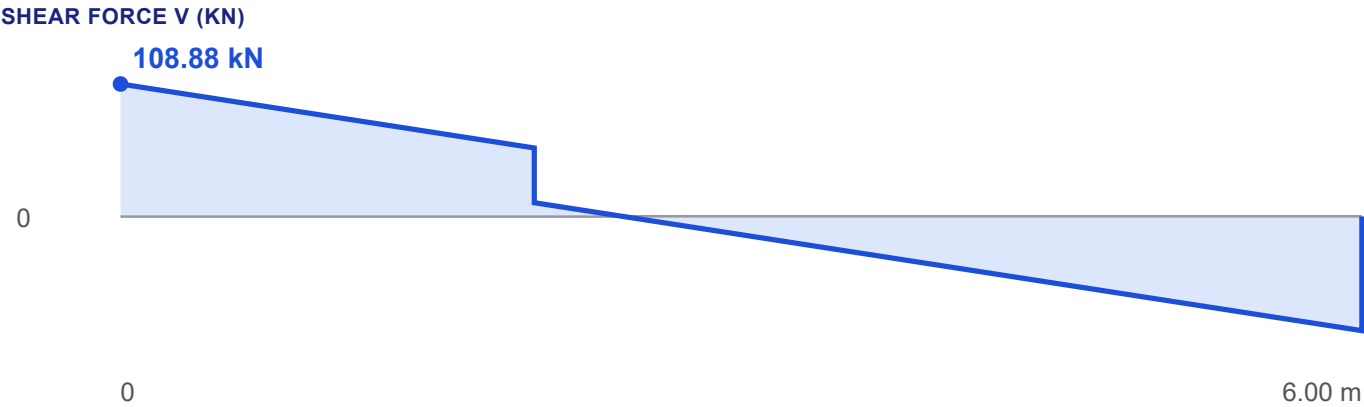
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (60.0 kg/m)	203 x 203 x 60 UC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	6.20, 17.10, 275, 0.00, 167.60, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E \omega_n \phi'' = 0.00 + 283.60 + 101.12 + 92.52 = 477.25 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(477.25/275)^2 + 3(33.07/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 283.60 + 101.12 + 92.52 = 477.25$, $\tau = 33.07 \rightarrow 3.055$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 264.40 + 4.64 = 269.04$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.010$; P3 web-flange junction, web: $\sigma_x = 0.00 + 245.19 + 4.64 = 249.83$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.854$; P4 web mid-depth: $\sigma_x = 0.00 + 4.64 = 4.64$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.031$ ($y_w = t_w/2 = 4.70 \text{ mm}$; $S_x/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \varepsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	X_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

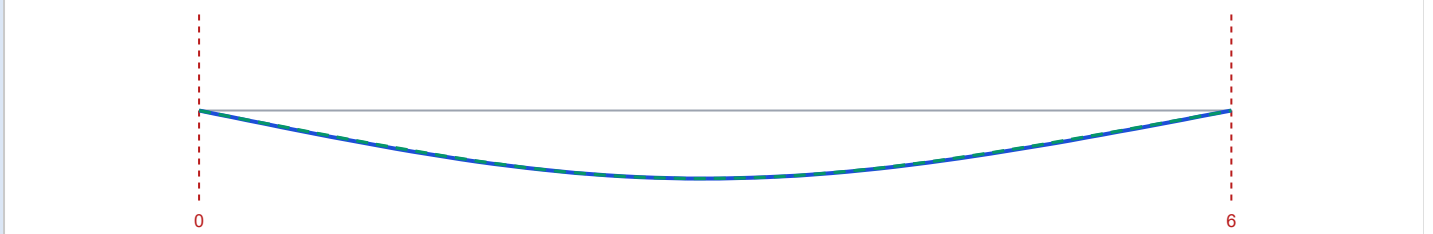
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	277.879 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.146	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	277.879 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W_{f,y}/M_{cr}}$	$\sqrt{656.0 \times 275 / 277.879}$	0.806	
$\chi_{LT} = Fn(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	0.806, 0.812, 0.34	0.814	Curve b
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.814, 0.806, 0.934, 0.967	0.842	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.842 \times 656.0 \times 275 \leq 180.400 =$	151.844 kN.m	
$M_{y,Ed}/M_{b,Rd}$	167.604 / 151.844	1.104	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
ϕ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.835 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315$ m: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	1.321	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.657 \times 0.753 \times 2.520$	1.245	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.116$ m: $166.313/146.844 + 1.000 \times 20.748/83.875 + 0.657 \times 0.753 \times 2.520 \times 8.969/41.348$	1.650	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260 / 3.660$ kN.m	2.943 / 2.568 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 2.943$ kN.m, V = 108.884 kN	58.60 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 58.60$ N/mm ² ; $V_{pl.Rd} = 352.225$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.839$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	108.884 / 295.686	0.368	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
Torq in Case 1 @ 2.664 m: $\theta_{max} \leq 2.00^\circ$	0.0557 rad = $3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	3.19°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.929	1.104	1.169	0.309	0.253	0.600	1.321	3.055	0.368	1.650	1.650

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.650

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 3.055 at x = 2.04 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with M_{z,tot}); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

UC 203 x 203 x 60, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at $e = +40$ mm (torque), $z_g = 0$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

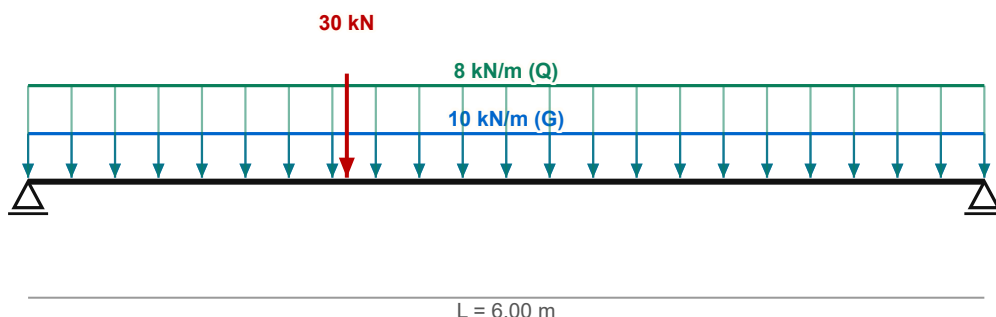
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

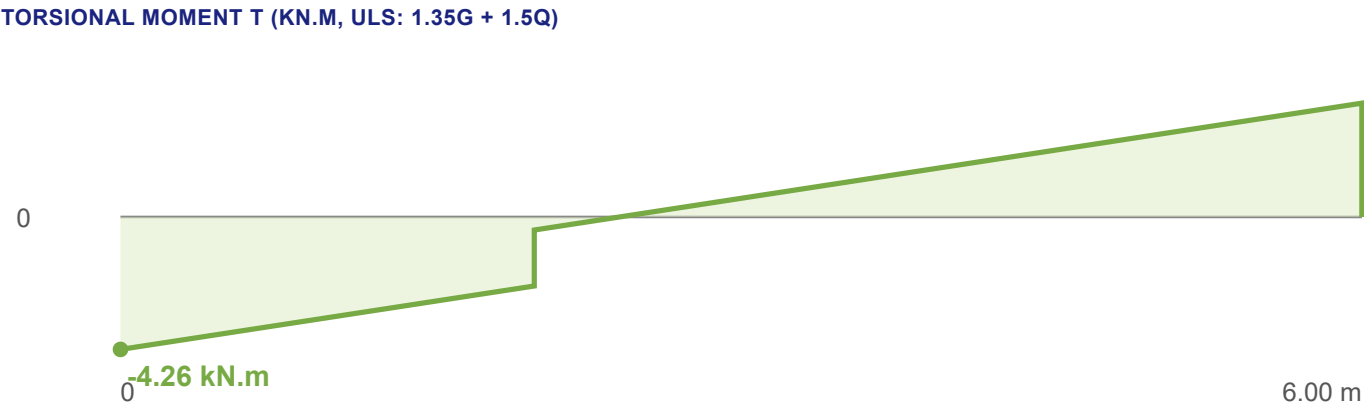
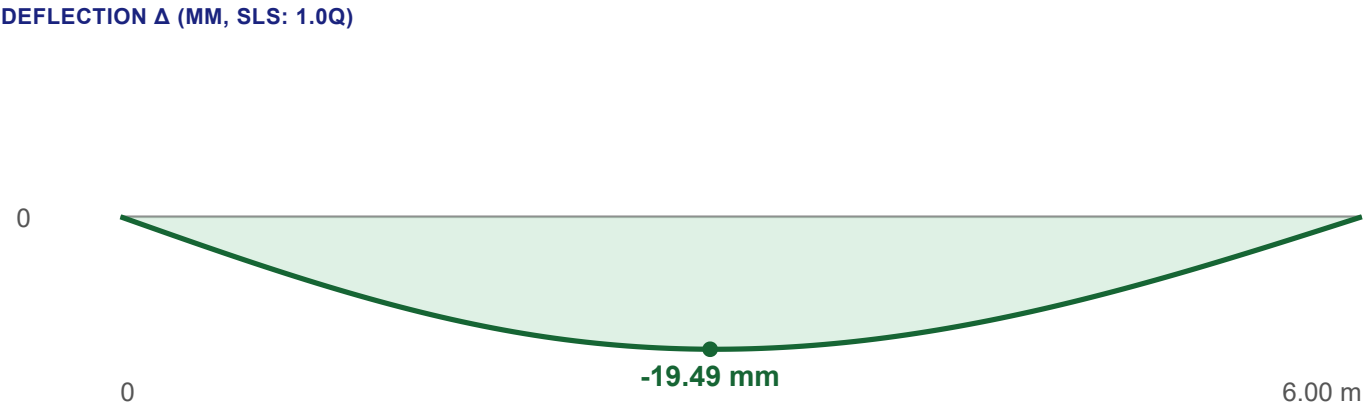
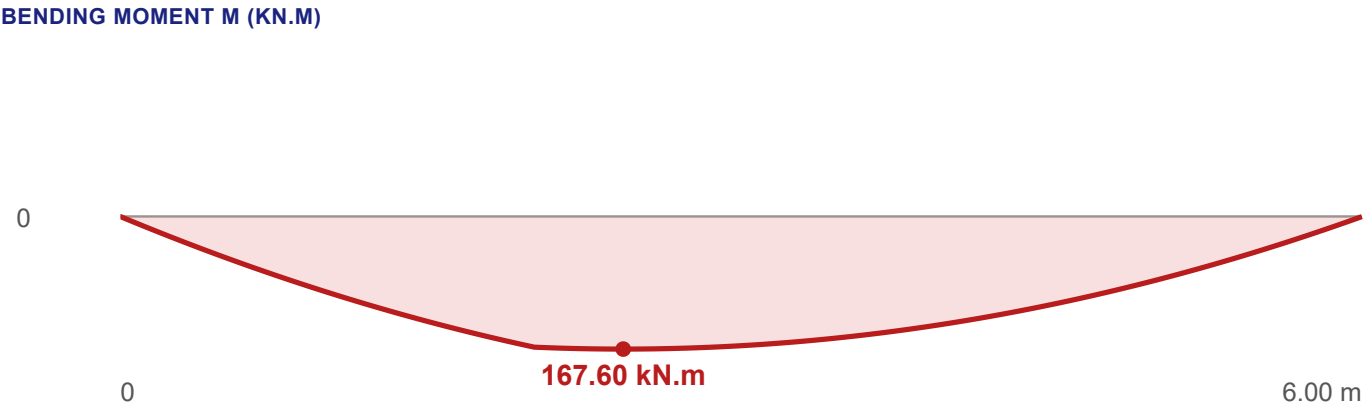
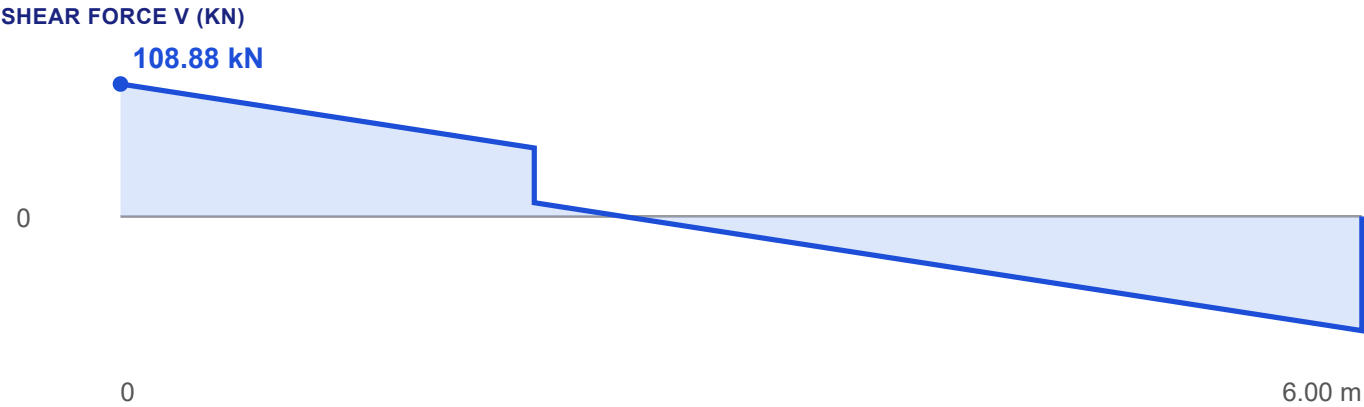
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.26	108.88	0.00	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6.00C	-3.66	-93.88	0.00	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (60.0 kg/m)	203 x 203 x 60 UC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	6.20, 17.10, 275, 0.00, 167.60, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E \omega_n \phi'' = 0.00 + 283.60 + 101.12 + 92.52 = 477.25 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(477.25/275)^2 + 3(33.07/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 283.60 + 101.12 + 92.52 = 477.25$, $\tau = 33.07 \rightarrow 3.055$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 264.40 + 4.64 = 269.04$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.010$; P3 web-flange junction, web: $\sigma_x = 0.00 + 245.19 + 4.64 = 249.83$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.854$; P4 web mid-depth: $\sigma_x = 0.00 + 4.64 = 4.64$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.031$ ($y_w = t_w/2 = 4.70 \text{ mm}$; $S_x/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \varepsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M _{max} =167.6 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.163, 6.000, 2060, 47.2, 0.197, 210000; z _g = 0 (load through the shear centre)	281.930 kN.m	SN003a
λ _{LT} = √W _{f,y} /M _{cr}	√ 656.0 x 275 / 281.930	0.800	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.800, 0.808, 0.34	0.817	Curve b
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
M _{b,Rd} = χW _{pl,y} ·f _y ≤ M _{c,y,Rd}	0.848 x 656.0 x 275 ≤ 180.400 =	152.975 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.109; χ _{LT,mod} = 0.652; M _{y,Ed} /M _{b,Rd,simp} = 1.425	117.633 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	167.604 / 152.975	1.096	Warning

Torsion Design

J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2.315 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 5.77	P385 App C
φ _{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	1.835 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	9.391 kN.m	
M _{z,Ed} = φ·M _{y,Ed}	max coincident	22.160 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.315 m: 167.433, 8.337, 21.602 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.348, M _{pl,z} = 83.875	1.321	Warning P385 3.1.2
k = k _w ·k _{zw} ·k _α	0.657 x 0.753 x 2.466	1.219	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.116 m: 166.313/147.421 + 1.000x20.748/83.875 + 0.657x0.753x2.466x8.969/41.348	1.640	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.260 / 3.660 kN.m	2.943 / 2.568 kN.m	

Torsion Shear Design @ 0 m

T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 2.943 kN.m, V = 108.884 kN	58.60 N/mm ²	P385
V _{pl,T,Rd} = √(1 - T _t /(1.25f _y /√3))·V _{pl,Rd}	@ x = 0 m: T _t = 58.60 N/mm ² ; V _{pl,Rd} = 352.225; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.839	295.686 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	108.884 / 295.686	0.368	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	19.49 mm	Warning
Torq in Case 1 @ 2.664 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0557 \text{ rad} = 3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	3.19°	> 2° advisory

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.321	3.055	0.368	1.640	1.640

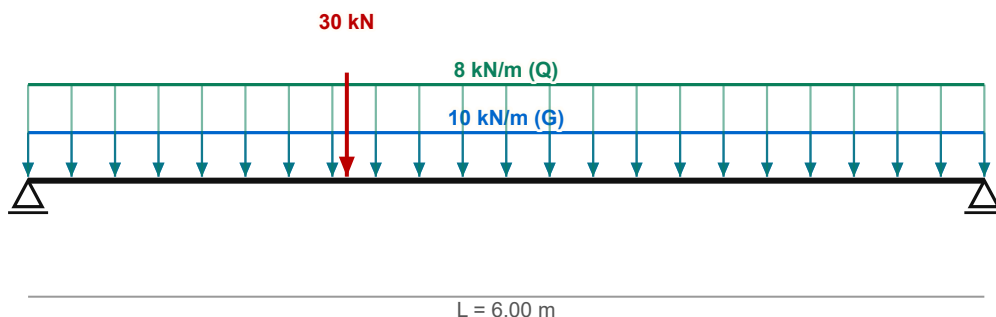
FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.640

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

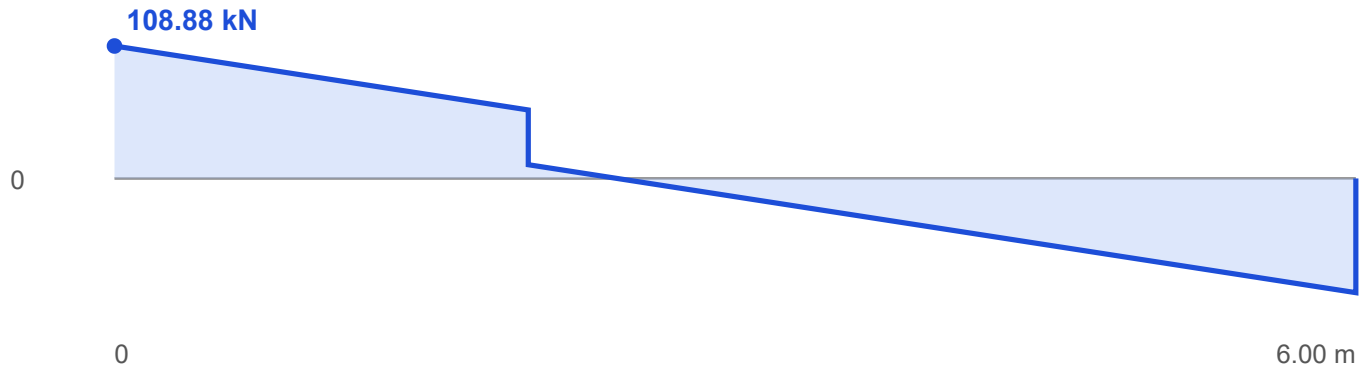
ADVISORY: Serviceability twist $\theta_{\text{max}} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

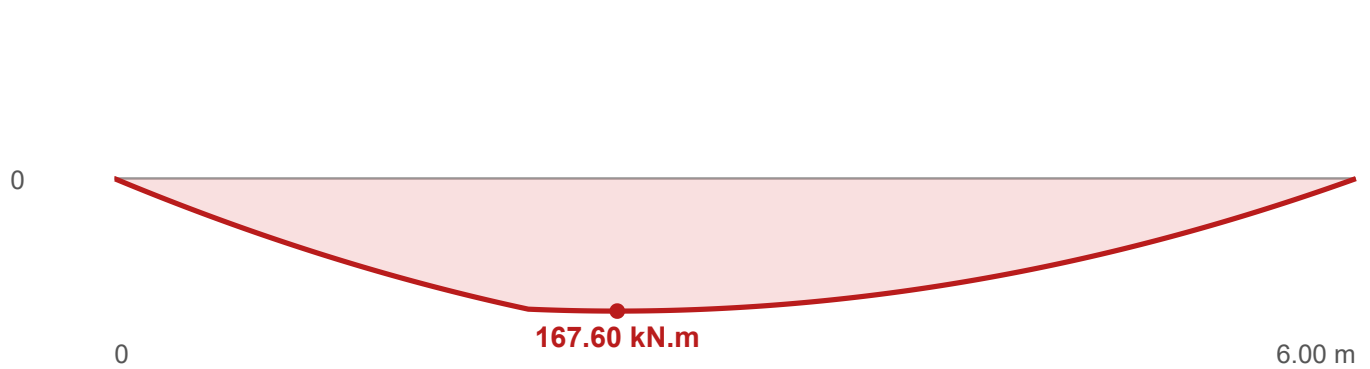
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 3.055 at x = 2.04 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



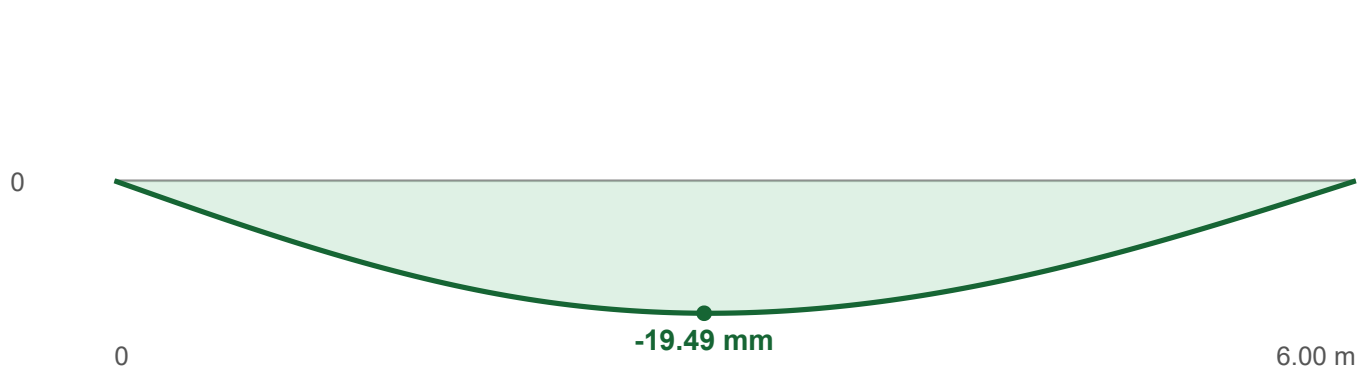
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

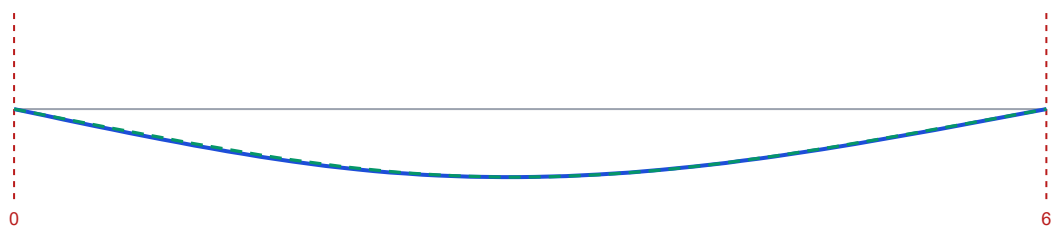
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	277.879 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.146	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN.m	1.163	Serna

Lateral Buckling Check $M_{b,Rd}$

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 105, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	219.624 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 219.624}$	0.906	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.906, 0.894, 0.34	0.756	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.756, 0.906, 0.934, 0.968	0.781	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.781 \times 656.0 \times 275 \leq 180.400 =$	140.966 kN.m	
$M_{y,Ed}/M_{b,Rd}$	167.604 / 140.966	1.189	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.189	1.169	0.309	0.253	0.600	1.189

FAIL — governing LTB $M_{Ed}/M_{b,Rd} = 1.189$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3t · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 1.169

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

UC 203 x 203 x 60, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +104.8 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

Member UC-L3t UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = 105 mm (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

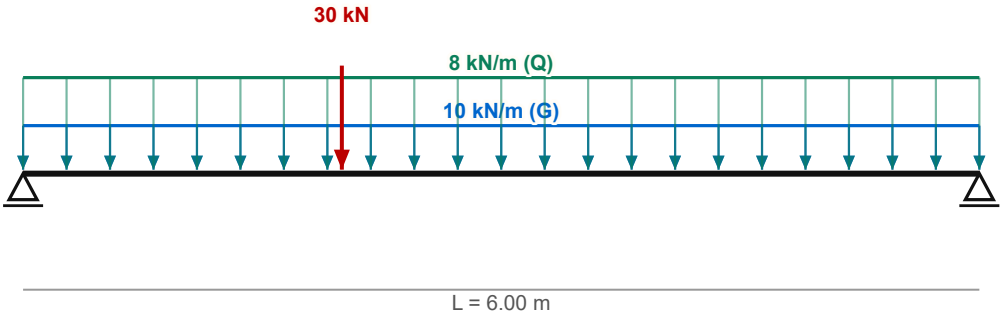
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

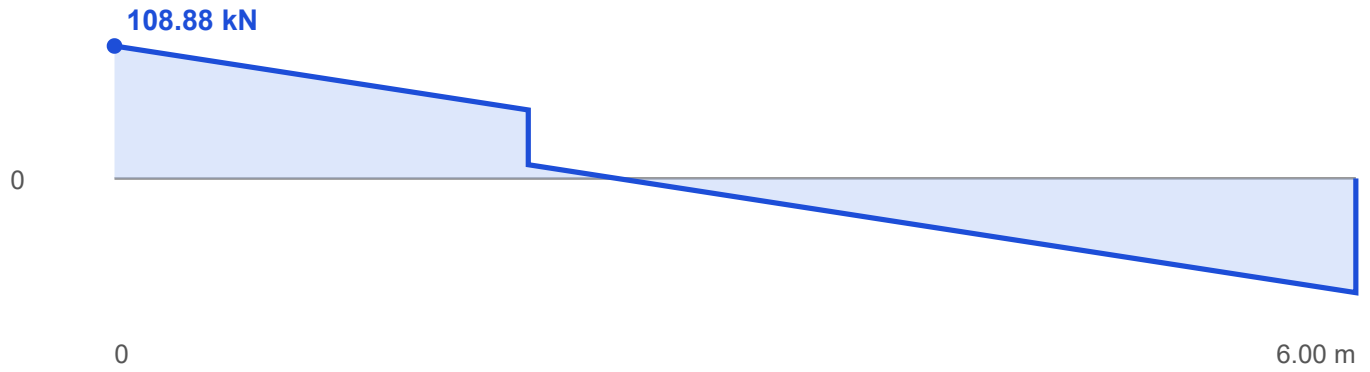
R_{min} ≥ 0

OK

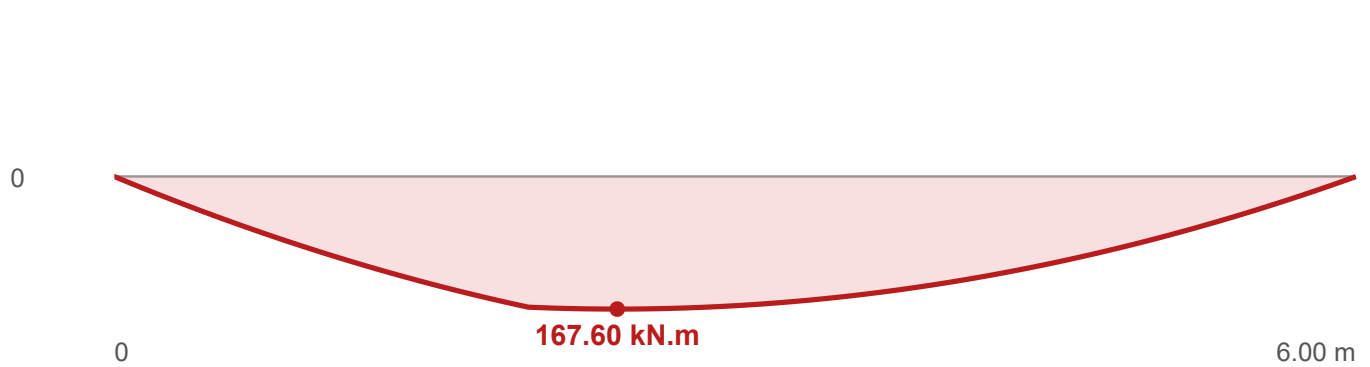
LOADING



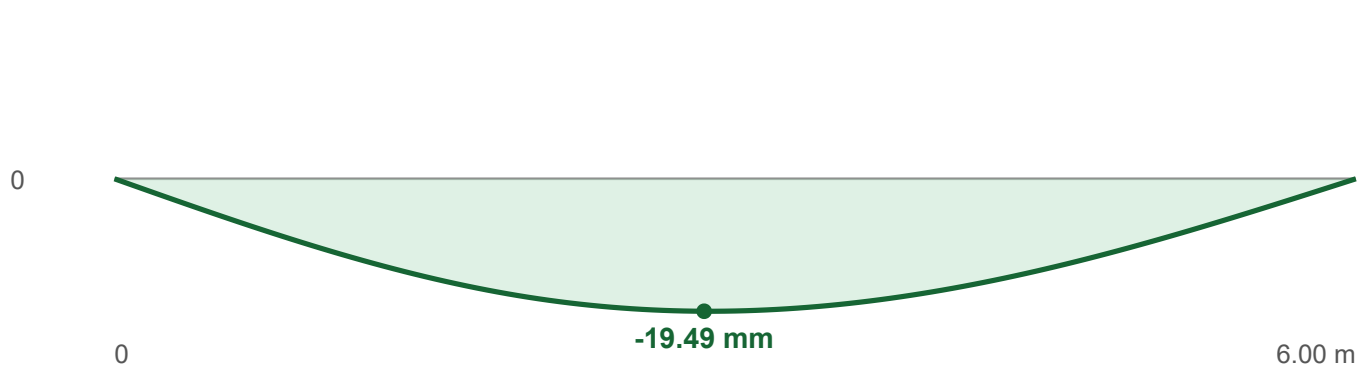
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343)} = 0.306$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 = 0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
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$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_y f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	19.49 mm	Warning
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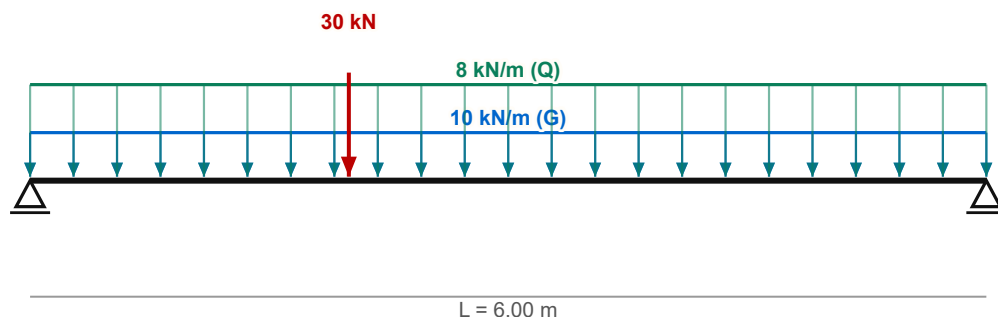
MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

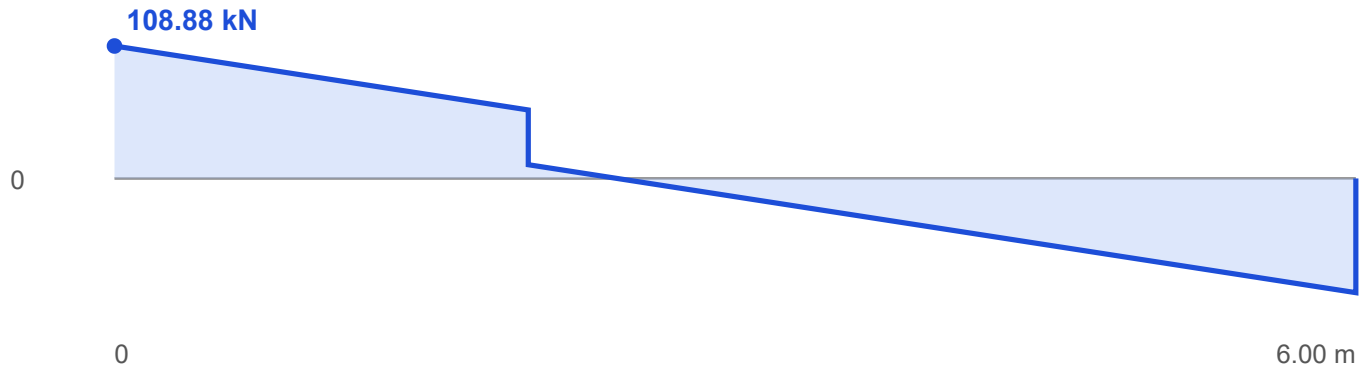
NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

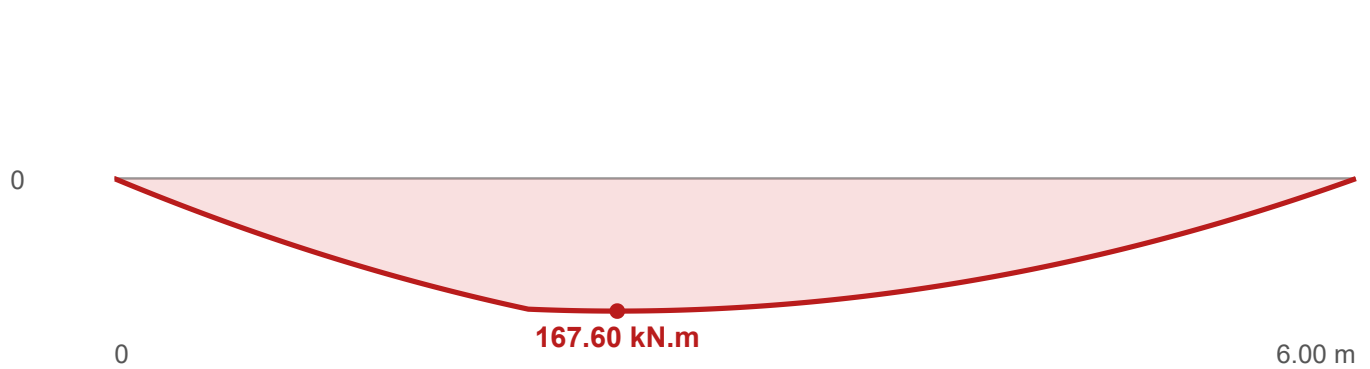
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



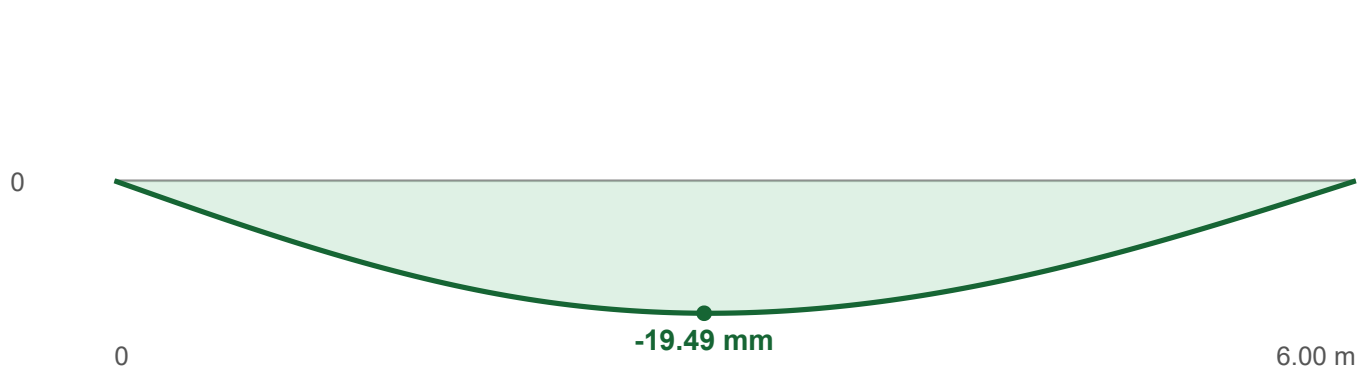
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

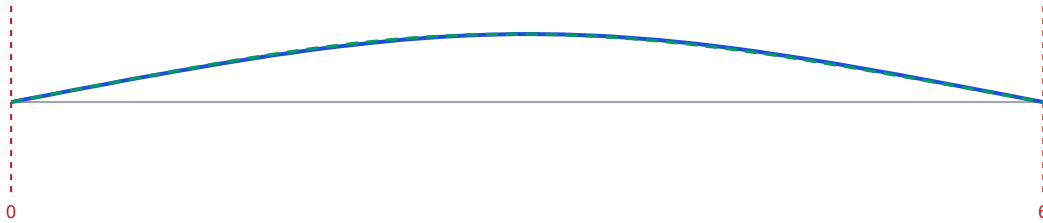
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	277.879 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.146	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN.m	1.163	Serna

Lateral Buckling Check $M_{b,Rd}$

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -105, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	350.653 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 350.653}$	0.717	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.717, 0.747, 0.34	0.861	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.861, 0.717, 0.934, 0.968	0.890	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.890 \times 656.0 \times 275 \leq 180.400 =$	160.527 kN.m	
$M_{y,Ed}/M_{b,Rd}$	167.604 / 160.527	1.044	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 3.02$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm **Warning**

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.929	1.044	1.169	0.309	0.253	0.600	1.044

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3b · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 1.169

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

UC 203 x 203 x 60, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -104.8 mm (-D/2, stabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

Member UC-L3b UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 0 mm z_g = -105 mm (kN/m)

Q UDL 8.000 0-6 m e = 0 mm z_g = -105 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = -105 mm (kN)

G SW 0.5886 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

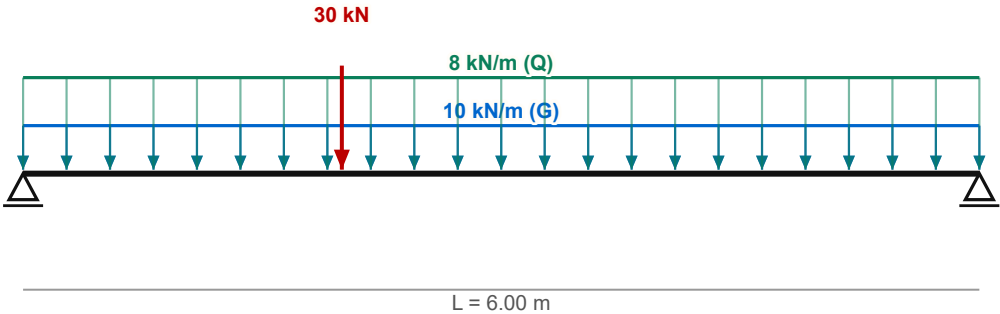
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

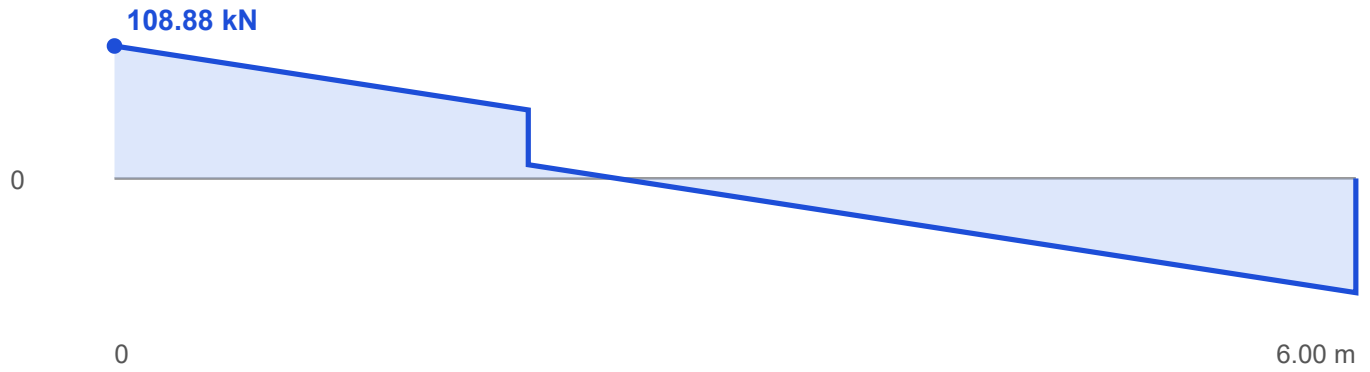
R_{min} ≥ 0

OK

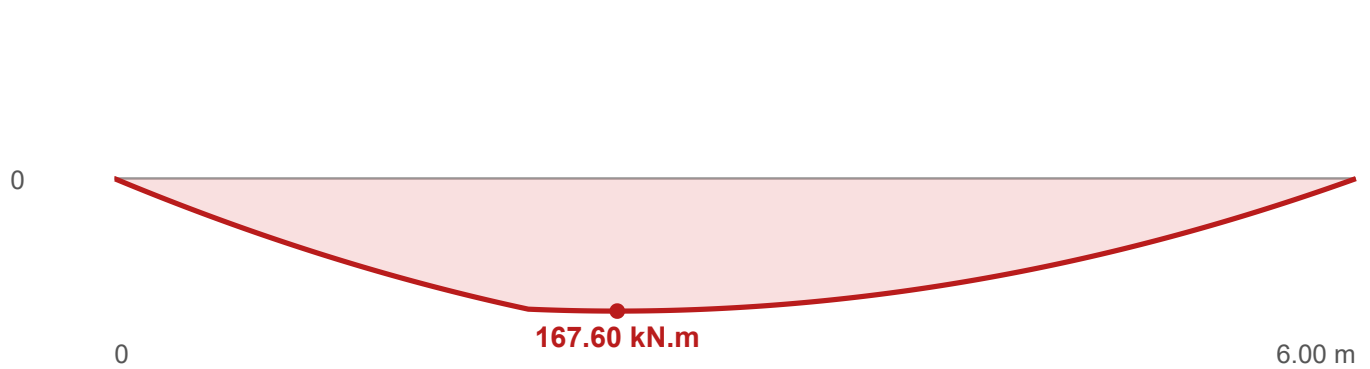
LOADING



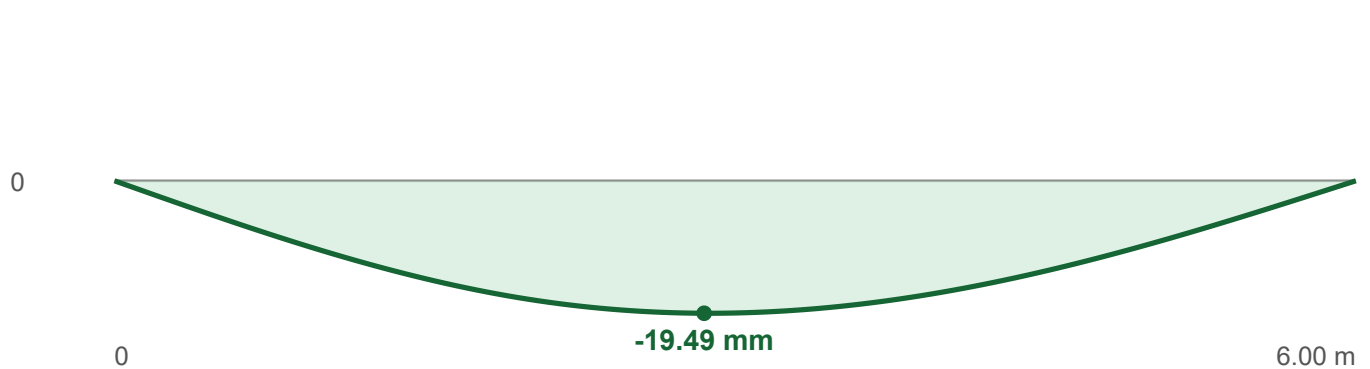
SHEAR FORCE V (kN)



BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check $M_{b,Rd}$

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = -105$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	281.930 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L4 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.927
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at e = +40 mm and z_g = +104.8 mm (top flange)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member UC-L4 UC 203 x 203 x 60
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

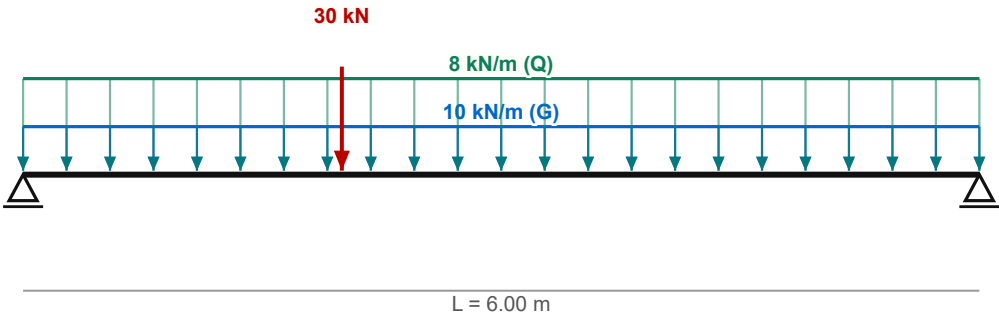
Member Loading and Member Forces

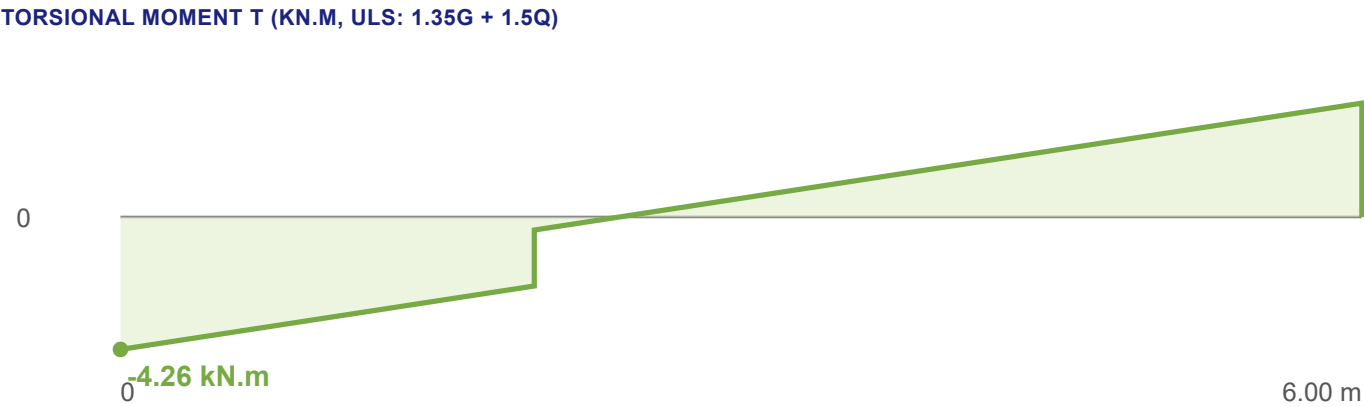
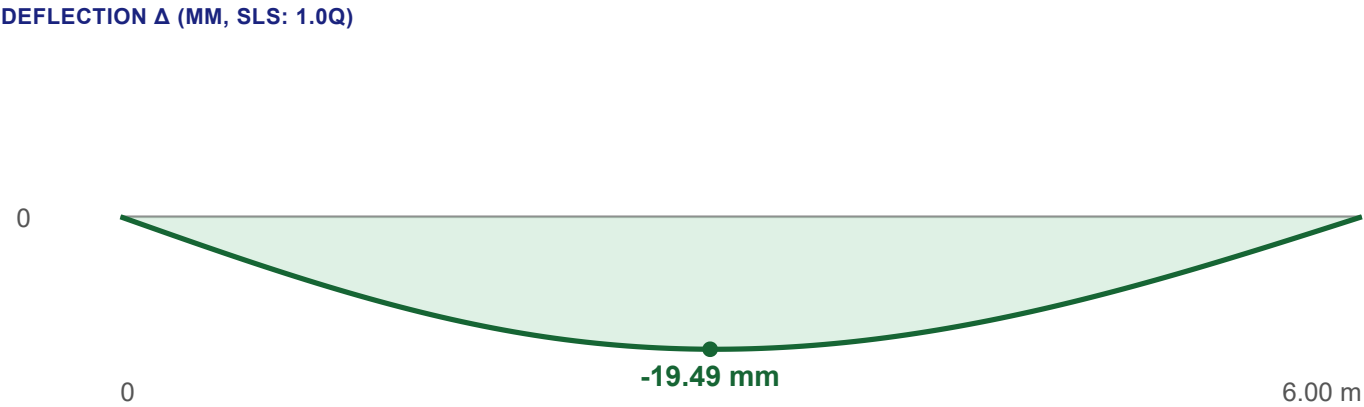
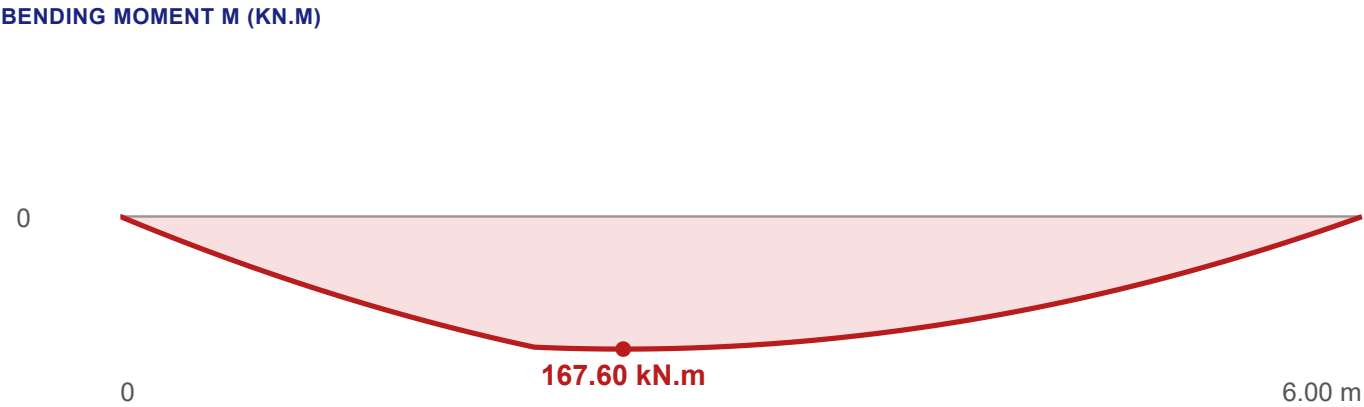
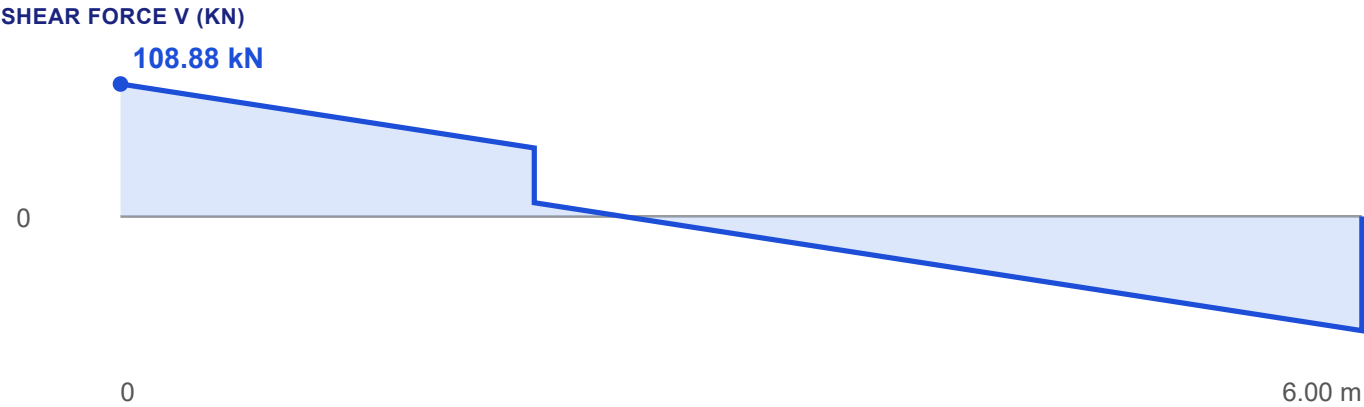
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)
G SW 0.5886 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.26	4.26	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6.00C	-3.66	-3.66	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 0.00, 167.60, 0.00	Class 1
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E \omega_n \phi'' = 0.00 + 283.60 + 101.12 + 92.52 = 477.25 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(477.25/275)^2 + 3(33.07/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 283.60 + 101.12 + 92.52 = 477.25$, $\tau = 33.07 \rightarrow 3.055$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 264.40 + 4.64 = 269.04$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.010$; P3 web-flange junction, web: $\sigma_x = 0.00 + 245.19 + 4.64 = 249.83$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.854$; P4 web mid-depth: $\sigma_x = 0.00 + 4.64 = 4.64$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.031$ ($y_w = t_w/2 = 4.70 \text{ mm}$; $S_x/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \varepsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.181	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

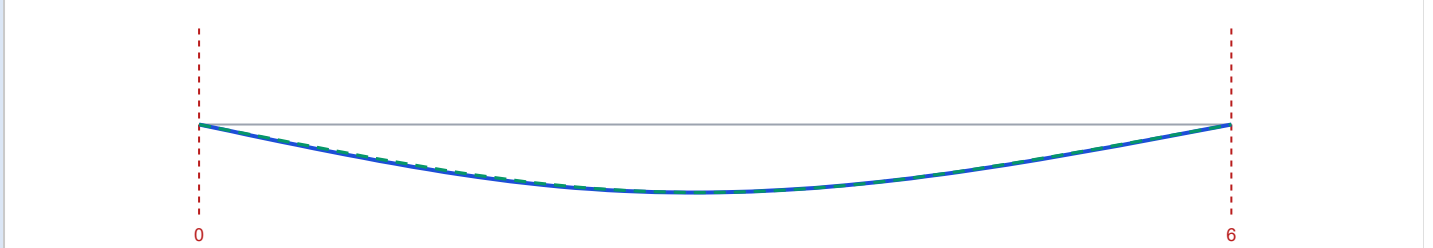
Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	277.879 / 242.476 kN.m (shape only: z _g = z _j = 0)	1.146	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M _{max} =167.6 kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 105, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	219.624 kN.m	converged
λ _{LT} = √W _{f,y} /M _{cr}	√ 656.0 x 275 / 219.624	0.906	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.906, 0.894, 0.34	0.756	Curve b
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.756, 0.906, 0.934, 0.968	0.781	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.781 x 656.0 x 275 ≤ 180.400 =	140.966 kN.m	
M _{y,Ed} /M _{b,Rd}	167.604 / 140.966	1.189	Warning

Buckled mode shape — critical eigenmode (normalised; twist φ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.84 m)



Torsion Design

J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 5.77	P385 App C
φ _{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	1.835 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	9.391 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	22.160 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.315 m: 167.433, 8.337, 21.602 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.348, M _{pl,z} = 83.875	1.321	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	0.655 x 0.761 x 4.222	2.102	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 165.178/136.428 + 1.000x20.068/83.875 + 0.655x0.761x4.222x9.391/41.348	1.927	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.260 / 3.660 kN.m	2.943 / 2.568 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 2.943$ kN.m, $V = 108.884$ kN	58.60 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 58.60$ N/mm ² ; $V_{pl.Rd} = 352.225$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.839$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	108.884 / 295.686	0.368	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
Torq in Case 1 @ 2.664 m: $\theta_{max} \leq 2.00^\circ$	0.0557 rad = $3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	3.19°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.929	1.189	1.169	0.309	0.253	0.600	1.321	3.055	0.368	1.927	1.927

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.927

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 3.055 at x = 2.04 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with M_{z,tot}); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member UC-L4 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.26	4.26	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6.00C	-3.66	-3.66	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

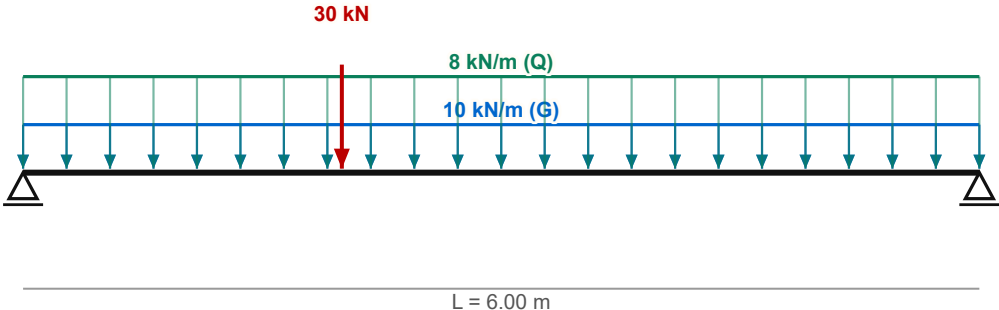
Uplift

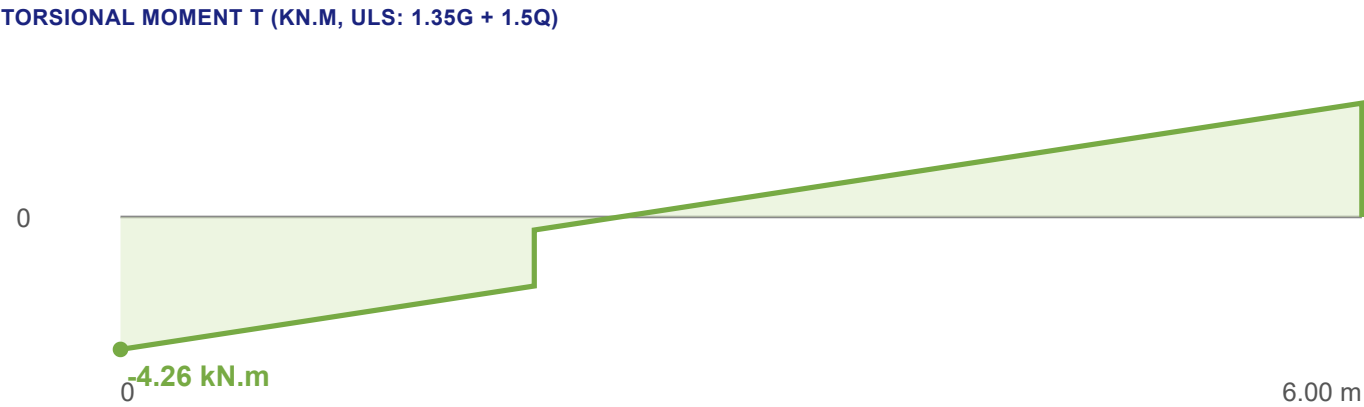
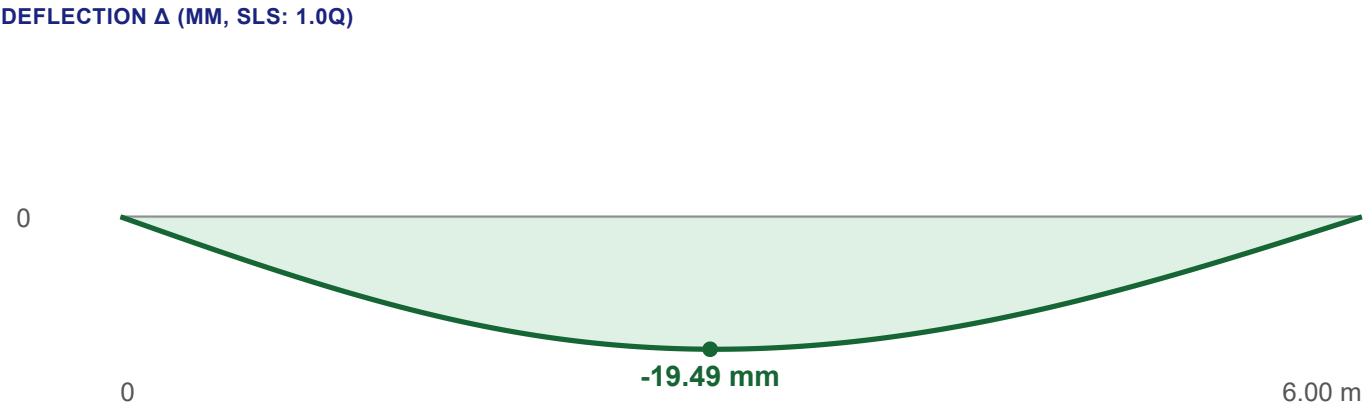
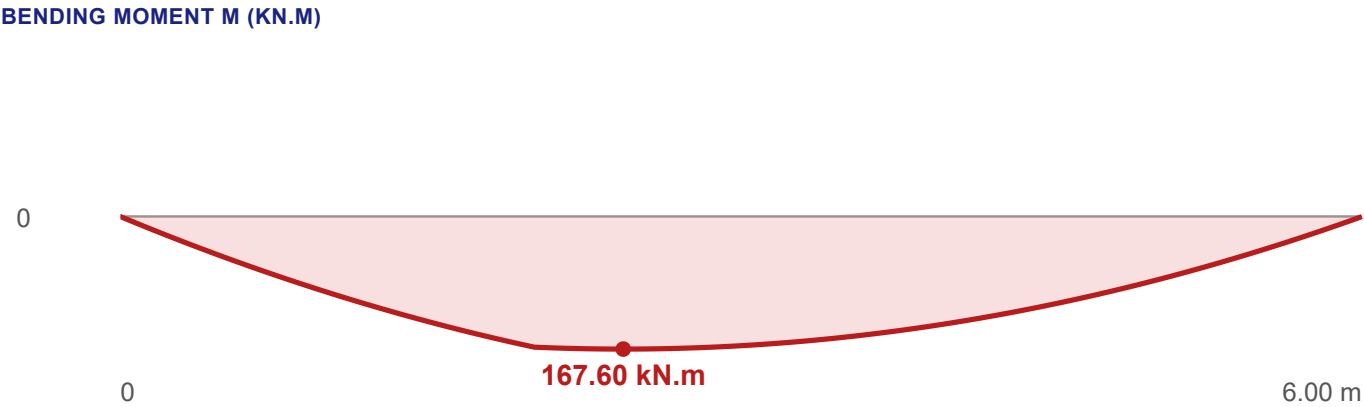
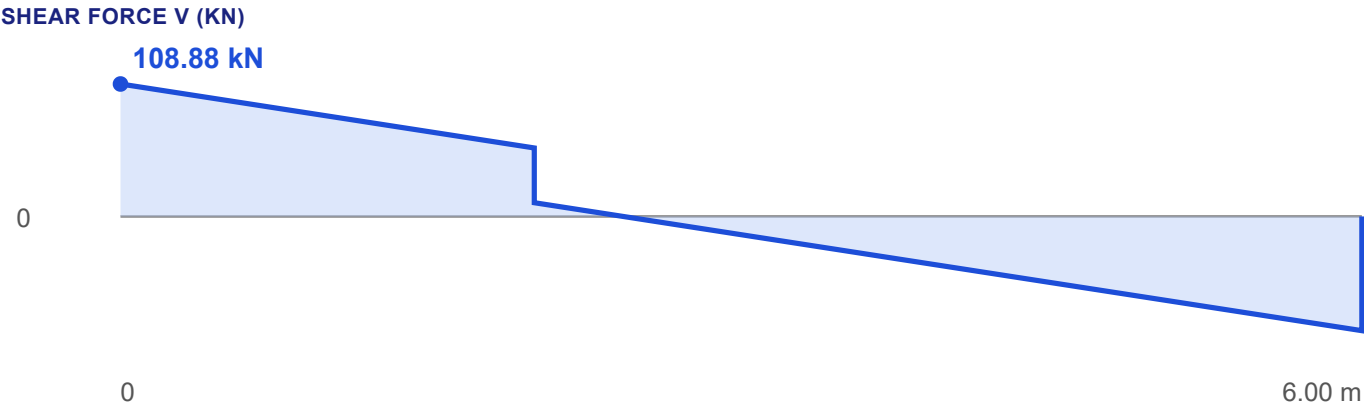
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (60.0 kg/m)	203 x 203 x 60 UC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	6.20, 17.10, 275, 0.00, 167.60, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E \omega_n \phi'' = 0.00 + 283.60 + 101.12 + 92.52 = 477.25 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(477.25/275)^2 + 3(33.07/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 283.60 + 101.12 + 92.52 = 477.25$, $\tau = 33.07 \rightarrow 3.055$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 264.40 + 4.64 = 269.04$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.010$; P3 web-flange junction, web: $\sigma_x = 0.00 + 245.19 + 4.64 = 249.83$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.854$; P4 web mid-depth: $\sigma_x = 0.00 + 4.64 = 4.64$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.031$ ($y_w = t_w/2 = 4.70 \text{ mm}$; $S_x/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894 ; 0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000 ; 1.000 \times 166.44$	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)

F_{Ed}/F_{Rd}	$F_{Ed} = 108.884 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 430.254 =$	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000)$ $= 0.253 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	X_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6 \text{ kN}\cdot\text{m}$		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105 \text{ mm}$ entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
$\bar{\lambda}_{LT}$ (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\bar{\lambda}_z\sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Torsion Design

J, H, a, Q_f, Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W_{n0}, S_{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2.315 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	1.835 kN.m ²	
$M_{w,Ed} = EI_w \varphi'' / (h - t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315 \text{ m}$: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	1.321	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.657 \times 0.753 \times 2.466$	1.219	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.116 \text{ m}$: 166.313/147.421 + $1.000 \times 20.748/83.875 +$ $0.657 \times 0.753 \times 2.466 \times 8.969/41.348$	1.640	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660 \text{ kN}\cdot\text{m}$	2.943 / 2.568 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 2.943$ kN.m, $V = 108.884$ kN	58.60 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 58.60$ N/mm ² ; $V_{pl.Rd} = 352.225$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.839$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	108.884 / 295.686	0.368	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
Torq in Case 1 @ 2.664 m: $\theta_{max} \leq 2.00^\circ$	0.0557 rad = $3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	3.19°	> 2° advisory

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.321	3.055	0.368	1.640	1.640

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.640

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

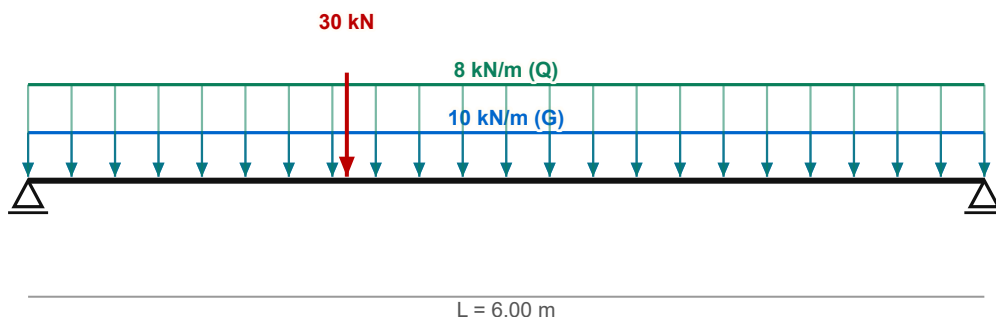
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

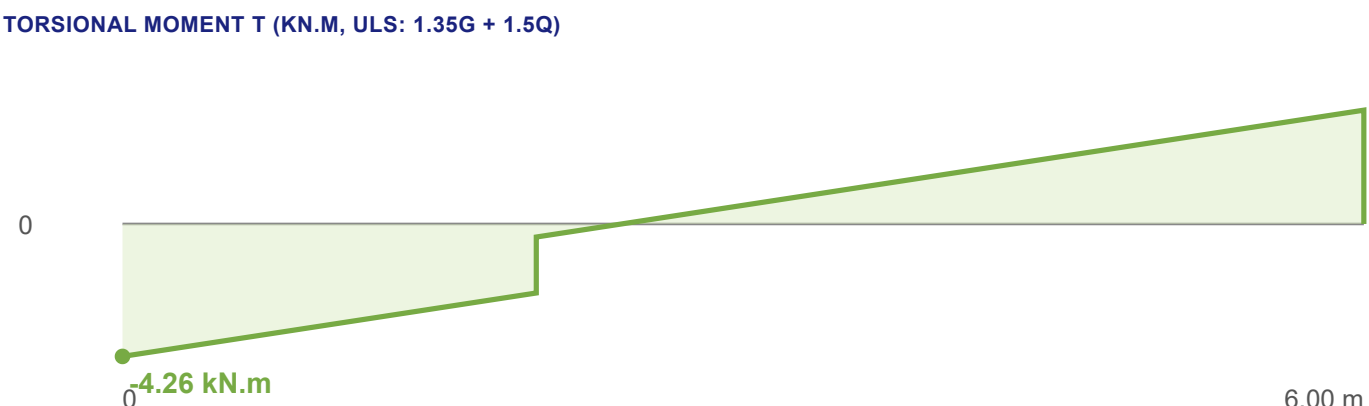
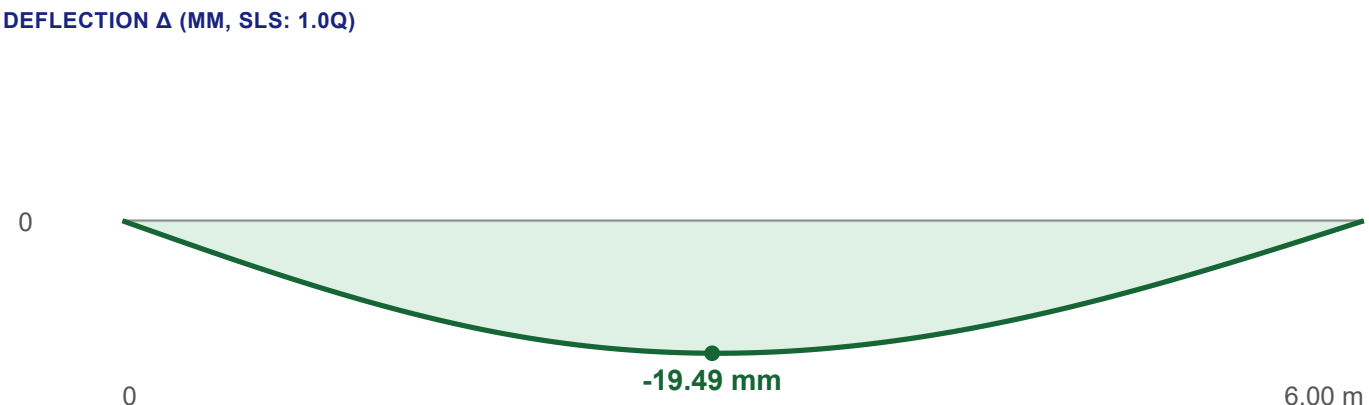
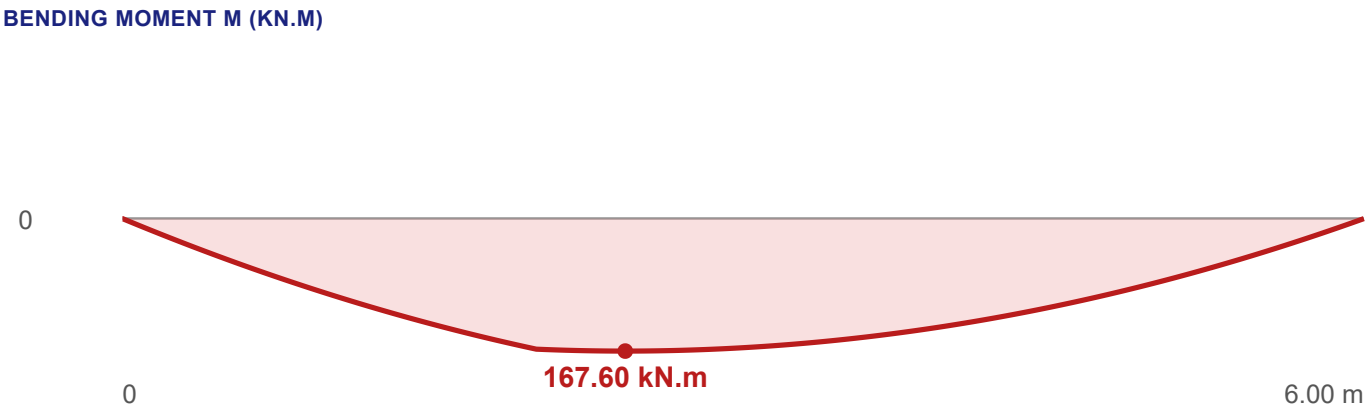
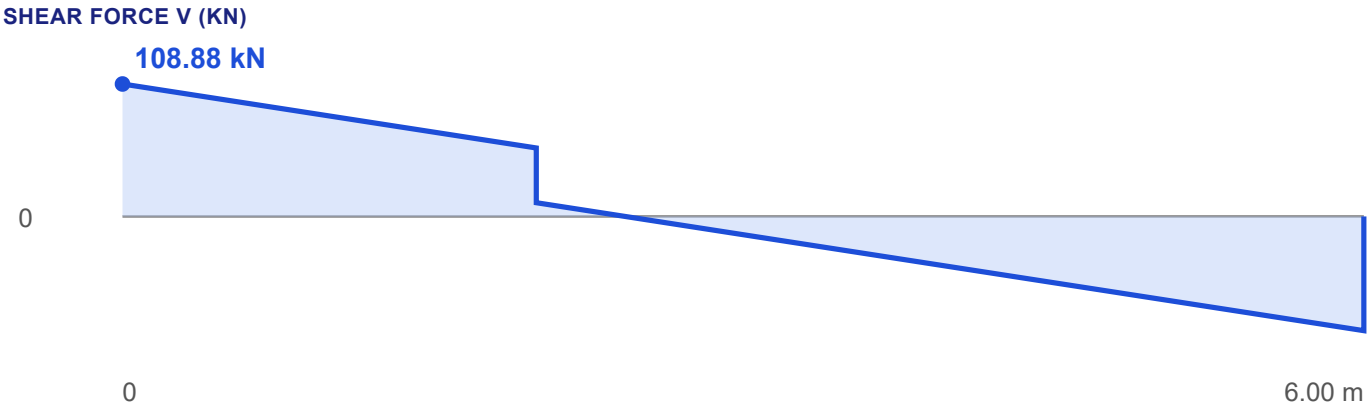
ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 3.055 at x = 2.04 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UC 203 x 203 x 60, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +104.8 mm





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (60.0 kg/m)	203 x 203 x 60 UC [S275]			
Class = Fn(c/t, d/t, fy, N, My, Mz)	6.20, 17.10, 275, 150.00, 167.60, 0.00			Class 1
	(Axial: Non-Slender)			
Web classified for	bending + compression: α = 0.680, ψ = -0.857; limits 50.5ε / 58.1ε / 108.5ε		Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1			

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	$275 \times 656.0/1$	180.400 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	$76.40 \times 275/1 =$ (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed}/N_{pl,Rd}$	$150.000 / 2101.000 =$	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$656.0, 22.184, 0.071$	656.0 cm^3	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	$656.0 \times 275/1$	180.400 kN.m	6.2.9.1(4)
$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$	$(167.604/180.400)^2 + (0)^1 =$	0.863	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 19.63 + 283.60 + 101.12 + 92.52 = 496.88 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(496.88/275)^2 + 3(33.07/275)^2 =$	3.308	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 283.60 + 101.12 + 92.52 = 496.88$, $\tau = 33.07 \rightarrow 3.308$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 264.40 + 4.64 = 288.67$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.155$; P3 web-flange junction, web: $\sigma_x = 19.63 + 245.19 + 4.64 = 269.46$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.989$; P4 web mid-depth: $\sigma_x = 19.63 + 4.64 = 24.27$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.038$ ($y_w = t_w/2 = 4.70 \text{ mm}$; $S_x/2 =$ half flange one side of the web)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20 / 14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.310 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.222	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.222	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.641	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.197	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y ·L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A·f _y /N _{cr})	√76.40x275/3523.45 (N _{cr,y} = π ² EI _y /L _{ey} ²)	0.771	
N _{b,y,Rd} = Area·χ·f _y /γ _{M1}	76.40x0.742x275/10/1 =	1558.850 kN	Curve b
L _{ez} = K _z ·L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A·f _y /N _{crz})	√76.40x275/1186.00 (N _{cr,z} = π ² EI _z /L _{ez} ²)	1.329	
N _{b,z,Rd} = Area·χ·f _y /γ _{M1}	76.40x0.377x275/10/1 =	791.578 kN	Curve c
L _{et} = K _t ·L _z ; λ _T ; N _{b,T,Rd}	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
N _{Ed} /N _{b,Rd}	150.000 / 791.578 = (min of N _{b,y,Rd} , N _{b,z,Rd})	0.189	OK

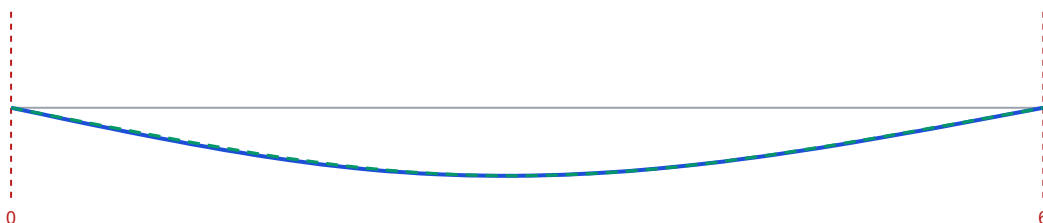
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = M _{cr} /M _{cr,uniform}	277.879 / 242.476 kN.m (shape only: z _g = z _j = 0)	1.146	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M _{max} =167.6 kN·m	1.163	Serna
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 163.33, ψ = 0.856, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 0.000 imposed + 22.160 twist φ·M _y = 22.160; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 163.33, ψ = 0.856, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 105, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	219.624 kN.m	converged
λ _{LT} = √(W·f _y /M _{cr})	√ 656.0 x 275 / 219.624	0.906	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.906, 0.894, 0.34	0.756	Curve b
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.756, 0.906, 0.934, 0.968	0.781	6.3.2.3
M _{b,Rd} = χW _{pl,y} ·f _y ≤ M _{c,y,Rd}	0.781 x 656.0 x 275 ≤ 180.400 =	140.966 kN.m	
M _{y,Ed} /M _{b,Rd}	167.604 / 140.966	1.189	Warning

Buckled mode shape — critical eigenmode (normalised; twist φ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.84 m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	167.604 / 140.966 ($M_{y,Rk} = 180.400$)	1.189	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(0.000 imposed + 22.160 twist $\phi \cdot M_y = 22.160$) / 83.875	0.264	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+(\chi_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\chi_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\chi_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\chi_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.189+0.759 \times 0.264$	1.551	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.189+1.265 \times 0.264$	1.683	Warning

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
ϕ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.835 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315$ m: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	1.321	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.655 \times 0.761 \times 4.222$	2.102	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2$ m: $165.178/136.428 + 1.000 \times 20.068/83.875 + 0.655 \times 0.761 \times 4.222 \times 9.391/41.348$	1.927	Warning EN 1993-6 A
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6

superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(\chi_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(\chi_{LT} M_{y,Rk}/\gamma_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 1.683$ (Eq 6.62 with $M_{z,tot} = 22.160$) + 0.478 (warping term of (A.1)) @ $x = 2$ m: $k_w = 0.655$, $k_{zw} = 0.761$, $k_\alpha = 4.222$, $M_w = 9.391$ kN.m) =	2.160	information only
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End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260 / 3.660$ kN.m	2.943 / 2.568 kN.m	
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Torsion Shear Design @ 0 m

T_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 2.943$ kN.m, $V = 108.884$ kN	58.60 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 58.60$ N/mm ² ; $V_{pl,Rd} = 352.225$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.839$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	$108.884 / 295.686$	0.368	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	19.49 mm	Warning
Torq in Case 1 @ 2.664 m: $\theta_{max} \leq 2.00^\circ$	0.0557 rad = $3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	3.19°	> 2° advisory

N _{Ed} /N _{Ed,Rd}	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/V _{pl}	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.071	0.863	0.189	1.189	1.551	1.683	1.169	0.309	0.929	1.189	0.253	0.641	1.321	3.308	0.368	1.927 3.308

FAIL — governing Elastic yield criterion (6.1) with torsion, $\kappa 6.2.7(5) = 3.308$

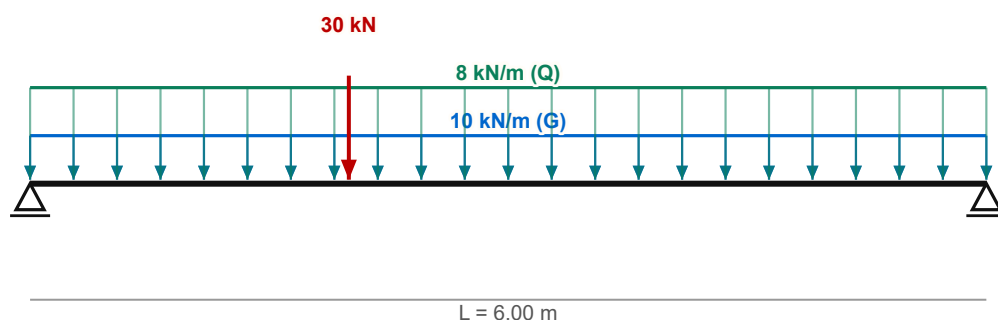
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

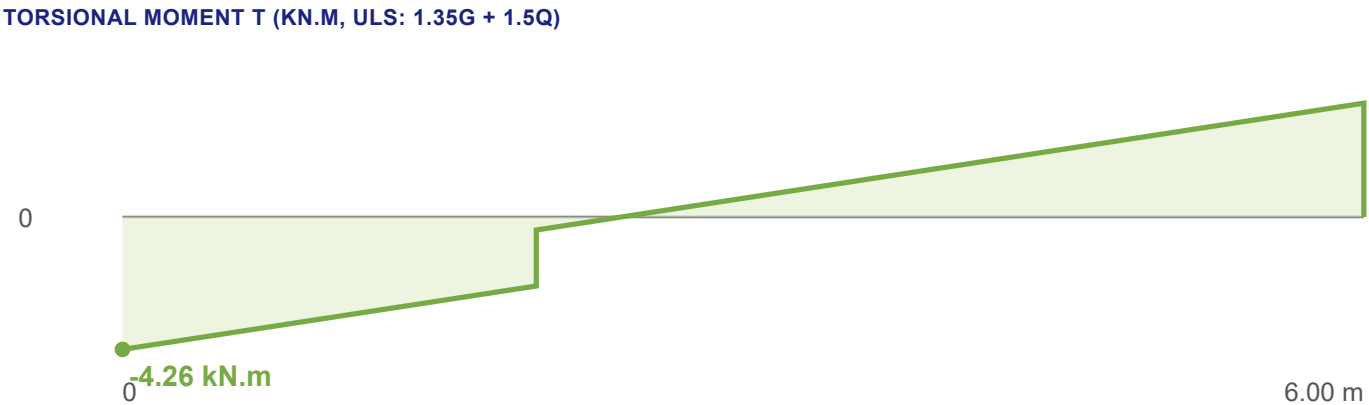
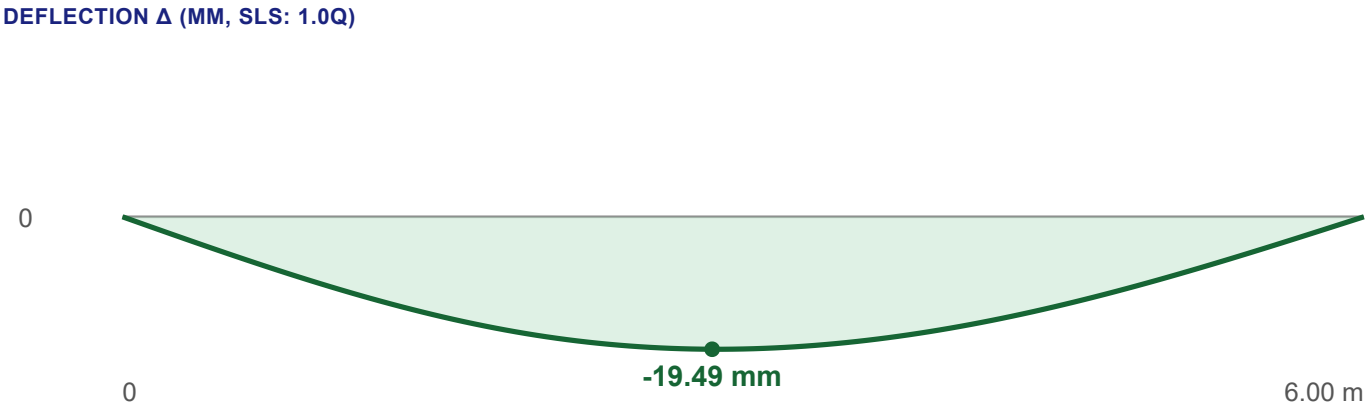
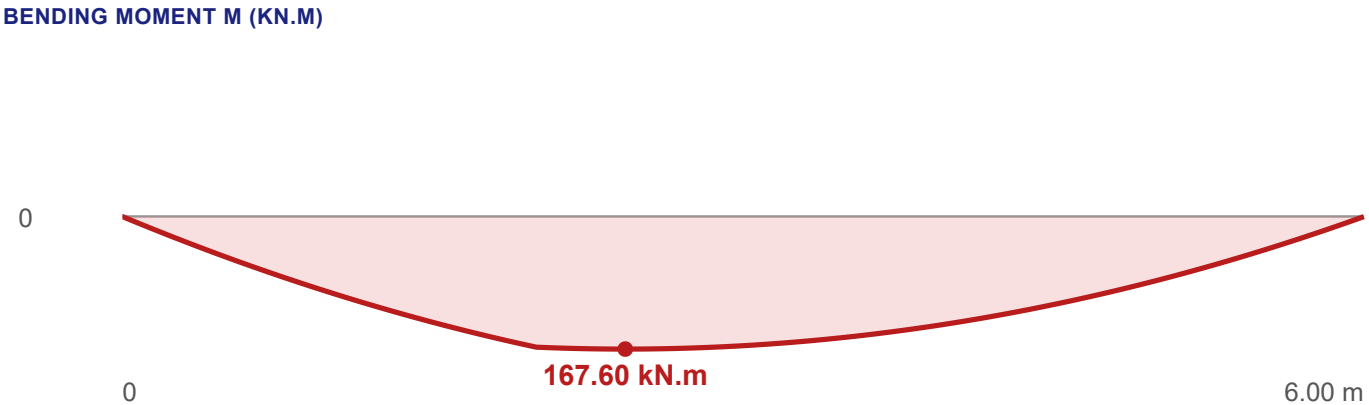
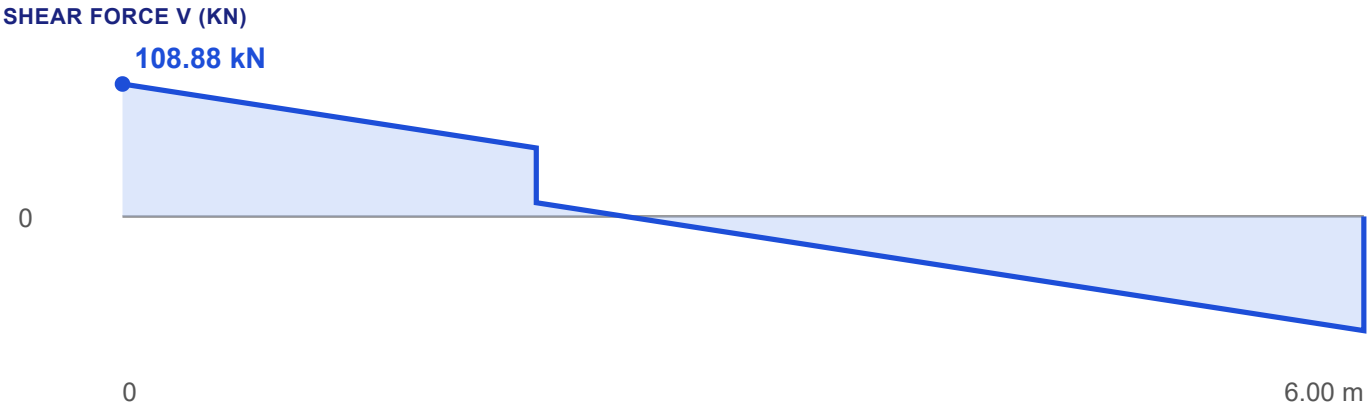
ADVISORY: Serviceability twist $\theta_{\max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/\gamma_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.683$ (Eq 6.62 with $M_{z,tot} = 22.16 \text{ kN}\cdot\text{m}$) + 0.478 (warping term of (A.1) at $x = 2.00 \text{ m}$, $k_w = 0.655$, $k_{zw} = 0.761$, $k_\alpha = 4.222$) = 2.160. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UC 203 x 203 x 60, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +104.8 mm





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (60.0 kg/m)	203 x 203 x 60 UC [S275]			
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	6.20, 17.10, 275, 150.00, 167.60, 0.00			Class 1
	(Axial: Non-Slender)			
Web classified for	bending + compression: α = 0.680, ψ = -0.857; limits 50.5ε / 58.1ε / 108.5ε		Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1			

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$108.884 / 352.225 =$	0.309	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$19.28 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.012 / 352.225 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$167.604 / 180.400 =$	0.929	OK
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$76.40 \times 275 / 1 = \text{(No bearing / block tearing design)}$	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2101.000 =$	0.071	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$656.0, 22.184, 0.071$	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$656.0 \times 275 / 1$	180.400 kN.m	6.2.9.1(4)
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(167.604 / 180.400)^2 + (0)^1 =$	0.863	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.041 m, P1 flange tip	@ x = 2.041 m (ULS: 1.35G + 1.5Q): $M_y = 165.624 \text{ kN.m}$, $V = 10.204 \text{ kN}$, $\phi = 0.12272 \text{ rad}$, $\phi' = 2.875e-5 \text{ rad/mm}$, $\phi'' = 4.362e-8 \text{ rad/mm}^2$, $\phi''' = 1.744e-11 \text{ rad/mm}^3$; $M_z = \phi M_y = 20.326 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 19.63 + 283.60 + 101.12 + 92.52 = 496.88 \text{ N/mm}^2$; $\tau = G t_f \phi' = 33.07 \text{ N/mm}^2$; $(496.88 / 275)^2 + 3(33.07 / 275)^2 =$	3.308	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 283.60 + 101.12 + 92.52 = 496.88$, $\tau = 33.07 \rightarrow 3.308$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 264.40 + 4.64 = 288.67$, $\tau = 1.68 + 33.07 + 1.89 = 36.64 \rightarrow 1.155$; P3 web-flange junction, web: $\sigma_x = 19.63 + 245.19 + 4.64 = 269.46$, $\tau = 5.06 + 21.89 = 26.96 \rightarrow 0.989$; P4 web mid-depth: $\sigma_x = 19.63 + 4.64 = 24.27$, $\tau = 5.75 + 21.89 = 27.64 \rightarrow 0.038$ ($y_w = t_w / 2 = 4.70 \text{ mm}$; $S_x / 2 =$ half flange one side of the web)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm}$ ($B = 205.80 \leq t_w + 30 \varepsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20 / 14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)

$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.310 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.222	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.222	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.641	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.197	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10 / 1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10 / 1 =$	791.578 kN	Curve c
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 791.578 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.189	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 22.160 twist $\phi \cdot M_y = 22.160$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 \text{ L}$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	$1.163, 6.000, 2060, 47.2, 0.197, 210000$; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	

χ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only) $M_{y,Ed}/M_{b,Rd}$	$(1/\sqrt{C_1})0.9\chi_z\sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$ 167.604 / 152.975	117.633 kN.m 1.096	P362 6.55 comparison Warning
Buckling Resistance			
$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	167.604 / 152.975 ($M_{y,Rk} = 180.400$)	1.096	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(0.000 imposed + 22.160 twist $\phi \cdot M_y = 22.160$) / 83.875	0.264	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+(\chi_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\chi_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq$ $1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\chi_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\chi_z = 1.329$, $U_{N,z} =$ 0.189 , $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.096+0.759 \times 0.264$	1.453	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.096+1.265 \times 0.264$	1.592	Warning
Torsion Design			
J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴		P385 App A
Torsion Bending Design @ 2.315 m			
T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 5.77	P385 App C
ϕ_{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	1.835 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	9.391 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.315 m: 167.433, 8.337, 21.602 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.348, M _{pl,z} = 83.875	1.321	Warning P385 3.1.2
k = k _w · k _{zw} · k _α	0.657 x 0.753 x 2.466	1.219	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} +$ $k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ x = 2.116 m: 166.313/147.421 + 1.000x20.748/83.875 + 0.657x0.753x2.466x8.969/41.348	1.640	Warning EN 1993-6 A
Basis with N _{Ed} / M _{z,Ed}	No expression in EN 1993-1-1 or EN 1993-6 combines N _{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross- section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} , and EN 1993-6 (A.1) with M _{z,Ed} = M _z + phi.M _y		EN 1993-1-1, EN 1993-6
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(\chi_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(\chi_{LT} M_{y,Rk}/\gamma_{M1}) +$ $k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_{\alpha} M_w/M_{f,Rd} = 1.592$ (Eq 6.62 with M _{z,tot} = 22.160) + 0.279 (warping term of (A.1)) @ x = 2 m: k _w = 0.655, k _{zw} = 0.761, k _α = 2.466, M _w = 9.391 kN.m) =	1.871	information only
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' – EI _w φ''' = 4.260 / 3.660 kN.m	2.943 / 2.568 kN.m	
Torsion Shear Design @ 0 m			
T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 2.943 kN.m, V = 108.884 kN	58.60 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: T _t = 58.60 N/mm ² ; V _{pl,Rd} = 352.225; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.839	295.686 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	108.884 / 295.686	0.368	OK
Deflection Check - Load Case 1 (SLS: 1.0Q)			
In-span δ ≤ Span/360	19.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	19.49 mm	Warning

Torq in Case 1 @ 2.664 m: $\theta_{\max} \leq 2.00^\circ$ 0.0557 rad = $3.19^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict) $3.19^\circ > 2^\circ$ advisory

N_Ed/N_Rd	N/N_Rd	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.071	0.863	0.189	1.096	1.453	1.592	1.169	0.309	0.929	1.096	0.253	0.641	1.321	3.308	0.368	1.640	3.308

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 3.308

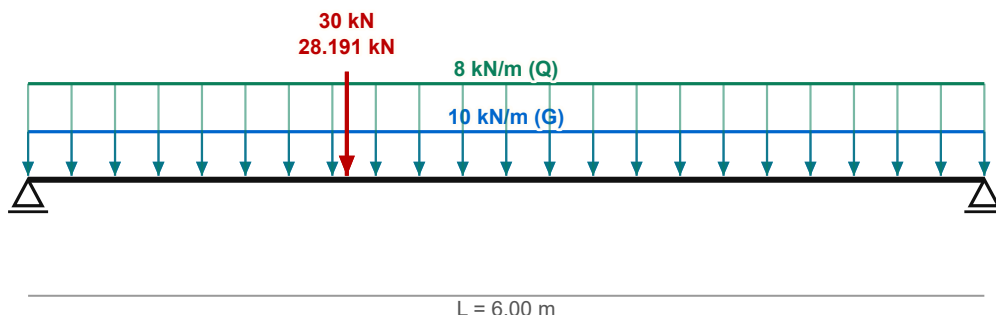
NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

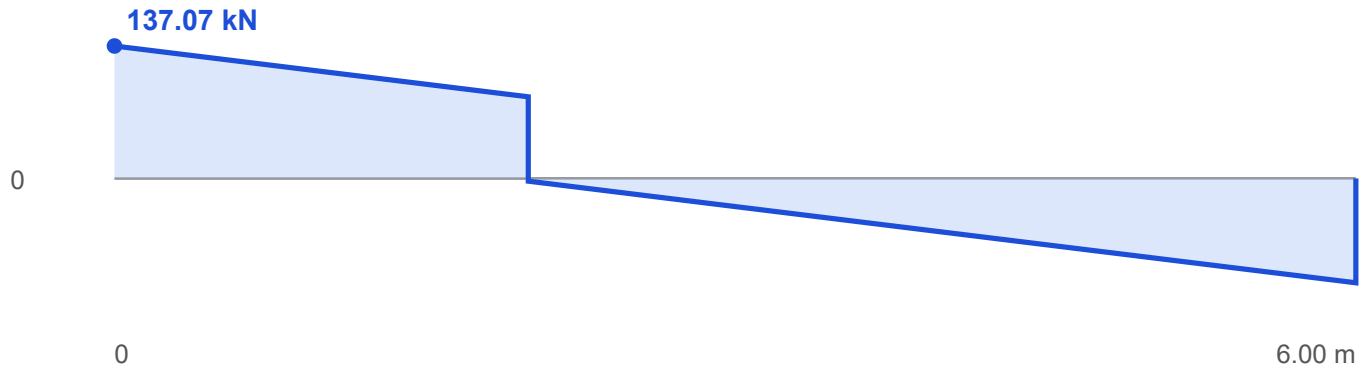
ADVISORY: Serviceability twist $\theta_{\max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

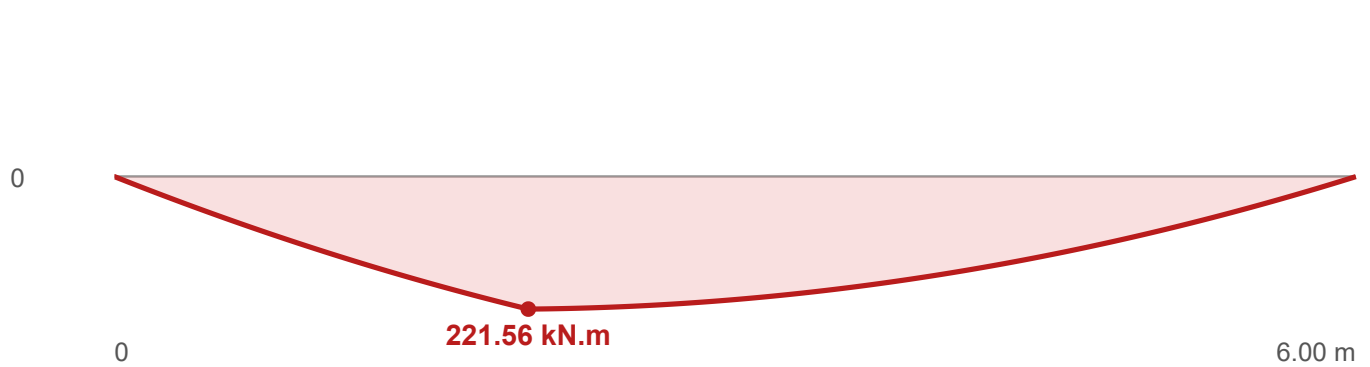
ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/\gamma_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/\gamma_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/\gamma_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 1.592$ (Eq 6.62 with $M_{z,tot} = 22.16$ kN·m) + 0.279 (warping term of (A.1) at $x = 2.00$ m, $k_w = 0.655$, $k_{zw} = 0.761$, $k_\alpha = 2.466$) = 1.871. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.



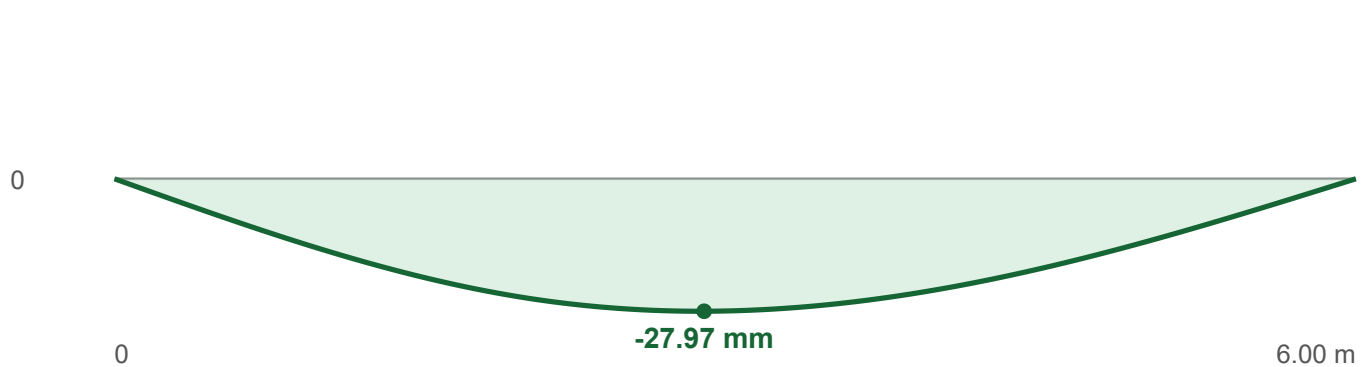
SHEAR FORCE V (kN)



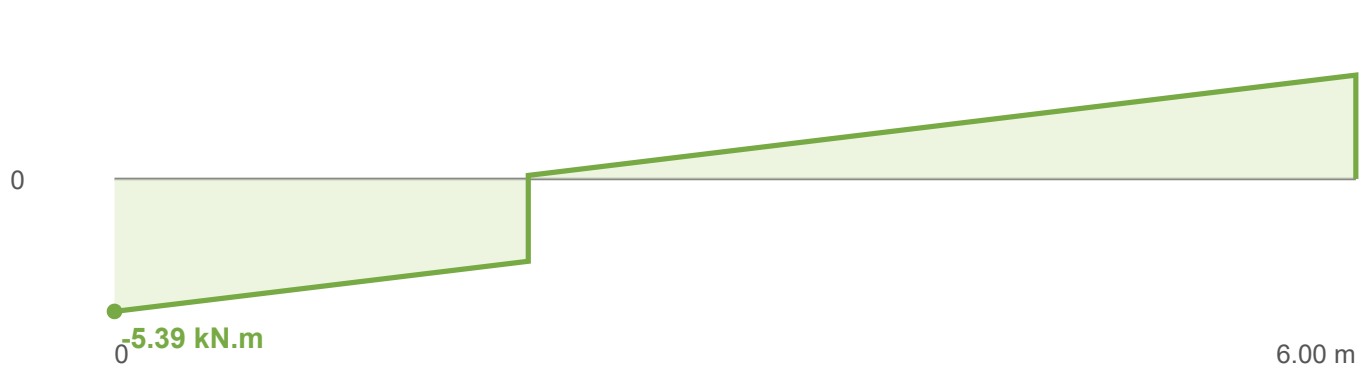
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 150.00, 221.56, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.680$, $\psi = -0.857$; limits $50.5\epsilon / 58.1\epsilon / 108.5\epsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2

Flange outstands under $M_y + M_z$ (+N)	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 413.8, tip 501.1 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 384.2, tip 296.9 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	137.075 / 352.225 =	0.389	OK
$h_w / t_w \leq 72 \epsilon / \eta$	19.28 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	2.801 / 352.225 =	0.008	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	221.560 / 180.400 =	1.228	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(221.560 / 180.400)^2 + (20.521 / 83.875)^1 =$	1.753	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 221.560$ kN.m, $V = 40.842$ kN, $\phi = 0.15797$ rad, $\phi' = 3.749e-5$ rad/mm, $\phi'' = 6.516e-8$ rad/mm ² , $\phi''' = 4.627e-11$ rad/mm ³ ; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 35.001 = 55.522$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + EW_{n0}\phi'' = 19.63 + 379.38 + 276.23 + 138.20 = 813.45$ N/mm ² ; $\tau = G_t \phi' = 43.12$ N/mm ² ; $(813.45/275)^2 + 3(43.12/275)^2 =$	8.823	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 379.38 + 276.23 + 138.20 = 813.45$, $\tau = 43.12 \rightarrow 8.823$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 353.70 + 12.67 = 386.00$, $\tau = 6.71 + 43.12 + 5.02 = 54.85 \rightarrow 2.090$; P3 web-flange junction, web: $\sigma_x = 19.63 + 328.00 + 12.67 = 360.30$, $\tau = 20.27 + 28.55 = 48.82 \rightarrow 1.811$; P4 web mid-depth: $\sigma_x = 19.63 + 12.67 = 32.30$, $\tau = 23.01 + 28.55 = 51.55 \rightarrow 0.119$ ($y_w = t_w/2 = 4.70$ mm; $S_f/2 =$ half flange one side of the web)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30\epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]			
s_s, c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)

$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90$ ≤ 100.00	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.075$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.319	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.319 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.376 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.268	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	137.075	ULS: 1.35G + 1.5Q	0.319	0.268	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.892	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	107.979	ULS: 1.35G + 1.5Q	0.251	0.220	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10 / 1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10 / 1 =$	791.578 kN	Curve c
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 791.578 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.189	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

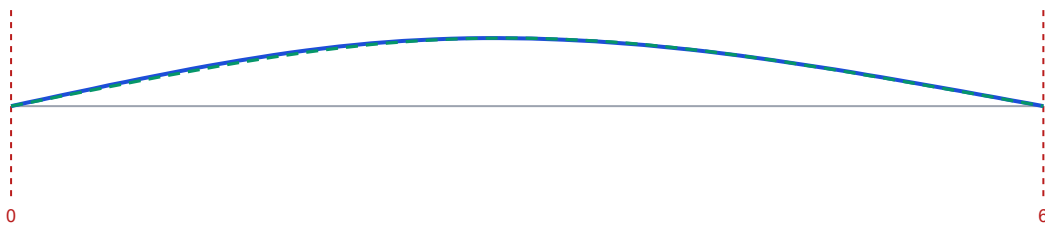
$C_1 = M_{cr} / M_{cr,uniform}$	$289.130 / 242.476$ kN.m (shape only: $z_g = z_j = 0$)	1.192	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 205.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=176.0$, $M(L/2)=205.6$, $M(3L/4)=132.4$, $M_{max}=221.6$ kN.m	1.216	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 205.61$, $\psi = 0.781$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 36.730 twist $\phi \cdot M_y = 57.251$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 205.61$, $\psi = 0.781$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e =$ portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
$M_{cr} =$ FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 105$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	226.932 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 226.932}$	0.892	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.892, 0.882, 0.34	0.765	Curve b

$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.765, 0.892, 0.916, 0.959	0.798	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.798 \times 656.0 \times 275 \leq 180.400 =$	143.956 kN.m	
$M_{y,Ed}/M_{b,Rd}$	221.560 / 143.956	1.539	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	221.560 / 143.956 ($M_{y,Rk} = 180.400$)	1.539	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(20.521 \text{ imposed} + 36.730 \text{ twist } \phi \cdot M_y = 57.251) / 83.875$	0.683	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\lambda_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.539+0.759 \times 0.683$	2.238	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.539+1.265 \times 0.683$	2.553	Warning

Torsion Design

J, H, a, Q_f, Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W_{n0}, S_{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\phi_{max} \text{ (ULS)}$	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
$B_{Ed} = EI_w \phi'' \text{ (max)}$	@ $x = 2$ m	2.696 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	13.796 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 217.419 \times 168.93e-3$ (@ $x = 2.465$ m) = $20.521 + 36.730$ (imposed constant $M_{z,Ed}$ + max coincident twist moment)	57.251 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2$ m: 221.560, 13.796, 55.522 kN.m ($M_{z,tot} = 20.521 + 35.001$); $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	2.504	Warning P385 3.1.2
$C_{mz} \text{ (EN 1993-6 A.1)}$	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.633 \times 0.338 \times 42.246$ ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 55.522/83.875$)	9.044	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2$ m: $221.560/137.995 + 1.000 \times 55.522/83.875 + 0.633 \times 0.338 \times 42.246 \times 13.796/41.348$ ($M_{z,tot} = 20.521$ imposed + 35.001 twist $\phi \cdot M_y$)	5.285	Warning EN 1993-6 A

Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi_i M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi_i M_y$				EN 1993-1-1, EN 1993-6																																
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 2.553$ (Eq 6.62 with $M_{z,tot} = 57.251$) + 3.100 (warping term of (A.1) @ $x = 1.643$ m: $k_w = 0.648$, $k_{zw} = 0.437$, $k_\alpha = 42.246$, $M_w = 10.715$ kN.m) =				5.653 information only																																
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 5.388 / 4.224$ kN.m				3.823 / 3.096 kN.m																																
Torsion Shear Design @ 0 m																																					
T_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.823$ kN.m, $V = 137.075$ kN				76.13 N/mm ² P385																																
$V_{pl.T,Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))}.V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 76.13$ N/mm ² ; $V_{pl,Rd} = 352.225$; $S_{mod} = V_{pl.T,Rd}/V_{pl,Rd} = 0.785$				276.531 kN 6.2.7(9)																																
$V_{Ed}/V_{pl.T,Rd}$	137.075 / 276.531				0.496 OK																																
Deflection Check - Load Case 1 (SLS: 1.0Q)																																					
In-span $\delta \leq \text{Span}/360$	27.97 $\leq$ 6000 / 360 = 16.67 mm @ $x = 2.85$ m				27.97 mm Warning																																
Torq in Case 1 @ 2.564 m: $\theta_{max} \leq 2.00^\circ$	0.0806 rad = 4.62° $\leq$ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)				4.62° > 2° advisory																																
<table><tr><td>$N_{Ed}/N_{pl,Rd}$</td><td>UNyz</td><td>UMyz</td><td>Ax+M_6.6.2</td><td>Ax+M_6.6.2</td><td>Deflection</td><td>V/Vpl</td><td>MA/Mc</td><td>LTB</td><td>F/F_Rd</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>LTB+T</td><td>Max</td></tr><tr><td>0.071</td><td>1.753</td><td>0.189</td><td>1.539</td><td>2.238</td><td>2.553</td><td>1.678</td><td>0.389</td><td>1.228</td><td>1.539</td><td>0.319</td><td>0.892</td><td>2.504</td><td>8.823</td><td>0.496</td><td>5.285 8.823</td></tr></table>						$N_{Ed}/N_{pl,Rd}$	UNyz	UMyz	Ax+M_6.6.2	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	0.071	1.753	0.189	1.539	2.238	2.553	1.678	0.389	1.228	1.539	0.319	0.892	2.504	8.823	0.496	5.285 8.823
$N_{Ed}/N_{pl,Rd}$	UNyz	UMyz	Ax+M_6.6.2	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max																						
0.071	1.753	0.189	1.539	2.238	2.553	1.678	0.389	1.228	1.539	0.319	0.892	2.504	8.823	0.496	5.285 8.823																						

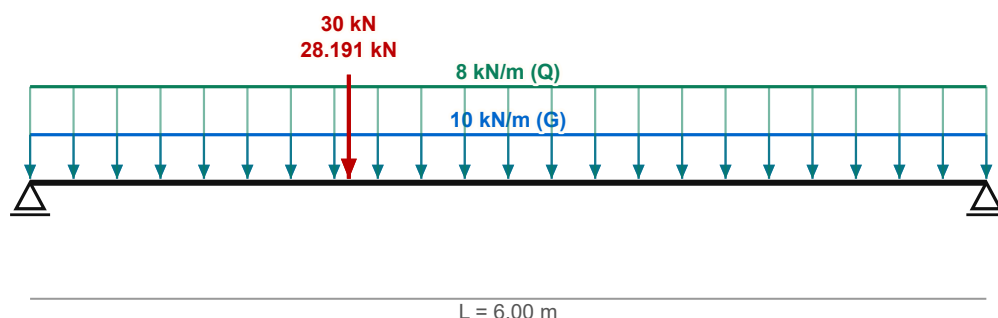
FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 8.823

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

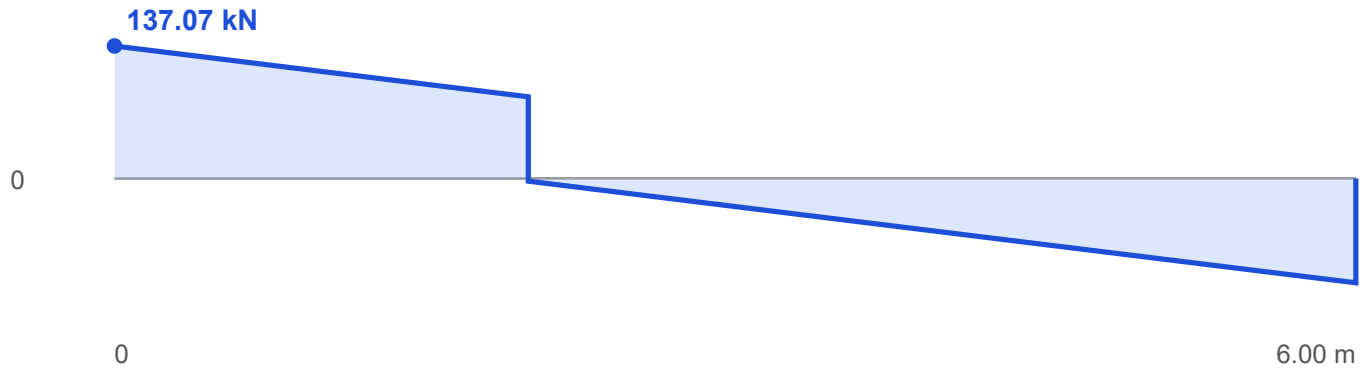
ADVISORY: Serviceability twist $\theta_{max} = 4.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

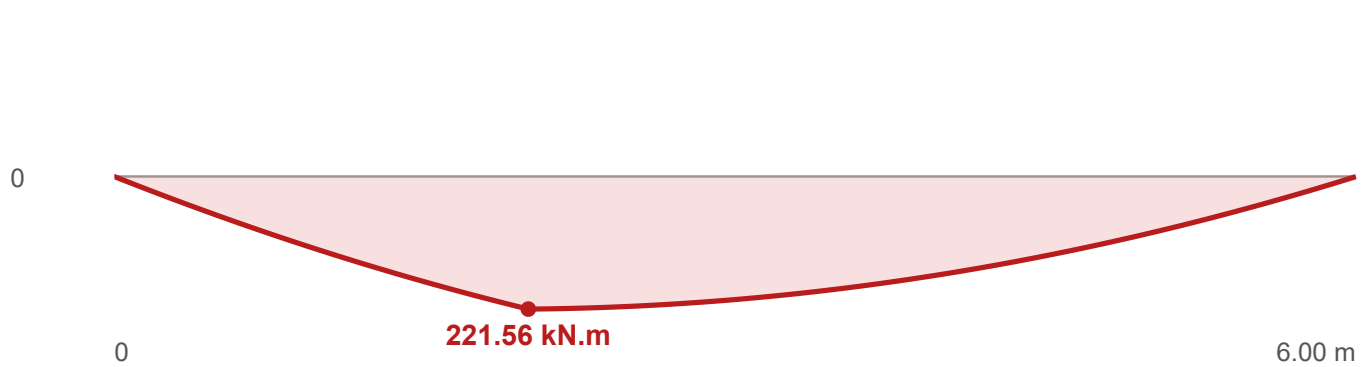
ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 2.553$ (Eq 6.62 with $M_{z,tot} = 57.25$ kN.m) + 3.100 (warping term of (A.1) at $x = 1.64$ m, $k_w = 0.648$, $k_{zw} = 0.437$, $k_\alpha = 42.246$) = 5.653. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.



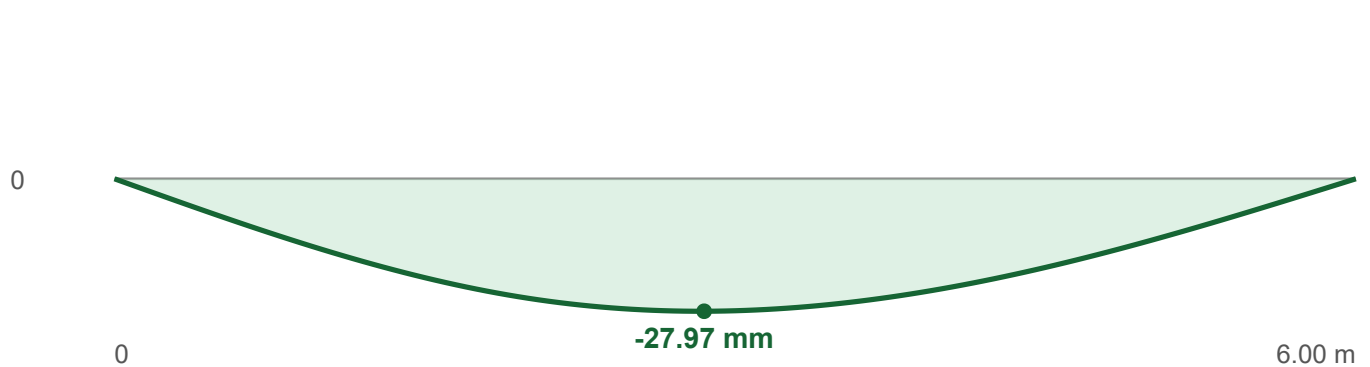
SHEAR FORCE V (kN)



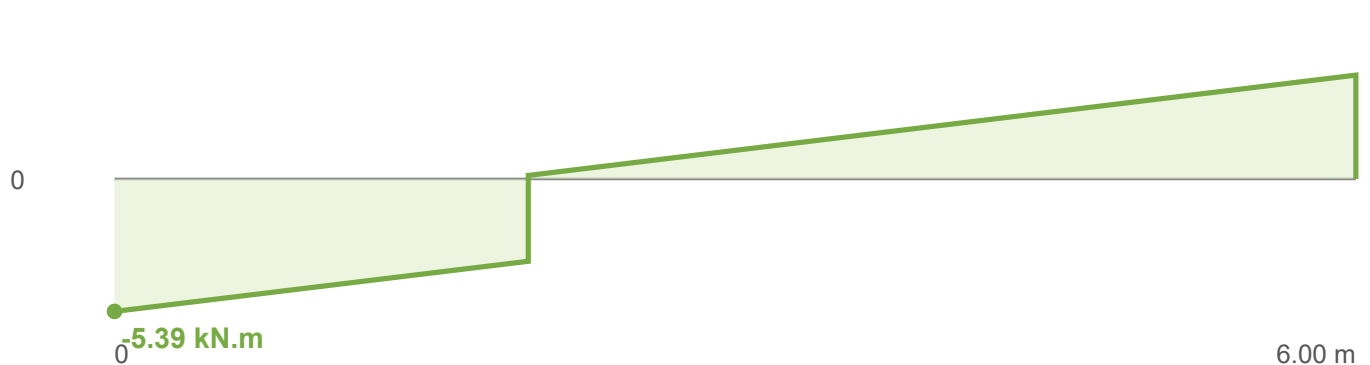
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 150.00, 221.56, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.680$, $\psi = -0.857$; limits $50.5\epsilon / 58.1\epsilon / 108.5\epsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2

Flange outstands under $M_y + M_z$ (+N)	$\sigma = N/A + M_y/W_{el,y} \pm M_z y/I_z$: outstand compressed by M_z : root 413.8, tip 501.1 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 384.2, tip 296.9 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	137.075 / 352.225 =	0.389	OK
$h_w / t_w \leq 72 \epsilon / \eta$	19.28 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	2.801 / 352.225 =	0.008	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	221.560 / 180.400 =	1.228	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(221.560 / 180.400)^2 + (20.521 / 83.875)^1 =$	1.753	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 221.560$ kN.m, $V = 40.842$ kN, $\phi = 0.15797$ rad, $\phi' = 3.749e-5$ rad/mm, $\phi'' = 6.516e-8$ rad/mm ² , $\phi''' = 4.627e-11$ rad/mm ³ ; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 35.001 = 55.522$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + E W_{n0} \phi'' = 19.63 + 379.38 + 276.23 + 138.20 = 813.45$ N/mm ² ; $\tau = G t_f \phi' = 43.12$ N/mm ² ; $(813.45/275)^2 + 3(43.12/275)^2 =$	8.823	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 379.38 + 276.23 + 138.20 = 813.45$, $\tau = 43.12 \rightarrow 8.823$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 353.70 + 12.67 = 386.00$, $\tau = 6.71 + 43.12 + 5.02 = 54.85 \rightarrow 2.090$; P3 web-flange junction, web: $\sigma_x = 19.63 + 328.00 + 12.67 = 360.30$, $\tau = 20.27 + 28.55 = 48.82 \rightarrow 1.811$; P4 web mid-depth: $\sigma_x = 19.63 + 12.67 = 32.30$, $\tau = 23.01 + 28.55 = 51.55 \rightarrow 0.119$ ($y_w = t_w/2 = 4.70$ mm; $S_f/2 =$ half flange one side of the web)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.		NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343)} = 0.306$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 137.075$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 430.254 =	0.319	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.319 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 + N_{Ed}/N_{pl,Rd} = 150.000/2101.000 = 0.071) = 0.376 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.268	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	137.075	ULS: 1.35G + 1.5Q	0.319	0.268	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.892	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	107.979	ULS: 1.35G + 1.5Q	0.251	0.220	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10/1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10/1 =$	791.578 kN	Curve c
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 791.578 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.189	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 205.6, 1.000, 300.000	1.216	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=176.0$, $M(L/2)=205.6$, $M(3L/4)=132.4$, $M_{max}=221.6$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 205.61$, $\psi = 0.781$, $\alpha_h = 0.000$	1.000	Table B.3

$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521 \text{ imposed} + 36.730 \text{ twist } \phi.M_y = 57.251; \psi = 1 \text{ for the imposed constant } M_z, \text{ the twist part taken at } C_{mz} = 1.0 \text{ (Table B.3 upper bound, conservative)}$	1.000	Table B.3
$C_{my} = 1.0 \text{ (diagram outside Table B.3)}$	$M_h = 0.00, M_s = 205.61, \psi = 0.781, \alpha_h = 0.000$	1.000	Table B.3
Lateral Buckling Check M.b.Rd			
$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	$1.216, 6.000, 2060, 47.2, 0.197, 210000; z_g = +105 \text{ mm entered — NOT applied: } C_2 \text{ not published for this moment diagram; } M_{cr} \text{ is at the shear centre (PASS blocked)}$	294.827 kN.m	BLOCKED
$\chi_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{656.0 \times 275 / 294.827}$	0.782	
$\chi_{LT} = \text{Fn}(\chi_{LT}, \Phi_{LT}, \alpha_{LT})$	$0.782, 0.794, 0.34$	0.827	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \chi_{LT}, k_c, f)$	$0.827, 0.782, 0.907, 0.953$	0.867	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.867 \times 656.0 \times 275 \leq 180.400 =$	156.441 kN.m	
$\chi_{LT} \text{ (P362 6.55 simplified) ; } M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\chi_z\sqrt{\beta_w} = 1.085; \chi_{LT,mod} = 0.674;$ $M_{y,Ed}/M_{b,Rd,simp} = 1.822$	121.608 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$221.560 / 156.441$	1.416	Warning
Buckling Resistance			
$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 1558.850 (N_{Rk} = 2101.000)$	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 791.578$	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	$221.560 / 156.441 (M_{y,Rk} = 180.400)$	1.416	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(20.521 \text{ imposed} + 36.730 \text{ twist } \phi.M_y = 57.251) / 83.875$	0.683	OK
$k_{zy} \text{ method}$	$M_{b,Rd} < M_{c,y,Rd} \text{ therefore susceptible to LTB, using Table B.2}$		Table B.2
$k_{yy} = C_{my}\{1+(\chi_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\chi_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\chi_z, 1) \cdot n_z/(C_{mLT}-0.25); \chi_z = 1.329, U_{N,z} = 0.189, C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.416+0.759 \times 0.683$	2.109	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.416+1.265 \times 0.683$	2.434	Warning
Torsion Design			
J, H, a, Q_f, Q_w	$47.2 \text{ cm}^4, 0.197 \text{ dm}^6, 1040 \text{ mm}, —, —$		P385 App A
W_{n0}, S_{w1}	$101 \text{ cm}^2, 734 \text{ cm}^4$		P385 App A
Torsion Bending Design @ 2 m			
$T_{Ed} \text{ (max)}$	combination ULS: $1.35G + 1.5Q$	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\phi_{max} \text{ (ULS)}$	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	$0.1701 \text{ rad} = 9.75^\circ$	
$B_{Ed} = EI_w \phi'' \text{ (max)}$	@ $x = 2 \text{ m}$	2.696 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	13.796 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi.M_{y,Ed}$	$20.521 + 217.419 \times 168.93e-3$ (@ $x = 2.465 \text{ m}$) = $20.521 + 36.730$ (imposed constant $M_{z,Ed}$ + max coincident twist moment)	57.251 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2 \text{ m}$: $221.560, 13.796, 55.522 \text{ kN.m}$ ($M_{z,tot} = 20.521 + 35.001$); $M_{pl,y} = 180.400, M_{pl,f} = 41.348, M_{pl,z} = 83.875$	2.504	Warning P385 3.1.2
$C_{mz} \text{ (EN 1993-6 A.1)}$	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.633 \times 0.338 \times 4.024$ ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 55.522/83.875$)	0.861	EN 1993-6 A

$M_y/M_{b,Rd} + C_{mz}M_{z,tot}/M_{z,Rk} + k_wk_{zw}k_\alpha M_w/M_{f,Rk}$	@ x = 2 m: 221.560/149.162 + 1.000x55.522/83.875 + 0.633x0.338x4.024x13.796/41.348 (M _{z,tot} = 20.521 imposed + 35.001 twist $\phi.M_y$)	2.435	Warning EN 1993-6 A																																
Basis with N _{Ed} / M _{z,Ed}	No expression in EN 1993-1-1 or EN 1993-6 combines N _{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi.M_y$ added to M _{z,Ed} , and EN 1993-6 (A.1) with M _{z,Ed} = M _z + $\phi.M_y$		EN 1993-1-1, EN 1993-6																																
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	N _{Ed} /(X _z N _{Rk} /Y _{M1}) + k _{zy} M _{y,Ed} /(X _{LT} M _{y,Rk} /Y _{M1}) + k _{zz} M _{z,tot} /(M _{z,Rk} /Y _{M1}) + k _w k _{zw} k _α M _w /M _{f,Rd} = 2.434 (Eq 6.62 with M _{z,tot} = 57.251) + 0.295 (warping term of (A.1) @ x = 1.643 m: k _w = 0.648, k _{zw} = 0.437, k _α = 4.024, M _w = 10.715 kN.m) =	2.729	information only																																
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	3.823 / 3.096 kN.m																																	
Torsion Shear Design @ 0 m																																			
T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.823 kN.m, V = 137.075 kN	76.13 N/mm²	P385																																
V _{pl,T,Rd} = √(1 - τ _t /(1.25f _y /√3)).V _{pl,Rd}	@ x = 0 m: τ _t = 76.13 N/mm²; V _{pl,Rd} = 352.225; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.785	276.531 kN	6.2.7(9)																																
V _{Ed} /V _{pl,T,Rd}	137.075 / 276.531	0.496	OK																																
Deflection Check - Load Case 1 (SLS: 1.0Q)																																			
In-span δ ≤ Span/360	27.97 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	27.97 mm	Warning																																
Torq in Case 1 @ 2.564 m: θ _{max} ≤ 2.00°	0.0806 rad = 4.62° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	4.62°	> 2° advisory																																
<table><tr><td>N_{Ed}/N_(pl,Rd)</td><td>UNyz</td><td>UMyz</td><td>Ax+M_6.6.1</td><td>Ax+M_6.6.2</td><td>Deflection</td><td>V/Vpl</td><td>MA/Mc</td><td>LTB</td><td>F/F_Rd</td><td>Web 7.2</td><td>Torsion</td><td>Yield 6.1</td><td>V+T</td><td>LTB+T</td><td>Max</td></tr><tr><td>0.071</td><td>1.753</td><td>0.189</td><td>1.416</td><td>2.109</td><td>2.434</td><td>1.678</td><td>0.389</td><td>1.228</td><td>1.416</td><td>0.319</td><td>0.892</td><td>2.504</td><td>8.823</td><td>0.496</td><td>2.435 8.823</td></tr></table>				N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	0.071	1.753	0.189	1.416	2.109	2.434	1.678	0.389	1.228	1.416	0.319	0.892	2.504	8.823	0.496	2.435 8.823
N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max																				
0.071	1.753	0.189	1.416	2.109	2.434	1.678	0.389	1.228	1.416	0.319	0.892	2.504	8.823	0.496	2.435 8.823																				

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 8.823

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 4.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 2.434$ (Eq 6.62 with $M_{z,tot} = 57.25$ kN.m) + 0.295 (warping term of (A.1) at x = 1.64 m, $k_w = 0.648$, $k_{zw} = 0.437$, $k_\alpha = 4.024$) = 2.729. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

UC-L7 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER
UC 203 x 203 x 60, 6 m SS, unrestrained; $L7 = L6 +$ "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member UC-L7 UC 203 x 203 x 60
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

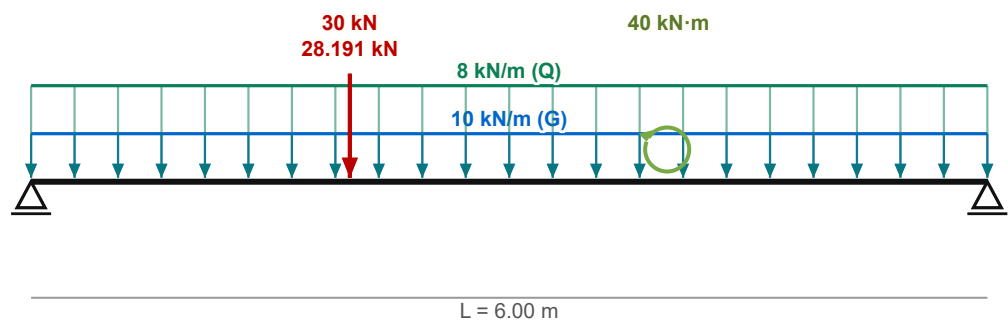
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 40$ mm $z_g = 105$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 105$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 105$ mm (kN)
Q PY 28.191 @ 2 m $e = 40$ mm $z_g = 105$ mm (kN)
Q M 40.000 @ 4 m (kN.m)
G SW 0.5886 kN/m 0–6 m (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.39	147.07	0.00	0.00	20.52	242.55	20.52	-31.88	147.07	—
	End 2, x = 6	150.00C	-4.22	-97.98	0.00	0.00	20.52	@ 2.274	@ —	@ 2.850	97.98	—

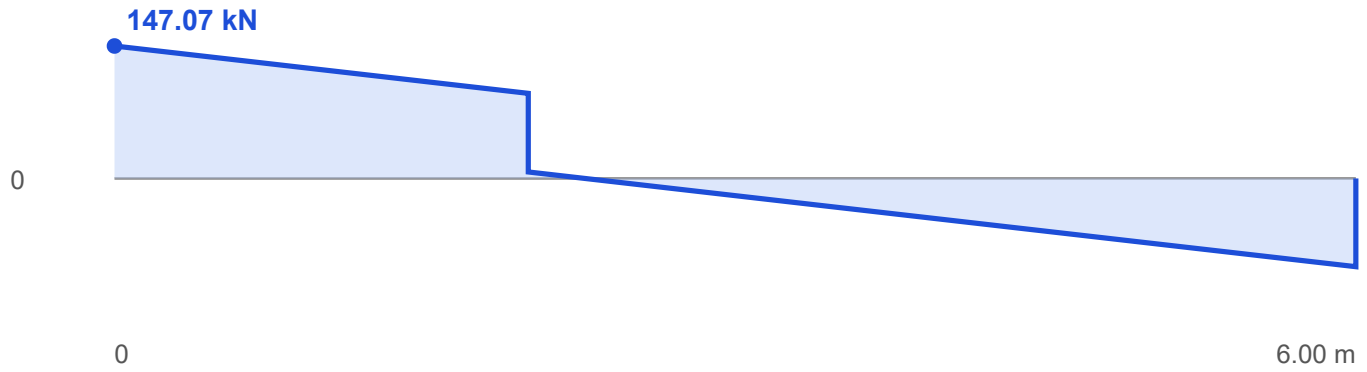
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

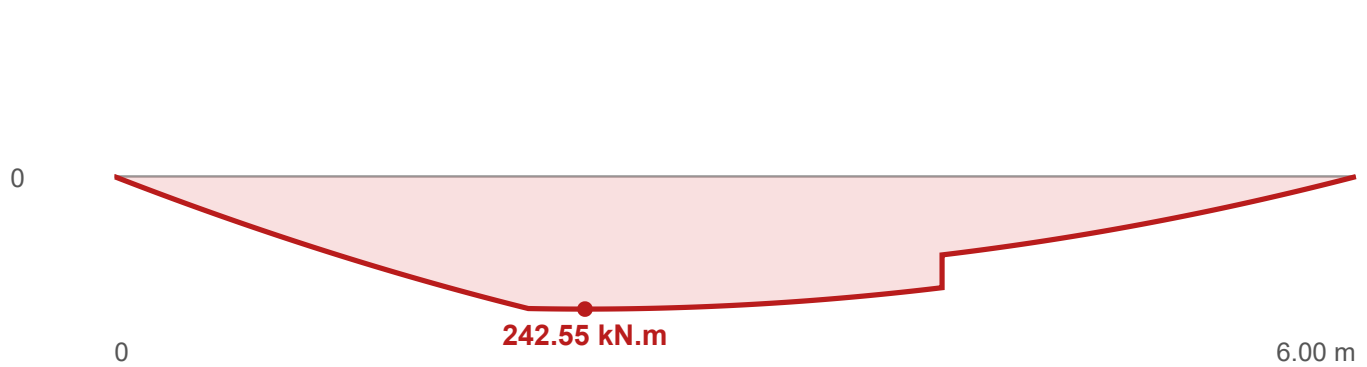
LOADING



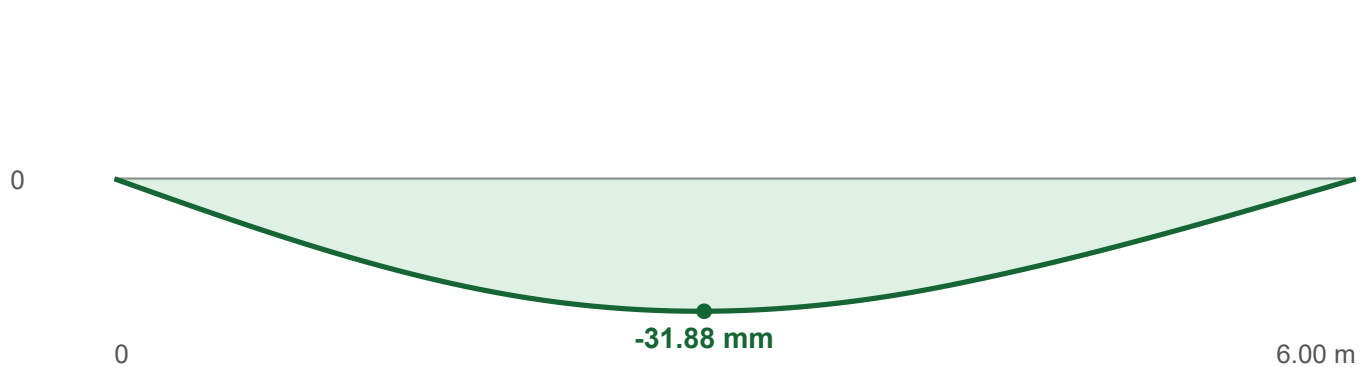
SHEAR FORCE V (kN)



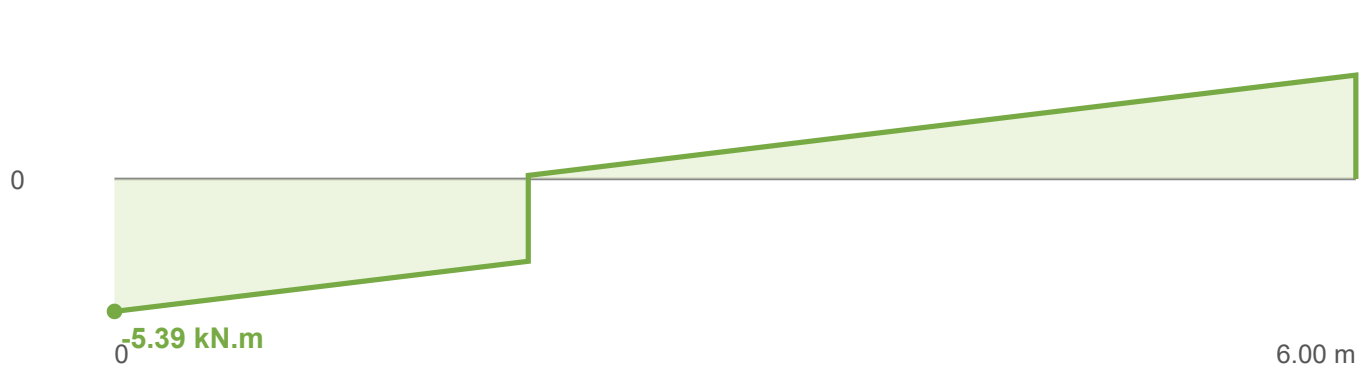
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 150.00, 242.55, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.680$, $\psi = -0.857$; limits $50.5\epsilon / 58.1\epsilon / 108.5\epsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2

Flange outstands under $M_y + M_z$ (+N)	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 449.7, tip 537.0 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 420.2, tip 332.9 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	147.075 / 352.225 =	0.418	OK
$h_w / t_w \leq 72 \epsilon / \eta$	19.28 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.006 / 352.225 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	242.546 / 180.400 =	1.344	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ x = 0.33 m (ULS: 1.35G + 1.5Q): V = 138.398, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 2100.999$, $n_v = 0.071$; $M_{v,y,Rd} = (W_{pl,y} - \rho A_v^2 / 4 t_w) f_y / \gamma_{M0} = 180.400$; $M_{v,z,Rd} = (W_{pl,z} - \rho A_v t_w / 4) f_y / \gamma_{M0} = 83.875$; $a_v = 0.235$; 6.2.9.1(4): no reduction ($N \leq 0.25 N_{V,Rd}$, $N \leq 0.5 h_w t_w (1 - \rho) f_y$); $M_{N,v,y,Rd} = 180.400$, $M_{N,v,z,Rd} = 83.875$, $\alpha = 2$, $\beta = 1$; $M_y = 47.103$, $M_z = 20.521$, $N = 150.000$ (plastic, 6.2.9.1 with $(1 - \rho) f_y$ on A_v)	0.313	OK 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(242.546 / 180.400)^2 + (20.521 / 83.875)^1 =$	2.052	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 241.560$ kN.m, V = 50.842 kN, $\phi = 0.15797$ rad, $\phi' = 3.749e-5$ rad/mm, $\phi'' = 6.516e-8$ rad/mm ² , $\phi''' = 4.627e-11$ rad/mm ³ ; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 38.160 = 58.681$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + E W_{n0} \phi'' = 19.63 + 413.63 + 291.95 + 138.20 = 863.41$ N/mm ² ; $\tau = G t_f \phi' = 43.12$ N/mm ² ; $(863.41 / 275)^2 + 3(43.12 / 275)^2 =$	9.931	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 413.63 + 291.95 + 138.20 = 863.41$, $\tau = 43.12 \rightarrow 9.931$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 385.63 + 13.39 = 418.65$, $\tau = 8.35 + 43.12 + 5.02 = 56.50 \rightarrow 2.444$; P3 web-flange junction, web: $\sigma_x = 19.63 + 357.60 + 13.39 = 390.63$, $\tau = 25.23 + 28.55 = 53.78 \rightarrow 2.132$; P4 web mid-depth: $\sigma_x = 19.63 + 13.39 = 33.02$, $\tau = 28.64 + 28.55 = 57.19 \rightarrow 0.144$ ($y_w = t_w / 2 = 4.70$ mm; $S_f / 2 =$ half flange one side of the web)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20 / 14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2/(2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275/4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1; L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000; 1.000 \times 166.44$	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/Y_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 147.075$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 =$	0.342	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.342 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 + N_{Ed}/N_{pl,Rd} = 150.000/2101.000 = 0.071) = 0.399 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.285	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	147.075	ULS: 1.35G + 1.5Q	0.342	0.285	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.955	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	97.979	ULS: 1.35G + 1.5Q	0.228	0.203	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); $c =$ distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y/N_{cr}}$	$\sqrt{76.40 \times 275/3523.45}$ ($N_{cr,y} = \pi^2 EI_y/L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y/Y_{M1}$	$76.40 \times 0.742 \times 275/10/1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y/N_{crz}}$	$\sqrt{76.40 \times 275/1186.00}$ ($N_{cr,z} = \pi^2 EI_z/L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y/Y_{M1}$	$76.40 \times 0.377 \times 275/10/1 =$	791.578 kN	Curve c
$L_{et} = K_t \cdot L_z; \lambda_T; N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 791.578 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.189	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

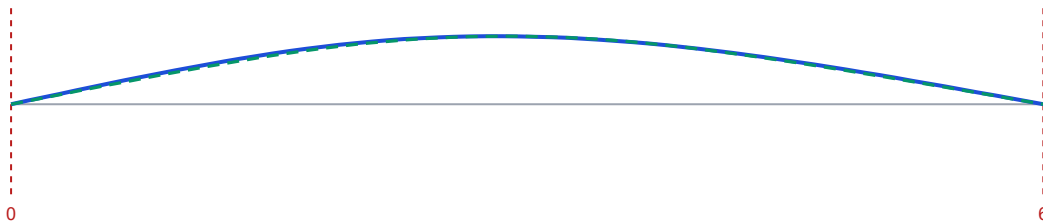
$C_1 = M_{cr}/M_{cr,uniform}$	$284.105 / 242.476$ kN.m (shape only: $z_g = z_j = 0$)	1.172	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 235.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=191.0, M(L/2)=235.6, M(3L/4)=117.4, M_{max}=242.5$ kN.m	1.213	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 235.61, \psi = 0.659, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 40.987 twist $\phi \cdot M_y = 61.508$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3

$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 235.61$, $\psi = 0.659$, $\alpha_h = 0.000$	1.000	Table B.3
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Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 105$, 0 mm; 1 combination(s), 3 solve(s)	228.573 kN.m	converged
$\chi_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{656.0 \times 275 / 228.573}$	0.888	
$\chi_{LT} = Fn(\chi_{LT}, \Phi_{LT}, \alpha_{LT})$	0.888, 0.879, 0.34	0.767	Curve b
$\chi_{LT,mod} = Fn(\chi_{LT}, \chi_{LT}, k_c, f)$	0.767, 0.888, 0.924, 0.963	0.797	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.797 \times 656.0 \times 275 \leq 180.400 =$	143.723 kN.m	
$M_{y,Ed}/M_{b,Rd}$	242.546 / 143.723	1.688	Warning

Buckled mode shape — critical eigenmode (normalised; twist φ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.86 m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	242.546 / 143.723 ($M_{y,Rk} = 180.400$)	1.688	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 40.987 twist $\varphi \cdot M_y = 61.508$) / 83.875	0.733	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1 + (\chi_y - 0.2)U_{N,y}\}$	$1.000\{1 + (0.771 - 0.2) \times 0.096\} \leq 1.000(1 + 0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1 + (2\chi_z - 0.6)U_{N,z}\}$	$1.000\{1 + (2 \times 1.329 - 0.6) \times 0.189\} \leq 1.000(1 + 1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\chi_z U_{N,z}/(C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.096 + 1.055 \times 1.688 + 0.759 \times 0.733$	2.433	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.189 + 0.975 \times 1.688 + 1.265 \times 0.733$	2.762	Warning

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.066 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 5.77	P385 App C
φ_{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	2.696 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h - t_f)$	max over span	13.796 kN.m	
$M_{z,tot} = M_{z,Ed} + \varphi \cdot M_{y,Ed}$	20.521 + 241.436 x 169.76e-3 (@ x = 2.564 m) = 20.521 + 40.987 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	61.508 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ x = 2.066 m: 241.980, 13.276, 59.315 kN.m ($M_{z,tot} = 20.521 + 38.794$); $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	2.827	Warning P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 228.573 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w \cdot k_{zw} \cdot k_\alpha \cdot M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 228.573 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi_i M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi_i M_y$				EN 1993-1-1, EN 1993-6											
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 5.388 / 4.224$ kN.m			3.823 / 3.096 kN.m												
Torsion Shear Design @ 0 m																
T_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.823$ kN.m, $V = 147.075$ kN			76.13 N/mm ²	P385											
$V_{pl.T.Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ $x = 0$ m: $T_t = 76.13$ N/mm ² ; $V_{pl.Rd} = 352.225$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.785$			276.531 kN	6.2.7(9)											
$V_{Ed}/V_{pl.T.Rd}$	147.075 / 276.531			0.532	OK											
Deflection Check - Load Case 1 (SLS: 1.0Q)																
In-span $\delta \leq \text{Span}/360$	31.88 $\leq$ 6000 / 360 = 16.67 mm @ $x = 2.85$ m			31.88 mm	Warning											
Torq in Case 1 @ 2.564 m: $\theta_{max} \leq 2.00^\circ$	0.0806 rad = 4.62° $\leq$ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)			4.62°	> 2° advisory											
$N_{Ed}/N_{pl.Rd}$	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.071	2.052	0.189	1.688	2.433	2.762	1.913	0.418	1.344	1.688	0.313	0.342	0.955	2.827	9.931	0.532	99.00099.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 4.6^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

UC-L7 · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 9.931

Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER

UC 203 x 203 x 60, 6 m SS, unrestrained; $L7 = L6 +$ "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member UC-L7 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40$ mm $z_g = 105$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 105$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 105$ mm (kN)

Q PY 28.191 @ 2 m $e = 40$ mm $z_g = 105$ mm (kN)

Q M 40.000 @ 4 m (kN.m)

G SW 0.5886 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

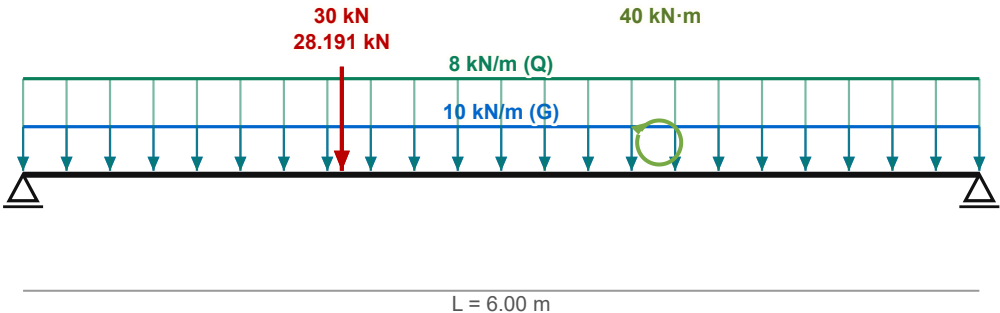
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.39	147.07	0.00	0.00	20.52	242.55	20.52	-31.88	147.07	—
	End 2, x = 6	150.00C	-4.22	-97.98	0.00	0.00	20.52	@ 2.274	@ —	@ 2.850	97.98	—

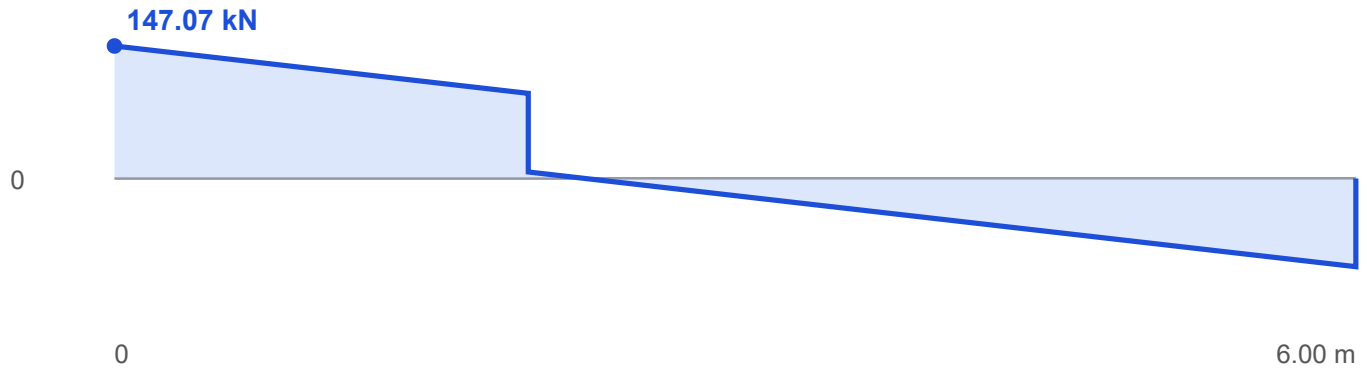
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

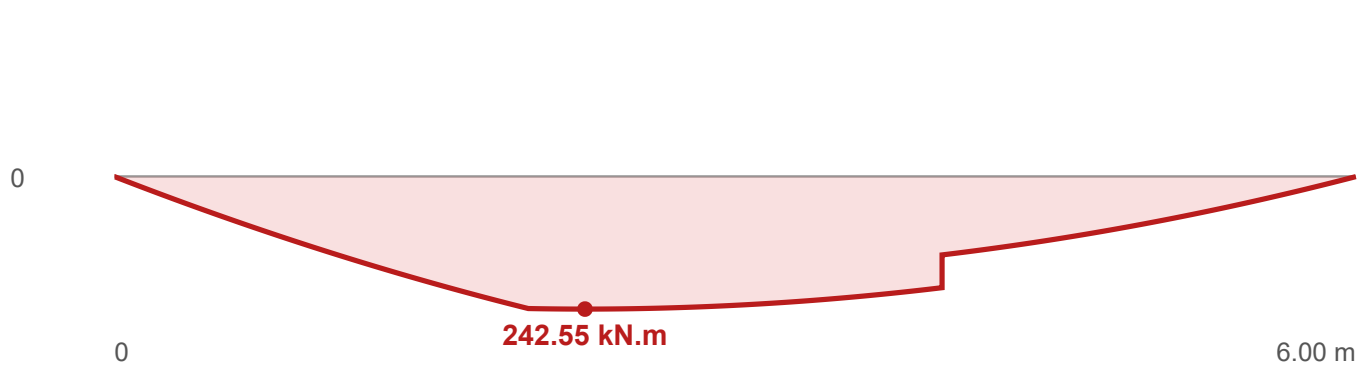
LOADING



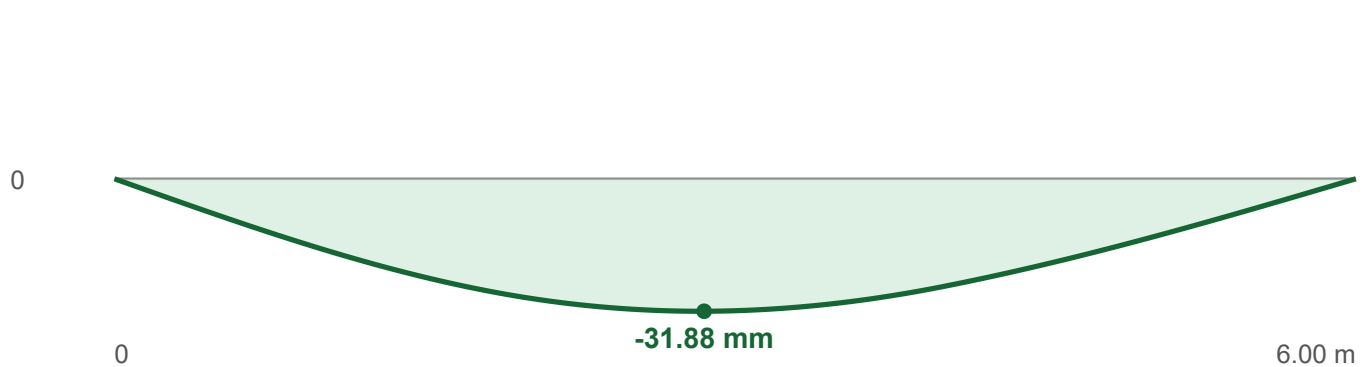
SHEAR FORCE V (kN)



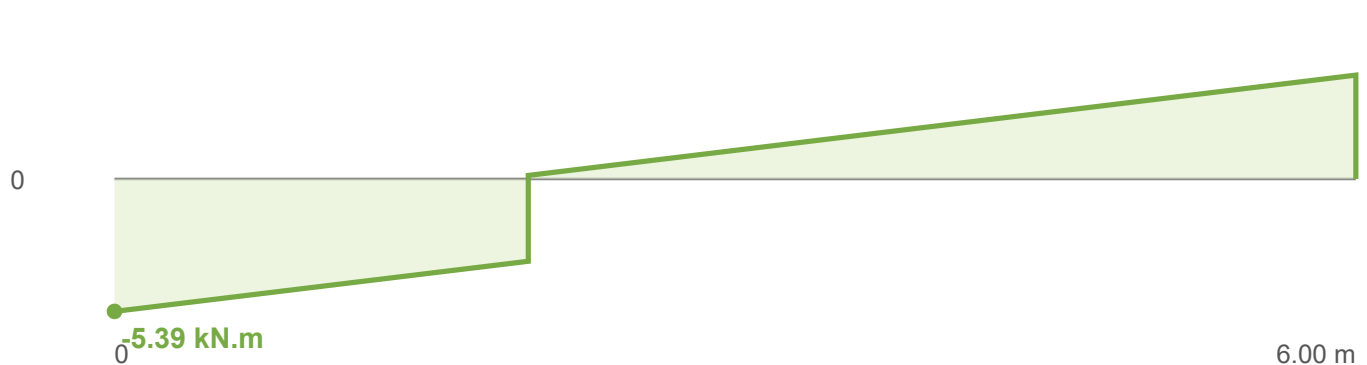
BENDING MOMENT M (kN.M)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 150.00, 242.55, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.680$, $\psi = -0.857$; limits $50.5\epsilon / 58.1\epsilon / 108.5\epsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2

Flange outstands under $M_y + M_z$ (+N)	$\sigma = N/A + M_y/W_{el,y} \pm M_z/l_z$: outstand compressed by M_z : root 449.7, tip 537.0 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha$, $10\epsilon/\alpha$, $21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 420.2, tip 332.9 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	147.075 / 352.225 =	0.418	OK
$h_w / t_w \leq 72 \epsilon / \eta$	19.28 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.006 / 352.225 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	242.546 / 180.400 =	1.344	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ x = 0.33 m (ULS: 1.35G + 1.5Q): V = 138.398, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 2100.999$, $n_v = 0.071$; $M_{v,y,Rd} = (W_{pl,y} - \rho A_v^2 / 4 t_w) f_y / \gamma_{M0} = 180.400$; $M_{v,z,Rd} = (W_{pl,z} - \rho A_v t_w / 4) f_y / \gamma_{M0} = 83.875$; $a_v = 0.235$; 6.2.9.1(4): no reduction ($N \leq 0.25 N_{V,Rd}$, $N \leq 0.5 h_w t_w (1 - \rho) f_y$); $M_{N,v,y,Rd} = 180.400$, $M_{N,v,z,Rd} = 83.875$, $\alpha = 2$, $\beta = 1$; $M_y = 47.103$, $M_z = 20.521$, $N = 150.000$ (plastic, 6.2.9.1 with $(1 - \rho) f_y$ on A_v)	0.313	OK 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(242.546 / 180.400)^2 + (20.521 / 83.875)^1 =$	2.052	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, P1 flange tip	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 241.560$ kN.m, V = 50.842 kN, $\phi = 0.15797$ rad, $\phi' = 3.749e-5$ rad/mm, $\phi'' = 6.516e-8$ rad/mm ² , $\phi''' = 4.627e-11$ rad/mm ³ ; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 38.160 = 58.681$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} + E W_{n0} \phi'' = 19.63 + 413.63 + 291.95 + 138.20 = 863.41$ N/mm ² ; $\tau = G t_f \phi' = 43.12$ N/mm ² ; $(863.41 / 275)^2 + 3(43.12 / 275)^2 =$	9.931	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 19.63 + 413.63 + 291.95 + 138.20 = 863.41$, $\tau = 43.12 \rightarrow 9.931$ (governs); P2 web-flange junction, flange: $\sigma_x = 19.63 + 385.63 + 13.39 = 418.65$, $\tau = 8.35 + 43.12 + 5.02 = 56.50 \rightarrow 2.444$; P3 web-flange junction, web: $\sigma_x = 19.63 + 357.60 + 13.39 = 390.63$, $\tau = 25.23 + 28.55 = 53.78 \rightarrow 2.132$; P4 web mid-depth: $\sigma_x = 19.63 + 13.39 = 33.02$, $\tau = 28.64 + 28.55 = 57.19 \rightarrow 0.144$ ($y_w = t_w / 2 = 4.70$ mm; $S_f / 2 =$ half flange one side of the web)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 147.075$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 430.254 =	0.342	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.342 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 + N_{Ed}/N_{pl,Rd} = 150.000/2101.000 = 0.071) = 0.399 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.285	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	147.075	ULS: 1.35G + 1.5Q	0.342	0.285	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.955	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	97.979	ULS: 1.35G + 1.5Q	0.228	0.203	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10/1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10/1 =$	791.578 kN	Curve c

$L_{et} = K_t \cdot L_z ; \bar{\lambda}_T ; N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 791.578 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.189	OK
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my			
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 235.6, 1.000, 300.000	1.213	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=191.0$, $M(L/2)=235.6$, $M(3L/4)=117.4$, $M_{max}=242.5$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 235.61$, $\psi = 0.659$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 40.987 twist $\phi \cdot M_y = 61.508$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 235.61$, $\psi = 0.659$, $\alpha_h = 0.000$	1.000	Table B.3
Lateral Buckling Check M.b.Rd			
$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.213, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	294.111 kN.m	BLOCKED
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 294.111}$	0.783	
$\chi_{LT} = Fn(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	0.783, 0.795, 0.34	0.826	Curve b
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.826, 0.783, 0.908, 0.954	0.866	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.866 \times 656.0 \times 275 \leq 180.400 =$	156.254 kN.m	
$\bar{\lambda}_{LT}$ (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\bar{\lambda}_z\sqrt{\beta_w} = 1.086$; $\chi_{LT,mod} = 0.673$; $M_{y,Ed}/M_{b,Rd,simp} = 1.998$	121.390 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	242.546 / 156.254	1.552	Warning
Buckling Resistance			
$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 1558.850$ ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 791.578$	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	$242.546 / 156.254$ ($M_{y,Rk} = 180.400$)	1.552	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(20.521 \text{ imposed} + 40.987 \text{ twist } \phi \cdot M_y = 61.508) / 83.875$	0.733	OK
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+(\bar{\lambda}_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\bar{\lambda}_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\bar{\lambda}_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\bar{\lambda}_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\bar{\lambda}_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.096 + 1.055 \times 1.552 + 0.759 \times 0.733$	2.291	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.189 + 0.975 \times 1.552 + 1.265 \times 0.733$	2.630	Warning
Torsion Design			
J, H, a, Q_f, Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W_{n0}, S_{w1}	101 cm ² , 734 cm ⁴		P385 App A
Torsion Bending Design @ 2.066 m			
T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
ϕ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	2.696 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	13.796 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 241.436 \times 169.76e-3$ (@ $x = 2.564$ m) = $20.521 + 40.987$ (imposed constant $M_{z,Ed}$ + max coincident twist moment)	61.508 kN.m	P385 3.1.2

$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ x = 2.066 m: 241.980, 13.276, 59.315 kN.m ($M_{z,tot} = 20.521 + 38.794$); $M_{pl,y} = 180.400$, $M_{pl,f} = 41.348$, $M_{pl,z} = 83.875$	2.827	Warning P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.633 \times 0.300 \times 5.704$ ($k_{zw} = 1 - M_{z,tot}/M_{z,Rk} = 1 - 58.681/83.875$)	1.085	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz}M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2 m: 241.560/149.069 + 1.000x58.681/83.875 + 0.633x0.300x5.704x13.796/41.348 ($M_{z,tot} = 20.521$ imposed + 38.160 twist $\phi \cdot M_y$)	2.682	Warning EN 1993-6 A
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6
superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only	$N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_w/M_{f,Rd} = 2.630$ (Eq 6.62 with $M_{z,tot} = 61.508$) + 0.396 (warping term of (A.1)) @ x = 1.469 m: $k_w = 0.654$, $k_{zw} = 0.465$, $k_\alpha = 5.704$, $M_w = 9.448$ kN.m) =	3.027	information only

End torques T_t St Venant part $GI_T \phi'$ at x = 0 / x = L; total $T = GI_T \phi' - EI_w \phi''' = 5.388 / 4.224$ kN.m

Torsion Shear Design @ 0 m

T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.823$ kN.m, $V = 147.075$ kN	76.13 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 76.13$ N/mm ² ; $V_{pl,Rd} = 352.225$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.785$	276.531 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	147.075 / 276.531	0.532	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$31.88 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	31.88 mm	Warning
Torq in Case 1 @ 2.564 m: $\theta_{max} \leq 2.00^\circ$	$0.0806 \text{ rad} = 4.62^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	4.62°	> 2° advisory

N_Ed/N_(pL_Rd)	UNyz	UMyz	Ax+M_6.4x+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max		
0.071	2.052	0.189	1.552	2.291	2.630	1.913	0.418	1.344	1.552	0.313	0.342	0.955	2.827	9.931	0.532	2.682	9.931

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 9.931

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 4.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: superposition of Eq 6.62 and (A.1) - not a Eurocode expression, information only: $N_{Ed}/(X_z N_{Rk}/Y_{M1}) + k_{zy} M_{y,Ed}/(X_{LT} M_{y,Rk}/Y_{M1}) + k_{zz} M_{z,tot}/(M_{z,Rk}/Y_{M1}) + k_w k_{zw} k_\alpha M_{w,Ed}/M_{f,Rd} = 2.630$ (Eq 6.62 with $M_{z,tot} = 61.51$ kN.m) + 0.396 (warping term of (A.1) at x = 1.47 m, $k_w = 0.654$, $k_{zw} = 0.465$, $k_\alpha = 5.704$) = 3.027. Not a utilisation; the verdict rests on (6.1), 6.61/6.62, (A.1), $V/V_{pl,T,Rd}$ and P385 3.1.2.

RHS 200 x 100 x 8.0 (HF), 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0, z_g = 0$)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

G SW 0.3443 kN/m 0-6 m (automatic)

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{\text{G, inf}} = 0.9$, EQU set A]

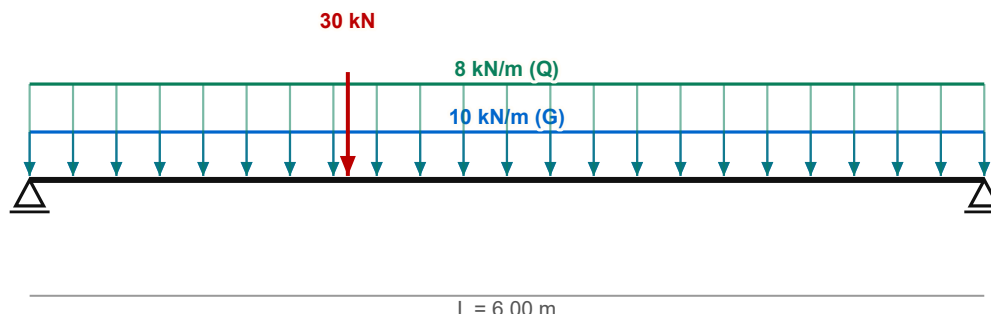
$\gamma_{G,inf}$ companions: 2 ULS Combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

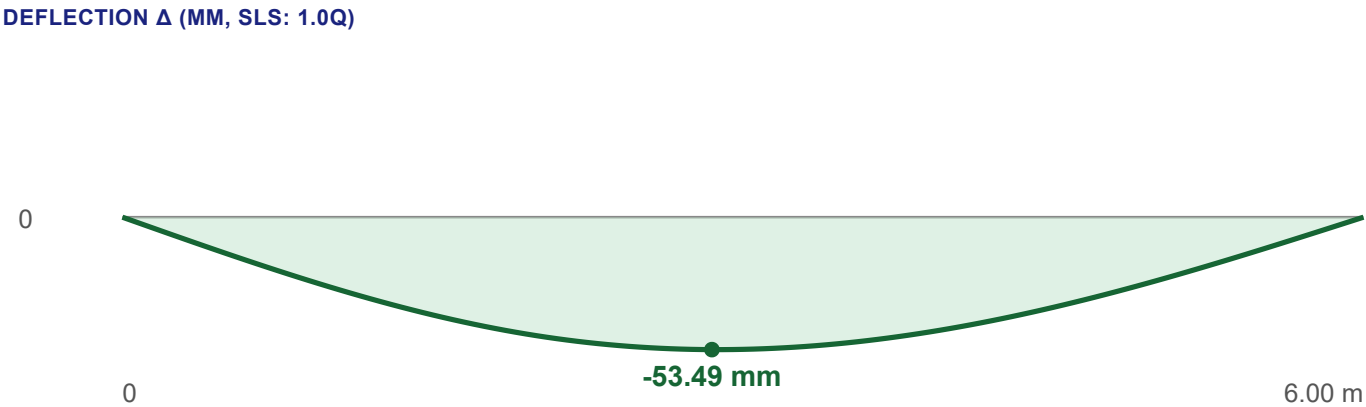
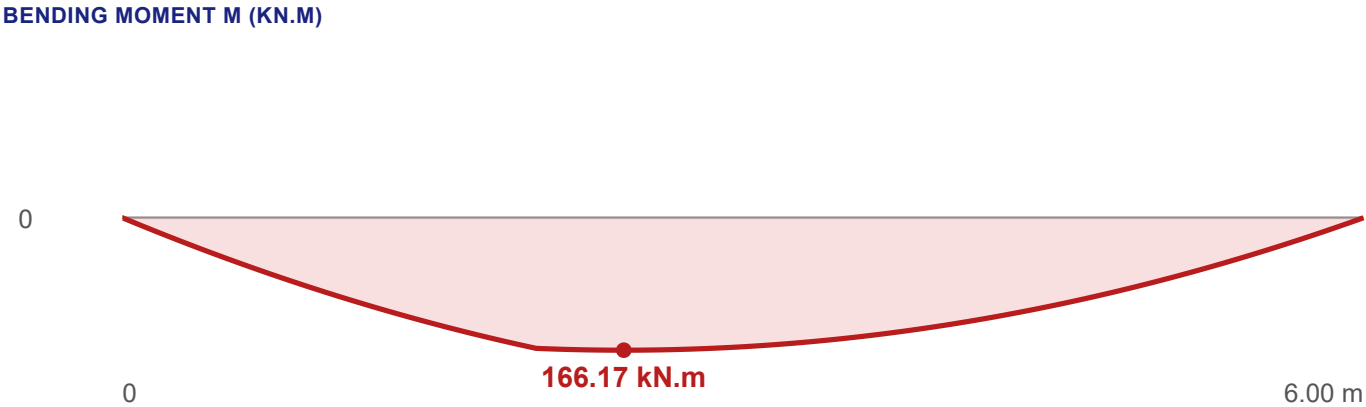
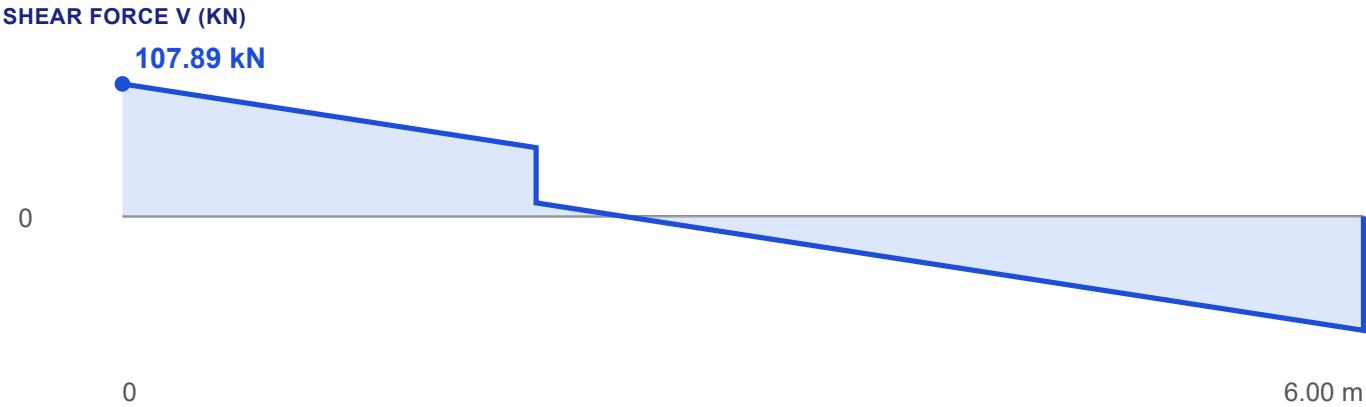
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6.00C	0.00	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.044 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017e3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

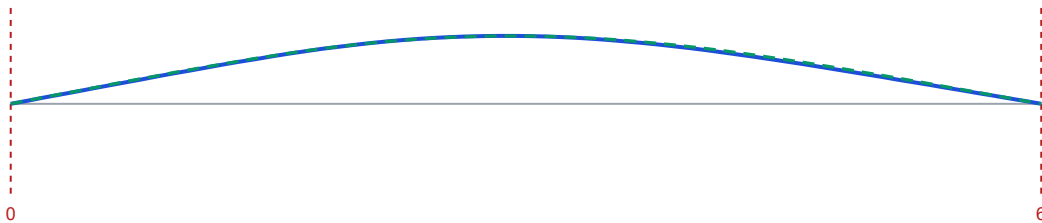
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	899.055 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 899.055}$	0.294	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.294 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 53.49 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0, z_g = 0$)

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L1 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: $1.35G + 1.5Q$

G UDL 10.000 0-6 m (kN/m)

Q UDL 8.000 0-6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: $1.35G + 1.5Q$ [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: $1.35G + 1.5Q$ [$\gamma_{G,inf} = 0.9$, EQU set A]

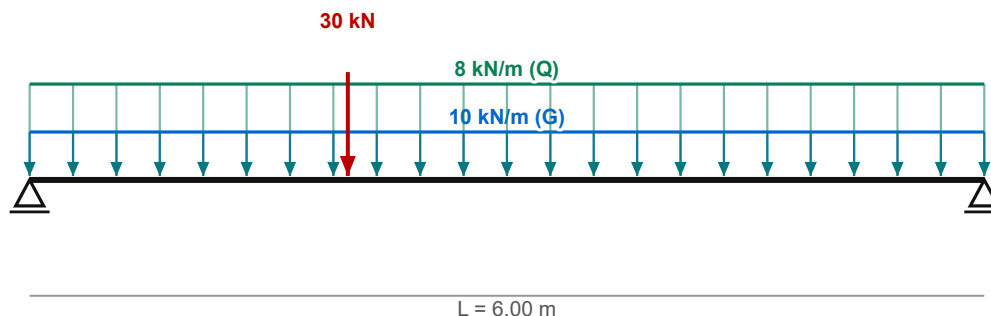
$\gamma_{G,inf}$ companions: 2 ULS Combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

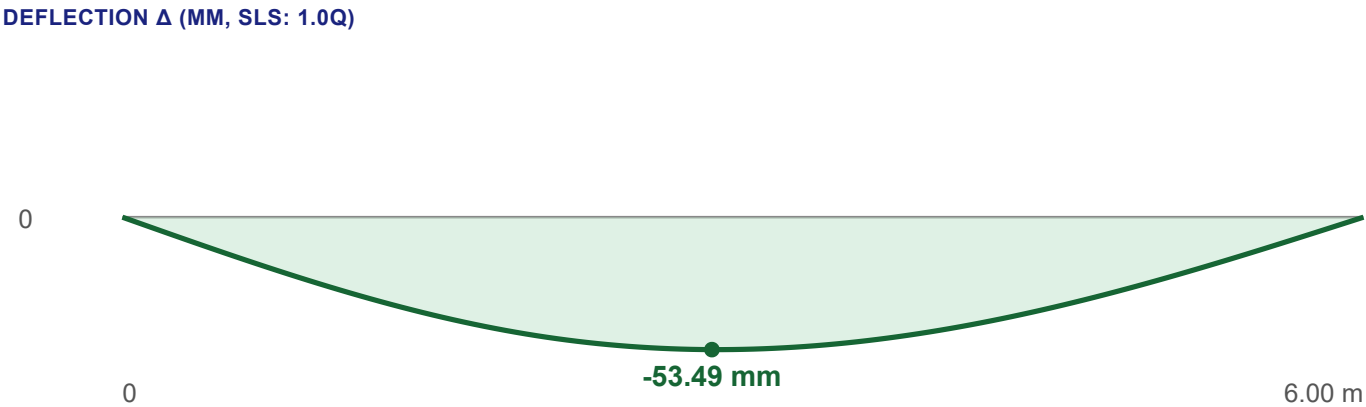
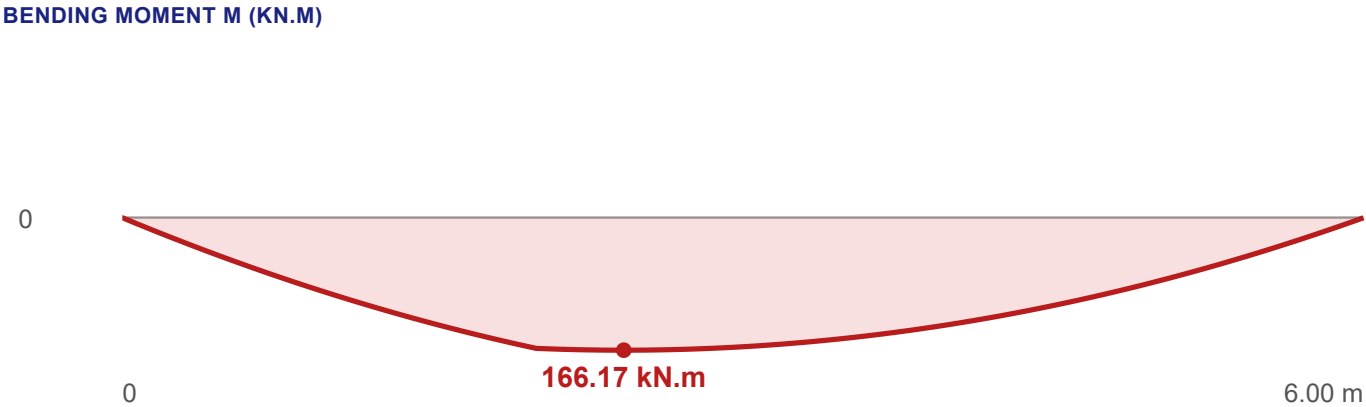
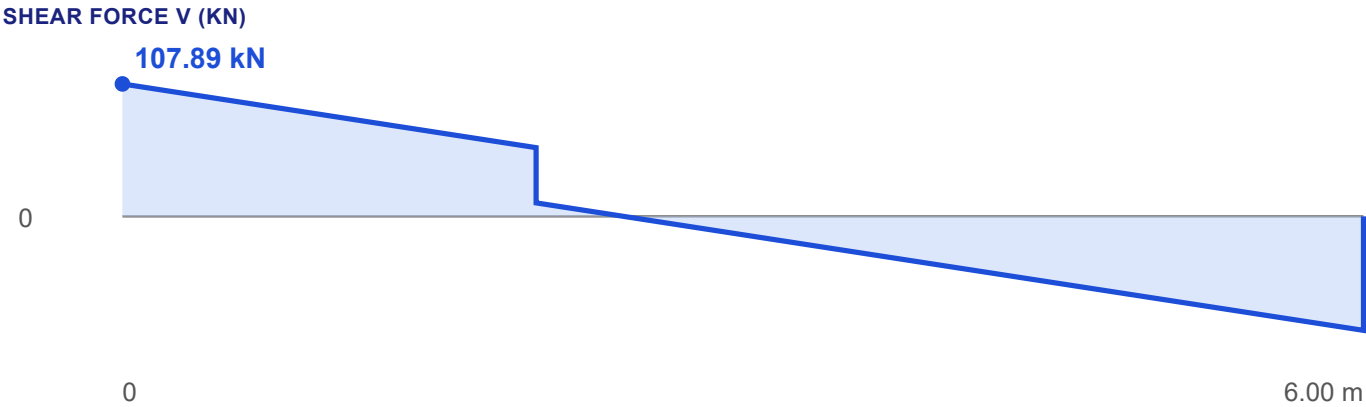
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.044 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017e3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.8, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	916.230 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member RHS-L2 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6 0.00C	0.00	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

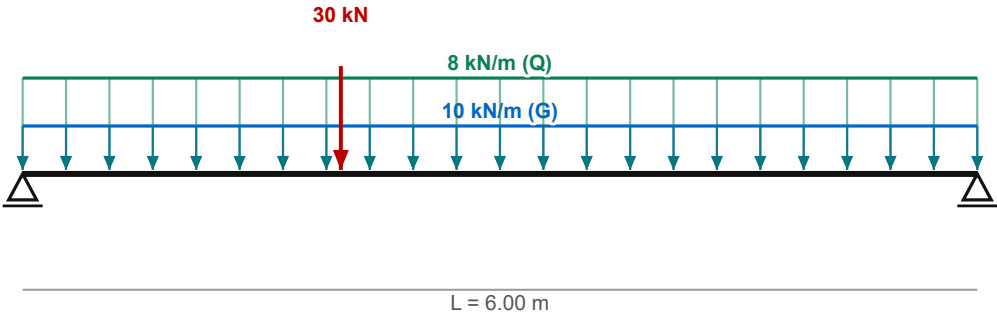
Uplift

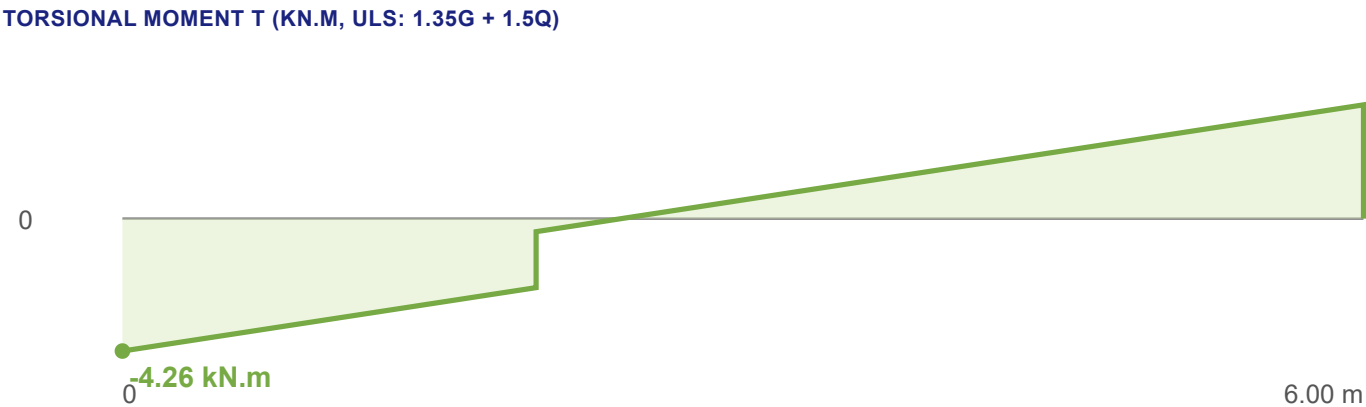
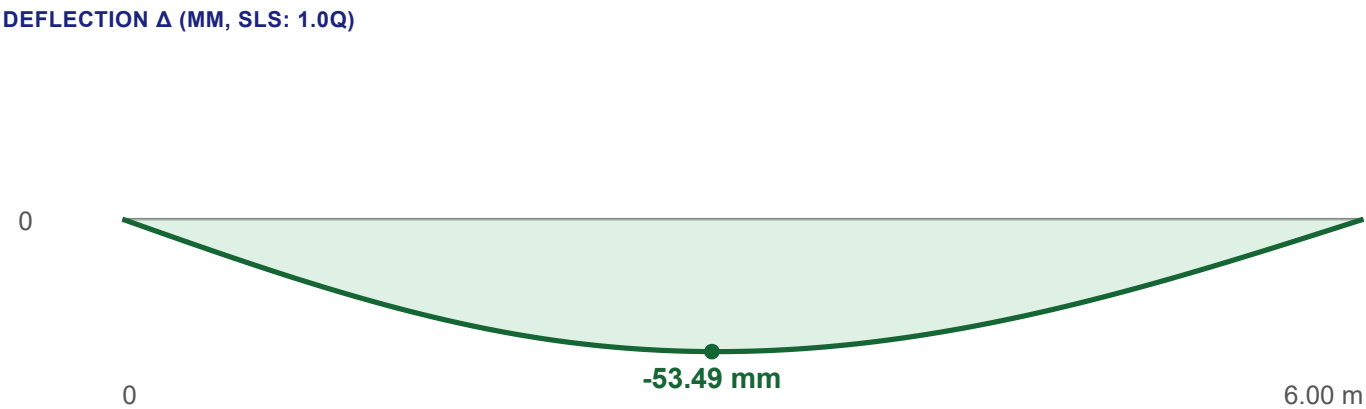
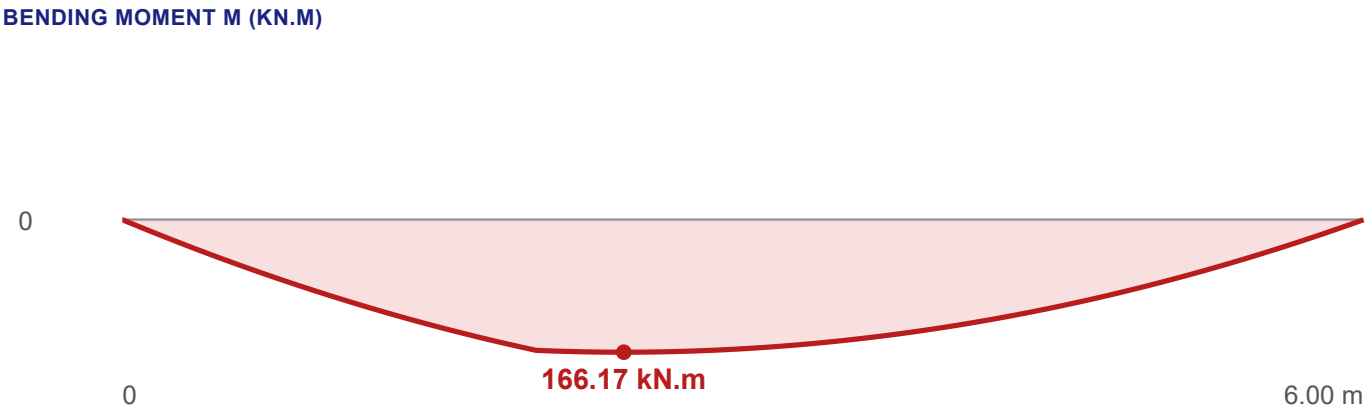
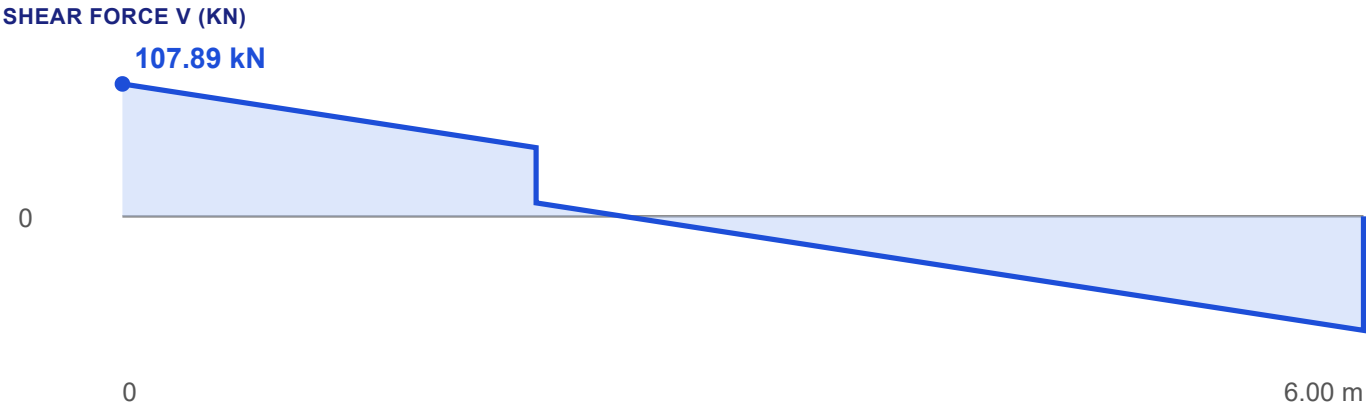
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200		
	B=100 t=8		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	9.50, 22.00, 275, 0.00, 166.17, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174$ kN.m, $V = 0.044$ kN, $T = 0.012$ kN.m, $\phi = 0.00450$ rad; $M_z = \phi M_y = 0.748$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 745.18 + 5.06 = 750.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06$ N/mm ² ; $(750.23/275)^2 + 3(0.06/275)^2 =$	7.443	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 745.18 + 5.06 = 750.23$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 7.443$ (governs); web mid-depth: $\sigma_x = 0.00 + 5.06 = 5.06$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 745.18 = 745.18$, $\tau = 0.05 \rightarrow 7.343$ ($Q_c = 35.3$, $Q_m = 72.2$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm ($d = 2000.00$ to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \times 10^3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.859 / 77.550 = 2.113) = 2.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.451	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

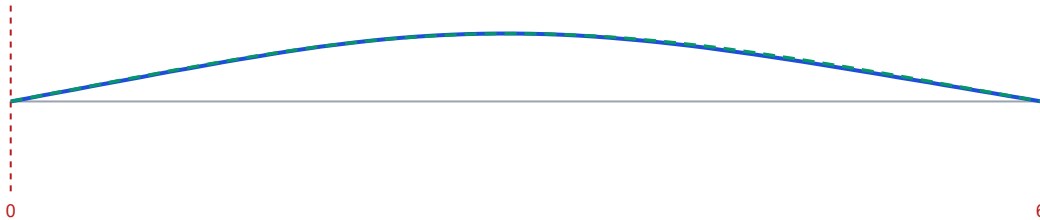
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k _c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	899.055 kN.m	converged
$\lambda_{LT} = \sqrt{W_{f,y}/M_{cr}}$	$\sqrt{282.0 \times 275 / 899.055}$	0.294	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.294 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /251	16.97 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
$(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t /(f _y /√3)] · V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 16.97 N/mm ² ; V _{pl,Rd} = 474.197; S _{mod} =	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	V _{pl,T,Rd} /V _{pl,Rd} = 0.893 107.894 / 423.507	0.255	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0020 rad = $0.11^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	7.443	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 7.443 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member RHS-L2 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6 0.00C		0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

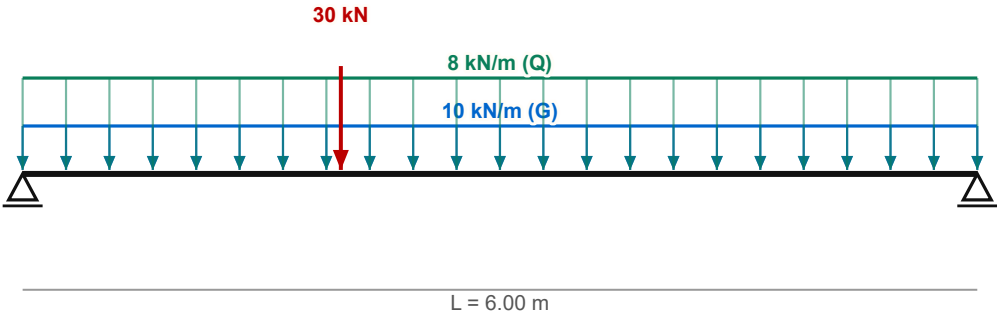
Uplift

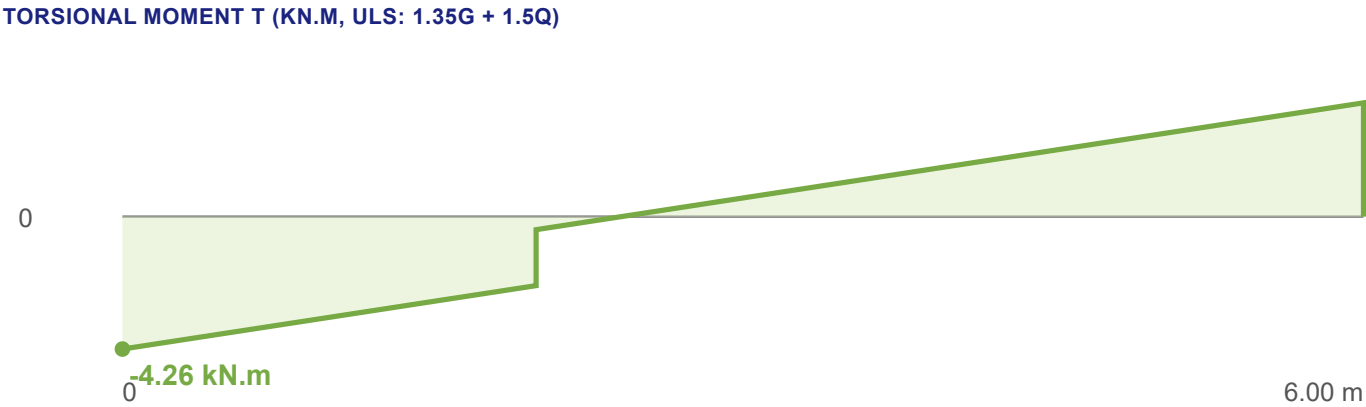
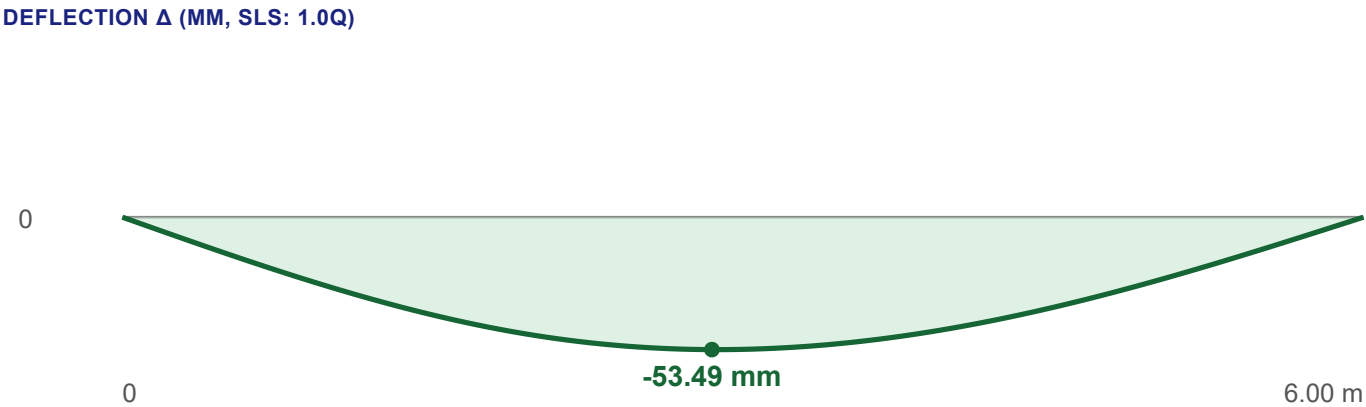
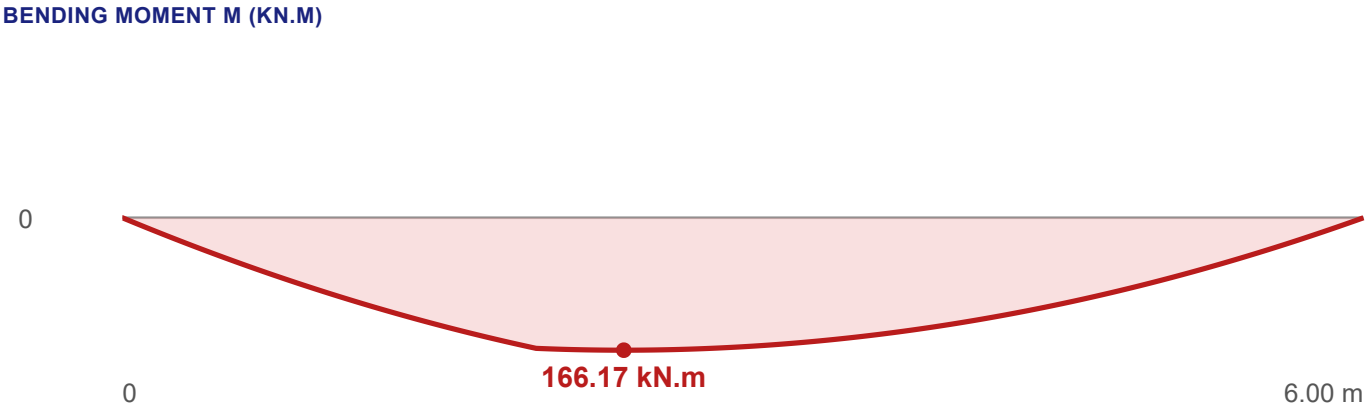
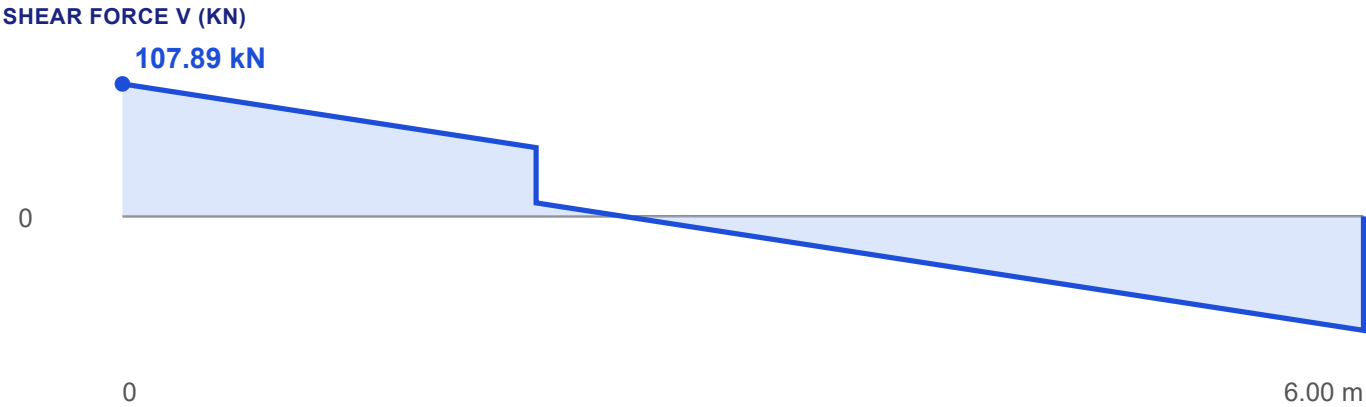
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200		
	B=100 t=8		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	9.50, 22.00, 275, 0.00, 166.17, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174$ kN.m, $V = 0.044$ kN, $T = 0.012$ kN.m, $\phi = 0.00450$ rad; $M_z = \phi M_y = 0.748$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 745.18 + 5.06 = 750.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06$ N/mm ² ; $(750.23/275)^2 + 3(0.06/275)^2 =$	7.443	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 745.18 + 5.06 = 750.23$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 7.443$ (governs); web mid-depth: $\sigma_x = 0.00 + 5.06 = 5.06$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 745.18 = 745.18$, $\tau = 0.05 \rightarrow 7.343$ ($Q_c = 35.3$, $Q_m = 72.2$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \times 10^3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.859 / 77.550 = 2.113) = 2.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.451	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.8, 1.000, 300.000	1.163	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.6, M(L/2)=161.8, M(3L/4)=110.1, M _{max} =166.2 kN·m		SN003a

Lateral Buckling Check M.b.Rd

M _{b,Rd} = M _{c,y,Rd}	closed hollow section, λ _{LT} = 0.291 ≤ 0.4: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.163, L _e = 6 m; z _g = 0 (load through the shear centre)	916.230 kN.m	SN003a
M _{y,Ed} /M _{b,Rd}	166.174 / 77.550	2.143	Warning

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /251	16.97 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} /(M _{pl,y,Rd} ·S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} ·S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - T _t /(f _y /√3)].V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, T _t = 16.97 N/mm ² ; V _{pl,Rd} = 474.197; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.893	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	107.894 / 423.507	0.255	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
Torq in Case 1 @ 2 m: θ _{max} ≤ 2.00°	0.0020 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	7.443	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down

forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 7.443 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

RHS-L3t · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +100 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L3t RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

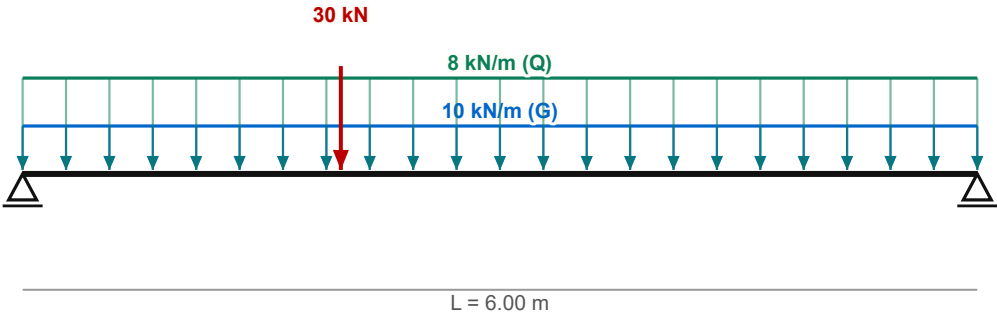
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

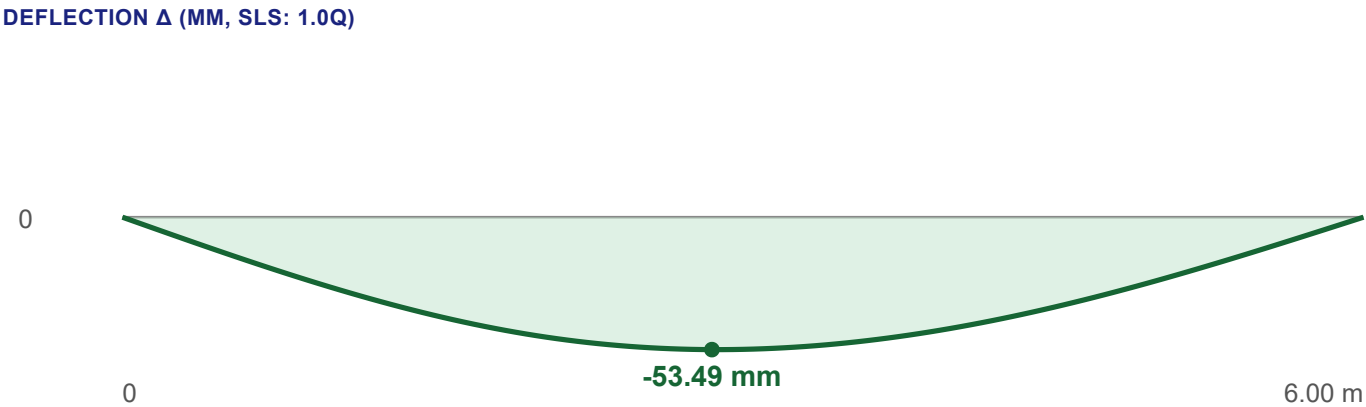
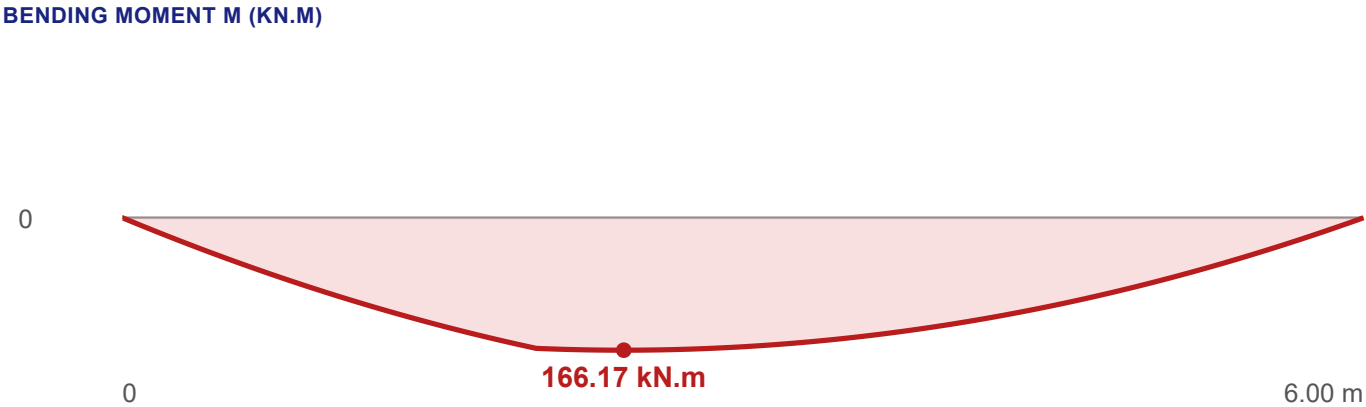
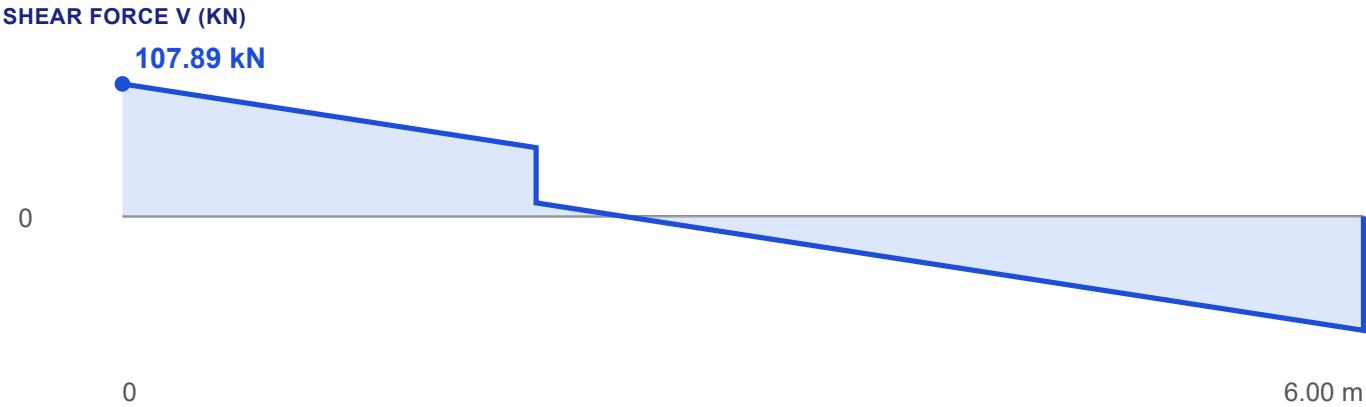
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	22.00 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017e3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

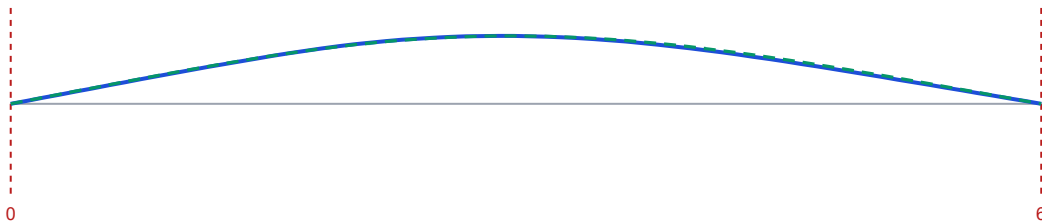
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	876.701 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 876.701}$	0.297	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.297 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 53.49 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L3t · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +100 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L3t RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

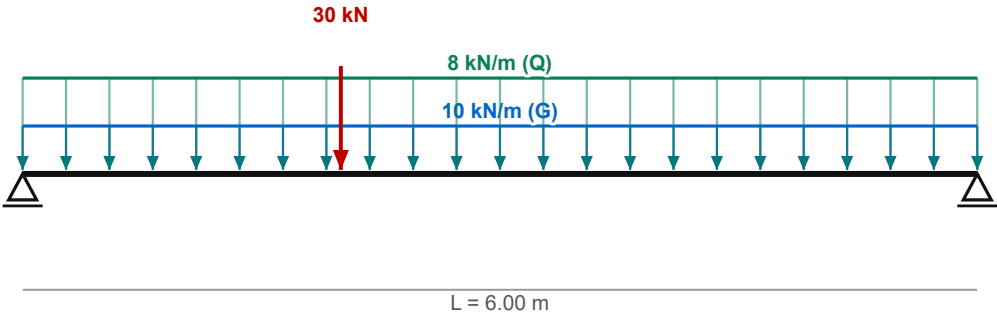
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

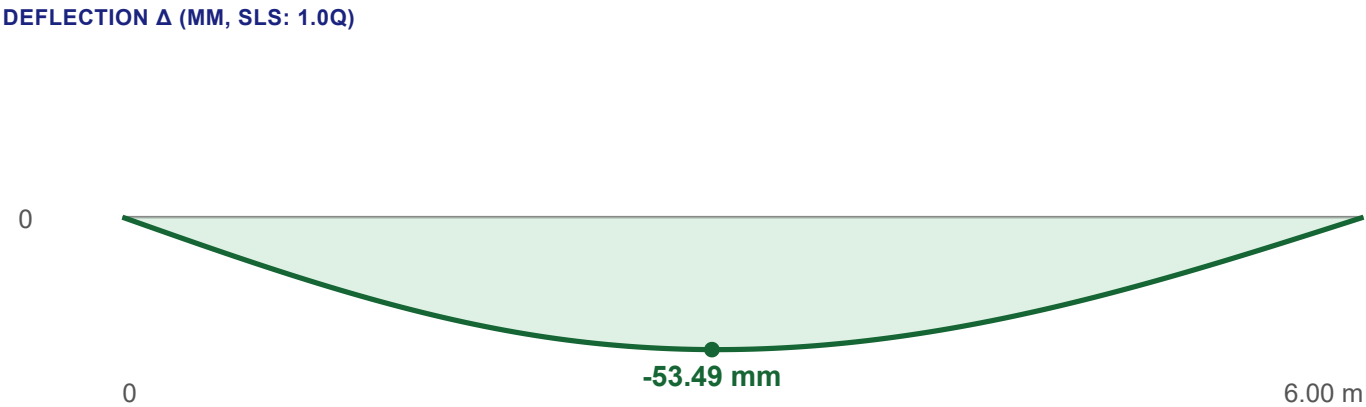
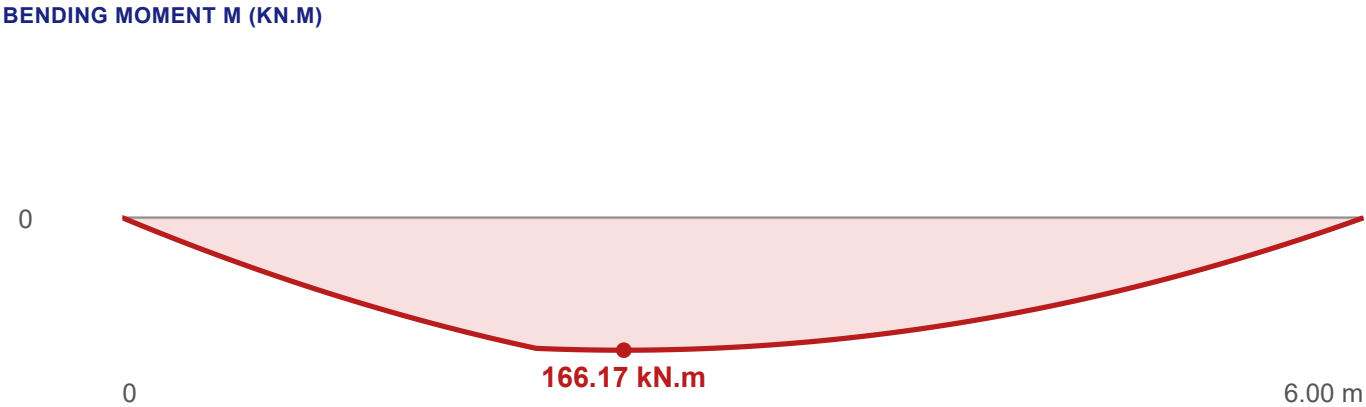
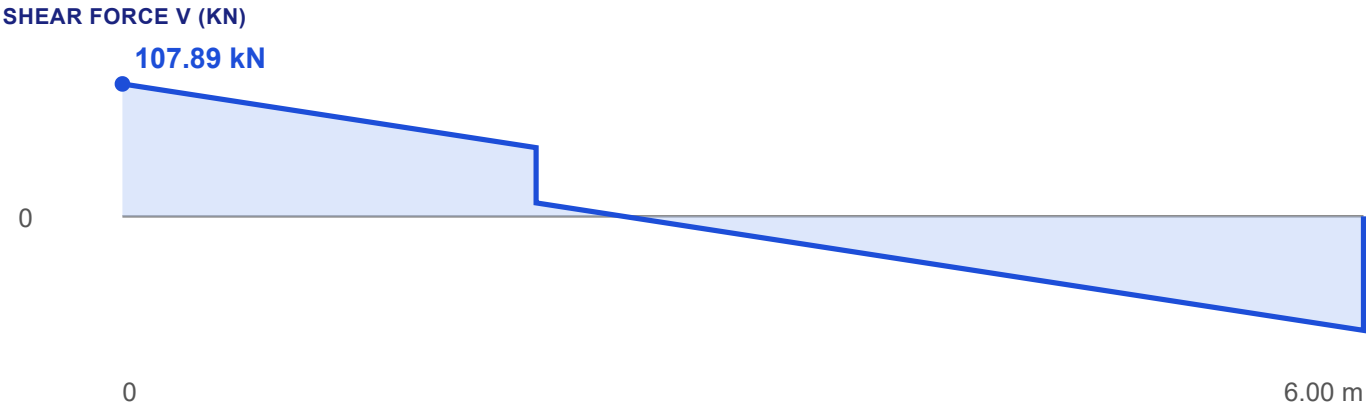
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	2986.7 mm ² x 275 / (√3 x 1)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	22.00 ≤ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3/176.00$	2974.017 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^3 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017)} = 0.298$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00/1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.8, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
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M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.163, L _e = 6 m; z _g = +100 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	916.230 kN.m	SN003a
M _{y.Ed} /M _{b.Rd}	166.174 / 77.550	2.143	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +100 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L3b · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -100 mm (-D/2, stabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L3b RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 0 mm z_g = -100 mm (kN/m)

Q UDL 8.000 0-6 m e = 0 mm z_g = -100 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = -100 mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

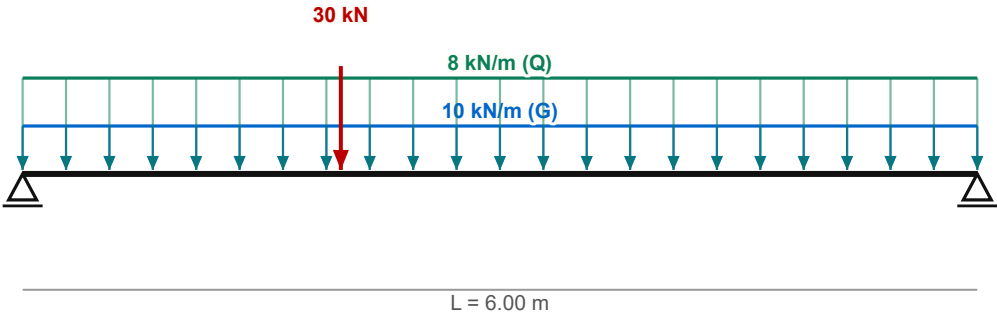
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

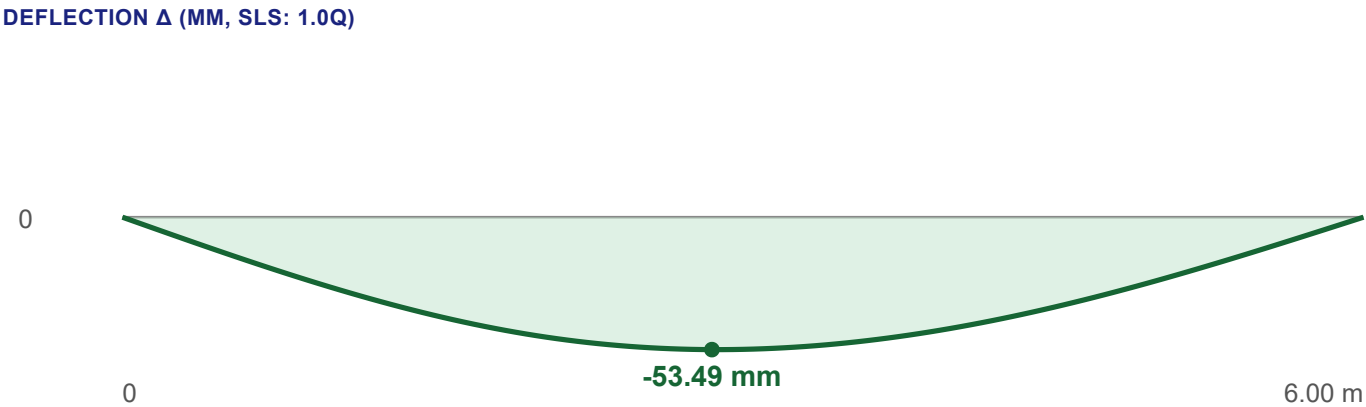
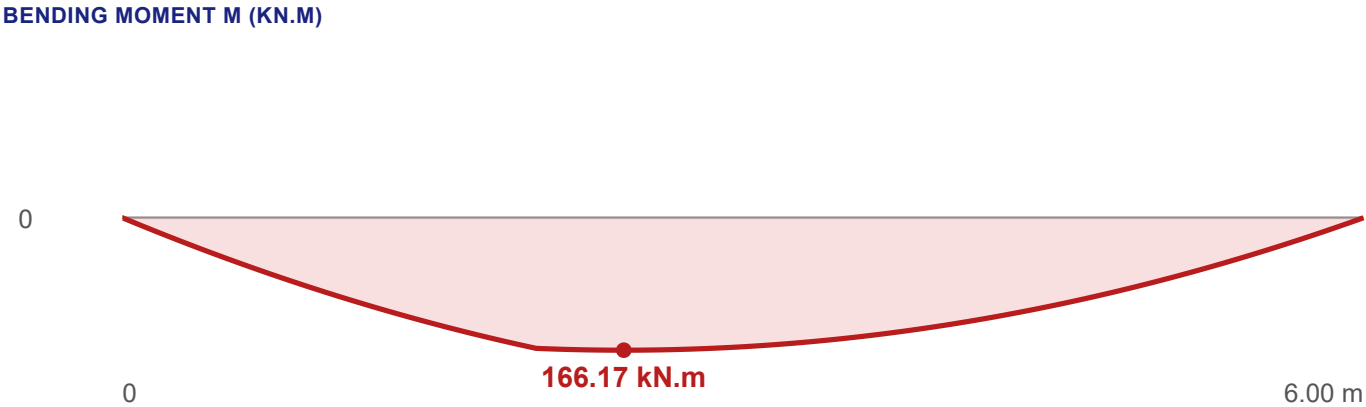
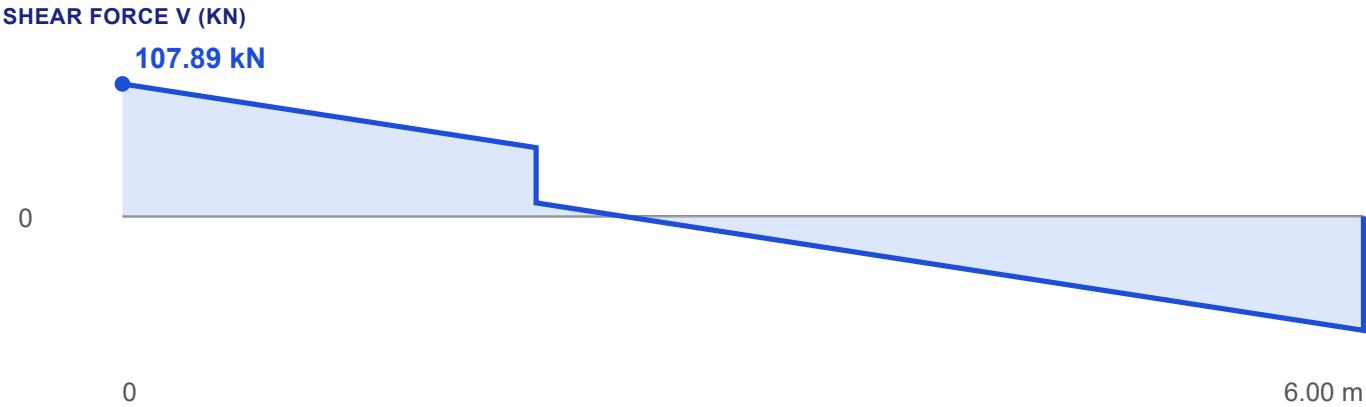
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.044 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017e3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

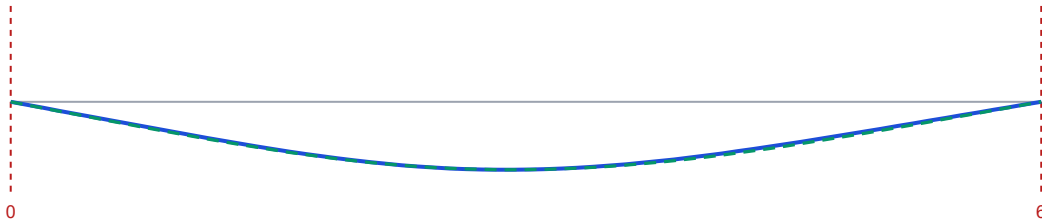
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -100, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	921.883 kN.m	converged
$\lambda_{LT} = \sqrt{W_y \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 921.883}$	0.290	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.290 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 53.49 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L3b · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -100$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L3b RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)

Q UDL 8.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -100$ mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

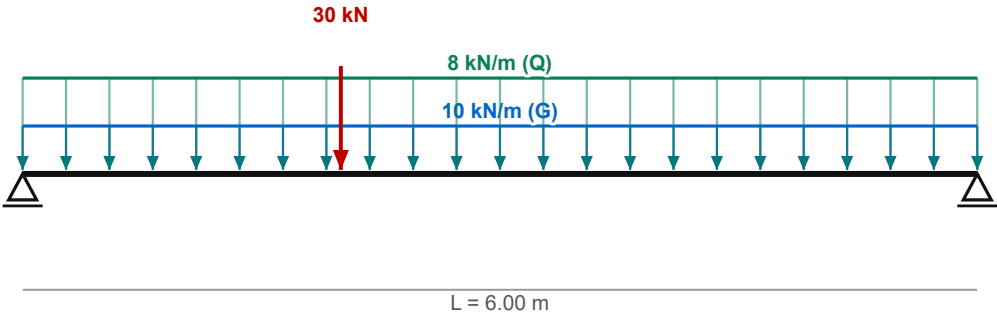
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

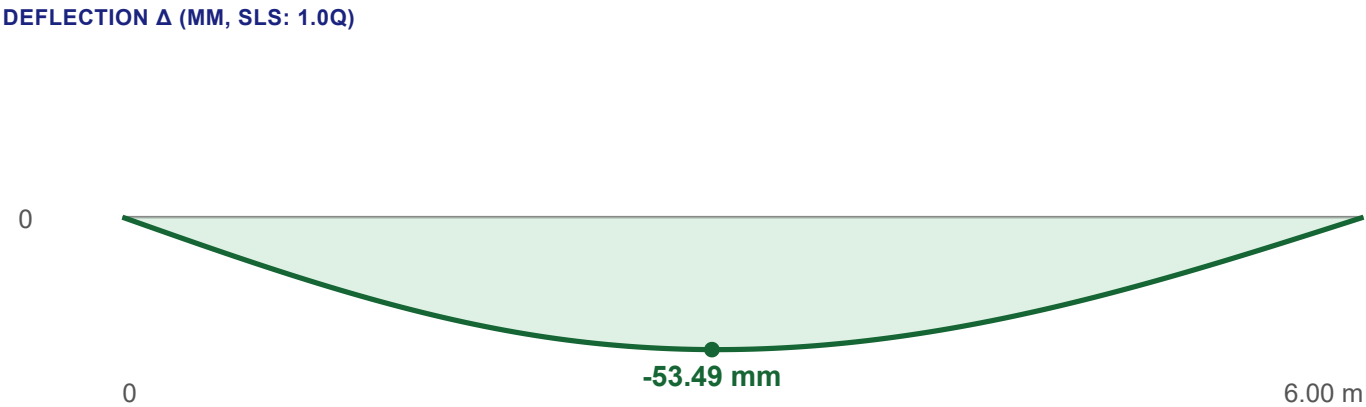
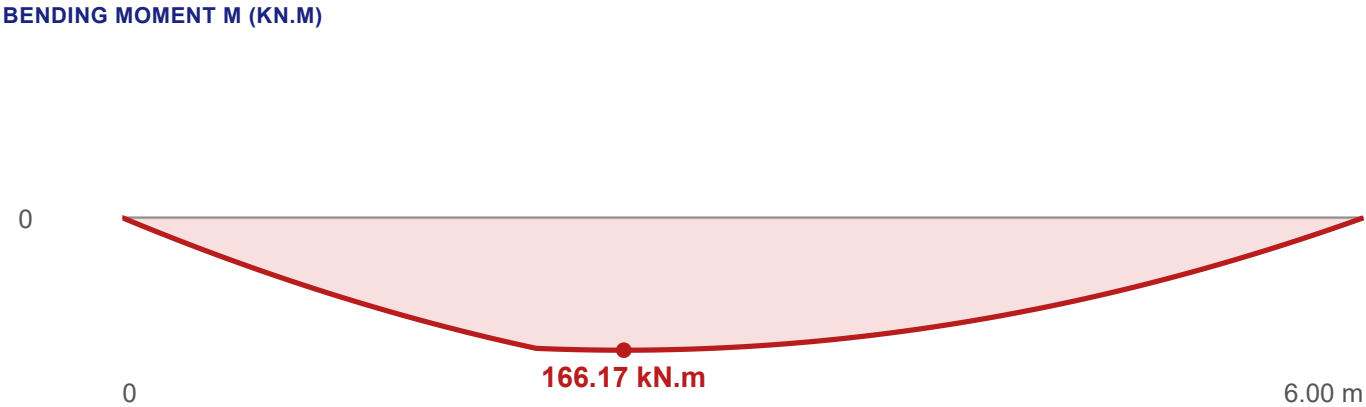
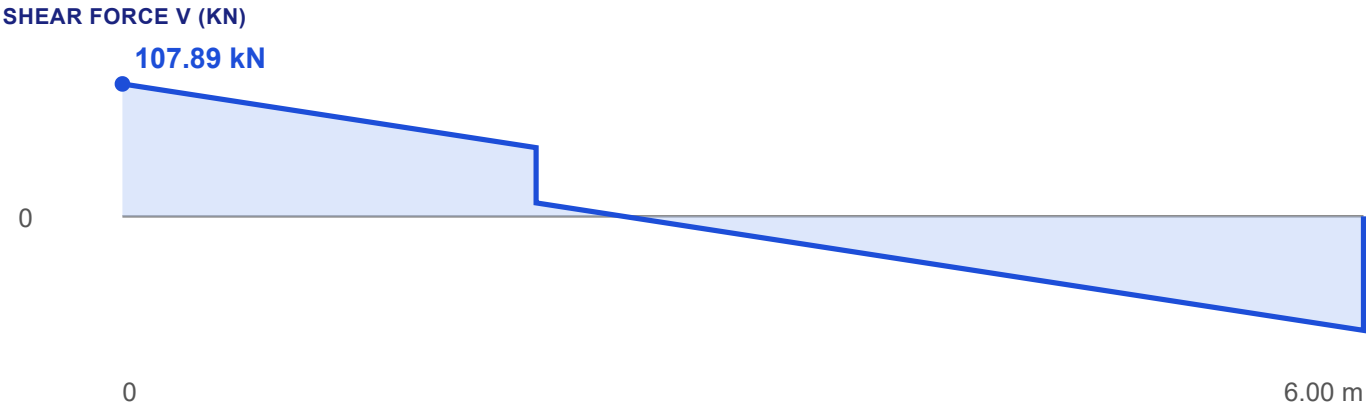
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = Fn(c/t, d/t, fy, N, My, Mz)	9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	22.00 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017e3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.8, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN·m		SN003a

Lateral Buckling Check $M_{b,Rd}$

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = -100$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	916.230 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member RHS-L4 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6 0.00C		0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

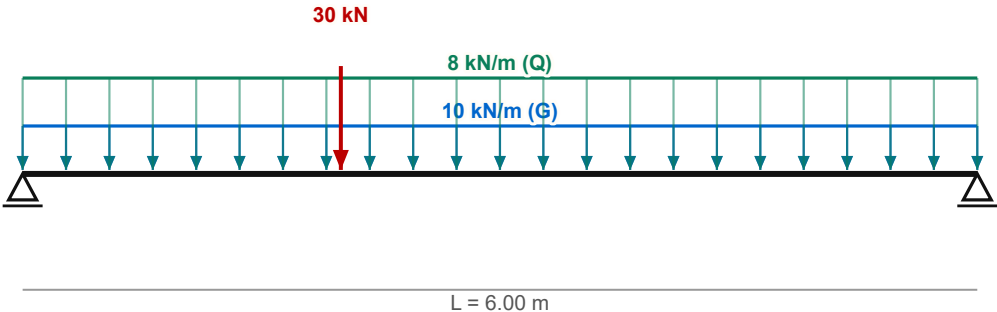
Uplift

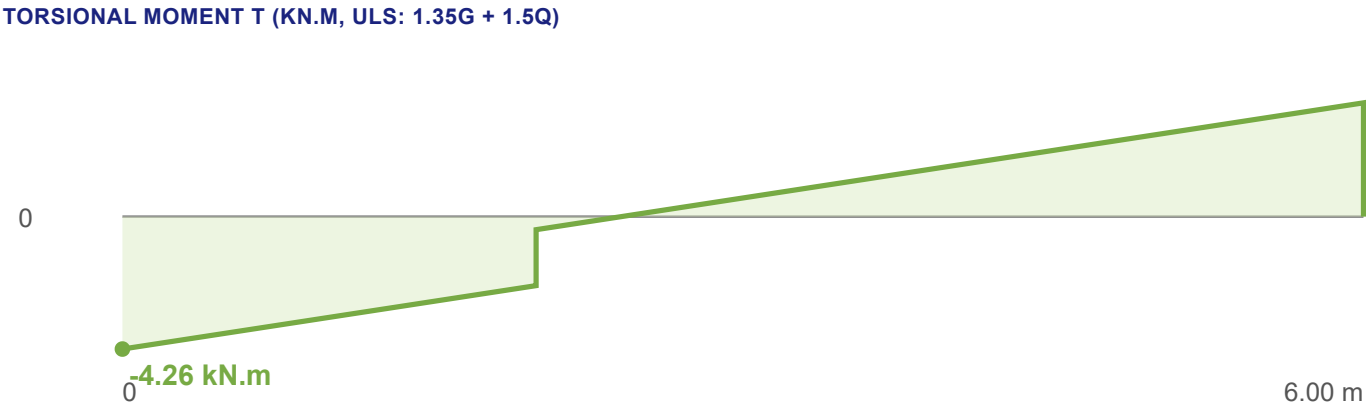
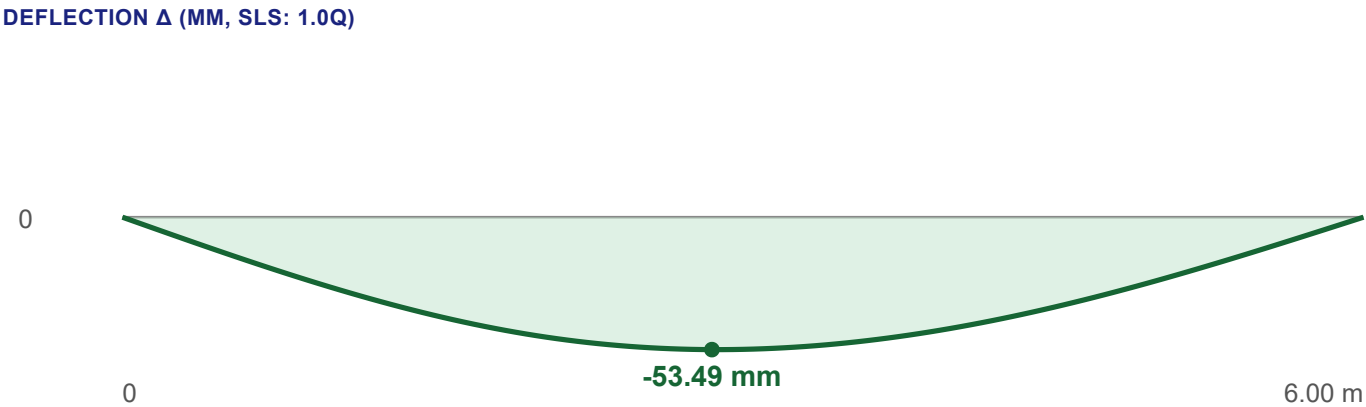
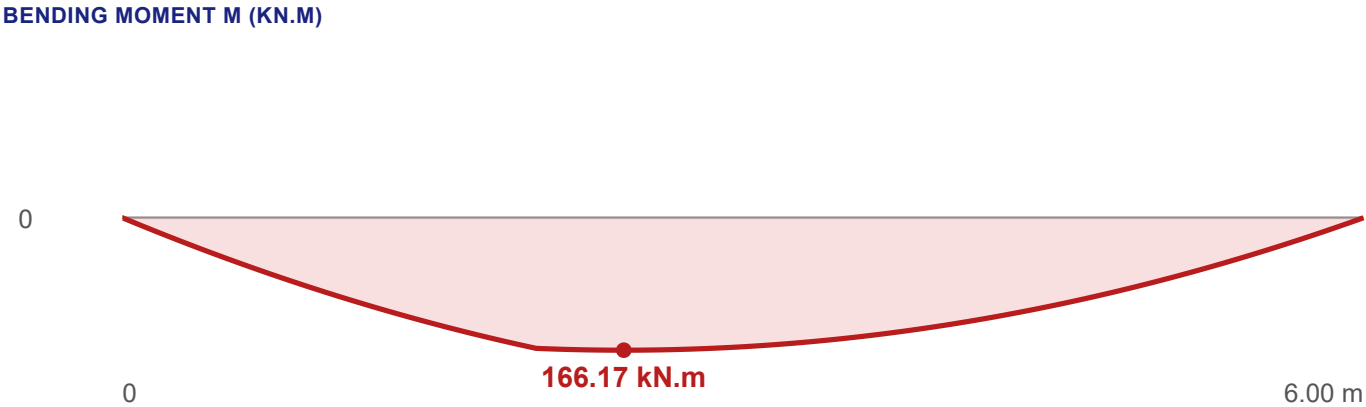
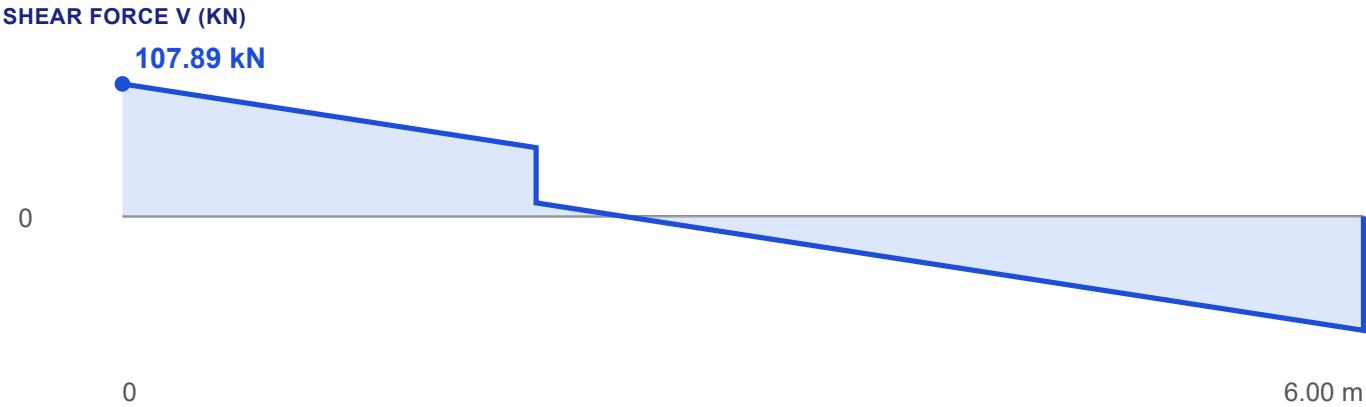
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200		
	B=100 t=8		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	9.50, 22.00, 275, 0.00, 166.17, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174$ kN.m, $V = 0.044$ kN, $T = 0.012$ kN.m, $\phi = 0.00450$ rad; $M_z = \phi M_y = 0.748$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 745.18 + 5.06 = 750.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06$ N/mm ² ; $(750.23/275)^2 + 3(0.06/275)^2 =$	7.443	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 745.18 + 5.06 = 750.23$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 7.443$ (governs); web mid-depth: $\sigma_x = 0.00 + 5.06 = 5.06$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 745.18 = 745.18$, $\tau = 0.05 \rightarrow 7.343$ ($Q_c = 35.3$, $Q_m = 72.2$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \times 10^3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.859 / 77.550 = 2.113) = 2.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.451	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

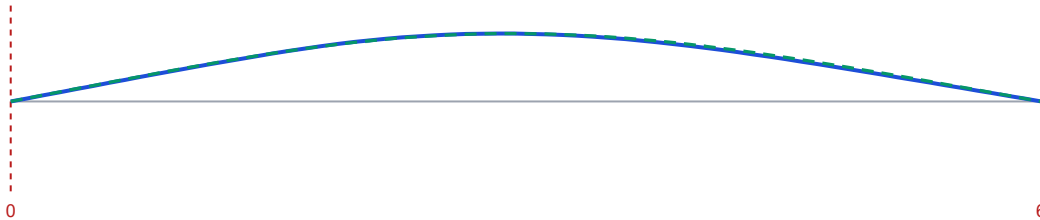
Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	899.055 / 787.606 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.6, M(L/2)=161.8, M(3L/4)=110.1, M _{max} =166.2 kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 100, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	876.701 kN.m	converged
λ _{LT} = √W _y ·f _y /M _{cr}	√ 282.0 x 275 / 876.701	0.297	
M _{b,Rd} = M _{c,y,Rd}	closed hollow section, λ _{LT} = 0.297 ≤ 0.4: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M _{y,Ed} /M _{b,Rd}	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist φ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.84 m)



Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /251	16.97 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
(M _{y,Ed} /(M _{pl,y,Rd} ·S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} ·S _{mod})) ^β	T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t /(f _y /√3)]·V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 16.97 N/mm ² ; V _{pl,Rd} = 474.197; S _{mod} =	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	V _{pl,T,Rd} /V _{pl,Rd} = 0.893 107.894 / 423.507	0.255	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0020 rad = $0.11^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	7.443	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 7.443 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member RHS-L4 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6.00C	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

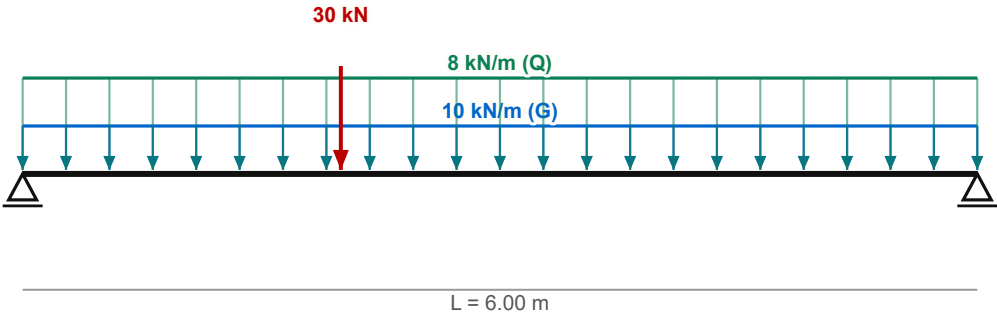
Uplift

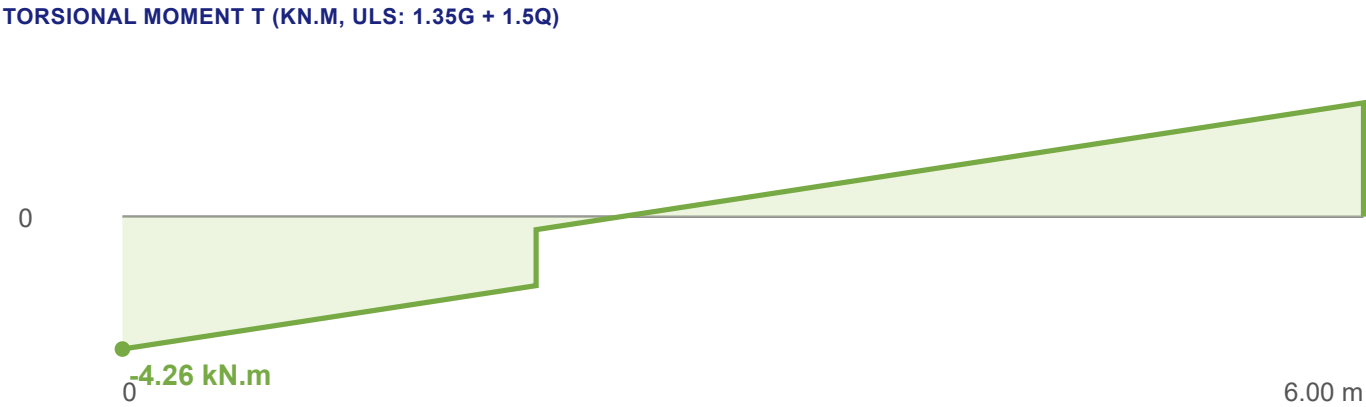
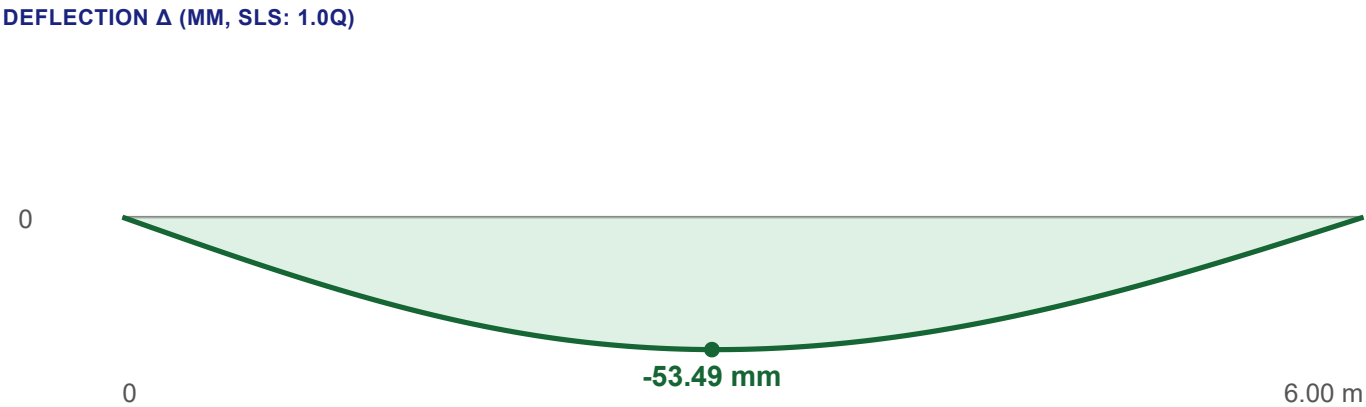
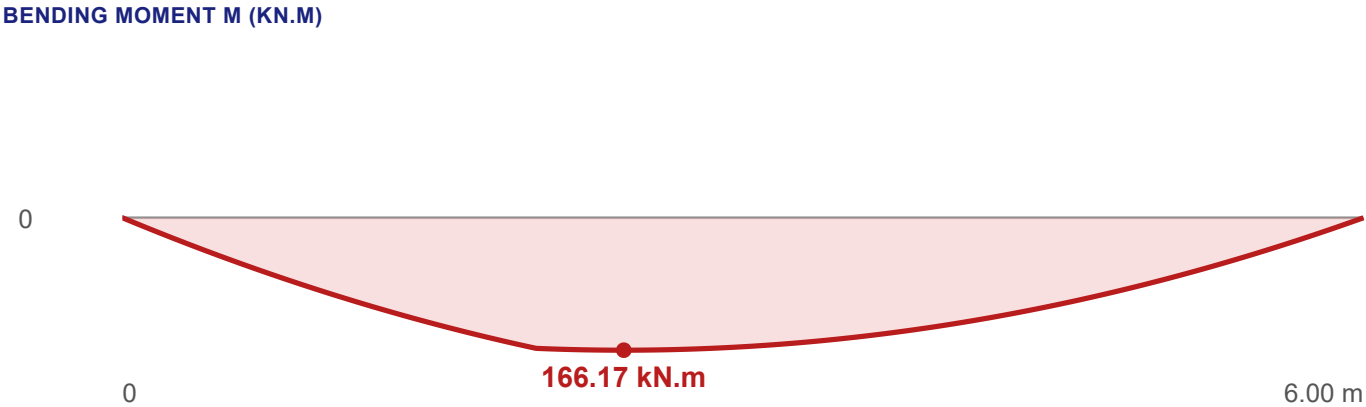
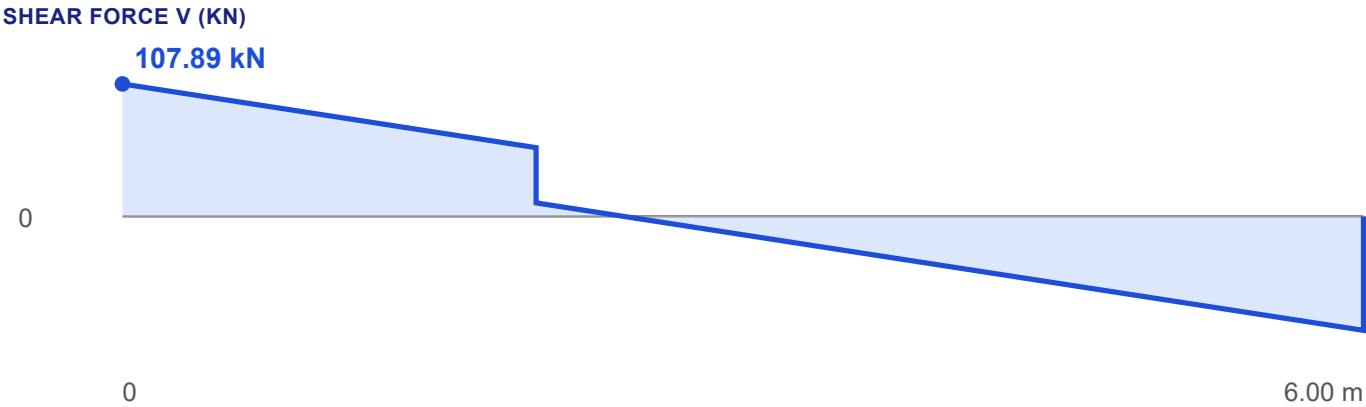
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200		
	B=100 t=8		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	9.50, 22.00, 275, 0.00, 166.17, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$d/t \leq 72 \epsilon / \eta$	22.00 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.044 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174$ kN.m, $V = 0.044$ kN, $T = 0.012$ kN.m, $\phi = 0.00450$ rad; $M_z = \phi M_y = 0.748$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 745.18 + 5.06 = 750.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06$ N/mm ² ; $(750.23/275)^2 + 3(0.06/275)^2 =$	7.443	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 745.18 + 5.06 = 750.23$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 7.443$ (governs); web mid-depth: $\sigma_x = 0.00 + 5.06 = 5.06$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 745.18 = 745.18$, $\tau = 0.05 \rightarrow 7.343$ ($Q_c = 35.3$, $Q_m = 72.2$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.341 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 163.859/77.550 = 2.113) = 2.032 \leq 1.4 \text{ (ULS: 1.35G + 1.5Q)} \quad 1.451 \quad \text{Warning 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) \quad 0.0, 0.0, 161.8, 1.000, 300.000 \quad 1.163 \quad \text{Serna}$$

C_1 basis: general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2 \text{ kN}\cdot\text{m}$ SN003a

Lateral Buckling Check M.b.Rd

$$M_{b,Rd} = M_{c,y,Rd} \quad \text{closed hollow section, } \lambda_{LT} = 0.291 \leq 0.4: \text{LTB may be ignored} \quad 77.550 \text{ kN}\cdot\text{m} \quad 6.3.2.2(4)$$

$$M_{cr} \text{ (information, } I_w = 0) \quad \text{SN003a with } C_1 = 1.163, L_e = 6 \text{ m; } z_g = +100 \text{ mm entered — NOT applied: } C_2 \text{ not published for this moment diagram; } M_{cr} \text{ is at the shear centre (PASS blocked)} \quad 916.230 \text{ kN}\cdot\text{m} \quad \text{SN003a}$$

$$M_{y,Ed}/M_{b,Rd} \quad 166.174 / 77.550 \quad 2.143 \quad \text{Warning}$$

Torsion Design

$$J, H, a, Q_f, Q_w \quad 1800 \text{ cm}^4, \text{ — dm}^6, \text{ — mm, —, —} \quad \text{P385 App A}$$

$$W_{n0}, S_{w1} \quad \text{—, —} \quad \text{P385 App A}$$

Torsion Bending Design @ 0 m

$$T_{Ed} \text{ (max)} \quad \text{combination ULS: 1.35G + 1.5Q} \quad 4.260 \text{ kN}\cdot\text{m}$$

$$W_t (= C) \quad \text{SCI P385 Table A.7/A.8; } I_t = 1800 \text{ cm}^4 \quad 251 \text{ cm}^3$$

$$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0}) \quad 275 \times 251 / (\sqrt{3} \times 1) \quad 39.852 \text{ kN}\cdot\text{m} \quad 6.2.7(7)$$

$$T_{Ed}/T_{Rd} \quad 4.260 / 39.852 \quad 0.107 \quad \text{OK}$$

$$T_{t,Ed} = T_{Ed}/W_t \quad 4.260 \times 10^3 / 251 \quad 16.97 \text{ N/mm}^2 \quad 6.2.7(7)$$

$$\text{Modified Local Capacity} \quad n/a - \text{not evaluated by beam-v03 (the cl 6.2.7 } T_{Ed}/T_{Rd}, V_{pl,T,Rd} \text{ with the 6.2.8(4) } p \text{ in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)} \quad \text{—} \quad \text{not evaluated}$$

Torsion Shear Design @ 0 m

$$V_{pl,T,Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd} \quad @ x = 0 \text{ m (ULS: 1.35G + 1.5Q): } T = 4.260 \text{ kN}\cdot\text{m, } T_t = 16.97 \text{ N/mm}^2; V_{pl,Rd} = 474.197; S_{mod} = \quad 423.507 \text{ kN} \quad 6.2.7(9) \text{ Eq 6.28}$$

$$V_{Ed}/V_{pl,T,Rd} \quad V_{pl,T,Rd}/V_{pl,Rd} = 0.893 \quad 107.894 / 423.507 \quad 0.255 \quad \text{OK}$$

Deflection Check - Load Case 1 (SLS: 1.0Q)

$$\text{In-span } \delta \leq \text{Span}/360 \quad 53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m} \quad 53.49 \text{ mm} \quad \text{Warning}$$

$$\text{Torq in Case 1 @ 2 m: } \theta_{max} \leq 2.00^\circ \quad 0.0020 \text{ rad} = 0.11^\circ \leq 2.00^\circ \text{ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)} \quad 0.11^\circ \quad \text{OK}$$

MA/Mc	M _{y,Ed} /M _{b,Rd}	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	7.443	0.255	2.143

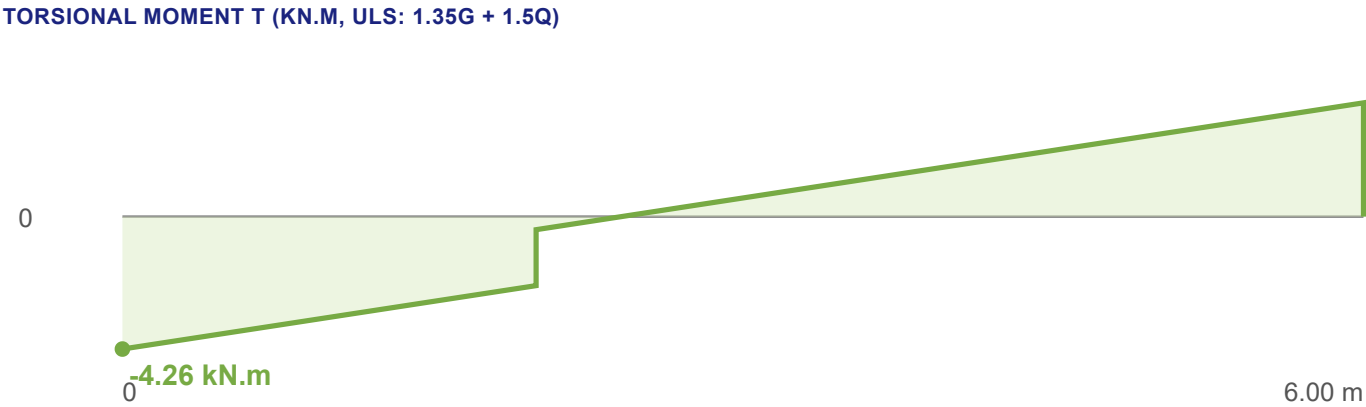
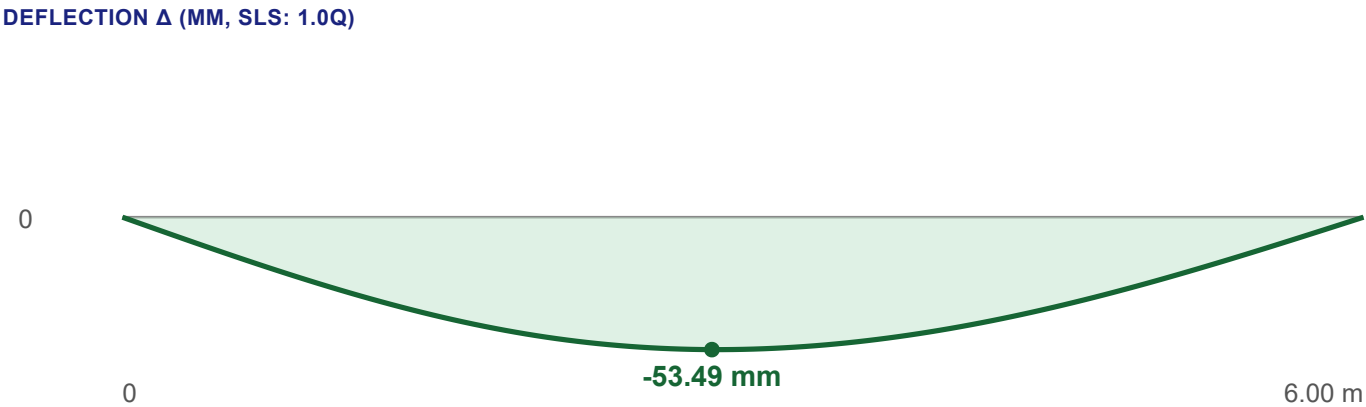
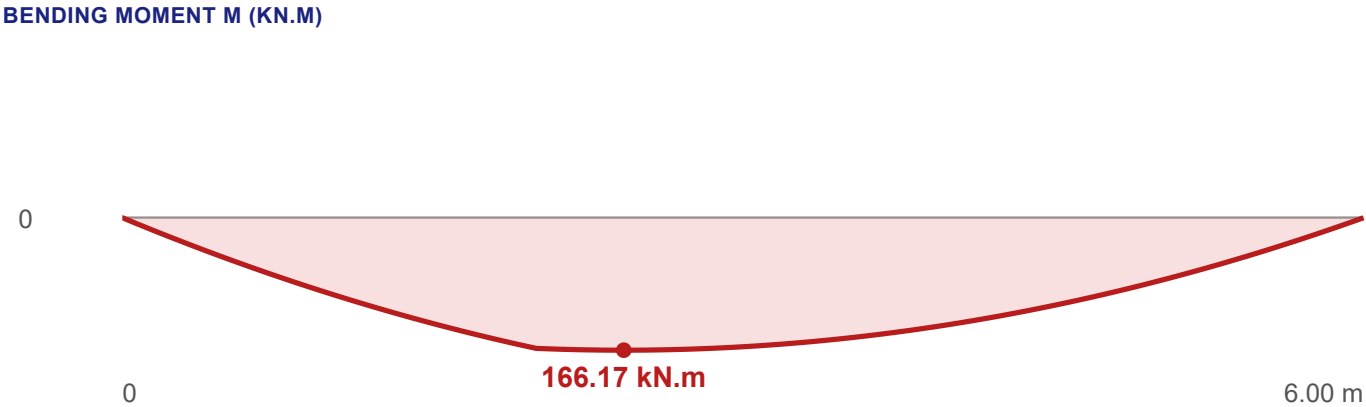
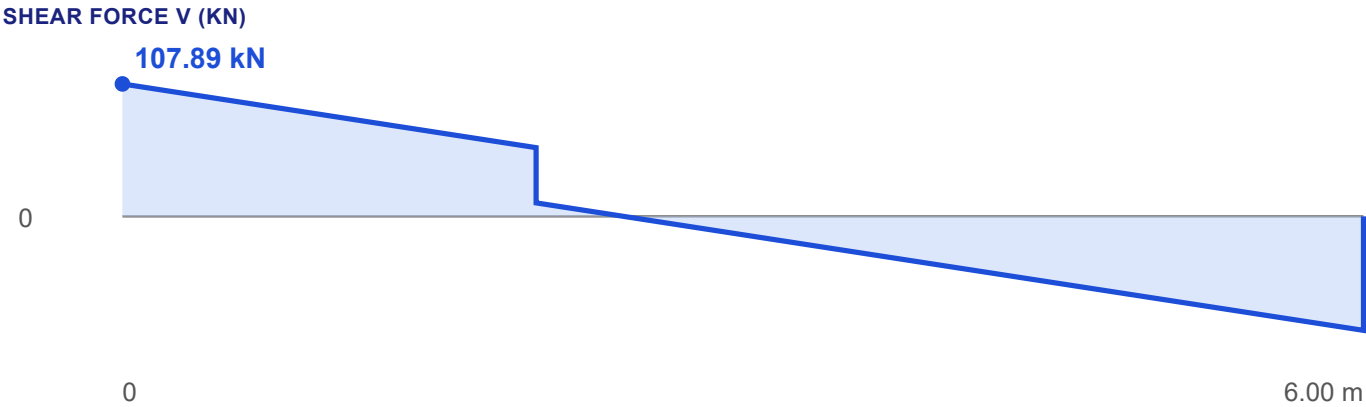
FAIL — governing Deflection = 3.210

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 7.443 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8			
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 166.17, 0.00 (Axial: Non-Slender)			Class 1
Web classified for	bending + compression: $\alpha = 0.597, \psi = -0.756$; limits 58.6ε / 67.5ε / 99.9ε	Class 1	Table 5.2	

Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.044 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	Warning
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$44.80 \times 275 / 1 = \text{(No bearing / block tearing design)}$	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 1232.000 =$	0.122	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$282.0, 29.867, 0.122$	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$282.0 \times 275 / 1$	77.550 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(166.174 / 77.550)^{1.69} + (0)^1 =$	3.621	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174 \text{ kN.m}$, $V = 0.044 \text{ kN}$, $T = 0.012 \text{ kN.m}$, $\phi = 0.00450 \text{ rad}$; $M_z = \phi M_y = 0.748 \text{ kN.m}$; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 33.48 + 745.18 + 5.06 = 783.72 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06 \text{ N/mm}^2$; $(783.72 / 275)^2 + 3(0.06 / 275)^2 =$	8.122	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 33.48 + 745.18 + 5.06 = 783.72$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 8.122$ (governs); web mid-depth: $\sigma_x = 33.48 + 5.06 = 38.54$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.020$; flange mid-width: $\sigma_x = 33.48 + 745.18 = 778.66$, $\tau = 0.05 \rightarrow 8.017$ ($Q_c = 35.3$, $Q_m = 72.2 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$176.00, 8.00, 8.00, 50.00 \text{ mm}$ ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \text{e}3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.859 / 77.550 + N_{Ed} / N_{pl,Rd} = 150.000 / 1232.000 = 2.235) = 2.129 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.521	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.216	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.521	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.195	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10 / 1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 368.021 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.408	OK

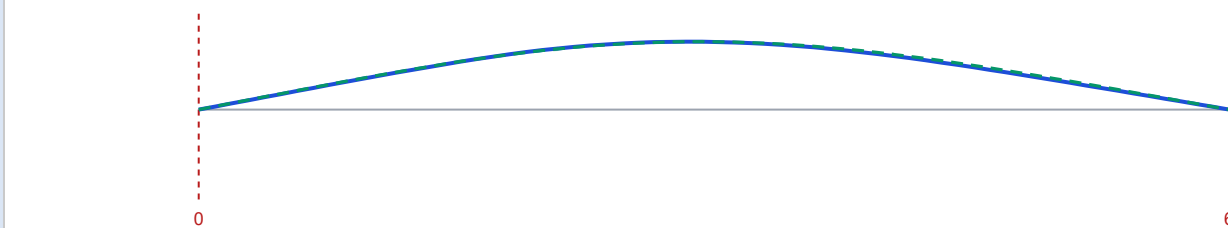
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 0.748 twist ϕ . $M_y = 0.748$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	876.701 kN.m	converged
$\lambda_{LT} = \sqrt{W_y \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 876.701}$	0.297	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.297 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.84 m)



Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 838.121$ ($N_{Rk} = 1232.000$)	0.179	OK
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$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	166.174 / 77.550 ($M_{y,Rk} = 77.550$)	2.143	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(0.000 imposed + 0.748 twist $\phi \cdot M_y = 0.748$) / 47.300	0.016	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.979-0.2) \times 0.179\} \leq 1.000(1+0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.702-0.2) \times 0.408\} \leq 1.000(1+0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6 x 1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.179 + 1.139 \times 2.143 + 1.326 \times 0.016$	2.641	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.408 + 0.684 \times 2.143 + 1.326 \times 0.016$	1.893	Warning

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3 γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /251	16.97 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} /(M _{pl,y,Rd} · S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} · S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
Basis with N _{Ed} / M _{z,Ed}	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with tau _t = T _{Ed} /W _t , the V _{Ed} shear flow and N/A + M _y /W _{el,y} + M _{z,tot} /W _{el,z} (information only: the Class 1/2 plastic route governs), T _{Ed} /T _{Rd} (6.2.7(1)), V _{Ed} /V _{pl,T,Rd} (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with V _{pl,T,Rd} , cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} ; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t /(f _y /√3)] · V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 16.97 N/mm ² ; V _{pl,Rd} = 474.197; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.893	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	107.894 / 423.507	0.255	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
Torq in Case 1 @ 2 m: θ _{max} ≤ 2.00°	0.0020 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.122	3.621	0.408	2.143	2.641	1.893	3.210	0.228	2.143	2.143	0.341	1.521	0.107	8.122	0.255	3.621

FAIL — governing Bending+axial cross-section (6.2.9) = 3.621

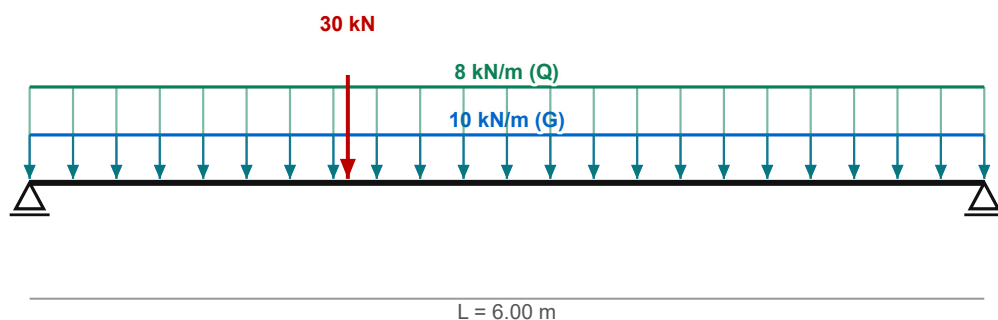
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

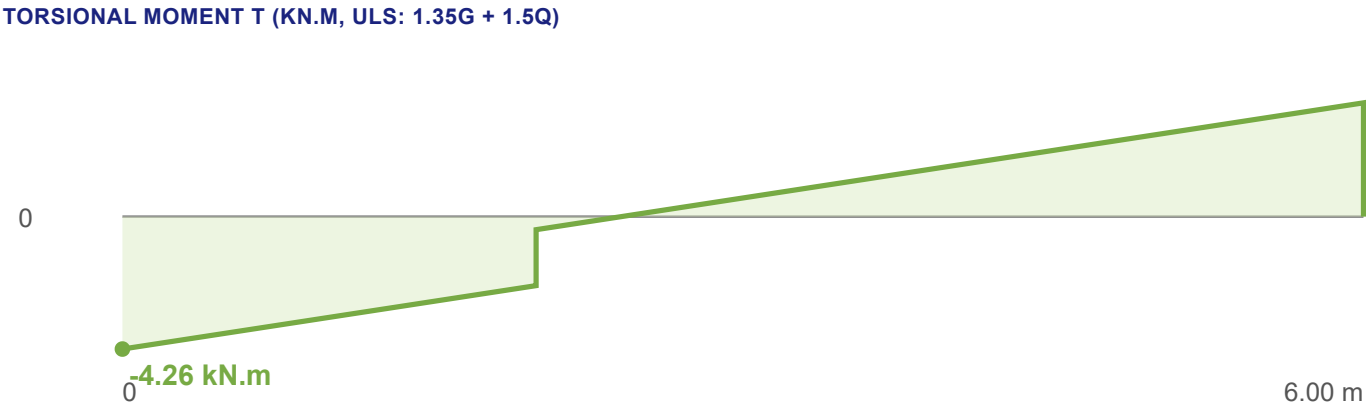
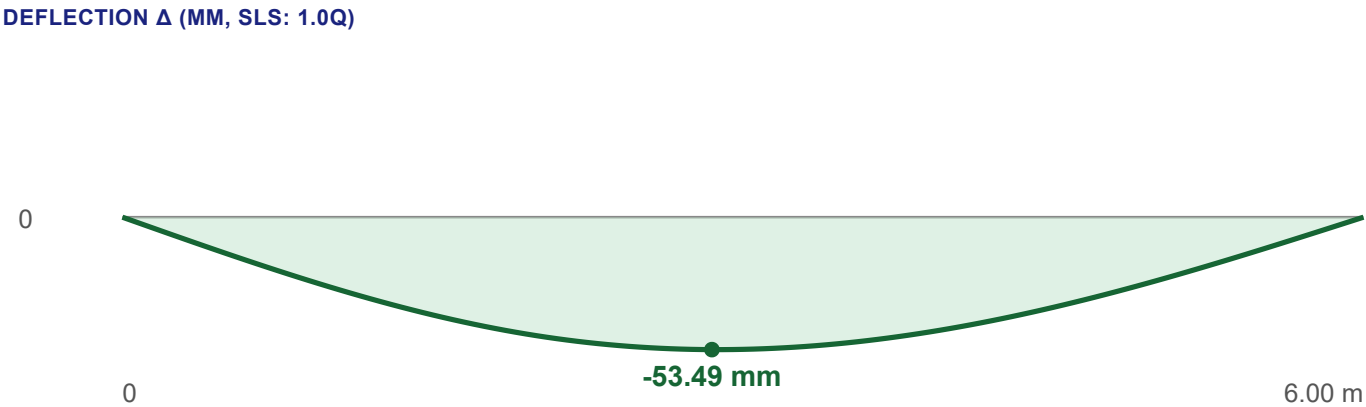
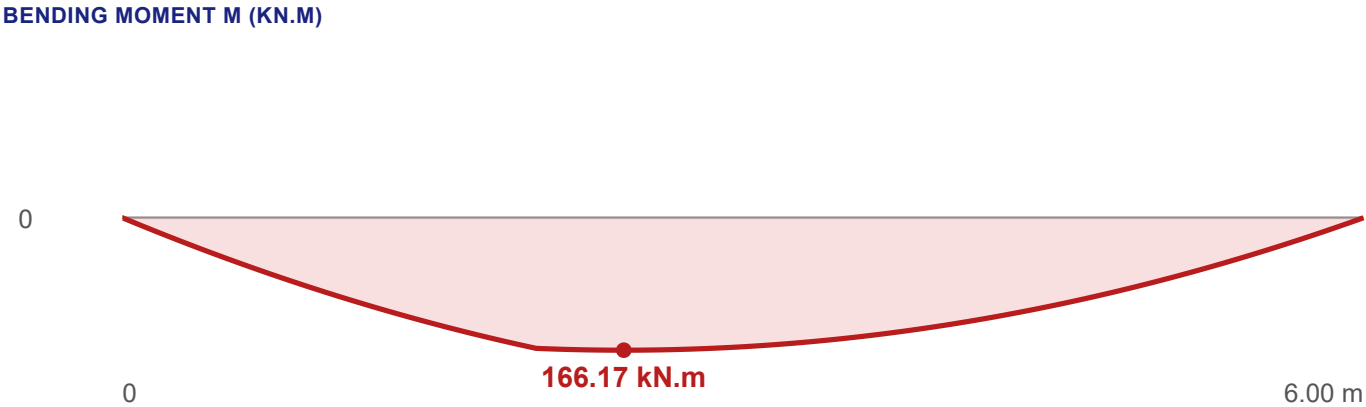
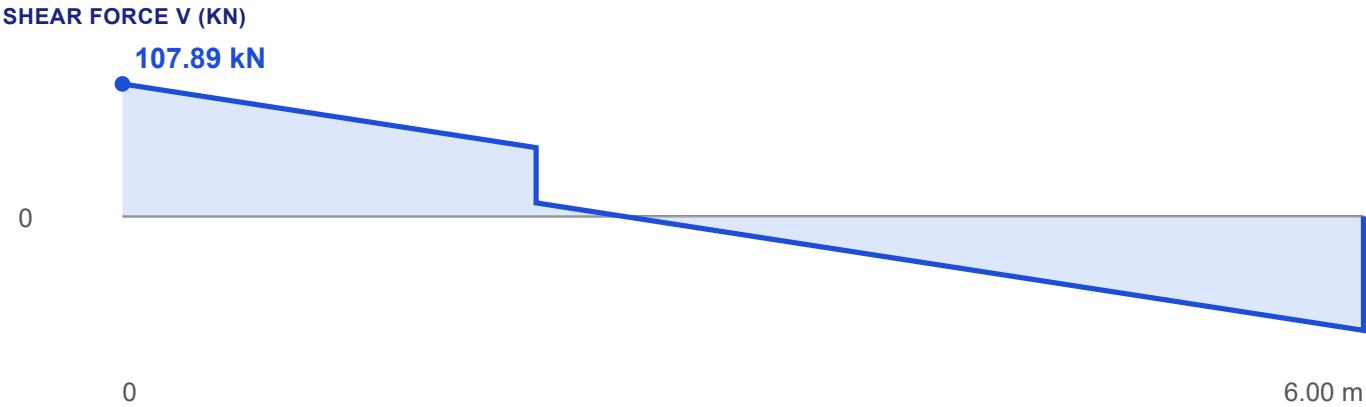
ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 8.122 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd}

(6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +100 mm





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8			
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 166.17, 0.00 (Axial: Non-Slender)			Class 1
Web classified for	bending + compression: $\alpha = 0.597, \psi = -0.756$; limits 58.6ε / 67.5ε / 99.9ε	Class 1	Table 5.2	

Auto Design Load Cases		1; SLS 1	
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.044 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	Warning
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$44.80 \times 275 / 1 =$ (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 1232.000 =$	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$282.0, 29.867, 0.122$	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	$282.0 \times 275 / 1$	77.550 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(166.174 / 77.550)^{1.69} + (0)^1 =$	3.621	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.424 m, corner	@ x = 2.424 m (ULS: 1.35G + 1.5Q): $M_y = 166.174 \text{ kN.m}$, $V = 0.044 \text{ kN}$, $T = 0.012 \text{ kN.m}$, $\phi = 0.00450 \text{ rad}$; $M_z = \phi M_y = 0.748 \text{ kN.m}$; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 33.48 + 745.18 + 5.06 = 783.72 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.05 + 0.01 = 0.06 \text{ N/mm}^2$; $(783.72 / 275)^2 + 3(0.06 / 275)^2 =$	8.122	> 1 information 6.2.7(5)
$\sigma_x, \tau, (6.1)$ at the 3 section points	corner: $\sigma_x = 33.48 + 745.18 + 5.06 = 783.72$, $\tau = 0.05 + 0.01 = 0.06 \rightarrow 8.122$ (governs); web mid-depth: $\sigma_x = 33.48 + 5.06 = 38.54$, $\tau = 0.05 + 0.02 = 0.07 \rightarrow 0.020$; flange mid-width: $\sigma_x = 33.48 + 745.18 = 778.66$, $\tau = 0.05 \rightarrow 8.017$ ($Q_c = 35.3$, $Q_m = 72.2 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.		NOT VERIFIED
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$176.00, 8.00, 8.00, 50.00 \text{ mm}$ ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00 / 8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \text{e}3)}$	0.193	6.4(1)

$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	0.5/0.193 = 2.588 $\rightarrow$ 1.000 ; 1.000 x 56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 56.00 x 8.00/1.0 per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	0.341 + 0.8 x ($M_{Ed}/M_{c,y,Rd} = 163.859/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 2.235$) = 2.129 $\leq$ 1.4 (ULS: 1.35G + 1.5Q)	1.521	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.216	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.521	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.195	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	44.80 x 0.680 x 275 / 10/1 =	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	44.80 x 0.299 x 275 / 10/1 =	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	150.000 / 368.021 = (min of $N_{b,y,Rd}$, $N_{b,z,Rd}$)	0.408	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.8, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 0.748 twist ϕ . $M_y = 0.748$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	916.230 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK

$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	166.174 / 77.550 ($M_{y,Rk} = 77.550$)	2.143	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	(0.000 imposed + 0.748 twist $\phi \cdot M_y = 0.748$) / 47.300	0.016	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my} \{1 + (\chi_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000 (1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz} \{1 + (\chi_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000 (1 + 0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6 x 1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	0.179 + 1.139 x 2.143 + 1.326 x 0.016	2.641	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	0.408 + 0.684 x 2.143 + 1.326 x 0.016	1.893	Warning

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t / (√3 γ _{M0})	275 x 251 / (√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} / T _{Rd}	4.260 / 39.852	0.107	OK
T _{t,Ed} = T _{Ed} / W _t	4.260 x 10 ³ / 251	16.97 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} / (M _{pl,y,Rd} · S _{mod})) ^α + (M _{z,Ed} / (M _{pl,z,Rd} · S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} / T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) ρ in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
Basis with N _{Ed} / M _{z,Ed}	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross- section with tau _t = T _{Ed} / W _t , the V _{Ed} shear flow and N/A + M _y / W _{el,y} + M _{z,tot} / W _{el,z} (information only: the Class 1/2 plastic route governs), T _{Ed} / T _{Rd} (6.2.7(1)), V _{Ed} / V _{pl,T,Rd} (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with V _{pl,T,Rd} , cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} ; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t / (f _y / √3)] · V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 16.97 N/mm ² ; V _{pl,Rd} = 474.197; S _{mod} = V _{pl,T,Rd} / V _{pl,Rd} = 0.893	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} / V _{pl,T,Rd}	107.894 / 423.507	0.255	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
Torq in Case 1 @ 2 m: θ _{max} ≤ 2.00°	0.0020 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/V _{pl}	MA/M _c	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.122	3.621	0.408	2.143	2.641	1.893	3.210	0.228	2.143	2.143	0.341	1.521	0.107	8.122	0.255	3.621

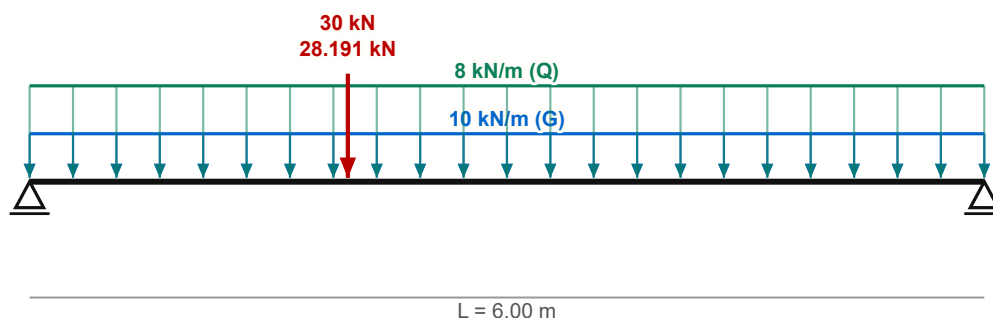
FAIL — governing Bending+axial cross-section (6.2.9) = 3.621

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +100 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

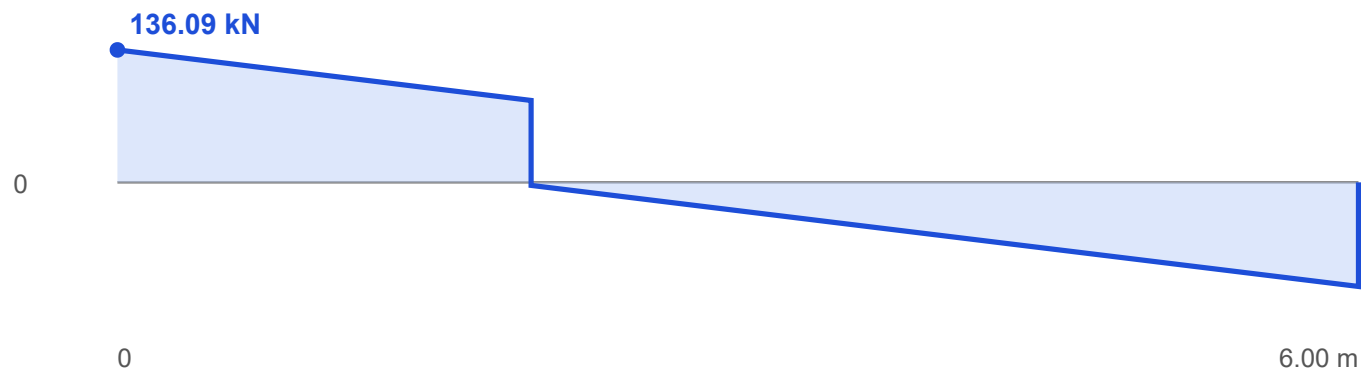
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

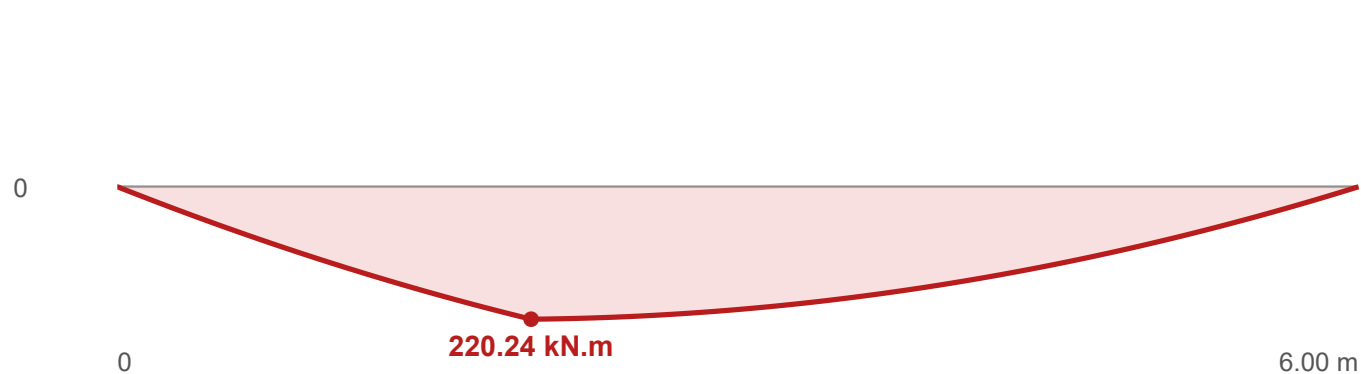
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 8.122 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



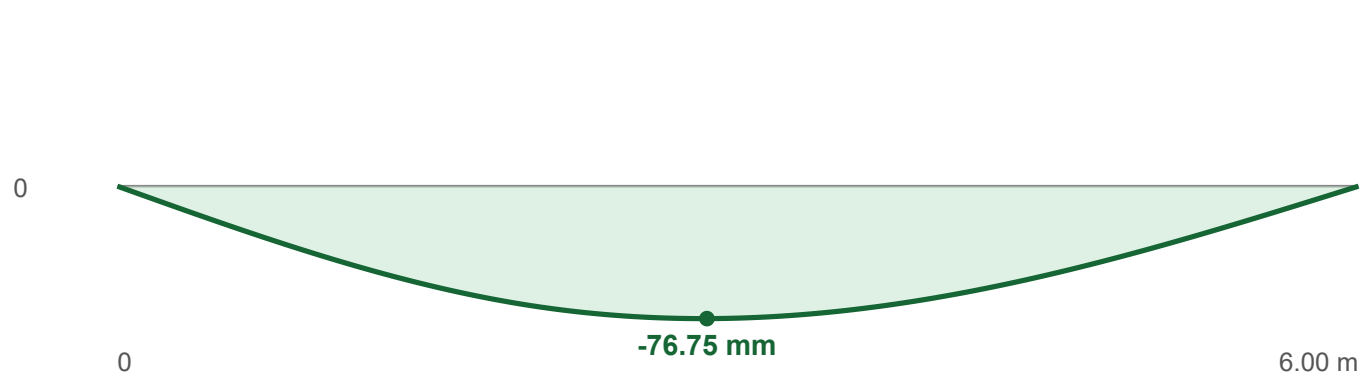
SHEAR FORCE V (kN)



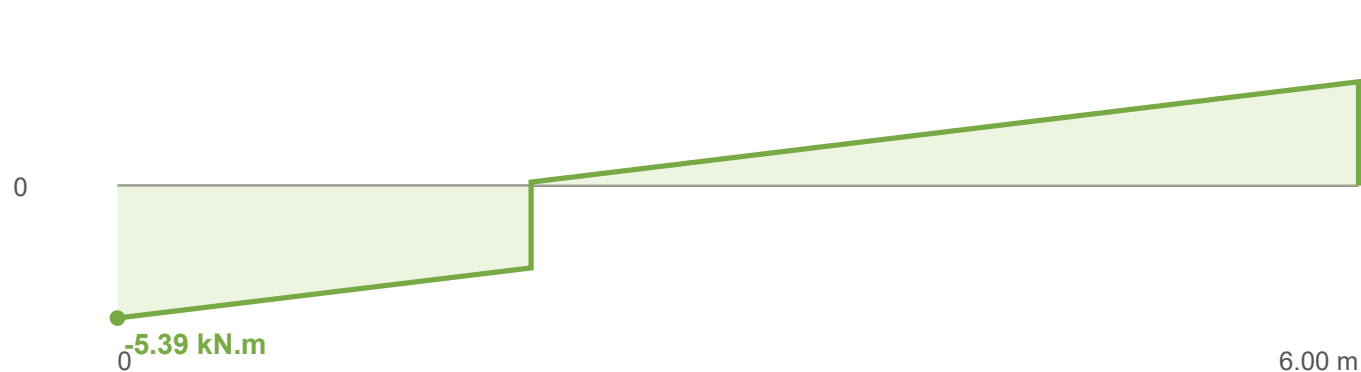
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200	
	B=100 t=8	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 220.24, 20.52	Class 1
	(Axial: Non-Slender)	
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$136.085 / 474.197 =$	0.287	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$3.131 / 474.197 =$	0.007	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$220.241 / 77.550 =$	2.840	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 237.099 = (A_{v,z} = 1493.3 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 172.0 / 1$	47.300 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$44.80 \times 275 / 1 = (\text{No bearing / block tearing design})$	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 1232.000 =$	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$282.0, 29.867, 0.122$	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	$282.0 \times 275 / 1$	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$172.0, 14.933, 0.122$	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	$172.0 \times 275 / 1$	47.300 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(220.241 / 77.550)^{1.69} + (20.521 / 47.300)^{1.69} =$	6.070	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, corner	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 220.241$ kN.m, $V = 84.156$ kN, $T = 3.348$ kN.m, $\phi = 0.00599$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.320 = 21.841$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 33.48 + 987.63 + 147.57 = 1168.68 \text{ N/mm}^2$; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 13.34 + 16.67 = 30.00 \text{ N/mm}^2$; $(1168.68/275)^2 + 3(30.00/275)^2 =$	18.096	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 33.48 + 987.63 + 147.57 = 1168.68$, $\tau = 13.34 + 16.67 = 30.00 \rightarrow 18.096$ (governs); web mid-depth: $\sigma_x = 33.48 + 147.57 = 181.05$, $\tau = 13.34 + 34.05 = 47.39 \rightarrow 0.523$; flange mid-width: $\sigma_x = 33.48 + 987.63 = 1021.11$, $\tau = 13.34 \rightarrow 13.794$ ($Q_c = 35.3$, $Q_m = 72.2 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$176.00, 8.00, 8.00, 50.00 \text{ mm}$ ($B/2 = 50.00 \leq t_w + 15\epsilon t_f = 118.93$); $f_{yw} = f_y = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\bar{\lambda}_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \times 10^3)}$	0.193	6.4(1)
$\chi_F = 0.5/\bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, $e = 40 \text{ mm}$); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)

F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 220.241/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 2.962) = 3.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.165	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	136.085	ULS: 1.35G + 1.5Q	0.258	0.254	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.165	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	106.990	ULS: 1.35G + 1.5Q	0.203	0.214	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10 / 1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z$; $\bar{\lambda}_T$; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 368.021 = (\text{min of } N_{b,y,Rd}, N_{b,z,Rd})$	0.408	OK

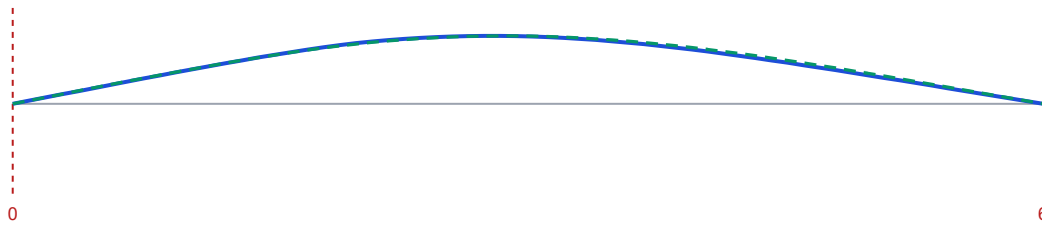
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr}/M_{cr,uniform}$	934.976 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.187	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 204.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.9$, $M(L/2)=204.1$, $M(3L/4)=131.3$, $M_{max}=220.2 \text{ kN}\cdot\text{m}$	1.217	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 1.320 twist $\varphi \cdot M_y = 21.841$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	910.839 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 910.839}$	0.292	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\bar{\lambda}_{LT} = 0.292 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	$220.241 / 77.550$	2.840	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.74$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	220.241 / 77.550 ($M_{y,Rk} = 77.550$)	2.840	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.320 twist $\phi \cdot M_y = 21.841$) / 47.300	0.462	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\chi_y-0.2)U_{N,y}\}$	$1.000\{1+(0.979-0.2) \times 0.179\} \leq 1.000(1+0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\chi_z-0.2)U_{N,z}\}$	$1.000\{1+(1.702-0.2) \times 0.408\} \leq 1.000(1+0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.179 + 1.139 \times 2.840 + 1.326 \times 0.462$	4.027	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.408 + 0.684 \times 2.840 + 1.326 \times 0.462$	2.961	Warning

Torsion Design

J, H, a, Q_f, Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0}, S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
$W_t (= C)$	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$5.388 / 39.852$	0.135	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 251$	21.46 N/mm ²	6.2.7(7)
Modified Local Capacity $(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 1.320$ (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.841 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $\tau_t = 21.46$ N/mm ² ; $V_{pl,Rd} = 474.197$; $S_{mod} =$	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.865$ $136.085 / 410.089$	0.332	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$76.75 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	76.75 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{max} \leq 2.00^\circ$	0.0030 rad = $0.17^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.17°	OK

N_Ed/N _{pl,Rd}	UNy	UMy	Ax+M _{6.6.1}	Ax+M _{6.6.2}	Deflection	V/V _{pl}	MA/M _c	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.122	6.070	0.408	2.840	4.027	2.961	4.605	0.287	2.840	2.840	0.662	2.165	0.135	18.096	0.332	6.070

FAIL — governing Biaxial bending + axial (6.2.9.1) = 6.070

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 18.096 at x = 2.00 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

RHS-L6 · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Biaxial bending + axial (6.2.9.1) = 6.070

Owner case: ... and axial load on beam, load applied at angle on beam
RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +100, factored by the combination) plus an entered CONSTANT M_{z,Ed} = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_{mz} = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.100 = 1.03 kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)

Includes Design for Torsion

Member RHS-L6 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

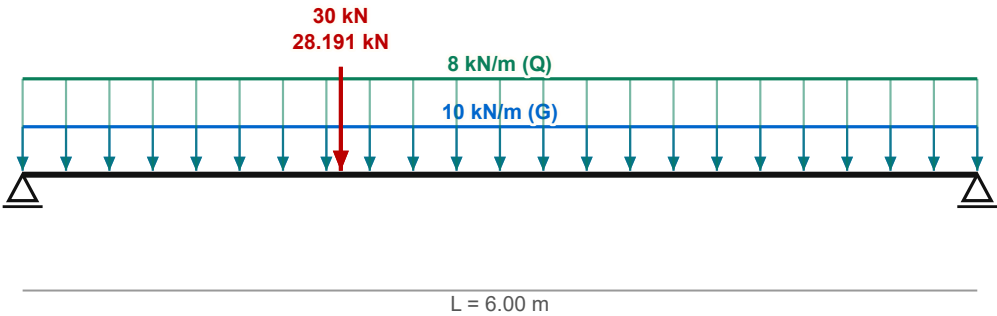
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)
Q PY 28.191 @ 2 m e = 40 mm z_g = 100 mm (kN)
G SW 0.3443 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-5.39	136.09	0.00	0.00	20.52	220.24	20.52	-76.75	136.09	—
	End 2, x = 6	150.00C	0.00	-106.99	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	106.99	—

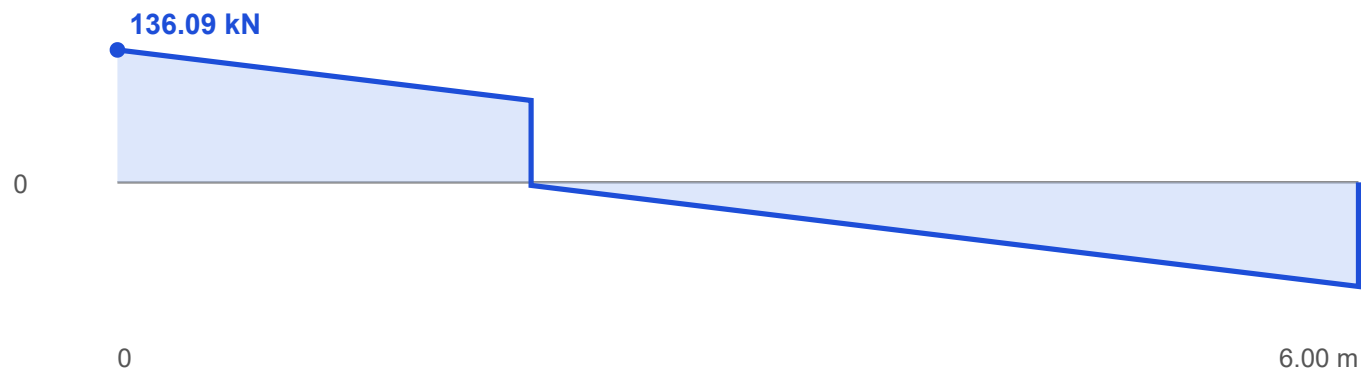
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

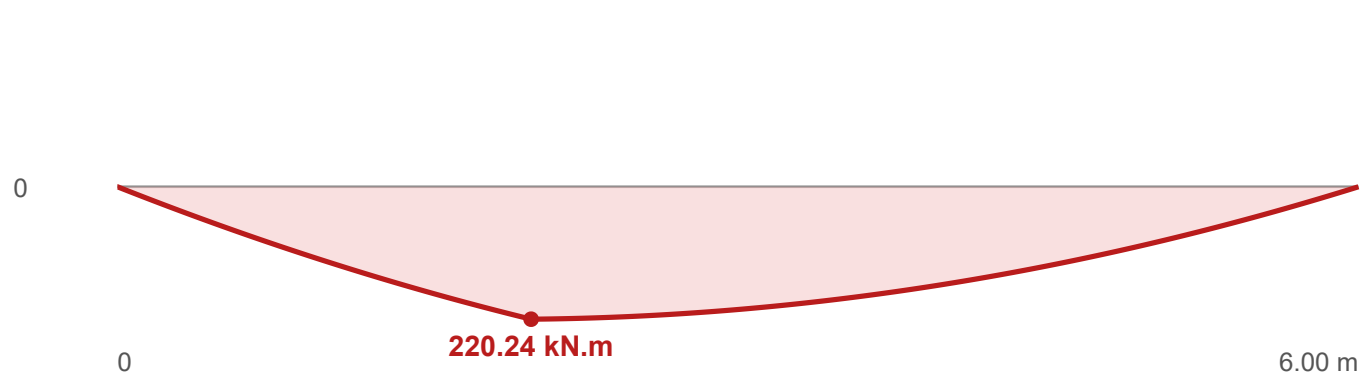
LOADING



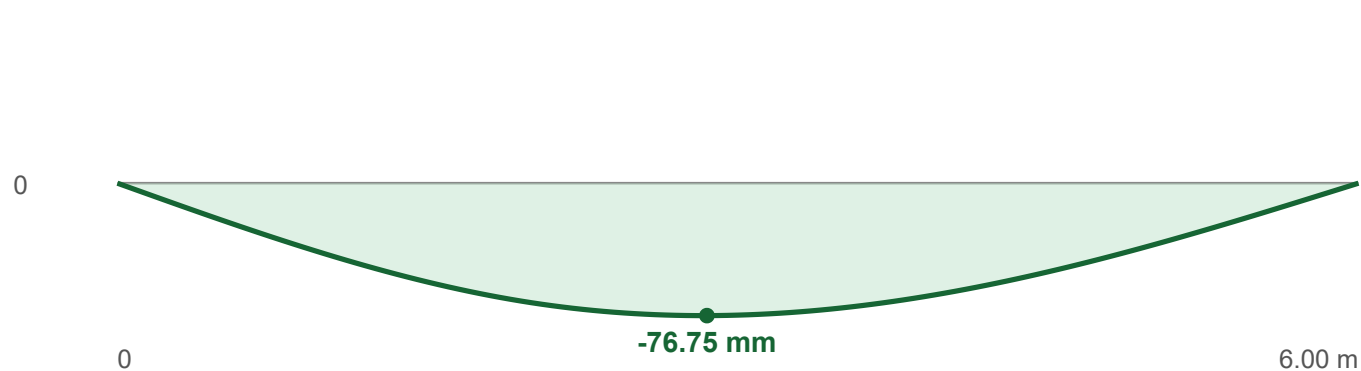
SHEAR FORCE V (kN)



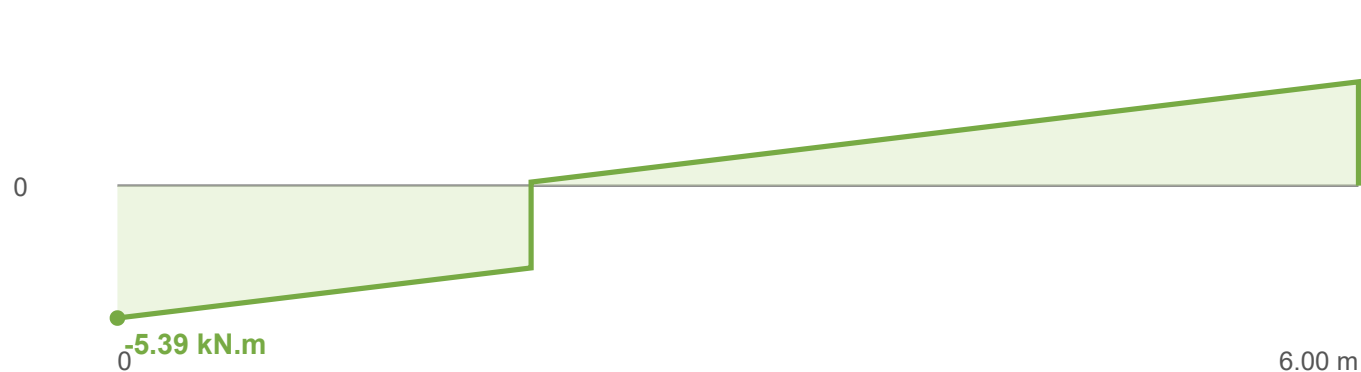
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 220.24, 20.52 (Axial: Non-Slender)	Class 1
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$136.085 / 474.197 =$	0.287	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$3.131 / 474.197 =$	0.007	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$220.241 / 77.550 =$	2.840	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 237.099 = (A_{v,z} = 1493.3 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	$275 \times 172.0 / 1$	47.300 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$44.80 \times 275 / 1 = (\text{No bearing / block tearing design})$	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 1232.000 =$	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$282.0, 29.867, 0.122$	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$282.0 \times 275 / 1$	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$172.0, 14.933, 0.122$	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	$172.0 \times 275 / 1$	47.300 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(220.241 / 77.550)^{1.69} + (20.521 / 47.300)^{1.69} =$	6.070	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, corner	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 220.241$ kN.m, $V = 84.156$ kN, $T = 3.348$ kN.m, $\phi = 0.00599$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.320 = 21.841$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 33.48 + 987.63 + 147.57 = 1168.68 \text{ N/mm}^2$; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 13.34 + 16.67 = 30.00 \text{ N/mm}^2$; $(1168.68/275)^2 + 3(30.00/275)^2 =$	18.096	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 33.48 + 987.63 + 147.57 = 1168.68$, $\tau = 13.34 + 16.67 = 30.00 \rightarrow 18.096$ (governs); web mid-depth: $\sigma_x = 33.48 + 147.57 = 181.05$, $\tau = 13.34 + 34.05 = 47.39 \rightarrow 0.523$; flange mid-width: $\sigma_x = 33.48 + 987.63 = 1021.11$, $\tau = 13.34 \rightarrow 13.794$ ($Q_c = 35.3$, $Q_m = 72.2 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 $N-M_y-M_z$ interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
NOT VERIFIED			
Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.			NOT VERIFIED
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$176.00, 8.00, 8.00, 50.00 \text{ mm}$ ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00 \text{ mm (default 0)}; c = 2000.00 \text{ mm (d = 2000.00 to the member end)}; k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3/176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000 \text{ (first pass } \lambda_F = 0.231 \leq 0.5 \text{ with } m_2 = 9.680, \text{ so } m_2 = 0); 0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275/3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1; L_{eff} = \chi_F \cdot l_y$	$0.5/0.193 = 2.588 \rightarrow 1.000; 1.000 \times 56.00$	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00/1.0 \text{ per web; load share to this web} = 0.935 \text{ (lever rule, } e = 40 \text{ mm)}; F_{Rd} \text{ for the load} = 131.795 \text{ kN}$	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the top flange, in compression) } / 131.795 =$	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 220.241/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 2.962) = 3.032 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	2.165	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	136.085	ULS: 1.35G + 1.5Q	0.258	0.254	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.165	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	106.990	ULS: 1.35G + 1.5Q	0.203	0.214	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{ held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87} \text{ (} N_{cr,y} = \pi^2 E I_y / L_{ey}^2 \text{)}$	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10/1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{ held in position at both ends, in direction at neither: } 1.0 \text{ L})$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46} \text{ (} N_{cr,z} = \pi^2 E I_z / L_{ez}^2 \text{)}$	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10/1 =$	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z; \lambda_T; N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 368.021 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.408	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	$0.0, 0.0, 204.1, 1.000, 300.000$	1.217	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.9, M(L/2)=204.1, M(3L/4)=131.3, M_{max}=220.2 \text{ kN}\cdot\text{m}$		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 204.13, \psi = 0.822, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521 \text{ imposed} + 1.320 \text{ twist } \phi \cdot M_y = 21.841; \psi = 1 \text{ for the imposed constant } M_z, \text{ the twist part taken at } C_{mz} = 1.0 \text{ (Table B.3 upper bound, conservative)}$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 204.13, \psi = 0.822, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b.Rd} = M_{c.y.Rd}$	closed hollow section, $\lambda_{LT} = 0.284 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.217$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	958.768 kN.m	SN003a
$M_{y.Ed}/M_{b.Rd}$	220.241 / 77.550	2.840	Warning

Buckling Resistance

$U_{N.y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N.z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M.y} = M_{y.Ed}/(X_{LT} \cdot M_{y.Rk}/\gamma_{M1})$	220.241 / 77.550 ($M_{y.Rk} = 77.550$)	2.840	Warning
$U_{M.z} = M_{z.Ed}/(M_{z.Rk}/\gamma_{M1})$	(20.521 imposed + 1.320 twist $\phi \cdot M_y = 21.841$) / 47.300	0.462	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b.Rd} = M_{c.y.Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N.y}\}$	$1.000\{1+(0.979-0.2)x0.179\} \leq 1.000(1+0.8x0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N.z}\}$	$1.000\{1+(1.702-0.2)x0.408\} \leq 1.000(1+0.8x0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.179+1.139x2.840+1.326x0.462$	4.027	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.408+0.684x2.840+1.326x0.462$	2.961	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t (= C)	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	5.388 / 39.852	0.135	OK
$T_{t.Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 251$	21.46 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
$(M_{y.Ed}/(M_{pl.y.Rd} \cdot S_{mod}))^\alpha + (M_{z.Ed}/(M_{pl.z.Rd} \cdot S_{mod}))^\beta$	T_{Ed}/T_{Rd} , $V_{pl.T.Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		
$M_{z,tot} = M_{z.Ed} + \phi \cdot M_{y.Ed}$	20.521 + 1.320 (imposed constant $M_{z.Ed}$ + max coincident twist moment of the St Venant solution)	21.841 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z.Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z.Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl.T.Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $T_t = 21.46$ N/mm ² ; $V_{pl.Rd} = 474.197$; $S_{mod} =$	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl.T.Rd}$	$V_{pl.T.Rd}/V_{pl.Rd} = 0.865$ 136.085 / 410.089	0.332	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$76.75 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	76.75 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{max} \leq 2.00^\circ$	0.0030 rad = $0.17^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.17°	OK

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.122	6.070	0.408	2.840	4.027	2.961	4.605	0.287	2.840	2.840	0.662	2.165	0.135	18.096	0.332	6.070

FAIL — governing Biaxial bending + axial (6.2.9.1) = 6.070

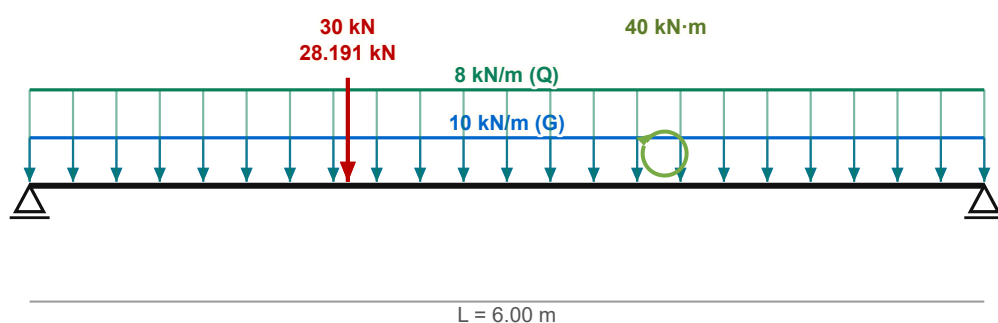
NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

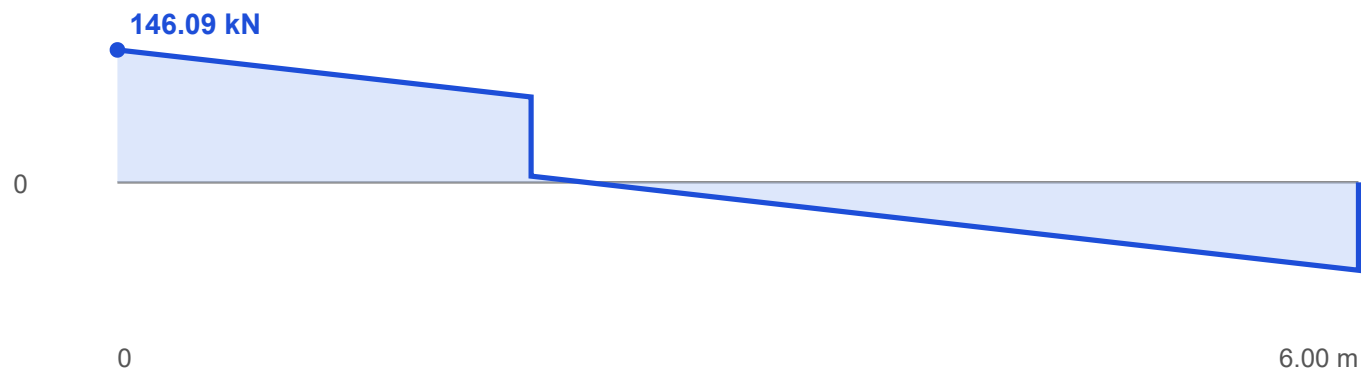
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

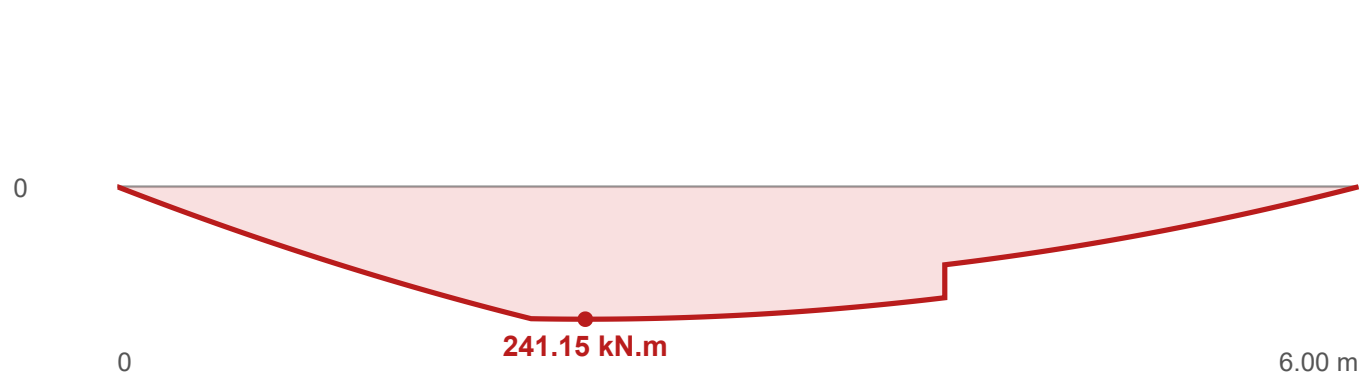
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 18.096 at $x = 2.00$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



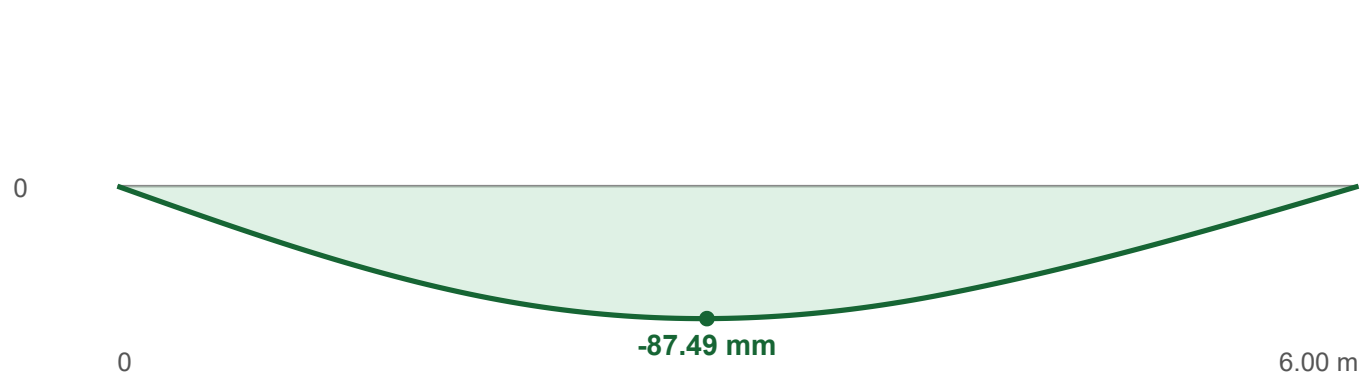
SHEAR FORCE V (kN)



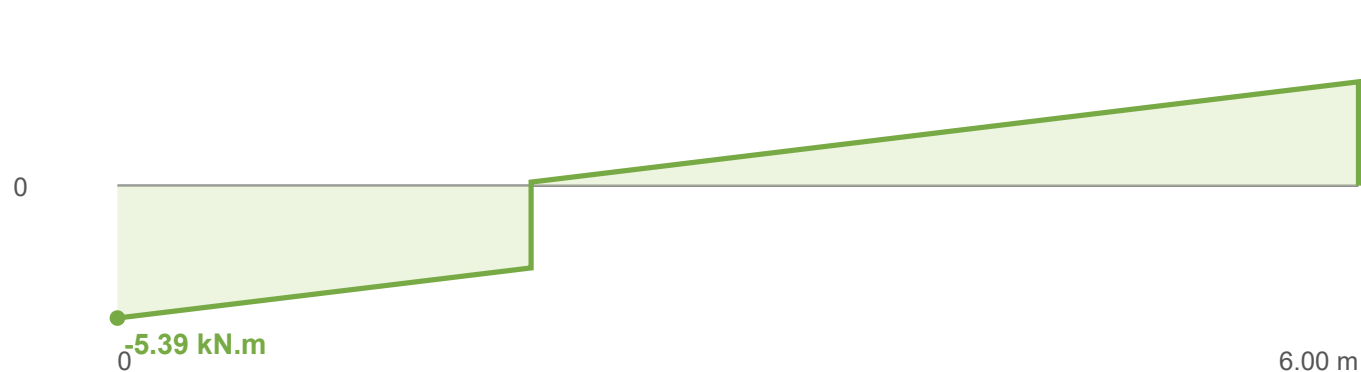
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200	
	B=100 t=8	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 241.15, 20.52	Class 1
	(Axial: Non-Slender)	
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$146.085 / 474.197 =$	0.308	OK
$d/t \leq 72 \epsilon / \eta$	$22.00 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.067 / 474.197 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$241.150 / 77.550 =$	3.110	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 237.099 = (A_{v,z} = 1493.3 \text{ mm}^2; \text{ no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 172.0 / 1$	47.300 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$44.80 \times 275 / 1 = (\text{No bearing / block tearing design})$	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 1232.000 =$	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$282.0, 29.867, 0.122$	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	$282.0 \times 275 / 1$	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$172.0, 14.933, 0.122$	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	$172.0 \times 275 / 1$	47.300 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(241.150 / 77.550)^{1.69} + (20.521 / 47.300)^{1.69} =$	7.033	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.262 m, corner	@ x = 2.262 m (ULS: 1.35G + 1.5Q): $M_y = 241.150 \text{ kN.m}$, $V = 0.067 \text{ kN}$, $T = 0.411 \text{ kN.m}$, $\phi = 0.00594 \text{ rad}$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.433 = 21.954 \text{ kN.m}$; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 33.48 + 1081.39 + 148.34 = 1263.21 \text{ N/mm}^2$; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 1.64 + 0.01 = 1.65 \text{ N/mm}^2$; $(1263.21/275)^2 + 3(1.65/275)^2 =$	21.100	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 33.48 + 1081.39 + 148.34 = 1263.21$, $\tau = 1.64 + 0.01 = 1.65 \rightarrow 21.100$ (governs); web mid-depth: $\sigma_x = 33.48 + 148.34 = 181.82$, $\tau = 1.64 + 0.03 = 1.66 \rightarrow 0.437$; flange mid-width: $\sigma_x = 33.48 + 1081.39 = 1114.87$, $\tau = 1.64 \rightarrow 16.436$ ($Q_c = 35.3$, $Q_m = 72.2 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 $N-M_y-M_z$ interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$176.00, 8.00, 8.00, 50.00 \text{ mm}$ ($B/2 = 50.00 \leq t_w + 15\epsilon t_f = 118.93$); $f_{yw} = f_y = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\bar{\lambda}_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855 \times 10^3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, $e = 40 \text{ mm}$); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)

F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the top flange, in compression) / } 131.795 =$	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 240.241/77.550 +$ $N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 3.220) = 3.238 \leq$ $1.4 \text{ (ULS: } 1.35G + 1.5Q)$	2.313	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	146.085	ULS: 1.35G + 1.5Q	0.277	0.267	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.313	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	96.990	ULS: 1.35G + 1.5Q	0.184	0.201	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10 / 1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z$; $\bar{\lambda}_T$; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 368.021 = (\text{min of } N_{b,y,Rd}, N_{b,z,Rd})$	0.408	OK

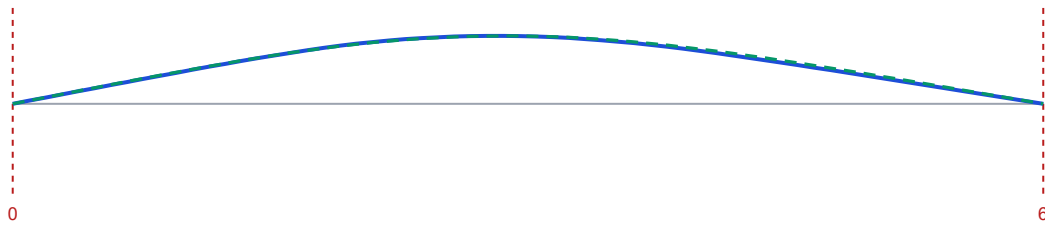
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr}/M_{cr,uniform}$	916.893 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.164	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 234.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.9$, $M(L/2)=234.1$, $M(3L/4)=116.3$, $M_{max}=241.1 \text{ kN}\cdot\text{m}$	1.214	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 1.440 twist $\phi \cdot M_y = 21.961$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 3 solve(s)	895.804 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 895.804}$	0.294	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\bar{\lambda}_{LT} = 0.294 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
$M_{y,Ed}/M_{b,Rd}$	241.150 / 77.550	3.110	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.86$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	241.150 / 77.550 ($M_{y,Rk} = 77.550$)	3.110	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.440 twist ϕ , $M_y = 21.961$) / 47.300	0.464	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1 + (\lambda_y - 0.2)U_{N,y}\}$	$1.000\{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000(1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1 + (\lambda_z - 0.2)U_{N,z}\}$	$1.000\{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000(1 + 0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.179 + 1.139 \times 3.110 + 1.326 \times 0.464$	4.338	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.408 + 0.684 \times 3.110 + 1.326 \times 0.464$	3.149	Warning

Torsion Design

J, H, a, Q_f, Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0}, S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
$W_t (= C)$	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$5.388 / 39.852$	0.135	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 251$	21.46 N/mm ²	6.2.7(7)
Modified Local Capacity $(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 1.440$ (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.961 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $\tau_t = 21.46$ N/mm ² ; $V_{pl,Rd} = 474.197$; $S_{mod} =$	410.089 kN	6.2.7(9) Eq 6.28
	$V_{pl,T,Rd}/V_{pl,Rd} = 0.865$		
$V_{Ed}/V_{pl,T,Rd}$	$146.085 / 410.089$	0.356	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.49 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	87.49 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{max} \leq 2.00^\circ$	0.0030 rad = $0.17^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.17°	OK

N_Ed/N _{pl,Rd}	Local	UNyz	UMyz	Ax+M _{6.61}	Ax+M _{6.62}	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
0.122	7.033	0.408	3.110	4.338	3.149	5.249	0.308	3.110	3.110	0.662	2.313	0.135	21.100	0.356	7.033

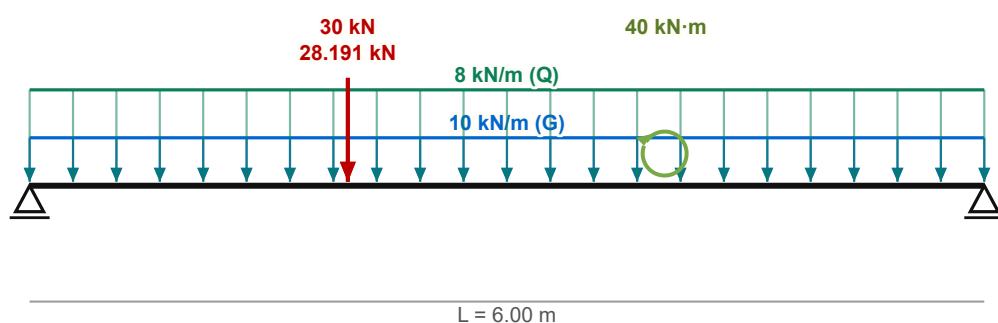
FAIL — governing Biaxial bending + axial (6.2.9.1) = 7.033

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

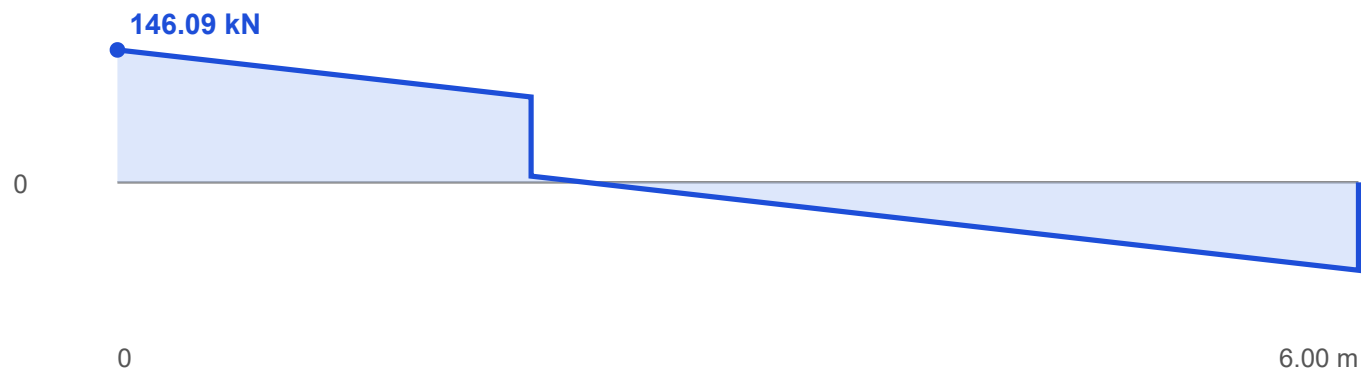
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

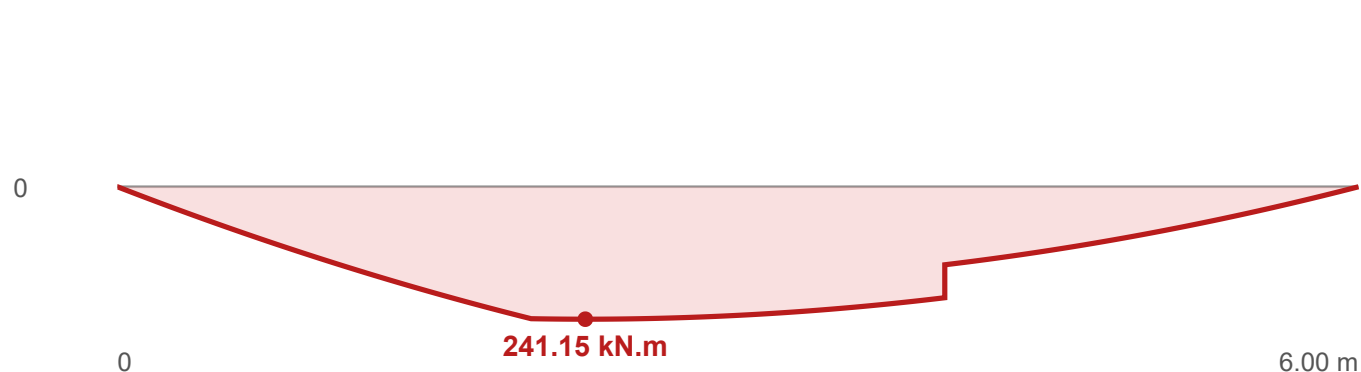
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 21.100 at x = 2.26 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



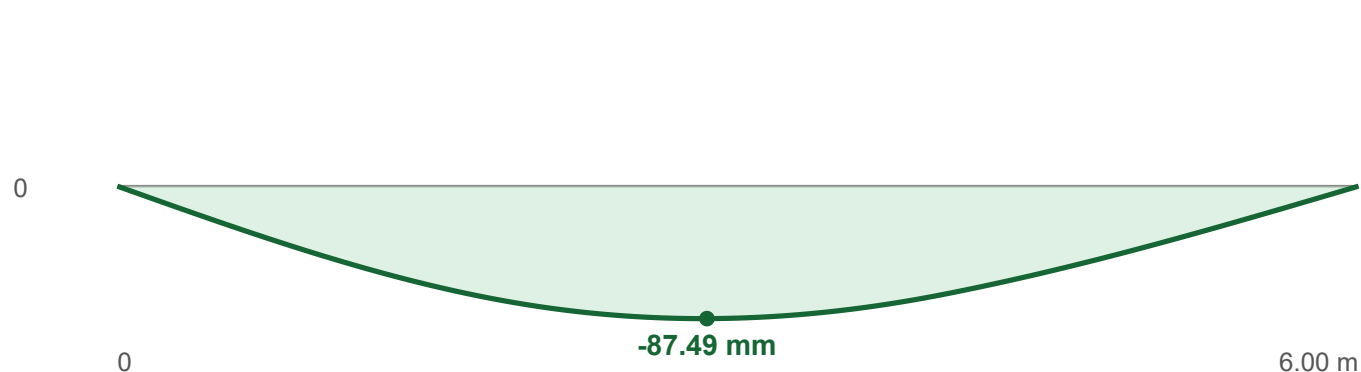
SHEAR FORCE V (kN)



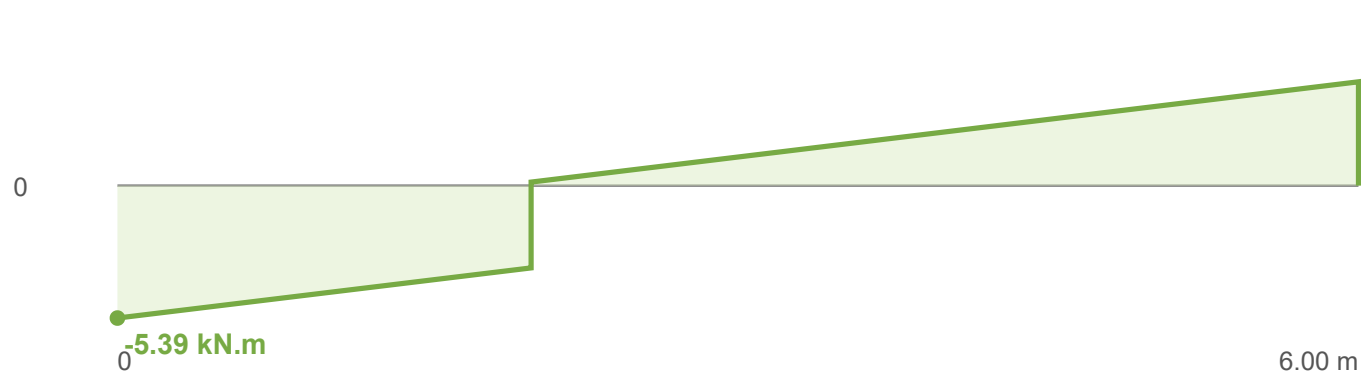
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200		
	B=100 t=8		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 241.15, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1	Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3} \times 1$)	474.197 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	146.085 / 474.197 =	0.308	OK
$d/t \leq 72 \epsilon / \eta$	22.00 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.067 / 474.197 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	241.150 / 77.550 =	3.110	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 237.099 = ($A_{v,z} = 1493.3$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 172.0 / 1	47.300 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	44.80 x 275 / 1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	282.0 x 275 / 1	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	172.0, 14.933, 0.122	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	172.0 x 275 / 1	47.300 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(241.150 / 77.550)^{1.69} + (20.521 / 47.300)^{1.69} =$	7.033	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.262 m, corner	@ x = 2.262 m (ULS: 1.35G + 1.5Q): $M_y = 241.150$ kN.m, $V = 0.067$ kN, $T = 0.411$ kN.m, $\phi = 0.00594$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.433 = 21.954$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 33.48 + 1081.39 + 148.34 = 1263.21$ N/mm ² ; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 1.64 + 0.01 = 1.65$ N/mm ² ; $(1263.21/275)^2 + 3(1.65/275)^2 =$	21.100	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 33.48 + 1081.39 + 148.34 = 1263.21$, $\tau = 1.64 + 0.01 = 1.65 \rightarrow 21.100$ (governs); web mid-depth: $\sigma_x = 33.48 + 148.34 = 181.82$, $\tau = 1.64 + 0.03 = 1.66 \rightarrow 0.437$; flange mid-width: $\sigma_x = 33.48 + 1081.39 = 1114.87$, $\tau = 1.64 \rightarrow 16.436$ ($Q_c = 35.3$, $Q_m = 72.2$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 $N-M_y-M_z$ interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15 \epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm ($d = 2000.00$ to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3/176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275/3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00/1.0$ per web; load share to this web = 0.935 (lever rule, $e = 40$ mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 240.241/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 3.220) = 3.238 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.313	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	146.085	ULS: 1.35G + 1.5Q	0.277	0.267	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.313	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	96.990	ULS: 1.35G + 1.5Q	0.184	0.201	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10/1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10/1 =$	368.021 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	$150.000 / 368.021 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.408	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 234.1, 1.000, 300.000	1.214	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.9$, $M(L/2)=234.1$, $M(3L/4)=116.3$, $M_{max}=241.1$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 1.440 twist $\phi \cdot M_y = 21.961$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.285 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.214$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	955.991 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	241.150 / 77.550	3.110	Warning

Buckling Resistance

$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	241.150 / 77.550 ($M_{y,Rk} = 77.550$)	3.110	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.440 twist $\phi \cdot M_y = 21.961$) / 47.300	0.464	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.979-0.2)x0.179\} \leq 1.000(1+0.8x0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.702-0.2)x0.408\} \leq 1.000(1+0.8x0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.179+1.139x3.110+1.326x0.464$	4.338	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.408+0.684x3.110+1.326x0.464$	3.149	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t (= C)	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	5.388 / 39.852	0.135	OK
$T_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 251$	21.46 N/mm ²	6.2.7(7)
Modified Local Capacity $(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 1.440 (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.961 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $T_t = 21.46$ N/mm ² ; $V_{pl,Rd} = 474.197$; $S_{mod} =$	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.865$ 146.085 / 410.089	0.356	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	87.49 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{max} \leq 2.00^\circ$	0.0030 rad = $0.17^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.17°	OK

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.122	7.033	0.408	3.110	4.338	3.149	5.249	0.308	3.110	3.110	0.662	2.313	0.135	21.100	0.356	7.033

FAIL — governing Biaxial bending + axial (6.2.9.1) = 7.033

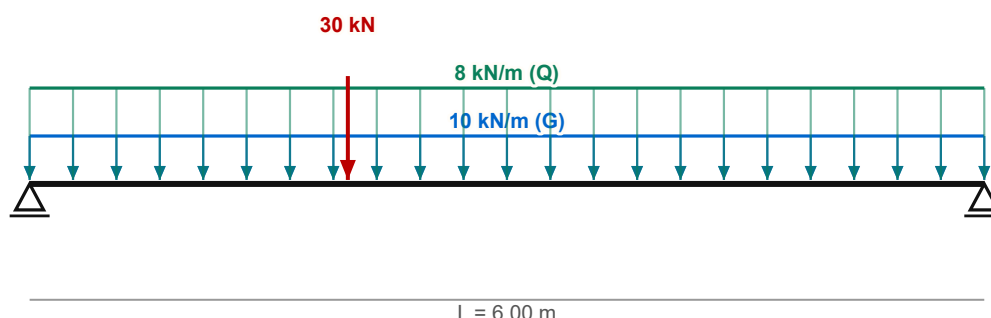
NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

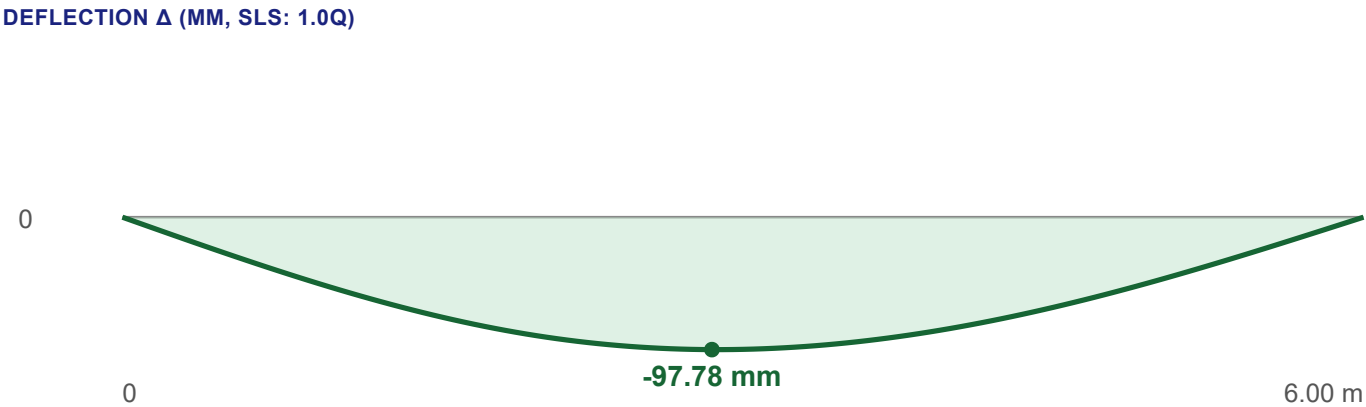
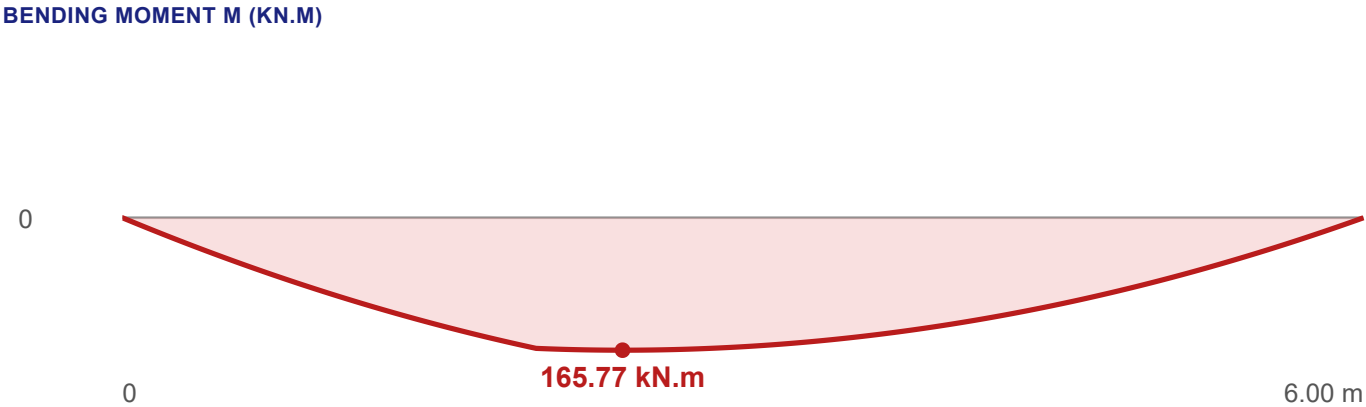
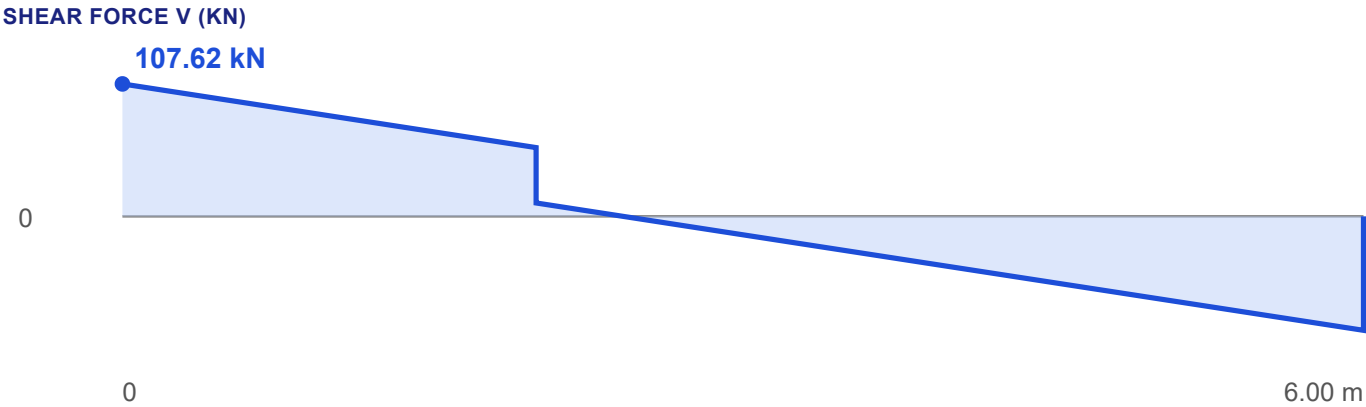
ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 21.100 at $x = 2.26$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1790 mm ² x 275 / ($\sqrt{3}$ x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

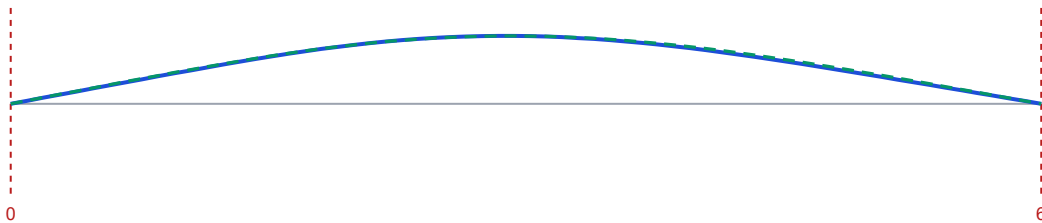
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	1190.076 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1190.076}$	0.211	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\bar{\lambda}_{LT} = 0.211 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

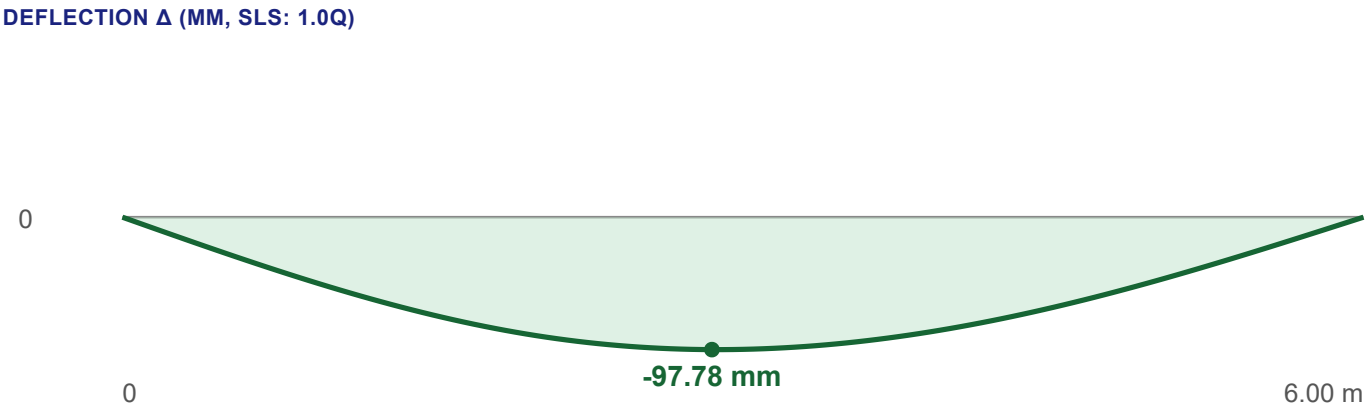
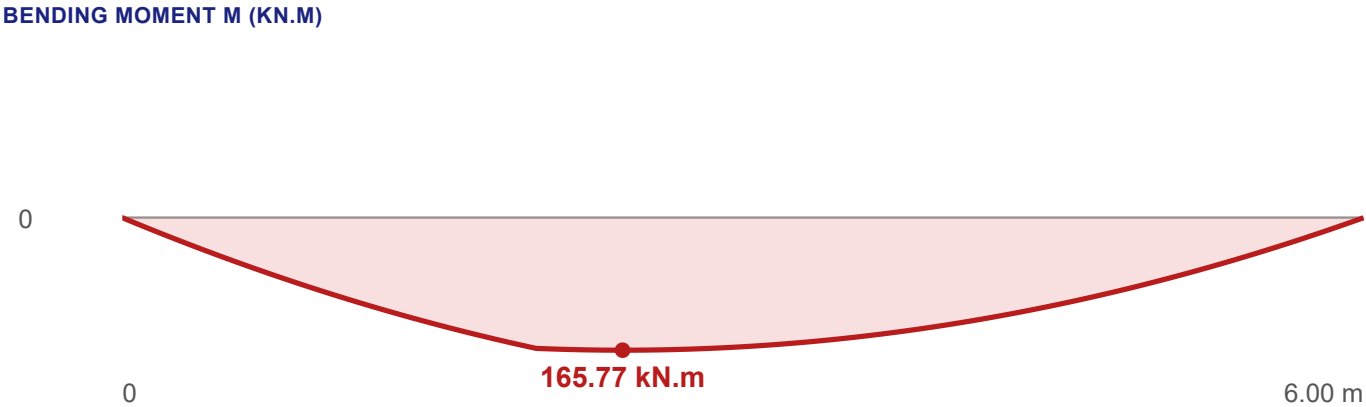
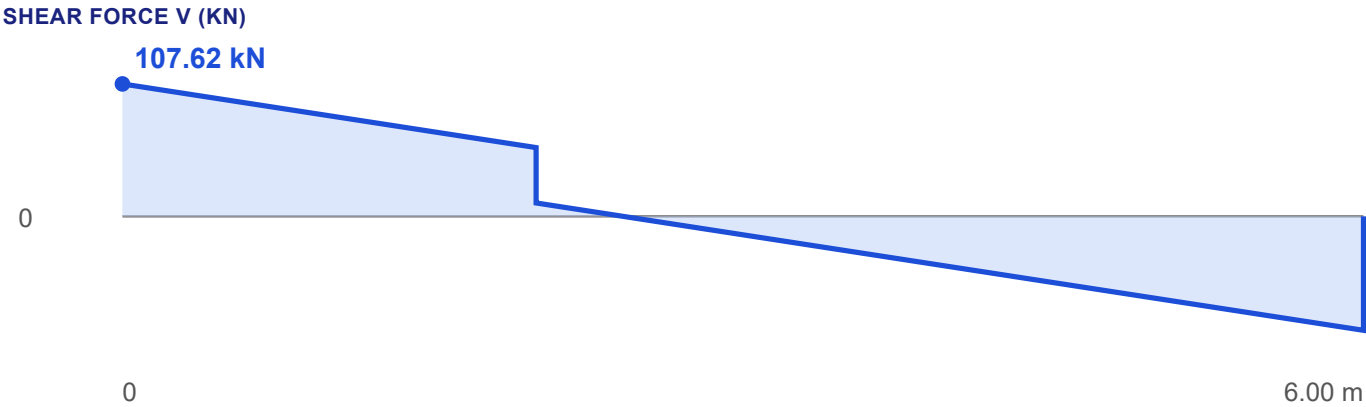
In-span $\delta \leq \text{Span}/360$ $97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 97.78 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	1790 mm ² x 275 / (√3 x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 ≤ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 ($e = 0$); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.4, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	$M_{(y,Ed)} / M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced

buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member SHS-L2 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6.00C	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

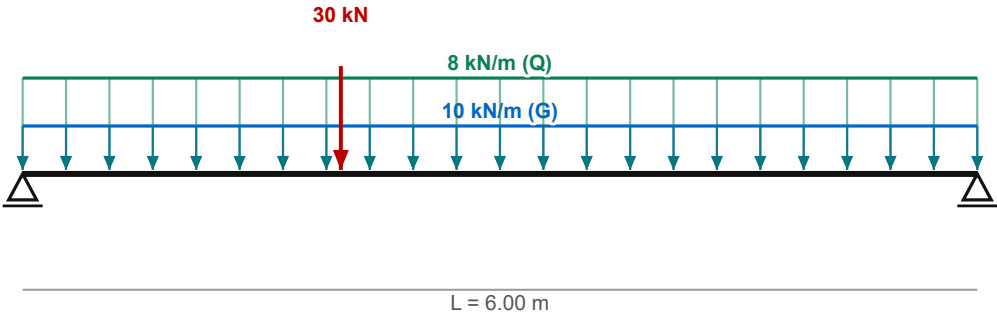
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

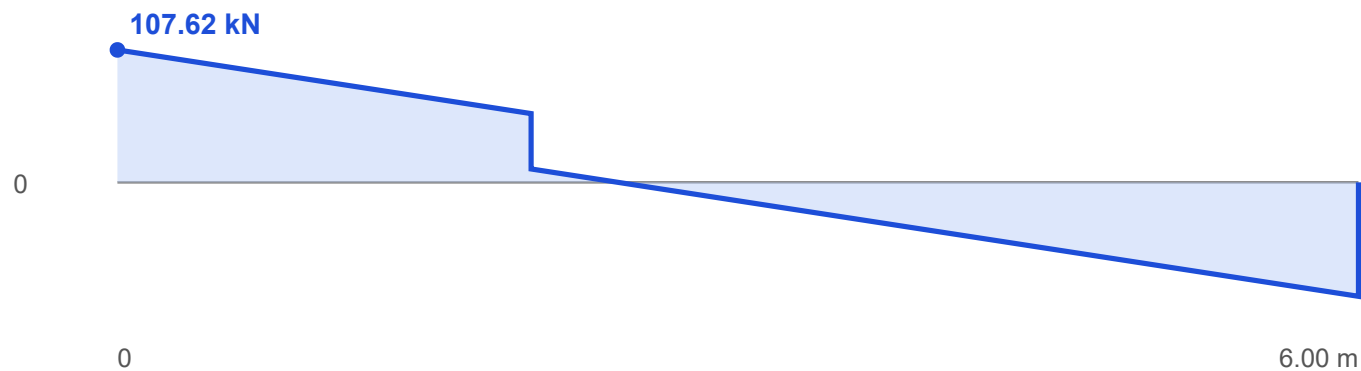
R_{min} ≥ 0

OK

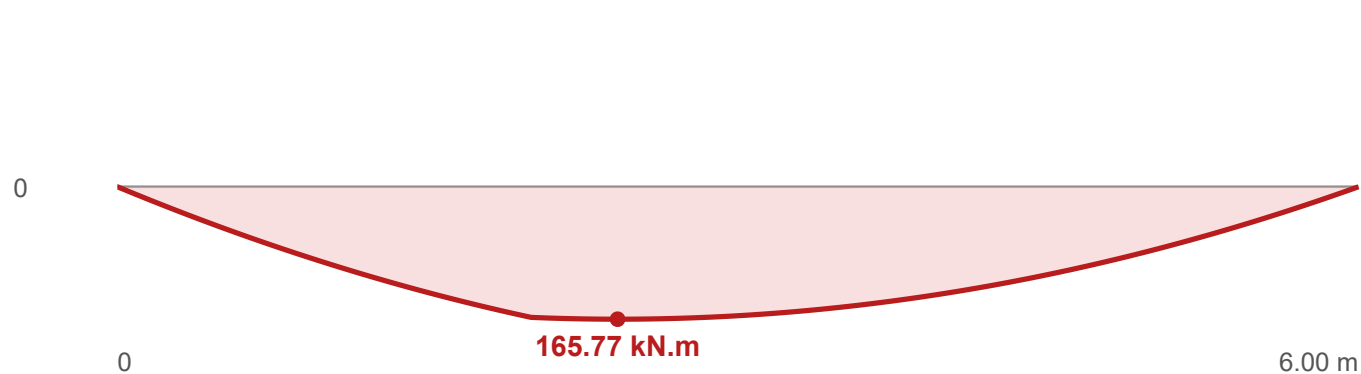
LOADING



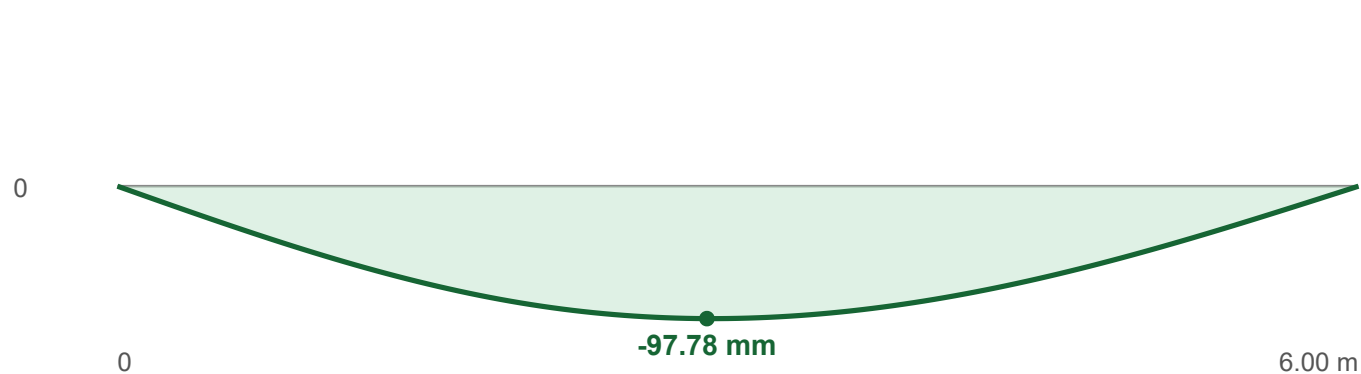
SHEAR FORCE V (kN)



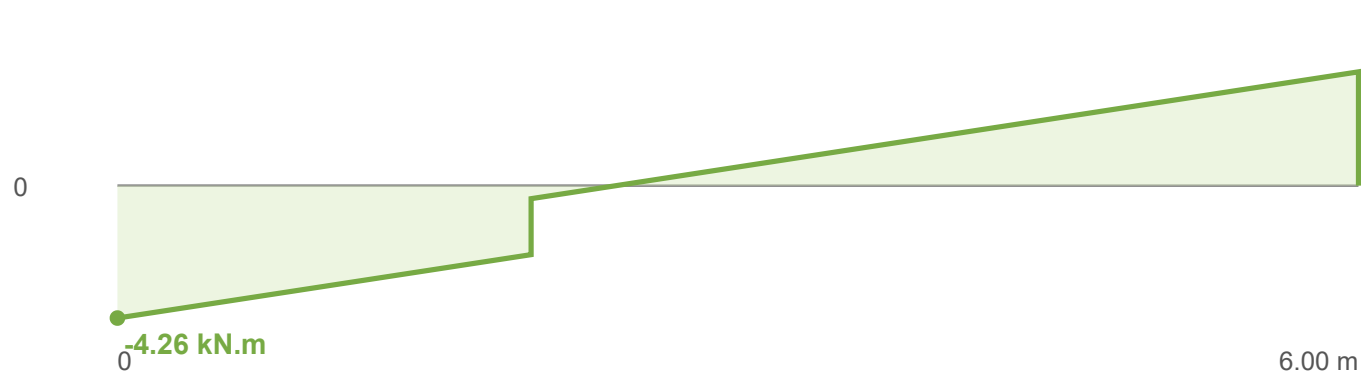
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150

B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 0.00, 165.77, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases 1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 1017.01 + 4.32 = 1021.33 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05 \text{ N/mm}^2$; $(1021.33/275)^2 + 3(0.05/275)^2 =$	13.793	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 1017.01 + 4.32 = 1021.33$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 13.793$ (governs); web mid-depth: $\sigma_x = 0.00 + 4.32 = 4.32$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 1017.01 = 1017.01$, $\tau = 0.03 \rightarrow 13.677$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00 \text{ mm}$ ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \times 10^3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.027	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

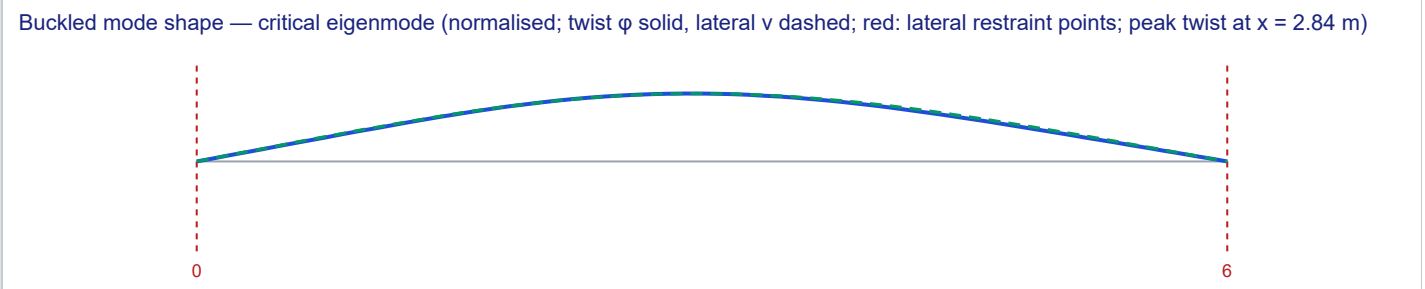
Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	1190.076 / 1042.431 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M _o , ψ, μ) (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.3, M(L/2)=161.4, M(3L/4)=109.8, M _{max} =165.8 kN·m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	1190.076 kN.m	converged
λ _{LT} = √W _f ·f _y /M _{cr}	√ 192.0 x 275 / 1190.076	0.211	
M _{b,Rd} = M _{c,y,Rd}	closed hollow section — not susceptible to LTB (λ _{LT} = 0.211 ≤ 0.4)	52.800 kN.m	6.3.2.1(2)
M _{y,Ed} /M _{b,Rd}	165.773 / 52.800	3.140	Warning



Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /((√3γ _{M0}))	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 38.105	0.112	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /240	17.75 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} /(M _{pl,y,Rd} ·S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} ·S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t /(f _y /√3)]·V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 17.75 N/mm ² ; V _{pl,Rd} = 284.201; S _{mod} =	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	V _{pl,T,Rd} /V _{pl,Rd} = 0.888 107.617 / 252.428	0.426	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0019 rad = $0.11^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	13.793	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 13.793 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member SHS-L2 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.2757 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6 0.00C		0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

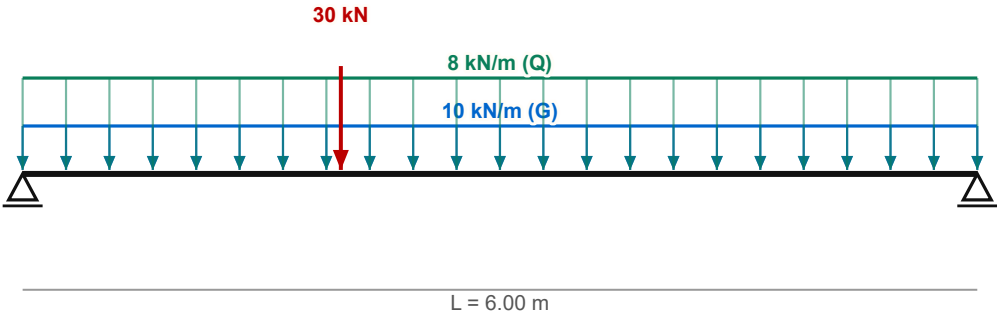
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

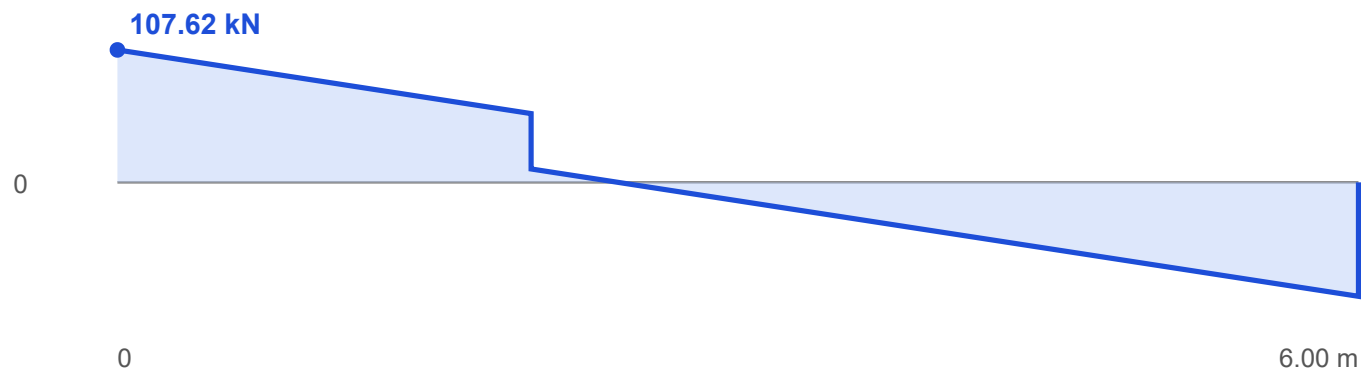
R_{min} ≥ 0

OK

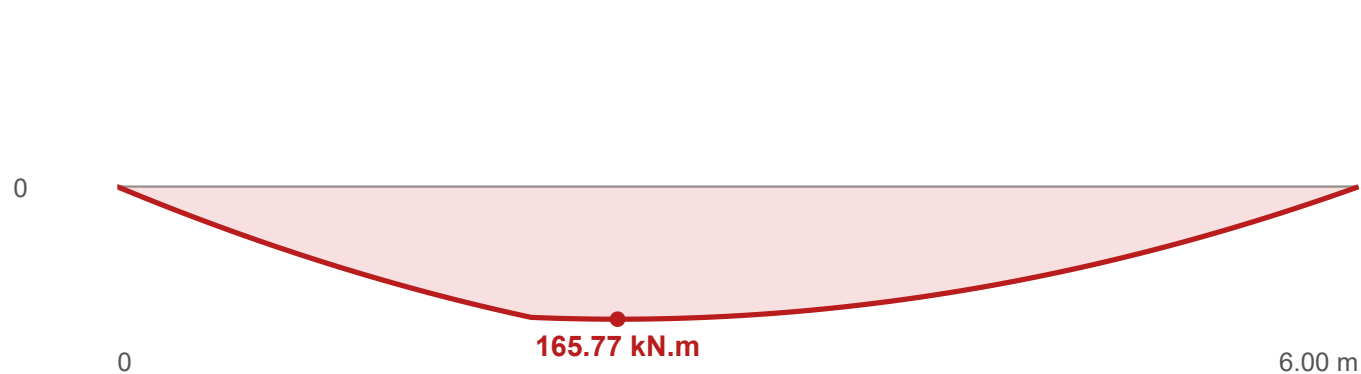
LOADING



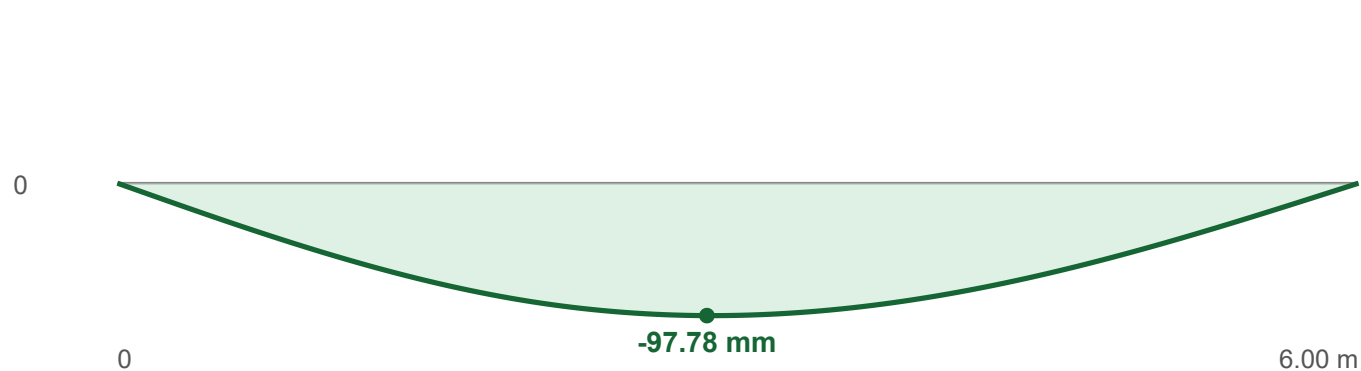
SHEAR FORCE V (kN)



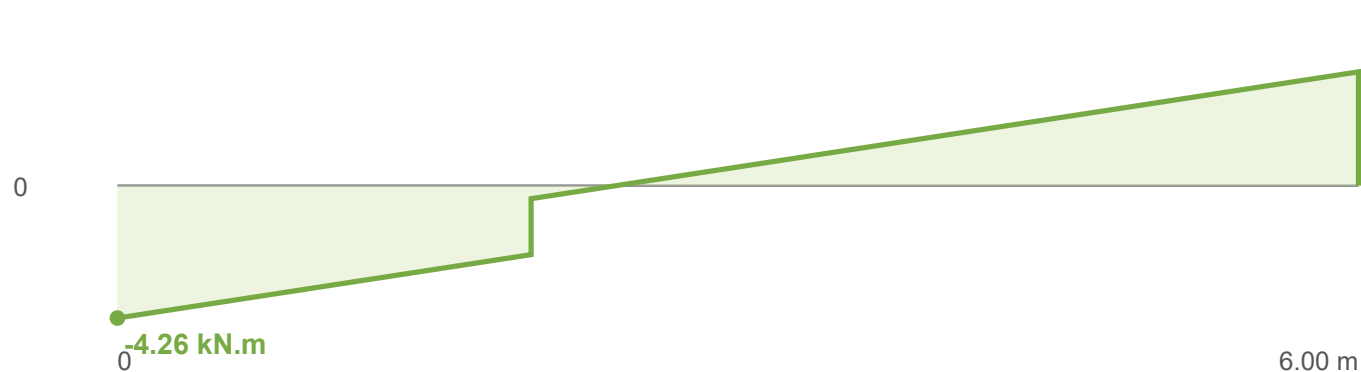
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150

B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 0.00, 165.77, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 1017.01 + 4.32 = 1021.33 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05 \text{ N/mm}^2$; $(1021.33/275)^2 + 3(0.05/275)^2 =$	13.793	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 1017.01 + 4.32 = 1021.33$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 13.793$ (governs); web mid-depth: $\sigma_x = 0.00 + 4.32 = 4.32$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 1017.01 = 1017.01$, $\tau = 0.03 \rightarrow 13.677$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00 \text{ mm}$ ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \times 10^3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.027	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.4, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{\max}=165.8$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	1212.847 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
$W_t (= C)$	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	275 x 240 / ($\sqrt{3} \times 1$)	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	4.260 / 38.105	0.112	OK
$T_{t,Ed} = T_{Ed}/W_t$	4.260 x 10 ³ /240	17.75 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
$(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m (ULS: 1.35G + 1.5Q): $T = 4.260$ kN.m, $T_t = 17.75$ N/mm ² ; $V_{pl,Rd} = 284.201$; $S_{mod} =$	252.428 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.888$ 107.617 / 252.428	0.426	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	97.78 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0019 rad = $0.11^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

MA/Mc	M _{(y,Ed)/M_(b,Rd)}	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	13.793	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down

forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 13.793 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

SHS-L3t · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 5.867

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +75$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L3t SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 0$ mm $z_g = 75$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 0$ mm $z_g = 75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 75$ mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

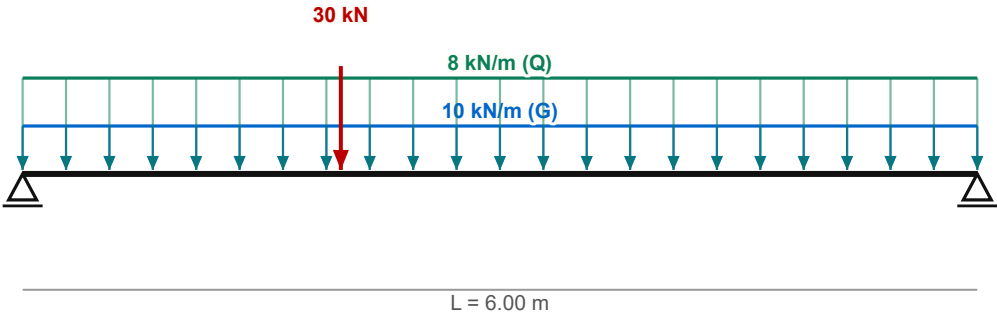
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

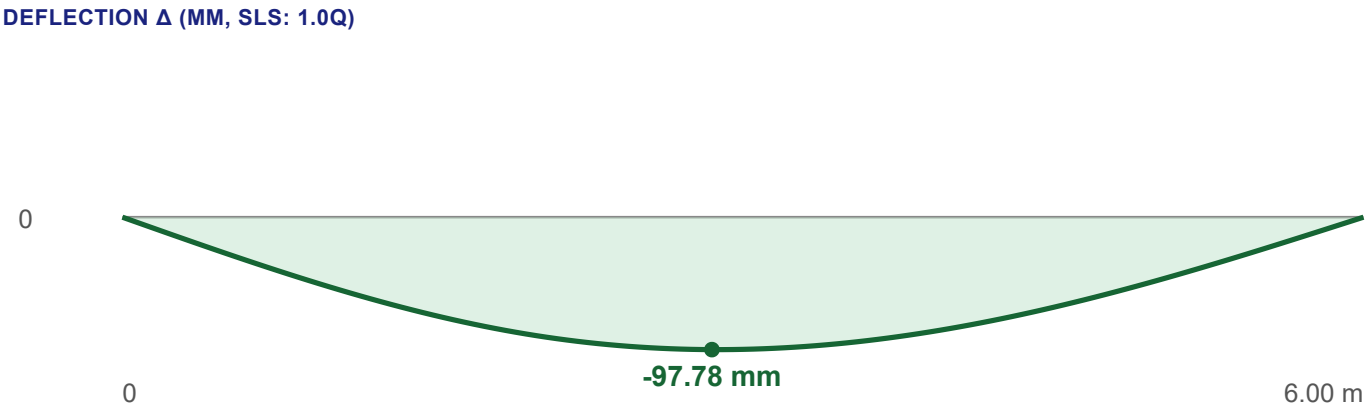
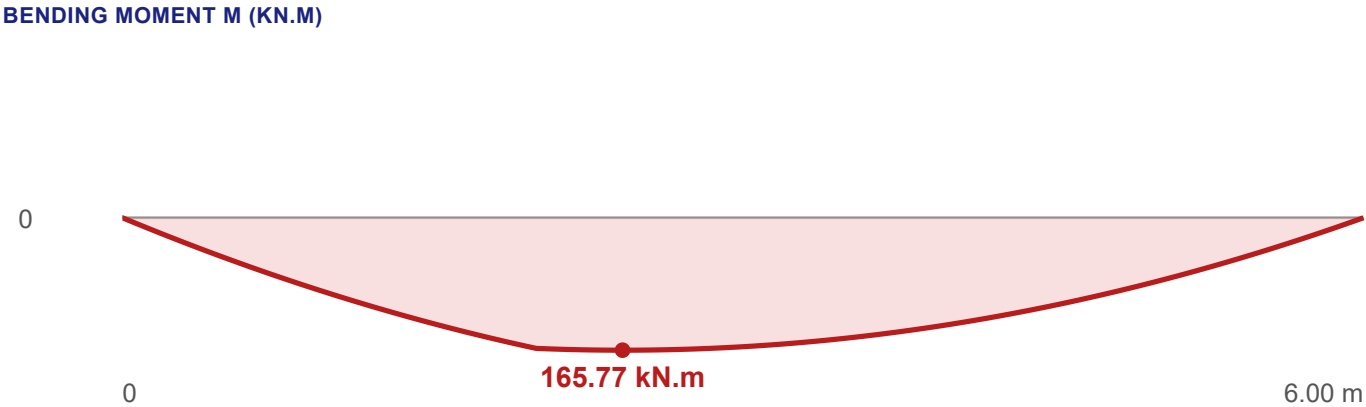
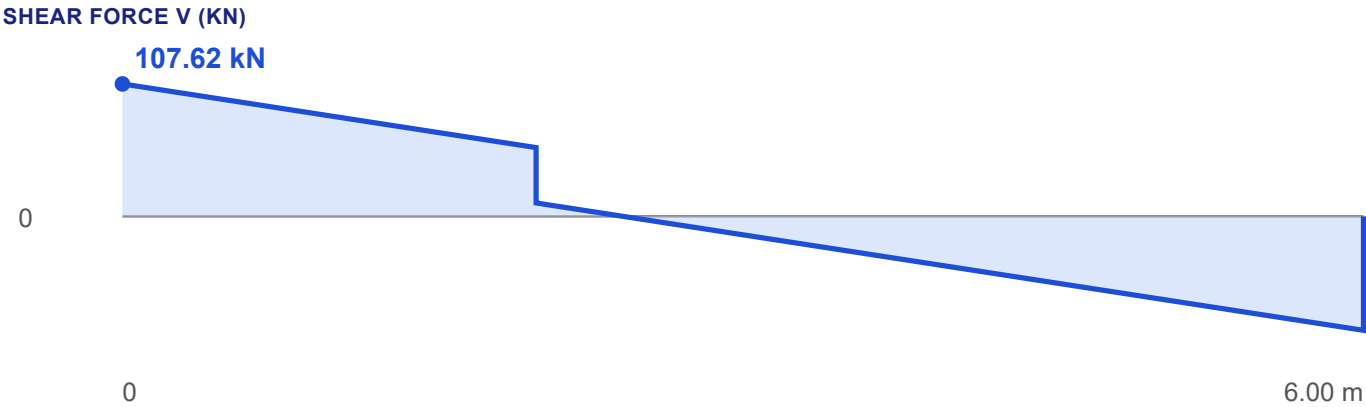
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1790 mm ² x 275 / ($\sqrt{3}$ x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

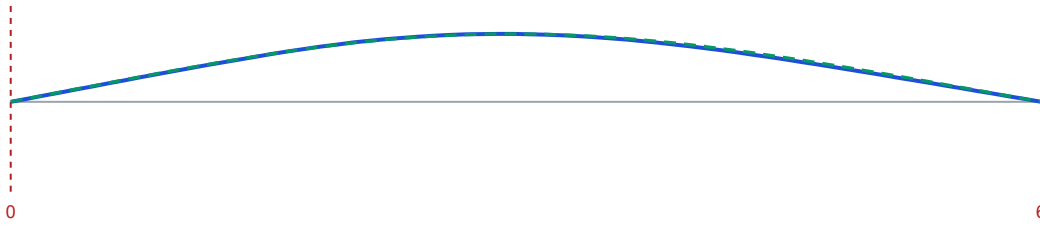
Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	1162.306 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1162.306}$	0.213	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\bar{\lambda}_{LT} = 0.213 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

97.78 mm

Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L3t · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 5.867

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +75$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L3t SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 0$ mm $z_g = 75$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 0$ mm $z_g = 75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 75$ mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

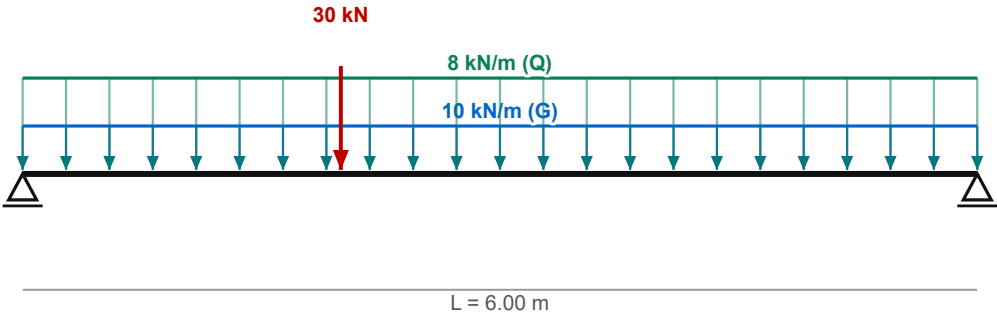
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

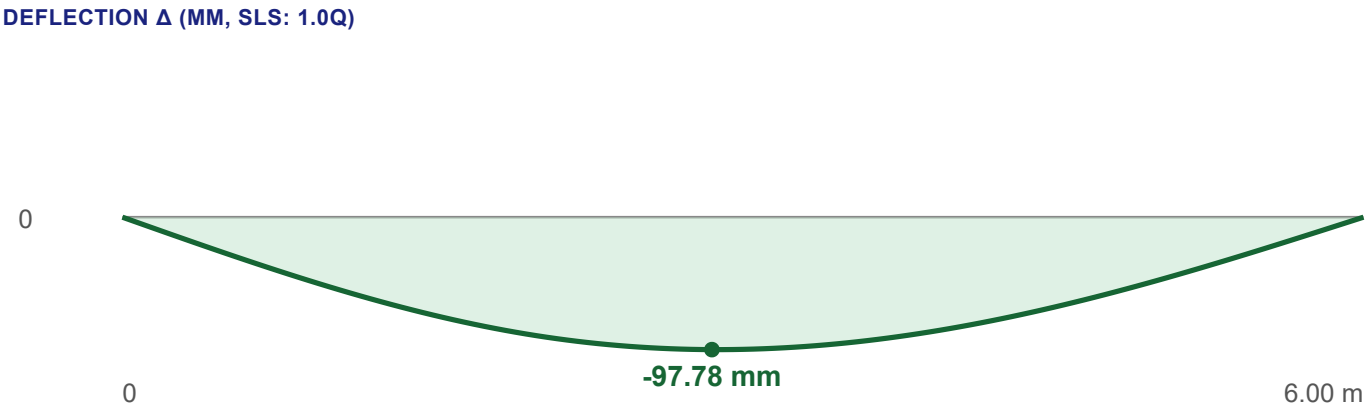
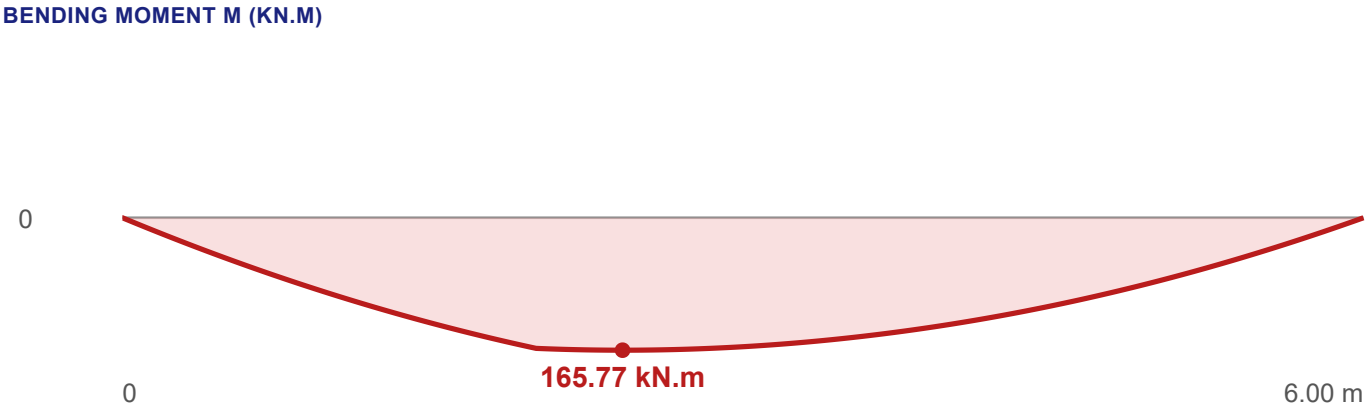
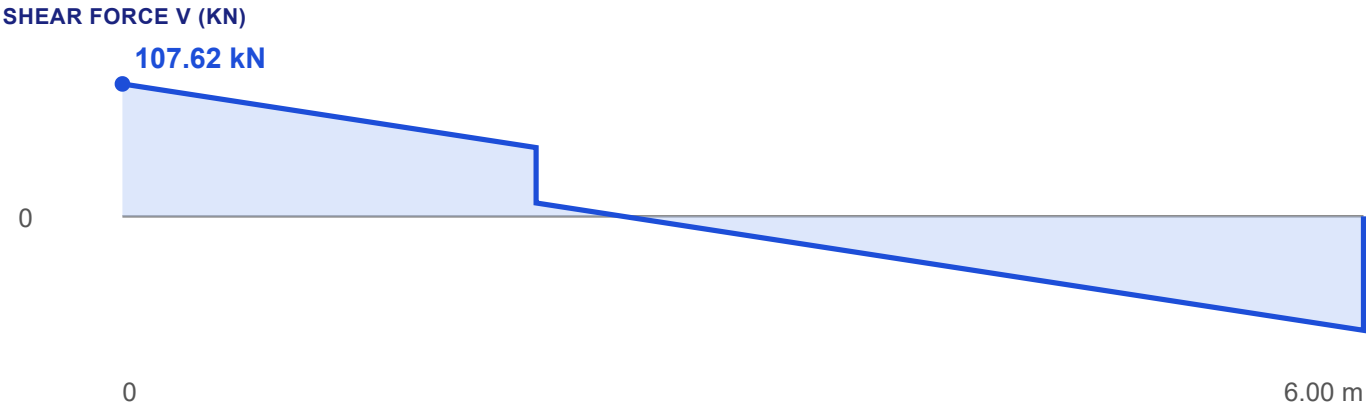
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	1790 mm ² x 275 / (√3 x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 ≤ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s , c; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.4, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
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M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.163, L _e = 6 m; z _g = +75 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	1212.847 kN.m	SN003a
M _{y.Ed} /M _{b.Rd}	165.773 / 52.800	3.140	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
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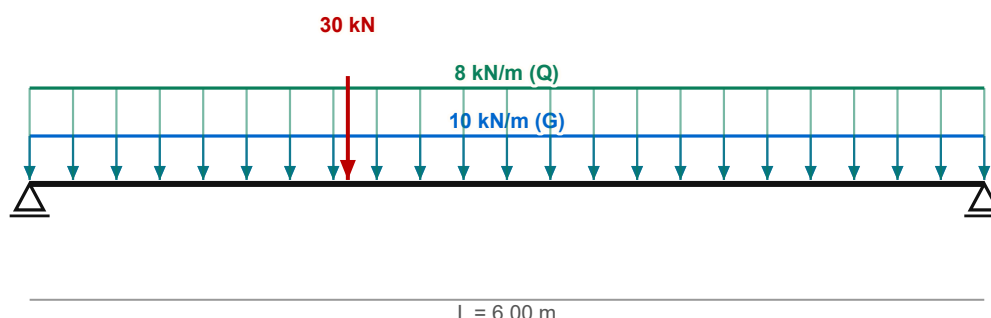
MA/Mc	M _{-(y.Ed)/M_{-(b.Rd)}}	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

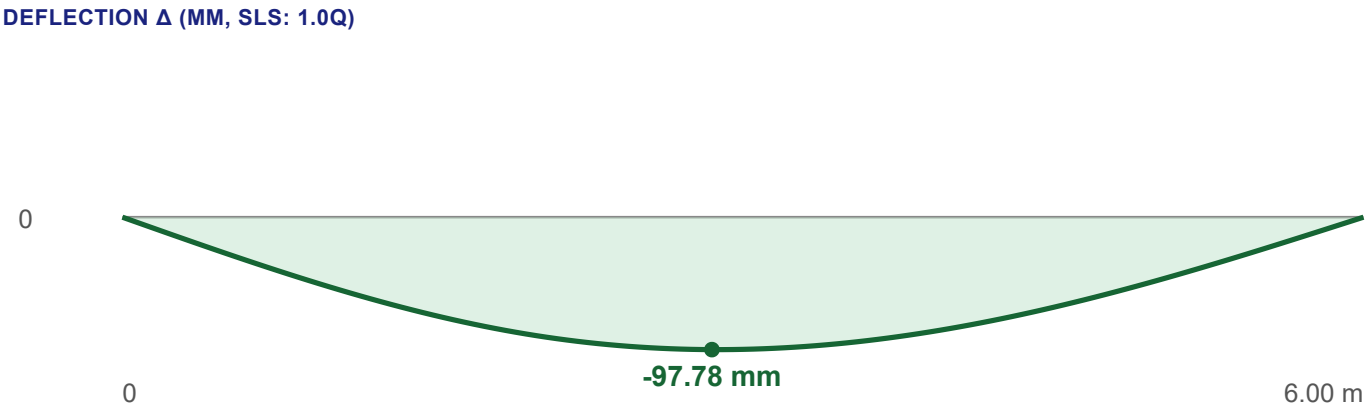
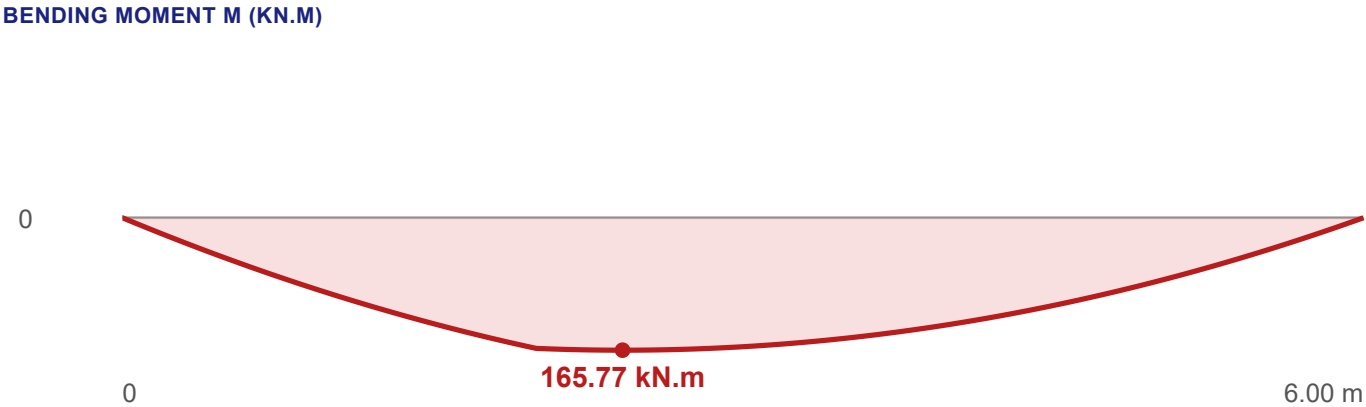
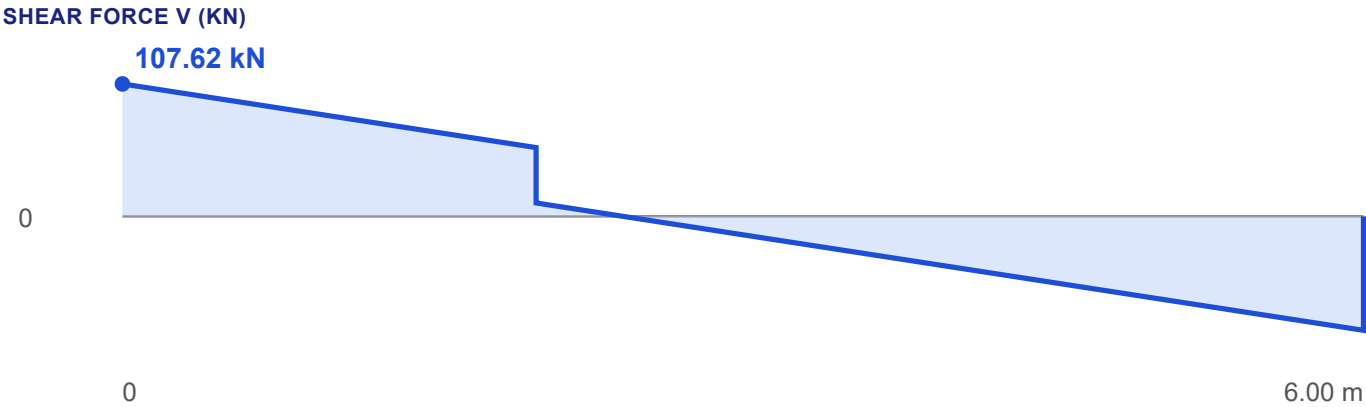
FAIL — governing Deflection = 5.867

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	1790 mm ² x 275 / (√3 x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 ≤ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

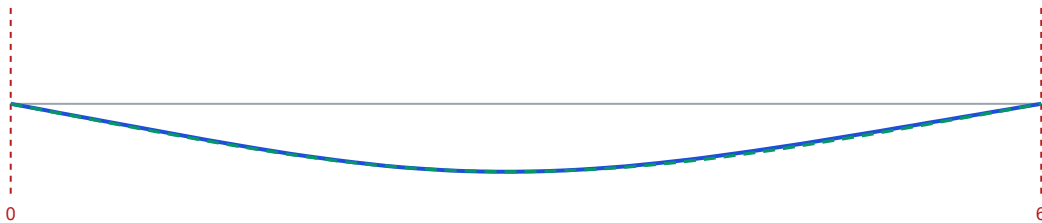
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -75, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	1218.400 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1218.400}$	0.208	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\bar{\lambda}_{LT} = 0.208 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 97.78 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L3b · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 5.867

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -75$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L3b SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q UDL 8.000 0-6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -75$ mm (kN)

G SW 0.2757 kN/m 0-6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

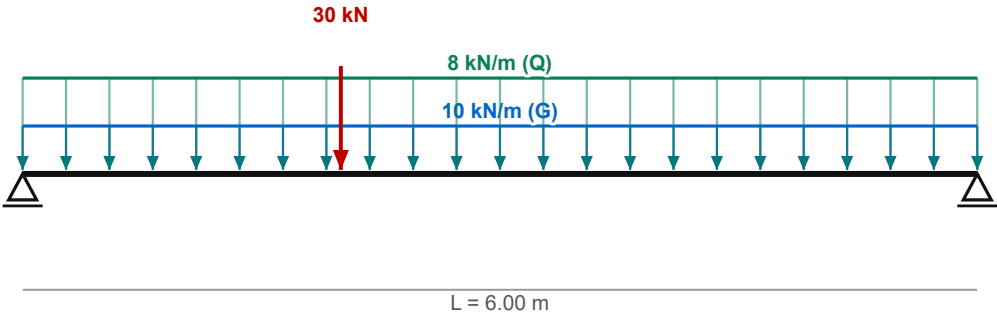
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

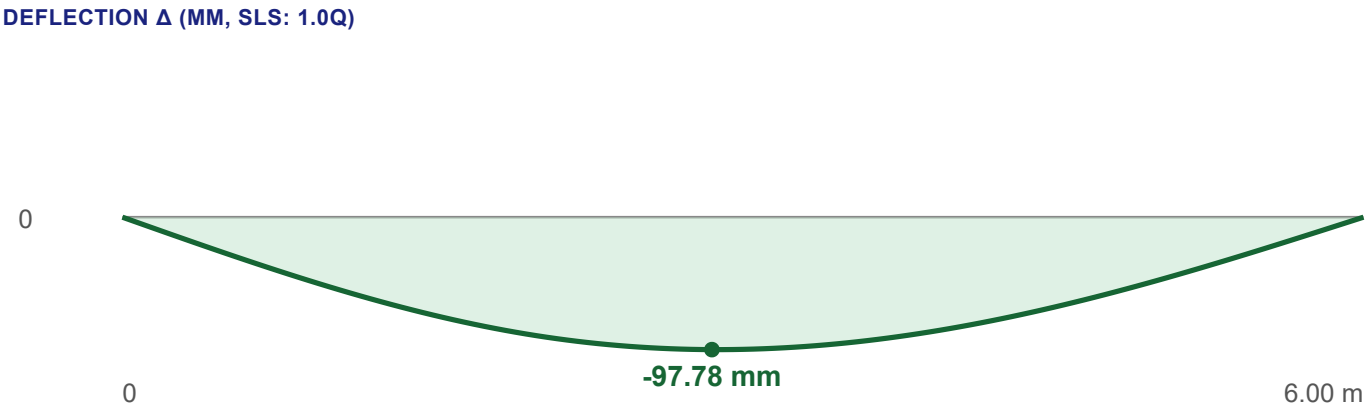
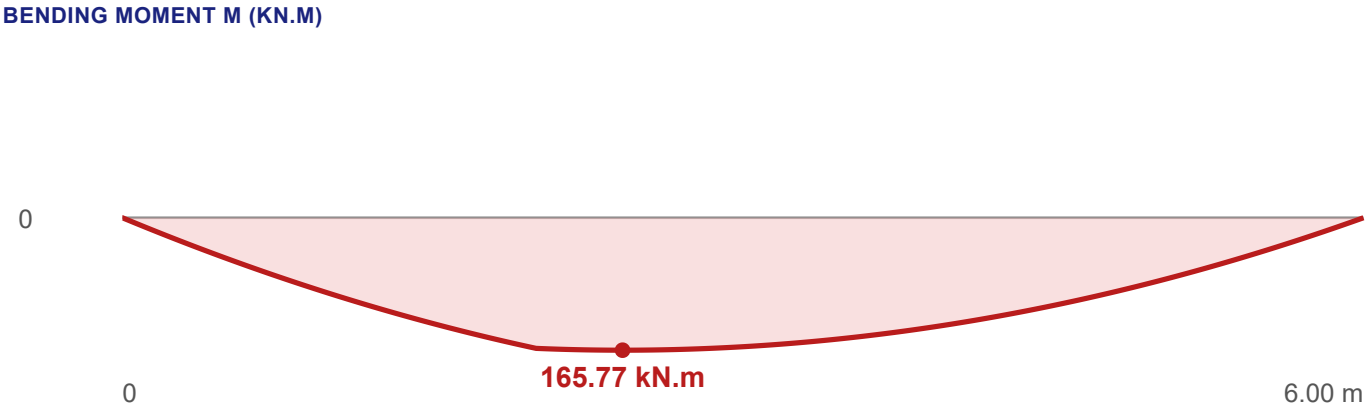
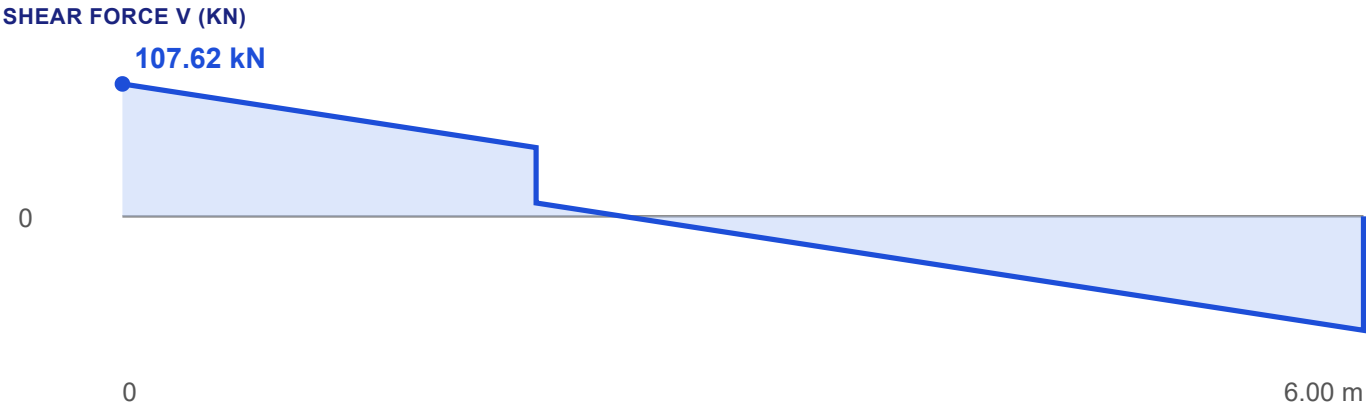
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, fy, N, My, Mz)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \times 1)$	1790 mm ² x 275 / ($\sqrt{3} \times 1$)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$d/t \leq 72 \epsilon / \eta$	20.81 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.058 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{m_1 + m_2}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{I_y \cdot t_w \cdot f_{yw} / F_{cr}}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.4, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = -75$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	$M_{y,Ed} / M_{b,Rd}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member SHS-L4 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)

G SW 0.2757 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6 0.00C		0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

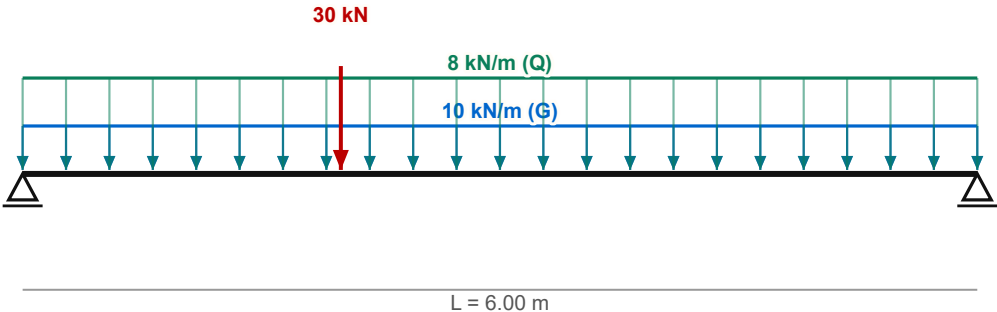
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

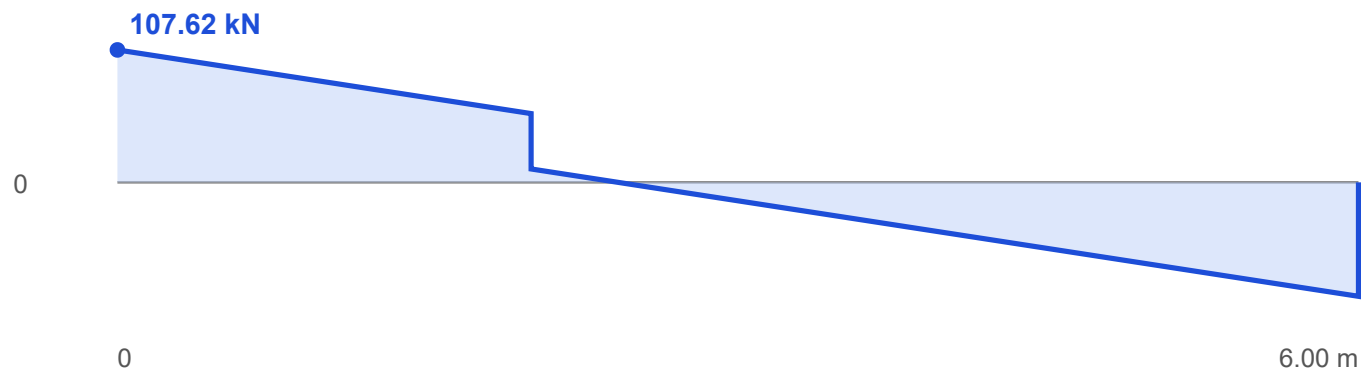
R_{min} ≥ 0

OK

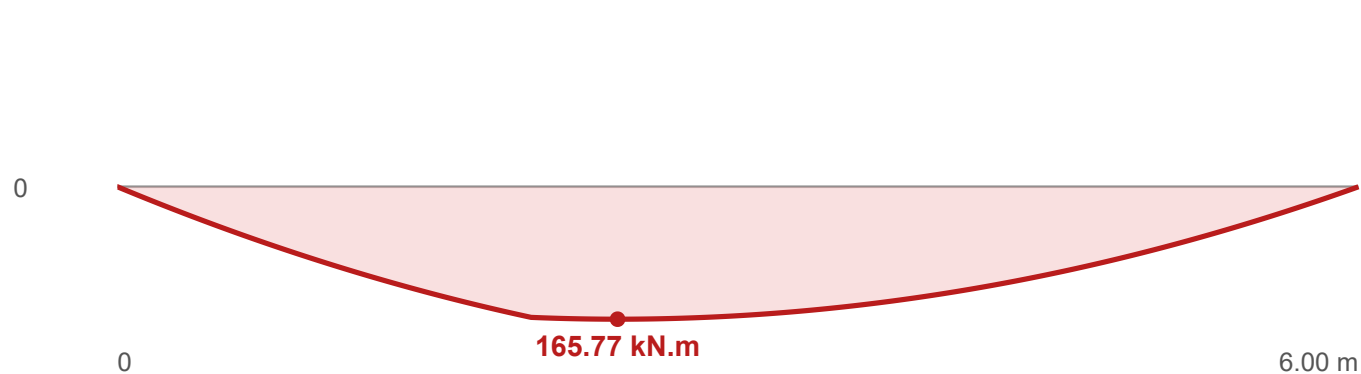
LOADING



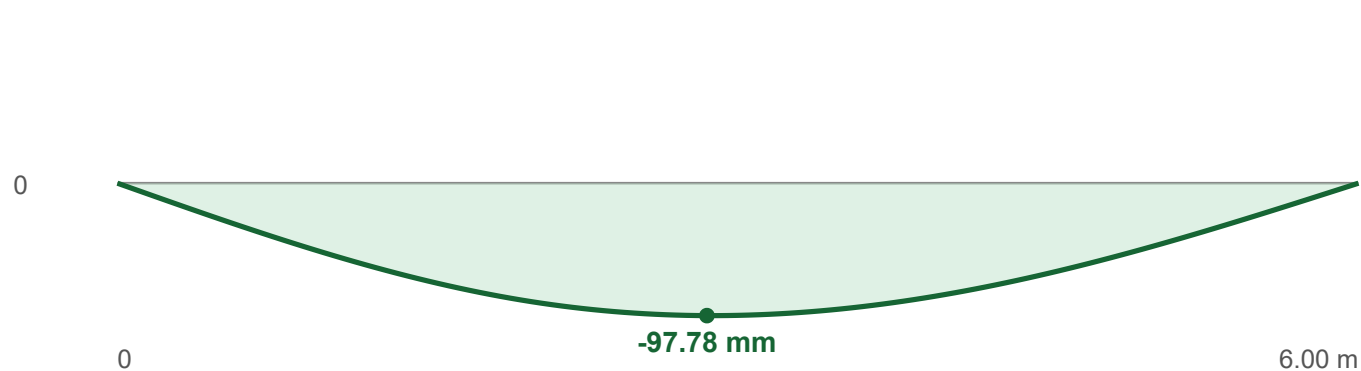
SHEAR FORCE V (kN)



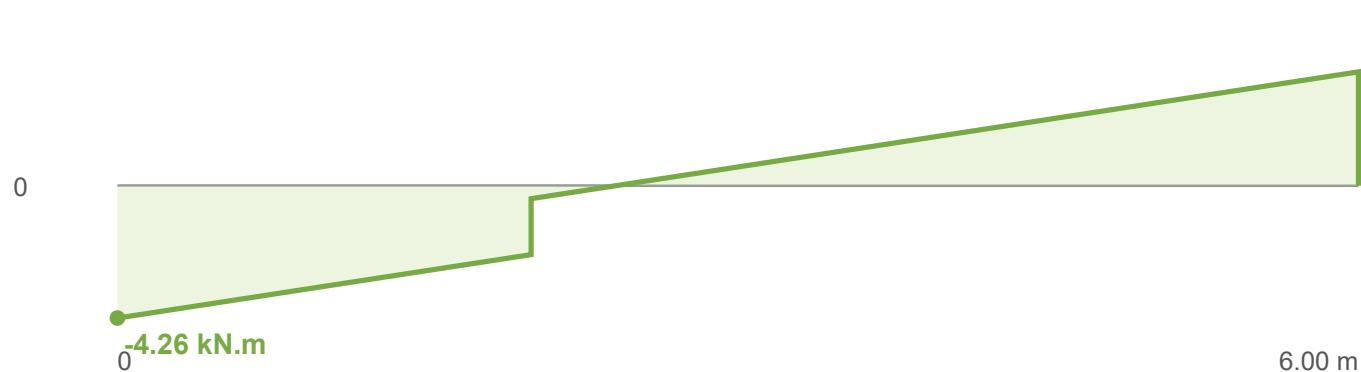
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150

B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 0.00, 165.77, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases 1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 1017.01 + 4.32 = 1021.33 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05 \text{ N/mm}^2$; $(1021.33/275)^2 + 3(0.05/275)^2 =$	13.793	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 1017.01 + 4.32 = 1021.33$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 13.793$ (governs); web mid-depth: $\sigma_x = 0.00 + 4.32 = 4.32$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 1017.01 = 1017.01$, $\tau = 0.03 \rightarrow 13.677$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00 \text{ mm}$ ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \times 10^3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40 \text{ mm}$); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.027	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

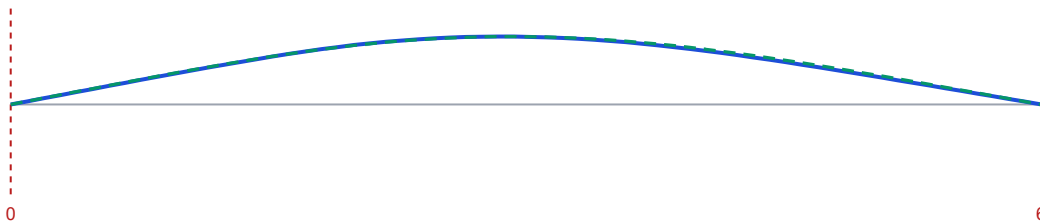
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k _c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	1162.306 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{192.0 \times 275 / 1162.306}$	0.213	
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.213 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed}/M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 38.105	0.112	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /240	17.75 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
(M _{y,Ed} /(M _{pl,y,Rd} ·S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} ·S _{mod})) ^β	T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t /(f _y /√3)]·V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 17.75 N/mm ² ; V _{pl,Rd} = 284.201; S _{mod} =	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	V _{pl,T,Rd} /V _{pl,Rd} = 0.888	0.426	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0019 rad = $0.11^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	13.793	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 13.793 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion

Member SHS-L4 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)

G SW 0.2757 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6 0.00C		0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

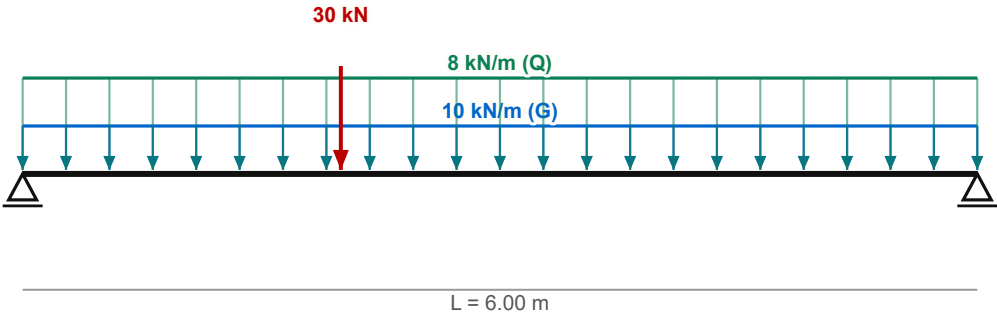
Uplift

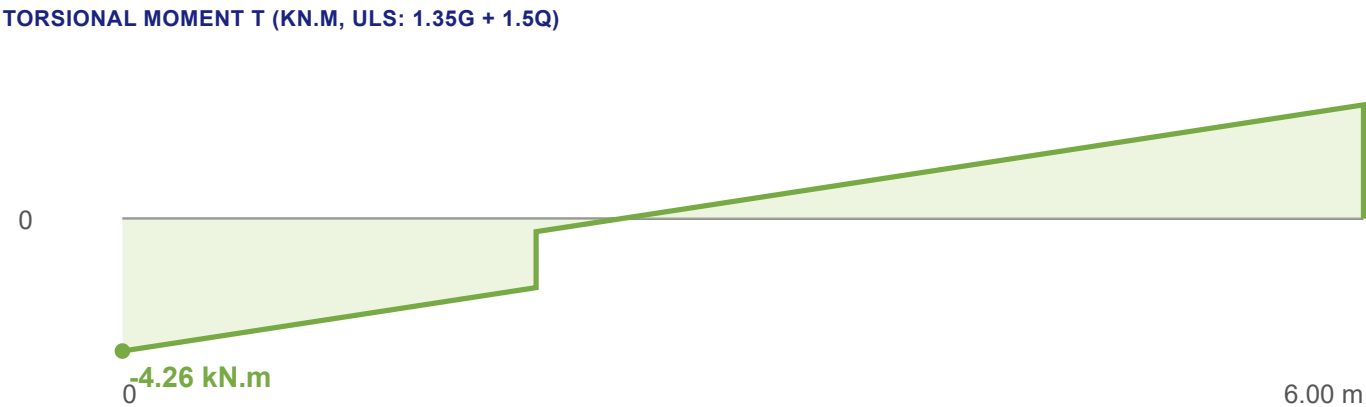
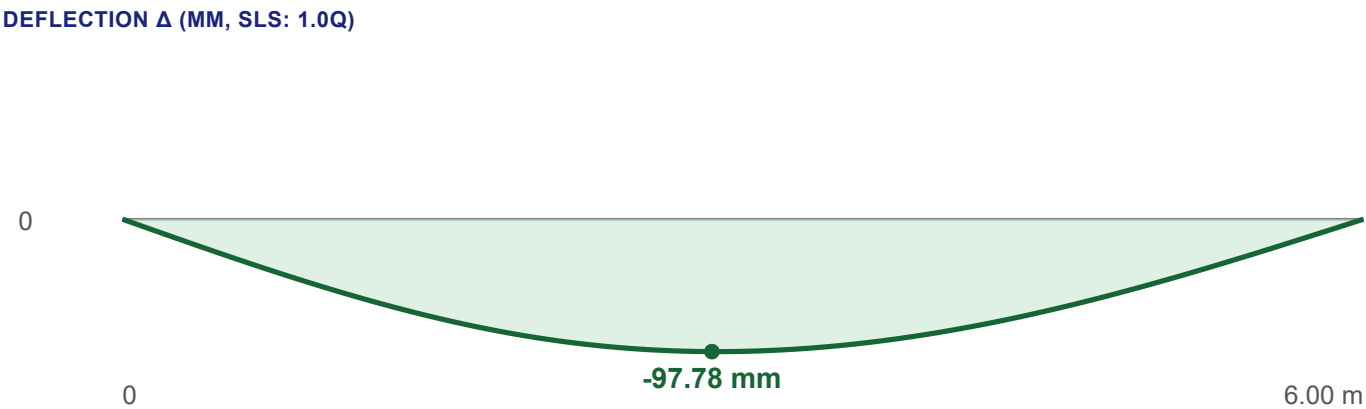
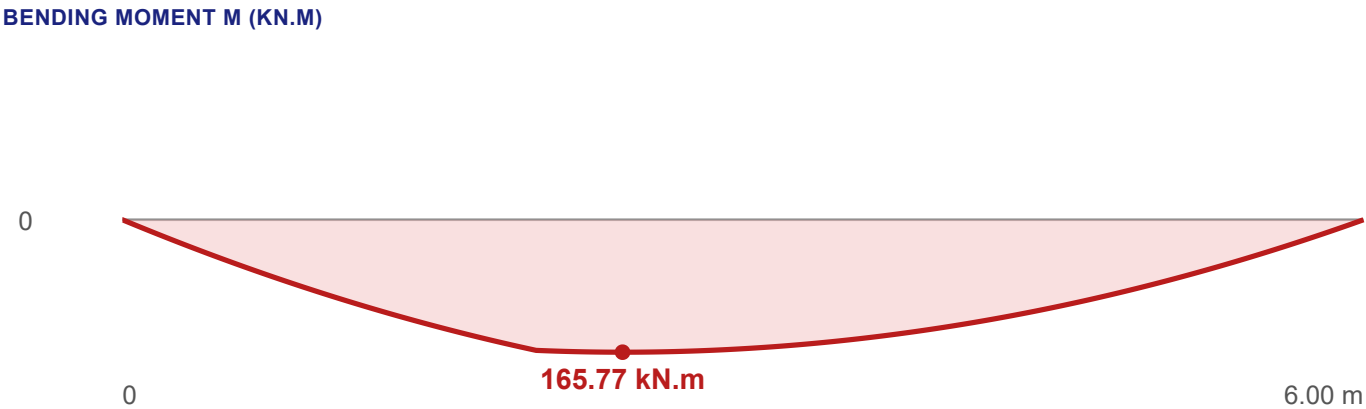
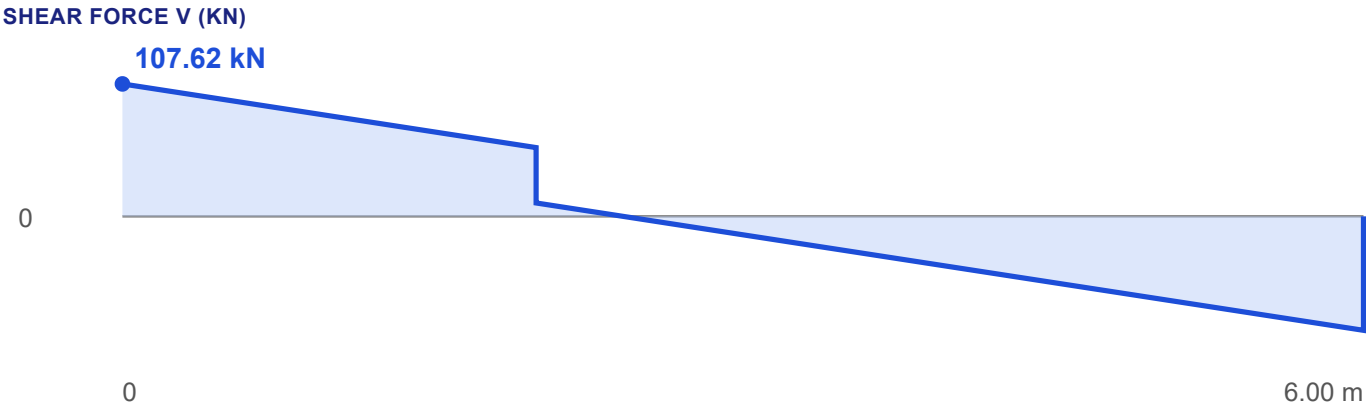
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150	
	B=150 t=6.3	
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 0.00, 165.77, 0.00	Class 1
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 0.00 + 1017.01 + 4.32 = 1021.33 \text{ N/mm}^2$; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05 \text{ N/mm}^2$; $(1021.33/275)^2 + 3(0.05/275)^2 =$	13.793	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 0.00 + 1017.01 + 4.32 = 1021.33$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 13.793$ (governs); web mid-depth: $\sigma_x = 0.00 + 4.32 = 4.32$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.000$; flange mid-width: $\sigma_x = 0.00 + 1017.01 = 1017.01$, $\tau = 0.03 \rightarrow 13.677$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00 \text{ mm}$ ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \text{e}3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40 \text{ mm}$); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK

$\eta_2 + 0.8\eta_1 \leq 1.4$
 $0.361 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 163.489/52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)
2.027
Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M₀, ψ, μ)
0.0, 0.0, 161.4, 1.000, 300.000
1.163
Serna

C₁ basis
general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.3, M(L/2)=161.4, M(3L/4)=109.8, M_{max}=165.8 kN·m
SN003a

Lateral Buckling Check M.b.Rd

M_{b,Rd} = M_{c,y,Rd}
closed hollow section — not susceptible to LTB (λ_{LT} = 0.209 ≤ 0.4)
52.800 kN.m
6.3.2.1(2)

M_{cr} (information, I_w = 0)
SN003a with C₁ = 1.163, L_e = 6 m; z_g = +75 mm entered — NOT applied: C₂ not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)
1212.847 kN.m
SN003a

M_{y,Ed}/M_{b,Rd}
165.773 / 52.800
3.140
Warning

Torsion Design

J, H, a, Q_f, Q_w
1910 cm⁴, — dm⁶, — mm, —, —
P385 App A

W_{n0}, S_{w1}
—, —
P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)
combination ULS: 1.35G + 1.5Q
4.260 kN.m

W_t (= C)
SCI P385 Table A.7/A.8; I_t = 1910 cm⁴
240 cm³

T_{Rd} = f_yW_t/(√3γ_{M0})
275 x 240/(√3 x 1)
38.105 kN.m
6.2.7(7)

T_{Ed}/T_{Rd}
4.260 / 38.105
0.112
OK

T_{t,Ed} = T_{Ed}/W_t
4.260 x 10³/240
17.75 N/mm²
6.2.7(7)

Modified Local Capacity (M_{y,Ed}/(M_{pl,y,Rd}·S_{mod}))^α + (M_{z,Ed}/(M_{pl,z,Rd}·S_{mod}))^β
n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd}, V_{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)
—
not evaluated

Torsion Shear Design @ 0 m

V_{pl,T,Rd} = [1 - τ_t/(f_y/√3)]·V_{pl,Rd}
@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ_t = 17.75 N/mm²; V_{pl,Rd} = 284.201; S_{mod} =
252.428 kN
6.2.7(9) Eq 6.28

V_{Ed}/V_{pl,T,Rd}
V_{pl,T,Rd}/V_{pl,Rd} = 0.888
107.617 / 252.428
0.426
OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360
97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m
97.78 mm
Warning

Torq in Case 1 @ 2 m: θ_{max} ≤ 2.00°
0.0019 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)
0.11°
OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	13.793	0.426	3.140

FAIL — governing Deflection = 5.867

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 13.793 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

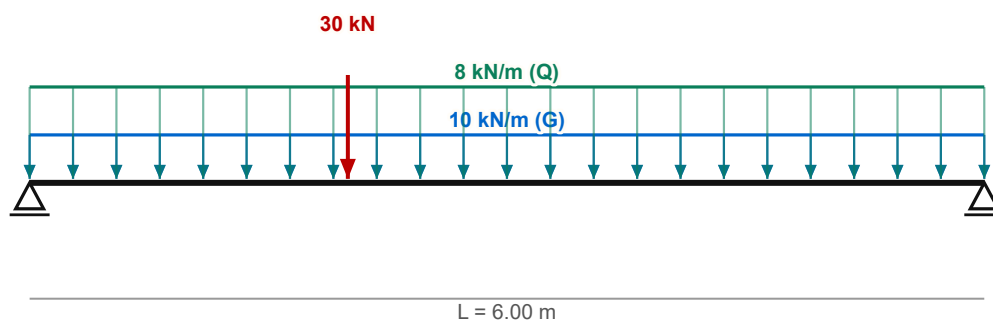
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN
value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all
loads at e = +40 mm, z_g = +75 mm

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

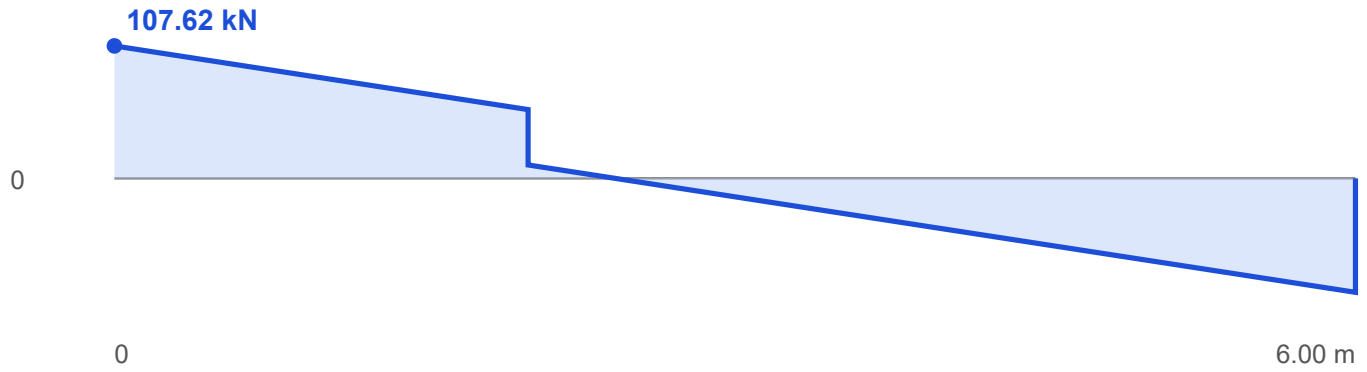
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	150.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

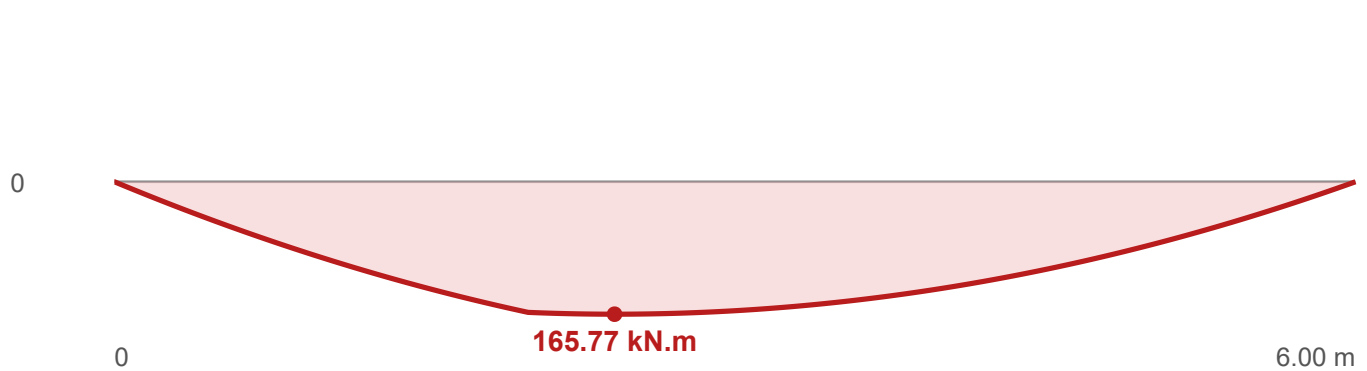
Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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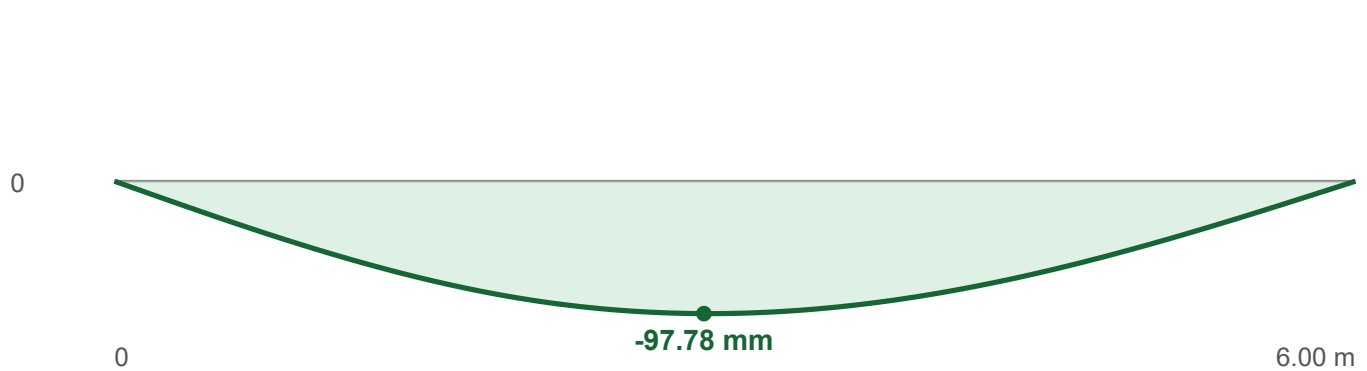
SHEAR FORCE V (kN)



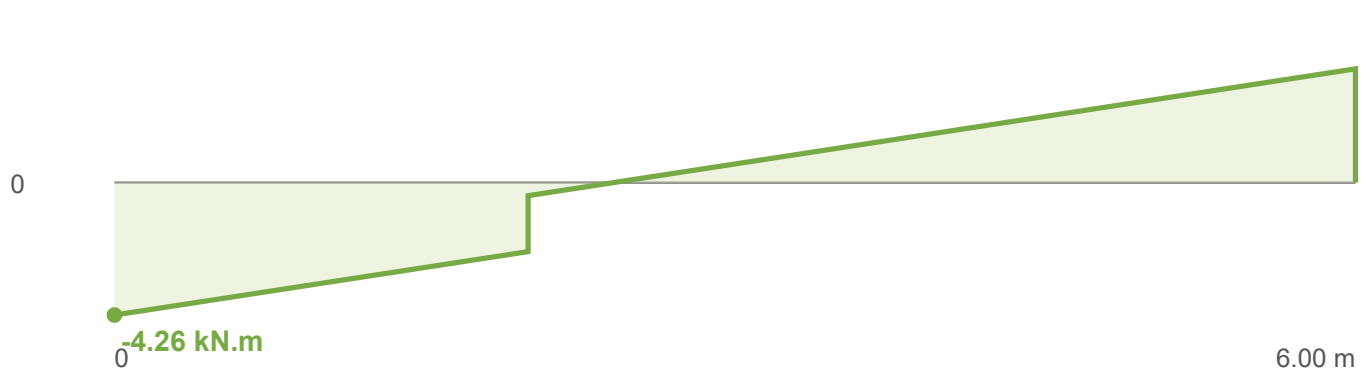
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	20.81, 20.81, 275, 150.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Web classified for	bending + compression: $\alpha = 0.665, \psi = -0.695$; limits 51.8ε / 59.6ε / 95.3ε	Class 1 Table 5.2

Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$35.80 \times 275 / 1 =$ (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 984.500 =$	0.152	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(165.773 / 52.800)^{1.7} + (0)^1 =$	7.031	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 41.90 + 1017.01 + 4.32 = 1063.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05$ N/mm ² ; $(1063.23 / 275)^2 + 3(0.05 / 275)^2 =$	14.948	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 41.90 + 1017.01 + 4.32 = 1063.23$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 14.948$ (governs); web mid-depth: $\sigma_x = 41.90 + 4.32 = 46.22$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.028$; flange mid-width: $\sigma_x = 41.90 + 1017.01 = 1058.91$, $\tau = 0.03 \rightarrow 14.827$ ($Q_c = 32.5$, $Q_m = 48.8$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00$ mm ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \times 10^3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 + N_{Ed} / N_{pl,Rd} = 150.000 / 984.500 = 3.249) = 2.960 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.114	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.229	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.114	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.209	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y ·L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A·f _y /N _{cr})	√(35.80x275/702.39 (N _{cr,y} = π²EI _y /L _{ey} ²))	1.181	
N _{b,y,Rd} = Area·χ·f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a
L _{ez} = K _z ·L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A·f _y /N _{crz})	√(35.80x275/702.39 (N _{cr,z} = π²EI _z /L _{ez} ²))	1.181	
N _{b,z,Rd} = Area·χ·f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a
L _{et} = K _t ·L _z ; λ _T ; N _{b,T,Rd}	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
N _{Ed} /N _{b,Rd}	150.000 / 533.439 = (min of N _{b,y,Rd} , N _{b,z,Rd})	0.281	OK

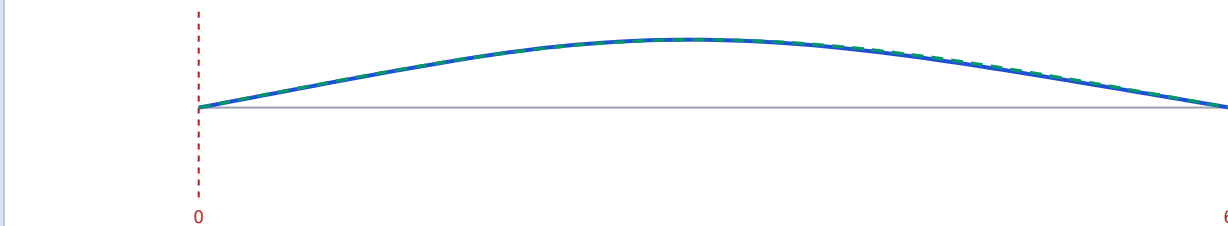
Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = M _{cr} /M _{cr,uniform}	1190.076 / 1042.431 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.3, M(L/2)=161.4, M(3L/4)=109.8, M _{max} =165.8 kN·m	1.163	Serna
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 161.42, ψ = 0.895, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 0.000 imposed + 0.704 twist φ. M _y = 0.704; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 161.42, ψ = 0.895, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 75, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	1162.306 kN.m	converged
λ _{LT} = √(W _y ·f _y /M _{cr})	√(192.0 x 275 / 1162.306)	0.213	
M _{b,Rd} = M _{c,y,Rd}	closed hollow section — not susceptible to LTB (λ _{LT} = 0.213 ≤ 0.4)	52.800 kN.m	6.3.2.1(2)
M _{y,Ed} /M _{b,Rd}	165.773 / 52.800	3.140	Warning

Buckled mode shape — critical eigenmode (normalised; twist φ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.84 m)



Buckling Resistance

U _{N,y} = N _{Ed} /(χ _y ·N _{Rk} /γ _{M1})	150.000 / 533.439 (N _{Rk} = 984.500)	0.281	OK
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$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	165.773 / 52.800 ($M_{y,Rk} = 52.800$)	3.140	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(0.000 imposed + 0.704 twist $\phi \cdot M_y = 0.704$) / 52.800	0.013	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6 x 1.225	0.735	Table B.1
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.281+1.225 \times 3.140+1.225 \times 0.013$	4.143	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$0.281+0.735 \times 3.140+1.225 \times 0.013$	2.605	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3 γ _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 38.105	0.112	OK
T _{t,Ed} = T _{Ed} /W _t	4.260 x 10 ³ /240	17.75 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} /(M _{pl,y,Rd} · S _{mod})) ^α + (M _{z,Ed} /(M _{pl,z,Rd} · S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} /T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
Basis with N _{Ed} / M _{z,Ed}	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with tau _t = T _{Ed} /W _t , the V _{Ed} shear flow and N/A + M _y /W _{el,y} + M _{z,tot} /W _{el,z} (information only: the Class 1/2 plastic route governs), T _{Ed} /T _{Rd} (6.2.7(1)), V _{Ed} /V _{pl,T,Rd} (6.2.7(9), Eq 6.2.8), 6.2.8(4)/6.2.9.1/6.2.10 with V _{pl,T,Rd} , cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} ; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - T _t /(f _y √3)] · V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, T _t = 17.75 N/mm ² ; V _{pl,Rd} = 284.201; S _{mod} =	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl,T,Rd}	V _{pl,T,Rd} /V _{pl,Rd} = 0.888 107.617 / 252.428	0.426	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
Torq in Case 1 @ 2 m: θ _{max} ≤ 2.00°	0.0019 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

N_Ed/N_(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	7.031	0.281	3.140	4.143	2.605	5.867	0.379	3.140	3.140	0.361	2.114	0.112	14.948	0.426	7.031

FAIL — governing Bending+axial cross-section (6.2.9) = 7.031

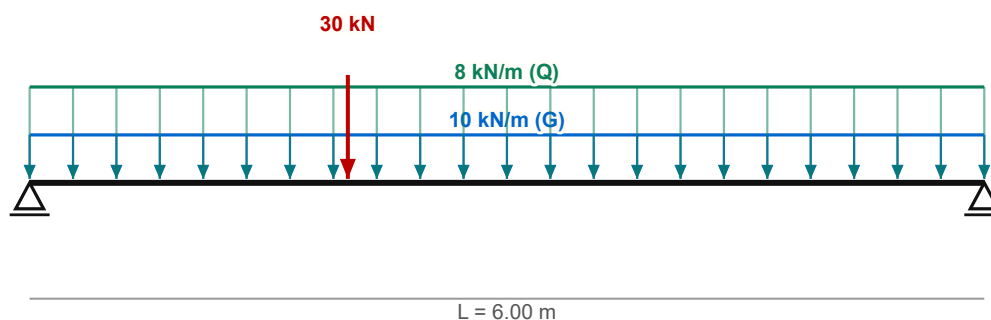
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

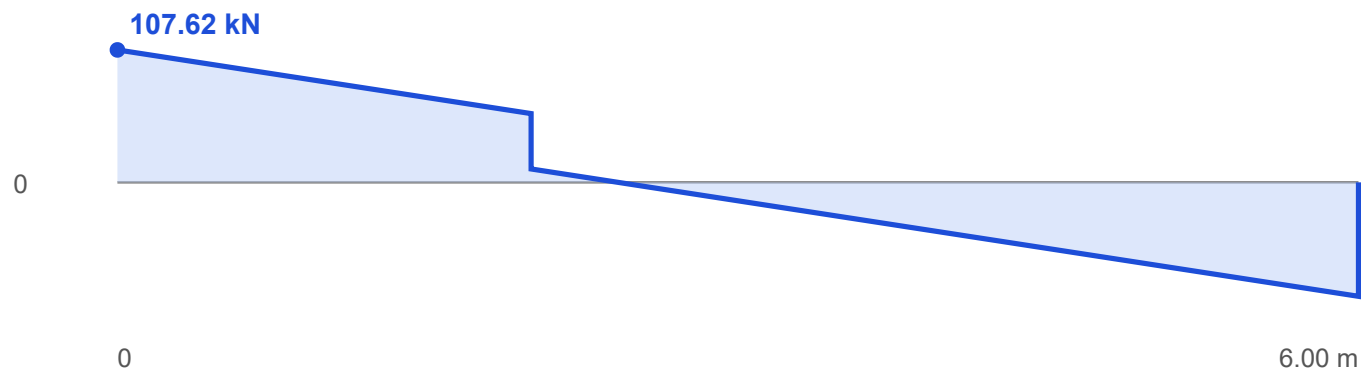
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 14.948 at x = 2.42 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd}

(6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

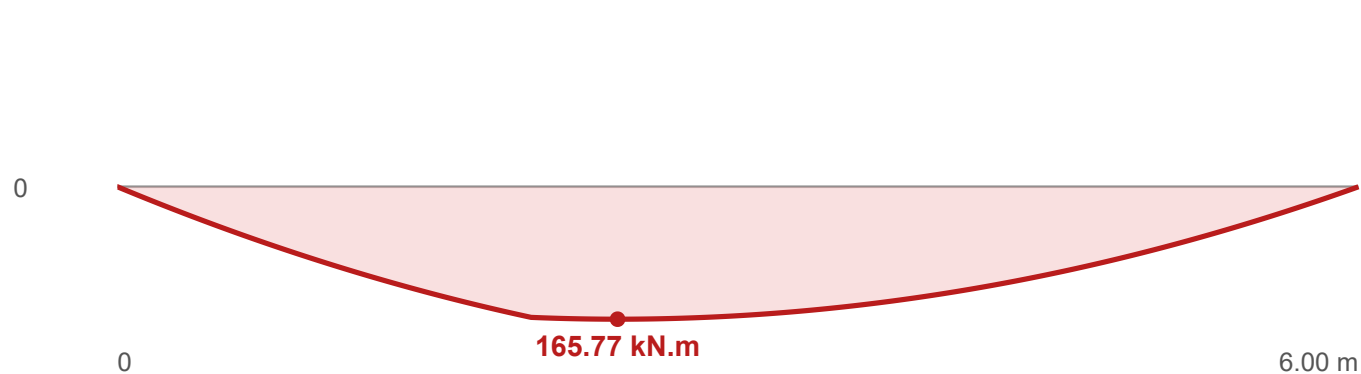
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN
value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all
loads at e = +40 mm, z_g = +75 mm



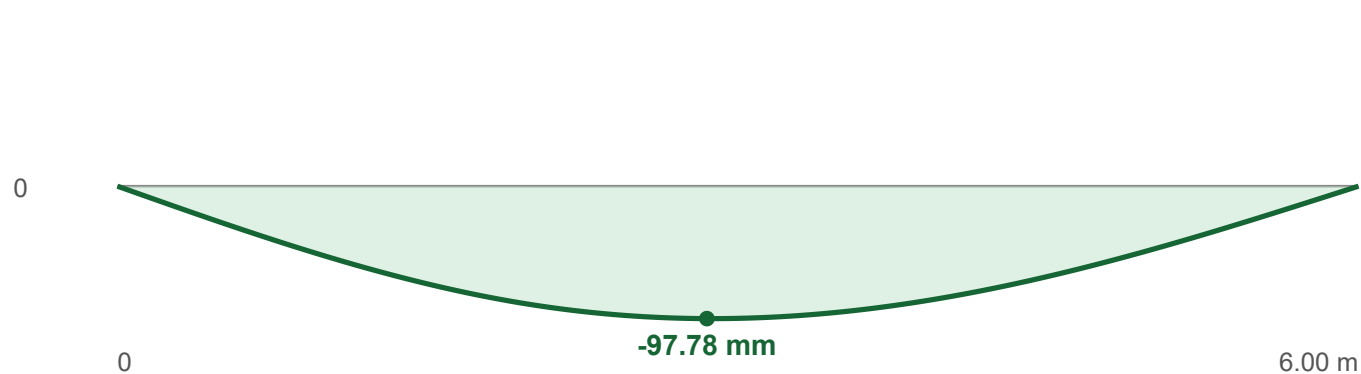
SHEAR FORCE V (kN)



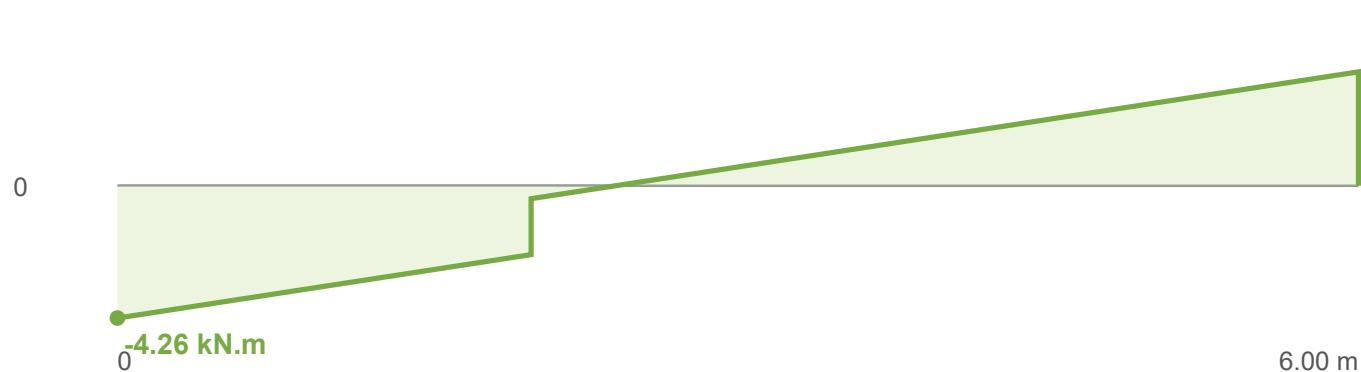
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	20.81, 20.81, 275, 150.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Web classified for	bending + compression: $\alpha = 0.665, \psi = -0.695$; limits 51.8ε / 59.6ε / 95.3ε	Class 1 Table 5.2

Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.058 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	Warning
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$35.80 \times 275 / 1 =$ (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 984.500 =$	0.152	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(165.773 / 52.800)^{1.7} + (0)^1 =$	7.031	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.418 m, corner	@ x = 2.418 m (ULS: 1.35G + 1.5Q): $M_y = 165.773$ kN.m, $V = 0.058$ kN, $T = 0.006$ kN.m, $\phi = 0.00424$ rad; $M_z = \phi M_y = 0.704$ kN.m; corner: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} = 41.90 + 1017.01 + 4.32 = 1063.23$ N/mm ² ; $\tau = T_{Ed} / W_t + V Q_c / (I_y t) = 0.03 + 0.02 = 0.05$ N/mm ² ; $(1063.23 / 275)^2 + 3(0.05 / 275)^2 =$	14.948	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 41.90 + 1017.01 + 4.32 = 1063.23$, $\tau = 0.03 + 0.02 = 0.05 \rightarrow 14.948$ (governs); web mid-depth: $\sigma_x = 41.90 + 4.32 = 46.22$, $\tau = 0.03 + 0.04 = 0.06 \rightarrow 0.028$; flange mid-width: $\sigma_x = 41.90 + 1017.01 = 1058.91$, $\tau = 0.03 \rightarrow 14.827$ ($Q_c = 32.5$, $Q_m = 48.8$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed} / T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed} / V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.		NOT VERIFIED
Web Transverse Forces (EN 1993-1-5 cl 6)			
Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00$ mm ($B/2 = 75.00 \leq t_w + 15 \epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2 t_f (1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222 \text{e}3)}$	0.212	6.4(1)

$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	0.5/0.212 = 2.359 $\rightarrow$ 1.000 ; 1.000 x 56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 56.07 x 6.30/1.0 per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	0.361 + 0.8 x ($M_{Ed}/M_{c,y,Rd} = 163.489/52.800 + N_{Ed}/N_{pl,Rd} = 150.000/984.500 = 3.249$) = 2.960 $\leq$ 1.4 (ULS: 1.35G + 1.5Q)	2.114	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.229	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.114	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.209	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	35.80 x 0.542 x 275 / 10/1 =	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	35.80 x 0.542 x 275 / 10/1 =	533.439 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed}/N_{b,Rd}$	150.000 / 533.439 = (min of $N_{b,y,Rd}$, $N_{b,z,Rd}$)	0.281	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.4, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 0.704 twist ϕ . $M_y = 0.704$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +75$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1212.847 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK

$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	165.773 / 52.800 ($M_{y,Rk} = 52.800$)	3.140	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	(0.000 imposed + 0.704 twist $\phi \cdot M_y = 0.704$) / 52.800	0.013	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my} \{1 + (\chi_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz} \{1 + (\chi_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6 x 1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	0.281 + 1.225 x 3.140 + 1.225 x 0.013	4.143	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	0.281 + 0.735 x 3.140 + 1.225 x 0.013	2.605	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t (= C)	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t / (√3 γ _{M0})	275 x 240 / (√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} / T _{Rd}	4.260 / 38.105	0.112	OK
T _{t,Ed} = T _{Ed} / W _t	4.260 x 10 ³ / 240	17.75 N/mm ²	6.2.7(7)
Modified Local Capacity (M _{y,Ed} / (M _{pl,y,Rd} · S _{mod})) ^α + (M _{z,Ed} / (M _{pl,z,Rd} · S _{mod})) ^β	n/a - not evaluated by beam-v03 (the cl 6.2.7 T _{Ed} / T _{Rd} , V _{pl,T,Rd} with the 6.2.8(4) ρ in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
Basis with N _{Ed} / M _{z,Ed}	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross- section with tau _t = T _{Ed} / W _t , the V _{Ed} shear flow and N/A + M _y / W _{el,y} + M _{z,tot} / W _{el,z} (information only: the Class 1/2 plastic route governs), T _{Ed} / T _{Rd} (6.2.7(1)), V _{Ed} / V _{pl,T,Rd} (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with V _{pl,T,Rd} , cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment phi.M _y added to M _{z,Ed} ; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

V _{pl,T,Rd} = [1 - τ _t / (f _y / √3)] · V _{pl,Rd}	@ x = 0 m (ULS: 1.35G + 1.5Q): T = 4.260 kN.m, τ _t = 17.75 N/mm ² ; V _{pl,Rd} = 284.201; S _{mod} = V _{pl,T,Rd} / V _{pl,Rd} = 0.888	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} / V _{pl,T,Rd}	107.617 / 252.428	0.426	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
Torq in Case 1 @ 2 m: θ _{max} ≤ 2.00°	0.0019 rad = 0.11° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.11°	OK

N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/V _{pl}	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	7.031	0.281	3.140	4.143	2.605	5.867	0.379	3.140	3.140	0.361	2.114	0.112	14.948	0.426	7.031

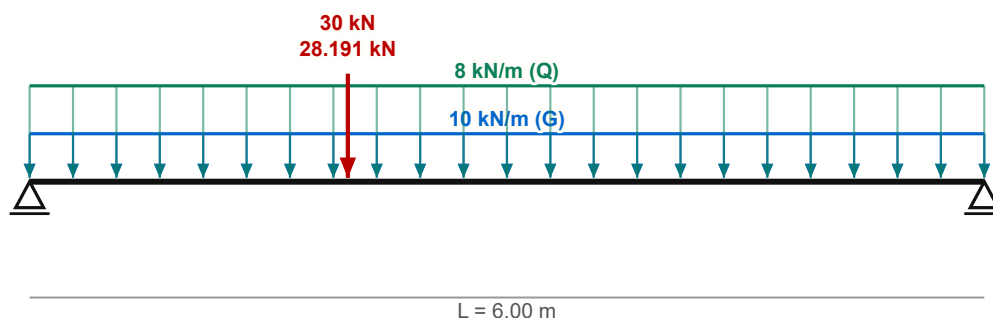
FAIL — governing Bending+axial cross-section (6.2.9) = 7.031

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

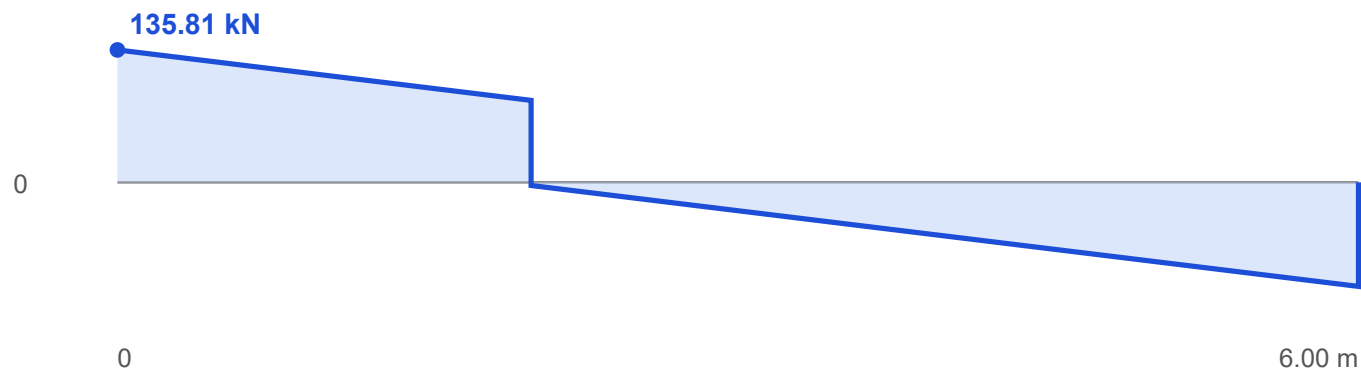
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

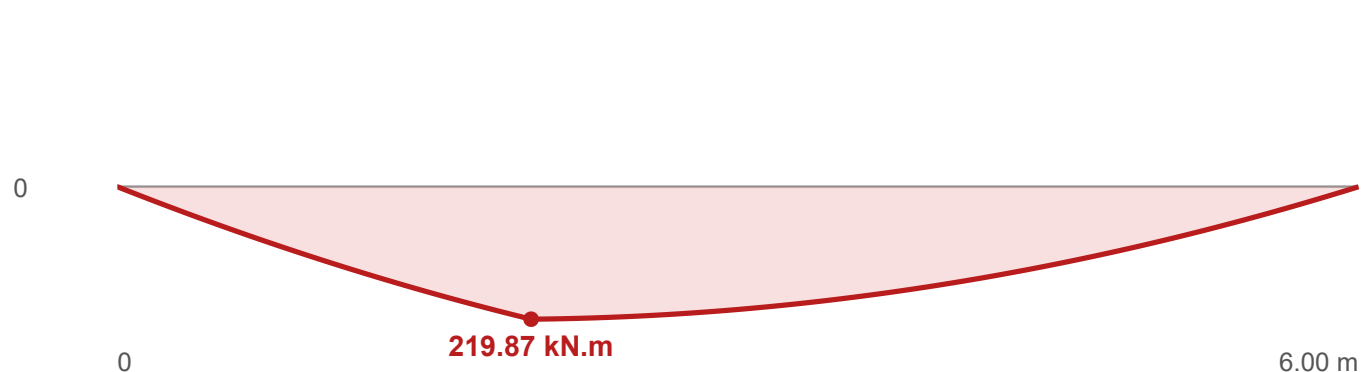
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 14.948 at $x = 2.42$ m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



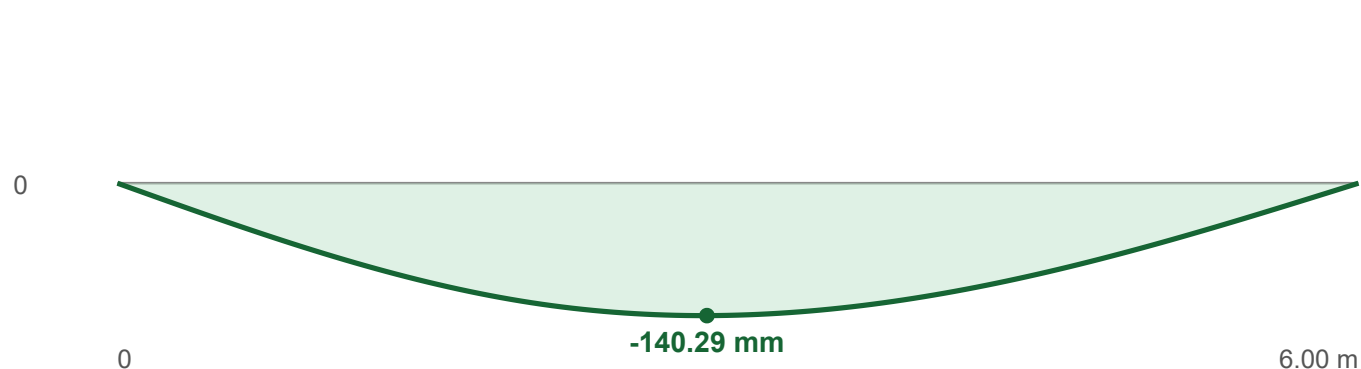
SHEAR FORCE V (kN)



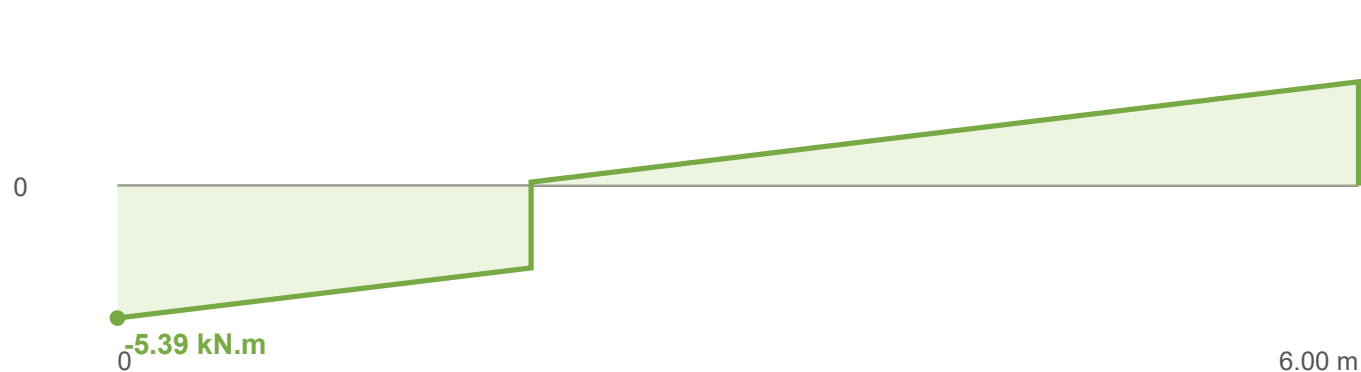
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150	
	B=150 t=6.3	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	20.81, 20.81, 275, 150.00, 219.87, 20.52	Class 1
	(Axial: Non-Slender)	
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$135.808 / 284.201 =$	0.478	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$3.223 / 284.201 =$	0.011	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$219.871 / 52.800 =$	4.164	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ x = 0.528 m (ULS: 1.35G + 1.5Q): V = 122.147, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 984.499$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,v,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 68.100$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	1.743	Warning 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 284.201 = (A_{v,z} = 1790 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$35.80 \times 275 / 1 =$ (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 984.500 =$	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(219.871 / 52.800)^{1.7} + (20.521 / 52.800)^{1.7} =$	11.579	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, corner	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 219.871$ kN.m, V = 84.063 kN, T = 3.348 kN.m, $\phi = 0.00565$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.241 = 21.762$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 41.90 + 1348.90 + 133.51 = 1524.31 \text{ N/mm}^2$; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 13.95 + 35.57 = 49.52 \text{ N/mm}^2$; $(1524.31/275)^2 + 3(49.52/275)^2 =$	30.822 > 1	information 6.2.7(5)
$\sigma_x, \tau, (6.1)$ at the 3 section points	corner: $\sigma_x = 41.90 + 1348.90 + 133.51 = 1524.31$, $\tau = 13.95 + 35.57 = 49.52 \rightarrow 30.822$ (governs); web mid-depth: $\sigma_x = 41.90 + 133.51 = 175.41$, $\tau = 13.95 + 53.36 = 67.31 \rightarrow 0.587$; flange mid-width: $\sigma_x = 41.90 + 1348.90 = 1390.80$, $\tau = 13.95 \rightarrow 25.586$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)	$f_y / \gamma_{M0} = 275$	
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$131.10, 6.30, 6.30, 75.00 \text{ mm}$ ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00 \text{ mm}$ (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)

$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40$ mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 219.871 / 52.800 + N_{Ed} / N_{pl,Rd} = 150.000 / 984.500 = 4.317) = 4.153 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.966	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	135.808	ULS: 1.35G + 1.5Q	0.251	0.266	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	2.966	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	106.712	ULS: 1.35G + 1.5Q	0.197	0.228	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y)$: held in position at both ends, in direction at neither: 1.0 L	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z)$: held in position at both ends, in direction at neither: 1.0 L	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 533.439 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.281	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

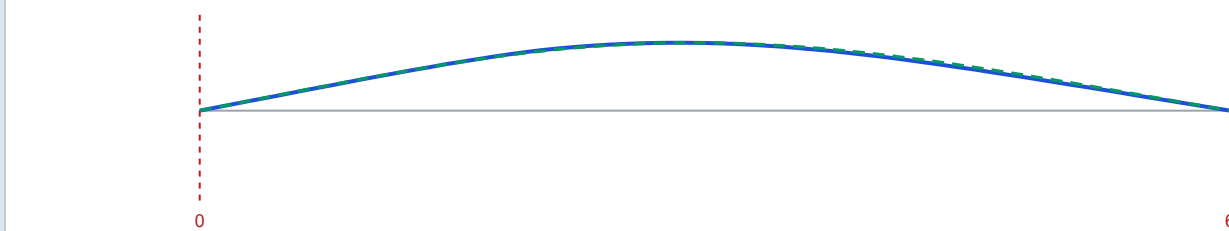
$C_1 = M_{cr} / M_{cr,uniform}$	$1237.844 / 1042.431$ kN.m (shape only: $z_g = z_j = 0$)	1.187	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 203.7, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.6$, $M(L/2)=203.7$, $M(3L/4)=131.0$, $M_{max}=219.9$ kN·m	1.218	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 1.241 twist $\phi \cdot M_y = 21.762$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_θ = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	1207.870 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1207.870}$	0.209	

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB (λ_{LT} = 0.209 ≤ 0.4)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed}/M_{b,Rd}$	219.871 / 52.800	4.164	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.74$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	219.871 / 52.800 ($M_{y,Rk} = 52.800$)	4.164	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.241 twist $\phi \cdot M_y = 21.762$) / 52.800	0.412	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.281 + 1.225 \times 4.164 + 1.225 \times 0.412$	5.887	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.281 + 0.735 \times 4.164 + 1.225 \times 0.412$	3.847	Warning

Torsion Design

J, H, a, Q_f , Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t (= C)	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³	
$T_{Rd} = f_y \cdot W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 240 / (\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$5.388 / 38.105$	0.141	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 240$	22.45 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
$(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 1.241 (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.762 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $\tau_t = 22.45$ N/mm ² ; $V_{pl,Rd} = 284.201$; $S_{mod} =$	244.018 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.859$ $135.808 / 244.018$	0.557	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$140.29 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	140.29 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0028 rad = $0.16^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.16°	OK
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N_Ed/N _{pl,Ed}	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/V _{pl}	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	11.579	0.281	4.164	5.887	3.847	8.417	0.478	4.164	4.164	1.743	0.699	2.966	0.141	30.822	0.557	11.579

FAIL — governing Biaxial bending + axial (6.2.9.1) = 11.579

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 30.822 at x = 2.00 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

SHS-L6 · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Biaxial bending + axial (6.2.9.1) = 11.579

Owner case: ... and axial load on beam, load applied at angle on beam
SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +75, factored by the combination) plus an entered CONSTANT M_{z,Ed} = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_{mz} = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.075 = 0.77 kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)

Includes Design for Torsion

Member SHS-L6 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

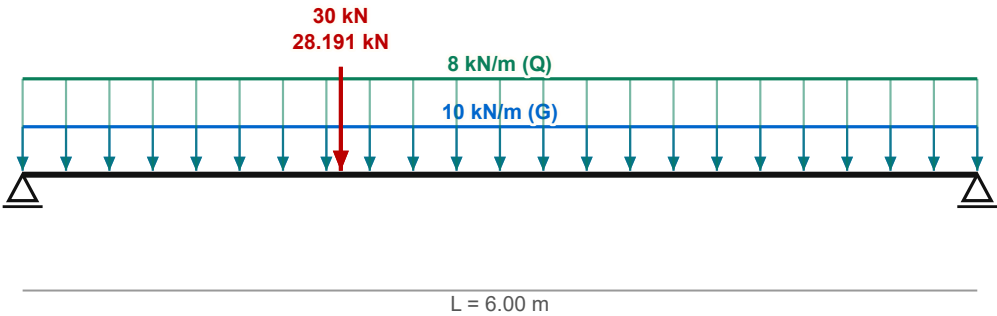
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)
Q PY 28.191 @ 2 m e = 40 mm z_g = 75 mm (kN)
G SW 0.2757 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-5.39	135.81	0.00	0.00	20.52	219.87	20.52	-140.29	135.81	—
	End 2, x = 6	150.00C	0.00	-106.71	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	106.71	—

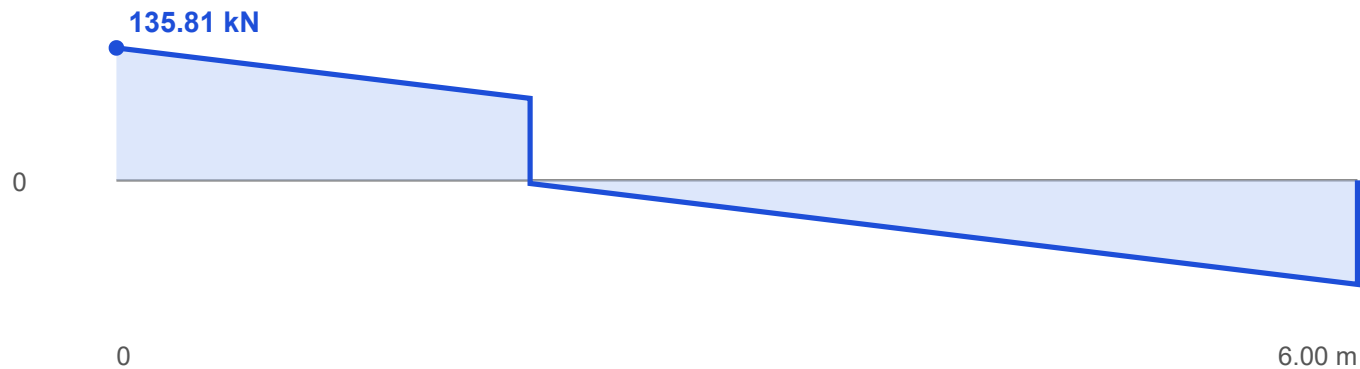
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

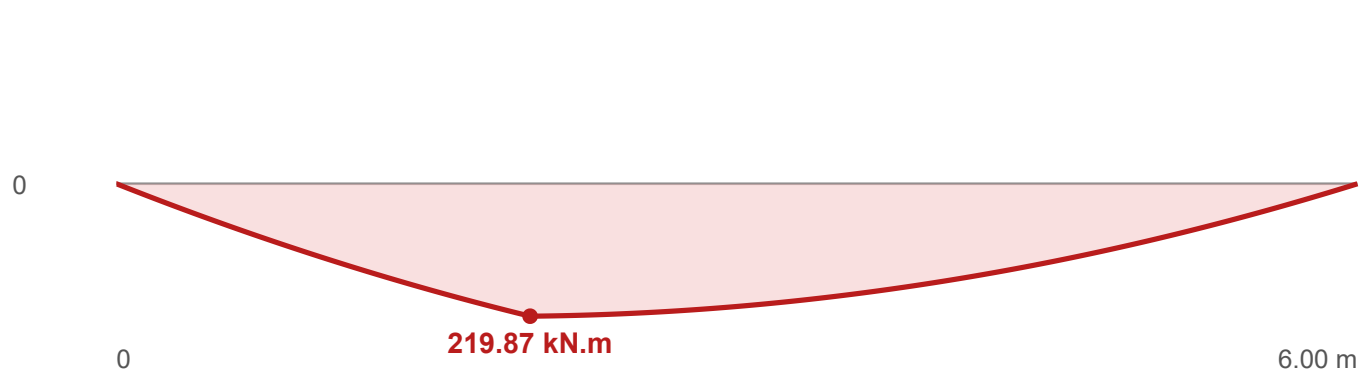
LOADING



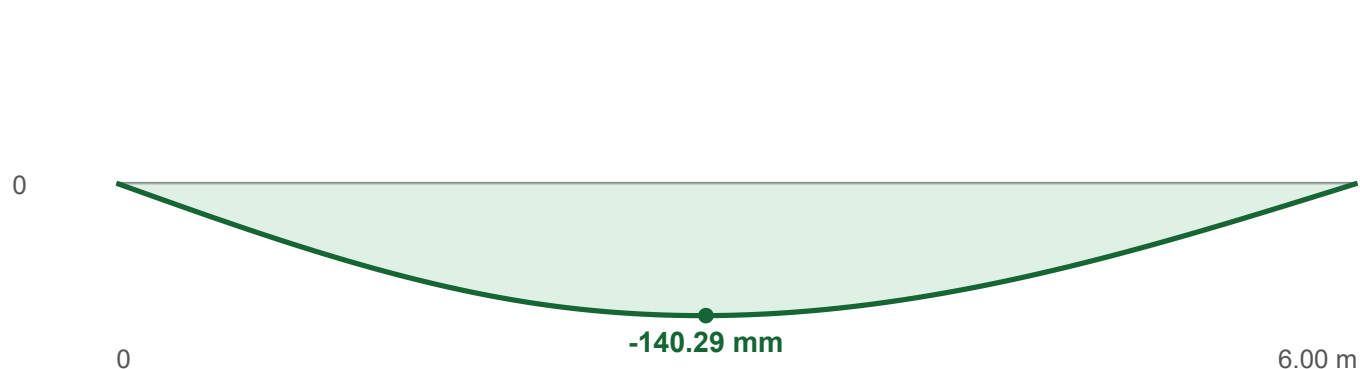
SHEAR FORCE V (kN)



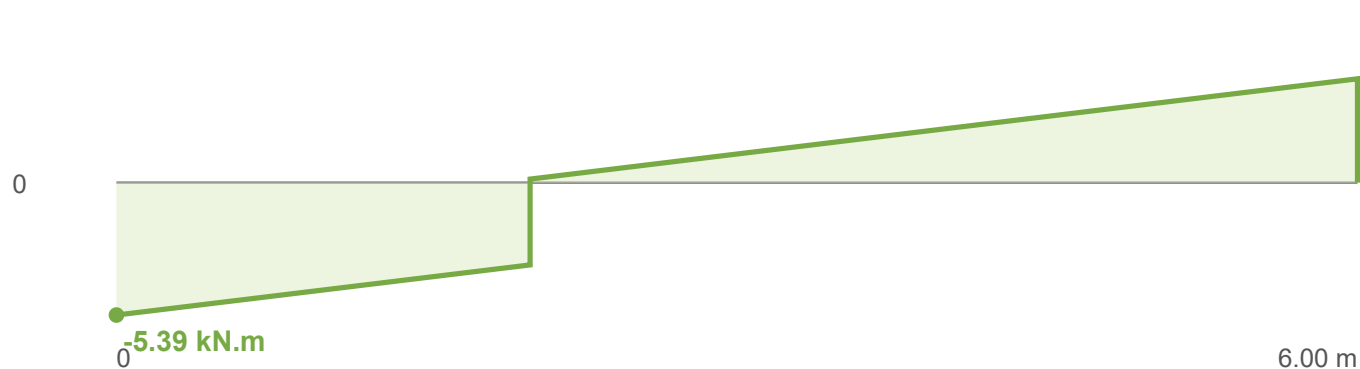
BENDING MOMENT M (kN.M)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	20.81, 20.81, 275, 150.00, 219.87, 20.52 (Axial: Non-Slender)	Class 1
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1790 mm ² x 275 / ($\sqrt{3} \times 1$)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	135.808 / 284.201 =	0.478	OK
$d/t \leq 72 \epsilon / \eta$	20.81 $\leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	3.223 / 284.201 =	0.011	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	219.871 / 52.800 =	4.164	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ x = 0.528 m (ULS: 1.35G + 1.5Q): V = 122.147, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 984.499$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,v,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 68.100$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	1.743	Warning 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0 / 1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275 / 1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275 / 1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	192.0 x 275 / 1	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(219.871 / 52.800)^{1.7} + (20.521 / 52.800)^{1.7} =$	11.579	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2 m, corner	@ x = 2 m (ULS: 1.35G + 1.5Q): $M_y = 219.871$ kN.m, $V = 84.063$ kN, $T = 3.348$ kN.m, $\phi = 0.00565$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.241 = 21.762$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 41.90 + 1348.90 + 133.51 = 1524.31$ N/mm ² ; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 13.95 + 35.57 = 49.52$ N/mm ² ; $(1524.31/275)^2 + 3(49.52/275)^2 =$	30.822 > 1	information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 41.90 + 1348.90 + 133.51 = 1524.31$, $\tau = 13.95 + 35.57 = 49.52 \rightarrow 30.822$ (governs); web mid-depth: $\sigma_x = 41.90 + 133.51 = 175.41$, $\tau = 13.95 + 53.36 = 67.31 \rightarrow 0.587$; flange mid-width: $\sigma_x = 41.90 + 1348.90 = 1390.80$, $\tau = 13.95 \rightarrow 25.586$ ($Q_c = 32.5$, $Q_m = 48.8$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 219.871 / 52.800 + N_{Ed} / N_{pl,Rd} = 150.000 / 984.500 = 4.317) = 4.153 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.966	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	135.808	ULS: 1.35G + 1.5Q	0.251	0.266	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	2.966	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	106.712	ULS: 1.35G + 1.5Q	0.197	0.228	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 1.0 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 1.0 =$	533.439 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 533.439 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.281	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 203.7, 1.000, 300.000	1.218	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.6$, $M(L/2)=203.7$, $M(3L/4)=131.0$, $M_{\max}=219.9$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3

$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521 \text{ imposed} + 1.241 \text{ twist } \phi.M_y = 21.762$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3
Lateral Buckling Check M.b.Rd			
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.204 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.218$, $L_e = 6 \text{ m}$; $z_g = +75 \text{ mm}$ entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1269.389 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	219.871 / 52.800	4.164	Warning
Buckling Resistance			
$U_{N,y} = N_{Ed}/(X_y.N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(X_z.N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(X_{LT}.M_{y,Rk}/\gamma_{M1})$	219.871 / 52.800 ($M_{y,Rk} = 52.800$)	4.164	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.241 twist $\phi.M_y = 21.762$) / 52.800	0.412	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2)0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2)0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny}+k_{yy}.U_{M,y}+k_{yz}.U_{M,z}$	$0.281+1.225 \times 4.164+1.225 \times 0.412$	5.887	Warning
$U_{Nz}+k_{zy}.U_{M,y}+k_{zz}.U_{M,z}$	$0.281+0.735 \times 4.164+1.225 \times 0.412$	3.847	Warning
Torsion Design			
J, H, a, Q_f, Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0}, S_{w1}	—, —		P385 App A
Torsion Bending Design @ 0 m			
$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
$W_t (= C)$	SCI P385 Table A.7/A.8; $I_t = 1910 \text{ cm}^4$	240 cm ³	
$T_{Rd} = f_y.W_t/(\sqrt{3}\gamma_{M0})$	275 x 240/($\sqrt{3} \times 1$)	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	5.388 / 38.105	0.141	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3/240$	22.45 N/mm ²	6.2.7(7)
Modified Local Capacity $(M_{y,Ed}/(M_{pl,y,Rd}.S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd}.S_{mod}))^\beta$	n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
$M_{z,tot} = M_{z,Ed} + \phi.M_{y,Ed}$	20.521 + 1.241 (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.762 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi.M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6
Torsion Shear Design @ 0 m			
$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})].V_{pl,Rd}$	@ x = 0 m (ULS: 1.35G + 1.5Q): $T = 5.388 \text{ kN.m}$, $\tau_t = 22.45 \text{ N/mm}^2$; $V_{pl,Rd} = 284.201$; $S_{mod} =$	244.018 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.859$ 135.808 / 244.018	0.557	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$140.29 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	140.29 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0028 \text{ rad} = 0.16^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.16°	OK

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/V _{pl}	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	11.579	0.281	4.164	5.887	3.847	8.417	0.478	4.164	4.164	1.743	0.699	2.966	0.141	30.822	0.557	11.579

FAIL — governing Biaxial bending + axial (6.2.9.1) = 11.579

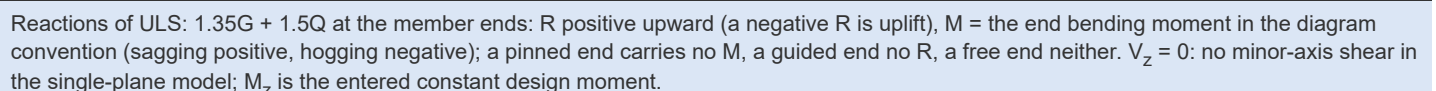
NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

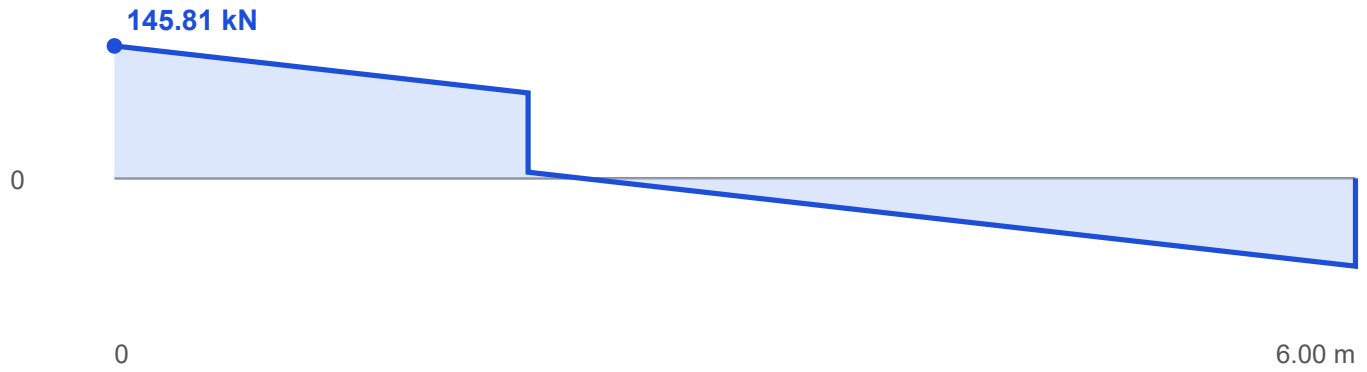
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

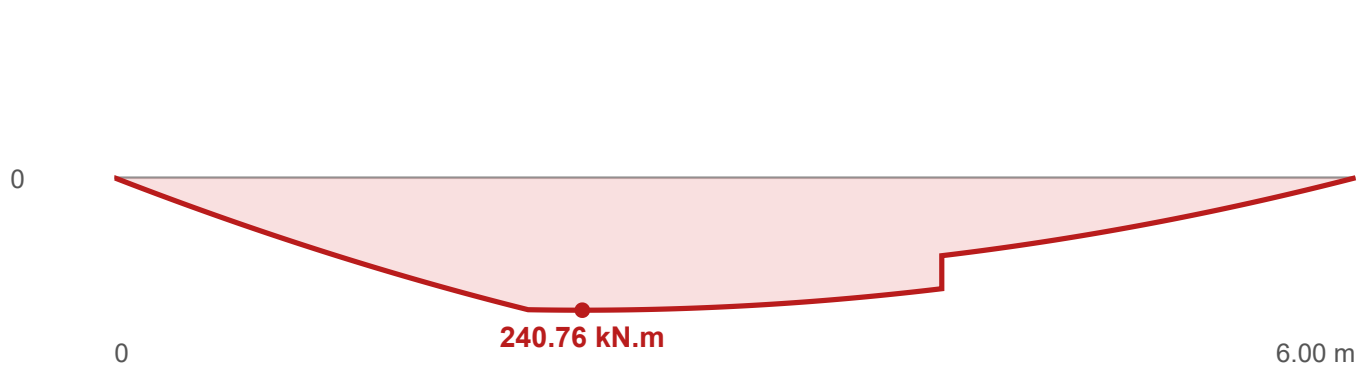
ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 30.822 at $x = 2.00 \text{ m}$ (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).



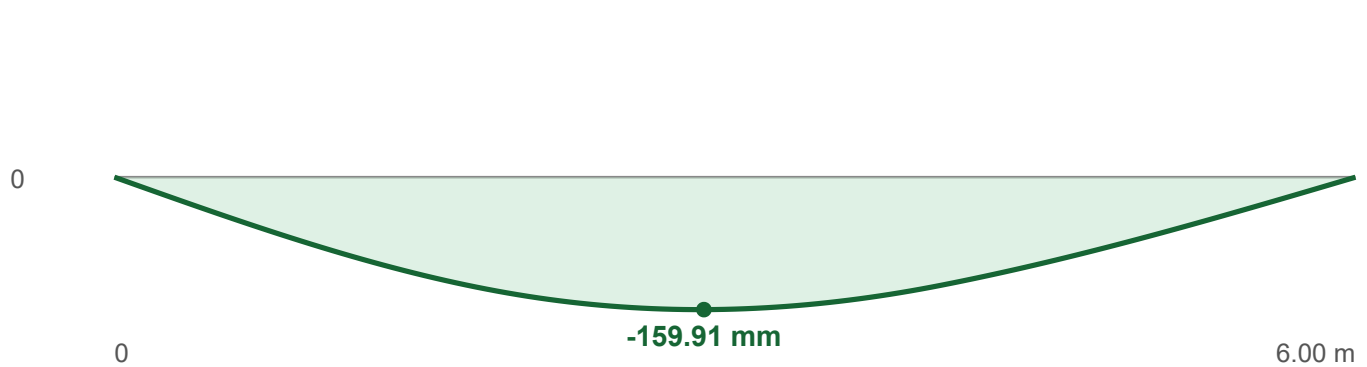
SHEAR FORCE V (kN)



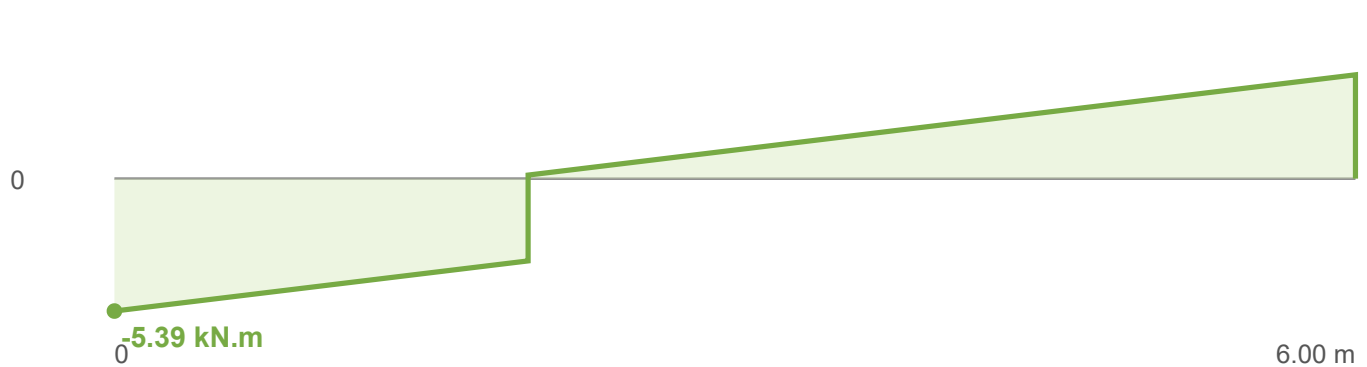
BENDING MOMENT M (kN.M)



DEFLECTION Δ (MM, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.M, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	20.81, 20.81, 275, 150.00, 240.76, 20.52 (Axial: Non-Slender)	Class 1
Web classified for	biaxial: uniform-compression web bound; limits $33\epsilon / 38\epsilon / 42\epsilon$	Class 1 Table 5.2

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1790 mm ² x 275 / ($\sqrt{3}$ x 1)	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	145.808 / 284.201 =	0.513	OK
$d/t \leq 72 \epsilon / \eta$	20.81 ≤ 72 x 0.924/1 = 66.56	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.002 / 284.201 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	240.758 / 52.800 =	4.560	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$ @ x	@ x = 0.918 m (ULS: 1.35G + 1.5Q): V = 122.057, ρ = 0.000; $N_{V,Rd} = (A - \rho A_v) f_y = 984.500$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,v,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 122.950$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	4.424	Warning 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(240.758/52.800)^{1.7} + (20.521/52.800)^{1.7} =$	13.483	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.262 m, corner	@ x = 2.262 m (ULS: 1.35G + 1.5Q): $M_y = 240.758$ kN.m, V = 0.002 kN, T = 0.411 kN.m, $\phi = 0.00560$ rad; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.348 = 21.869$ kN.m; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 41.90 + 1477.04 + 134.17 = 1653.11$ N/mm ² ; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 1.71 + 0.00 = 1.71$ N/mm ² ; $(1653.11/275)^2 + 3(1.71/275)^2 =$	36.136	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 41.90 + 1477.04 + 134.17 = 1653.11$, $\tau = 1.71 + 0.00 = 1.71 \rightarrow 36.136$ (governs); web mid-depth: $\sigma_x = 41.90 + 134.17 = 176.07$, $\tau = 1.71 + 0.00 = 1.71 \rightarrow 0.410$; flange mid-width: $\sigma_x = 41.90 + 1477.04 = 1518.94$, $\tau = 1.71 \rightarrow 30.508$ ($Q_c = 32.5$, $Q_m = 48.8$ cm ³ : the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M _y -M _z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)

$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40$ mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 239.871 / 52.800 + N_{Ed} / N_{pl,Rd} = 150.000 / 984.500 = 4.695) = 4.456 \leq 1.4$ (ULS: 1.35G + 1.5Q)	3.183	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	145.808	ULS: 1.35G + 1.5Q	0.270	0.280	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	3.183	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	96.712	ULS: 1.35G + 1.5Q	0.179	0.215	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	$1 \times 6 = (y-y)$: held in position at both ends, in direction at neither: 1.0 L	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z)$: held in position at both ends, in direction at neither: 1.0 L	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 533.439 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.281	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

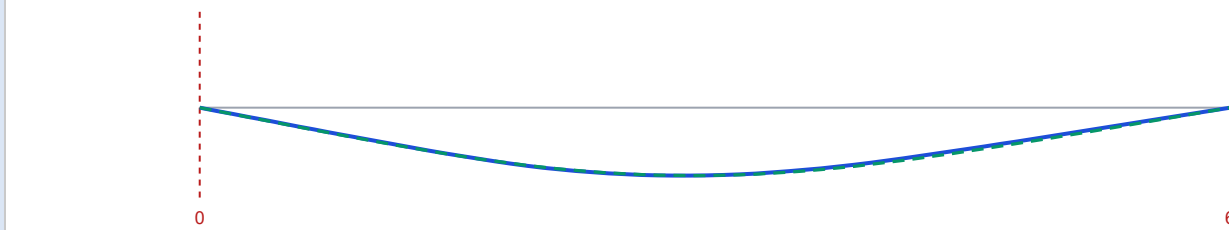
$C_1 = M_{cr} / M_{cr,uniform}$	$1213.717 / 1042.431$ kN.m (shape only: $z_g = z_j = 0$)	1.164	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 233.7, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.6$, $M(L/2)=233.7$, $M(3L/4)=116.0$, $M_{max}=240.8$ kN·m	1.214	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 20.521$ imposed + 1.355 twist $\phi \cdot M_y = 21.876$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 3 solve(s)	1187.543 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1187.543}$	0.211	

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.211 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
$M_{y,Ed}/M_{b,Rd}$	240.758 / 52.800	4.560	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.86$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	240.758 / 52.800 ($M_{y,Rk} = 52.800$)	4.560	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.355 twist $\phi \cdot M_y = 21.876$) / 52.800	0.414	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2) \times 0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zy}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.281 + 1.225 \times 4.560 + 1.225 \times 0.414$	6.374	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.281 + 0.735 \times 4.560 + 1.225 \times 0.414$	4.140	Warning

Torsion Design

J, H, a, Q_f , Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t (= C)	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³	
$T_{Rd} = f_y \cdot W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 240 / (\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$5.388 / 38.105$	0.141	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3 / 240$	22.45 N/mm ²	6.2.7(7)
Modified Local Capacity	n/a - not evaluated by beam-v03 (the cl 6.2.7	—	not evaluated
$(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha + (M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)		
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 1.355$ (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.876 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6

Torsion Shear Design @ 0 m

$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T = 5.388$ kN.m, $\tau_t = 22.45$ N/mm ² ; $V_{pl,Rd} = 284.201$; $S_{mod} =$	244.018 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$V_{pl,T,Rd}/V_{pl,Rd} = 0.859$ $145.808 / 244.018$	0.598	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$159.91 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	159.91 mm	Warning
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Torq in Case 1 @ 2 m: $\theta_{\max} \leq 2.00^\circ$	0.0028 rad = $0.16^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.16°	OK
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N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	13.483	0.281	4.560	6.374	4.140	9.595	0.513	4.560	4.560	4.424	0.699	3.183	0.141	36.136	0.598	13.483

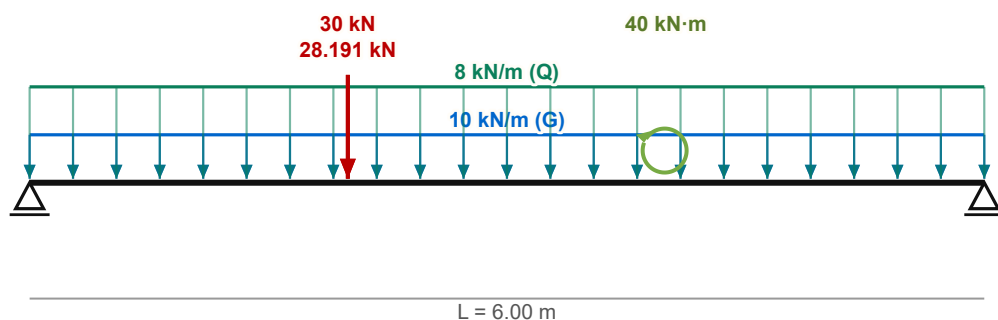
FAIL — governing Biaxial bending + axial (6.2.9.1) = 13.483

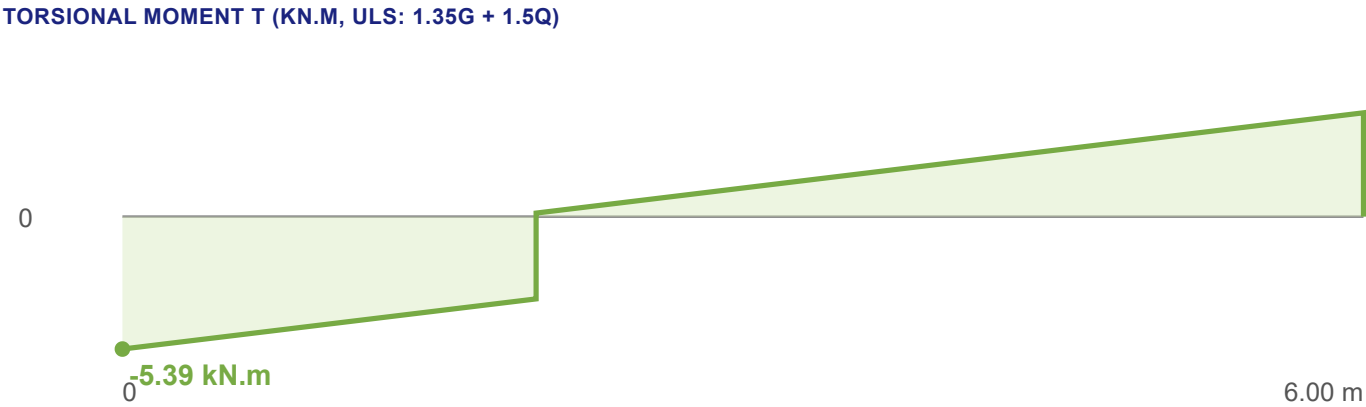
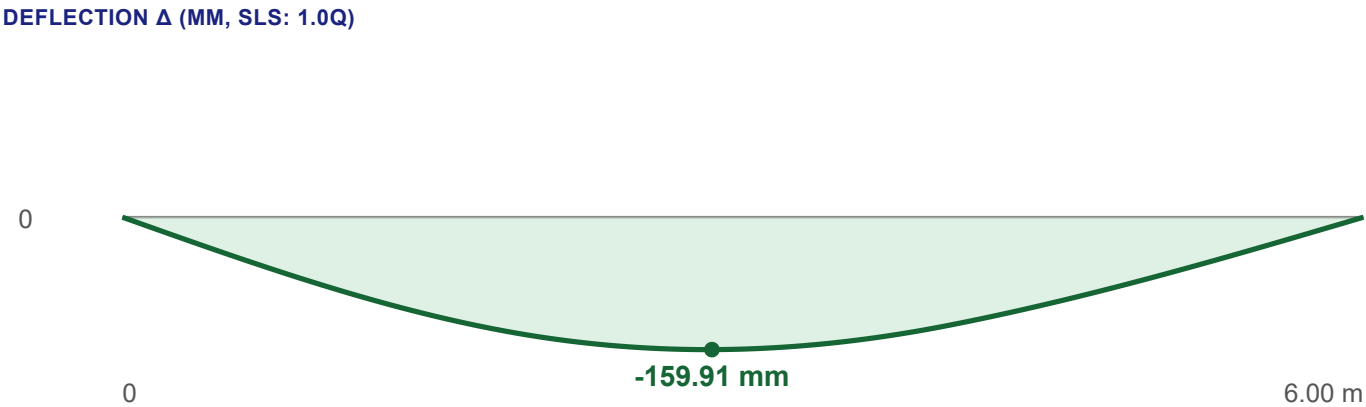
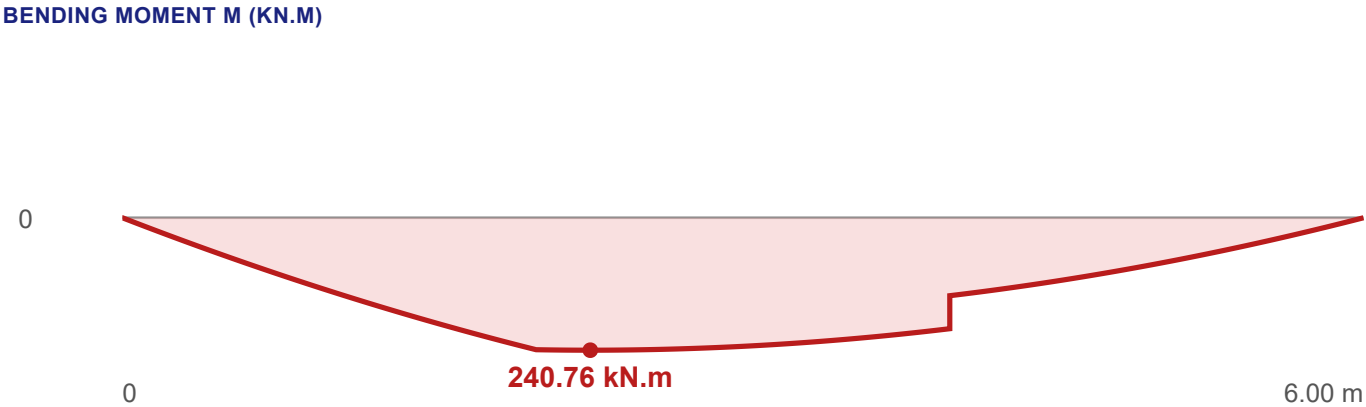
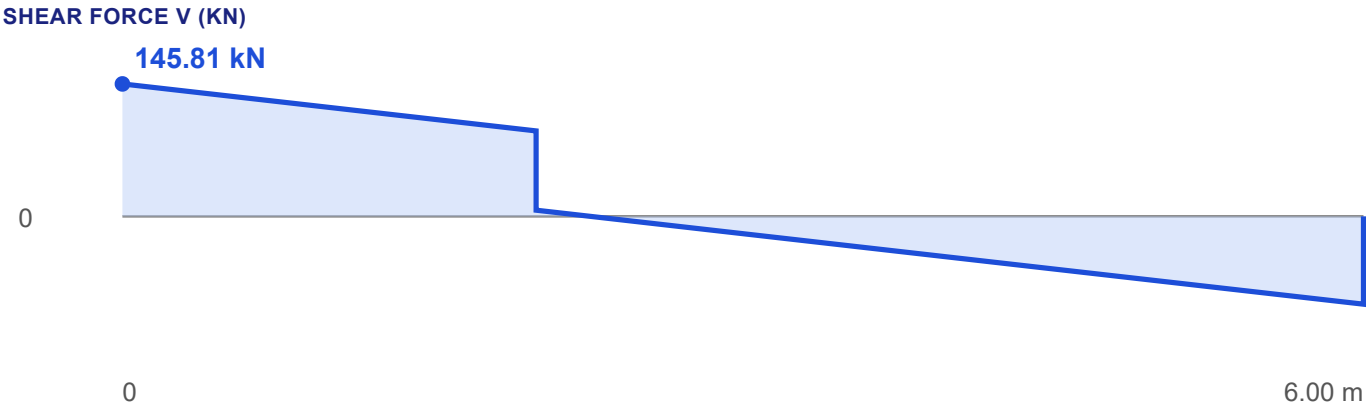
ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 36.136 at x = 2.26 m (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), rho from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150			
	B=150 t=6.3			
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 150.00, 240.76, 20.52			Class 1
	(Axial: Non-Slender)			
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε		Class 1	Table 5.2

Auto Design Load Cases	1; SLS 1		
Shear Capacity Check			
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$145.808 / 284.201 =$	0.513	OK
$d/t \leq 72 \epsilon / \eta$	$20.81 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)
Local Capacity Check			
$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.002 / 284.201 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$240.758 / 52.800 =$	4.560	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ $x = 0.918 \text{ m}$ (ULS: $1.35G + 1.5Q$): $V = 122.057$, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 984.500$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800 ; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800 ; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,v,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 122.950$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	4.424	Warning 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 284.201 = (A_{v,z} = 1790 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$35.80 \times 275 / 1 = (\text{No bearing / block tearing design})$	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 984.500 =$	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$192.0, 17.900, 0.152$	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	$192.0 \times 275 / 1$	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(240.758 / 52.800)^{1.7} + (20.521 / 52.800)^{1.7} =$	13.483	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.262 m, corner	@ $x = 2.262 \text{ m}$ (ULS: $1.35G + 1.5Q$): $M_y = 240.758 \text{ kN.m}$, $V = 0.002 \text{ kN}$, $T = 0.411 \text{ kN.m}$, $\phi = 0.00560 \text{ rad}$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 1.348 = 21.869 \text{ kN.m}$; corner: $\sigma_x = N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z} = 41.90 + 1477.04 + 134.17 = 1653.11 \text{ N/mm}^2$; $\tau = T_{Ed}/W_t + VQ_c/(I_y t) = 1.71 + 0.00 = 1.71 \text{ N/mm}^2$; $(1653.11/275)^2 + 3(1.71/275)^2 =$	36.136	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 3 section points	corner: $\sigma_x = 41.90 + 1477.04 + 134.17 = 1653.11$, $\tau = 1.71 + 0.00 = 1.71 \rightarrow 36.136$ (governs); web mid-depth: $\sigma_x = 41.90 + 134.17 = 176.07$, $\tau = 1.71 + 0.00 = 1.71 \rightarrow 0.410$; flange mid-width: $\sigma_x = 41.90 + 1477.04 = 1518.94$, $\tau = 1.71 \rightarrow 30.508$ ($Q_c = 32.5$, $Q_m = 48.8 \text{ cm}^3$: the V_{Ed} shear flow round the closed mid-line, Eq 6.20)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1)/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 $N-M_y-M_z$ interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.		NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm ($d = 2000.00$ to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40$ mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 239.871 / 52.800 + N_{Ed} / N_{pl,Rd} = 150.000 / 984.500 = 4.695) = 4.456 \leq 1.4$ (ULS: 1.35G + 1.5Q)	3.183	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	145.808	ULS: 1.35G + 1.5Q	0.270	0.280	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	3.183	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	96.712	ULS: 1.35G + 1.5Q	0.179	0.215	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 1.0 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 1.0 =$	533.439 kN	Curve a
$L_{et} = K_t \cdot L_z$; λ_T ; $N_{b,T,Rd}$	n/a - not evaluated by beam-v03 (cl 6.3.1.4 torsional / torsional-flexural buckling is evaluated for channels only)	—	not evaluated
$N_{Ed} / N_{b,Rd}$	$150.000 / 533.439 = (\min \text{ of } N_{b,y,Rd}, N_{b,z,Rd})$	0.281	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 233.7, 1.000, 300.000	1.214	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4) = 189.6$, $M(L/2) = 233.7$, $M(3L/4) = 116.0$, $M_{\max} = 240.8$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3

$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M_{z,Ed} = 20.521 \text{ imposed} + 1.355 \text{ twist } \phi.M_y = 21.876$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3
Lateral Buckling Check M.b.Rd			
$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.204 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.214$, $L_e = 6 \text{ m}$; $z_g = +75 \text{ mm}$ entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1265.547 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	240.758 / 52.800	4.560	Warning
Buckling Resistance			
$U_{N,y} = N_{Ed}/(X_y.N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(X_z.N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(X_{LT}.M_{y,Rk}/\gamma_{M1})$	240.758 / 52.800 ($M_{y,Rk} = 52.800$)	4.560	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 1.355 twist $\phi.M_y = 21.876$) / 52.800	0.414	OK
k_{zy} method	not susceptible to torsional deformation (closed section, fully restrained, or $M_{b,Rd} = M_{c,y,Rd}$), using Table B.1		Table B.1
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2)0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2)0.281\} \leq 1.000(1+0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny}+k_{yy}.U_{M,y}+k_{yz}.U_{M,z}$	$0.281+1.225 \times 4.560+1.225 \times 0.414$	6.374	Warning
$U_{Nz}+k_{zy}.U_{M,y}+k_{zz}.U_{M,z}$	$0.281+0.735 \times 4.560+1.225 \times 0.414$	4.140	Warning
Torsion Design			
J, H, a, Q_f, Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0}, S_{w1}	—, —		P385 App A
Torsion Bending Design @ 0 m			
$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
$W_t (= C)$	SCI P385 Table A.7/A.8; $I_t = 1910 \text{ cm}^4$	240 cm ³	
$T_{Rd} = f_y.W_t/(\sqrt{3}\gamma_{M0})$	$275 \times 240/(\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$5.388 / 38.105$	0.141	OK
$\tau_{t,Ed} = T_{Ed}/W_t$	$5.388 \times 10^3/240$	22.45 N/mm ²	6.2.7(7)
Modified Local Capacity $(M_{y,Ed}/(M_{pl,y,Rd} \cdot S_{mod}))^\alpha +$ $(M_{z,Ed}/(M_{pl,z,Rd} \cdot S_{mod}))^\beta$	n/a - not evaluated by beam-v03 (the cl 6.2.7 T_{Ed}/T_{Rd} , $V_{pl,T,Rd}$ with the 6.2.8(4) p in 6.2.9.1/6.2.10 checks are the verdict basis; the elastic (6.1) check is printed for information)	—	not evaluated
$M_{z,tot} = M_{z,Ed} + \phi.M_{y,Ed}$	$20.521 + 1.355$ (imposed constant $M_{z,Ed}$ + max coincident twist moment of the St Venant solution)	21.876 kN.m	P385 3.1.2 / 5.2.1(3)
Basis with $N_{Ed} / M_{z,Ed}$	EN 1993-1-1 6.2.7(7): warping neglected for the hollow section; the verification is 6.2.7(5)/(6.1) at the cross-section with $\tau_{t,t} = T_{Ed}/W_t$, the V_{Ed} shear flow and $N/A + M_y/W_{el,y} + M_{z,tot}/W_{el,z}$ (information only: the Class 1/2 plastic route governs), T_{Ed}/T_{Rd} (6.2.7(1)), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), 6.2.8(4)/6.2.9.1/6.2.10 with $V_{pl,T,Rd}$, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi.M_y$ added to $M_{z,Ed}$; EN 1993-6 (A.1) does not apply to a closed section		EN 1993-1-1, EN 1993-6
Torsion Shear Design @ 0 m			
$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})].V_{pl,Rd}$	@ x = 0 m (ULS: 1.35G + 1.5Q): $T = 5.388 \text{ kN.m}$, $\tau_t = 22.45 \text{ N/mm}^2$; $V_{pl,Rd} = 284.201$; $S_{mod} =$	244.018 kN	6.2.7(9) Eq 6.28
	$V_{pl,T,Rd}/V_{pl,Rd} = 0.859$		
$V_{Ed}/V_{pl,T,Rd}$	$145.808 / 244.018$	0.598	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$159.91 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	159.91 mm	Warning
Torq in Case 1 @ 2 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0028 \text{ rad} = 0.16^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.16°	OK

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/V _{pl}	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max	
0.152	13.483	0.281	4.560	6.374	4.140	9.595	0.513	4.560	4.560	4.424	0.699	3.183	0.141	36.136	0.598	13.483

FAIL — governing Biaxial bending + axial (6.2.9.1) = 13.483

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 36.136 at $x = 2.26 \text{ m}$ (corner): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 hollow section: the cross-section is verified by the plastic route the code gives - T_{Ed}/T_{Rd} (6.2.7(1))/(7), warping neglected), $V_{Ed}/V_{pl,T,Rd}$ (6.2.7(9), Eq 6.28), ρ from $V_{pl,T,Rd}$ (6.2.8(4)) and the 6.2.9.1/6.2.10 N-M_y-M_z interaction; the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

Beam & Beam-Portion (Member) (FAIL)

Member PFC-L1 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.2296 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	0.00	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

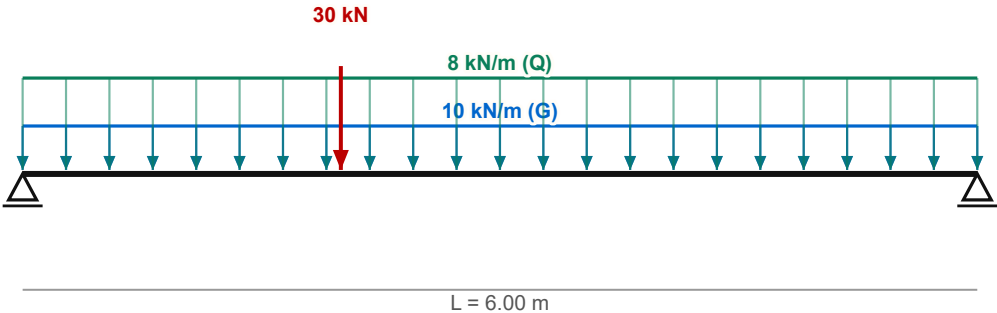
Uplift

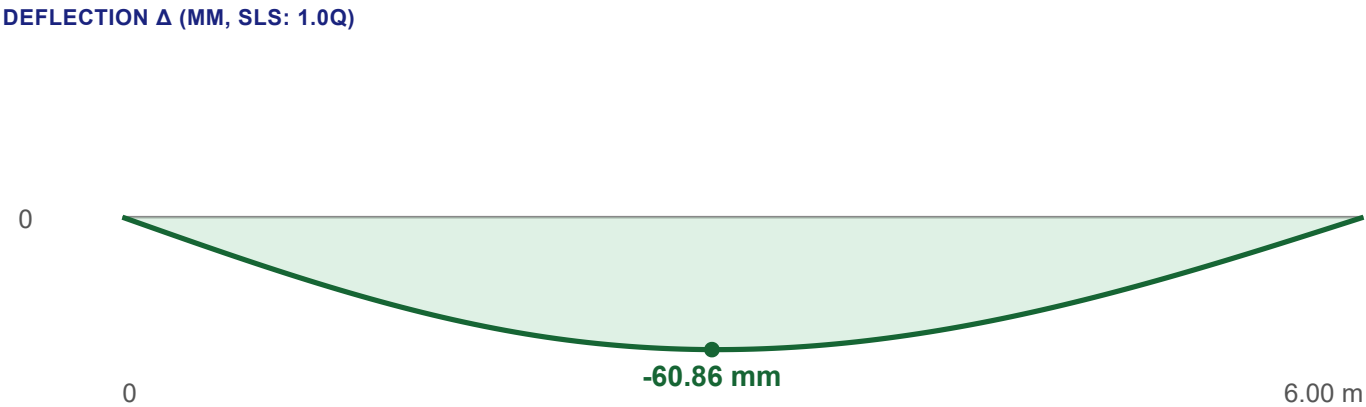
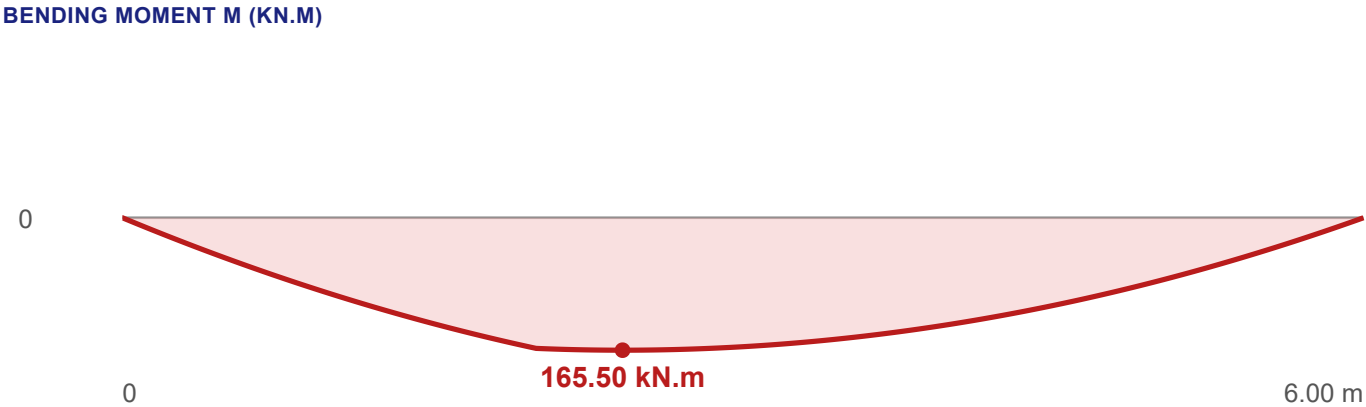
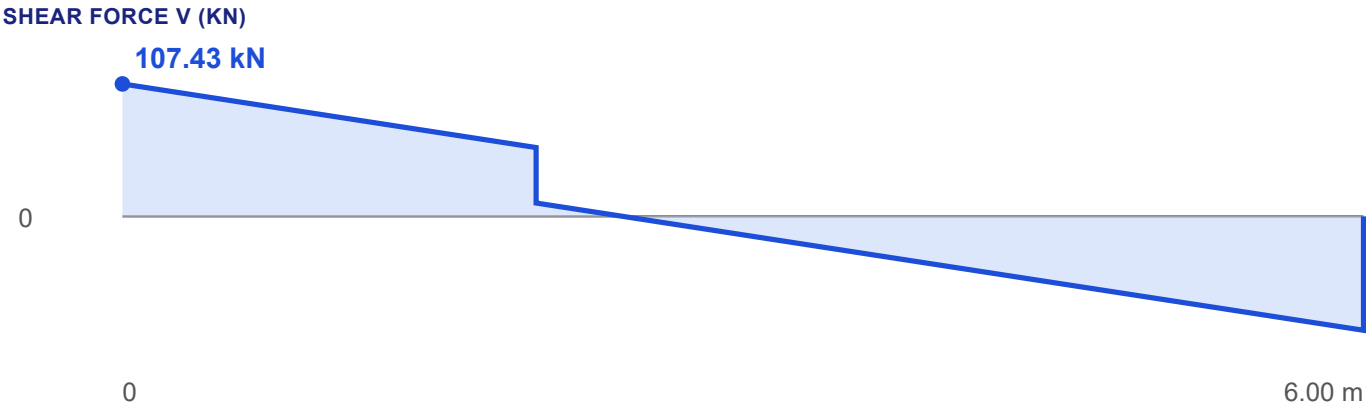
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 0.00, 165.50, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

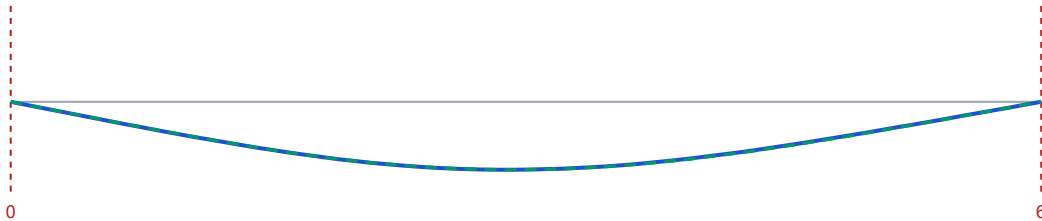
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 60.86 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.651	6.475	3.652	0.505	0.452	1.666	6.475

FAIL — governing LTB $M_{Ed}/M_{b,Rd} = 6.475$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member PFC-L1 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

G SW 0.2296 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	0.00	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

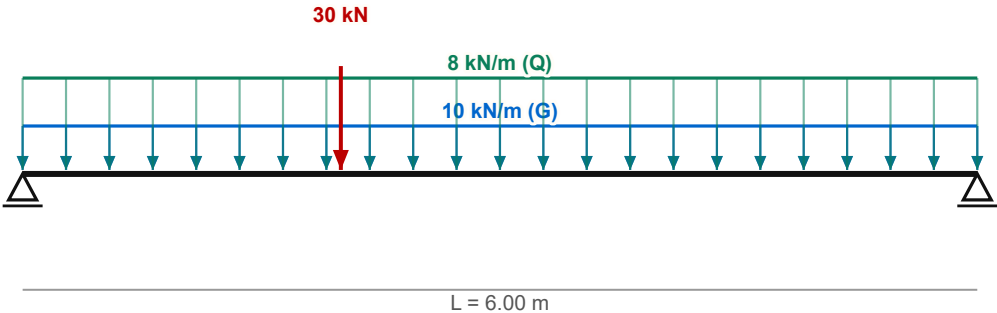
Uplift

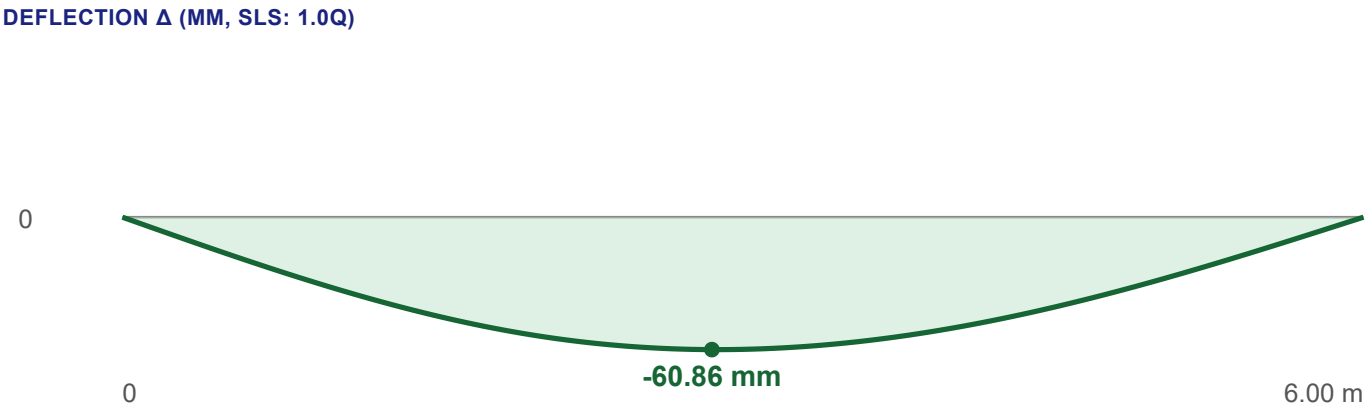
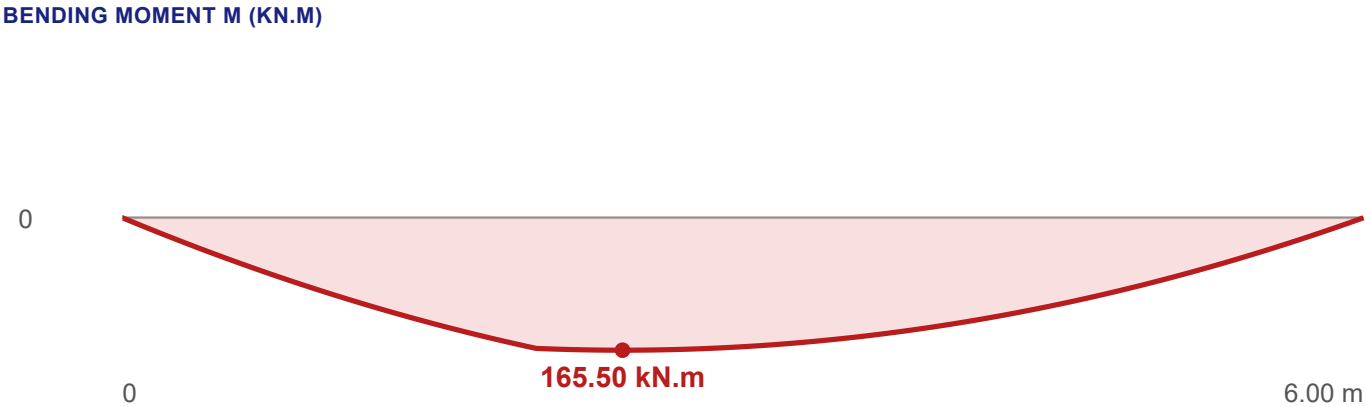
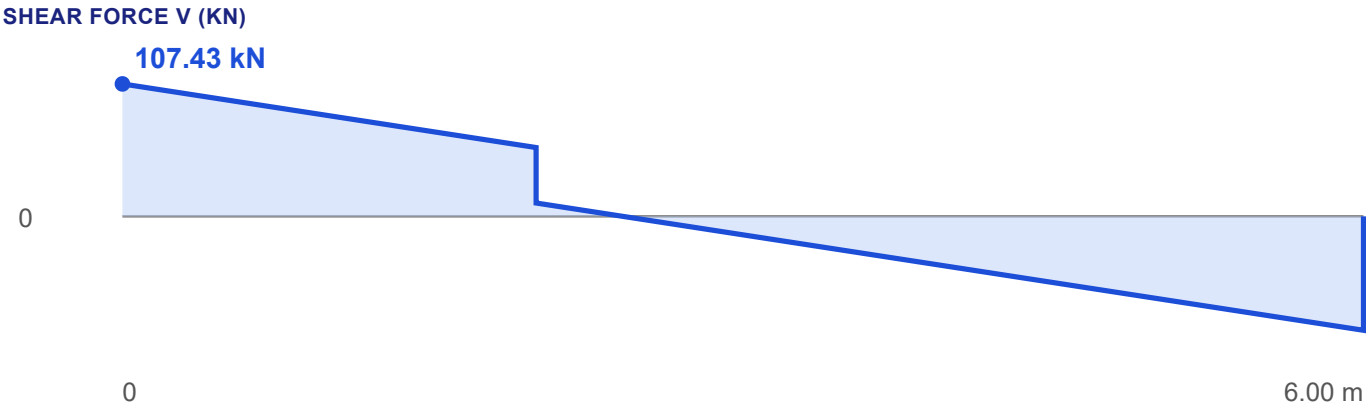
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 0.00, 165.50, 0.00		Class 1
	(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
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$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F · E · t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
l _e = k _F · E · t _w ² / (2f _{yw} · h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² / (2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(l _y · t _w · f _{yw} / F _{cr})	√(144.19 x 6.00 x 275 / 1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F · l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} · L _{eff} · t _w / γ _{M1}	275 x 144.19 x 6.00 / 1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/k	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} · f _y (k chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _f · λ _{LT} ⁻²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{cr} route (load through the shear centre)	M _{cr} = 35.680, λ _̄ = 1.323, χ = 0.402, f = 0.983, χ _{mod} = 0.409	25.538 kN.m	SN003a (z _y = 0)
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
2.651	18.794	3.652	0.505	0.452	1.666	18.794

FAIL — governing LTB M_{Ed}/M_{b,Rd} = 18.794

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member PFC-L2 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6.00C	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

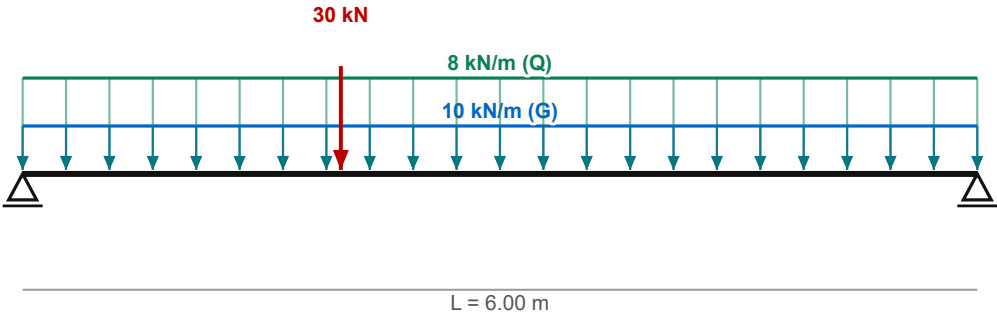
Uplift

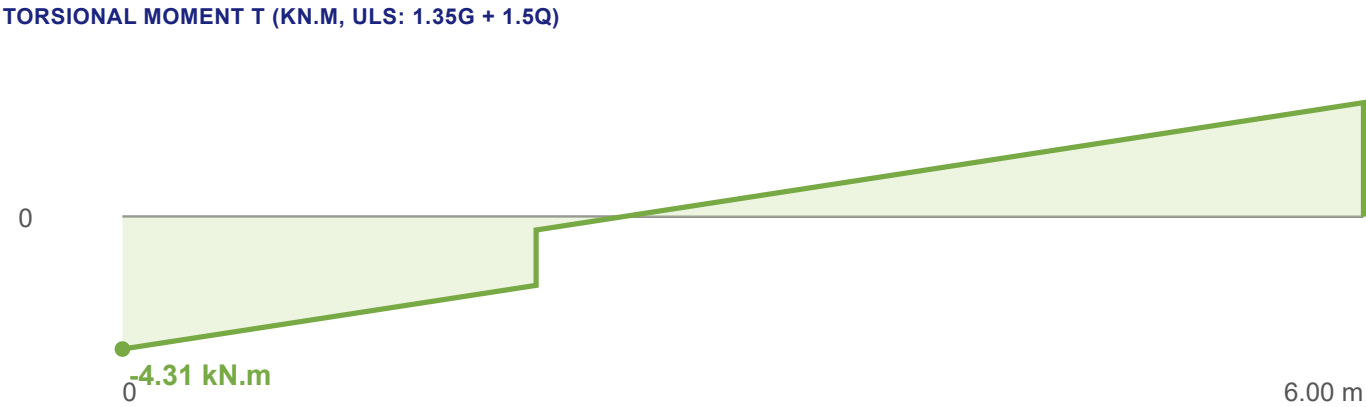
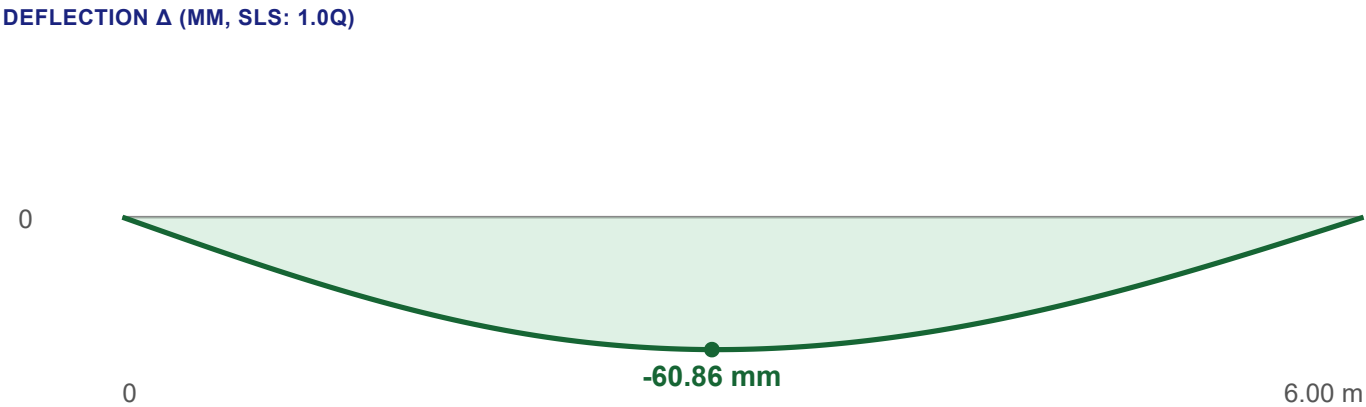
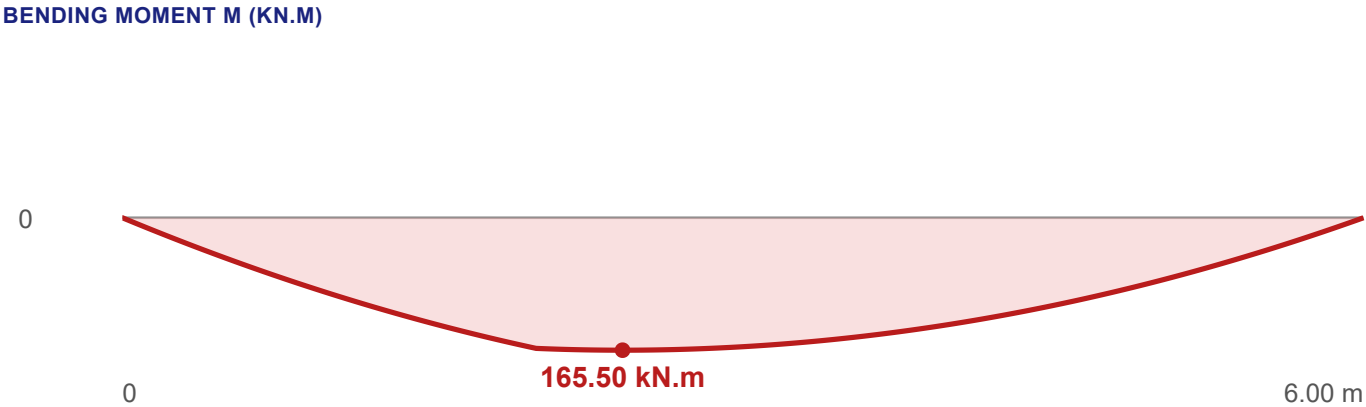
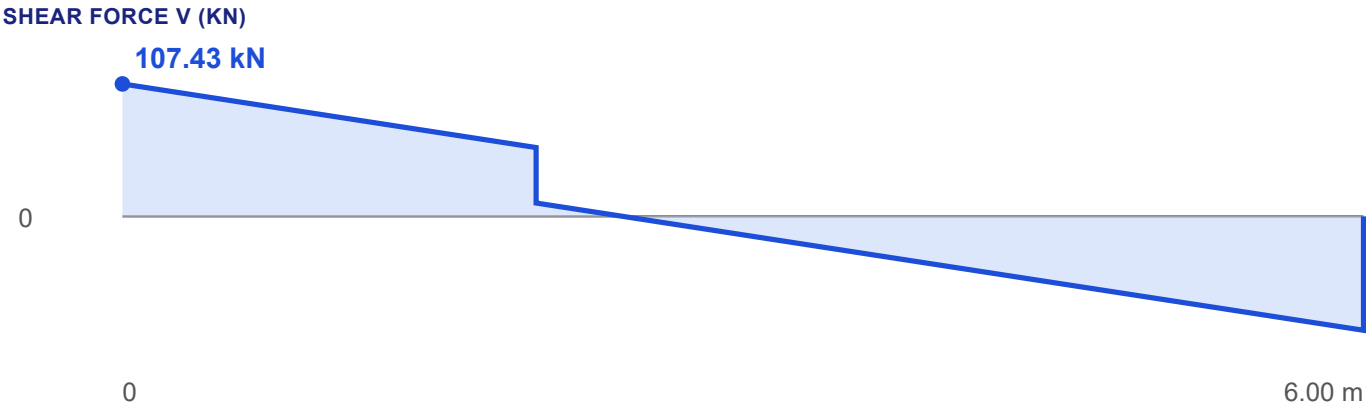
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

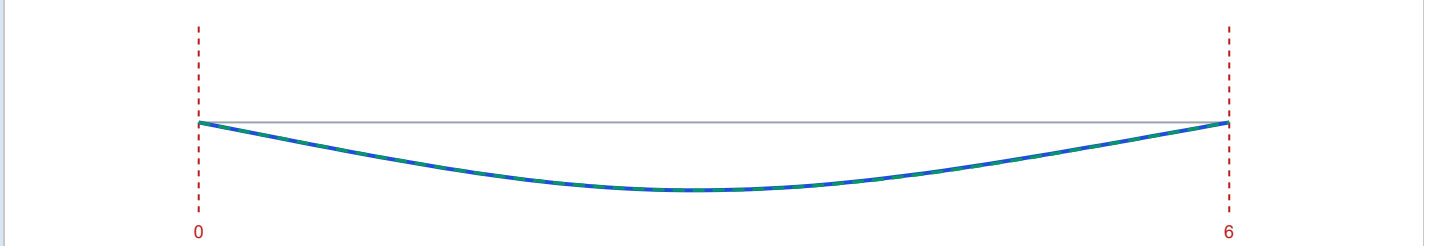
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 35.819 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	M _{y,Ed} /M _{b,Rd}	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G, \text{inf}} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.50Q [$y_{G, inf} = 0.9$, EQU set A]

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

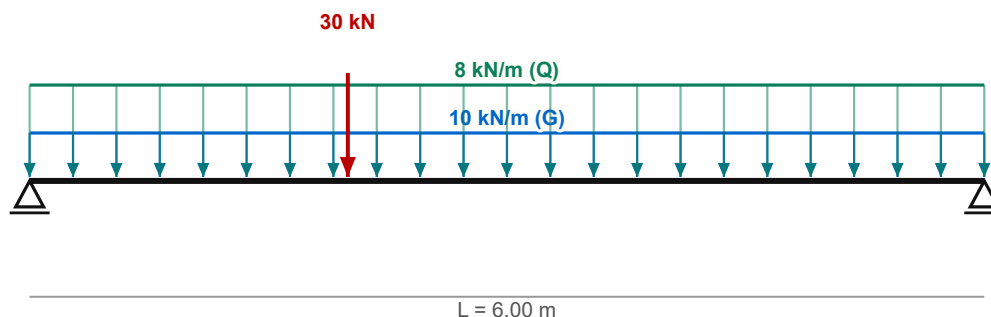
Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)

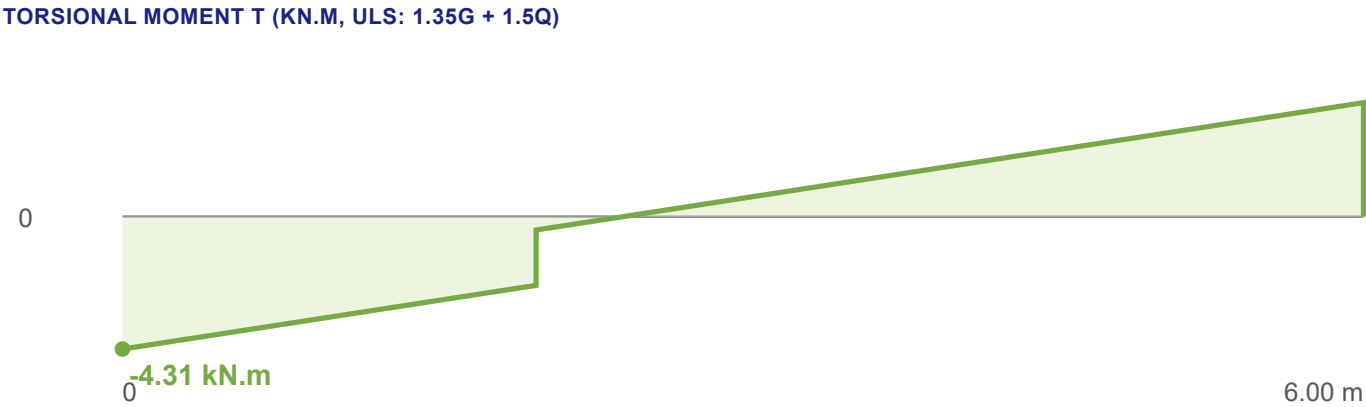
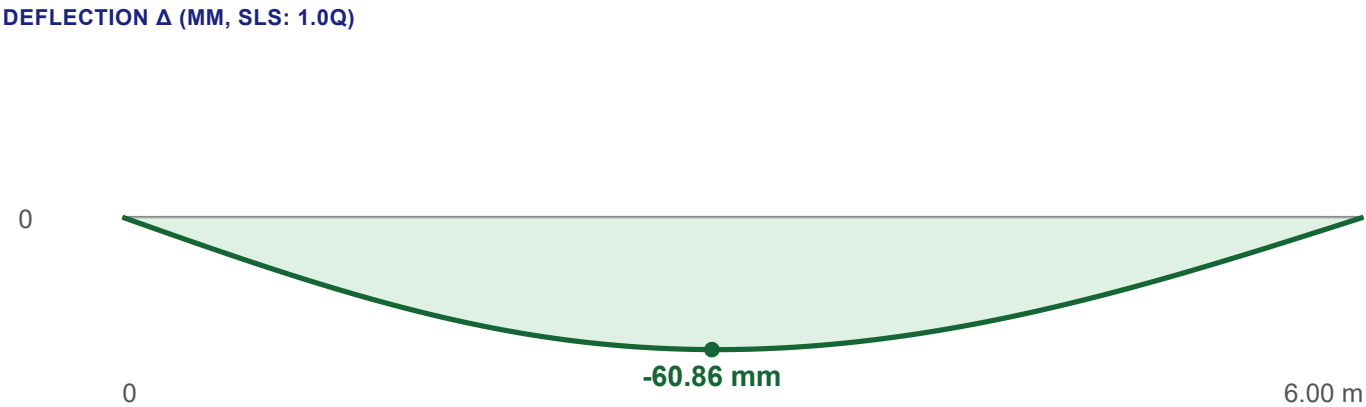
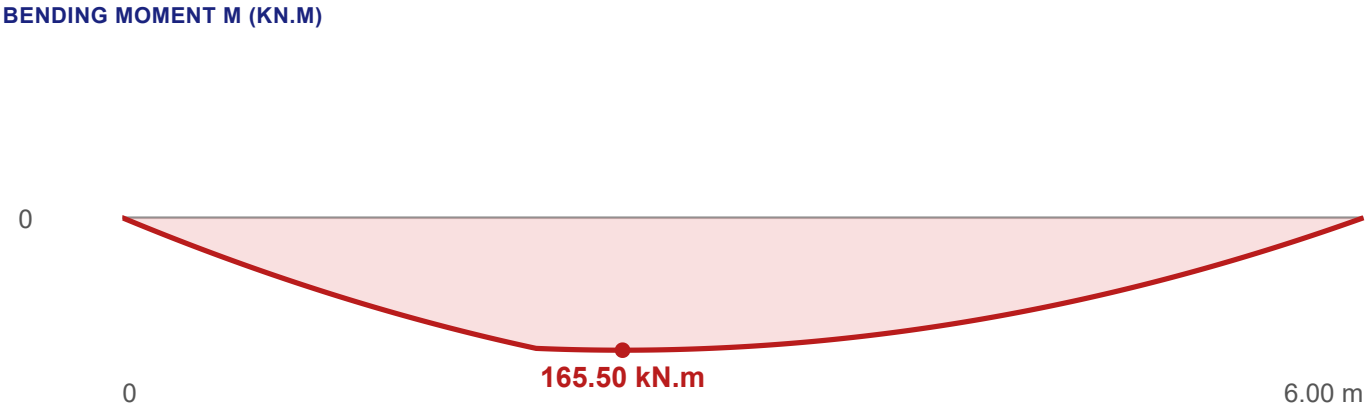
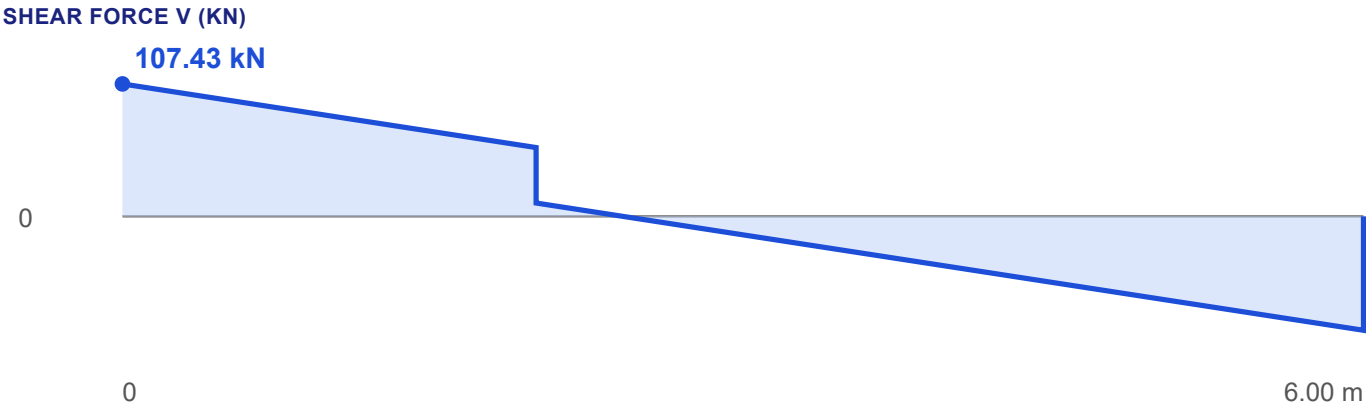
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	4.31	107.43	0.00	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6.00C	-3.71	-92.43	0.00	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; $M_{\perp}$ is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the	—	not used
C_1 basis	P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/k$ without C_1 channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5 \text{ kN}\cdot\text{m}$		P362

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e/i_z)/k$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = E I _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
k = k _w · k _{zw} · k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part G I _T φ' at x = 0 / x = L; total T = G I _T φ' - E I _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T _t , T _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.771 kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
V _{pl,T,Rd} = [√(1 - T _t /(1.25 f _y /√3)) - T _w /(f _y /√3)] · V _{pl,Rd}	@ x = 0 m: T _t = 195.04 (T _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.099	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M_(y,Ed)/M_(b,Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\text{max}} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L2w · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L2 twin "load through the web line": $e = +29.1$ mm = P385 Table A.3 e_0 (shear centre to the web line; the centroid is at $e_{sc} = 50.9$ mm), $z_g = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L2w PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

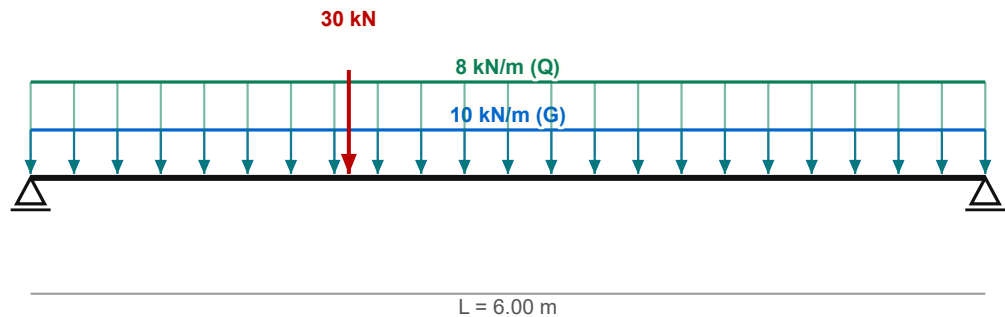
Member Loading and Member Forces

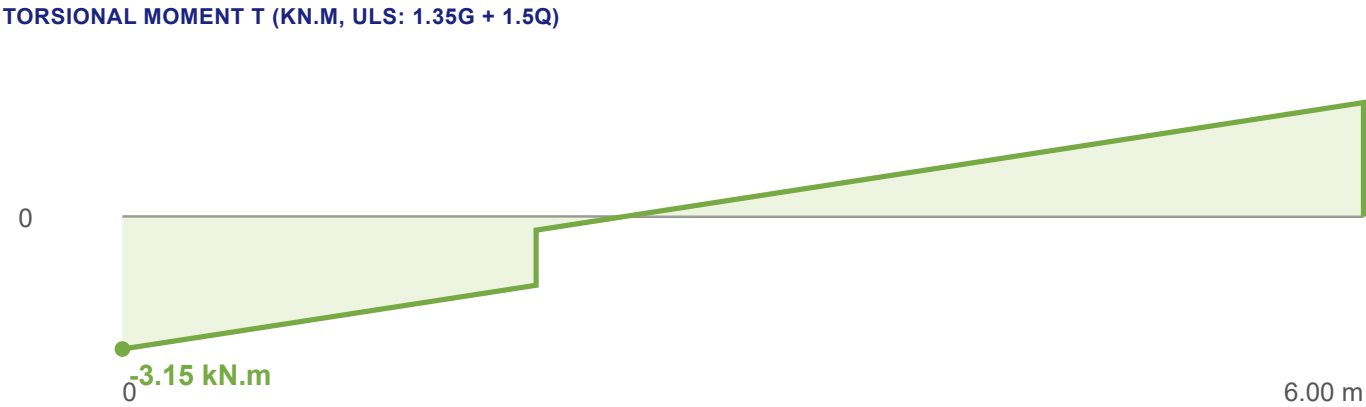
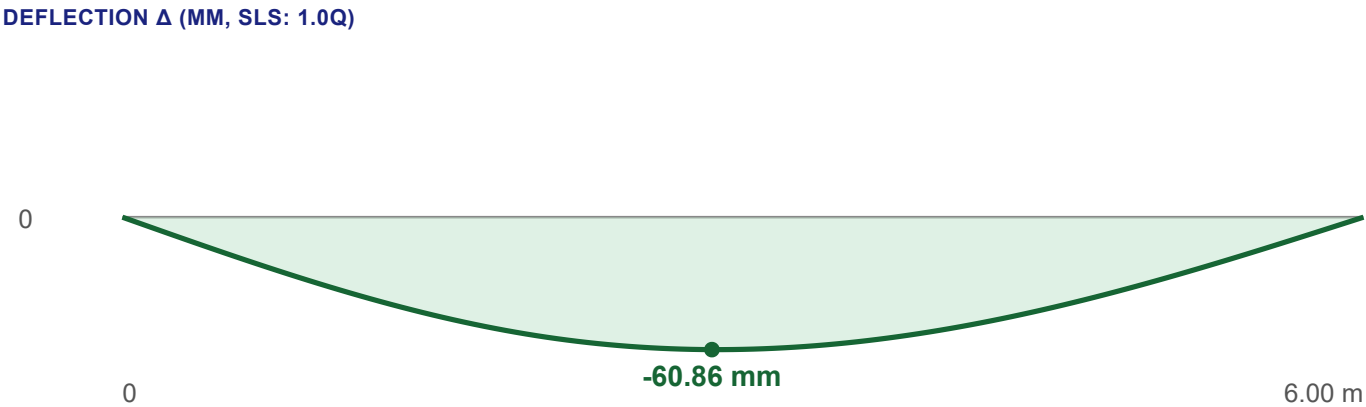
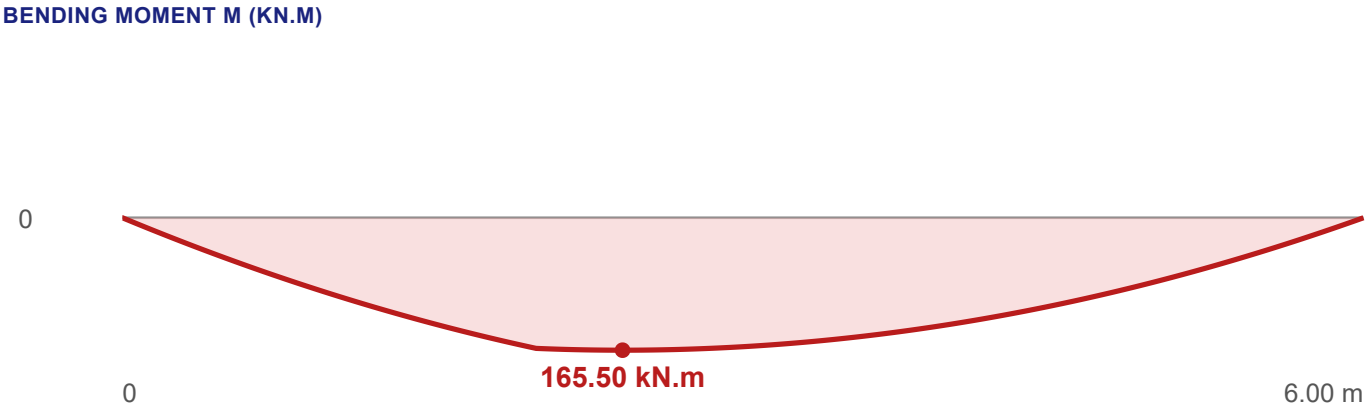
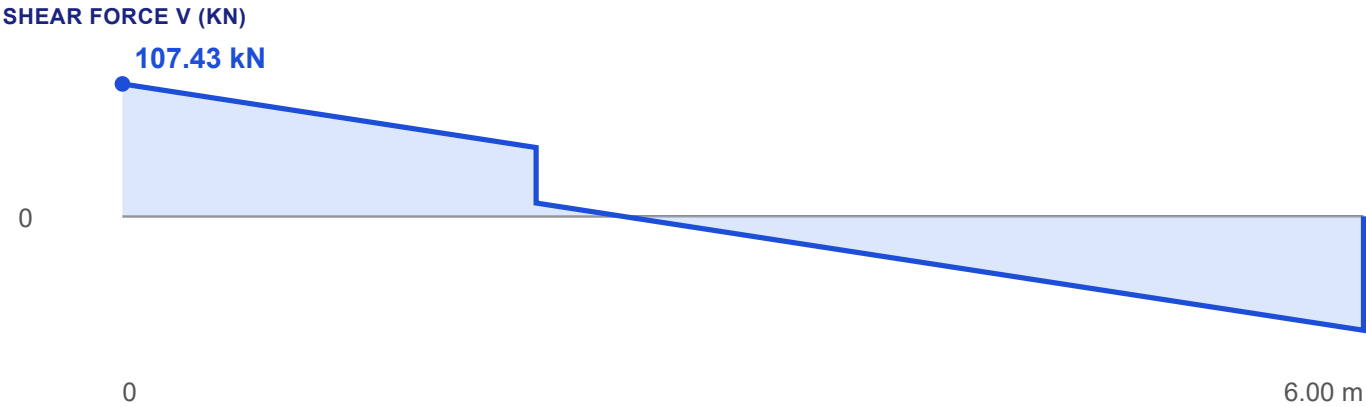
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 29$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 29$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 29$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)
 $V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]
 $V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.15	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-2.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\phi = 0.48330$ rad, $\phi' = 2.342e-5$ rad/mm, $\phi'' = 1.351e-7$ rad/mm ² , $\phi''' = 1.117e-10$ rad/mm ³ ; $M_z = \phi M_y = 79.975$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.27 + 2366.13 + 114.38 = 3324.78$ N/mm ² ; $\tau = G_t \phi' = 23.72$ N/mm ² ; $(3324.78/275)^2 + 3(23.72/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 2366.13 + 114.38 = 3324.78$, $\tau = 23.72 \rightarrow 146.193$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 343.86 = 1135.36$, $\tau = 0.24 + 23.72 + 2.16 = 26.12 \rightarrow 17.072$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1162.19 + 77.20 = 2030.89$, $\tau = 0.42 + 23.72 + 1.17 = 25.31 \rightarrow 54.564$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1162.19 + 77.20 = 1978.12$, $\tau = 0.89 + 11.38 + 2.44 = 14.71 \rightarrow 51.750$; P4 web mid-depth: $\sigma_x = 0.00 + 1162.19 = 1162.19$, $\tau = 1.12 + 11.38 + 1.81 = 14.31 \rightarrow 17.868$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y/\gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

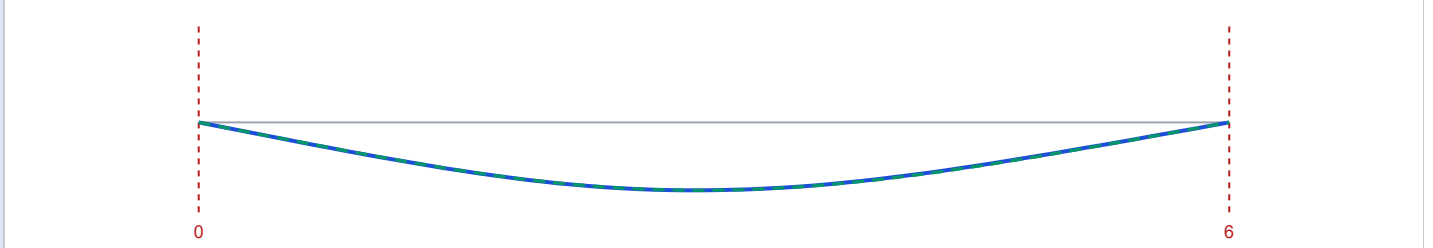
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	3.146 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.4854 rad = 27.81°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.498 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	2.657 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	80.134 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 1.690, 79.767 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	12.165	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 35.819 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: M _{y,Ed} reaches M _{cr} = 35.819 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi'$ $- EI_w \phi''' = 3.146 / 2.710 \text{ kN.m}$	2.754 / 2.339 kN.m

Torsion Shear Design @ 0 m

T_t, T_w at the V-T station	@ $x = 0 \text{ m}$ (ULS: 1.35G + 1.5Q): $T_t = 2.754 \text{ kN.m}$, $V = 142.45 / 3.82 \text{ N/mm}^2$ $= 107.430 \text{ kN}$	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $T_t = 142.45$ ($T_w = 3.82$) N/mm^2 ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.507$	107.911 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 107.911	0.996 OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2028 \text{ rad} = 11.62^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	11.62°	> 2° advisory

MA/Mc	$M_{(y.Ed)}/M_{(b.Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	0.452	1.666	12.165	146.193	0.996	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 11.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 146.193 at $x = 2.46 \text{ m}$ (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L2w · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L2 twin "load through the web line": $e = +29.1$ mm = P385 Table A.3 e_0 (shear centre to the web line; the centroid is at $e_{sc} = 50.9$ mm), $z_g = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L2w PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

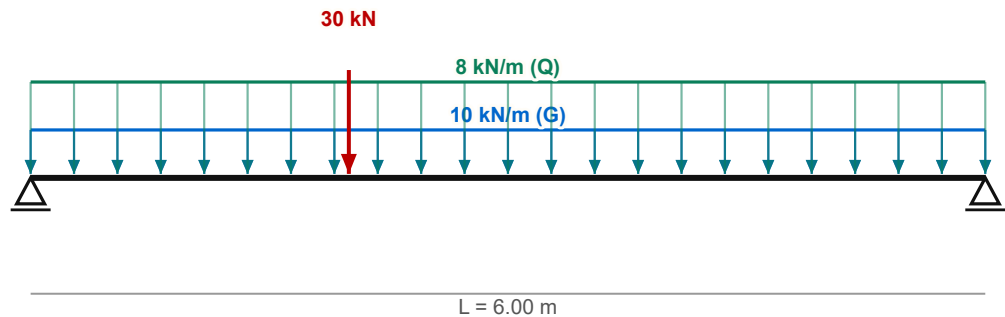
Member Loading and Member Forces

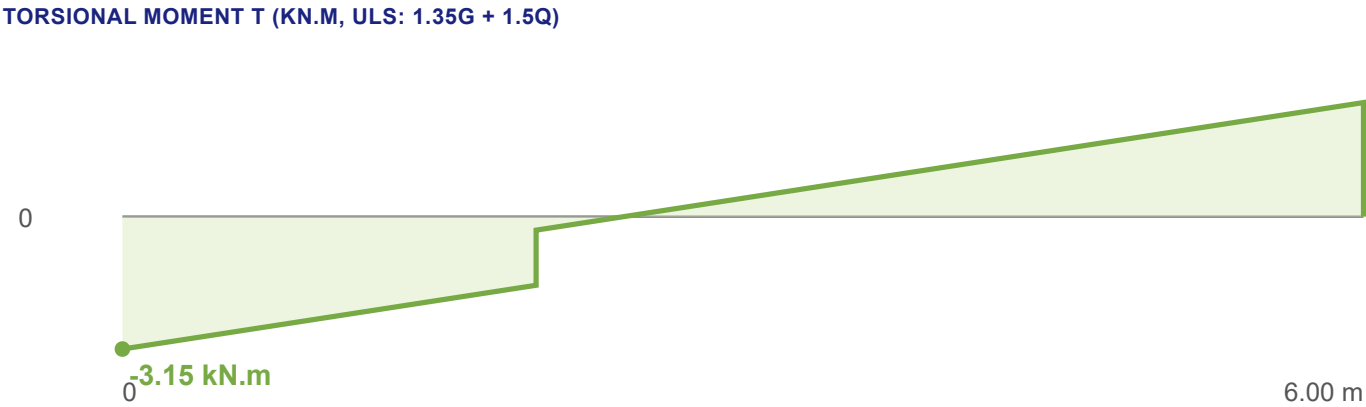
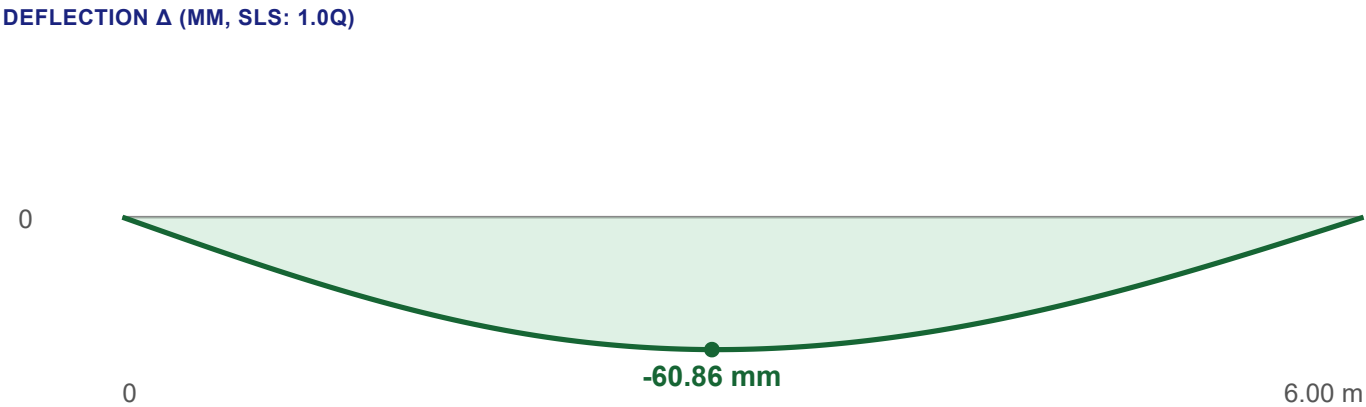
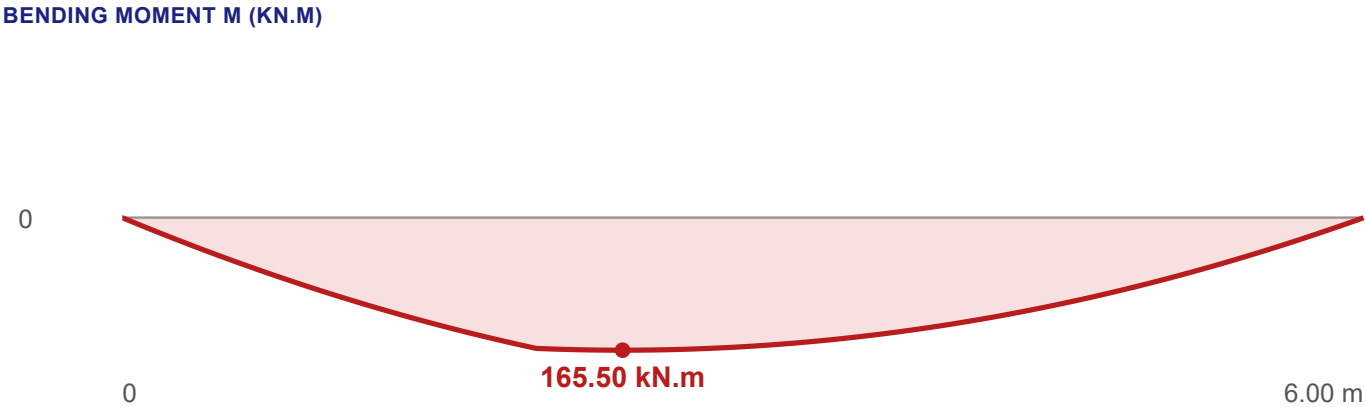
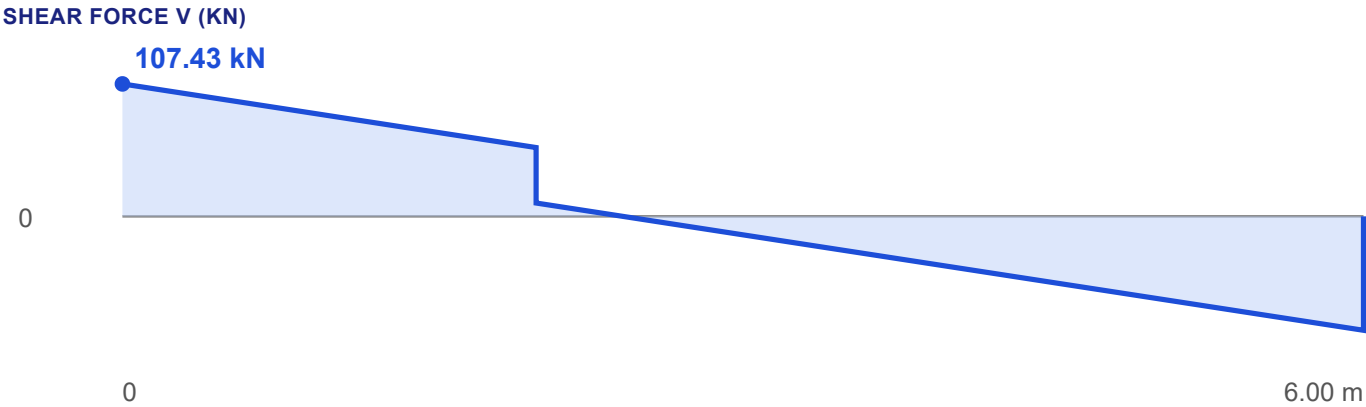
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 29$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 29$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 29$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)
 $V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]
 $V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.15	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-2.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\phi = 0.48330$ rad, $\phi' = 2.342e-5$ rad/mm, $\phi'' = 1.351e-7$ rad/mm ² , $\phi''' = 1.117e-10$ rad/mm ³ ; $M_z = \phi M_y = 79.975$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.27 + 2366.13 + 114.38 = 3324.78$ N/mm ² ; $\tau = G_t \phi' = 23.72$ N/mm ² ; $(3324.78/275)^2 + 3(23.72/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 2366.13 + 114.38 = 3324.78$, $\tau = 23.72 \rightarrow 146.193$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 343.86 = 1135.36$, $\tau = 0.24 + 23.72 + 2.16 = 26.12 \rightarrow 17.072$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1162.19 + 77.20 = 2030.89$, $\tau = 0.42 + 23.72 + 1.17 = 25.31 \rightarrow 54.564$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1162.19 + 77.20 = 1978.12$, $\tau = 0.89 + 11.38 + 2.44 = 14.71 \rightarrow 51.750$; P4 web mid-depth: $\sigma_x = 0.00 + 1162.19 = 1162.19$, $\tau = 1.12 + 11.38 + 1.81 = 14.31 \rightarrow 17.868$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		146.193	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
X _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _y .f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	3.146 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.4854 rad = 27.81°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.498 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	2.657 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	80.134 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 1.690, 79.767 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	12.165	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 3.146 / 2.710 kN.m	2.754 / 2.339 kN.m	

Torsion Shear Design @ 0 m

τ _t , τ _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 2.754 kN.m, V = 107.430 kN	142.45 / 3.82 N/mm ²	P385
V _{pl,T,Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)].V _{pl,Rd}	@ x = 0 m: τ _t = 142.45 (τ _w = 3.82) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.507	107.911 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 107.911	0.996	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2028 \text{ rad} = 11.62^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	11.62°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	12.165	146.193	0.996	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\text{max}} = 11.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 146.193 at $x = 2.46 \text{ m}$ (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L3t · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +100$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L3t PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

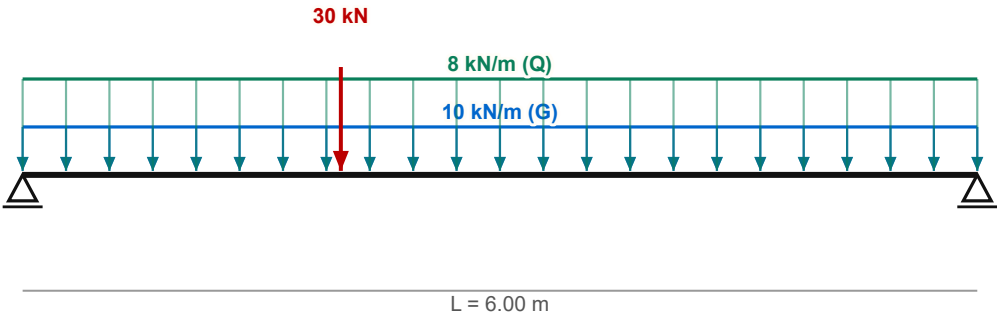
Member Loading and Member Forces

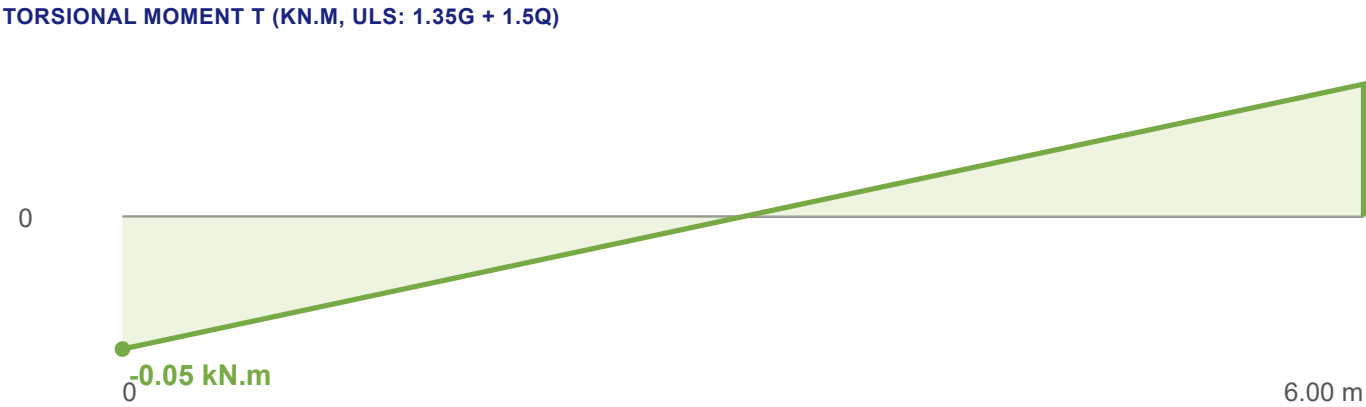
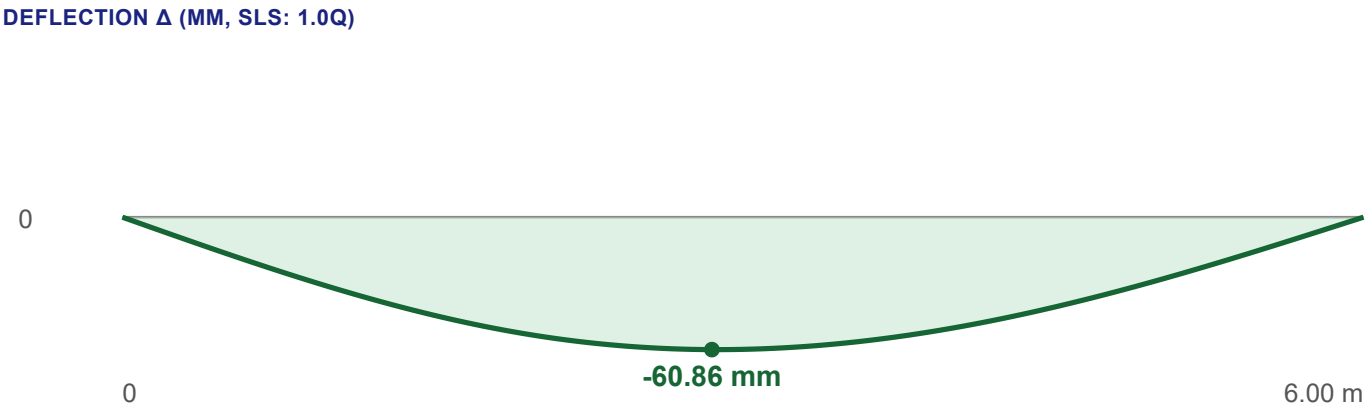
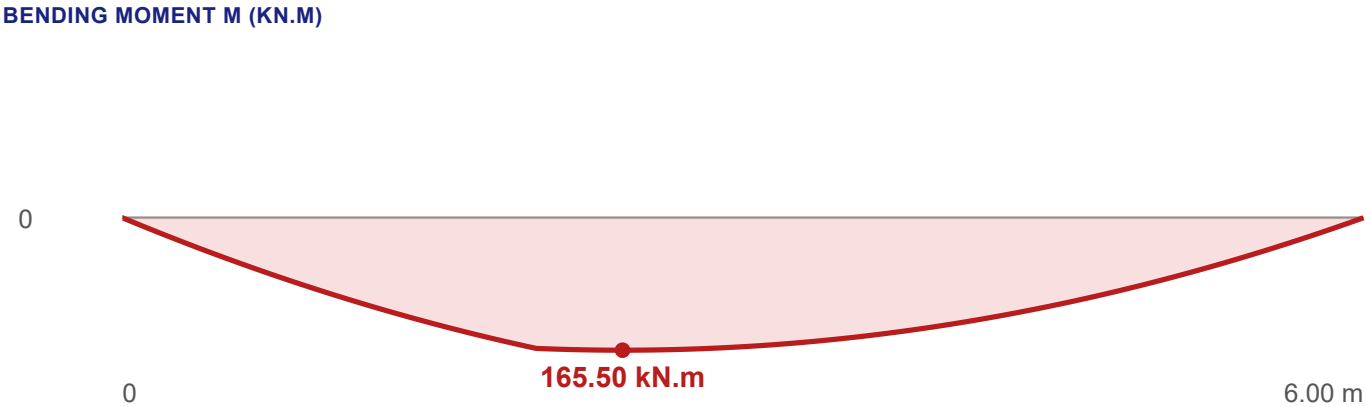
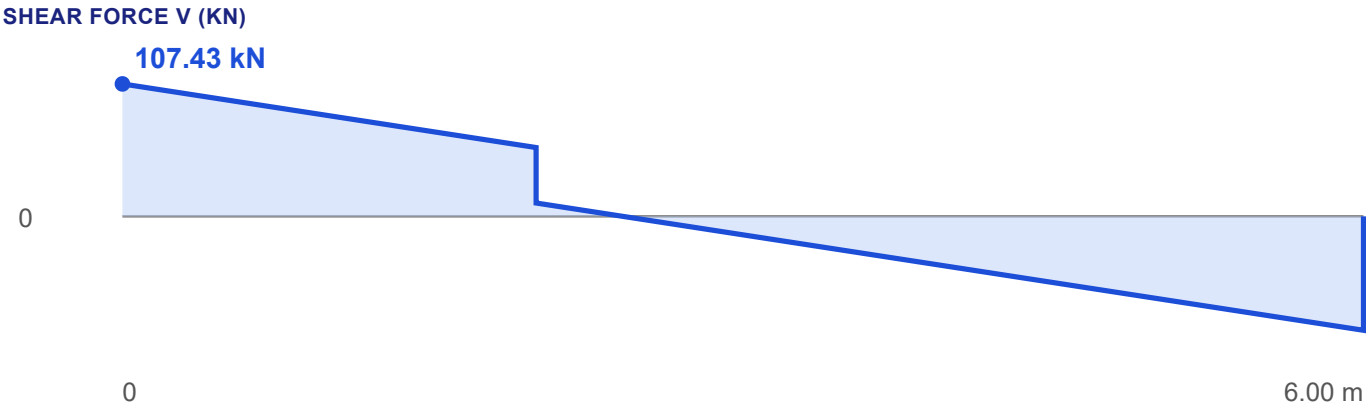
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 0$ mm $z_g = 100$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 0$ mm $z_g = 100$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 100$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]
 $V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.05	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-0.05	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
Auto Design Load Cases	1; SLS 1	(Axial: Non-Slender)

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	$275 \times 227.0/1$	62.425 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.44 m, P1 flange tip	@ x = 2.44 m (ULS: 1.35G + 1.5Q): $M_y = 165.498$ kN.m, $V = 0.542$ kN, $\phi = 0.00689$ rad, $\phi' = 9.356e-7$ rad/mm, $\phi'' = 1.667e-9$ rad/mm ² , $\phi''' = 2.102e-14$ rad/mm ³ ; $M_z = \phi M_y = 1.141$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$ N/mm ² ; $\tau = G t_f \phi' = 0.95$ N/mm ² ; $(879.54/275)^2 + 3(0.95/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$, $\tau = 0.95 \rightarrow 10.229$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.60 + 4.91 = 796.51$, $\tau = 0.11 + 0.95 + 0.00 = 1.06 \rightarrow 8.389$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.60 + 16.58 + 0.95 = 809.14$, $\tau = 0.19 + 0.95 + 0.00 = 1.14 \rightarrow 8.657$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.83 + 16.58 + 0.95 = 756.36$, $\tau = 0.41 + 0.45 + 0.00 = 0.86 \rightarrow 7.565$; P4 web mid-depth: $\sigma_x = 0.00 + 16.58 = 16.58$, $\tau = 0.51 + 0.45 + 0.00 = 0.97 \rightarrow 0.004$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		10.229	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$
			6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

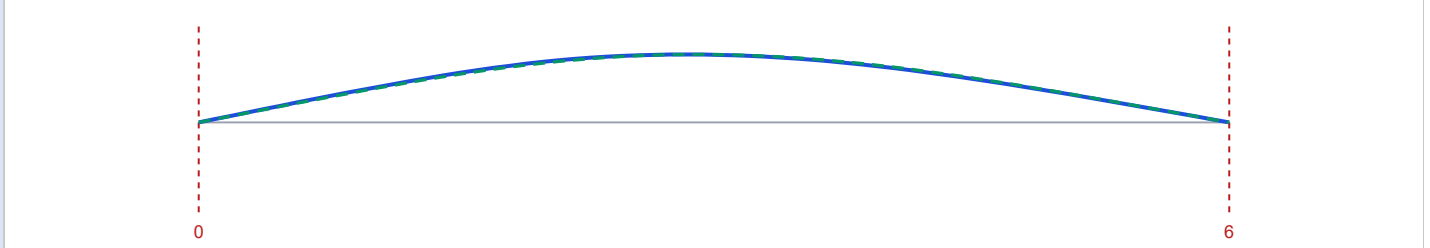
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	30.940 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 30.940}$	1.420	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.420, 1.644, 0.76	0.366	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.368 \times 227.0 \times 275 \leq 62.425 =$	22.992 kN.m	
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 22.992$	7.198	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2.988$ m	0.004 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	0.020 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	1.166 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 0.020, 1.137 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	7.101	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 30.940 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 30.940 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_p b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 0.047 / 0.047$ kN.m	0.040 / 0.040 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 0.040$ kN.m, $V = 107.430$ kN	2.05 / 0.08 N/mm ² P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 2.05$ ($T_w = 0.08$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.994$	211.552 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 211.552	0.508 OK
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 0 m: $\theta_{max} \leq 2.00^\circ$	0.0000 rad = $0.00^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.00° OK

MA/Mc	$M_{(y.Ed)}/M_{(b.Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	7.198	3.652	0.505	0.452	1.666	7.101	10.229	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 10.229 at $x = 2.44$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L3t · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +100$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L3t PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

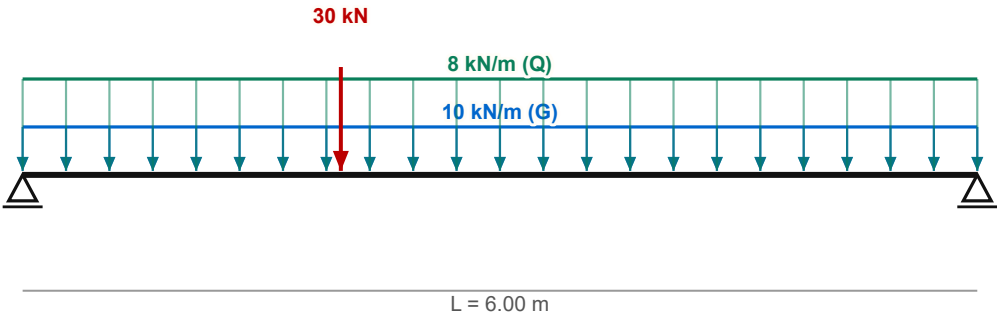
Member Loading and Member Forces

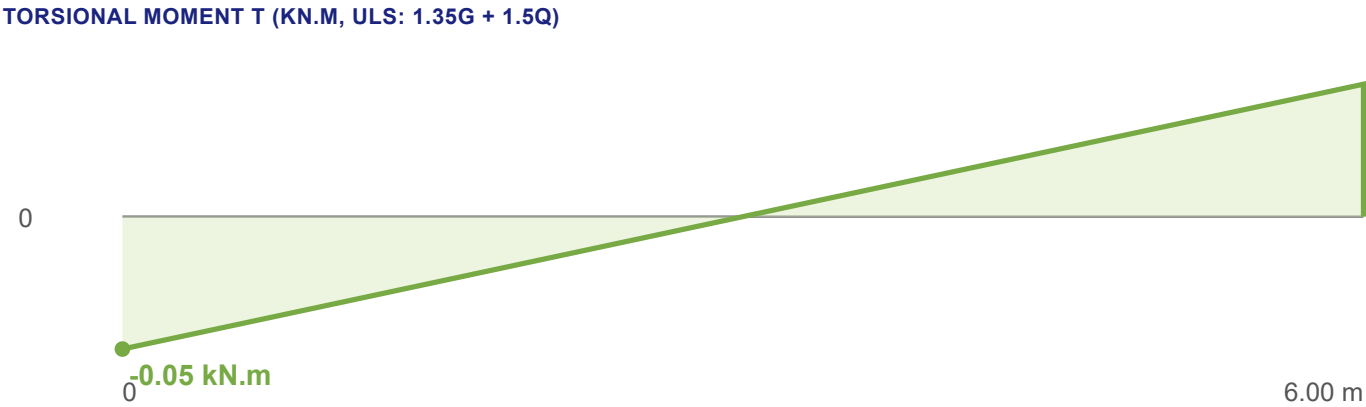
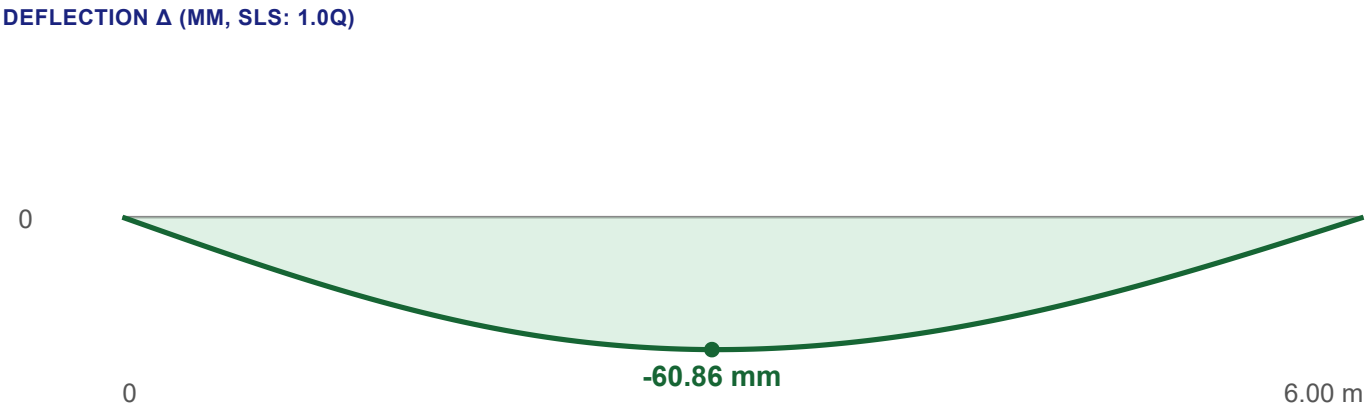
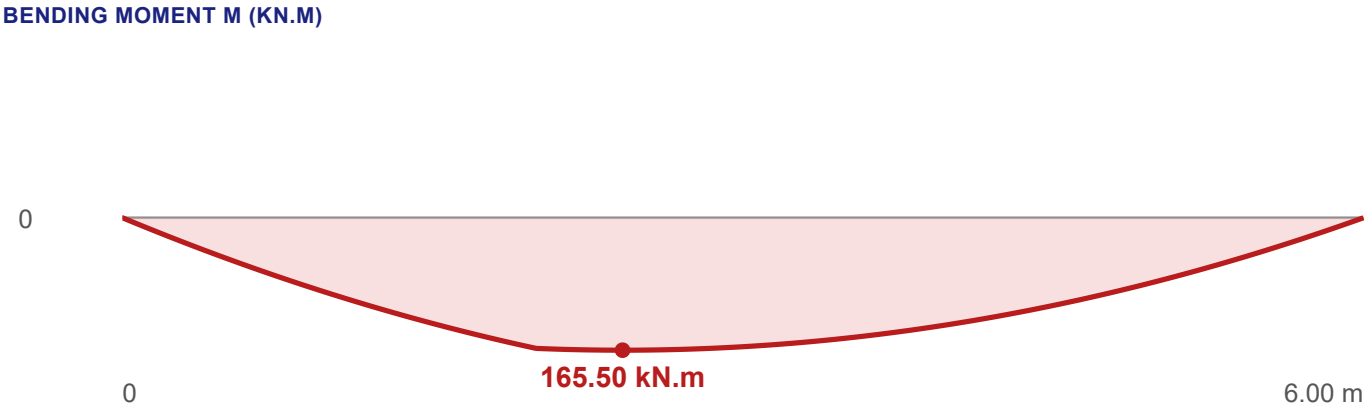
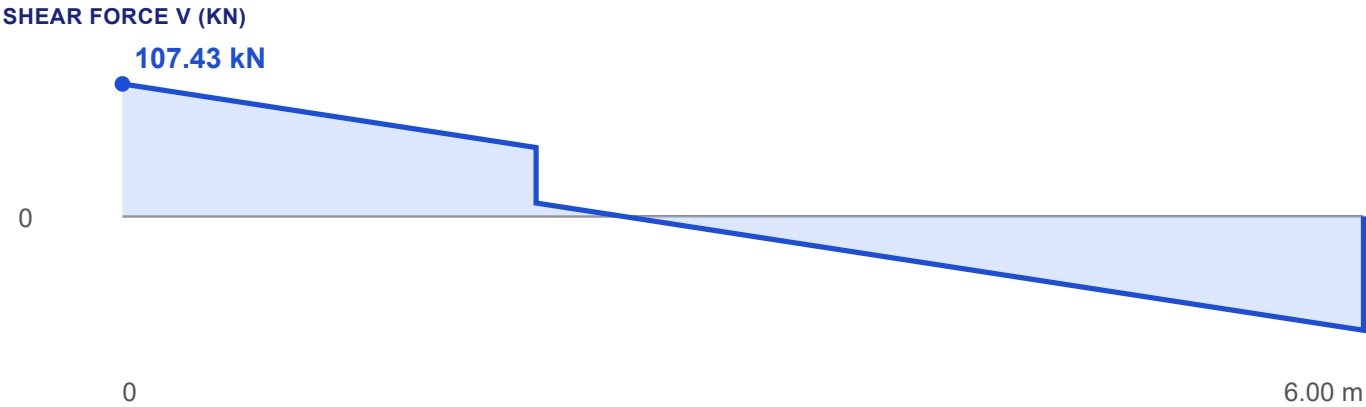
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 0$ mm $z_g = 100$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 0$ mm $z_g = 100$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 100$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]
 $V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.05	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-0.05	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.44 m, P1 flange tip	@ x = 2.44 m (ULS: 1.35G + 1.5Q): $M_y = 165.498$ kN.m, $V = 0.542$ kN, $\phi = 0.00689$ rad, $\phi' = 9.356e-7$ rad/mm, $\phi'' = 1.667e-9$ rad/mm², $\phi''' = 2.102e-14$ rad/mm³; $M_z = \phi M_y = 1.141$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$ N/mm²; $\tau = G t_f \phi' = 0.95$ N/mm²; $(879.54/275)^2 + 3(0.95/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$, $\tau = 0.95 \rightarrow 10.229$ (governs); P1b flange at $W_{n0} = 0$: $\sigma_x = 0.00 + 791.60 + 4.91 = 796.51$, $\tau = 0.11 + 0.95 + 0.00 = 1.06 \rightarrow 8.389$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.60 + 16.58 + 0.95 = 809.14$, $\tau = 0.19 + 0.95 + 0.00 = 1.14 \rightarrow 8.657$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.83 + 16.58 + 0.95 = 756.36$, $\tau = 0.41 + 0.45 + 0.00 = 0.86 \rightarrow 7.565$; P4 web mid-depth: $\sigma_x = 0.00 + 16.58 = 16.58$, $\tau = 0.51 + 0.45 + 0.00 = 0.97 \rightarrow 0.004$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377)} = 0.433$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
Z _g	Z _g = +100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L _E switch); the shear-centre M _{cr} route is not offered	load height	
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W.f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = EI _w φ" (max)	@ x = 2.988 m	0.004 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	0.020 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	1.166 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 0.020, 1.137 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	7.101	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _b ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' – EI _w φ''' = 0.047 / 0.047 kN.m	0.040 / 0.040 kN.m	

Torsion Shear Design @ 0 m

τ _t , τ _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 0.040 kN.m, V = 107.430 kN	2.05 / 0.08 N/mm ²	P385
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$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 2.05$ ($\tau_w = 0.08$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.994$	211.552 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 211.552	0.508	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 0 m: $\theta_{max} \leq 2.00^\circ$	0.0000 rad = $0.00^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.00°	OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	7.101	10.229	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 10.229 at x = 2.44 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L3b · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -100$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L3b PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

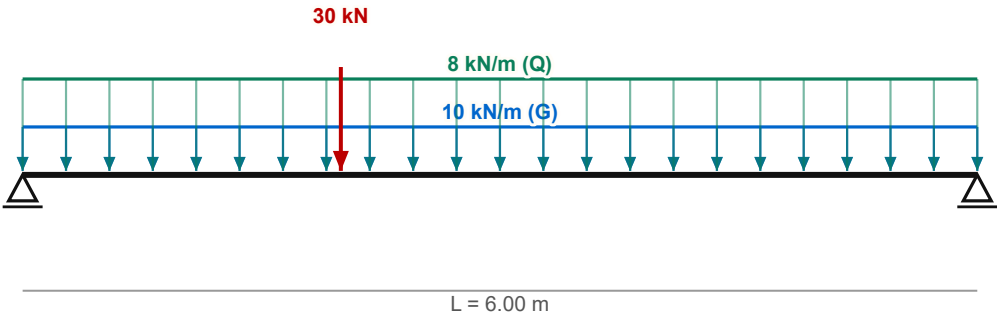
Member Loading and Member Forces

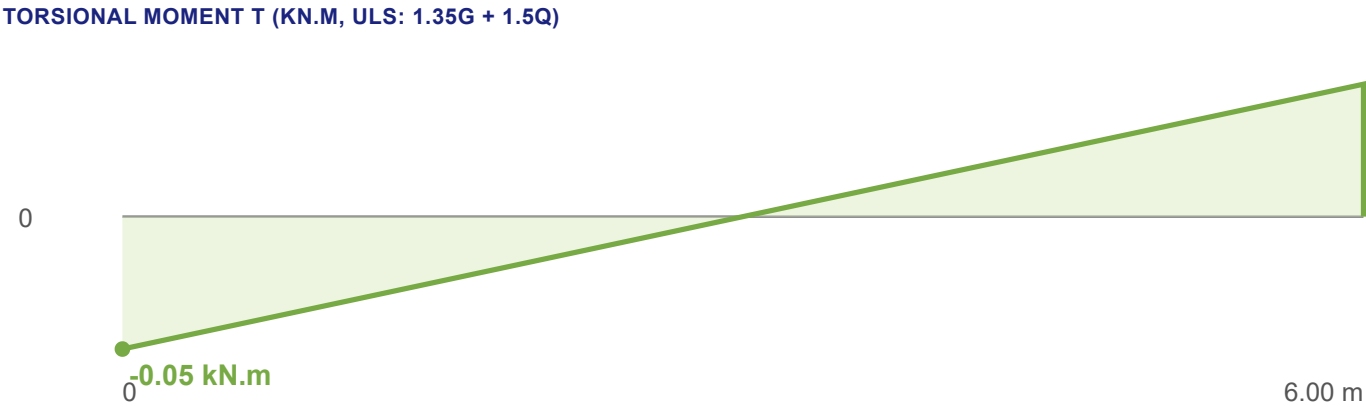
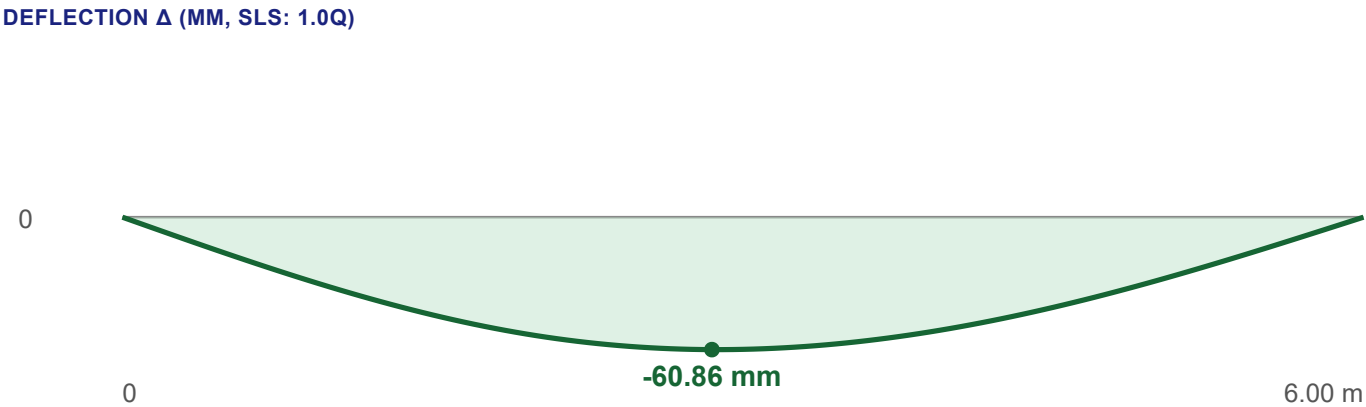
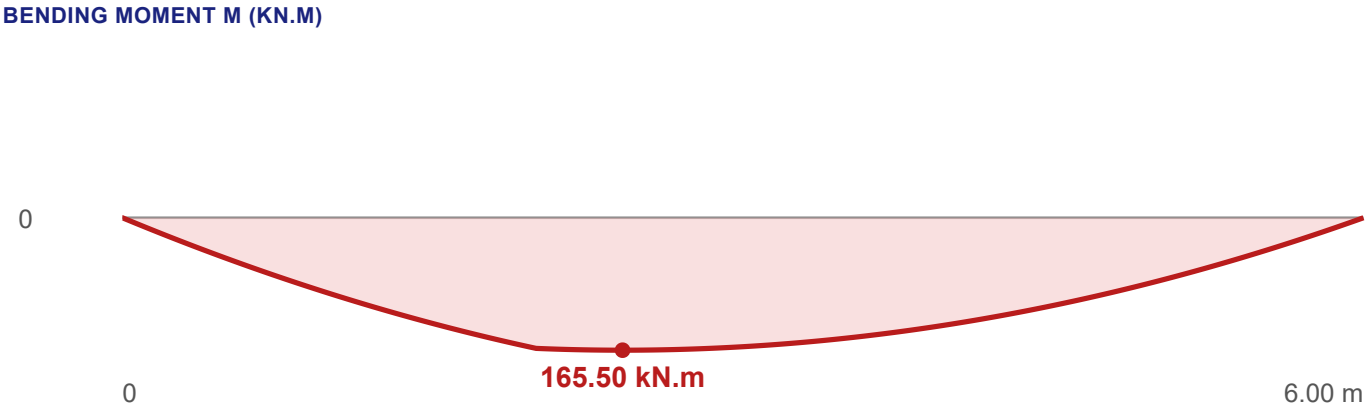
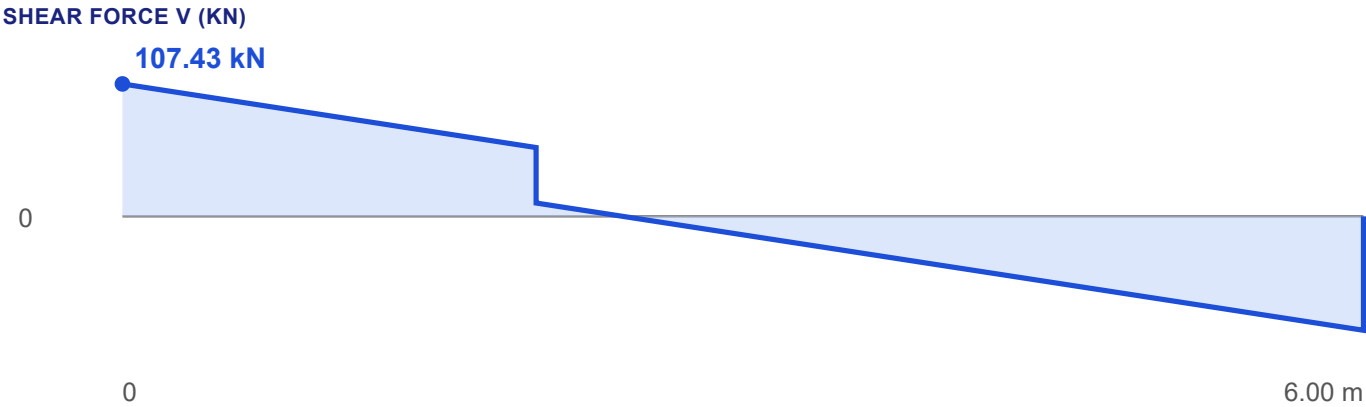
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -100$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C	0.00	0.05	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6 0.00C	-0.05	-92.43	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	$275 \times 227.0/1$	62.425 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.44 m, P1 flange tip	@ x = 2.44 m (ULS: 1.35G + 1.5Q): $M_y = 165.498$ kN.m, $V = 0.542$ kN, $\phi = 0.00689$ rad, $\phi' = 9.356e-7$ rad/mm, $\phi'' = 1.667e-9$ rad/mm ² , $\phi''' = 2.102e-14$ rad/mm ³ ; $M_z = \phi M_y = 1.141$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$ N/mm ² ; $\tau = G t_f \phi' = 0.95$ N/mm ² ; $(879.54/275)^2 + 3(0.95/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$, $\tau = 0.95 \rightarrow 10.229$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.60 + 4.91 = 796.51$, $\tau = 0.11 + 0.95 + 0.00 = 1.06 \rightarrow 8.389$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.60 + 16.58 + 0.95 = 809.14$, $\tau = 0.19 + 0.95 + 0.00 = 1.14 \rightarrow 8.657$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.83 + 16.58 + 0.95 = 756.36$, $\tau = 0.41 + 0.45 + 0.00 = 0.86 \rightarrow 7.565$; P4 web mid-depth: $\sigma_x = 0.00 + 16.58 = 16.58$, $\tau = 0.51 + 0.45 + 0.00 = 0.97 \rightarrow 0.004$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		10.229	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$
			6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

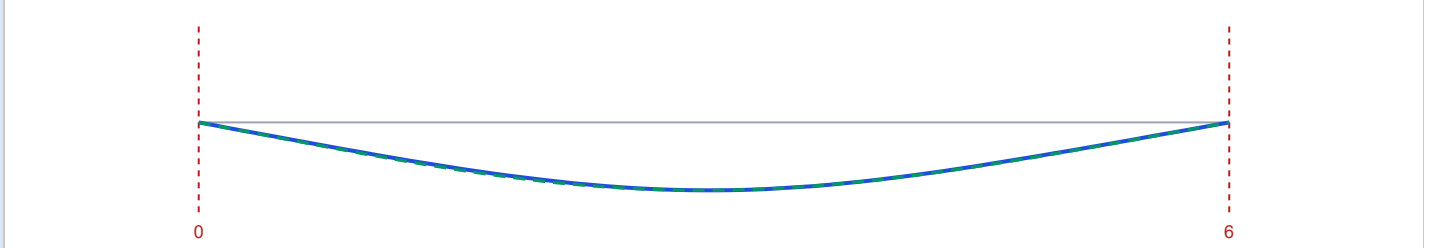
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -100$, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	41.391 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 41.391}$	1.228	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.228, 1.380, 0.76	0.442	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.442, 1.228, 0.935, 0.979	0.452	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.452 \times 227.0 \times 275 \leq 62.425 =$	28.204 kN.m	
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 28.204$	5.868	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2.988$ m	0.004 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	0.020 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	1.166 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 0.020, 1.137 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	7.101	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 41.391 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 41.391 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_p b^2 f_y / 4$) [verify]			EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi'$ – $EI_w \phi''' = 0.047 / 0.047$ kN.m	0.040 / 0.040 kN.m		
Torsion Shear Design @ 0 m				
T_t, τ_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 0.040$ kN.m, $V = 107.430$ kN	2.05 / 0.08 N/mm ²		P385
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 2.05$ ($\tau_w = 0.08$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.994$	211.552 kN		6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 211.552	0.508		OK
Deflection Check - Load Case 1 (SLS: 1.0Q)				
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm		Warning
Torq in Case 1 @ 0 m: $\theta_{max} \leq 2.00^\circ$	0.0000 rad = $0.00^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.00°		OK

MA/Mc	$M_{(y.Ed)}/M_{(b.Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	5.868	3.652	0.505	0.452	1.666	7.101	10.229	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 10.229 at $x = 2.44$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L3b · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -100$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L3b PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

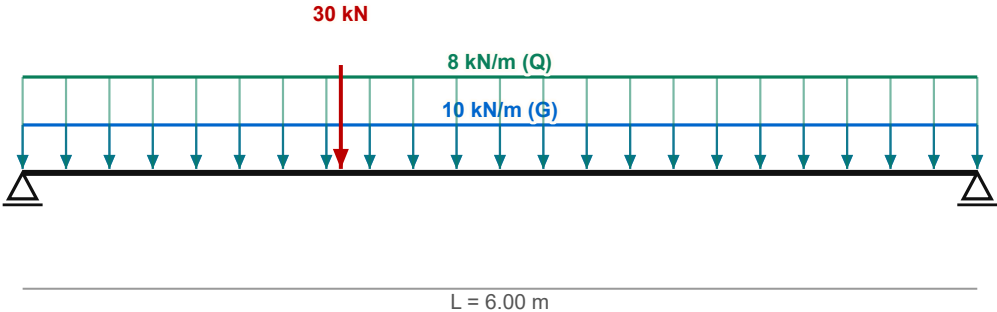
Member Loading and Member Forces

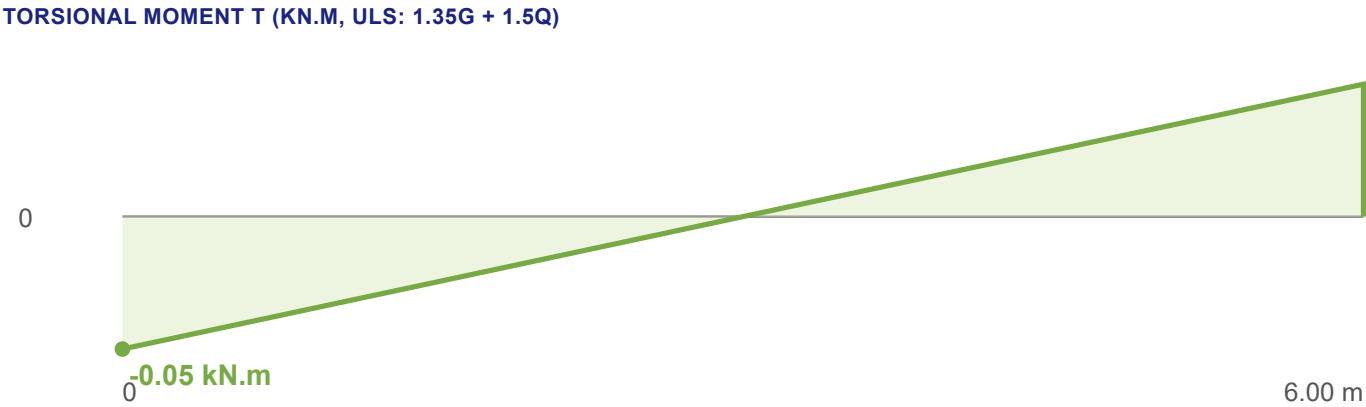
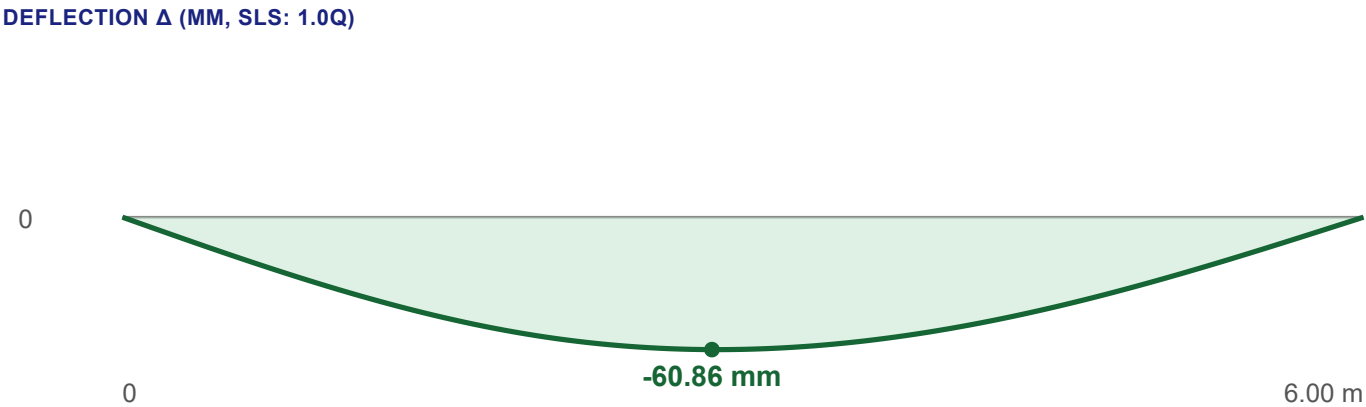
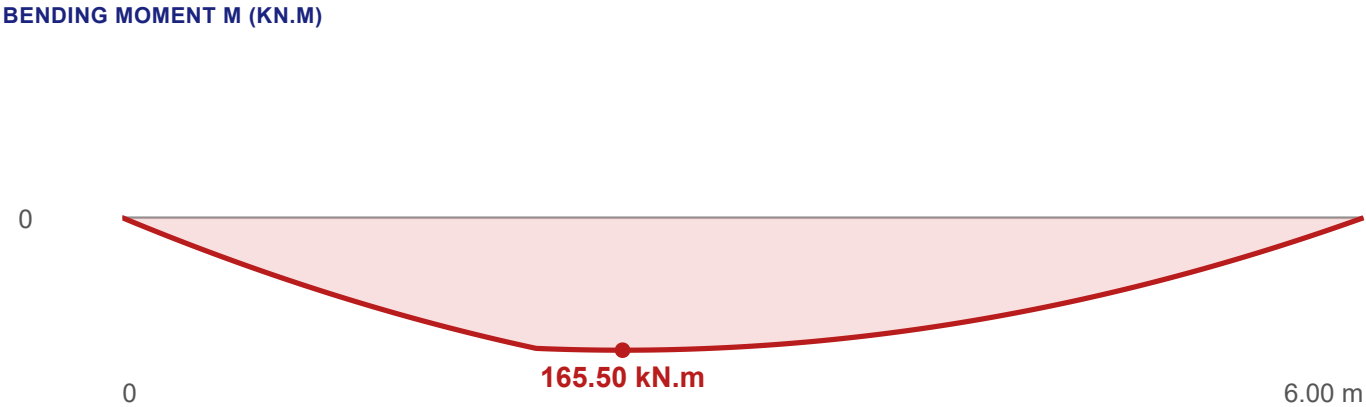
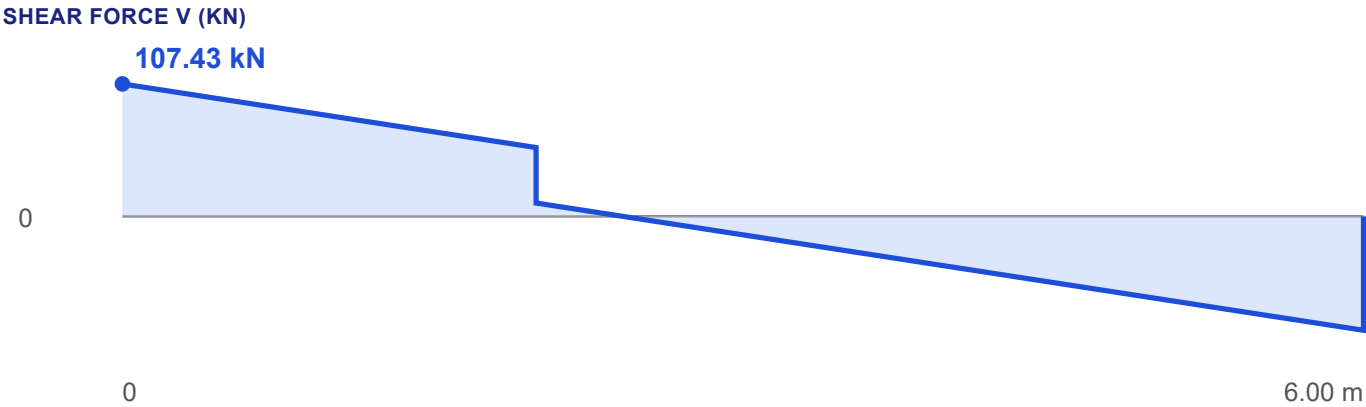
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -100$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.05	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-0.05	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y}/\gamma_{M0}$	$275 \times 227.0/1$	62.425 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.44 m, P1 flange tip	@ x = 2.44 m (ULS: 1.35G + 1.5Q): $M_y = 165.498$ kN.m, $V = 0.542$ kN, $\phi = 0.00689$ rad, $\phi' = 9.356e-7$ rad/mm, $\phi'' = 1.667e-9$ rad/mm ² , $\phi''' = 2.102e-14$ rad/mm ³ ; $M_z = \phi M_y = 1.141$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$ N/mm ² ; $\tau = G t_f \phi' = 0.95$ N/mm ² ; $(879.54/275)^2 + 3(0.95/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.38 + 33.76 + 1.41 = 879.54$, $\tau = 0.95 \rightarrow 10.229$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.60 + 4.91 = 796.51$, $\tau = 0.11 + 0.95 + 0.00 = 1.06 \rightarrow 8.389$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.60 + 16.58 + 0.95 = 809.14$, $\tau = 0.19 + 0.95 + 0.00 = 1.14 \rightarrow 8.657$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.83 + 16.58 + 0.95 = 756.36$, $\tau = 0.41 + 0.45 + 0.00 = 0.86 \rightarrow 7.565$; P4 web mid-depth: $\sigma_x = 0.00 + 16.58 = 16.58$, $\tau = 0.51 + 0.45 + 0.00 = 0.97 \rightarrow 0.004$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		10.229	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$
			6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
Z _g	Z _g = -100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L _E switch); the shear-centre M _{cr} route is not offered		load height
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
X _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W.f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = EI _w φ" (max)	@ x = 2.988 m	0.004 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	0.020 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	1.166 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 0.020, 1.137 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	7.101	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _b ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 0.047 / 0.047 kN.m	0.040 / 0.040 kN.m	

Torsion Shear Design @ 0 m

τ _t , τ _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 0.040 kN.m, V = 107.430 kN	2.05 / 0.08 N/mm ²	P385
--	--	-------------------------------	------

$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 2.05$ ($\tau_w = 0.08$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.994$	211.552 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 211.552	0.508	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 0 m: $\theta_{max} \leq 2.00^\circ$	0.0000 rad = $0.00^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.00°	OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	7.101	10.229	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 10.229 at x = 2.44 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member PFC-L4 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)

— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C		4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6 0.00C		-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

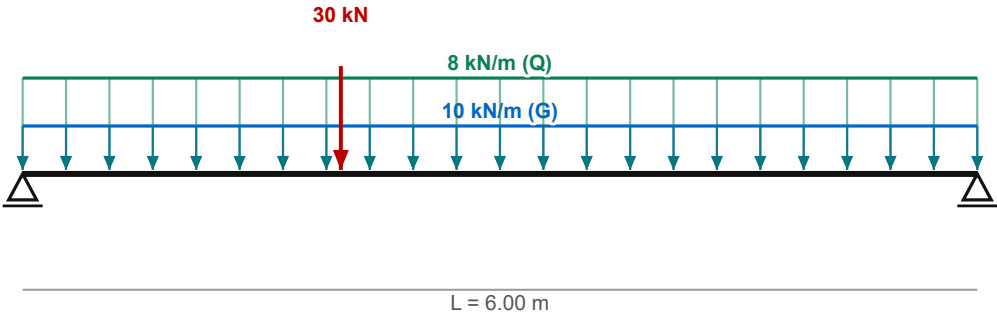
Uplift

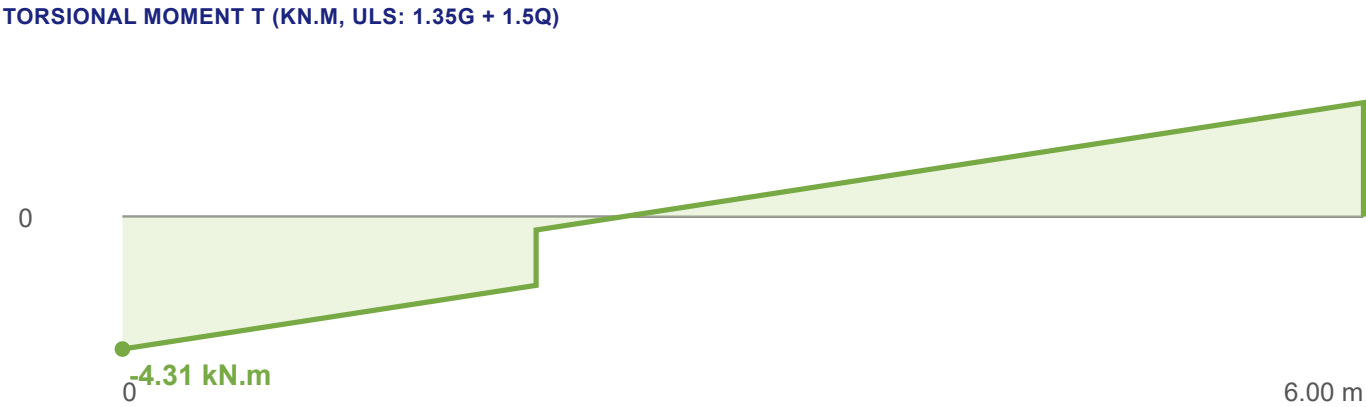
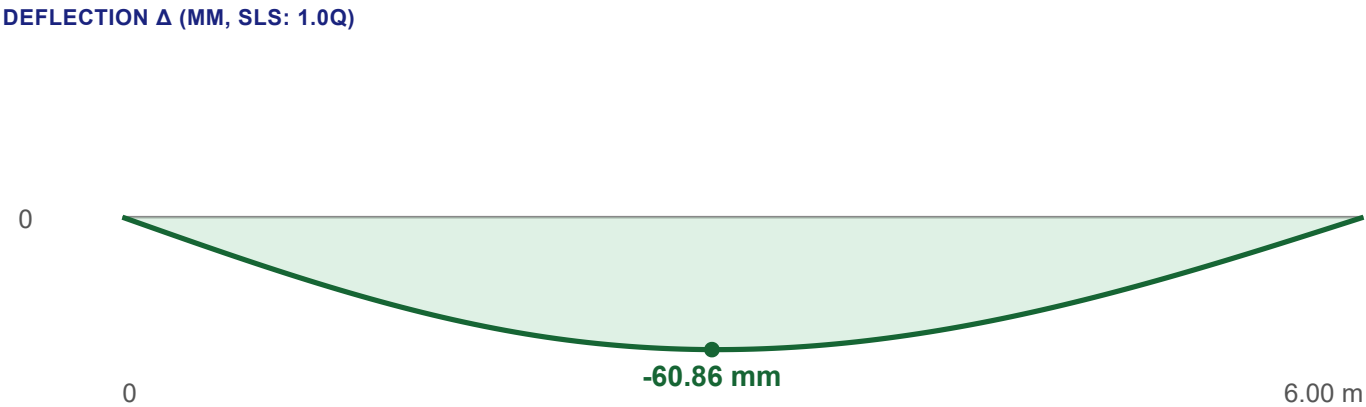
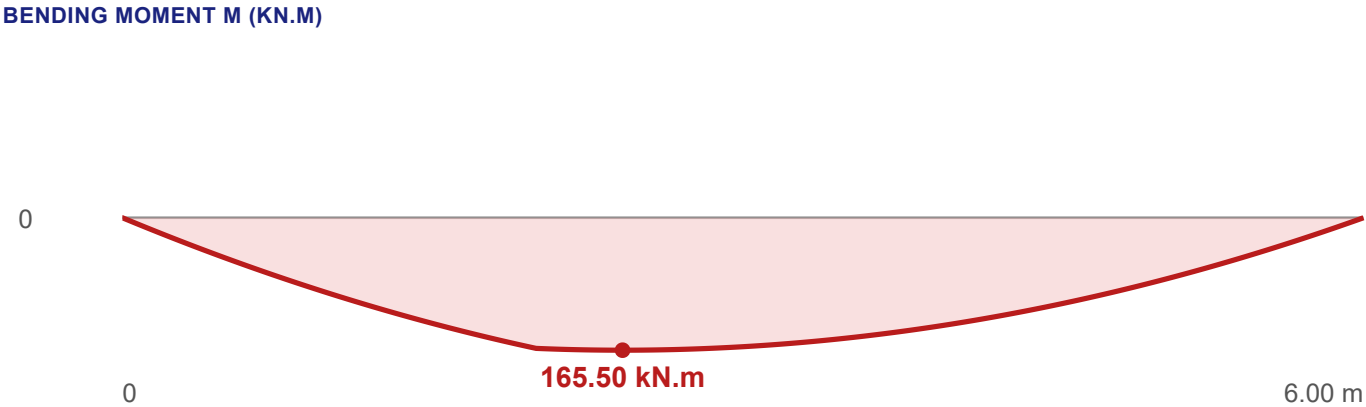
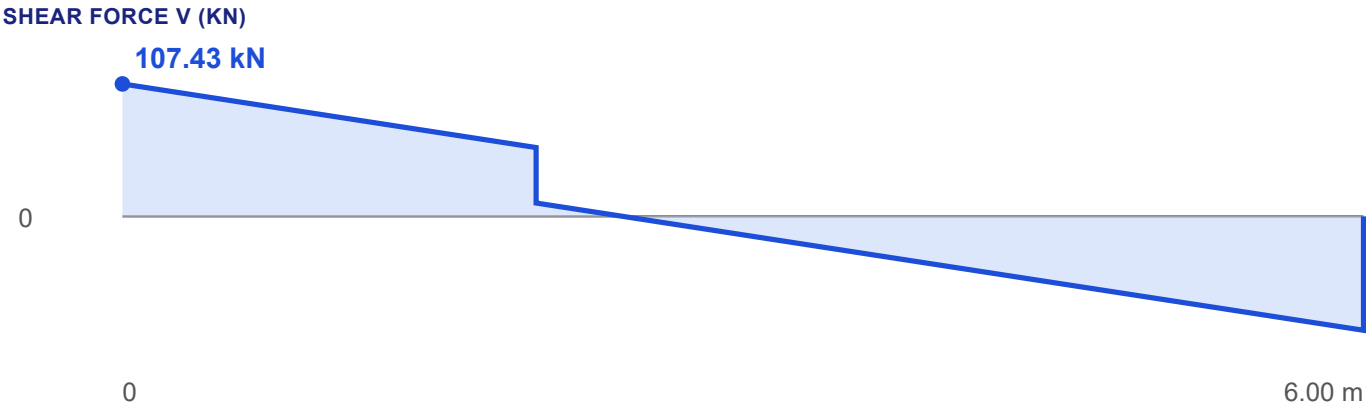
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

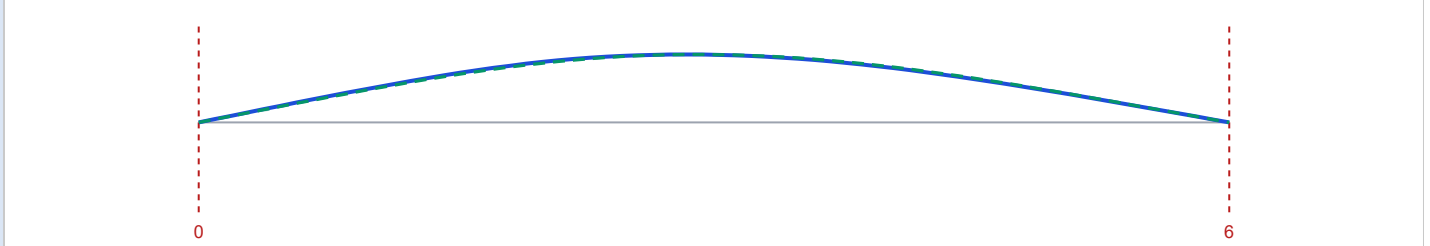
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	30.940 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 30.940}$	1.420	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.420, 1.644, 0.76	0.366	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.368 \times 227.0 \times 275 \leq 62.425 =$	22.992 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 22.992	7.198	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 30.940 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 30.940 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	M _{y,Ed} /M _{b,Rd}	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	7.198	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L4 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

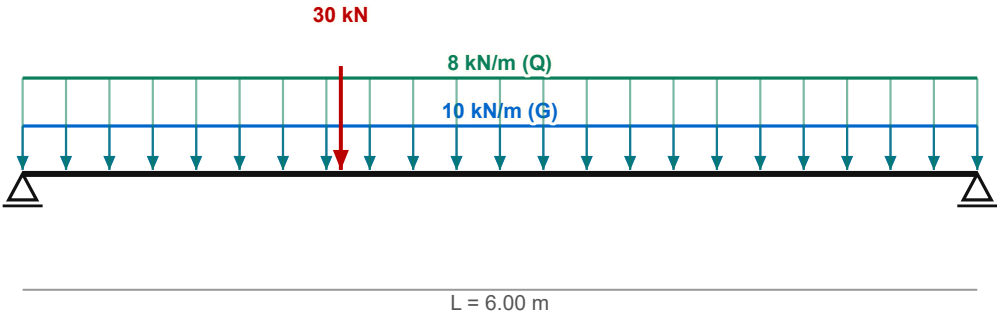
Member Loading and Member Forces

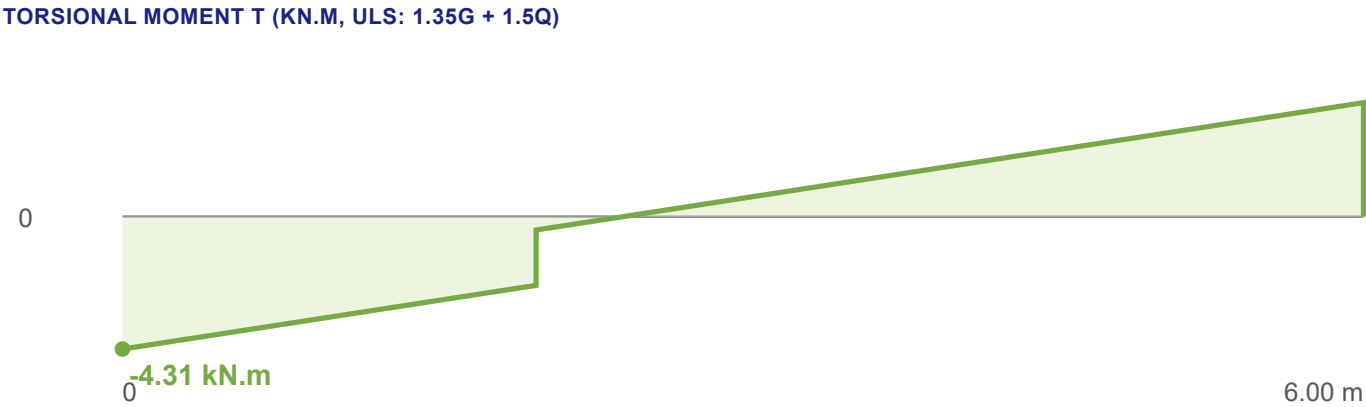
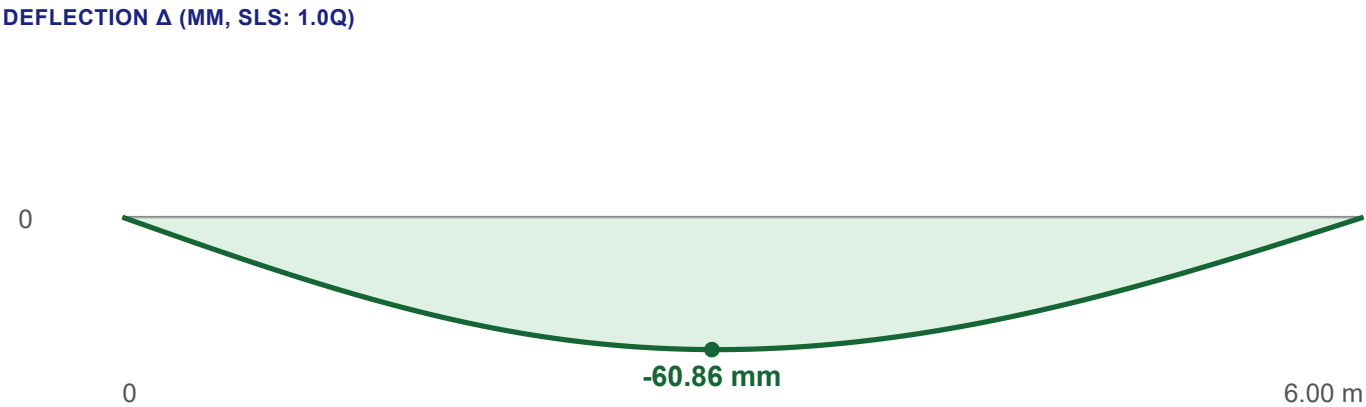
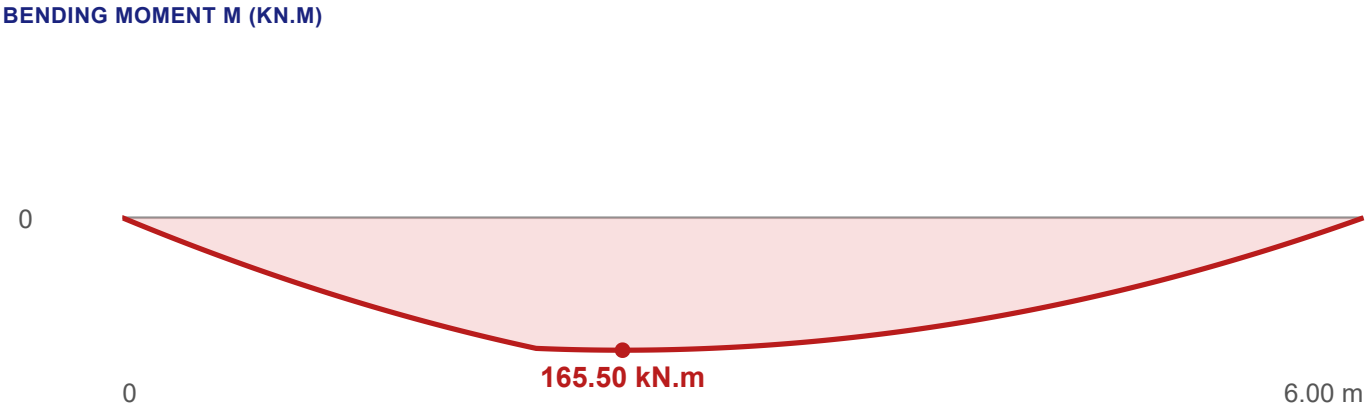
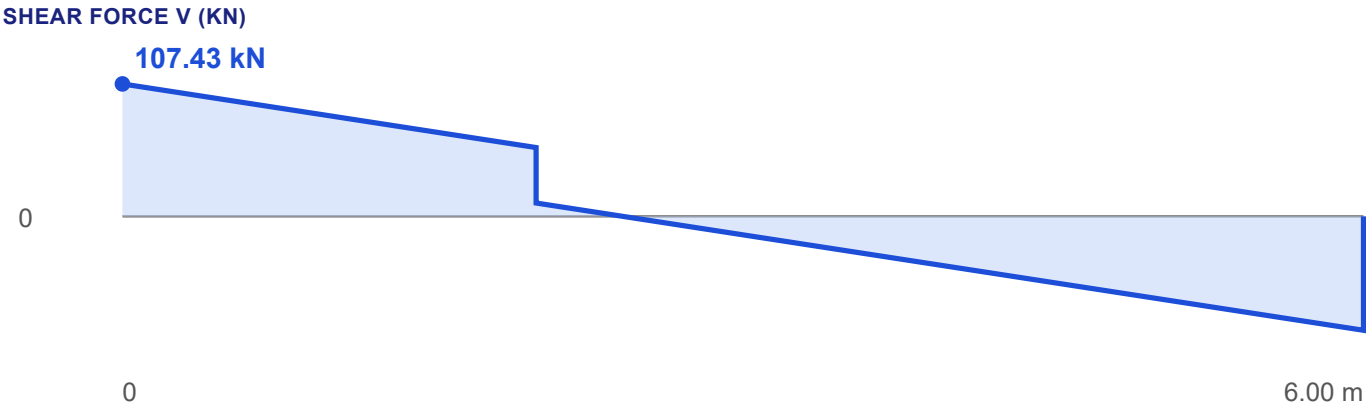
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the	—	not used
C_1 basis	P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C_1 channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		P362

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
Z_g	$Z_g = +100$ mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered	load height	
$\lambda_{LT} = (L_e/i_z)/\kappa$	(6000/23.9)/96 (S275)	2.615	P362 channel
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	165.504 / 8.806	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 9.128$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y/4$) [verify]		
End torques T_t	St Venant part $GI_t \phi'$ at $x = 0 / x = L$; total $T = GI_t \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

τ_t , τ_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² $= 107.430$ kN		P385
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$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.099$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	0.2787 rad = $15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with M_{z,tot}); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: M_{y,Ed} reaches the elastic critical moment M_{cr} (M_{y,Ed} >= M_{cr}): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L5 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
PFC 200 x 75 x 23, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +100 mm

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L5 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

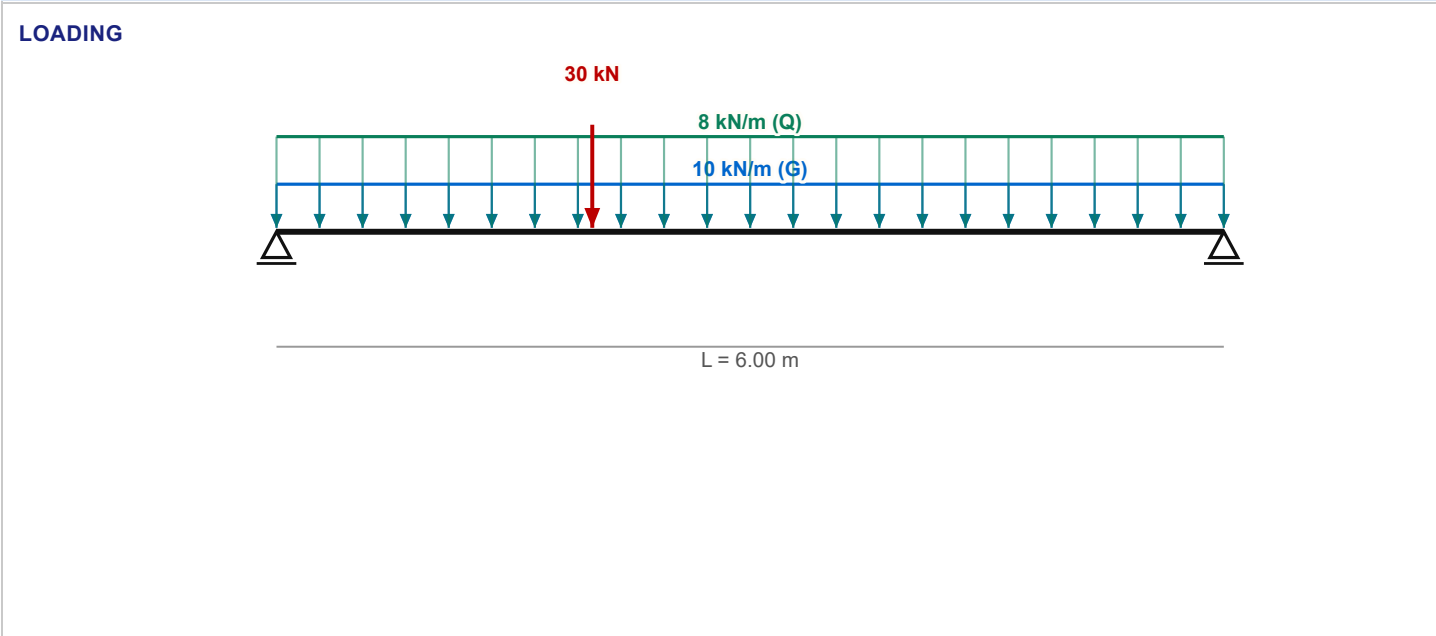
End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

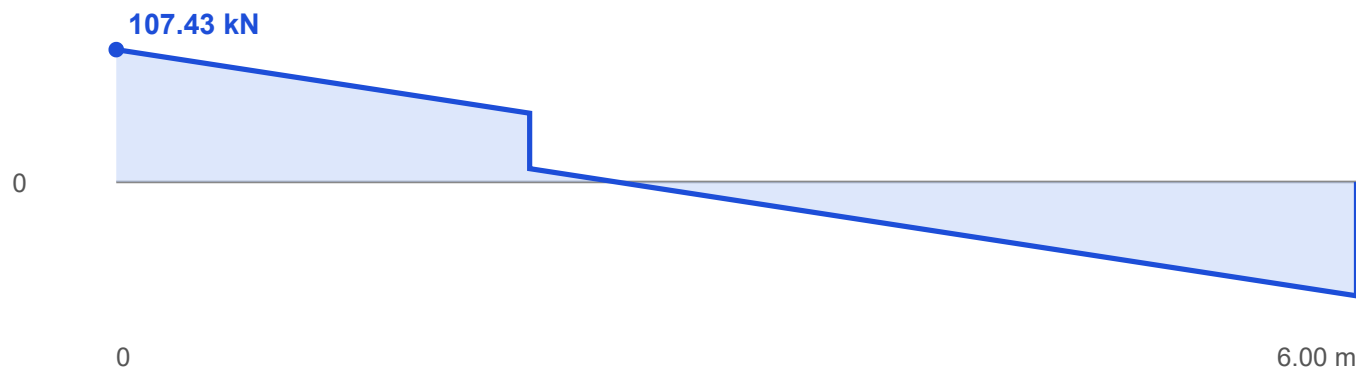
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 — G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	150.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

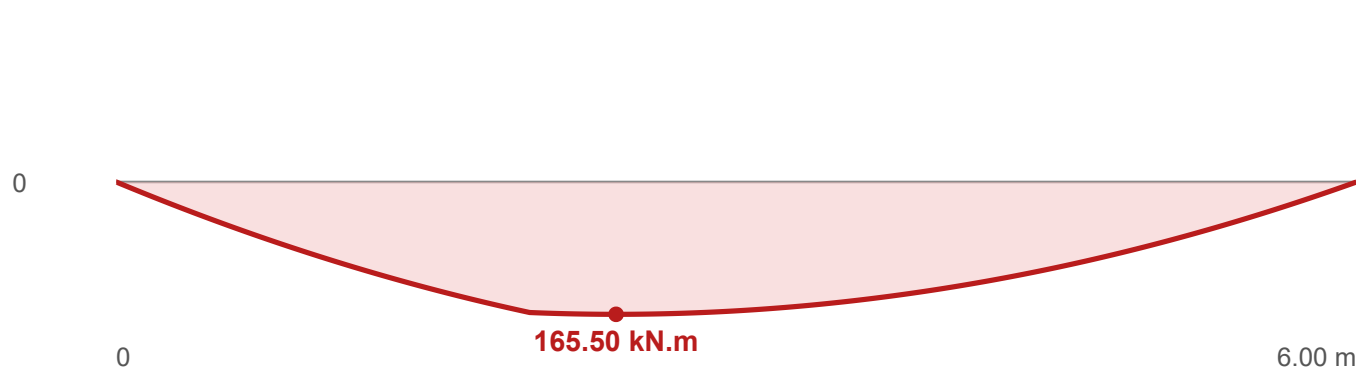
Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK



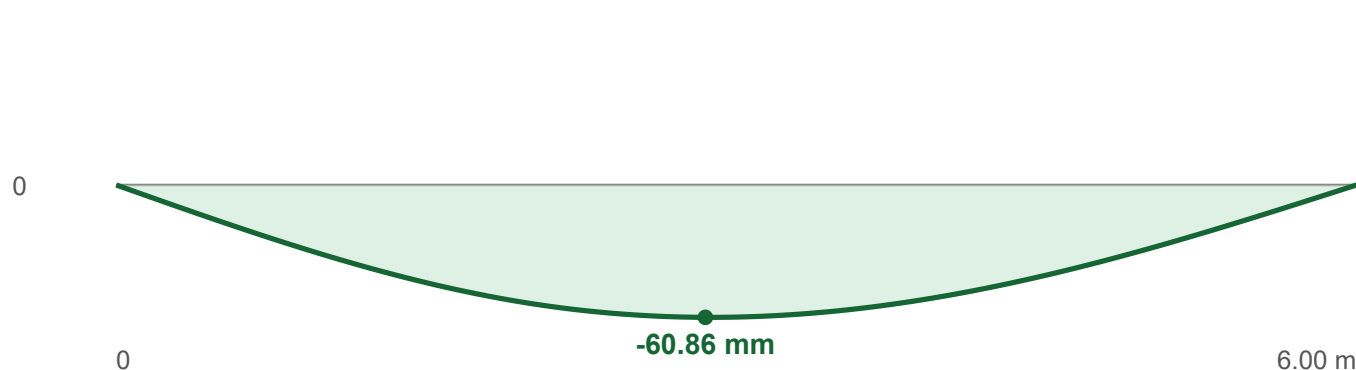
SHEAR FORCE V (kN)



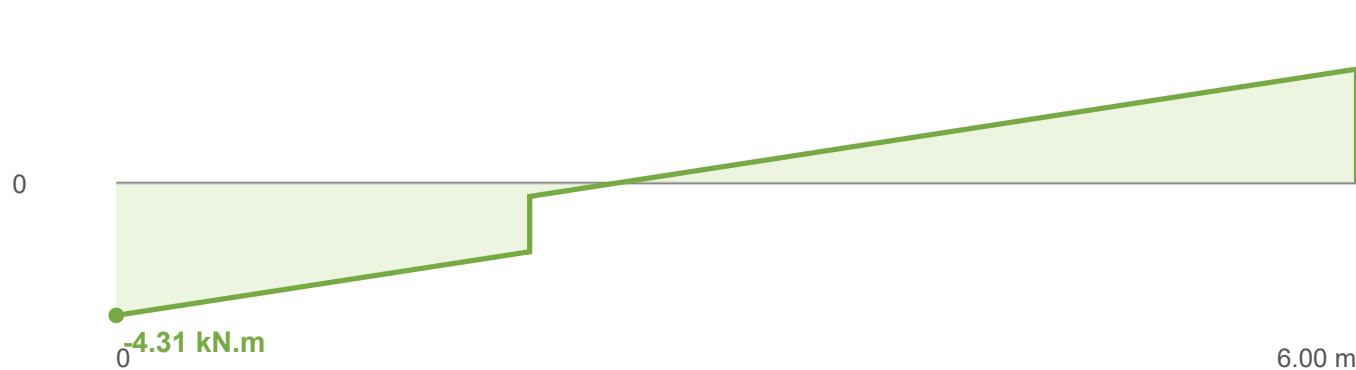
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 150.00, 165.50, 0.00		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.801$, $\psi = -0.635$; limits 42.1ε / 48.5ε / 91.2ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3}$ x 1)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	29.90 x 275 / 1 = (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 822.250 =	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	0.182 + 165.504 / 62.425 + 0 =	2.834	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\phi = 0.66174$ rad, $\phi' = 3.186e-5$ rad/mm, $\phi'' = 1.851e-7$ rad/mm ² , $\phi''' = 1.535e-10$ rad/mm ³ ; $M_z = \phi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 50.17 + 844.27 + 3239.73 + 156.69 = 4290.86$ N/mm ² ; $\tau = G t_f \phi' = 32.26$ N/mm ² ; $(4290.86/275)^2 + 3(32.26/275)^2 =$	243.499	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 50.17 + 844.27 + 3239.73 + 156.69 = 4290.86$, $\tau = 32.26 \rightarrow 243.499$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 50.17 + 791.50 + 470.81 = 1312.48$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 22.828$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 791.50 + 1591.28 + 105.76 = 2538.71$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 85.270$; P3 web-flange junction, web: $\sigma_x = 50.17 + 738.73 + 1591.28 + 105.76 = 2485.94$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 81.733$; P4 web mid-depth: $\sigma_x = 50.17 + 1591.28 = 1641.45$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 35.642$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; S_{w1}, S_{w2}, S_{w3} = P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	275 x 75.00 / (275 x 6.00) = 12.500; 0.02 x (175.00 / 12.50) ² = 3.920 if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	0.9 x 5.429 x 210000 x 6.00 ³ / 175.00	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	5.429 x 210000 x 6.00 ² / (2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	0.5 / 0.433 = 1.154 $\rightarrow 1.000$; 1.000 x 144.19	144.19 mm	6.4(1)

$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 144.19 x 6.00/1.0	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.597 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.427	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.427	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.770	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.382	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10 / 1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10 / 1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\bar{\lambda}_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10 / 1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
$N_{Ed} / N_{b,T,Rd}$	$150.000 / 408.410 =$ (the lower of χ_T and the flexural χ feeds $U_{N,y}$, $U_{N,z}$)	0.367	OK
$N_{Ed} / N_{b,Rd}$	$150.000 / 83.643 =$ (min of $N_{b,y,Rd}$, $N_{b,z,Rd}$, $N_{b,T,Rd}$)	1.793	Warning

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

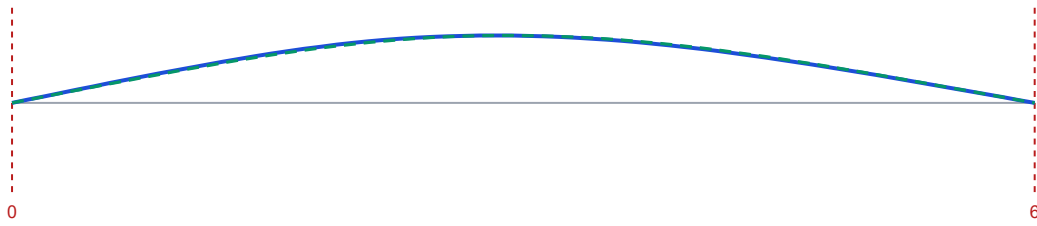
$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.14$, $\psi = 0.892$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 109.717 twist $\phi \cdot M_y = 109.717$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.14$, $\psi = 0.892$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	30.940 kN.m	converged

$\lambda_{LT} = \sqrt{W_{fy}/M_{cr}}$	$\sqrt{227.0 \times 275 / 30.940}$	1.420	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.420, 1.644, 0.76	0.366	Curve d
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.368 \times 227.0 \times 275 \leq 62.425 =$	22.992 kN.m	
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 22.992$	7.198	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.84$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 408.410$ ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 83.643$ (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	$165.504 / 22.992$ ($M_{y,Rk} = 62.425$)	7.198	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$(0.000 \text{ imposed} + 109.717 \text{ twist } \phi \cdot M_y = 109.717) / 9.295$	11.804	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 7.198 + 2.076 \times 11.804$	33.422	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$1.793 + 0.880 \times 7.198 + 2.076 \times 11.804$	32.636	Warning

Torsion Design

J, H, a, Q_f, Q_w	$11.6 \text{ cm}^4, 0.0107 \text{ dm}^6, 489 \text{ mm}, \text{---}, \text{---}$		P385 App A
W_{n0}, S_{w1}	$40.3 \text{ cm}^2, 115 \text{ cm}^4$; $W_{n2} = 27.2 \text{ cm}^2$, $S_{w2} = 62.5$, $S_{w3} = 46.3 \text{ cm}^4$, $e_0 = 29.1$, $e_{sc} = 50.9 \text{ mm}$		P385 App A

Torsion Bending Design @ 2.415 m

$T_{Ed} \text{ (max)}$	combination ULS: $1.35G + 1.5Q$	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
$\phi_{max} \text{ (ULS)}$	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	$0.6646 \text{ rad} = 38.08^\circ$	
$B_{Ed} = EI_w \phi'' \text{ (max)}$	@ $x = 2 \text{ m}$	0.683 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415 \text{ m}$: $165.503, 2.315, 109.220 \text{ kN.m}$; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty$ ($M_{y,Ed} \geq M_{cr} = 30.940 \text{ kN.m}$)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 30.940 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y/4$) [verify]		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6

End torques T_t	St Venant part $GI_T\phi'$ at $x = 0 / x = L$; total $T = GI_T\phi' - EI_w\phi''' = 4.307 / 3.707$ kN.m																
Torsion Shear Design @ 0 m																	
T_t, τ_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN										P385						
$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.099$										6.2.7(9)						
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 20.965										Warning						
Deflection Check - Load Case 1 (SLS: 1.0Q)																	
In-span $\delta \leq \text{Span}/360$	60.86 $\leq$ 6000 / 360 = 16.67 mm @ $x = 2.85$ m										Warning						
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	0.2787 rad = 15.97° $\leq$ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)										> 2° advisory						
N _{Ed} /N _{pl.Rd}	UNyz	UMyz	Ax+M _{6.0}	Ax+M _{6.0}	Deflection V/Vpl	MA/Mc	LTB	M-V	N _{b.T}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	
0.182	2.834	1.793	11.804	33.422	32.636	6.52	0.505	2.651	7.198	2.662	0.367	0.452	1.770	0.367	243.499	124.99	0.00243.49

FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 243.499

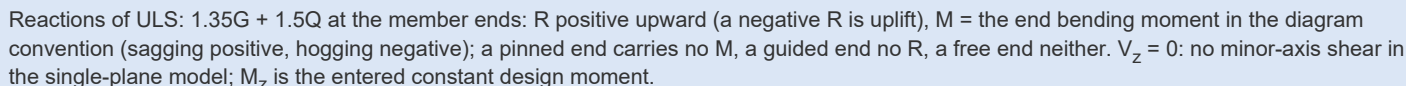
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

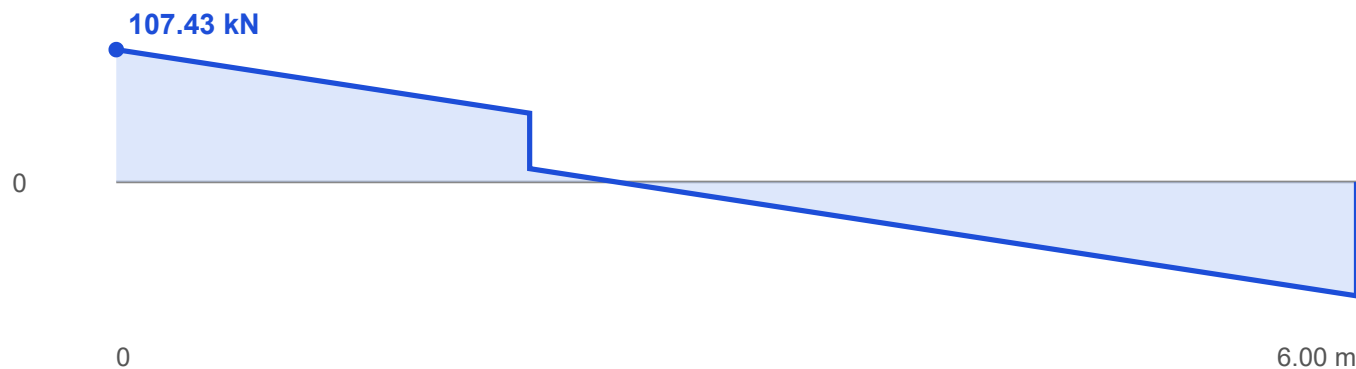
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

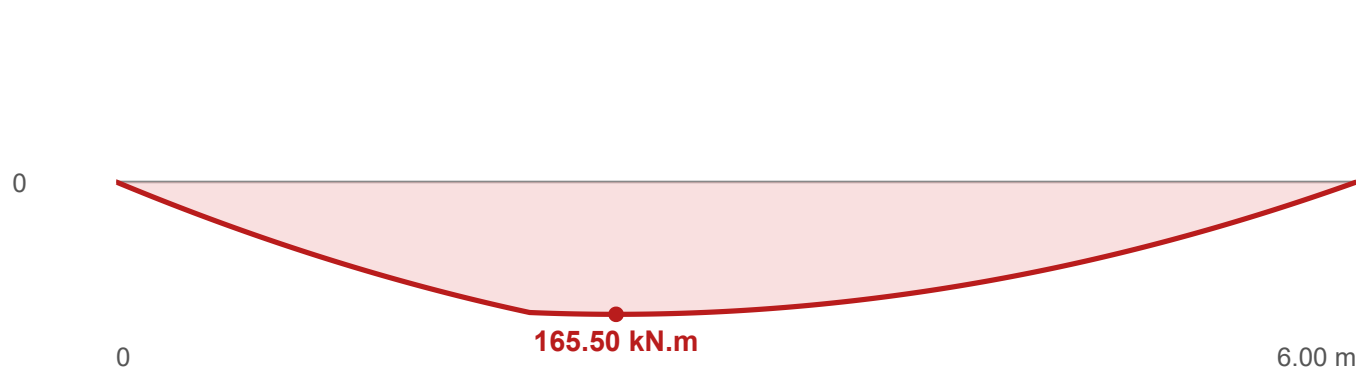
PFC 200 x 75 x 23, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +100 mm



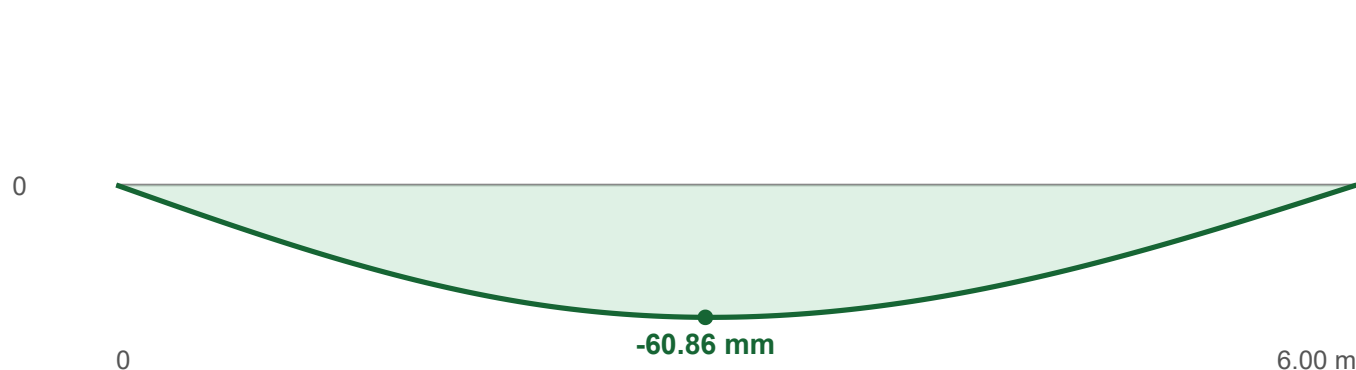
SHEAR FORCE V (kN)



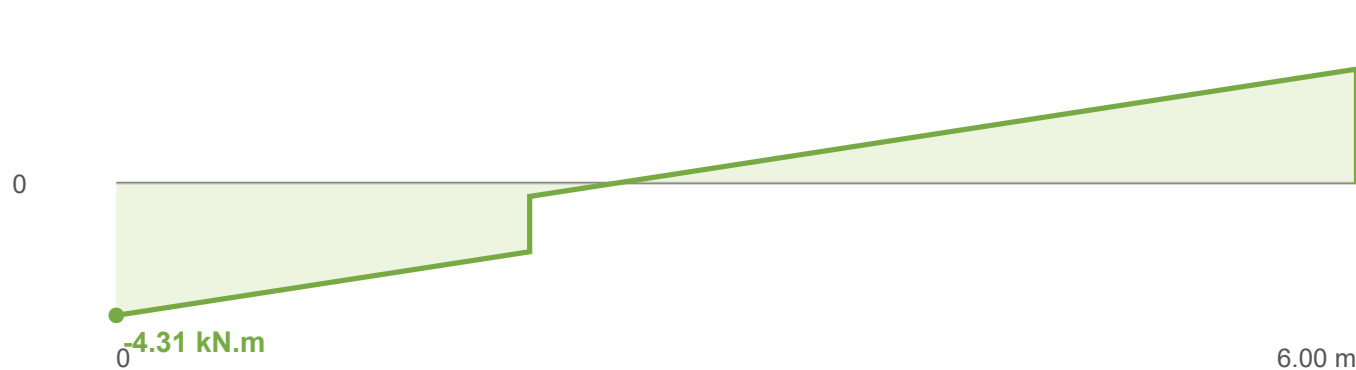
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 150.00, 165.50, 0.00		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.801$, $\psi = -0.635$; limits 42.1ε / 48.5ε / 91.2ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd} @ x$	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$29.90 \times 275 / 1 =$ (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 822.250 =$	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd} \text{ (channel, no } M_N \text{ reduction)}$	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	$0.182 + 165.504 / 62.425 + 0 =$	2.834	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476 \text{ kN.m}$, $V = 1.185 \text{ kN}$, $\phi = 0.66174 \text{ rad}$, $\phi' = 3.186e-5 \text{ rad/mm}$, $\phi'' = 1.851e-7 \text{ rad/mm}^2$, $\phi''' = 1.535e-10 \text{ rad/mm}^3$; $M_z = \phi M_y = 109.503 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 50.17 + 844.27 + 3239.73 + 156.69 = 4290.86 \text{ N/mm}^2$; $\tau = G t_f \phi' = 32.26 \text{ N/mm}^2$; $(4290.86 / 275)^2 + 3(32.26 / 275)^2 =$	243.499	Warning 6.2.7(5)
$\sigma_x, \tau, (6.1) \text{ at the 5 section points}$	P1 flange tip: $\sigma_x = 50.17 + 844.27 + 3239.73 + 156.69 = 4290.86$, $\tau = 32.26 \rightarrow 243.499$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 50.17 + 791.50 + 470.81 = 1312.48$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 22.828$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 791.50 + 1591.28 + 105.76 = 2538.71$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 85.270$; P3 web-flange junction, web: $\sigma_x = 50.17 + 738.73 + 1591.28 + 105.76 = 2485.94$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 81.733$; P4 web mid-depth: $\sigma_x = 50.17 + 1591.28 = 1641.45$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 35.642$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99 \text{ mm}$ from the toe, $y_1 = 7.31 \text{ mm}$; $S_{w1}, S_{w2}, S_{w3} = \text{P385 Table A.2 points 1, 2, 3}$)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)

$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 144.19 x 6.00/1.0	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.597 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.427	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.427	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.770	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.382	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10 / 1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\bar{\lambda}_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10 / 1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\bar{\lambda}_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10 / 1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
$N_{Ed} / N_{b,T,Rd}$	$150.000 / 408.410 =$ (the lower of χ_T and the flexural χ feeds $U_{N,y}, U_{N,z}$)	0.367	OK
$N_{Ed} / N_{b,Rd}$	$150.000 / 83.643 =$ (min of $N_{b,y,Rd}, N_{b,z,Rd}, N_{b,T,Rd}$)	1.793	Warning

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets $\bar{\lambda}_{LT} = (L_e/i_z)/\kappa$ without C_1	—	not used
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5$ kN·m		P362
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.14, \psi = 0.892, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M_{z,Ed} = 0.000$ imposed + 109.717 twist ϕ . $M_y = 109.717$; $\psi = 1$ for the imposed constant M_z , the twist part taken at $C_{mz} = 1.0$ (Table B.3 upper bound, conservative)	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.14, \psi = 0.892, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	1 x 6 =	6 m
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z_g	$z_g = +100$ mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered		load height
$\lambda_{LT} = (L_e/i_z)/\kappa$	(6000/23.9)/96 (S275)	2.615	P362 channel
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W_{f,y}/\lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	165.504 / 8.806	18.794	Warning

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	165.504 / 8.806 ($M_{y,Rk} = 62.425$)	18.794	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(0.000 imposed + 109.717 twist ϕ . $M_y = 109.717$) / 9.295	11.804	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.367+1.188 \times 18.794+2.076 \times 11.804$	47.196	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$1.793+0.880 \times 18.794+2.076 \times 11.804$	42.845	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty$ ($M_{y,Ed} \geq M_{cr} = 9.128$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b b^2 f_y / 4$) [verify]		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T_t, T_w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, V = 195.04 / 5.22 N/mm ² = 107.430 kN	P385
$V_{pl.T.Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ x = 0 m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 20.965	5.124 Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	0.2787 rad = $15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

N_Ed/N_(dEd)	UNyz	UMyz	Ax+M_6.0k+M_6.0k	Deflection V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.182	2.834	1.793	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794	18.794

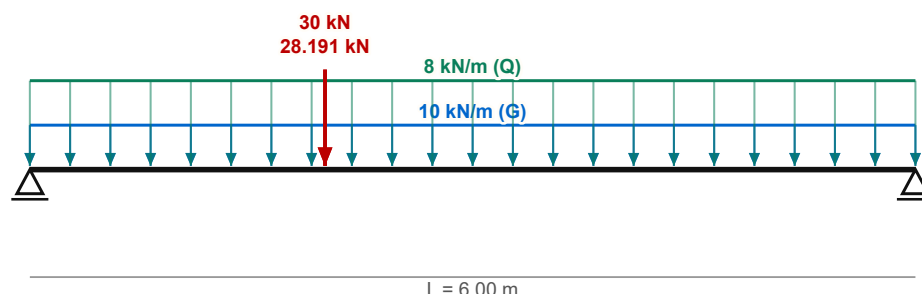
FAIL — governing Elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 243.499

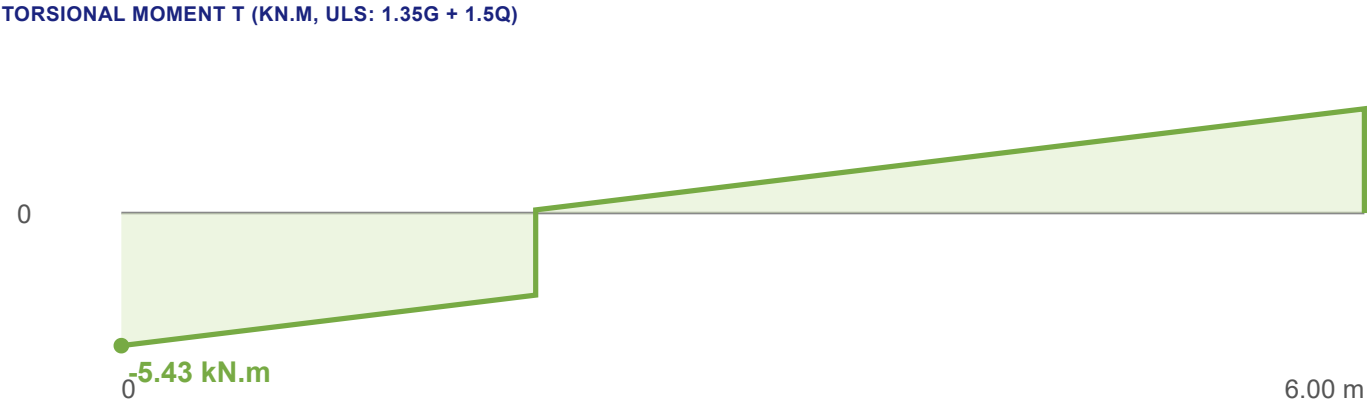
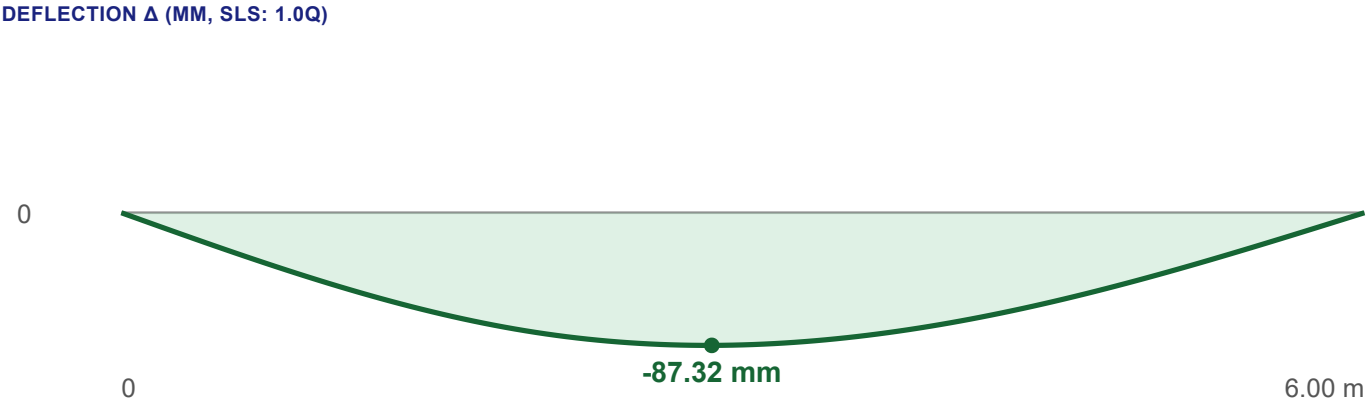
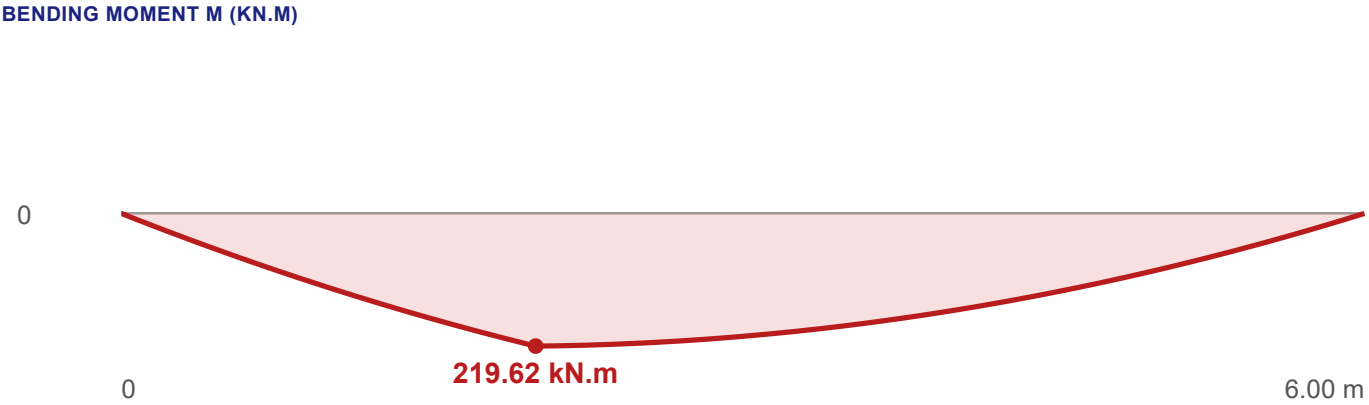
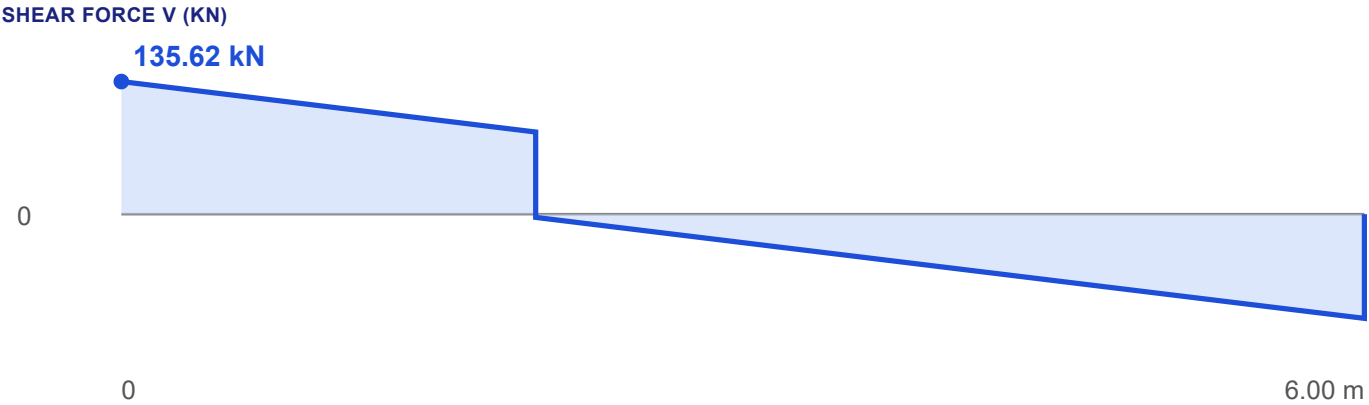
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 150.00, 219.62, 20.52	(Axial: Non-Slender)	Class 1
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$135.621 / 212.754 =$	0.637	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$3.286 / 212.754 =$	0.015	High Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$219.622 / 62.425 =$	3.518	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 3.286 > V_{pl,Rd} = 0.000$: pure shear governs	3285542509.905	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 308.016 = (A_{v,z} = 1940 \text{ mm}^2; \text{ no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 60.6 / 1$	16.665 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$29.90 \times 275 / 1 = (\text{No bearing / block tearing design})$	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 822.250 =$	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	$0.182 + 219.622 / 62.425 + 20.521 / 16.665 =$	4.932	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.141 m, P1 flange tip	@ x = 2.141 m (ULS: 1.35G + 1.5Q): $M_y = 218.901 \text{ kN.m}$, $V = 6.927 \text{ kN}$, $\varphi = 0.83929 \text{ rad}$, $\varphi' = 1.094 \text{e-4 rad/mm}$, $\varphi'' = 3.934 \text{e-7 rad/mm}^2$, $\varphi''' = 5.793 \text{e-10 rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \varphi M_y = 20.521 + 183.721 = 204.242 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_{z,tot} / W_{el,z} + E W_{n0} \varphi'' = 50.17 + 1116.84 + 6042.66 + 332.98 = 7542.65 \text{ N/mm}^2$; $\tau = G t \varphi' = 110.73 \text{ N/mm}^2$; $(7542.65 / 275)^2 + 3(110.73 / 275)^2 =$	752.771	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 50.17 + 1116.84 + 6042.66 + 332.98 = 7542.65$, $\tau = 110.73 \rightarrow 752.771$ (governs); P1b flange at $W_{n0} = 0$: $\sigma_x = 50.17 + 1047.04 + 878.14 = 1975.35$, $\tau = 1.42 + 110.73 + 11.19 = 123.34 \rightarrow 52.200$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 1047.04 + 2968.01 + 224.74 = 4289.96$, $\tau = 2.48 + 110.73 + 6.08 = 119.29 \rightarrow 243.920$; P3 web-flange junction, web: $\sigma_x = 50.17 + 977.24 + 2968.01 + 224.74 = 4220.16$, $\tau = 5.18 + 53.15 + 12.67 = 71.00 \rightarrow 235.701$; P4 web mid-depth: $\sigma_x = 50.17 + 2968.01 = 3018.18$, $\tau = 6.53 + 53.15 + 9.39 = 69.07 \rightarrow 120.644$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99 \text{ mm}$ from the toe, $y_1 = 7.31 \text{ mm}$; $S_{w1}, S_{w2}, S_{w3} = \text{P385 Table A.2 points 1, 2, 3})$		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 135.621 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.570	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.570 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.716 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q\text{)} \quad 0.511 \quad \text{OK 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	135.621	ULS: 1.35G + 1.5Q	0.570	0.511	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.448	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	106.525	ULS: 1.35G + 1.5Q	0.448	0.424	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y · L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A · f _y / N _{cr})	√(29.90x275/1128.42 (N _{cr,y} = π²EI _y /L _{ey} ²))	0.852	
N _{b,y,Rd} = Area · χ _{f,y} / γ _{M1}	29.90x0.629x275/10/1 =	517.570 kN	Curve c
L _{ez} = K _z · L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A · f _y / N _{crz})	√(29.90x275/97.87 (N _{cr,z} = π²EI _z /L _{ez} ²))	2.892	
N _{b,z,Rd} = Area · χ _{f,y} / γ _{M1}	29.90x0.102x275/10/1 =	83.643 kN	Curve c
i ₀ ² = i _y ² + i _z ² + y ₀ ²	81.10² + 23.90² + 50.90² (y ₀ = e _{sc} = shear centre to centroid, SCI P385 Table A.3 (Blue Book e ₀ + c _y))	9739.2 mm²	6.3.1.4
N _{cr,T} = (GI _T + π²EI _w /L _T ²) / i ₀ ²	(81000 x 11.6e4 + π² x 210000 x 0.0107e12/6000²) / 9739.2 (I _T , I _w : SCI P385 Table A.3; L _T = 6 m = L _{cr,y} (no intermediate twist restraint))	1028.010 kN	6.3.1.4
N _{cr,TF} = (N _{cr,y} + N _{cr,T}) / 2β [1 - √(1 - 4βN _{cr,y} N _{cr,T} / (N _{cr,y} + N _{cr,T})²)]	N _{cr,y} = 1128.425 (flexure about the axis of symmetry y-y, L _{ey}), β = 1 - (y ₀ /i ₀)² = 0.734; N _{cr} = min(N _{cr,T} , N _{cr,TF}) = 709.070 (TF mode)	709.070 kN	6.3.1.4
λ _T = √(A · f _y / N _{cr})	√(29.90x275/709.070)	1.077	
N _{b,T,Rd} = Area · χ _T · f _y / γ _{M1}	29.90x0.497x275/10/1 = (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
N _{Ed} / N _{b,T,Rd}	150.000 / 408.410 = (the lower of χ _T and the flexural χ feeds U _{N,y} , U _{N,z})	0.367	OK
N _{Ed} / N _{b,Rd}	150.000 / 83.643 = (min of N _{b,y,Rd} , N _{b,z,Rd} , N _{b,T,Rd})	1.793	Warning

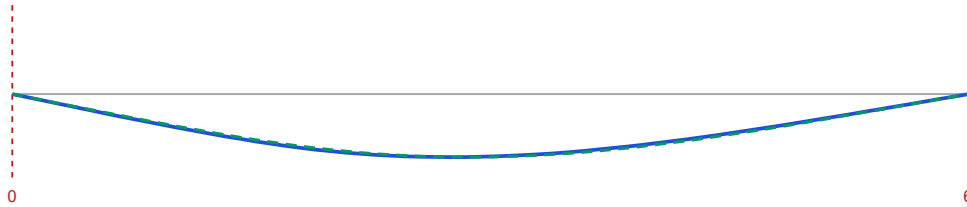
Equivalent Uniform Moment Factors C₁, C_{mLT}, C_{mz}, and C_{my}

C ₁ = M _{cr} / M _{cr,uniform}	37.280 / 31.304 kN.m (shape only: z _g = z _j = 0)	1.191	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 203.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=174.4, M(L/2)=203.4, M(3L/4)=130.8, M _{max} =219.6 kN·m	1.218	Serna
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.43, ψ = 0.817, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 20.521 imposed + 185.324 twist φ. M _y = 205.845; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.43, ψ = 0.817, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 100, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	32.041 kN.m	converged
λ _{LT} = √(W _f · f _y / M _{cr})	√(227.0 x 275 / 32.041)	1.396	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.396, 1.609, 0.76	0.374	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.374, 1.396, 0.916, 0.988	0.379	6.3.2.3
M _{b,Rd} = χW _{pl,y} · f _y ≤ M _{c,y,Rd}	0.379 x 227.0 x 275 ≤ 62.425 =	23.659 kN.m	
M _{y,Ed} / M _{b,Rd}	219.622 / 23.659	9.283	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.74$ m)



Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	219.622 / 23.659 ($M_{y,Rk} = 62.425$)	9.283	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 185.324 twist ϕ , $M_y = 205.845$) / 9.295	22.146	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 9.283 + 2.076 \times 22.146$	57.368	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$1.793 + 0.880 \times 9.283 + 2.076 \times 22.146$	55.941	Warning

Torsion Design

J , H , a , Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	5.850 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 217.302 x 852.84e-3 (@ $x = 2.315$ m) = 20.521 + 185.324 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	205.845 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2$ m: 219.622, 5.850, 200.534 kN.m ($M_{z,tot} = 20.521 + 180.013$); $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	25.621	Warning P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 32.041$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$ (A.1) scope	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 32.041$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
Basis with N_{Ed} / $M_{z,Ed}$	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y/4$) [verify] No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0$ / $x = L$; total $T = GI_T \phi' - EI_w \phi''' = 5.435$ / 4.271 kN.m	4.870 / 3.764 kN.m	

Torsion Shear Design @ 0 m

τ_t , τ_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 4.870$ kN.m, $V = 135.621$ kN	251.90 / 5.50 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.000$	0.000 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	135.621 / 0.000	∞	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.32 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	87.32 mm	Warning
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Torq in Case 1 @ 2.34 m: $\theta_{\max} \leq 2.00^\circ$ 0.4098 rad = 23.48° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict) 23.48° > 2° advisory

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/V _{pl}	MA/Mc	LTB	M-V	N _{b,T}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max	
0.182	4.932	1.793	22.146	57.368	55.941	5.239	0.637	3.518	9.283	3285542509.905	2.367	0.570	2.448	0.367	752.779	99.000	99.000	328554

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 3285542509.905

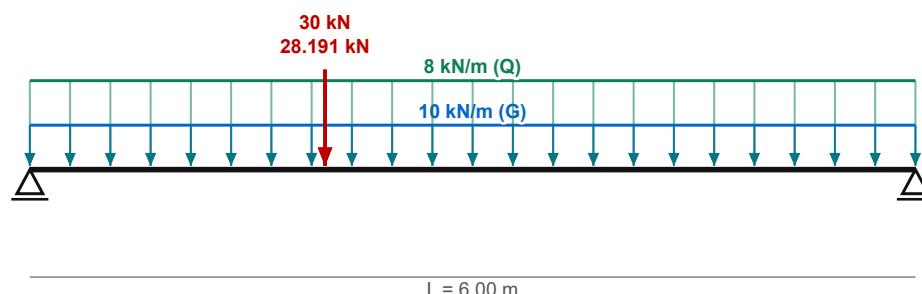
ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

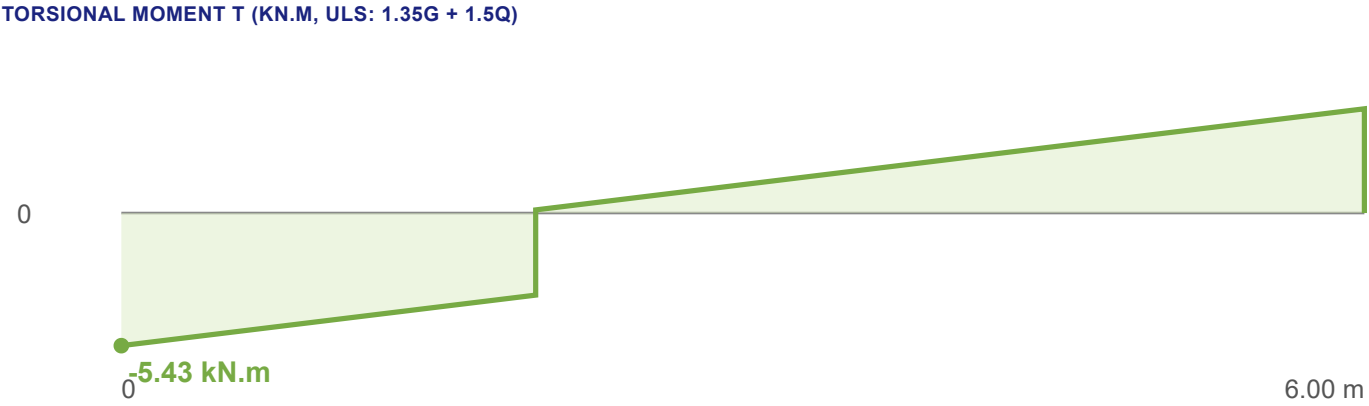
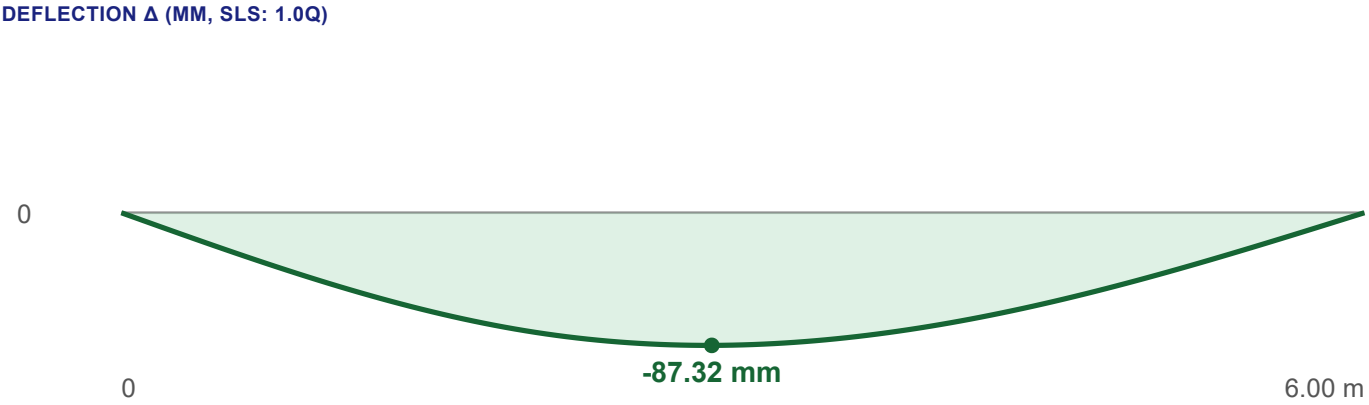
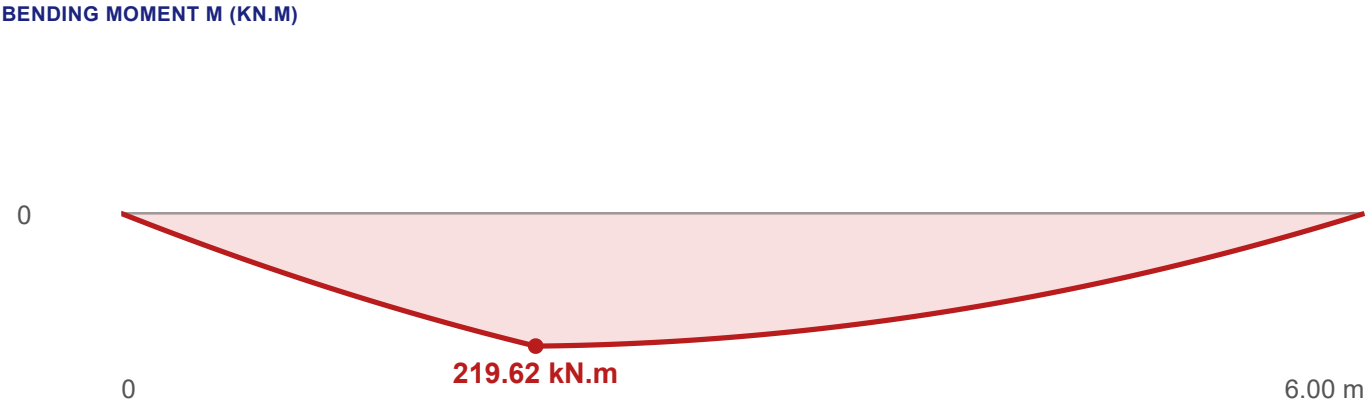
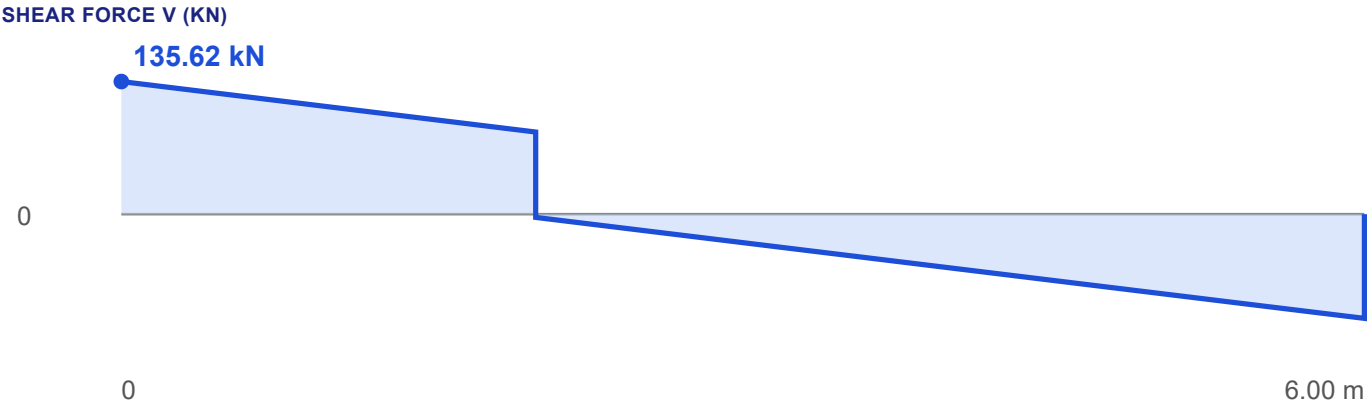
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 23.5^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)			
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 150.00, 219.62, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε		Class 1 Table 5.2
Auto Design Load Cases	1; SLS 1		

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$135.621 / 212.754 =$	0.637	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$3.286 / 212.754 =$	0.015	High Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$219.622 / 62.425 =$	3.518	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 3.286 > V_{pl,Rd} = 0.000$: pure shear governs	3285542509.905	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 308.016 = (A_{v,z} = 1940 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 60.6 / 1$	16.665 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$29.90 \times 275 / 1 = (\text{No bearing / block tearing design})$	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 822.250 =$	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	$0.182 + 219.622 / 62.425 + 20.521 / 16.665 =$	4.932	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.141 m, P1 flange tip	@ x = 2.141 m (ULS: 1.35G + 1.5Q): $M_y = 218.901 \text{ kN.m}$, $V = 6.927 \text{ kN}$, $\varphi = 0.83929 \text{ rad}$, $\varphi' = 1.094\text{e-}4 \text{ rad/mm}$, $\varphi'' = 3.934\text{e-}7 \text{ rad/mm}^2$, $\varphi''' = 5.793\text{e-}10 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \varphi M_y = 20.521 + 183.721 = 204.242 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_{z,tot} / W_{el,z} + E W_{n0} \varphi'' = 50.17 + 1116.84 + 6042.66 + 332.98 = 7542.65 \text{ N/mm}^2$; $\tau = G t \varphi' = 110.73 \text{ N/mm}^2$; $(7542.65 / 275)^2 + 3(110.73 / 275)^2 =$	752.771	Warning 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 50.17 + 1116.84 + 6042.66 + 332.98 = 7542.65$, $\tau = 110.73 \rightarrow 752.771$ (governs); P1b flange at $W_{n0} = 0$: $\sigma_x = 50.17 + 1047.04 + 878.14 = 1975.35$, $\tau = 1.42 + 110.73 + 11.19 = 123.34 \rightarrow 52.200$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 1047.04 + 2968.01 + 224.74 = 4289.96$, $\tau = 2.48 + 110.73 + 6.08 = 119.29 \rightarrow 243.920$; P3 web-flange junction, web: $\sigma_x = 50.17 + 977.24 + 2968.01 + 224.74 = 4220.16$, $\tau = 5.18 + 53.15 + 12.67 = 71.00 \rightarrow 235.701$; P4 web mid-depth: $\sigma_x = 50.17 + 2968.01 = 3018.18$, $\tau = 6.53 + 53.15 + 9.39 = 69.07 \rightarrow 120.644$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99 \text{ mm}$ from the toe, $y_1 = 7.31 \text{ mm}$; S_{w1}, S_{w2}, S_{w3} = P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(5)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 135.621 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.570	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.570 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.716 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q\text{)} \quad 0.511 \quad \text{OK 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	135.621	ULS: 1.35G + 1.5Q	0.570	0.511	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.448	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	106.525	ULS: 1.35G + 1.5Q	0.448	0.424	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y · L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A · f _y / N _{cr})	√(29.90x275/1128.42 (N _{cr,y} = π ² EI _y /L _{ey} ²))	0.852	
N _{b,y,Rd} = Area · χ _{f,y} / γ _{M1}	29.90x0.629x275/10/1 =	517.570 kN	Curve c
L _{ez} = K _z · L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A · f _y / N _{crz})	√(29.90x275/97.87 (N _{cr,z} = π ² EI _z /L _{ez} ²))	2.892	
N _{b,z,Rd} = Area · χ _{f,y} / γ _{M1}	29.90x0.102x275/10/1 =	83.643 kN	Curve c
i ₀ ² = i _y ² + i _z ² + y ₀ ²	81.10 ² + 23.90 ² + 50.90 ² (y ₀ = e _{sc} = shear centre to centroid, SCI P385 Table A.3 (Blue Book e ₀ + c _y))	9739.2 mm ²	6.3.1.4
N _{cr,T} = (GI _T + π ² EI _w /L _T ²)/i ₀ ²	(81000 x 11.6e4 + π ² x 210000 x 0.0107e12/6000 ²)/9739.2 (I _T , I _w : SCI P385 Table A.3; L _T = 6 m = L _{cr,y} (no intermediate twist restraint))	1028.010 kN	6.3.1.4
N _{cr,TF} = (N _{cr,y} + N _{cr,T})/2β [1 - √(1 - 4βN _{cr,y} N _{cr,T} /(N _{cr,y} + N _{cr,T}) ²)]	N _{cr,y} = 1128.425 (flexure about the axis of symmetry y-y, L _{ey}), β = 1 - (y ₀ /i ₀) ² = 0.734; N _{cr} = min(N _{cr,T} , N _{cr,TF}) = 709.070 (TF mode)	709.070 kN	6.3.1.4
λ _T = √(A · f _y / N _{cr})	√(29.90x275/709.070)	1.077	
N _{b,T,Rd} = Area · χ _T · f _y / γ _{M1}	29.90x0.497x275/10/1 = (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
N _{Ed} /N _{b,T,Rd}	150.000 / 408.410 = (the lower of χ _T and the flexural χ feeds U _{N,y} , U _{N,z})	0.367	OK
N _{Ed} /N _{b,Rd}	150.000 / 83.643 = (min of N _{b,y,Rd} , N _{b,z,Rd} , N _{b,T,Rd})	1.793	Warning

Equivalent Uniform Moment Factors C₁, C_{mLT}, C_{mz}, and C_{my}

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 203.4, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=174.4, M(L/2)=203.4, M(3L/4)=130.8, M _{max} =219.6 kN·m		P362
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.43, ψ = 0.817, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 20.521 imposed + 185.324 twist φ · M _y = 205.845; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.43, ψ = 0.817, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
z _g	z _g = +100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L _E switch); the shear-centre M _{cr} route is not offered		load height
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} · f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _{f,y} /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	219.622 / 8.806	24.940	Warning

Buckling Resistance

$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	219.622 / 8.806 ($M_{y,Rk} = 62.425$)	24.940	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	(20.521 imposed + 185.324 twist $\phi \cdot M_y = 205.845$) / 9.295	22.146	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.367 + 1.188 \times 24.940 + 2.076 \times 22.146$	75.965	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$1.793 + 0.880 \times 24.940 + 2.076 \times 22.146$	69.726	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
B _{Ed} = EI _w ϕ'' (max)	@ x = 2 m	1.097 kN.m ²	
M _{w,Ed} = EI _w $\phi''/(h-t_f)$	max over span	5.850 kN.m	
M _{z,tot} = M _{z,Ed} + $\phi \cdot M_{y,Ed}$	20.521 + 217.302 x 852.84e-3 (@ x = 2.315 m) = 20.521 + 185.324 (imposed constant M _{z,Ed} + max coincident twist moment)	205.845 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ x = 2 m: 219.622, 5.850, 200.534 kN.m (M _{z,tot} = 20.521 + 180.013); M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	25.621	Warning P385 3.1.2
C _{mz} (EN 1993-6 A.1)	C _{mz} = 1.0: the imposed M _z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
k = k _w · k _{zw} · k _{α}	x x ∞ ($M_{y,Ed} \geq M_{cr} = 9.128$ kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _{z,tot} /M _{z,Rk} + k _w k _{zw} k _{α} M _w /M _{f,Rk} (A.1) scope	k _{α} unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr} EN 1993-6 A
Basis with N _{Ed} / M _{z,Ed}	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _b ² f _y /4) [verify]		EN 1993-1-1, EN 1993-6
End torques T _t	No expression in EN 1993-1-1 or EN 1993-6 combines N _{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to M _{z,Ed} , and EN 1993-6 (A.1) with M _{z,Ed} = M _z + $\phi \cdot M_y$		
	St Venant part GI _T ϕ' at x = 0 / x = L; total T = GI _T ϕ' – EI _w $\phi'' = 5.435$ / 4.271 kN.m	4.870 / 3.764 kN.m	

Torsion Shear Design @ 0 m

T _t , T _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 4.870 kN.m, V = 251.90 / 5.50 N/mm ²		P385
V _{pl,T,Rd} = [$\sqrt{1 - T_t/(1.25f_y/\sqrt{3})} - T_w/(f_y/\sqrt{3})$].V _{pl,Rd}	@ x = 0 m: T _t = 251.90 (T _w = 5.50) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.000	0.000 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	135.621 / 0.000	∞	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	87.32 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	87.32 mm	Warning
Torq in Case 1 @ 2.34 m: $\theta_{max} \leq 2.00^\circ$	0.4098 rad = 23.48° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	23.48°	> 2° advisory

N _{Ed} /N _{pl,Rd}	UNyz	UMyz	Ax+M ₆ /A ₆	Ax+M ₆ /A ₆	Deflection	V/V _{pl}	MA/Mc	LTB	M-V	N _{b,T}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.182	4.932	1.793	24.940	75.965	69.726	2.239	0.637	3.518	24.940	3285542509.905	2.448	0.367	752.779	99.000	99.000	328554	

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 3285542509.905

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 23.5^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L7 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)
Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER
PFC 200 x 75 x 23, 6 m SS, unrestrained; $L7 = L6 +$ "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term ($C2 z_g$); only forces with a lever arm do

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L7 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork) — simply supported

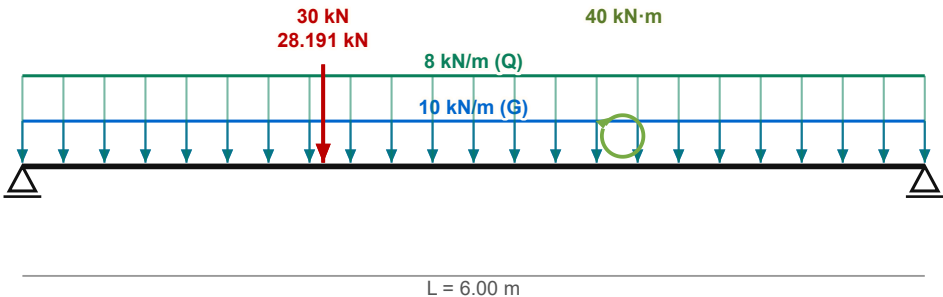
Member Loading and Member Forces

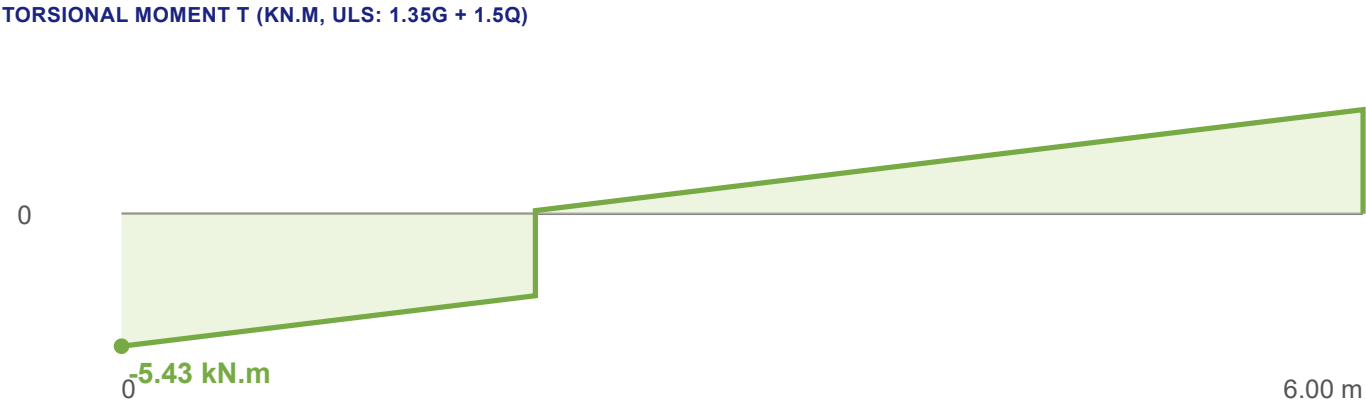
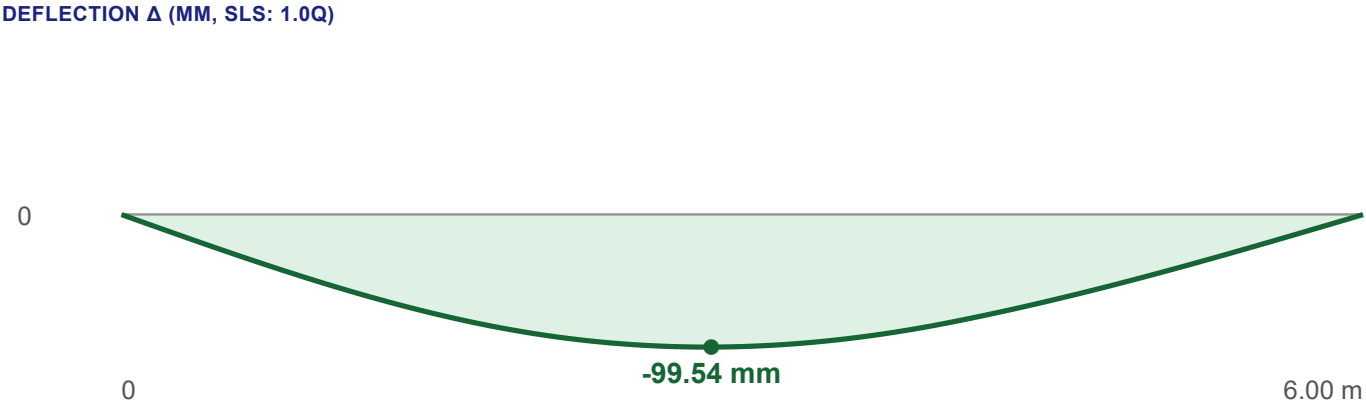
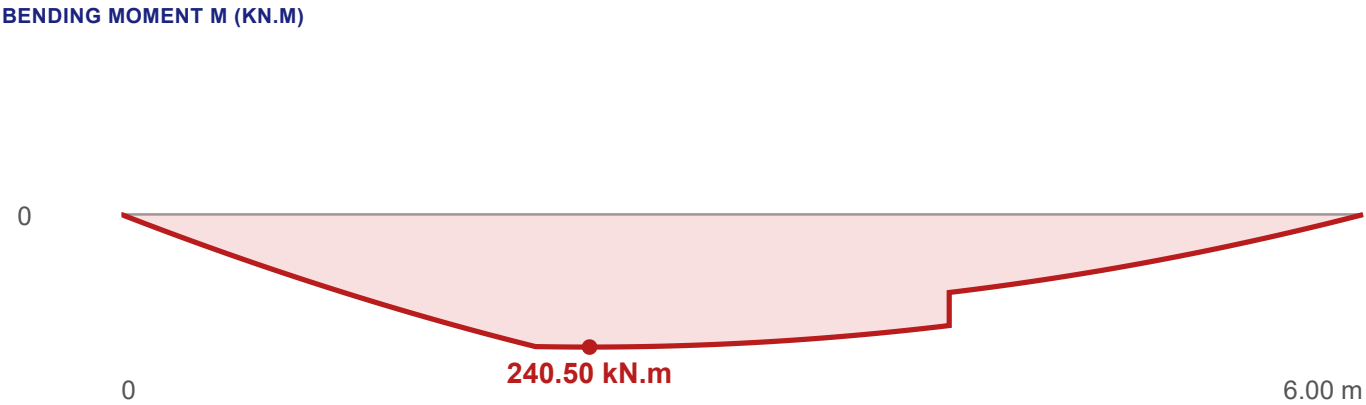
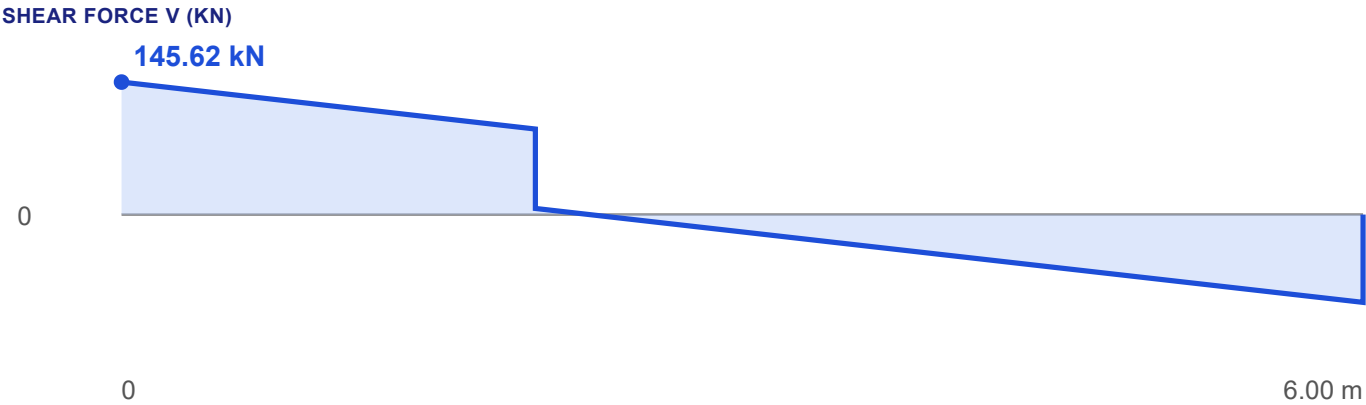
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm $z_g = 100$ mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm $z_g = 100$ mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm $z_g = 100$ mm (kN)
Q PY 28.191 @ 2 m e = 40 mm $z_g = 100$ mm (kN)
Q M 40.000 @ 4 m (kN.m)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
 $Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]
 $Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered $Y_{G,inf}$, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.43	145.62	0.00	0.00	20.52	240.50	20.52	-99.54	145.62	—
	End 2, x = 6	150.00C	-4.27	-96.53	0.00	0.00	20.52	@ 2.262	@ —	@ 2.850	96.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} ≥ 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]			
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 150.00, 240.50, 20.52			Class 1
	(Axial: Non-Slender)			
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε		Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1			

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$145.621 / 212.754 =$	0.684	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.048 / 212.754 =$	0.000	High Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$240.495 / 62.425 =$	3.853	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2.262 m: $V_{Ed} = 0.048 > V_{pl,Rd} = 0.000$: pure shear governs	47749441.757	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 308.016 = (A_{v,z} = 1940 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 60.6 / 1$	16.665 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$29.90 \times 275 / 1 =$ (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 822.250 =$	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	$0.182 + 240.495 / 62.425 + 20.521 / 16.665 =$	5.266	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.29 m, P1 flange tip	@ x = 2.29 m (ULS: 1.35G + 1.5Q): $M_y = 240.483 \text{ kN.m}$, $V = 0.782 \text{ kN}$, $\phi = 0.85153 \text{ rad}$, $\phi' = 5.644\text{e-}5 \text{ rad/mm}$, $\phi'' = 3.189\text{e-}7 \text{ rad/mm}^2$, $\phi''' = 4.269\text{e-}10 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 204.779 = 225.300 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_{z,tot} / W_{el,z} + E W_{n0} \phi'' = 50.17 + 1226.96 + 6665.68 + 269.87 = 8212.67 \text{ N/mm}^2$; $\tau = G t_f \phi' = 57.14 \text{ N/mm}^2$; $(8212.67 / 275)^2 + 3(57.14 / 275)^2 =$	892.003	Warning
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 50.17 + 1226.96 + 6665.68 + 269.87 = 8212.67$, $\tau = 57.14 \rightarrow 892.003$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 50.17 + 1150.27 + 968.68 = 2169.12$, $\tau = 0.16 + 57.14 + 8.25 = 65.55 \rightarrow 62.386$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 1150.27 + 3274.03 + 182.14 = 4656.61$, $\tau = 0.28 + 57.14 + 4.48 = 61.91 \rightarrow 286.882$; P3 web-flange junction, web: $\sigma_x = 50.17 + 1073.59 + 3274.03 + 182.14 = 4579.92$, $\tau = 0.58 + 27.43 + 9.34 = 37.35 \rightarrow 277.420$; P4 web mid-depth: $\sigma_x = 50.17 + 3274.03 = 3324.19$, $\tau = 0.74 + 27.43 + 6.92 = 35.08 \rightarrow 146.168$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99 \text{ mm}$ from the toe, $y_1 = 7.31 \text{ mm}$; $S_{w1}, S_{w2}, S_{w3} = \text{P385 Table A.2 points 1, 2, 3}$)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 145.621 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.612	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.612 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.758 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.541 \quad \text{OK 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	145.621	ULS: 1.35G + 1.5Q	0.612	0.541	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.631	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	96.525	ULS: 1.35G + 1.5Q	0.406	0.394	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y ·L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A·f _y /N _{cr})	√(29.90x275/1128.42 (N _{cr,y} = π ² EI _y /L _{ey} ²))	0.852	
N _{b,y,Rd} = Area·χ _f ·f _y /γ _{M1}	29.90x0.629x275/10/1 =	517.570 kN	Curve c
L _{ez} = K _z ·L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A·f _y /N _{crz})	√(29.90x275/97.87 (N _{cr,z} = π ² EI _z /L _{ez} ²))	2.892	
N _{b,z,Rd} = Area·χ _f ·f _y /γ _{M1}	29.90x0.102x275/10/1 =	83.643 kN	Curve c
i ₀ ² = i _y ² + i _z ² + y ₀ ²	81.10 ² + 23.90 ² + 50.90 ² (y ₀ = e _{sc} = shear centre to centroid, SCI P385 Table A.3 (Blue Book e ₀ + c _y))	9739.2 mm ²	6.3.1.4
N _{cr,T} = (GI _T + π ² EI _w /L _T ²)/i ₀ ²	(81000 x 11.6e4 + π ² x 210000 x 0.0107e12/6000 ²)/9739.2 (I _T , I _w : SCI P385 Table A.3; L _T = 6 m = L _{cr,y} (no intermediate twist restraint))	1028.010 kN	6.3.1.4
N _{cr,TF} = (N _{cr,y} + N _{cr,T})/2β [1 - √(1 - 4βN _{cr,y} N _{cr,T} /(N _{cr,y} + N _{cr,T}) ²)]	N _{cr,y} = 1128.425 (flexure about the axis of symmetry y-y, L _{ey}), β = 1 - (y ₀ /i ₀) ² = 0.734; N _{cr} = min(N _{cr,T} , N _{cr,TF}) = 709.070 (TF mode)	709.070 kN	6.3.1.4
λ _T = √(A·f _y /N _{cr})	√(29.90x275/709.070)	1.077	
N _{b,T,Rd} = Area·χ _T ·f _y /γ _{M1}	29.90x0.497x275/10/1 = (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
N _{Ed} /N _{b,T,Rd}	150.000 / 408.410 = (the lower of χ _T and the flexural χ feeds U _{N,y} , U _{N,z})	0.367	OK
N _{Ed} /N _{b,Rd}	150.000 / 83.643 = (min of N _{b,y,Rd} , N _{b,z,Rd} , N _{b,T,Rd})	1.793	Warning

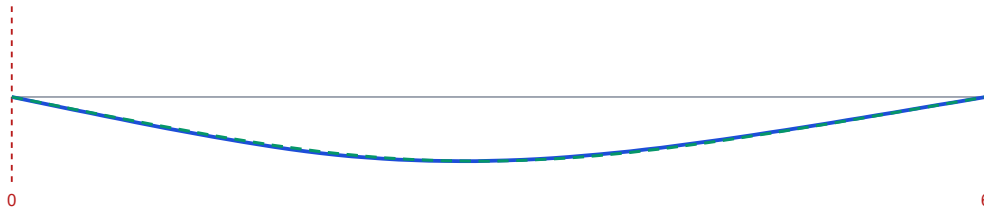
Equivalent Uniform Moment Factors C₁, C_{mLT}, C_{mz}, and C_{my}

C ₁ = M _{cr} /M _{cr,uniform}	36.575 / 31.304 kN.m (shape only: z _g = z _j = 0)	1.168	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 233.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=189.4, M(L/2)=233.4, M(3L/4)=115.8, M _{max} =240.5 kN·m	1.214	Serna
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.43, ψ = 0.695, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 20.521 imposed + 205.657 twist φ·M _y = 226.178; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.43, ψ = 0.695, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	63 elements, 0.000 %; z _g = 100, 0 mm; 1 combination(s), 3 solve(s)	31.947 kN.m	converged
λ _{LT} = √(W·f _y /M _{cr})	√(227.0 x 275 / 31.947)	1.398	
χ _{LT} = Fn(λ _{LT} , φ _{LT} , α _{LT})	1.398, 1.612, 0.76	0.374	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.374, 1.398, 0.925, 0.989	0.378	6.3.2.3
M _{b,Rd} = χW _{pl,y} ·f _y ≤ M _{c,y,Rd}	0.378 x 227.0 x 275 ≤ 62.425 =	23.577 kN.m	
M _{y,Ed} /M _{b,Rd}	240.495 / 23.577	10.201	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.76$ m)



Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	240.495 / 23.577 ($M_{y,Rk} = 62.425$)	10.201	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	(20.521 imposed + 205.657 twist ϕ , $M_y = 226.178$) / 9.295	24.333	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.852 \times 0.367\} \leq 1.000 (1 + 0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 2.892 \times 1.793\} \leq 1.000 (1 + 0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 10.201 + 2.076 \times 24.333$	63.000	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$1.793 + 0.880 \times 10.201 + 2.076 \times 24.333$	61.290	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.241 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w,Ed} = EI_w \phi'' / (h - t_f)$	max over span	5.850 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	20.521 + 240.078 x 856.62e-3 (@ $x = 2.44$ m) = 20.521 + 205.657 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	226.178 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2.241$ m: 240.490, 4.090, 224.533 kN.m ($M_{z,tot} = 20.521 + 204.012$); $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	29.161	Warning P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty$ ($M_{y,Ed} \geq M_{cr} = 31.947$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 31.947$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi'' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	

Torsion Shear Design @ 0 m

T_t , T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 4.870$ kN.m, $V = 145.621$ kN	251.90 / 5.50 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25 f_y / \sqrt{3}))} - T_w/(f_y / \sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 251.90$ ($T_w = 5.50$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd} / V_{pl,Rd} = 0.000$	0.000 kN	6.2.7(9)

$V_{Ed}/V_{pl.T.Rd}$	145.621 / 0.000	∞	Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)			
In-span $\delta \leq \text{Span}/360$	$99.54 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	99.54 mm	Warning
Torq in Case 1 @ 2.34 m: $\theta_{\max} \leq 2.00^\circ$	$0.4098 \text{ rad} = 23.48^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	23.48°	> 2° advisory
N_Ed/N_(pLd)	UNyz	UMyz	Ax+M_6.61
0.182	5.266	1.793	24.33363.00061.2905.972
Deflection	V/Vpl	MA/Mc	LTB
0.684	3.853	10.201477494	41.757
M-V	N_b.T	F/F_Rd	Web 7.2
0.612	2.631	0.367	892.0099.00099.000477494
Torsion	Yield 6.1	V+T	LTB+T
Max			

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 47749441.757

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 23.5^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-L7 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)
Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER
PFC 200 x 75 x 23, 6 m SS, unrestrained; L7 = L6 + "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-L7 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork) — simply supported

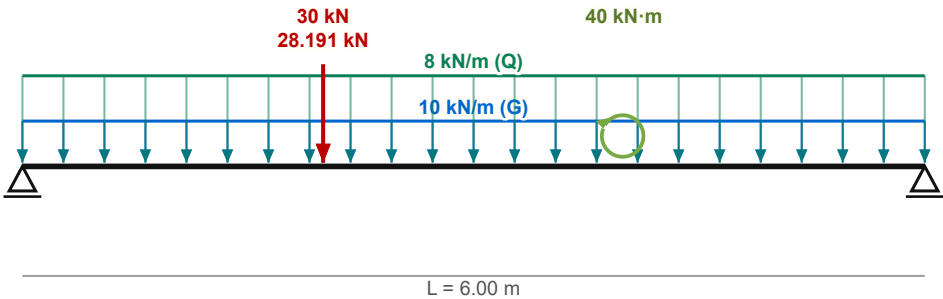
Member Loading and Member Forces

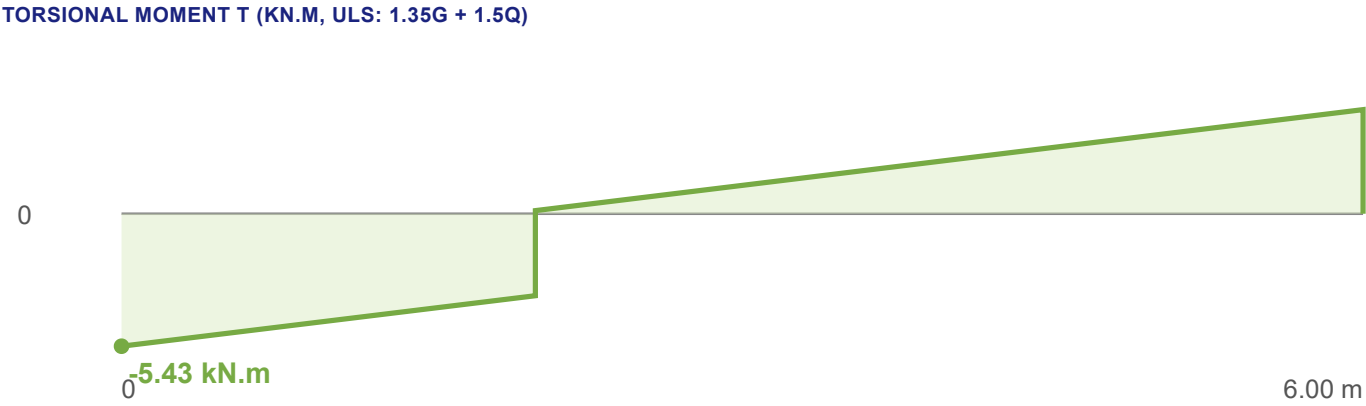
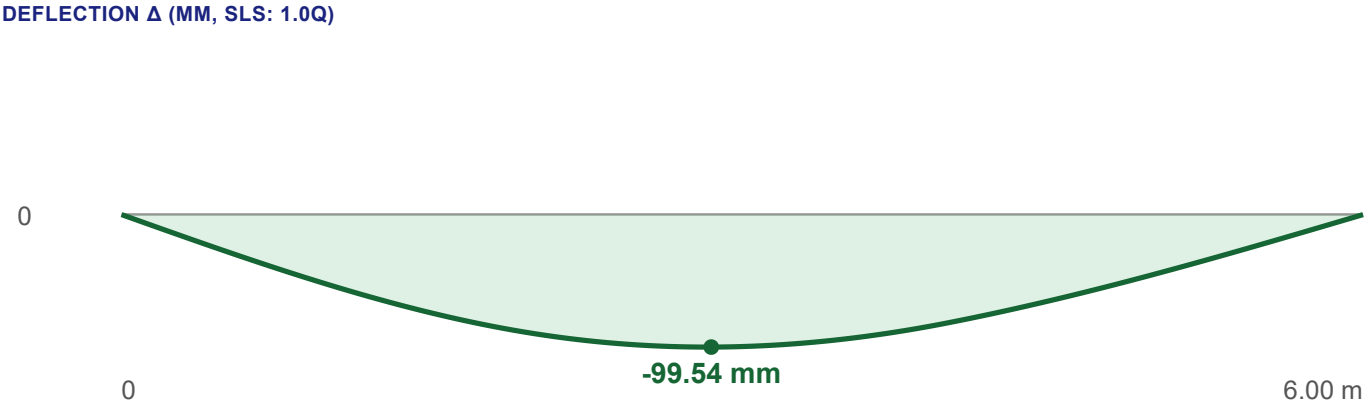
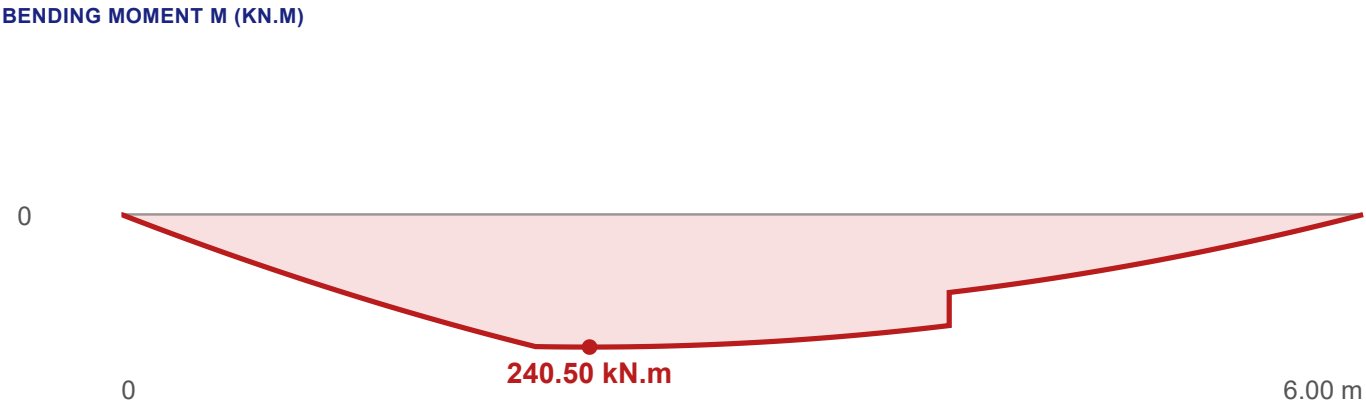
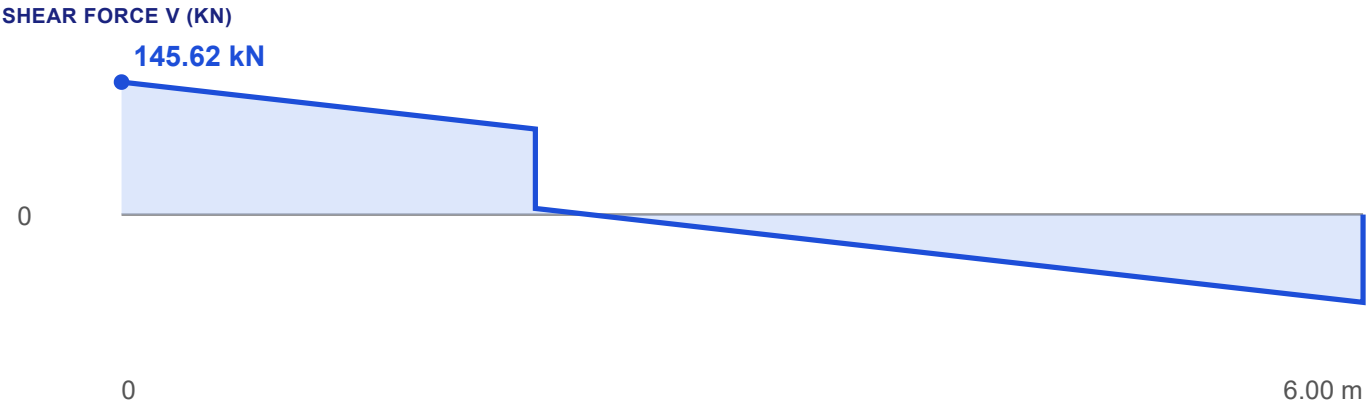
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm $z_g = 100$ mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm $z_g = 100$ mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm $z_g = 100$ mm (kN)
Q PY 28.191 @ 2 m e = 40 mm $z_g = 100$ mm (kN)
Q M 40.000 @ 4 m (kN.m)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
 $Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]
 $Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered $Y_{G,inf}$, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	5.43	145.62	0.00	0.00	20.52	240.50	20.52	-99.54	145.62	—
	End 2, x = 6	150.00C	-4.27	-96.53	0.00	0.00	20.52	@ 2.262	@ —	@ 2.850	96.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} ≥ 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)				
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]			
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 150.00, 240.50, 20.52			Class 1
	(Axial: Non-Slender)			
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε		Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1			

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$145.621 / 212.754 =$	0.684	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Local Capacity Check

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.048 / 212.754 =$	0.000	High Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$240.495 / 62.425 =$	3.853	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2.262 m: $V_{Ed} = 0.048 > V_{pl,Rd} = 0.000$: pure shear governs	47749441.757	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 308.016 = (A_{v,z} = 1940 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 60.6 / 1$	16.665 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$29.90 \times 275 / 1 =$ (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 822.250 =$	0.182	OK
$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed} / M_{c,y,Rd} + M_{z,Ed} / M_{c,z,Rd}$	$0.182 + 240.495 / 62.425 + 20.521 / 16.665 =$	5.266	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.29 m, P1 flange tip	@ x = 2.29 m (ULS: 1.35G + 1.5Q): $M_y = 240.483 \text{ kN.m}$, $V = 0.782 \text{ kN}$, $\phi = 0.85153 \text{ rad}$, $\phi' = 5.644\text{e-}5 \text{ rad/mm}$, $\phi'' = 3.189\text{e-}7 \text{ rad/mm}^2$, $\phi''' = 4.269\text{e-}10 \text{ rad/mm}^3$; $M_{z,tot} = M_{z,Ed} + \phi M_y = 20.521 + 204.779 = 225.300 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_{z,tot} / W_{el,z} + E W_{n0} \phi'' = 50.17 + 1226.96 + 6665.68 + 269.87 = 8212.67 \text{ N/mm}^2$; $\tau = G t_f \phi' = 57.14 \text{ N/mm}^2$; $(8212.67 / 275)^2 + 3(57.14 / 275)^2 =$	892.003	Warning
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 50.17 + 1226.96 + 6665.68 + 269.87 = 8212.67$, $\tau = 57.14 \rightarrow 892.003$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 50.17 + 1150.27 + 968.68 = 2169.12$, $\tau = 0.16 + 57.14 + 8.25 = 65.55 \rightarrow 62.386$; P2 web-flange junction, flange: $\sigma_x = 50.17 + 1150.27 + 3274.03 + 182.14 = 4656.61$, $\tau = 0.28 + 57.14 + 4.48 = 61.91 \rightarrow 286.882$; P3 web-flange junction, web: $\sigma_x = 50.17 + 1073.59 + 3274.03 + 182.14 = 4579.92$, $\tau = 0.58 + 27.43 + 9.34 = 37.35 \rightarrow 277.420$; P4 web mid-depth: $\sigma_x = 50.17 + 3274.03 = 3324.19$, $\tau = 0.74 + 27.43 + 6.92 = 35.08 \rightarrow 146.168$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99 \text{ mm}$ from the toe, $y_1 = 7.31 \text{ mm}$; $S_{w1}, S_{w2}, S_{w3} = \text{P385 Table A.2 points 1, 2, 3}$)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): verdict-binding	Class 1/2 open section with N_{Ed} : EN 1993-1-1 6.2.7(6) admits the plastic resistance for bending + torsion only and no expression of EN 1993-1-1 or SCI P385 combines N_{Ed} with the bimoment plastically, so the elastic verification 6.2.7(5)/(6.1) is the cross-section check (verdict-binding)		6.2.7(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 145.621 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.612	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.612 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.758 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.541 \quad \text{OK 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	145.621	ULS: 1.35G + 1.5Q	0.612	0.541	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.631	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	96.525	ULS: 1.35G + 1.5Q	0.406	0.394	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y ·L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √(A·f _y /N _{cr})	√(29.90x275/1128.42 (N _{cr,y} = π²EI _y /L _{ey} ²)	0.852	
N _{b,y,Rd} = Area·χ _y ·f _y /γ _{M1}	29.90x0.629x275/10/1 =	517.570 kN	Curve c
L _{ez} = K _z ·L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A·f _y /N _{crz})	√(29.90x275/97.87 (N _{cr,z} = π²EI _z /L _{ez} ²)	2.892	
N _{b,z,Rd} = Area·χ _z ·f _y /γ _{M1}	29.90x0.102x275/10/1 =	83.643 kN	Curve c
i ₀ ² = i _y ² + i _z ² + y ₀ ²	81.10² + 23.90² + 50.90² (y ₀ = e _{sc} = shear centre to centroid, SCI P385 Table A.3 (Blue Book e ₀ + c _y))	9739.2 mm²	6.3.1.4
N _{cr,T} = (GI _T + π²EI _w /L _T ²)/i ₀ ²	(81000 x 11.6e4 + π² x 210000 x 0.0107e12/6000²)/9739.2 (I _T , I _w : SCI P385 Table A.3; L _T = 6 m = L _{cr,y} (no intermediate twist restraint))	1028.010 kN	6.3.1.4
N _{cr,TF} = (N _{cr,y} + N _{cr,T})/2β [1 - √(1 - 4βN _{cr,y} N _{cr,T} /(N _{cr,y} + N _{cr,T} ²))]	N _{cr,y} = 1128.425 (flexure about the axis of symmetry y-y, L _{ey}), β = 1 - (y ₀ /i ₀)² = 0.734; N _{cr} = min(N _{cr,T} , N _{cr,TF}) = 709.070 (TF mode)	709.070 kN	6.3.1.4
λ _T = √(A·f _y /N _{cr})	√(29.90x275/709.070)	1.077	
N _{b,T,Rd} = Area·χ _T ·f _y /γ _{M1}	29.90x0.497x275/10/1 = (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
N _{Ed} /N _{b,T,Rd}	150.000 / 408.410 = (the lower of χ _T and the flexural χ feeds U _{N,y} , U _{N,z})	0.367	OK
N _{Ed} /N _{b,Rd}	150.000 / 83.643 = (min of N _{b,y,Rd} , N _{b,z,Rd} , N _{b,T,Rd})	1.793	Warning

Equivalent Uniform Moment Factors C₁, C_{mLT}, C_{mz}, and C_{my}

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 233.4, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=189.4, M(L/2)=233.4, M(3L/4)=115.8, M _{max} =240.5 kN·m		P362
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.43, ψ = 0.695, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M _{z,Ed} = 20.521 imposed + 205.657 twist φ·M _y = 226.178; ψ = 1 for the imposed constant M _z , the twist part taken at C _{mz} = 1.0 (Table B.3 upper bound, conservative)	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.43, ψ = 0.695, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
Z _g	Z _g = +100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L _E switch); the shear-centre M _{cr} route is not offered		load height
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} ·f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _y ·f _y /λ _{LT} ²	227.0 x 275 / 2.615²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	240.495 / 8.806	27.310	Warning

Buckling Resistance

$U_{N,y} = N_{Ed} / (X_y \cdot N_{Rk} / Y_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed} / (X_z \cdot N_{Rk} / Y_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed} / (X_{LT} \cdot M_{y,Rk} / Y_{M1})$	240.495 / 8.806 ($M_{y,Rk} = 62.425$)	27.310	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / Y_{M1})$	(20.521 imposed + 205.657 twist ϕ . $M_y = 226.178$) / 9.295	24.333	Warning
k_{zy} method	$M_{b,Rd} < M_{c,y,Rd}$ therefore susceptible to LTB, using Table B.2		Table B.2
$k_{yy} = C_{my} \{1 + 0.6 \lambda_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.852 \times 0.367\} \leq 1.000 (1 + 0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 2.892 \times 1.793\} \leq 1.000 (1 + 0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 27.310 + 2.076 \times 24.333$	83.322	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$1.793 + 0.880 \times 27.310 + 2.076 \times 24.333$	76.354	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.241 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w,Ed} = EI_w \phi'' / (h - t_f)$	max over span	5.850 kN.m	
$M_{z,tot} = M_{z,Ed} + \phi \cdot M_{y,Ed}$	$20.521 + 240.078 \times 856.62e-3$ (@ $x = 2.44$ m) = 20.521 + 205.657 (imposed constant $M_{z,Ed}$ + max coincident twist moment)	226.178 kN.m	P385 3.1.2
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_{z,tot}/M_{pl,z}$	@ $x = 2.241$ m: 240.490, 4.090, 224.533 kN.m ($M_{z,tot} = 20.521 + 204.012$); $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	29.161	Warning P385 3.1.2
C_{mz} (EN 1993-6 A.1)	$C_{mz} = 1.0$: the imposed M_z is a constant diagram ($\psi = 1$, Table B.3); the twist-induced part carries no proxy	1.000	Table B.3
$k = k_w \cdot k_{zw} \cdot k_a$	$x \propto (M_{y,Ed} \geq M_{cr} = 9.128$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_{z,tot}/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	k_a unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]		
Basis with $N_{Ed} / M_{z,Ed}$	No expression in EN 1993-1-1 or EN 1993-6 combines N_{Ed} with warping torsion at member level; the verification is EN 1993-1-1 6.2.7(5)/(6.1) at the cross-section with all stresses, cl 6.3.3 (6.61/6.62) with the second-order minor-axis moment $\phi \cdot M_y$ added to $M_{z,Ed}$, and EN 1993-6 (A.1) with $M_{z,Ed} = M_z + \phi \cdot M_y$		EN 1993-1-1, EN 1993-6
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	

Torsion Shear Design @ 0 m

T_t , T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 4.870$ kN.m, $V = 145.621$ kN	251.90 / 5.50 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.000$	0.000 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	145.621 / 0.000	∞	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$99.54 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	99.54 mm	Warning
Torq in Case 1 @ 2.34 m: $\theta_{max} \leq 2.00^\circ$	0.4098 rad = $23.48^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	23.48°	> 2° advisory

N_Ed/N_Rd	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	Yield 6.1	V+T	LTB+T	Max
0.182	5.266	1.793	27.310	83.322	76.354	5.972	0.684	3.853	27.310	4774941.757	0.612	2.631	0.367	892.009	9.000	99.000	4774941.757

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 47749441.757

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 23.5^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

Beam & Beam-Portion (Member)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member UB-R1 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm (kN)

G SW 0.8044 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0 0.00C	4.26	4.26	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6 0.00C	-3.66	-3.66	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

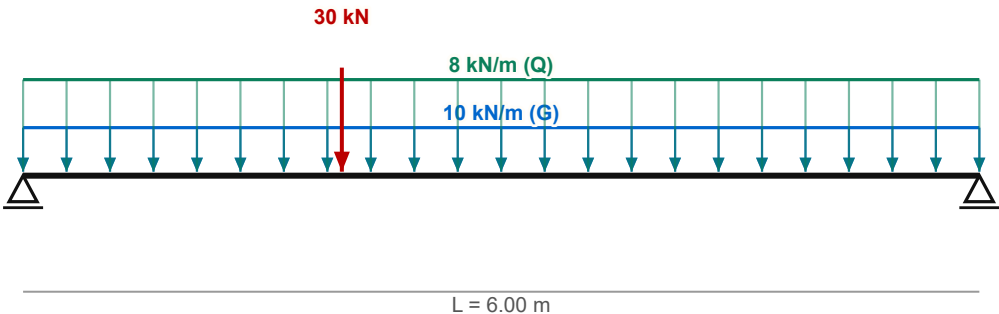
Uplift

no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

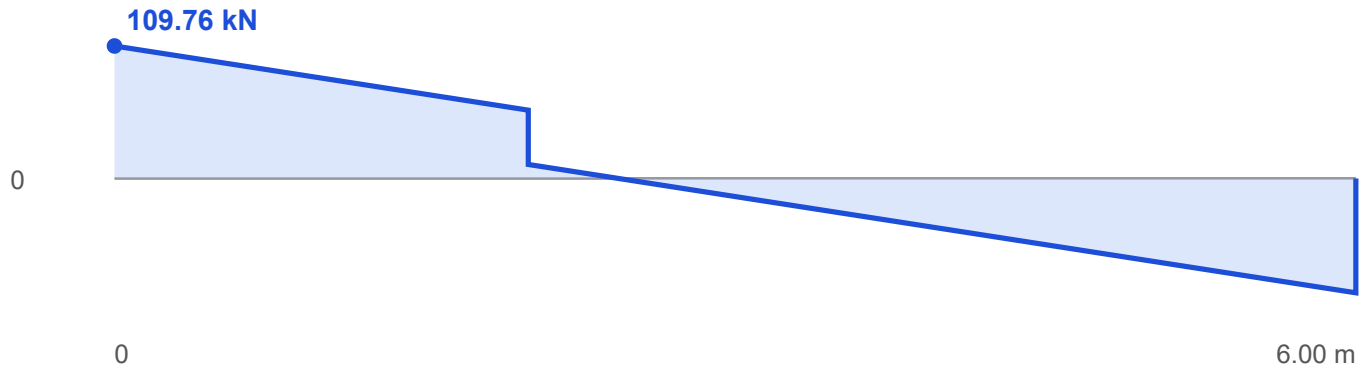
R_{min} ≥ 0

OK

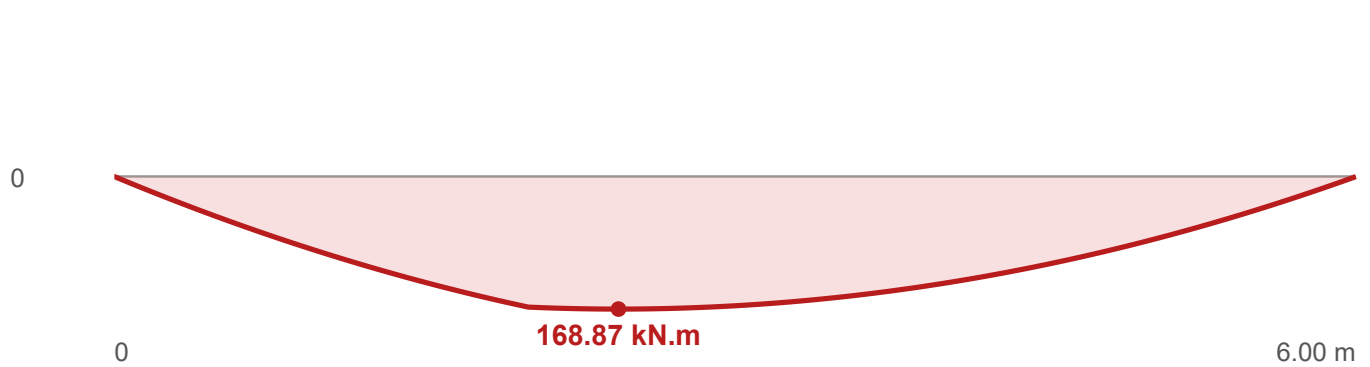
LOADING



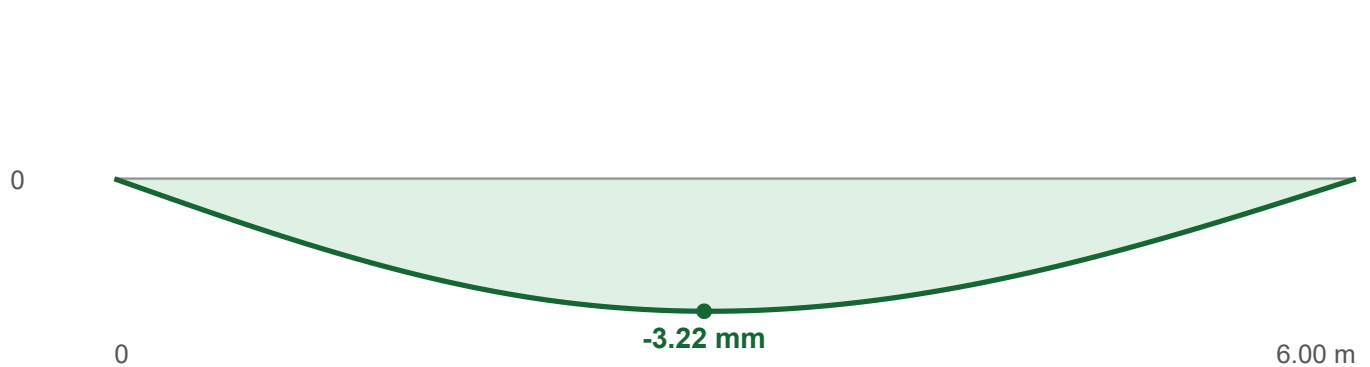
SHEAR FORCE V (kN)



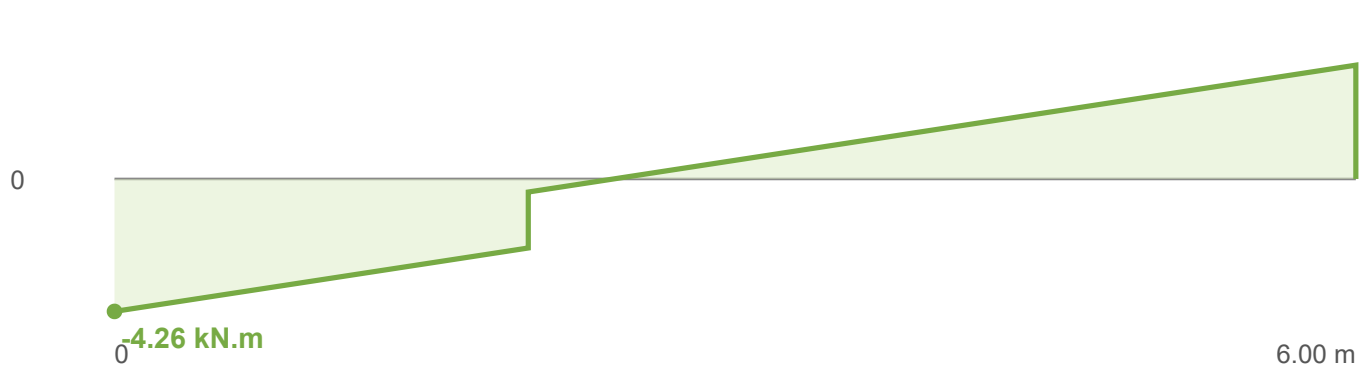
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd - Fully Restrained Beam

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

M_{b,Rd} = M_{c,y,Rd}Fully Restrained503.250 kN.m

Torsion Design

J, H, a, Q_f, Q_w69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, —P385 App A

W_{n0}, S_{w1}212 cm², 1620 cm⁴P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)combination ULS: 1.35G + 1.5Q4.260 kN.m

Torsion analysisSCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)L/a = 3.23P385 App C

φ_{max} (ULS)L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)0.0605 rad = 3.47°

B_{Ed} = EI_wφ'' (max)@ x = 2 m3.435 kN.m²

M_{w,Ed} = EI_wφ''/(h-t_f)max over span7.736 kN.m

M_{z,Ed} = φ.M_{y,Ed}max coincident10.129 kN.m

(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z}@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 40.255, M_{pl,z} = 83.6000.415OK P385 3.1.2

End torques T_tSt Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m1.902 / 1.724 kN.m

Torsion Shear Design @ 0 m

T_t at the V-T station@ x = 0 m (ULS: 1.35G + 1.5Q): T_t = 1.902 kN.m, V = 109.758 kN27.22 N/mm²P385

V_{pl,T.Rd} = √(1 - T_t/(1.25f_y/√3)).V_{pl,Rd}@ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260; S_{mod} = V_{pl,T.Rd}/V_{pl,Rd} = 0.929702.492 kN6.2.7(9)

V_{Ed}/V_{pl,T.Rd}109.758 / 702.4920.156OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/3603.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m3.22 mmOK

Torq in Case 1 @ 2.788 m: θ_{max} ≤ 2.00°0.0252 rad = 1.44° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)1.44°OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
0.336	0.336	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.415

PASS — governing Bending+torsion cross-section (P385 3.1.2) = 0.415

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R2 unrestrained between fork ends ($U_y + R_x$ held at both ends, warping free) - the reference for R3..R11

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

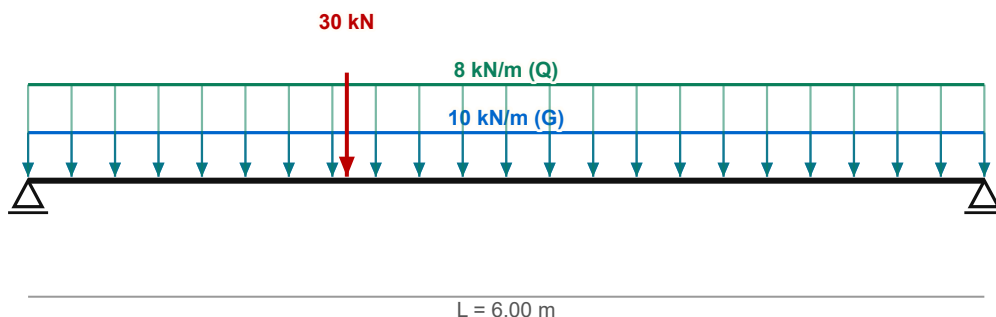
$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.26	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-3.66	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

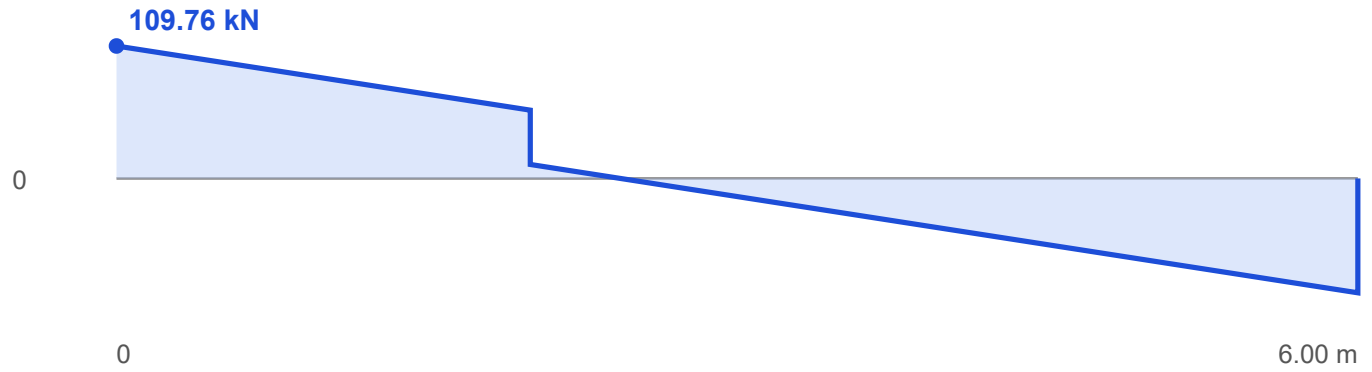
Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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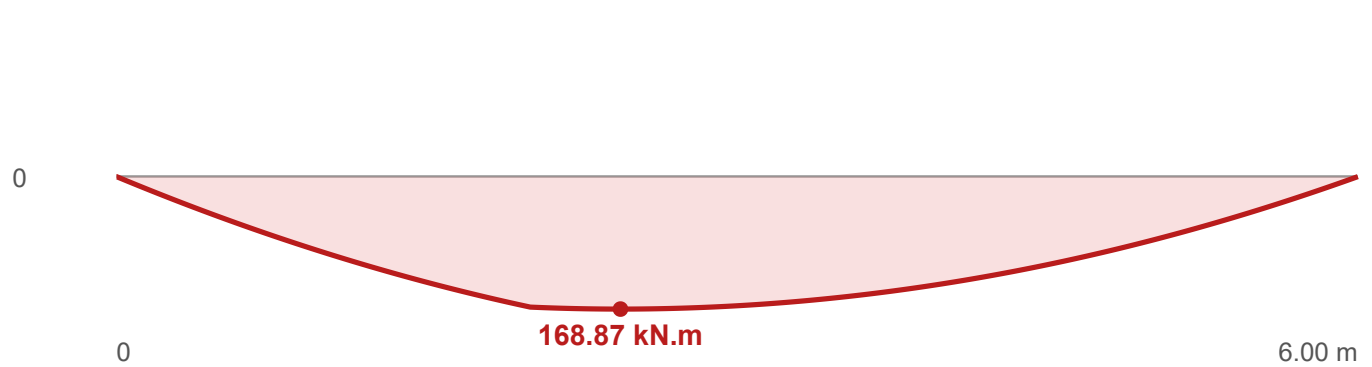
LOADING



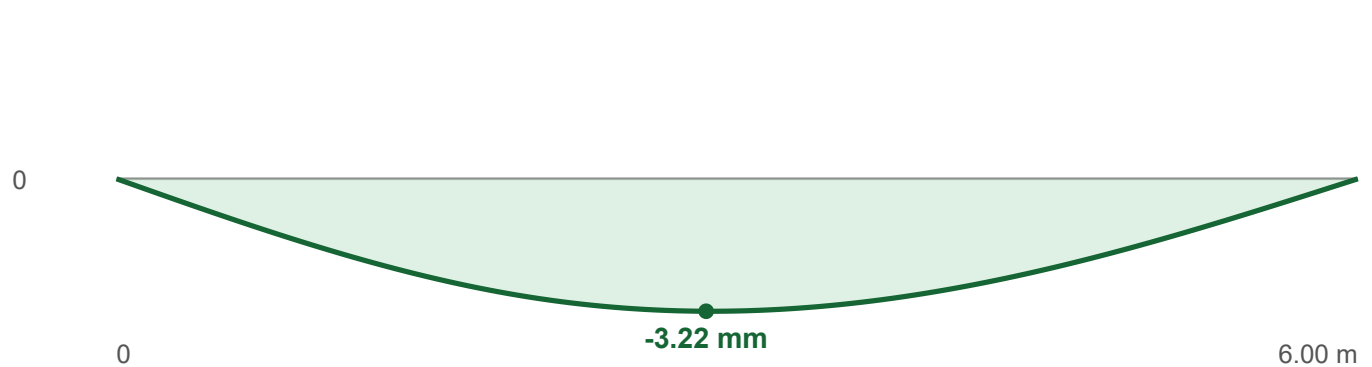
SHEAR FORCE V (kN)



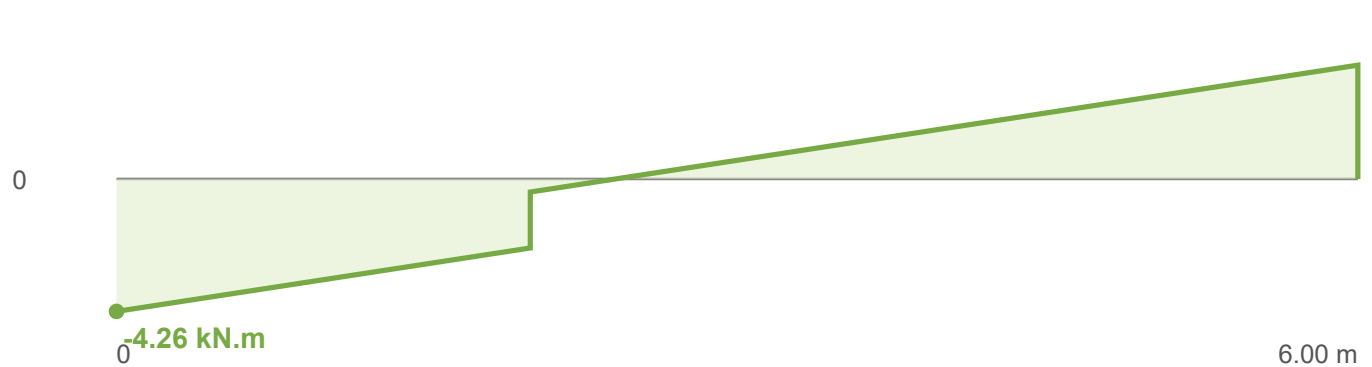
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038\text{e-}5 \text{ rad/mm}$, $\phi'' = 1.724\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.103\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$ P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
σ_x, τ , (6.1) at the 4 section points	$f_y / \gamma_{M0} = 275$		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

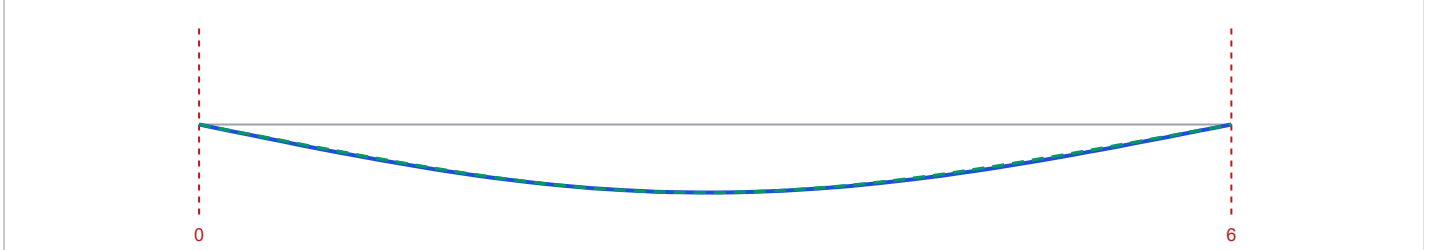
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.93 m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w ϕ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.753	1.025	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ x = 2.365 m: 168.802/282.919 + 1.000x9.873/83.600 + 0.663x0.882x1.753x7.403/40.255	0.903	OK EN 1993-6 A
End torques T _t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total T = $GI_T \phi' - EI_w \phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902 \text{ kN.m}$, $V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

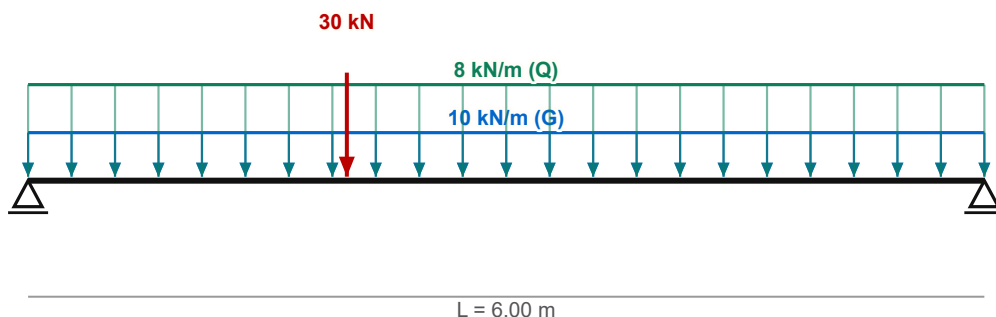
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.903	0.903

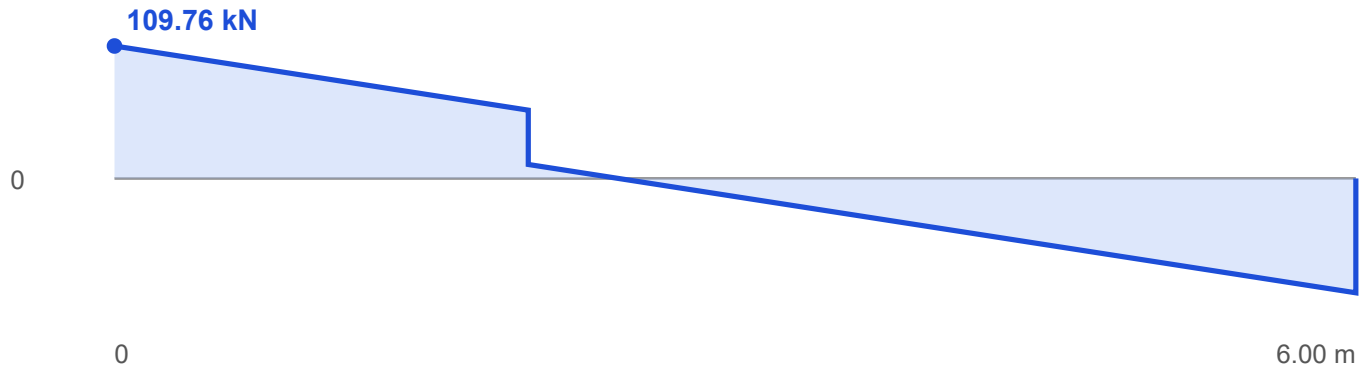
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.903

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

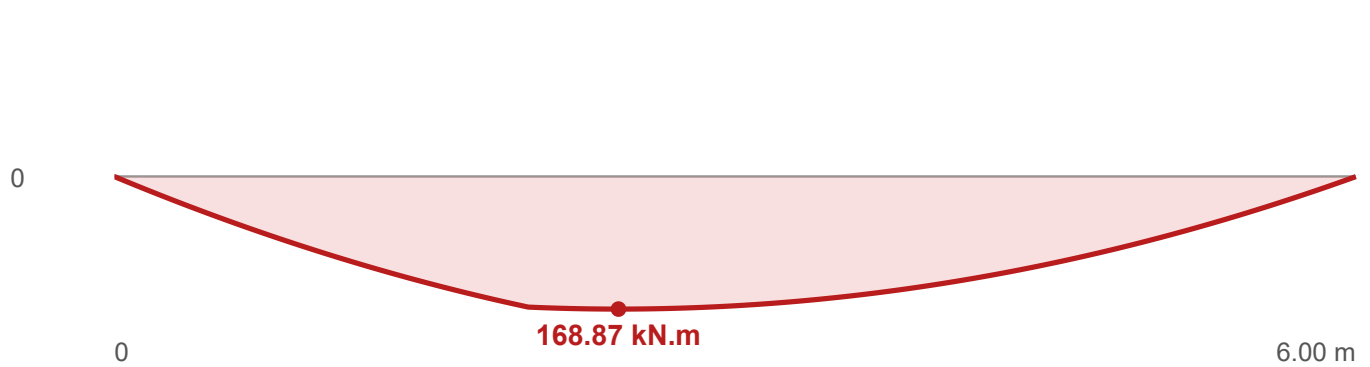
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



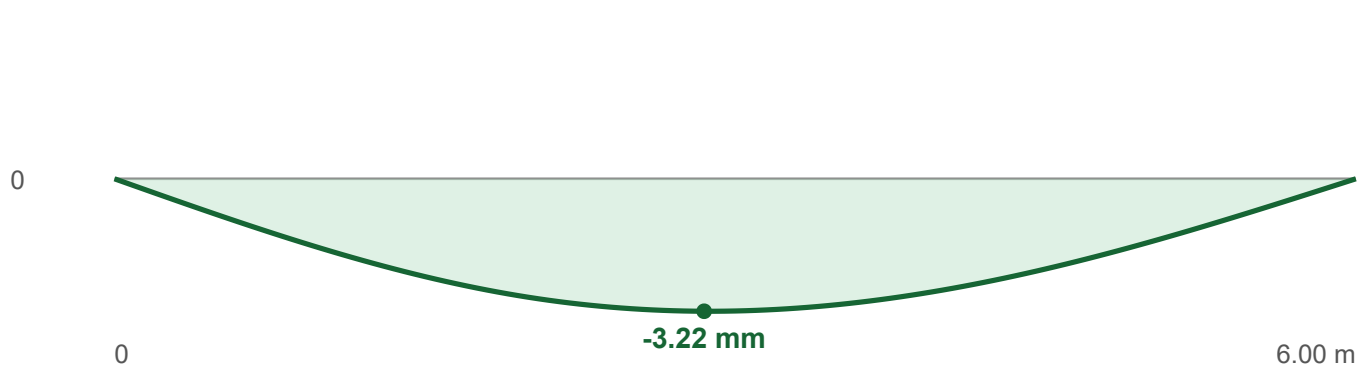
SHEAR FORCE V (kN)



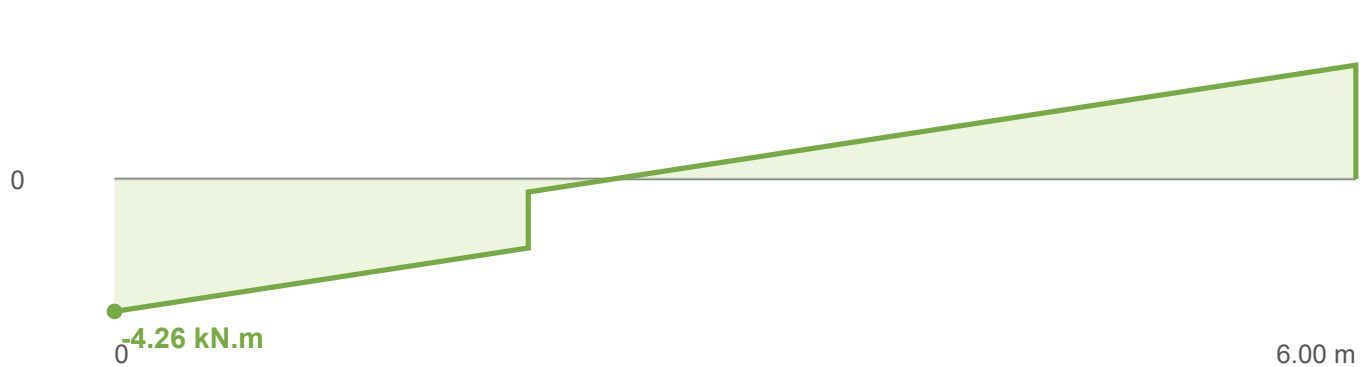
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000)$ $= 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	10.129 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 40.255, M _{pl,z} = 83.600	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.736	1.015	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.365 m: 168.802/285.039 + 1.000x9.873/83.600 + 0.663x0.882x1.736x7.403/40.255	0.897	OK EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.260 / 3.660 kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 1.902 kN.m, V = 109.758 kN	27.22 N/mm ²	P385
V _{pl,T,Rd} = √(1 - T _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 27.22 N/mm ² ; V _{pl,Rd} = 756.260; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.929	702.492 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

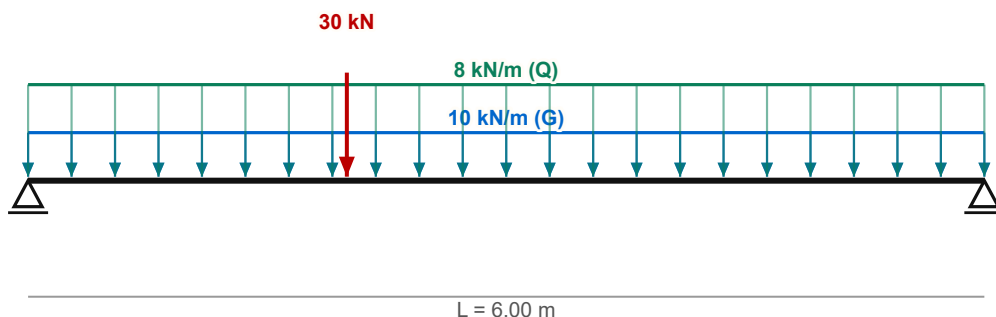
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

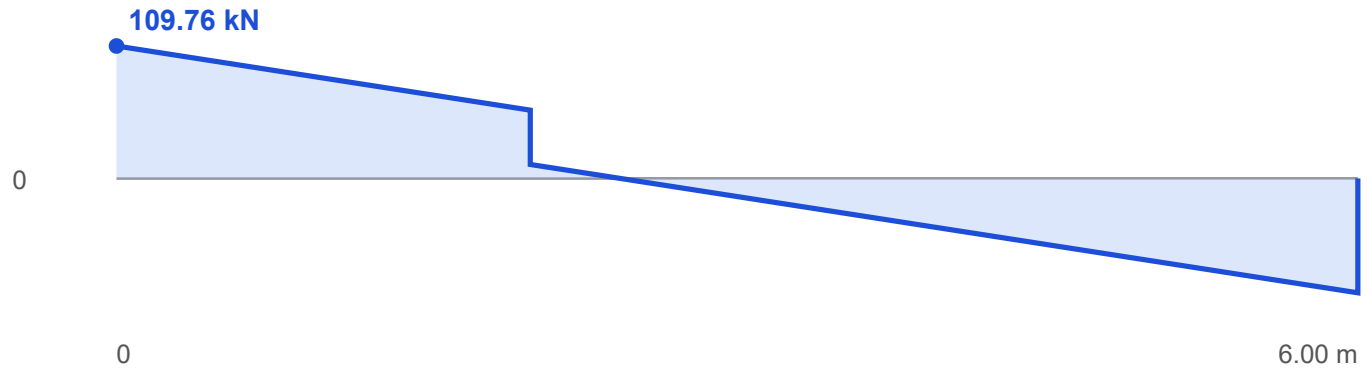
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

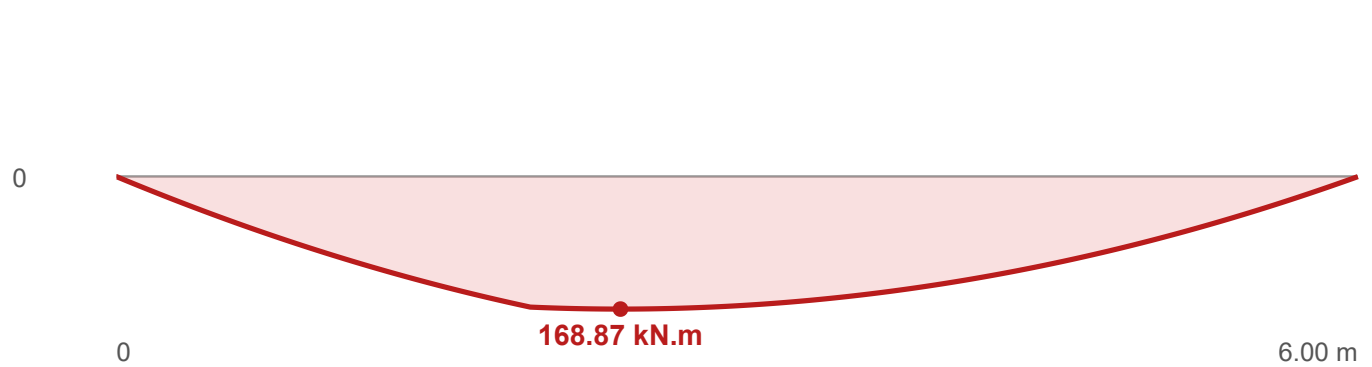
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



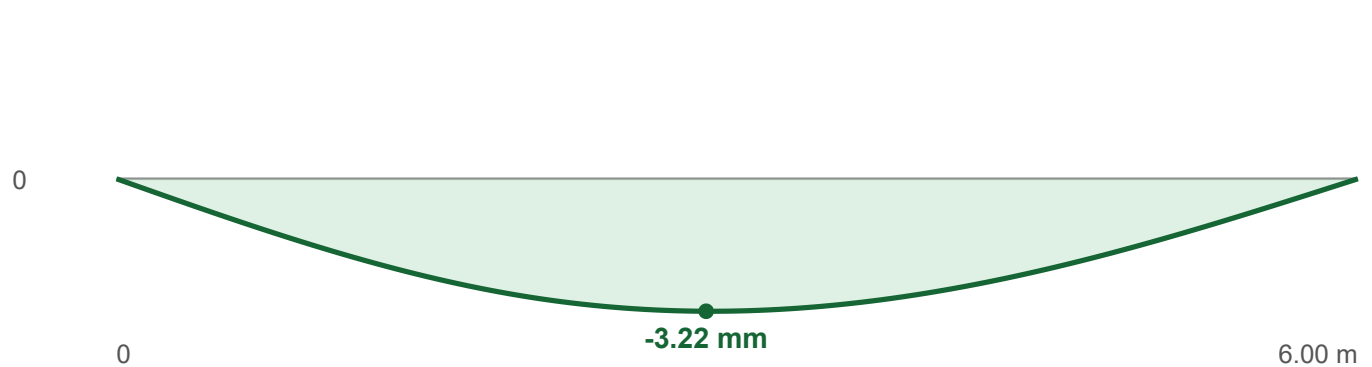
SHEAR FORCE V (kN)



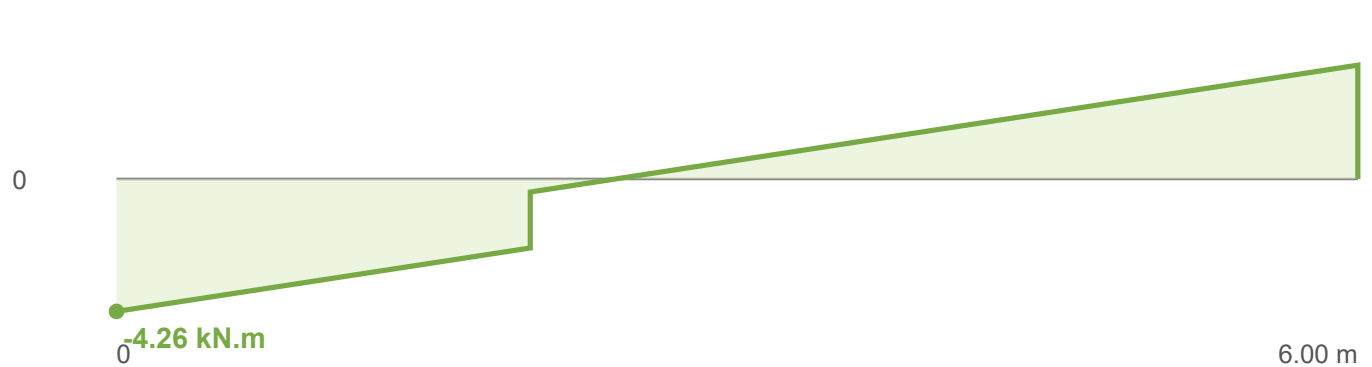
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000)$ $= 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

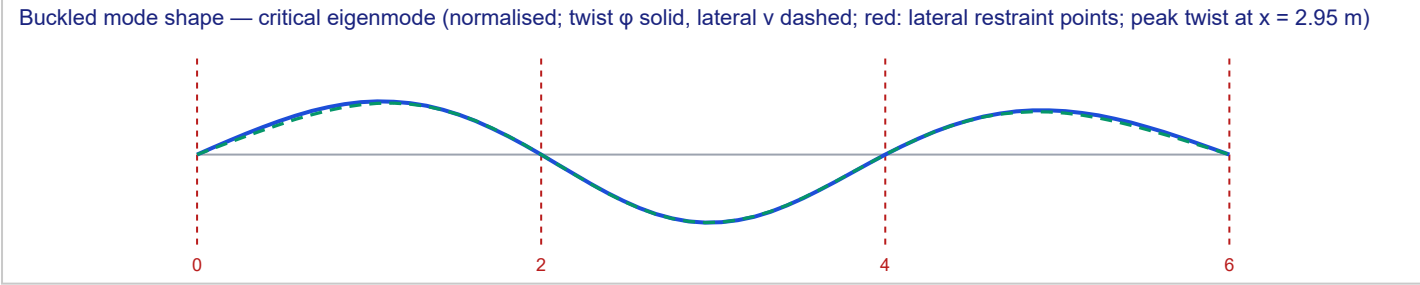
Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform} (critical bay)	bay between restraints containing the eigenmode peak (x = 2.952 m); whole-member ratio 1.376	1.036	NA 2.18
C ₁ = fn(M ₁ , M ₂ , M _o , ψ, μ) (standard, comparison)	136.3, 166.3, 13.3, 0.820, 0.080 [segment 2–4 m] — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=168.8, M(L/2)=164.6, M(3L/4)=153.8, M _{max} =168.9 kN·m	1.035	Serna

Lateral Buckling Check M.b.Rd

Governing bay 2–4 m (isolated, fork ends)	M _{y,Ed} = 168.868, M _{cr} = 2355.701, λ = 0.462, χ = 0.965	485.737 kN.m	bay by bay
L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0; lateral restraint points at x = 0, 2, 4, 6 m	2 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	63 elements, 0.001 %; 1 combination(s), 2 solve(s) + 1 cached	3128.975 kN.m (whole member)	merged
λ _{LT} = √W _{f,y} /M _{cr}	√ 1830.0 x 275 / 2355.701	0.462	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.462, 0.595, 0.49	0.965	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.965, 0.462, —, 1.000	0.965	isolated bay: f = 1
M _{b,Rd} = χW _{pl,y} ·f _y ≤ M _{c,y,Rd}	0.965 x 1830.0 x 275 ≤ 503.250 =	485.737 kN.m	
M _{y,Ed} /M _{b,Rd}	168.868 / 485.737	0.348	OK



Lateral Restraint Portions (bay by bay, fork ends)

Portion	From – To (m)	L _e (m)	M _{y,Ed} (kN.m)	M _{cr} (kN.m)	λ _{LT}	χ _{LT}	M _{b,Rd} (kN.m)	M _{y,Ed} /M _{b,Rd}	
1	0 – 2	2	166.344	3763.377	0.366	1.000	503.250	0.331	
2	2 – 4	2	168.868	2355.701	0.462	0.965	485.737	0.348	governs
3	4 – 6	2	136.344	3675.482	0.370	1.000	503.250	0.271	

χ_{LT} per portion without the f-factor. Whole-member M_{cr} = 3128.975 kN.m. intermediately restrained member: lateral-torsional buckling checked BAY BY BAY between the lateral restraints. Each bay is solved in isolation with fork ends, its own M_{cr} and its own peak moment; the worst bay governs. Governing bay 2–4 m: M_{Ed} = 168.9, M_{cr} = 2355.7, M_{b,Rd} = 485.7 kN·m, utilisation 0.35. (Whole-member eigen M_{cr} = 3129.0 kN·m retained for reference.)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	166.344	ULS: 1.35G + 1.5Q	361.617	9.040	restraint design force, advisory
4	lateral restraint 2	136.344	ULS: 1.35G + 1.5Q	296.399	7.410	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ.M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 40.255, M_{pl,z} = 83.600 0.415 OK P385 3.1.2
k = k_w.k_{zw}.k_α 0.663 x 0.881 x 1.057 0.618 EN 1993-6 A
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_αM_w/M_{f,Rk} @ x = 2.415 m: 168.862/502.958 + 1.000x9.943/83.600 + 0.663x0.881x1.057x7.356/40.255 0.568 OK EN 1993-6 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m

Torsion Shear Design @ 0 m

T_t at the V-T station @ x = 0 m (ULS: 1.35G + 1.5Q): T_t = 1.902 kN.m, V = 109.758 kN 27.22 N/mm² P385
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)).V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260; S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.929 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

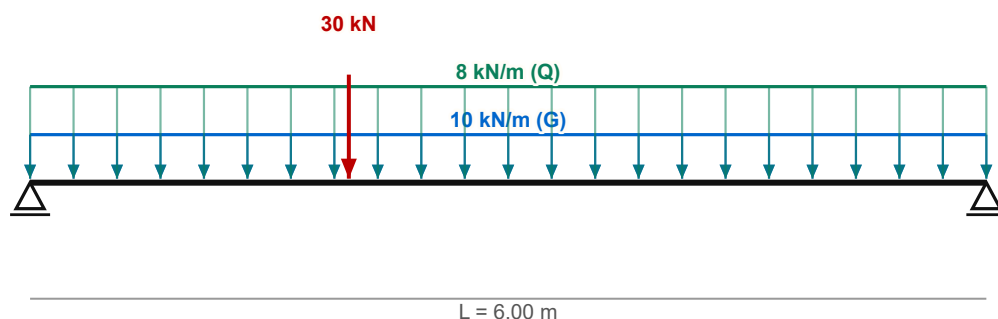
In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK
Torq in Case 1 @ 2.788 m: θ_{max} ≤ 2.00° 0.0252 rad = 1.44° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict) 1.44° OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.348	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.568	0.568

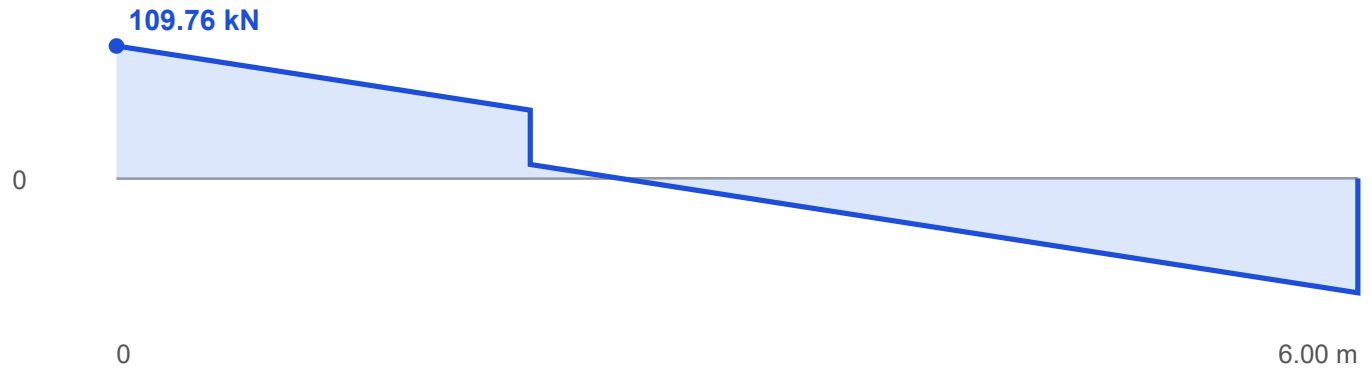
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.568

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

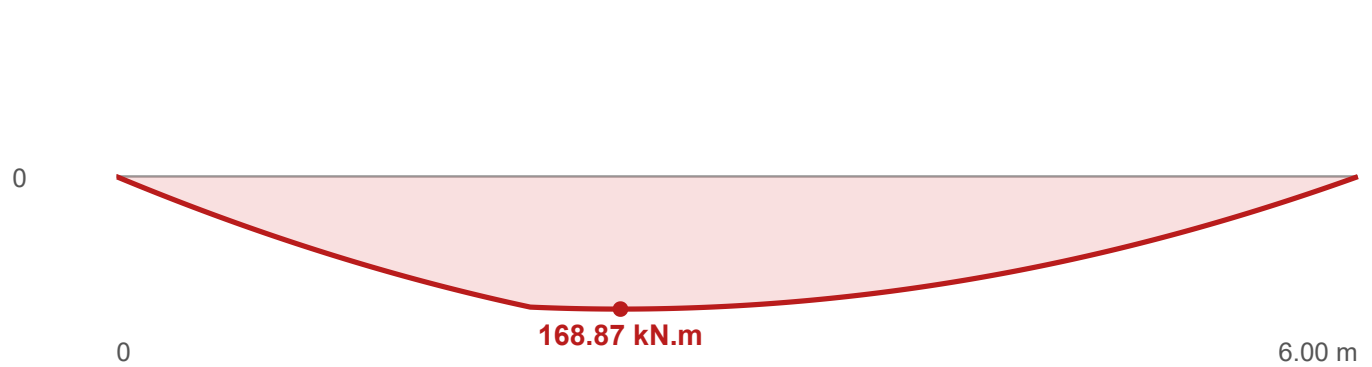
ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



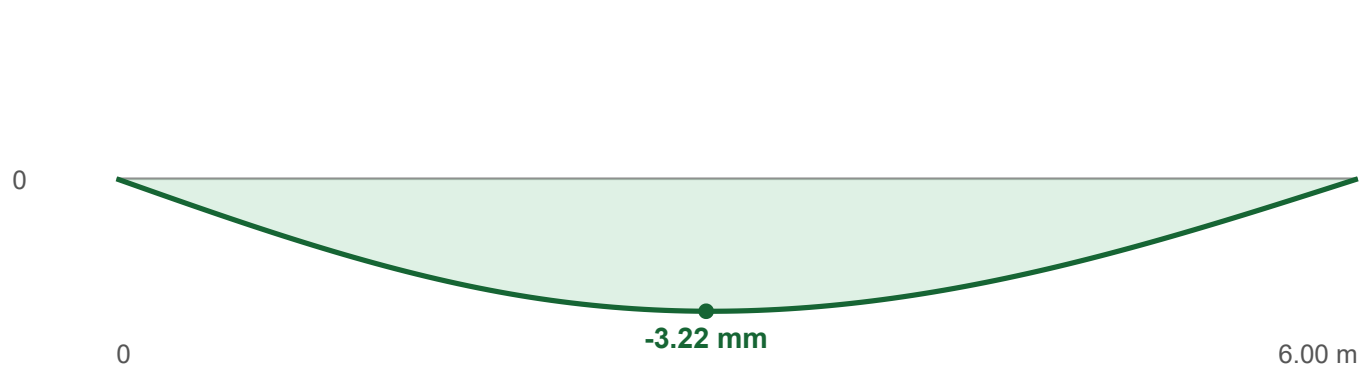
SHEAR FORCE V (kN)



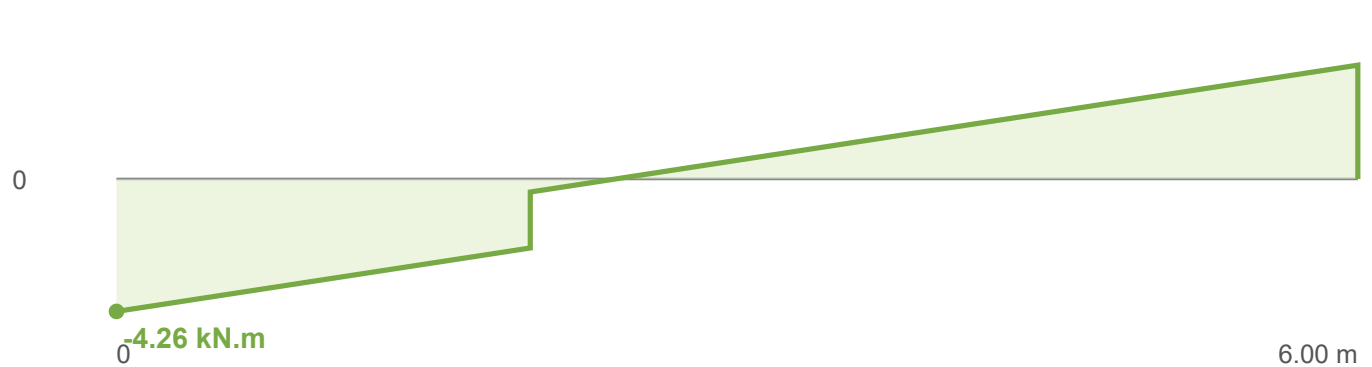
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038\text{e-}5 \text{ rad/mm}$, $\phi'' = 1.724\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.103\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	166.344	ULS: 1.35G + 1.5Q	361.617	9.040	restraint design force, advisory
4	lateral restraint 2	136.344	ULS: 1.35G + 1.5Q	296.399	7.410	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w φ'' (max)	@ $x = 2$ m	3.435 kN.m ²	

$M_{w.Ed} = E I_w \varphi'' / (h - t_f)$	max over span	7.736 kN.m	
$M_{z.Ed} = \varphi \cdot M_{y.Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	0.663 x 0.882 x 1.736	1.015	EN 1993-6 A
$M_y/M_{b.Rd} + C_{mz} M_z/M_{z.Rk} + k_w k_{zw} k_\alpha M_w/M_{f.Rk}$	@ x = 2.365 m: 168.802/285.039 + 1.000x9.873/83.600 + 0.663x0.882x1.736x7.403/40.255	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, $V = 109.758$ kN	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22$ N/mm ² ; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	0.0252 rad = $1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

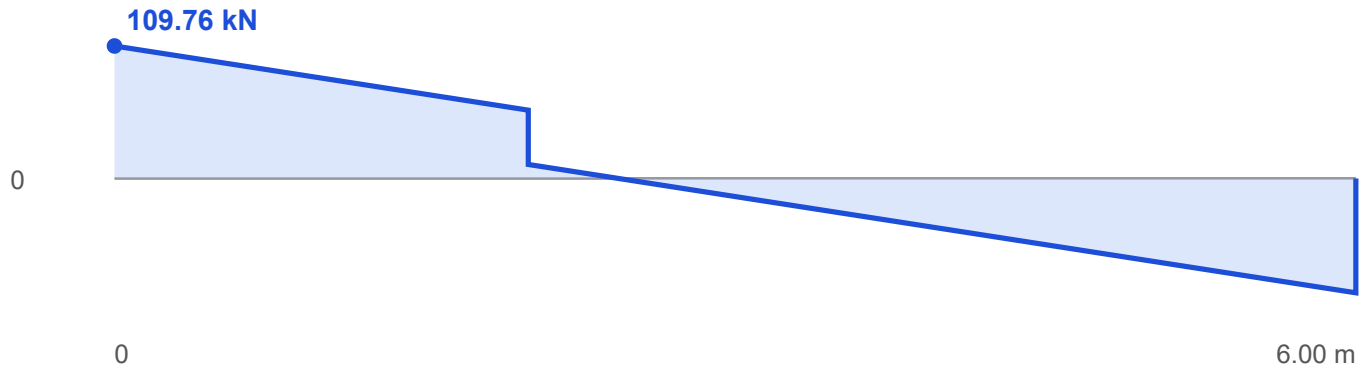
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

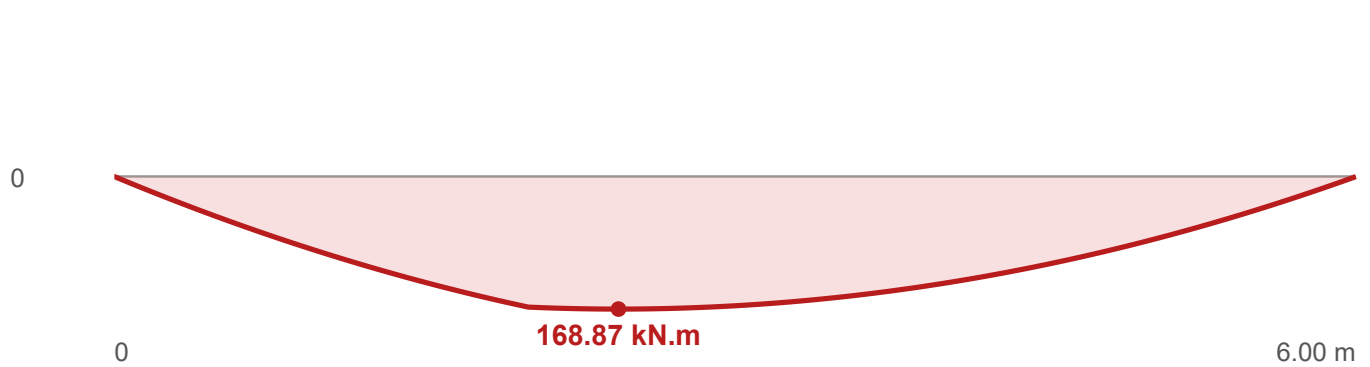
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: Standard (closed-form) M_{cr} : the member is treated as ONE segment of length $L_E = 1.00 \times L$ with C_1 from the whole-member moment diagram; intermediate lateral restraints are not applied on this route (conservative on L_E). Use the FE eigenvalue method for the bay-by-bay M_{cr} , or verify each bay separately.

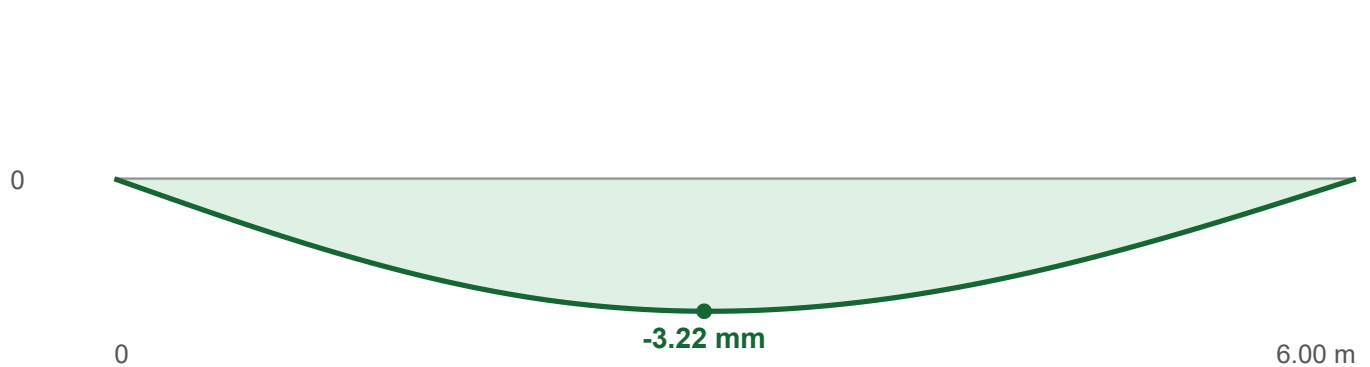
SHEAR FORCE V (kN)



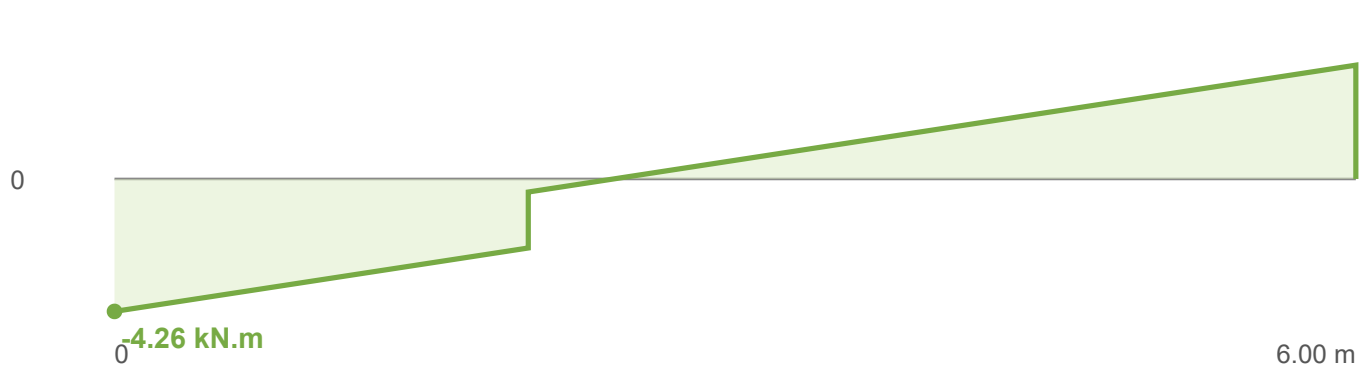
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.425 m, P1 flange tip	@ x = 2.425 m (ULS: 1.35G + 1.5Q): $M_y = 168.867 \text{ kN.m}$, $V = 0.287 \text{ kN}$, $\phi = 0.01961 \text{ rad}$, $\phi' = 3.659e-6 \text{ rad/mm}$, $\phi'' = 9.042e-9 \text{ rad/mm}^2$, $\phi''' = 5.110e-13 \text{ rad/mm}^3$; $M_z = \phi M_y = 3.311 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.89 + 16.89 + 40.26 = 162.04 \text{ N/mm}^2$; $\tau = G t_f \phi' = 4.74 \text{ N/mm}^2$; $(162.04/275)^2 + 3(4.74/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.89 + 16.89 + 40.26 = 162.04$, $\tau = 4.74 \rightarrow 0.348$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 101.05 + 0.88 = 101.92$, $\tau = 0.02 + 4.74 + 0.11 = 4.87 \rightarrow 0.138$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.41 + 0.88 = 98.28$, $\tau = 0.05 + 2.93 = 2.99 \rightarrow 0.128$; P4 web mid-depth: $\sigma_x = 0.00 + 0.88 = 0.88$, $\tau = 0.07 + 2.93 = 3.01 \rightarrow 0.000$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_{\bar{F}} > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_{\bar{F}} > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_{\bar{F}} = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_{\bar{F}} \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

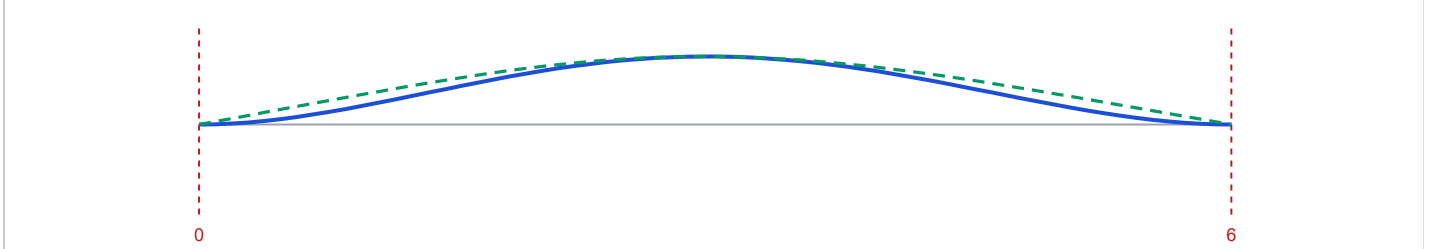
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	669.516 / 606.993 kN.m (shape only: $z_g = z_j = 0$)	1.103	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m [$k = k_w = 1$: clamped / warping-fixed end(s) taken as forks]	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 $v, \phi, \phi' = 0$; End 2 $v, \phi, \phi' = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	669.516 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 669.516}$	0.867	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.867, 0.896, 0.49	0.722	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.722, 0.867, 0.952, 0.976	0.739	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.739 \times 1830.0 \times 275 \leq 503.250 =$	371.977 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 371.977	0.454	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.45 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	$EI_w \phi'''' - GI_T \phi'' = m_t(x)$: warping-torsion FE (240 elements); twist restrained at $x = 0$ m ($\phi = 0, \phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0, \phi' = 0$: warping fixed); closed forms not applicable: warping-fixed end; 2 FE solve(s)	mesh error 0.000 %	FE (≤ 0.5 %)
ϕ_{max} (ULS)	warping-torsion FE, twist restrained at $x = 0$ m ($\phi = 0, \phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0, \phi' = 0$: warping fixed)	0.0204 rad = 1.17°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 0$ m	4.007 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	9.024 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	3.406 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.45$ m: 168.866, 3.937, 3.326 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.250	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_{\alpha}$	0.680 x 0.960 x 1.337	0.874	EN 1993-6 A

$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_{\alpha}M_w/M_{f,Rk}$	@ x = 2.45 m: 168.866/363.160 + 1.000x3.326/83.600 + 0.680x0.960x1.337x3.937/40.255	0.590	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T\phi'$ at x = 0 / x = L; total T = $GI_T\phi'$ - $EI_w\phi''' = 4.380 / 3.540$ kN.m	0.000 / 0.000 kN.m	

Torsion Shear Design @ 0 m

T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 0.000$ kN.m, V = 109.758 kN	0.00 N/mm²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))}.V_{pl,Rd}$	@ x = 0 m: $T_t = 0.00$ N/mm²; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 1.000$	756.260 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 756.260	0.145	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.75 m: $\theta_{max} \leq 2.00^\circ$	$0.0086 \text{ rad} = 0.49^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.49°	OK

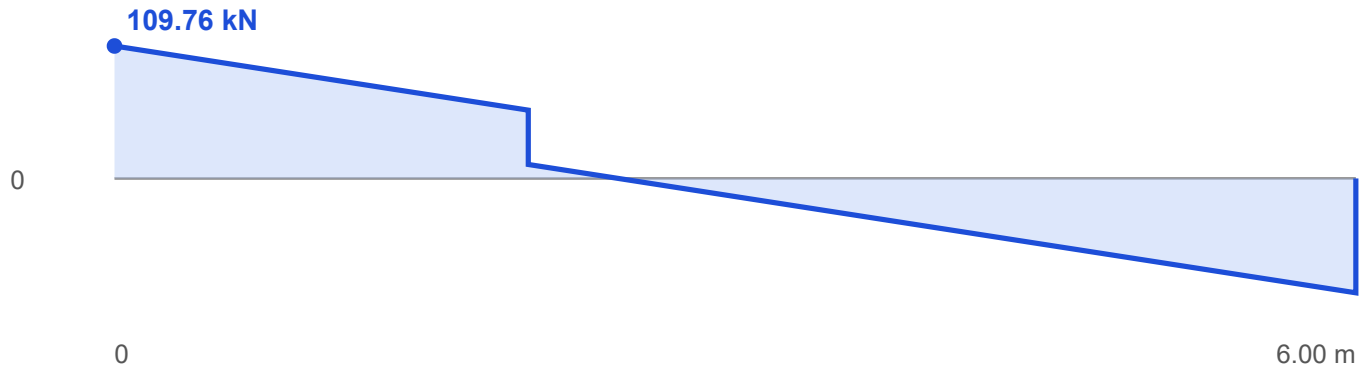
MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.454	0.193	0.145	0.251	0.257	0.250	0.348	0.145	0.590	0.590

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.590

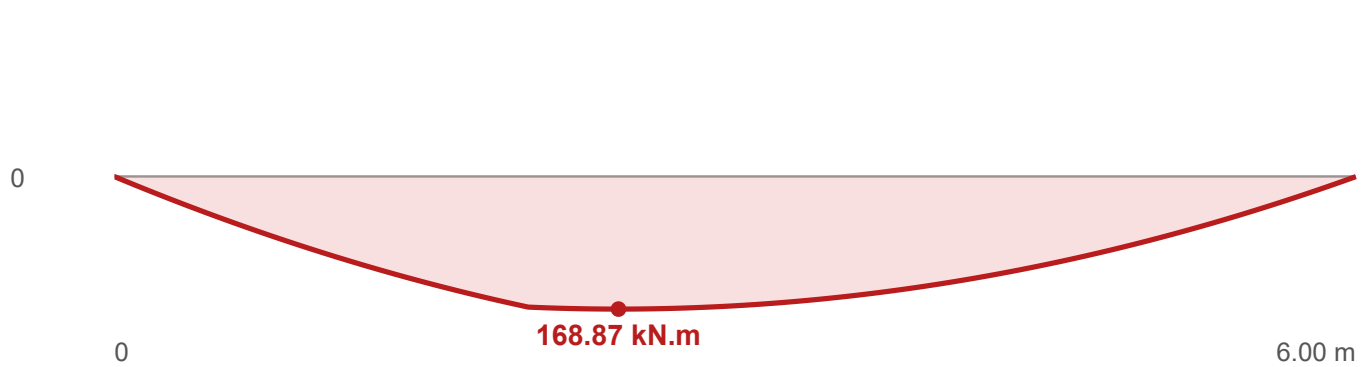
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

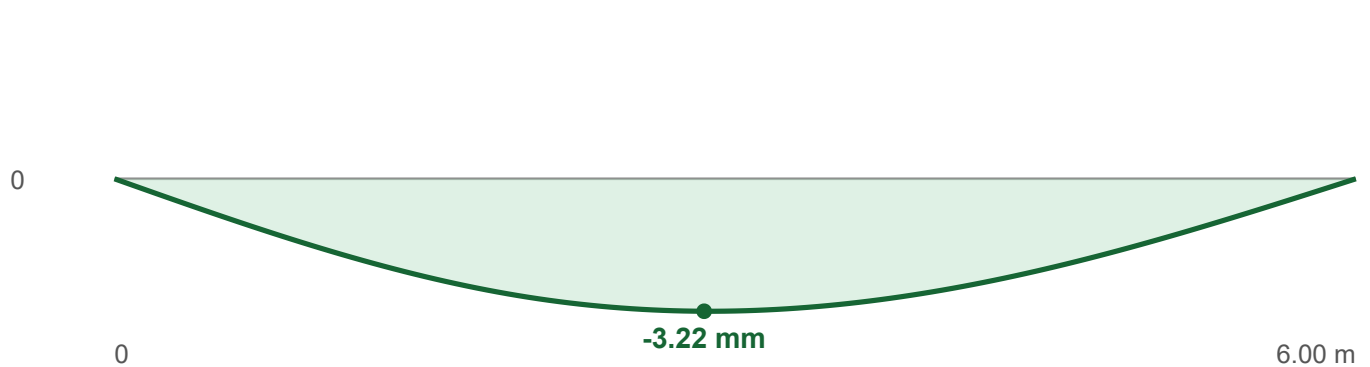
SHEAR FORCE V (kN)



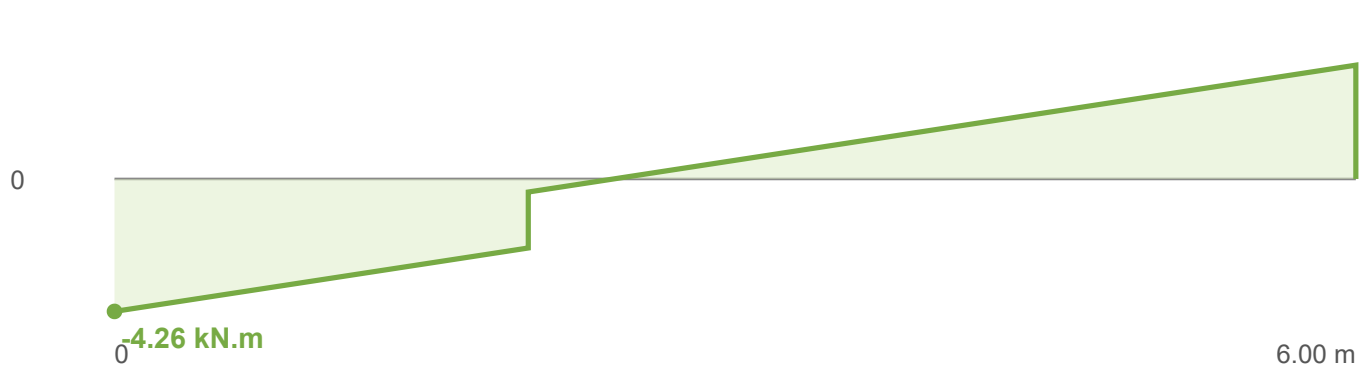
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.425 m, P1 flange tip	@ x = 2.425 m (ULS: 1.35G + 1.5Q): $M_y = 168.867 \text{ kN.m}$, $V = 0.287 \text{ kN}$, $\phi = 0.01961 \text{ rad}$, $\phi' = 3.659e-6 \text{ rad/mm}$, $\phi'' = 9.042e-9 \text{ rad/mm}^2$, $\phi''' = 5.110e-13 \text{ rad/mm}^3$; $M_z = \phi M_y = 3.311 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.89 + 16.89 + 40.26 = 162.04 \text{ N/mm}^2$; $\tau = G t_f \phi' = 4.74 \text{ N/mm}^2$; $(162.04/275)^2 + 3(4.74/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.89 + 16.89 + 40.26 = 162.04$, $\tau = 4.74 \rightarrow 0.348$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 101.05 + 0.88 = 101.92$, $\tau = 0.02 + 4.74 + 0.11 = 4.87 \rightarrow 0.138$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.41 + 0.88 = 98.28$, $\tau = 0.05 + 2.93 = 2.99 \rightarrow 0.128$; P4 web mid-depth: $\sigma_x = 0.00 + 0.88 = 0.88$, $\tau = 0.07 + 2.93 = 3.01 \rightarrow 0.000$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_{\bar{F}} > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_{\bar{F}} > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_{\bar{F}} = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_{\bar{F}} \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.45 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	$EI_w\varphi'''' - GI_T\varphi'' = m_t(x)$: warping-torsion FE (240 elements); twist restrained at $x = 0$ m ($\varphi = 0$, $\varphi' = 0$: warping fixed), $x = 6$ m ($\varphi = 0$, $\varphi' = 0$: warping fixed); closed forms not applicable: warping-fixed end; 0 FE solve(s) + 2 cached	mesh error 0.000 %	FE (≤ 0.5 %)
$\varphi_{\max}$ (ULS)	warping-torsion FE, twist restrained at $x = 0$ m ($\varphi = 0$, $\varphi' = 0$: warping fixed), $x = 6$ m ($\varphi = 0$, $\varphi' = 0$: warping fixed)	0.0204 rad = 1.17°	
B _{Ed} = $EI_w\varphi''$ (max)	@ $x = 0$ m	4.007 kN.m ²	
M _{w,Ed} = $EI_w\varphi''/(h-t_f)$	max over span	9.024 kN.m	
M _{z,Ed} = $\varphi \cdot M_{y,Ed}$	max coincident	3.406 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.45$ m: 168.866, 3.937, 3.326 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.250	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.680 \times 0.960 \times 1.736$	1.134	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.425$ m: $168.867/285.039 + 1.000 \times 3.311/83.600 + 0.680 \times 0.960 \times 1.736 \times 3.943/40.255$	0.743	OK EN 1993-6 A
End torques T _t	St Venant part $GI_T\varphi'$ at $x = 0 / x = L$; total $T = GI_T\varphi' - EI_w\varphi''' = 4.380 / 3.540$ kN.m	0.000 / 0.000 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 0.000$ kN.m, $V = 109.758$ kN	0.00 N/mm ²	P385
-----------------------------	---	------------------------	------

$V_{pl.T.Rd} = \sqrt[3]{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 0.00 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; S_{mod}	756.260 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	$= V_{pl.T.Rd}/V_{pl.Rd} = 1.000$ 109.758 / 756.260	0.145	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.75 m: $\theta_{max} \leq 2.00^\circ$	$0.0086 \text{ rad} = 0.49^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	0.49°	OK

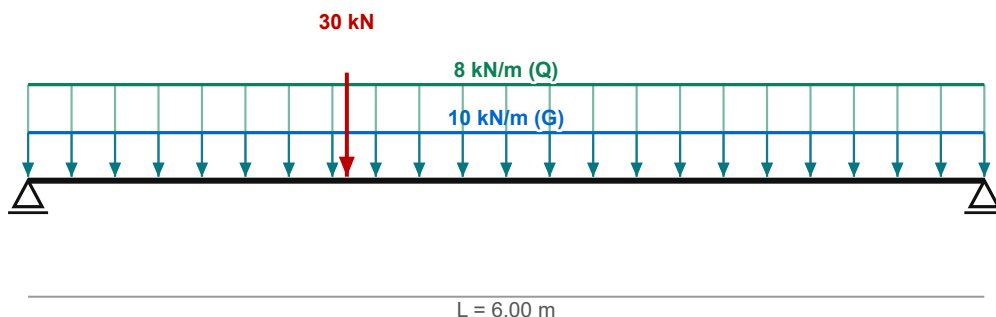
MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.250	0.348	0.145	0.743	0.743

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.743

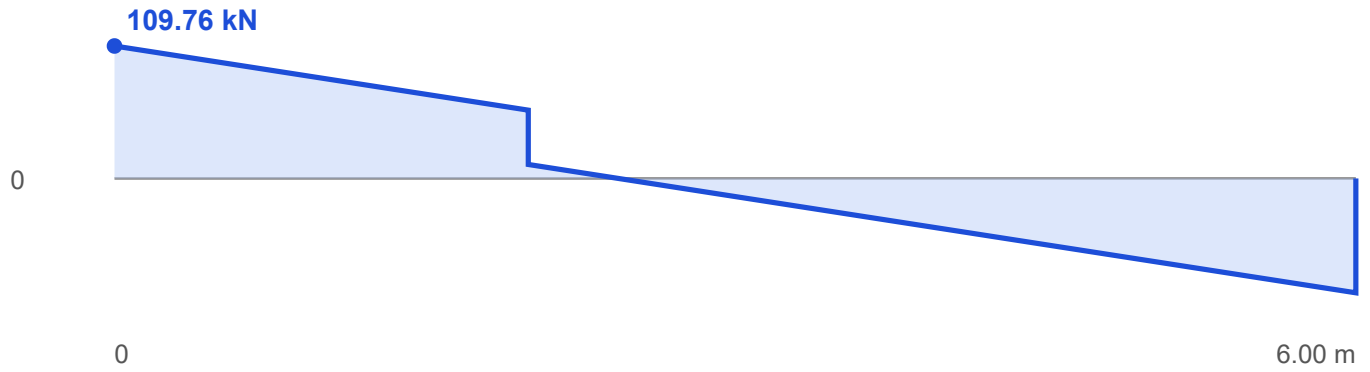
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

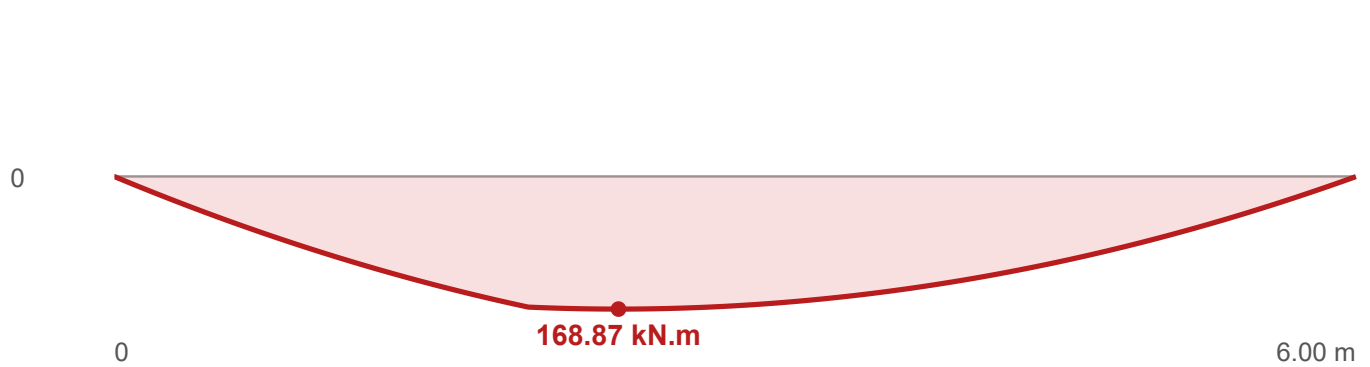
ADVISORY: Standard (closed-form) M_{cr} : warping-fixed ends (End 1 and End 2) taken as fork ends, $k = k_w = 1$ (conservative: the SN003a $k = 0.5 / k_w = 0.5$ rows are not applied; the FE eigen route uses $v' = 0 / \phi' = 0$ and gives the higher M_{cr}).



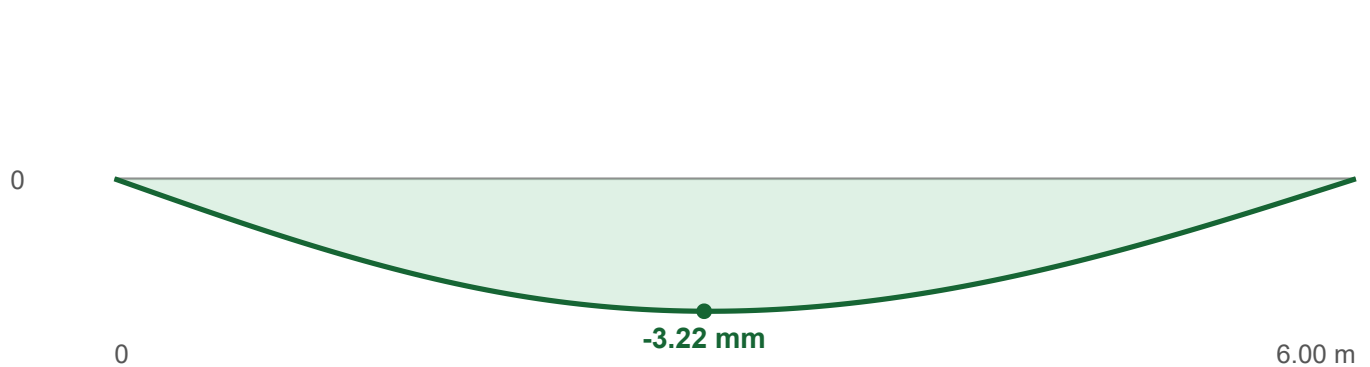
SHEAR FORCE V (kN)



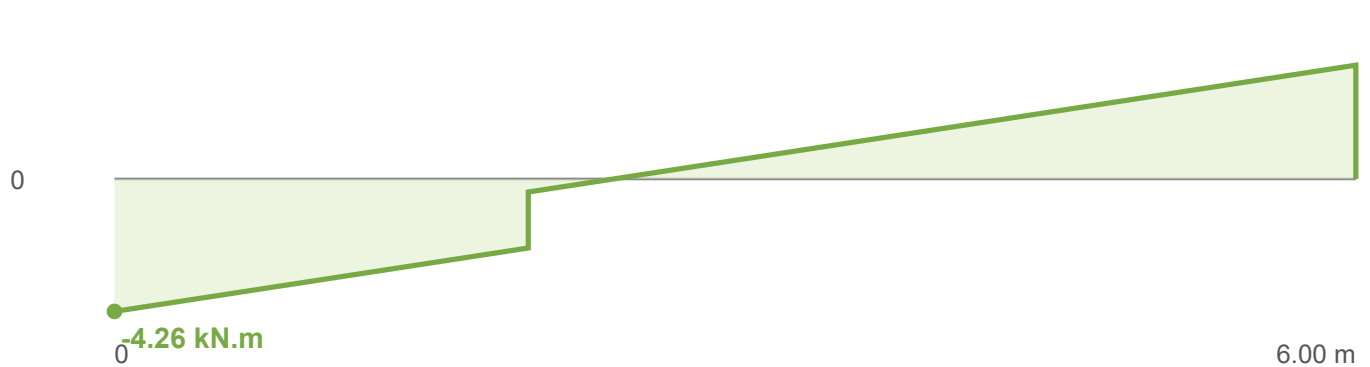
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038\text{e-}5 \text{ rad/mm}$, $\phi'' = 1.724\text{e-}8 \text{ rad/mm}^2$, $\phi''' = 2.103\text{e-}12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$ P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
σ_x, τ , (6.1) at the 4 section points	$f_y / \gamma_{M0} = 275$		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

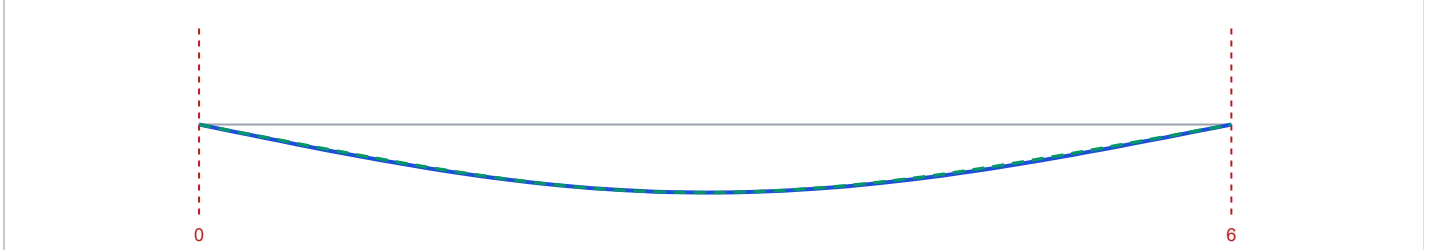
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.93 m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w ϕ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.753	1.025	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.365 m: 168.802/282.919 + 1.000x9.873/83.600 + 0.663x0.882x1.753x7.403/40.255	0.903	OK EN 1993-6 A
End torques T _t	St Venant part G I _T ϕ' at x = 0 / x = L; total T = G I _T ϕ' - E I _w $\phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902 \text{ kN.m}$, $V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

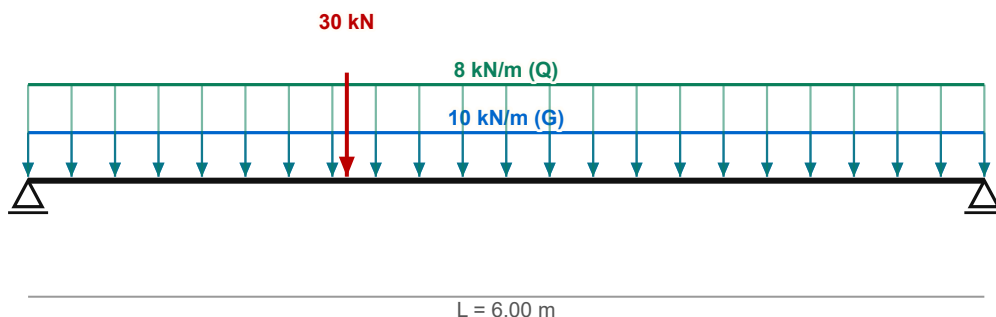
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm}$ @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.903	0.903

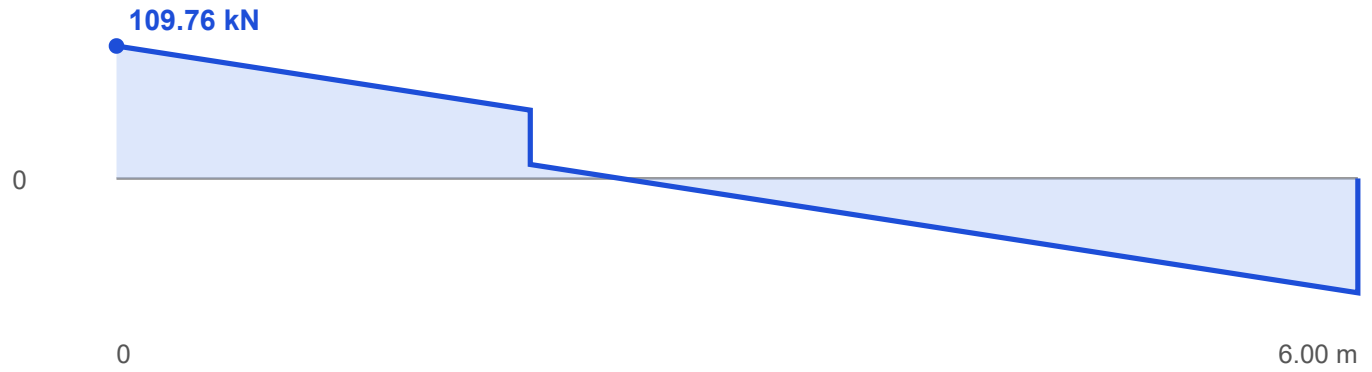
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.903

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

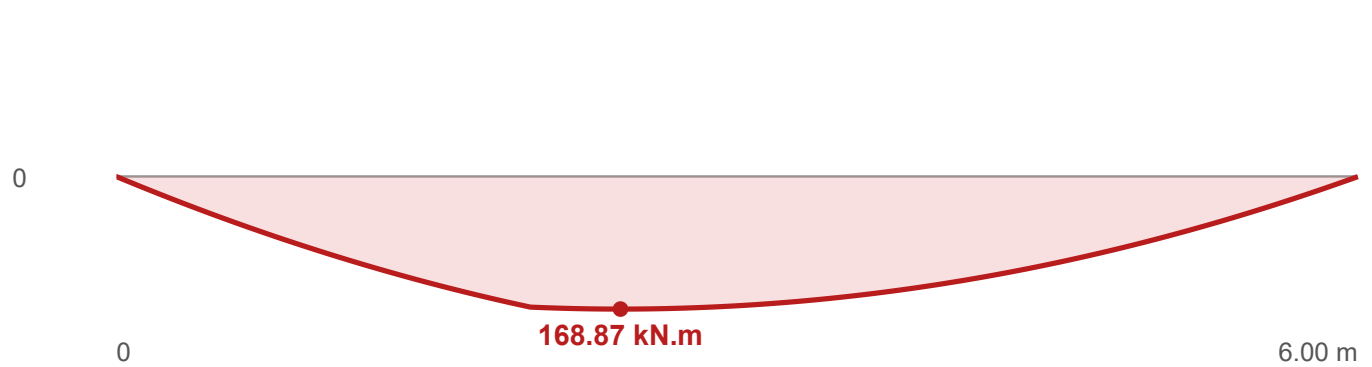
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



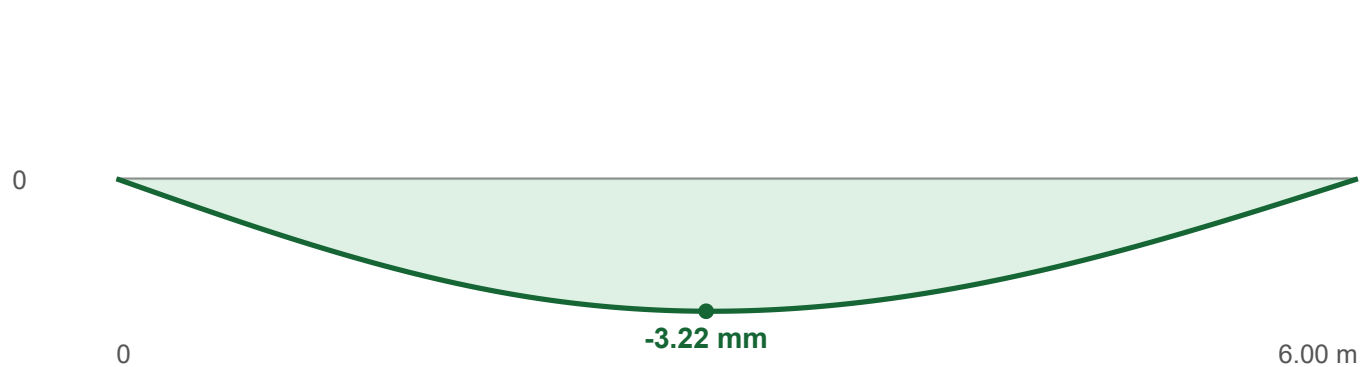
SHEAR FORCE V (kN)



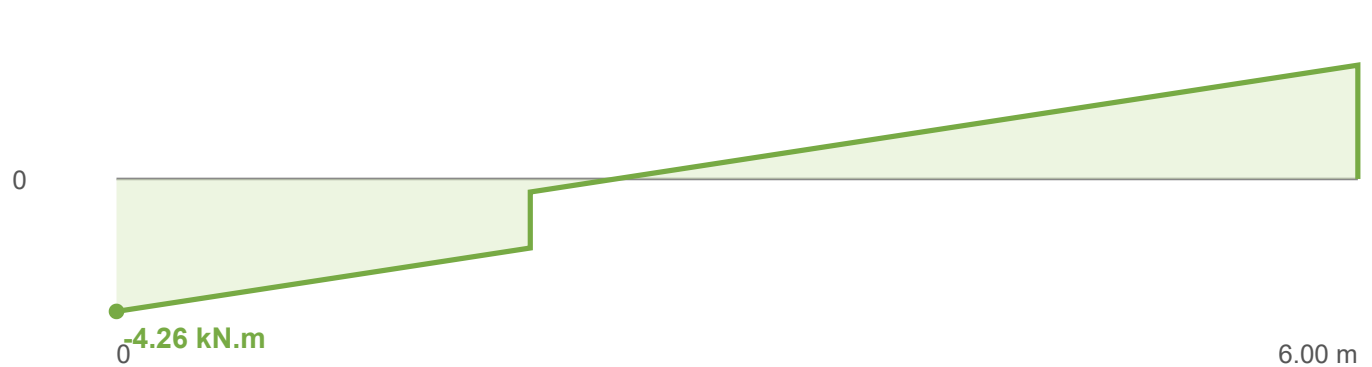
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
$\varphi_{\max}$ (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ·M _{y,Ed}	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
k = k _w ·k _{zw} ·k _α	$0.663 \times 0.882 \times 1.736$	1.015	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ x = 2.365 m: $168.802/285.039 + 1.000 \times 9.873/83.600 + 0.663 \times 0.882 \times 1.736 \times 7.403/40.255$	0.897	OK EN 1993-6 A
End torques T _t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T _t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 1.902 kN.m, V = 109.758 kN	27.22 N/mm ²	P385
V _{pl,T,Rd} = $\sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: T _t = 27.22 N/mm ² ; V _{pl,Rd} = 756.260; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.929	702.492 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

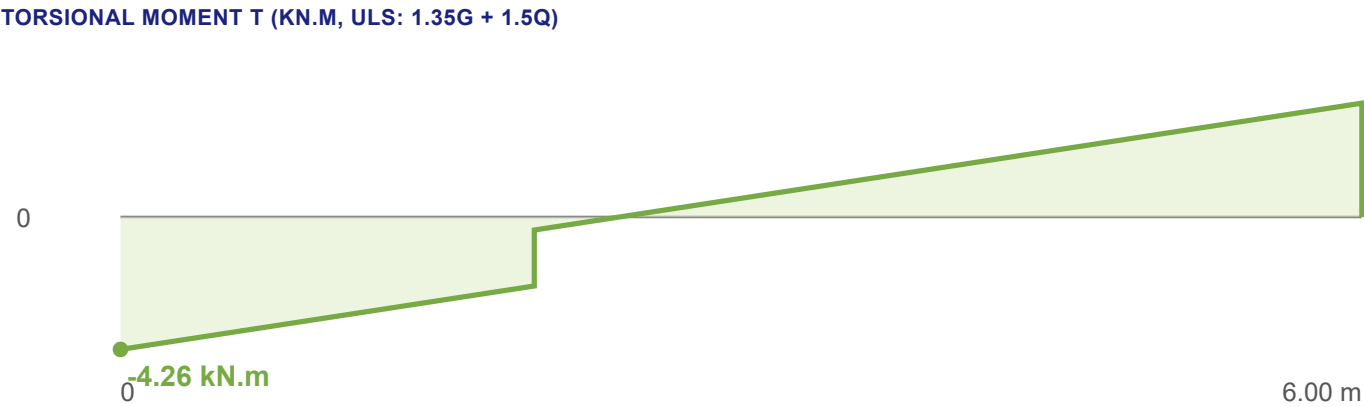
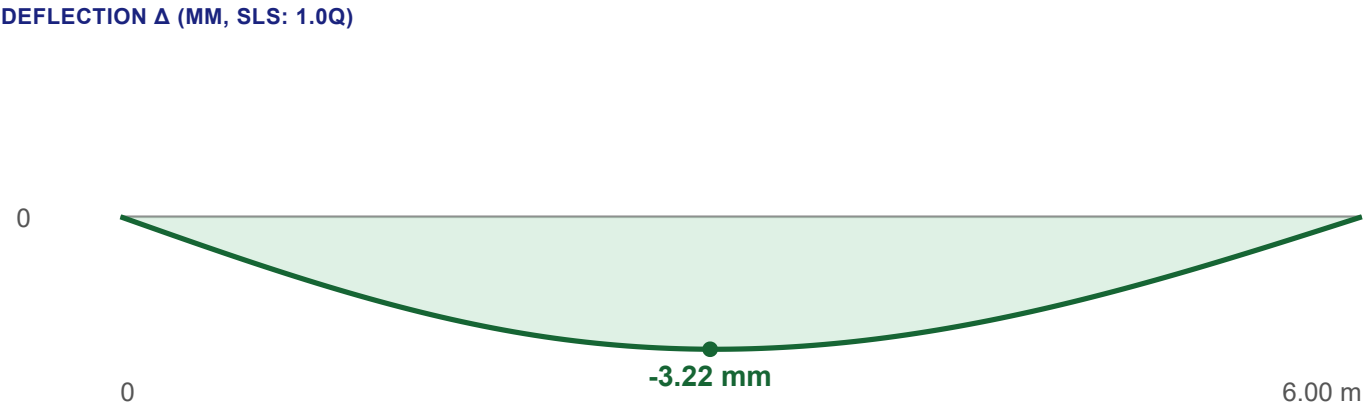
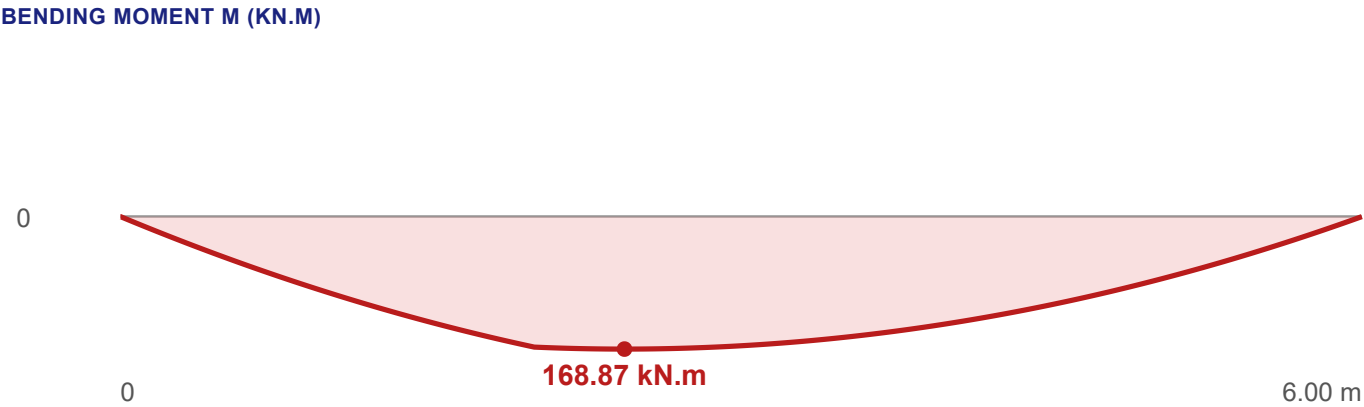
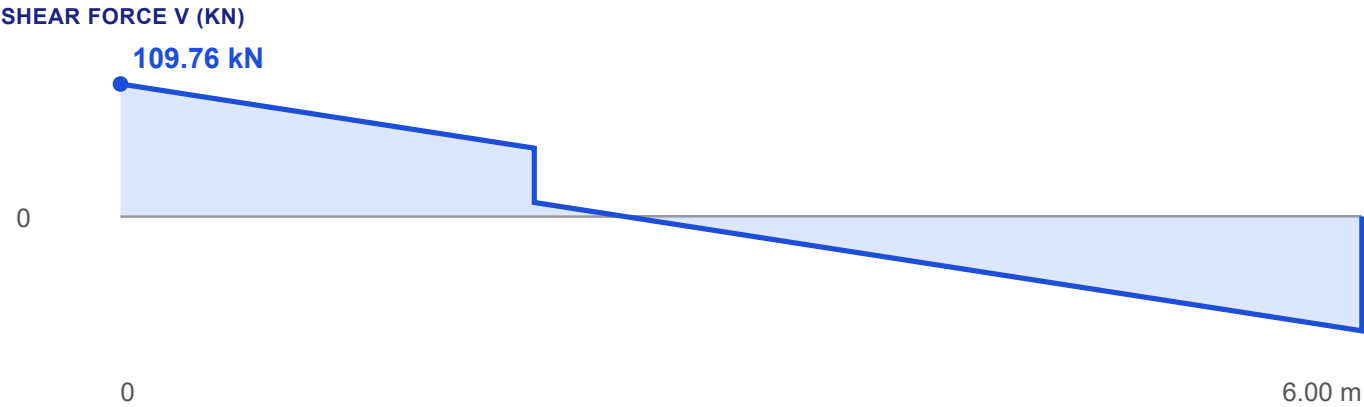
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R6 "DESTABILIZING LOADS" as the switch alone with every $z_g = 0$: the eigen route BLOCKS this contradictory input (load height is carried by z_g), the standard route applies $L_E = 1.2 L$ (BS 5950 practice) - kept to record the refusal; the modelled destabilising case is L_{3t} / L_4 ($z_g = +D/2$)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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The diagram shows a horizontal beam of length $L = 6.00 \text{ m}$ supported by a pin support on the left and a roller support on the right. A point load of 30 kN is applied downwards at the midpoint. Two distributed loads are applied over the entire length of the beam: a green load of 8 kN/m labeled (Q) and a blue load of 10 kN/m labeled (G).



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

NOT VERIFIED

Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833 ; 0.833 \times 192.79$	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) \quad 0.179 \quad \text{OK 7.2}$$

$$= 0.251 \leq 1.4 \quad (\text{ULS: } 1.35G + 1.5Q)$$

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm ⁴)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

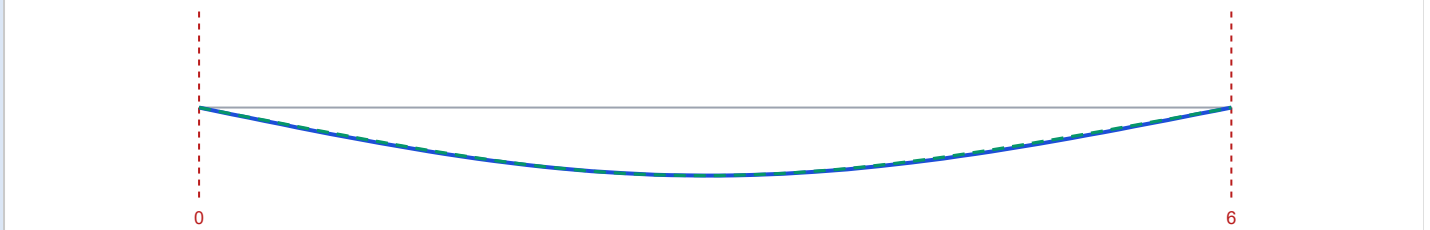
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	3.435 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.882 \times 1.753$	1.025	EN 1993-6 A

$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2.365 m: 168.802/282.919 + 1.000x9.873/83.600 + 0.663x0.882x1.753x7.403/40.255	0.903	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T\phi'$ at x = 0 / x = L; total T = $GI_T\phi'$ - $EI_w\phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, V = 109.758 kN	27.22 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	0.0252 rad = $1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

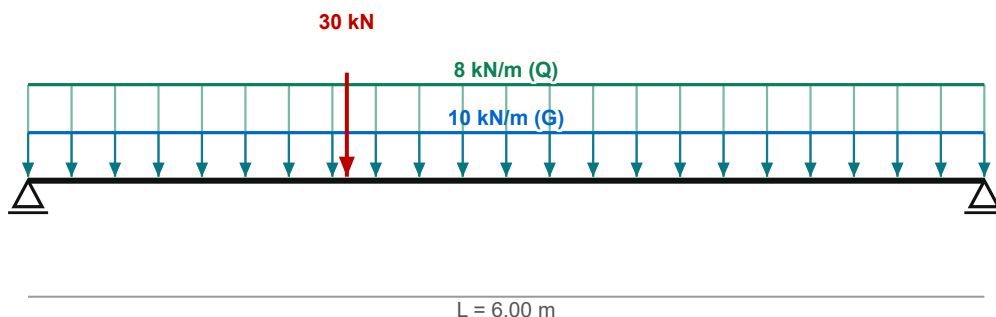
MA/Mc	$M_{(y.Ed)}/M_{(b.Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.903	0.903

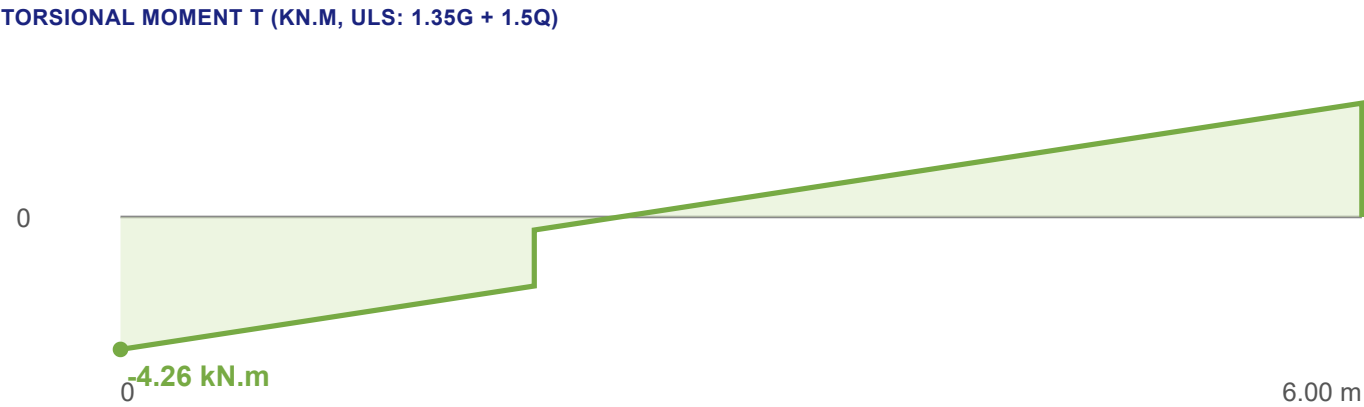
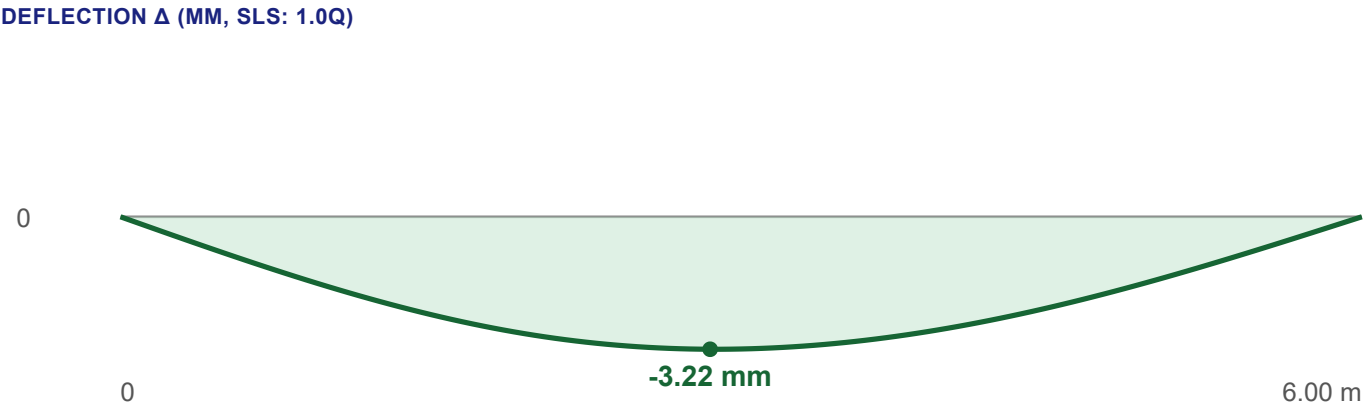
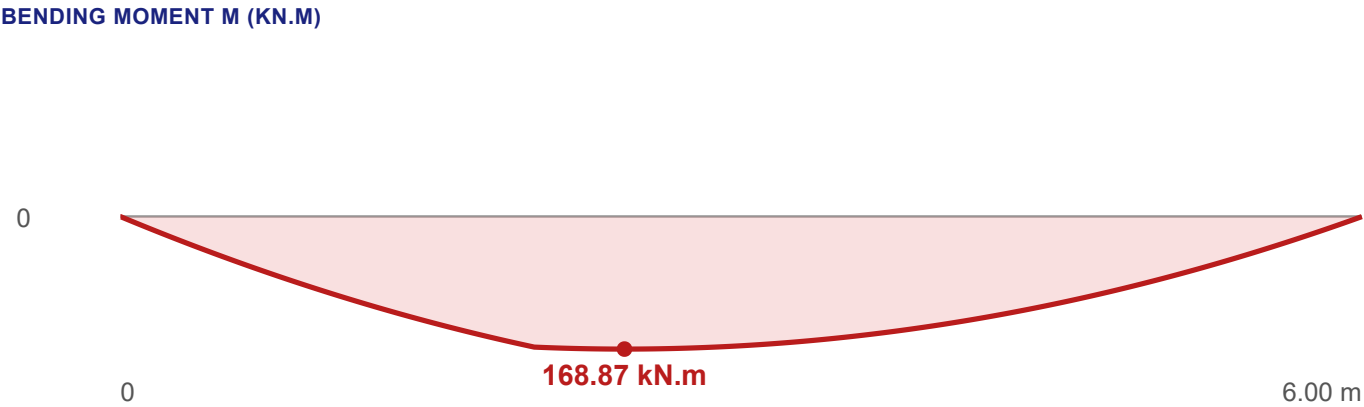
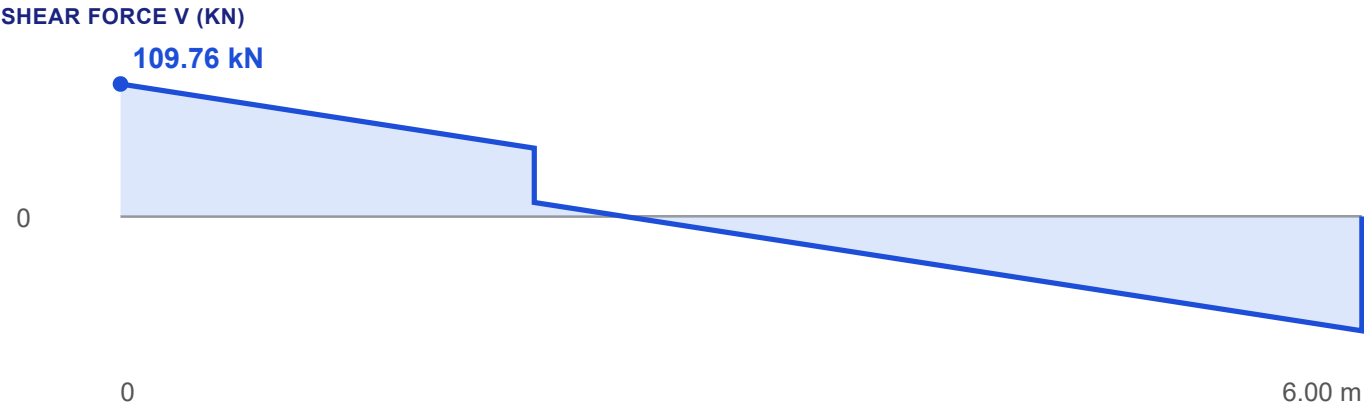
NOT VERIFIED — governing LTB+torsion (EN 1993-6 Annex A) = 0.903

NOT COVERED: Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (82.0 kg/m)	457 x 191 x 82 UB [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 0.00, 168.87, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1.2 L$	1.2 x 6 =	7.2 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 7.200, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	306.318 kN.m	SN003a
$\lambda_{LT} = \sqrt[3]{W_{pl,y}/M_{cr}}$	$\sqrt[3]{1830.0 \times 275 / 306.318}$	1.282	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.282, 1.332, 0.49	0.483	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.483, 1.282, 0.928, 0.981	0.493	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.493 \times 1830.0 \times 275 \leq 503.250 =$	248.097 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.637$; $\chi_{LT,mod} = 0.341$; $M_{y,Ed}/M_{b,Rd,simp} = 0.983$	171.787 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 248.097	0.681	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0} , S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
$\varphi_{\max}$ (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.883 \times 2.229$	1.304	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2.315 m: $168.675/243.284 + 1.000 \times 9.793/83.600 + 0.663 \times 0.883 \times 2.229 \times 7.449/40.255$	1.052	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, V = 109.758 kN	27.22 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

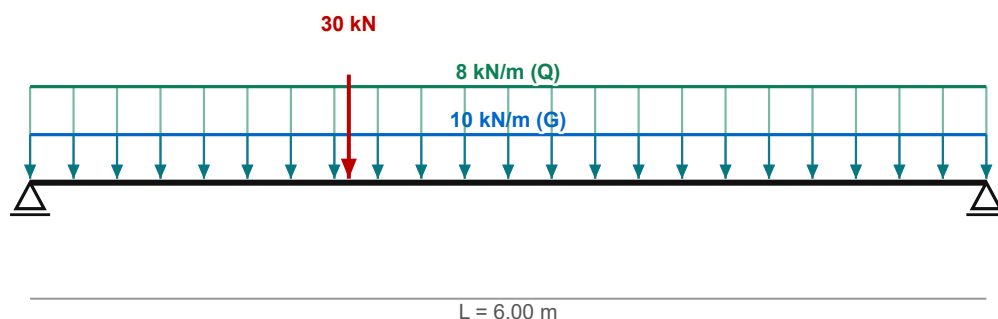
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.681	0.193	0.145	0.251	0.257	0.415	0.710	0.156	1.052	1.052

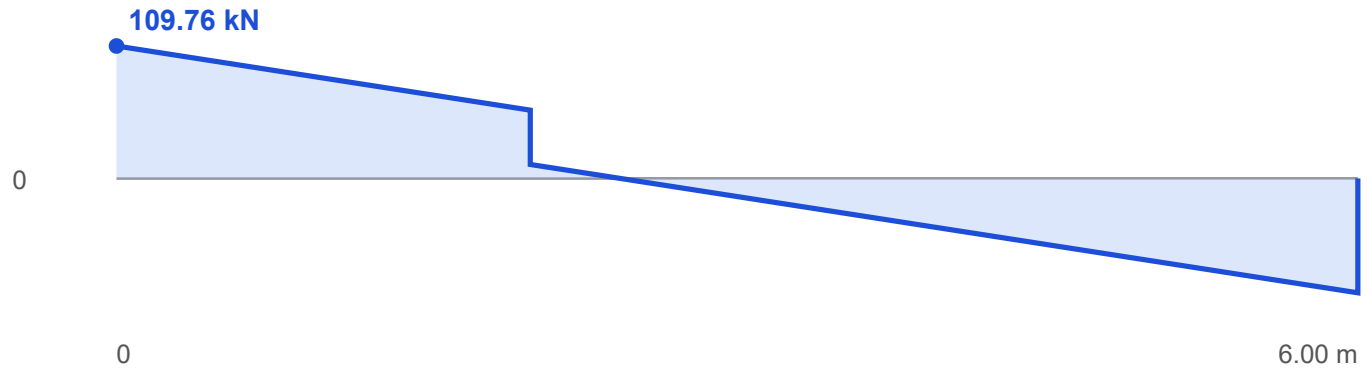
FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.052

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

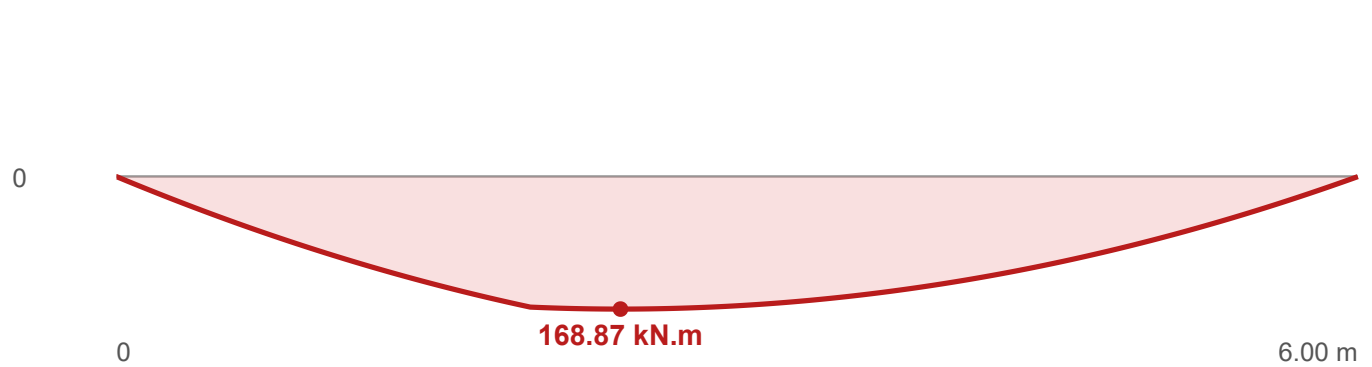
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



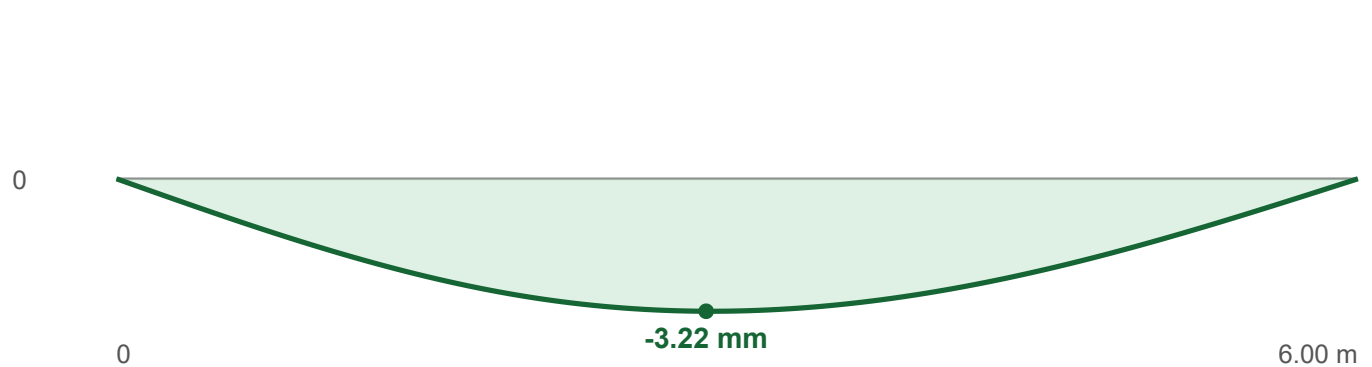
SHEAR FORCE V (kN)



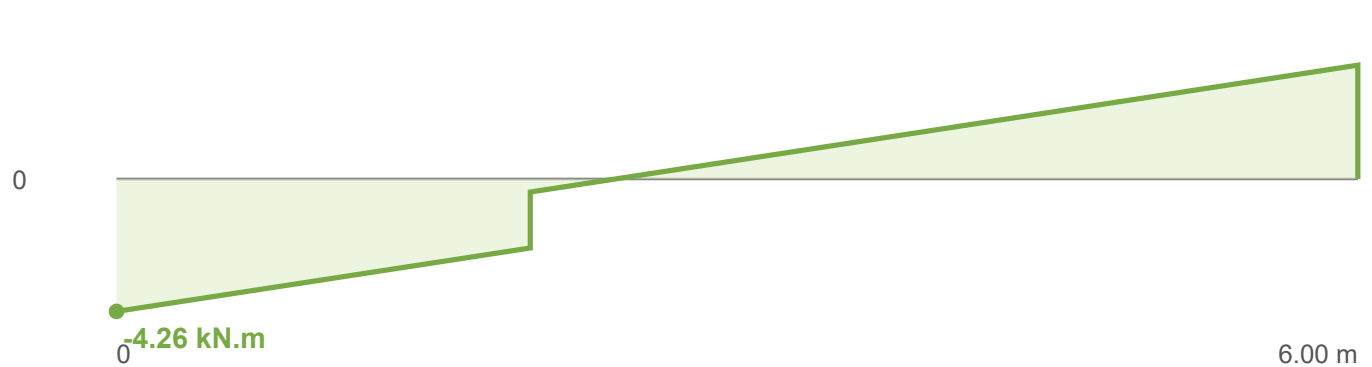
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

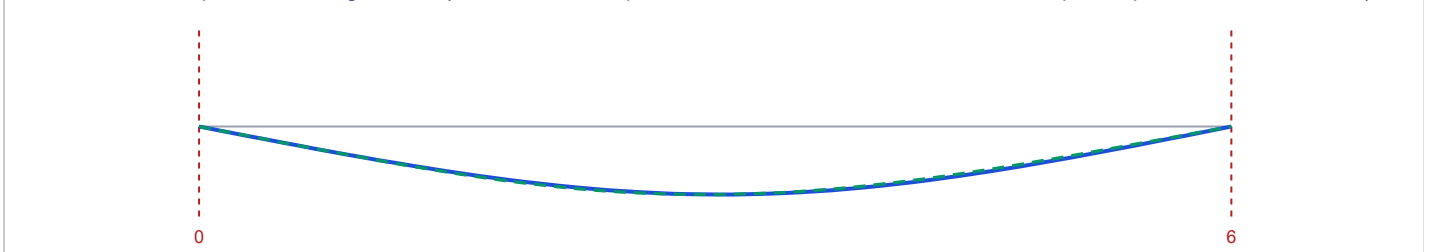
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -230$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	542.681 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 542.681}$	0.963	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.963, 0.986, 0.49	0.662	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.662, 0.963, 0.934, 0.969	0.683	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.683 \times 1830.0 \times 275 \leq 503.250 =$	343.812 kN.m	
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 343.812$	0.491	OK

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 3.02$ m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = $E I_w \phi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
M _{w,Ed} = $E I_w \phi''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_{\alpha}$	$0.663 \times 0.881 \times 1.452$	0.849	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ $x = 2.39$ m: $168.840/333.034 + 1.000 \times 9.909/83.600 + 0.663 \times 0.881 \times 1.452 \times 7.379/40.255$	0.781	OK EN 1993-6 A
End torques T _t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total T = $GI_T \phi' - E I_w \phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902 \text{ kN.m}$, $V = 109.758 \text{ kN}$	27.22 N/mm ²	P385
$V_{pl.T.Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl.Rd} = 756.260$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

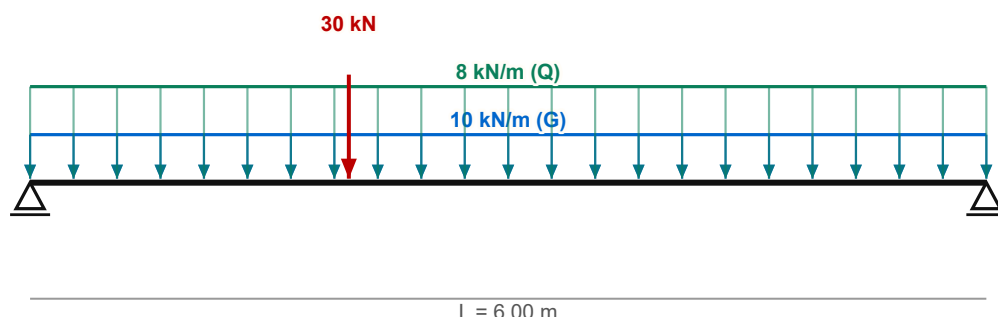
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm}$ @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.491	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.781	0.781

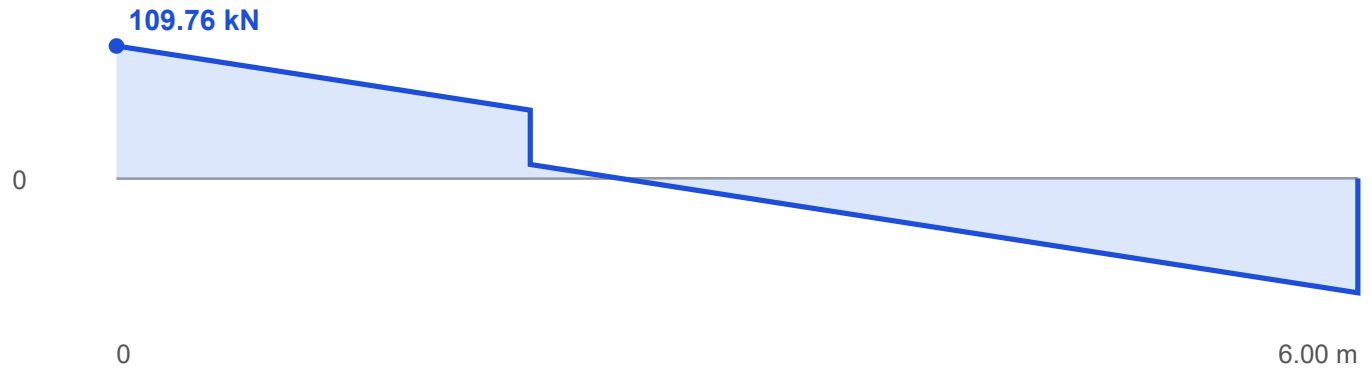
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.781

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

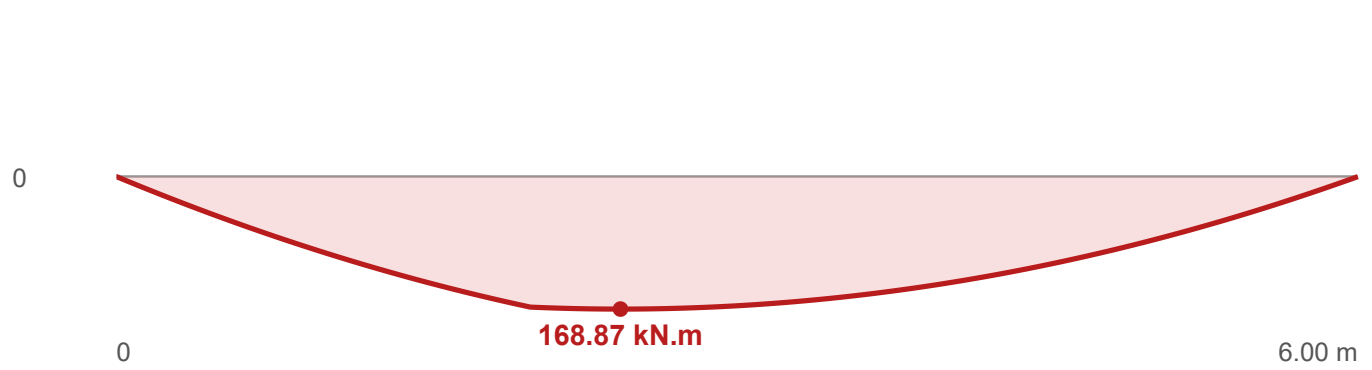
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



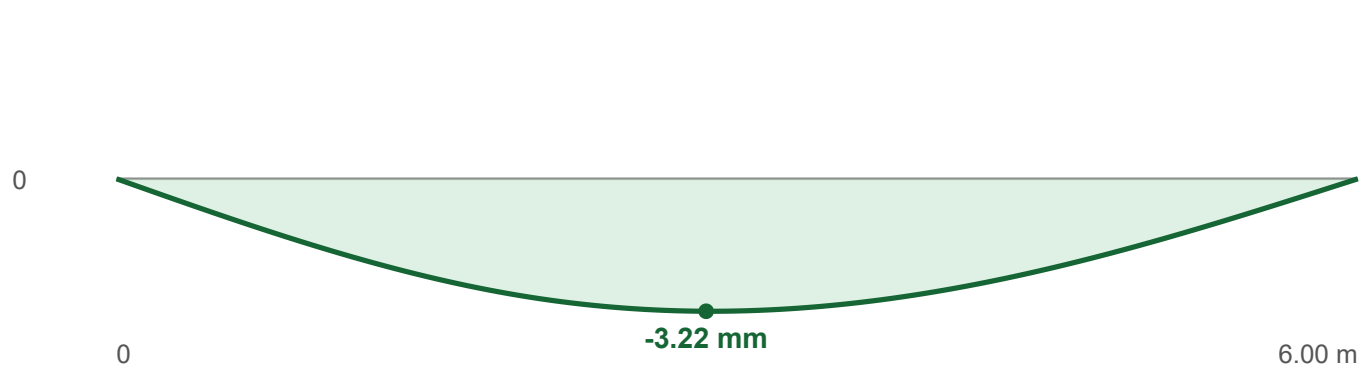
SHEAR FORCE V (kN)



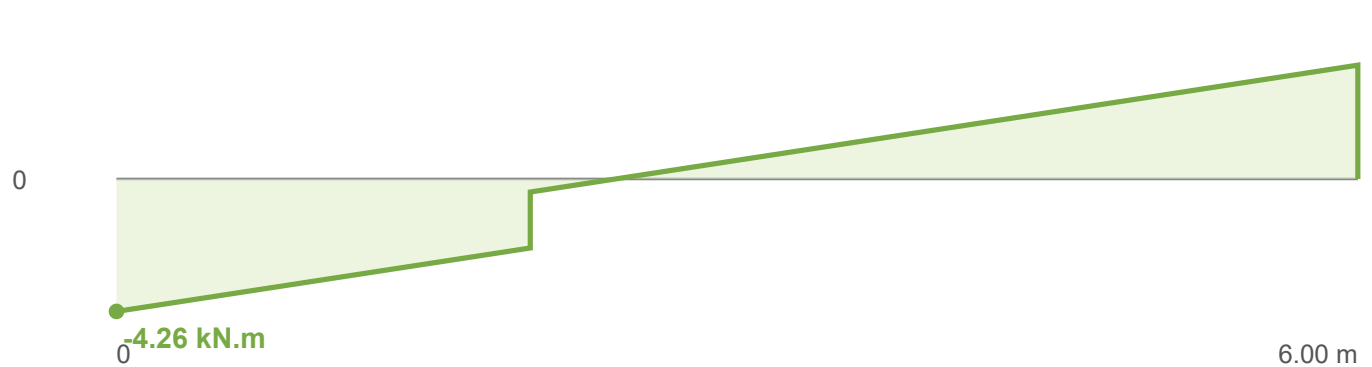
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = -230$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$168.868 / 293.432$	0.575	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0} , S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
φ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.882 \times 1.736$	1.015	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2.365 m: 168.802/285.039 + 1.000x9.873/83.600 + 0.663x0.882x1.736x7.403/40.255	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

τ_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, V = 109.758 kN	27.22 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)

$V_{Ed}/V_{pl.T.Rd}$

109.758 / 702.492

0.156

OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

3.22 mm

OK

Torq in Case 1 @ 2.788 m: $\theta_{\max} \leq 2.00^\circ$

$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)

1.44°

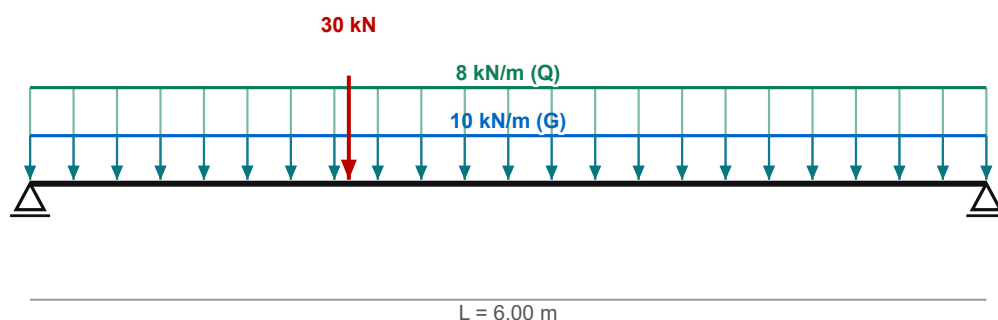
OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.897	0.897

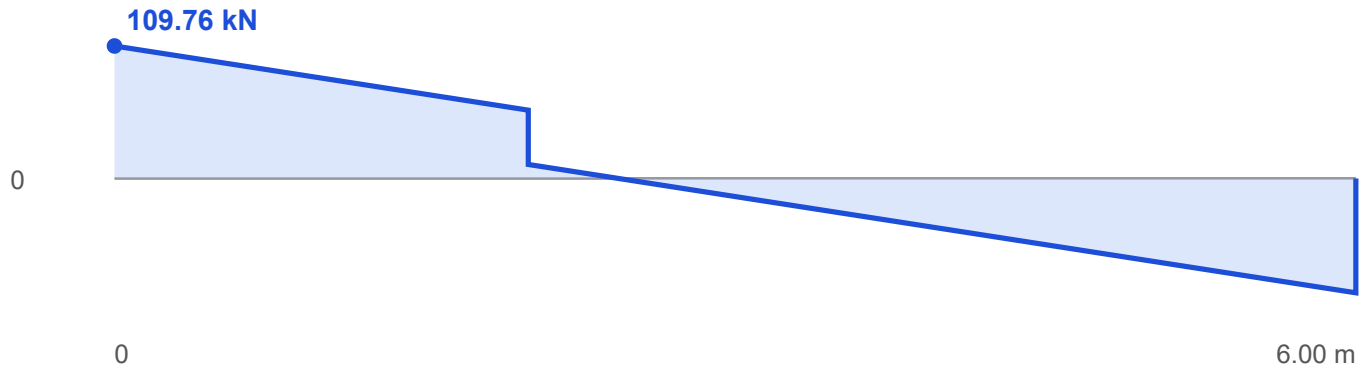
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

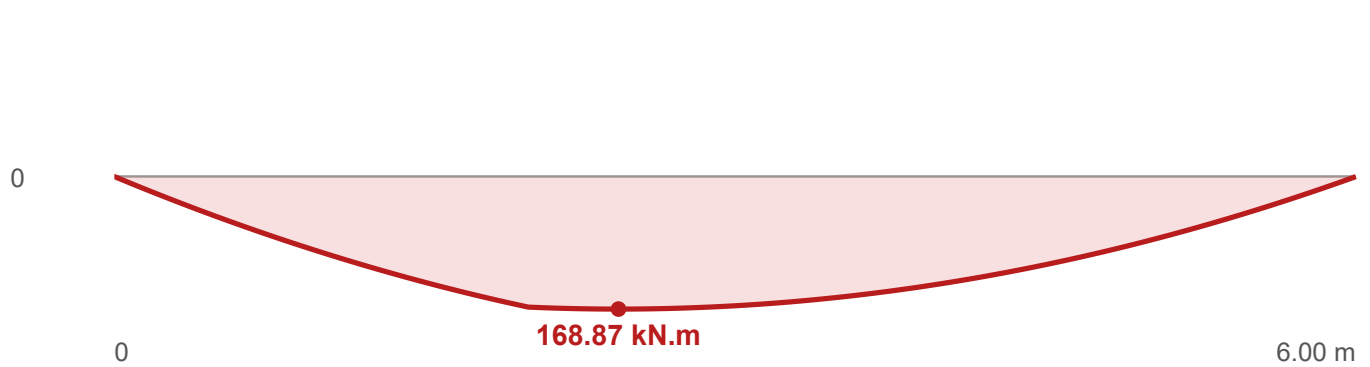
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



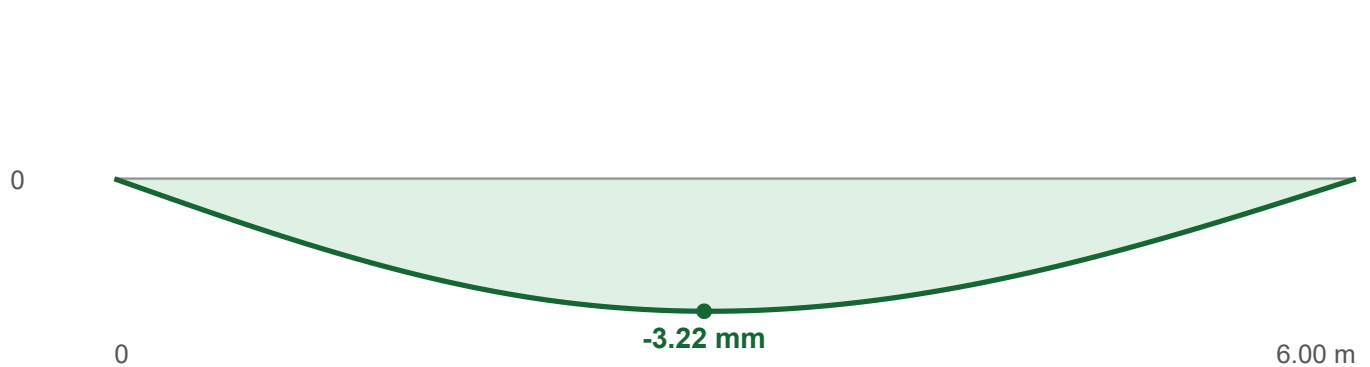
SHEAR FORCE V (kN)



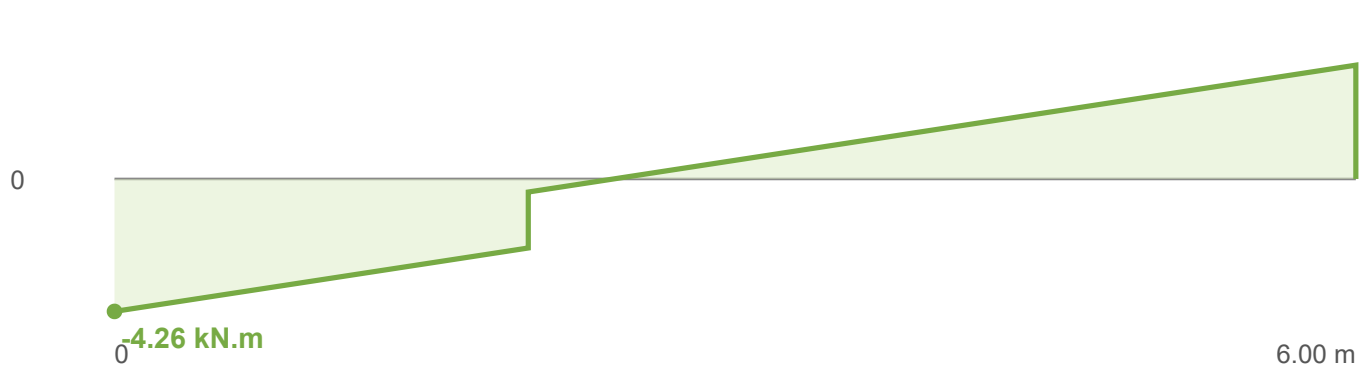
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd} \text{ (at max M)}$	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$ P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
σ_x, τ , (6.1) at the 4 section points	$f_y / \gamma_{M0} = 275$		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

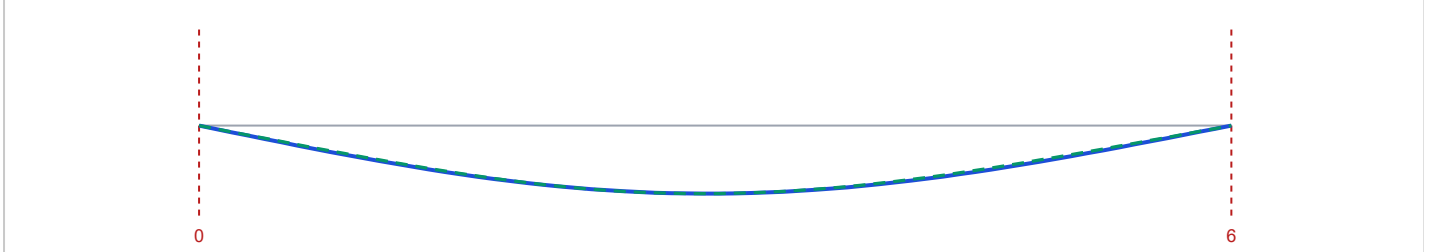
$C_1 = M_{cr}/M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k _c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Advisory	The L_e factor does not affect EC3 LTB on the eigen route; the buckling length is set by the end restraints and the restraint positions. It overrides the strut lengths of both axes (the end-fixity defaults are printed with them).	advisory
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Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at x = 2.93 m)



Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\phi = 0$, warping free)	L/a = 3.23	P385 App C
ϕ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = E I _w ϕ'' (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = E I _w $\phi'''/(h-t_f)$	max over span	7.736 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
k = k _w · k _{zw} · k _α	0.663 x 0.882 x 1.753	1.025	EN 1993-6 A

$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_{\alpha}M_w/M_{f,Rk}$	@ x = 2.365 m: 168.802/282.919 + 1.000x9.873/83.600 + 0.663x0.882x1.753x7.403/40.255	0.903	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T\phi'$ at x = 0 / x = L; total T = $GI_T\phi'$ - $EI_w\phi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, V = 109.758 kN	27.22 N/mm ²	P385
$V_{pl.T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))}.V_{pl,Rd}$	@ x = 0 m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl.T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl.T,Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

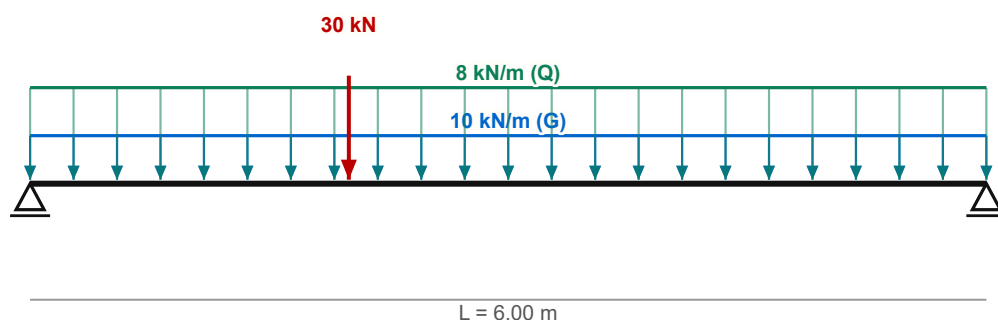
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{max} \leq 2.00^\circ$	0.0252 rad = $1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	$M_{(y.Ed)}/M_{(b.Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.903	0.903

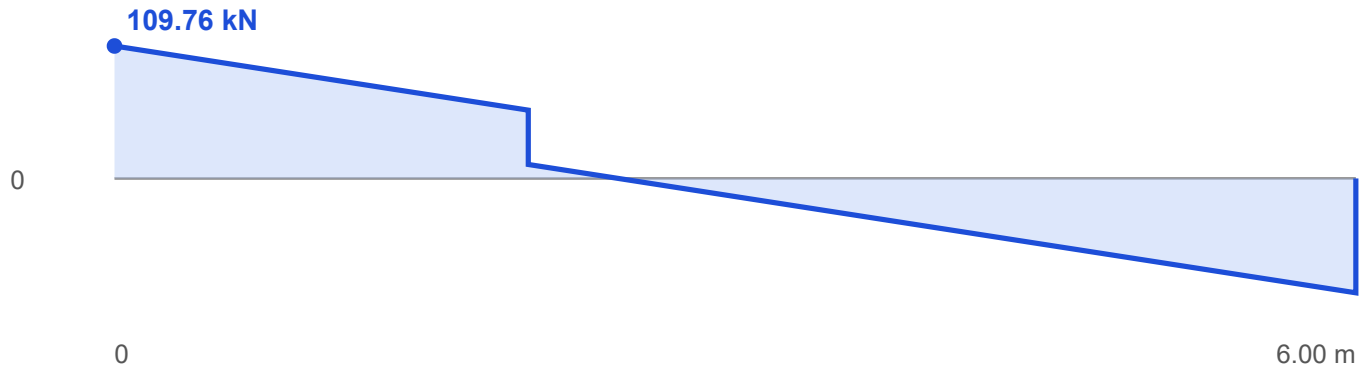
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.903

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

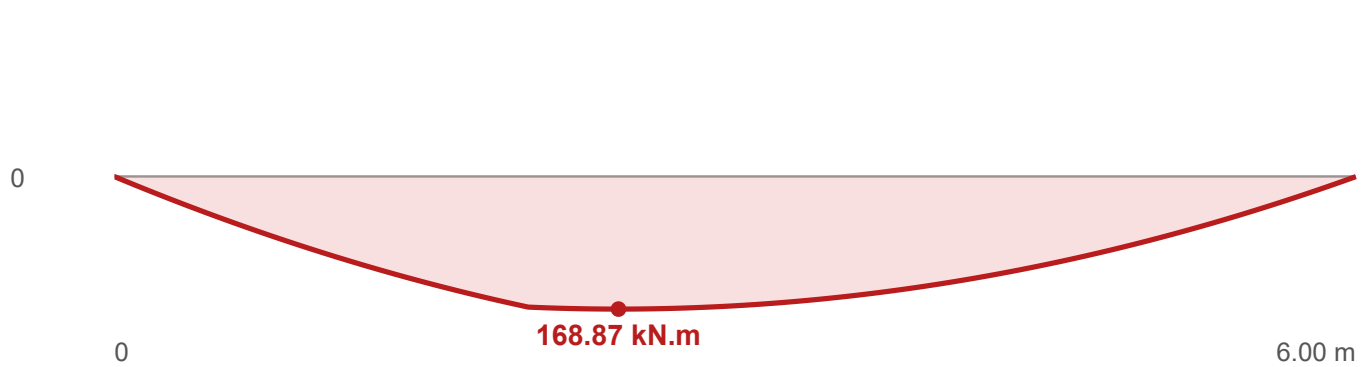
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



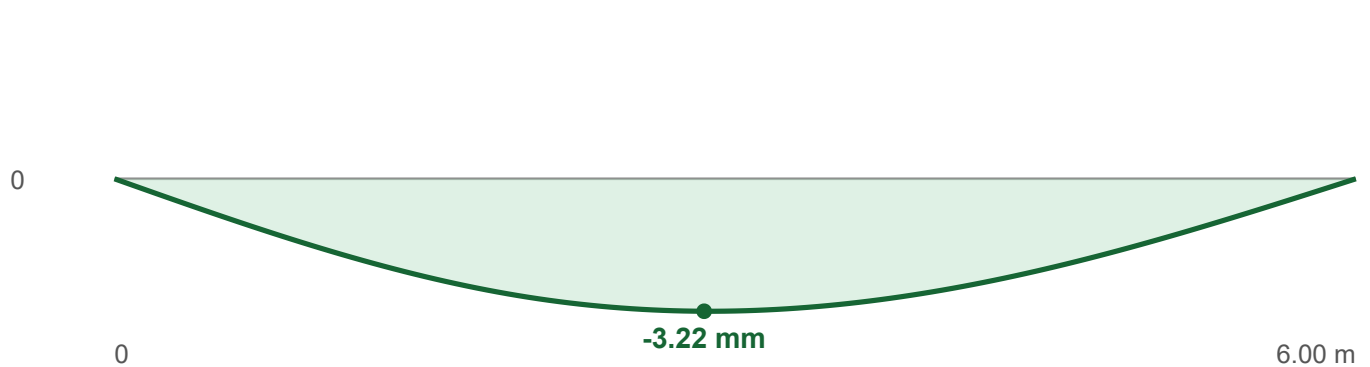
SHEAR FORCE V (kN)



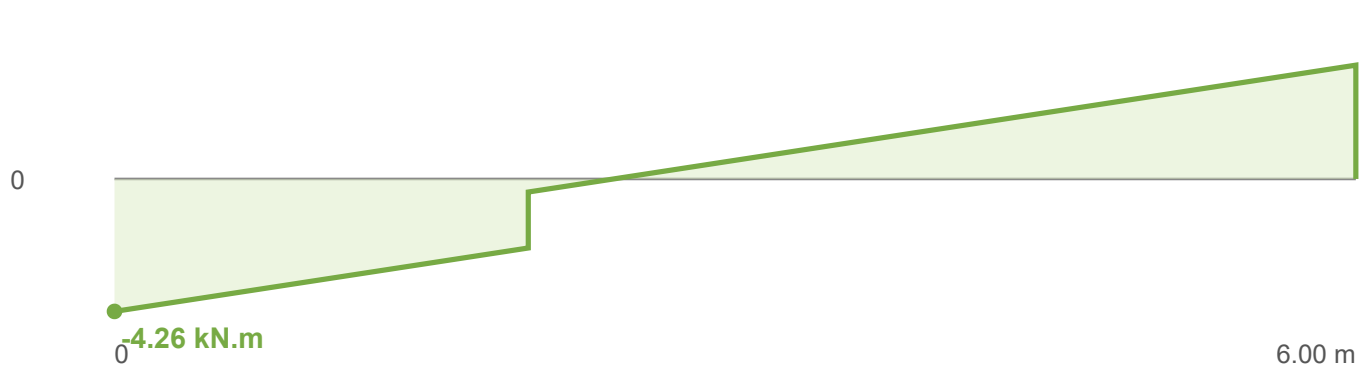
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{\max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 0.85 L$	$0.85 \times 6 =$	5.1 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 5.100, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	510.582 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 510.582}$	0.993	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.993, 1.015, 0.49	0.643	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.643, 0.993, 0.928, 0.966	0.666	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.666 \times 1830.0 \times 275 \leq 503.250 =$	335.069 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.159$; $\chi_{LT,mod} = 0.562$; $M_{y,Ed}/M_{b,Rd,simp} = 0.597$	282.726 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 335.069	0.504	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W_{n0} , S_{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
$\varphi_{\max}$ (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 40.255$, $M_{pl,z} = 83.600$	0.415	OK P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$0.663 \times 0.882 \times 1.494$	0.874	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ x = 2.365 m: 168.802/323.840 + $1.000 \times 9.873/83.600 +$ $0.663 \times 0.882 \times 1.494 \times 7.403/40.255$	0.800	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	

Torsion Shear Design @ 0 m

T_t at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 1.902$ kN.m, V = 109.758 kN	27.22 N/mm ²	P385
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.929$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

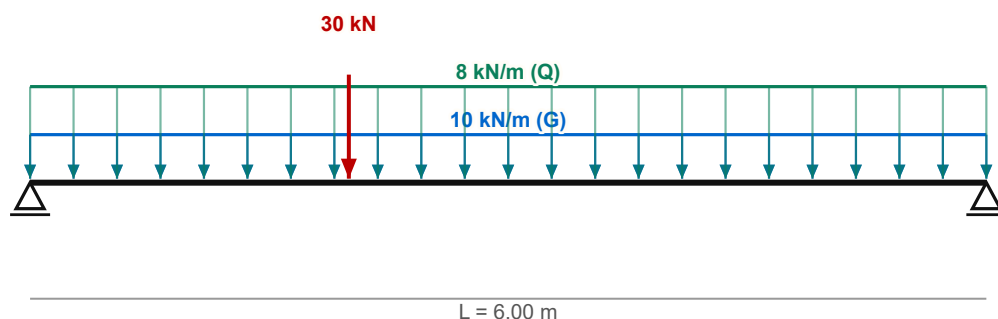
In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
Torq in Case 1 @ 2.788 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.0252 \text{ rad} = 1.44^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	1.44°	OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
0.336	0.504	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.800	0.800

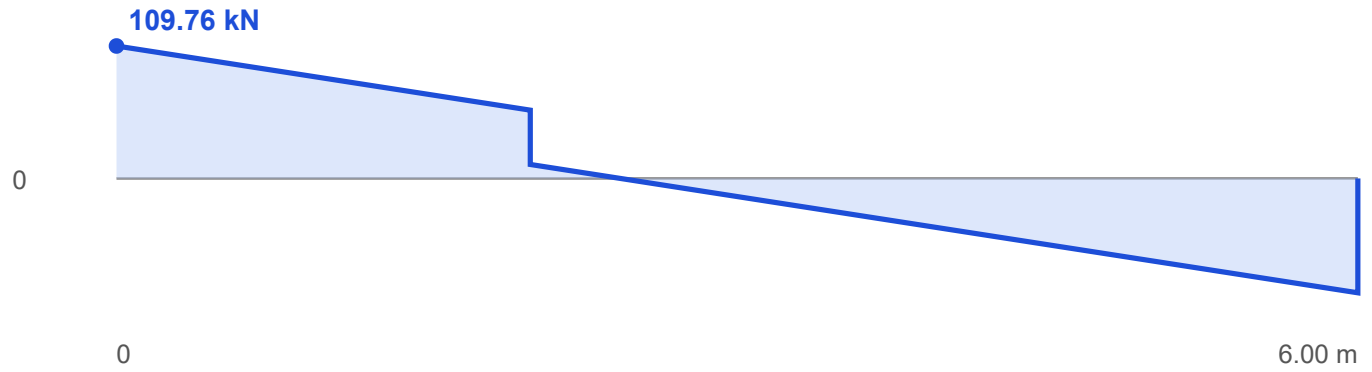
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.800

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

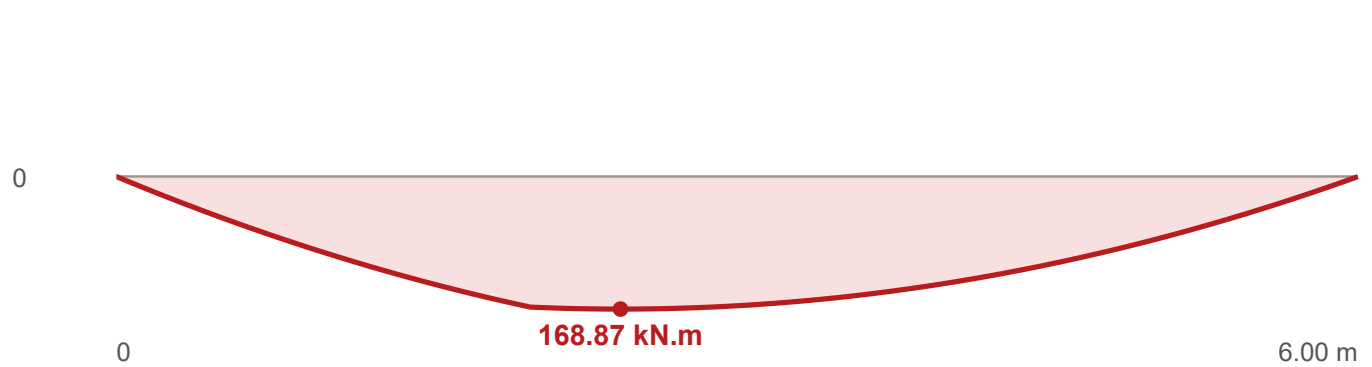
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.



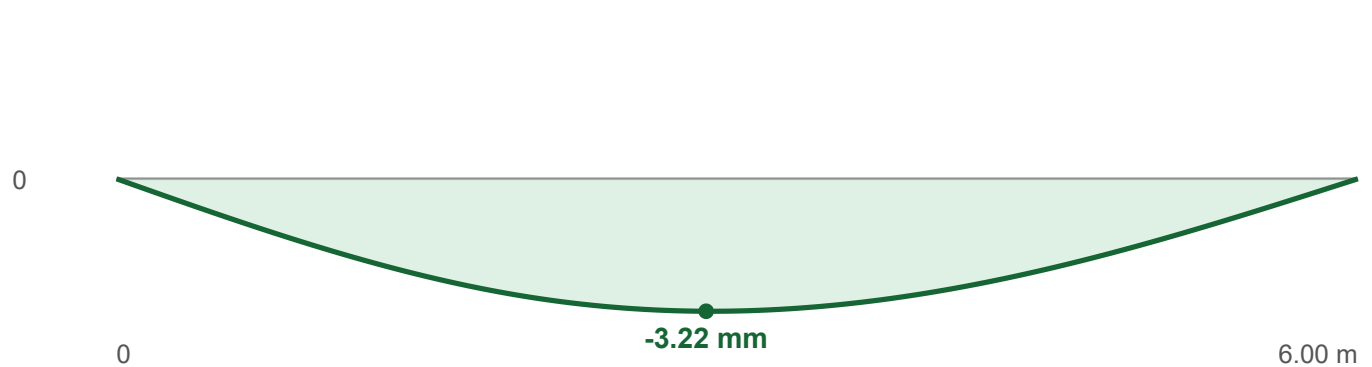
SHEAR FORCE V (kN)



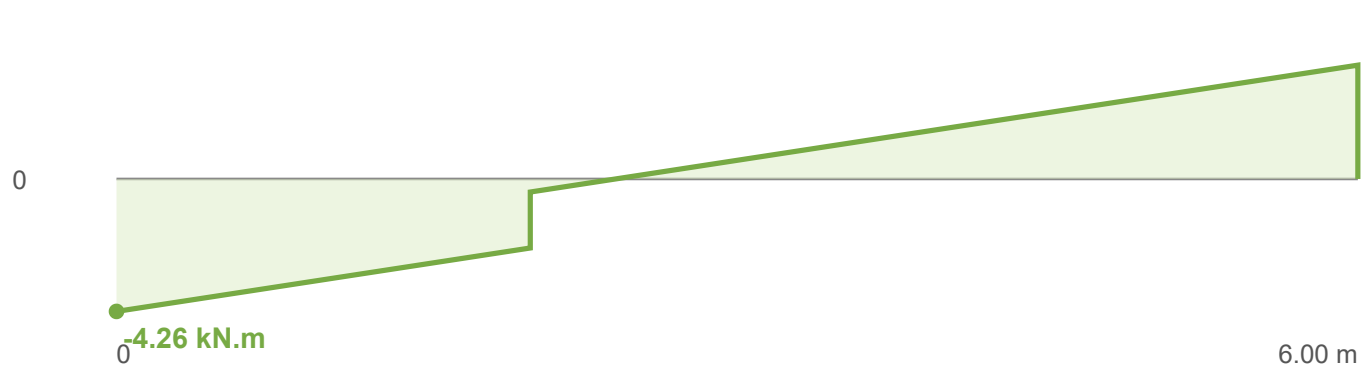
BENDING MOMENT M (kN.m)



DEFLECTION Δ (mm, SLS: 1.0Q)



TORSIONAL MOMENT T (kN.m, ULS: 1.35G + 1.5Q)



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases

1; SLS 1

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$109.758 / 756.260 =$	0.145	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$43.23 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd - Fully Restrained Beam

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	$0.006 / 756.260 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$168.868 / 503.250 =$	0.336	OK
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.241 m, P1 flange tip	@ x = 2.241 m (ULS: 1.35G + 1.5Q): $M_y = 168.362 \text{ kN.m}$, $V = 5.188 \text{ kN}$, $\phi = 0.05733 \text{ rad}$, $\phi' = 1.038e-5 \text{ rad/mm}$, $\phi'' = 1.724e-8 \text{ rad/mm}^2$, $\phi''' = 2.103e-12 \text{ rad/mm}^3$; $M_z = \phi M_y = 9.653 \text{ kN.m}$; P1 flange tip: $\sigma_x = N/A + M_y / W_{el,y} + M_z / W_{el,z} + E W_{n0} \phi'' = 0.00 + 104.57 + 49.25 + 76.75 = 230.57 \text{ N/mm}^2$; $\tau = G t_f \phi' = 13.45 \text{ N/mm}^2$; $(230.57/275)^2 + 3(13.45/275)^2 =$		
σ_x, τ , (6.1) at the 4 section points	P1 flange tip: $\sigma_x = 0.00 + 104.57 + 49.25 + 76.75 = 230.57$, $\tau = 13.45 \rightarrow 0.710$ (governs); P2 web-flange junction, flange: $\sigma_x = 0.00 + 100.74 + 2.56 = 103.30$, $\tau = 0.30 + 13.45 + 0.45 = 14.19 \rightarrow 0.149$; P3 web-flange junction, web: $\sigma_x = 0.00 + 97.11 + 2.56 = 99.67$, $\tau = 0.96 + 8.32 = 9.28 \rightarrow 0.135$; P4 web mid-depth: $\sigma_x = 0.00 + 2.56 = 2.56$, $\tau = 1.28 + 8.32 = 9.60 \rightarrow 0.004$ ($y_w = t_w/2 = 4.95 \text{ mm}$; $S_f/2 =$ half flange one side of the web)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30 \varepsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_{\bar{F}} > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$I_e = k_F E t_w^3 / (2 f_{yw} h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_{\bar{F}} > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_{\bar{F}} = \sqrt{(I_y t_w f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_{\bar{F}} \leq 1$; $L_{eff} = \chi_F I_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

M_{b,Rd} = M_{c,y,Rd}Fully Restrained503.250 kN.m

Torsion Design

J, H, a, Q_f, Q_w69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, —P385 App A

W_{n0}, S_{w1}212 cm², 1620 cm⁴P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)combination ULS: 1.35G + 1.5Q4.260 kN.m

Torsion analysisSCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)L/a = 3.23P385 App C

φ_{max} (ULS)L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)0.0605 rad = 3.47°

B_{Ed} = EI_wφ" (max)@ x = 2 m3.435 kN.m²

M_{w,Ed} = EI_wφ"/(h-t_f)max over span7.736 kN.m

M_{z,Ed} = φ.M_{y,Ed}max coincident10.129 kN.m

(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z}@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 40.255, M_{pl,z} = 83.6000.415OK P385 3.1.2

End torques T_tSt Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m1.902 / 1.724 kN.m

Torsion Shear Design @ 0 m

T_t at the V-T station@ x = 0 m (ULS: 1.35G + 1.5Q): T_t = 1.902 kN.m, V = 109.758 kN27.22 N/mm²P385

V_{pl,T.Rd} = √(1 - T_t/(1.25f_y/√3)).V_{pl,Rd}@ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260; S_{mod} = V_{pl,T.Rd}/V_{pl,Rd} = 0.929702.492 kN6.2.7(9)

V_{Ed}/V_{pl,T.Rd}109.758 / 702.4920.156OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/3603.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m3.22 mmOK

Torq in Case 1 @ 2.788 m: θ_{max} ≤ 2.00°0.0252 rad = 1.44° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)1.44°OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
0.336	0.336	0.193	0.145	0.251	0.257	0.415	0.710	0.156	0.415

PASS — governing Bending+torsion cross-section (P385 3.1.2) = 0.415

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R10 · UB 457 x 191 x 82 · M_{cr} method: refused · verdict **REFUSED**

Owner case: BOTTOM FLANGE RESTREAINED

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R10 "BOTTOM FLANGE RESTREAINED" = tension-flange restraint only: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Tension-flange (bottom flange of a sagging beam) restraint is not modelled: the tool's restraints (end flags $U_y / R_x / R_z /$ warping and the intermediate lateral restraints v, v', ϕ, ϕ') act at the shear centre and have no height in the cross-section, so a restraint that holds the tension flange laterally while the compression flange is free cannot be described; it needs a restraint-height term in the LTB eigen model (lateral constraint at $z = -h/2$, which couples v and ϕ) or the tension-flange-restraint method of BS 5950-1 Annex G / SCI P093. Treating it as unrestrained (R2) is the conservative bound; treating it as fully restrained (R9) is unsafe.

Owner case: BOTTOM FLANGE RESTREAINED

UB-R11 · UB 457 x 191 x 82 · M_{cr} method: refused · verdict **REFUSED**

Owner case: PARTIALLY RESTRAINED

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R11 "PARTIALLY RESTRAINED" = elastic (spring) restraint: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Elastic (partial) restraint is not modelled: every restraint in the tool is rigid or absent (each of v , v' , ϕ , ϕ' is held or free), so a lateral or torsional spring of finite stiffness (a purlin, a sheeting rail, a slab with a stiffness) cannot be entered; it needs lateral / rotational spring stiffness terms added to the LTB eigen element matrices (and to the warping-torsion FE) with a stiffness input per restraint and per end, plus the EN 1993-1-1 BB.2 minimum-stiffness test to decide whether a spring counts as a full restraint. The bounds are R2 (no restraint) and R3 (rigid restraints at the same positions).

Owner case: PARTIALLY RESTRAINED

Beam & Beam-Portion (Member) (FAIL)

Includes Design for Torsion with Span Warping, Ends Free to Warp

Member PFC-R1 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm (kN)

G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

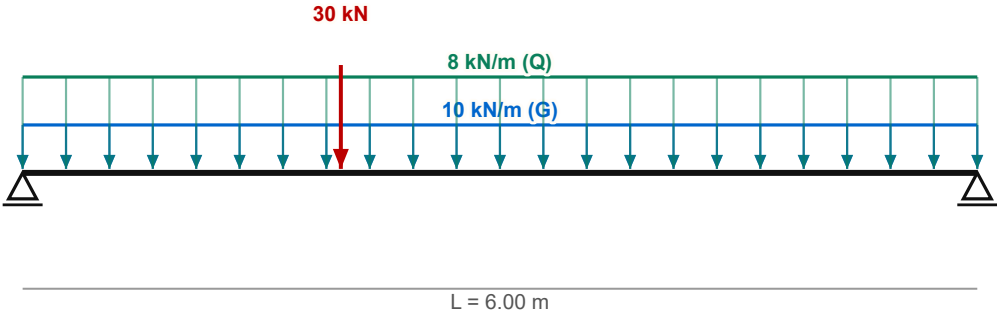
Uplift

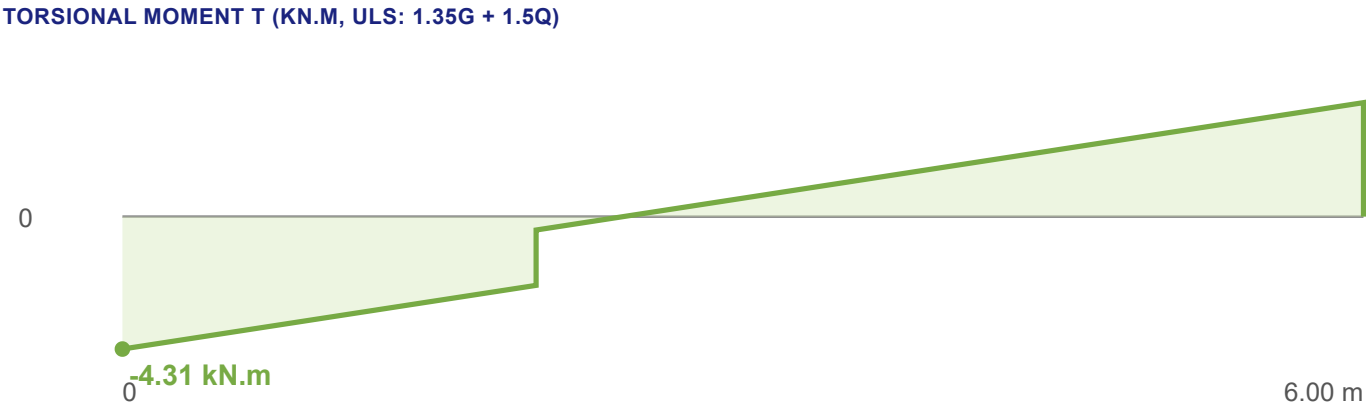
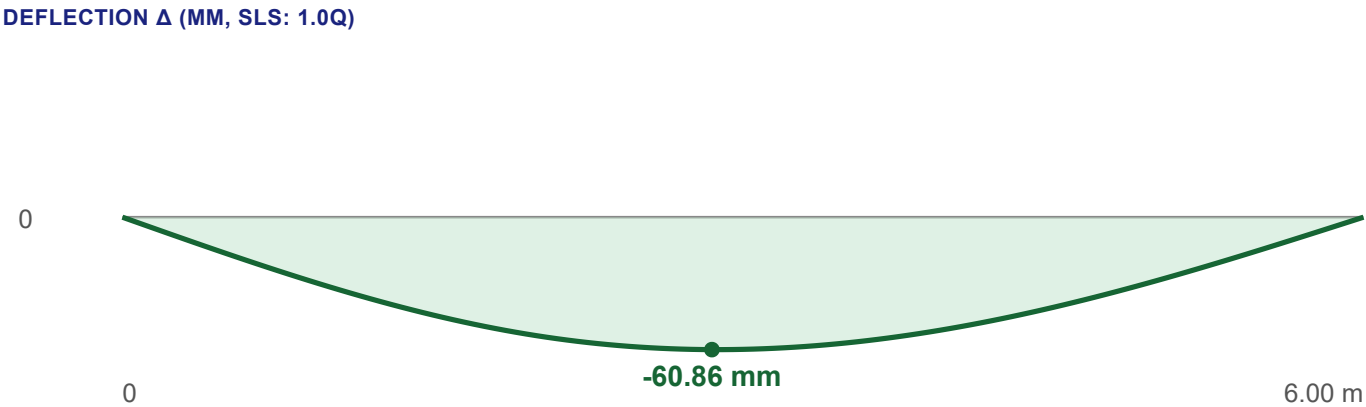
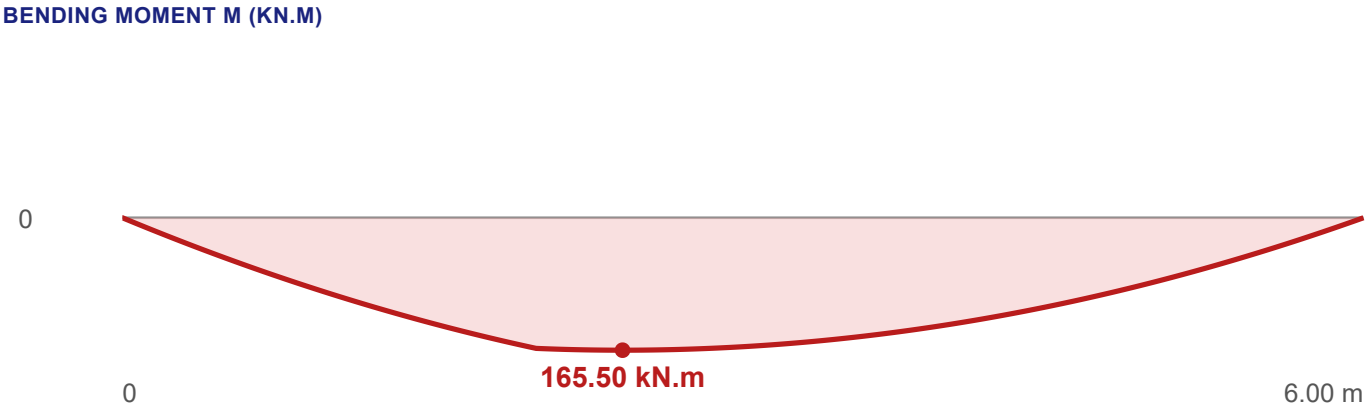
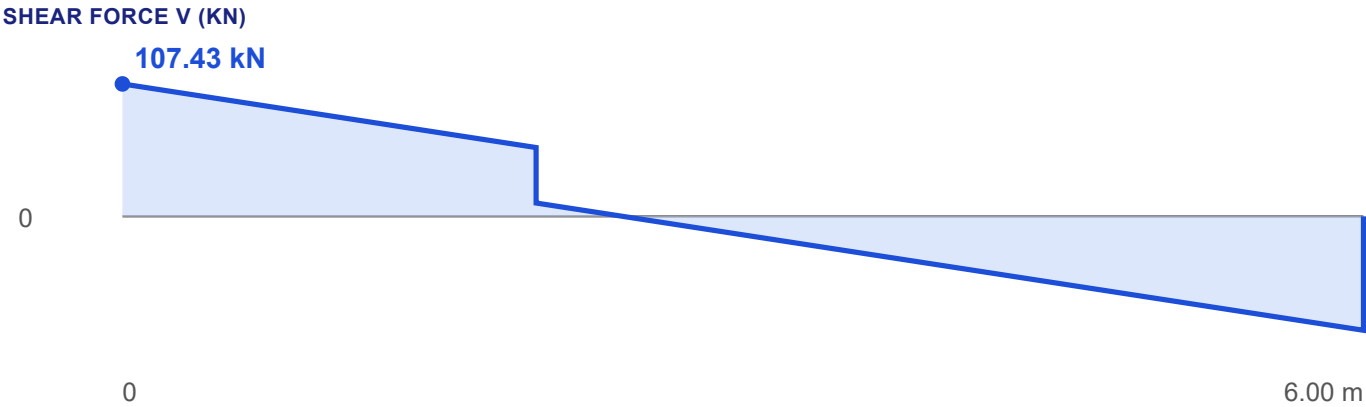
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd - Fully Restrained Beam

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$ Fully Restrained 62.425 kN.m

Torsion Design

J, H, a, Q_f, Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
W_{n0}, S_{w1} 40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 12.27 P385 App C
φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 0.683 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 3.645 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 109.717 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665 14.062 **Warning** P385 3.1.2
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.307 / 3.707 kN.m 3.771 / 3.200 kN.m

Torsion Shear Design @ 0 m

T_t, τ_w at the V-T station @ x = 0 m (ULS: 1.35G + 1.5Q): T_t = 3.771 kN.m, V = 195.04 / 5.22 N/mm² P385
= 107.430 kN
V_{pl,T,Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)] · V_{pl,Rd} @ x = 0 m: τ_t = 195.04 (τ_w = 5.22) N/mm²; V_{pl,Rd} = 20.965 kN 6.2.7(9)
= 212.754; S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099
V_{Ed}/V_{pl,T,Rd} 107.430 / 20.965 5.124 **Warning**

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 60.86 mm **Warning**
Torq in Case 1 @ 2.465 m: θ_{max} ≤ 2.00° 0.2787 rad = 15.97° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict) 15.97° **> 2° advisory**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.651	2.651	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	14.062

FAIL — governing Bending+torsion cross-section (P385 3.1.2) = 14.062

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385); check that the rotation is acceptable for whatever the member supports.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

PFC-R2 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: (reference) unrestrained between fork ends - the base of the restraint variants
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$ mm); R2 unrestrained between fork ends ($U_y + R_x$ held at both ends, warping free) - the reference for R3..R11

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R2 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

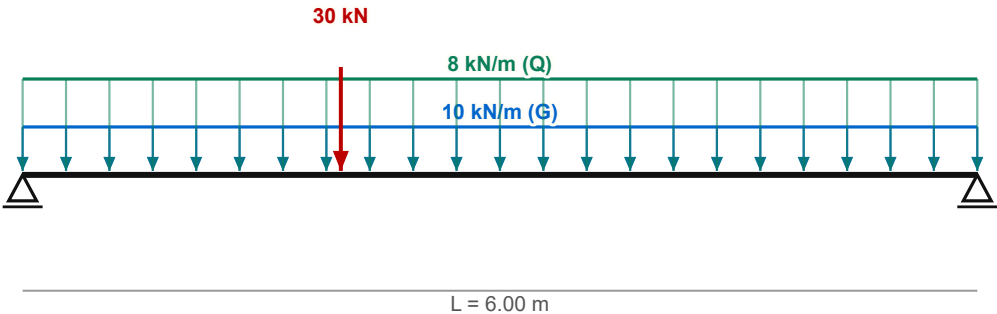
Member Loading and Member Forces

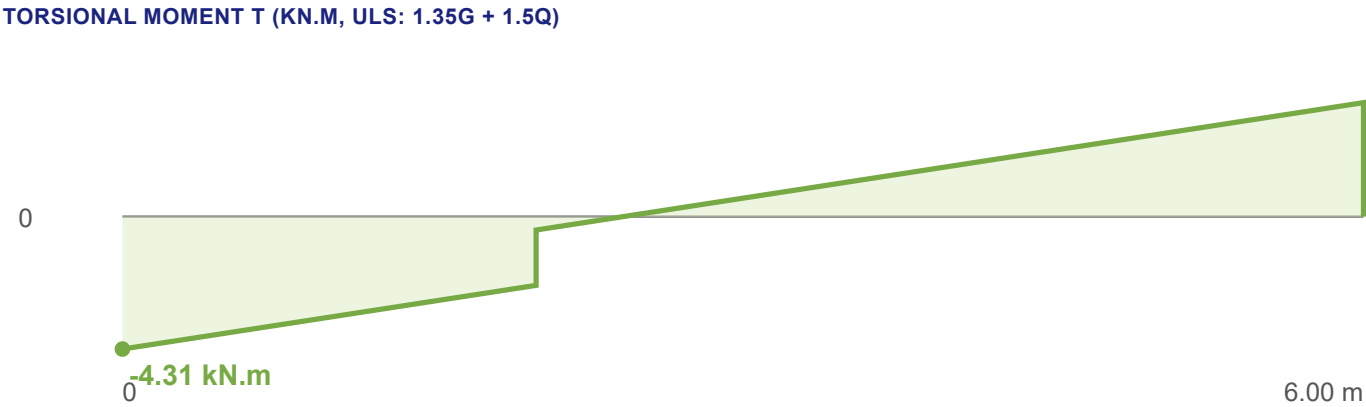
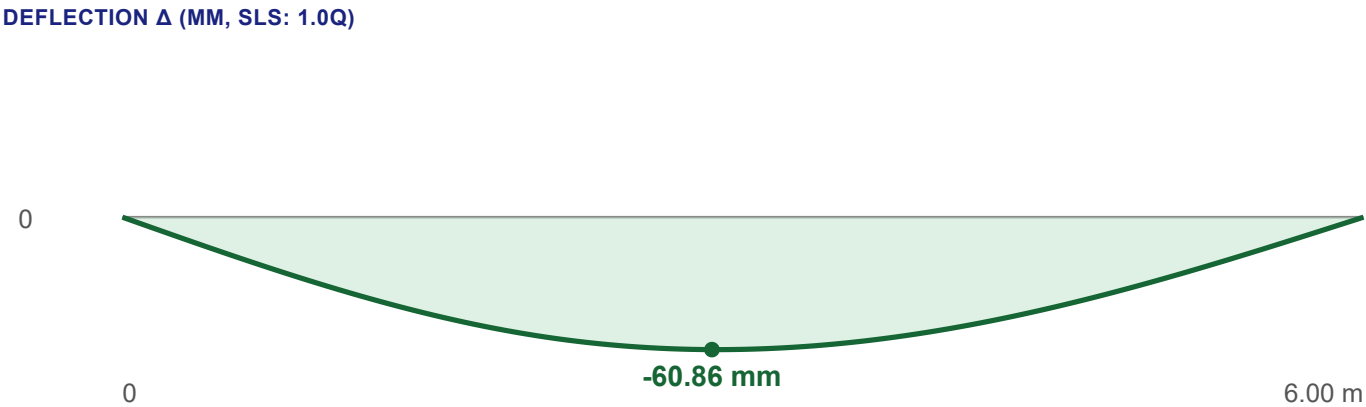
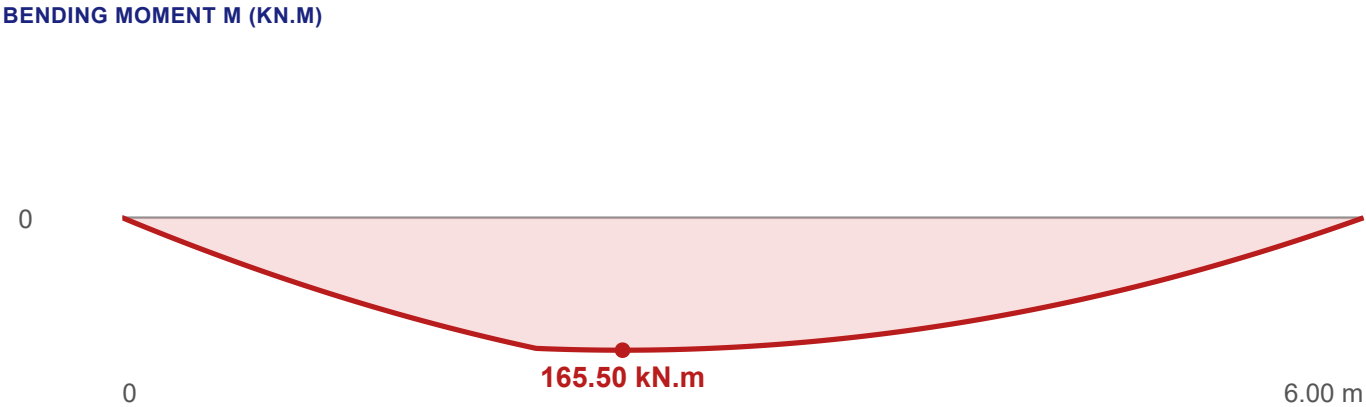
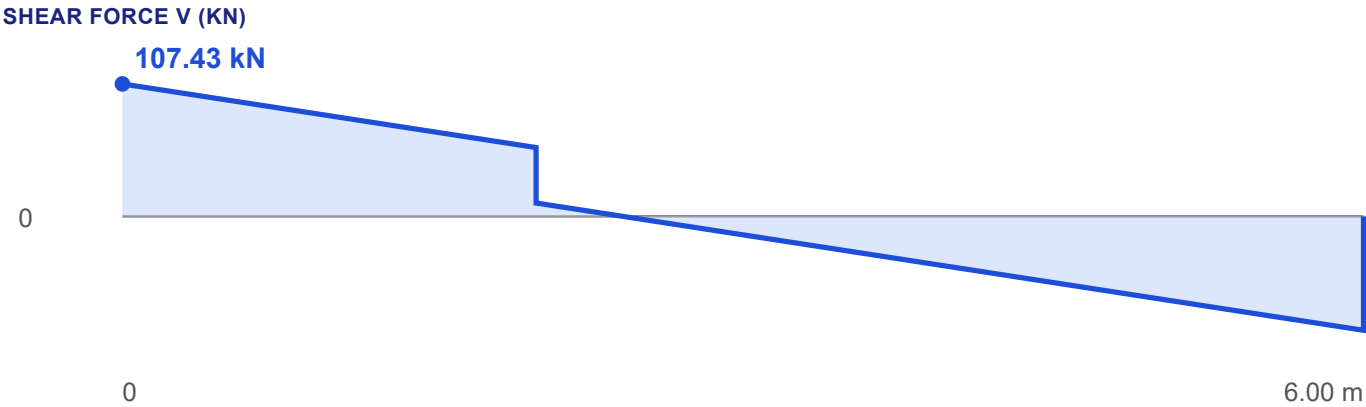
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]
 $V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

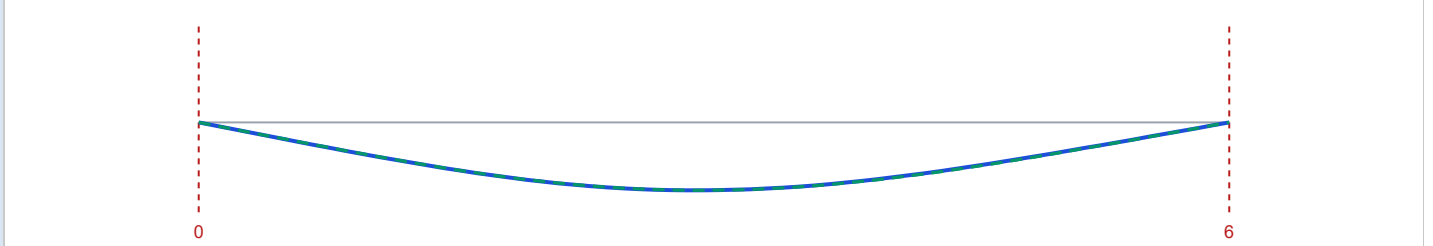
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 35.819 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_p b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	M _y (y.Ed)/M _b (b.Ed)	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

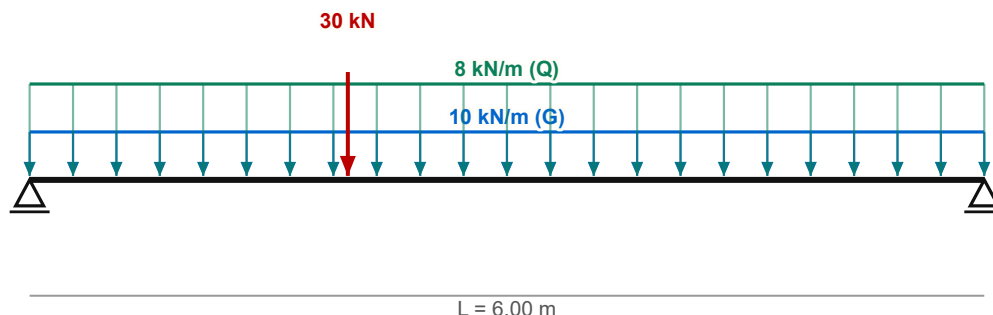
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

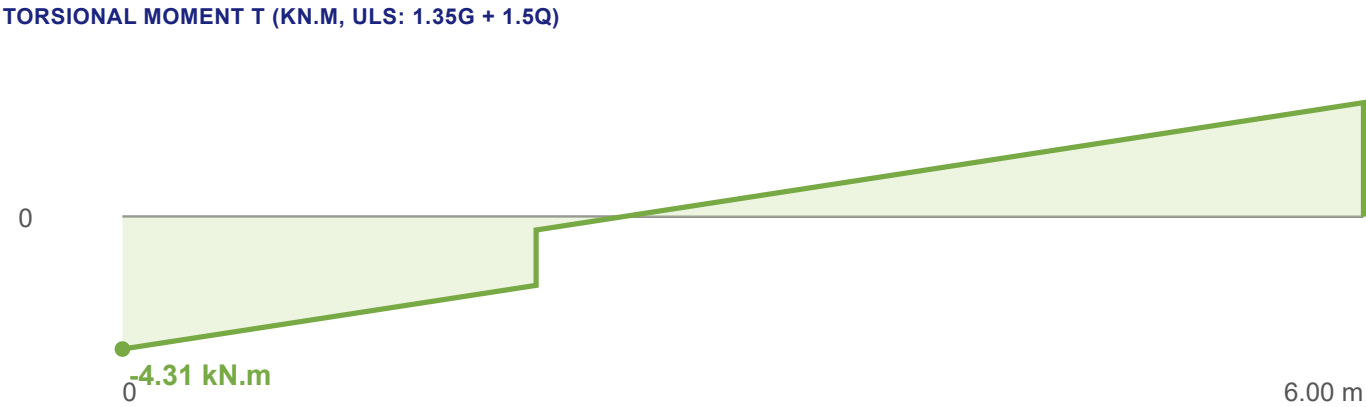
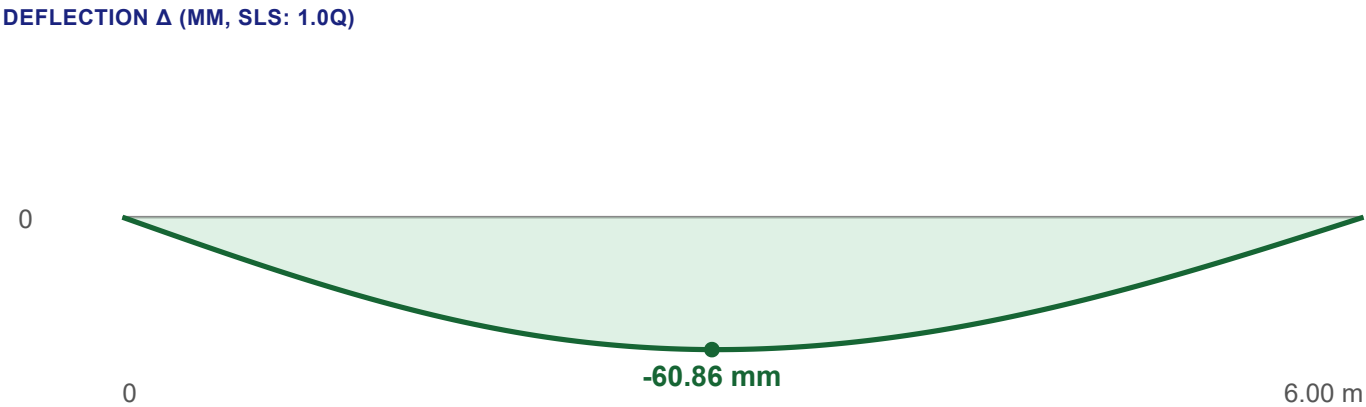
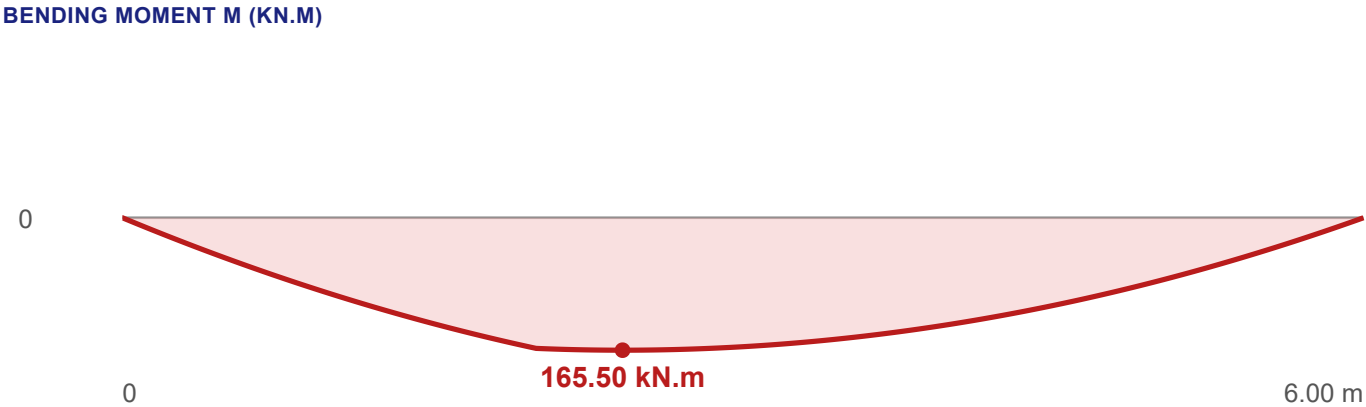
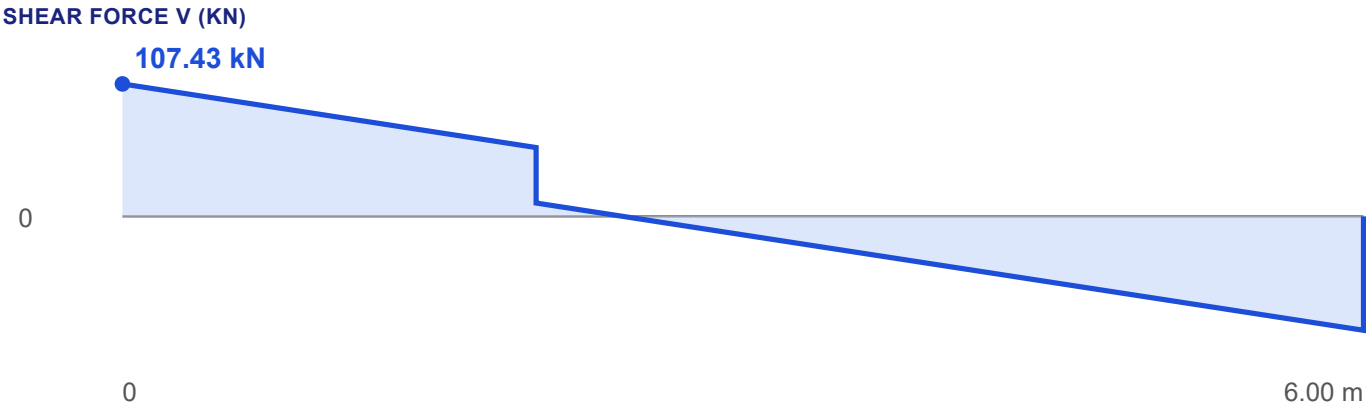
ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

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$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C_1	—	not used
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5 \text{ kN}\cdot\text{m}$		P362

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e/i_z)/\kappa$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = E I _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
k = k _w · k _{zw} · k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f ² · f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part G I _T φ' at x = 0 / x = L; total T = G I _T φ' - E I _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T _t , T _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.771 kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{1 - T_t/(1.25f_y/\sqrt{3})} - T_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: T _t = 195.04 (T _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.099	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M_(y,Ed)/M_(b,Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

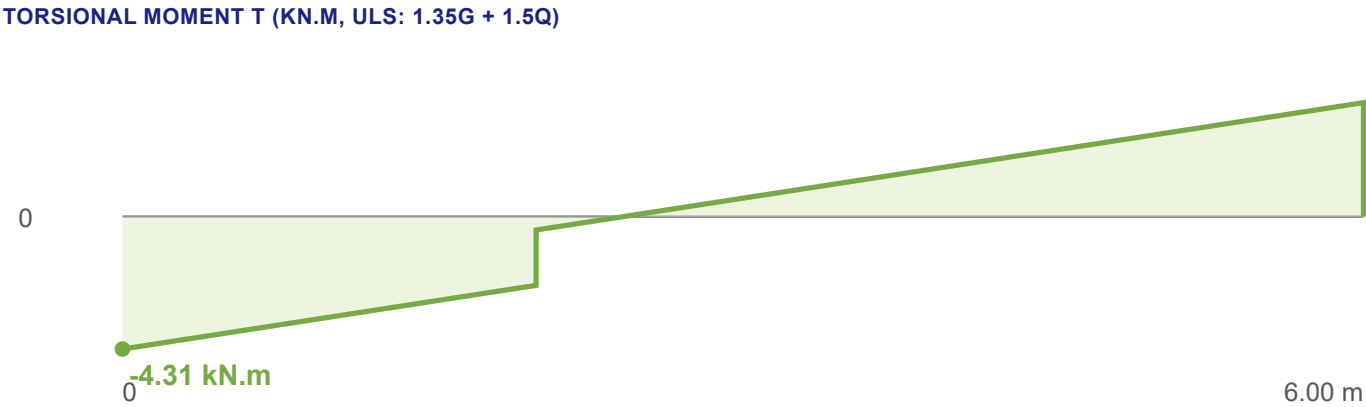
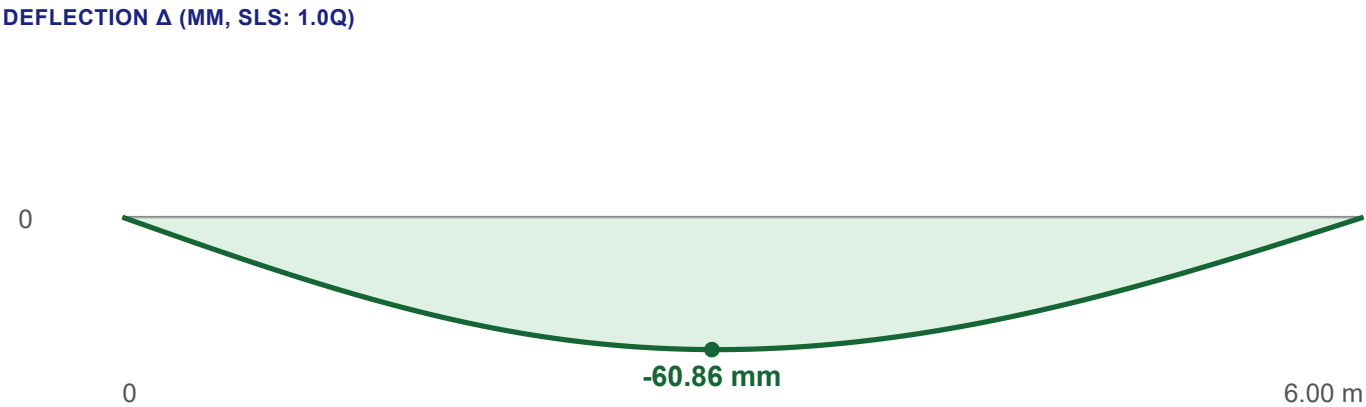
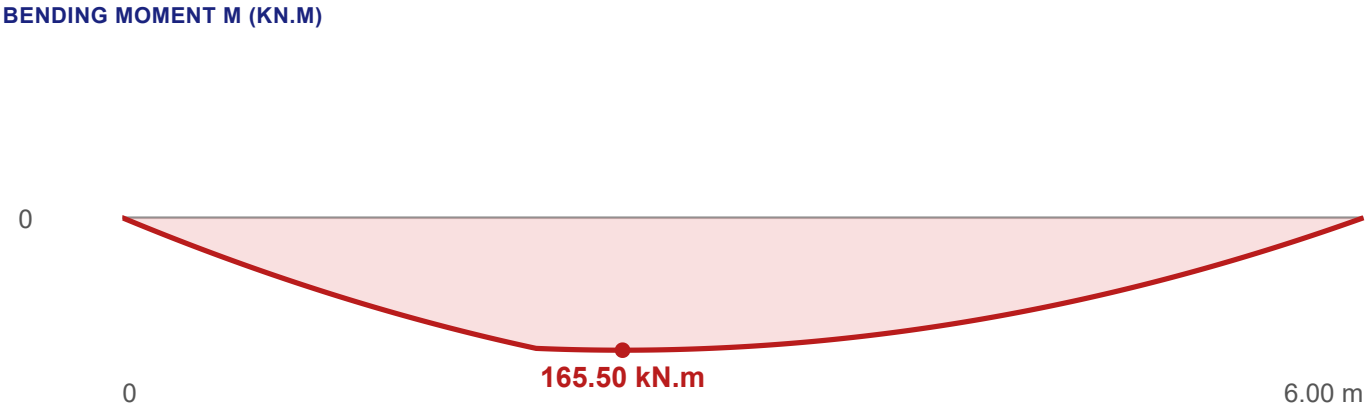
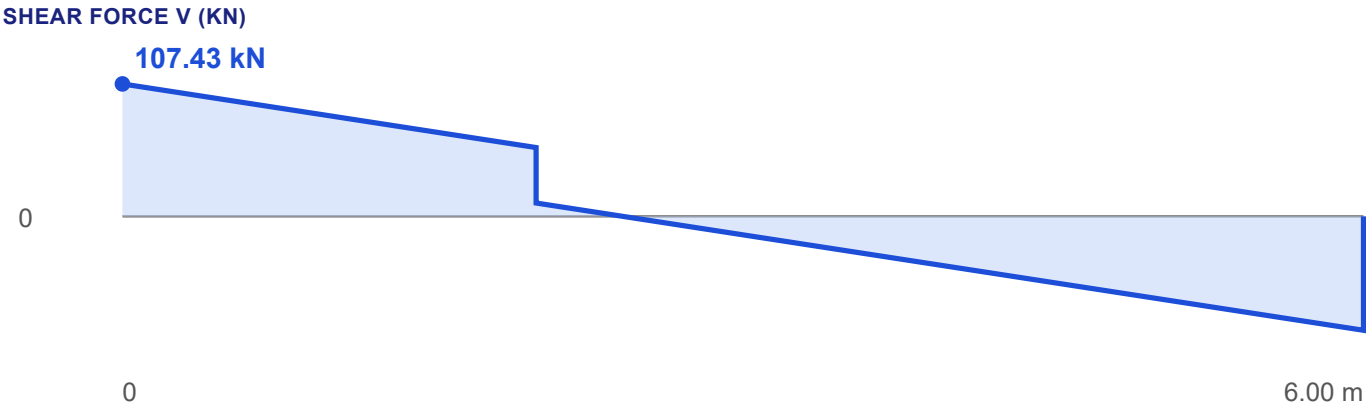
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\text{max}} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

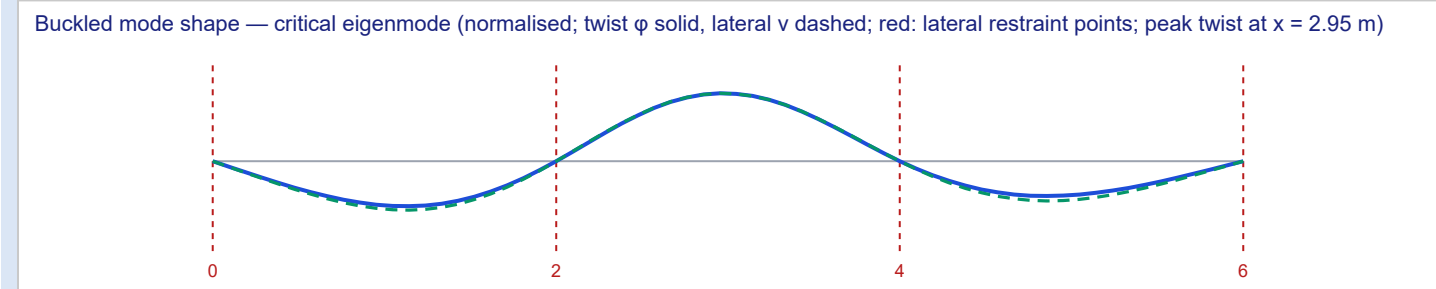
Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform} (critical bay)	bay between restraints containing the eigenmode peak (x = 2.952 m); whole-member ratio 1.345	1.037	NA 2.18
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	133.2, 163.2, 12.9, 0.816, 0.079 [segment 2–4 m] — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=165.4, M(L/2)=161.1, M(3L/4)=150.4, M _{max} =165.5 kN·m	1.036	Serna

Lateral Buckling Check M.b.Rd

Governing bay 2–4 m (isolated, fork ends)	M _{y,Ed} = 165.503, M _{cr} = 118.953, λ = 0.724, χ = 0.742	46.302 kN.m	bay by bay
L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0; lateral restraint points at x = 0, 2, 4, 6 m	2 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	63 elements, 0.001 %; 1 combination(s), 2 solve(s) + 1 cached	154.263 kN.m (whole member)	verged
λ _{LT} = √W _f y/M _{cr}	√ 227.0 x 275 / 118.953	0.724	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.724, 0.820, 0.76	0.742	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.742, 0.724, —, 1.000	0.742	isolated bay: f = 1
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.742 x 227.0 x 275 ≤ 62.425 =	46.302 kN.m	
M _{y,Ed} /M _{b,Rd}	165.503 / 46.302	3.574	Warning



Lateral Restraint Portions (bay by bay, fork ends)

Portion	From – To (m)	L _e (m)	M _{y,Ed} (kN.m)	M _{cr} (kN.m)	λ _{LT}	χ _{LT}	M _{b,Rd} (kN.m)	M _{y,Ed} /M _{b,Rd}	
1	0 – 2	2	163.240	188.358	0.576	0.855	53.375	3.058	
2	2 – 4	2	165.503	118.953	0.724	0.742	46.302	3.574	governs
3	4 – 6	2	133.240	183.987	0.582	0.850	53.040	2.512	

χ_{LT} per portion without the f-factor. Whole-member M_{cr} = 154.263 kN.m. intermediately restrained member: lateral-torsional buckling checked BAY BY BAY between the lateral restraints. Each bay is solved in isolation with fork ends, its own M_{cr} and its own peak moment; the worst bay governs. Governing bay 2–4 m: M_{Ed} = 165.5, M_{cr} = 119.0, M_{b,Rd} = 46.3 kN·m, utilisation 3.57. (Whole-member eigen M_{cr} = 154.3 kN·m retained for reference.)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	163.240	ULS: 1.35G + 1.5Q	816.199	20.405	restraint design force, advisory
4	lateral restraint 2	133.240	ULS: 1.35G + 1.5Q	666.199	16.655	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
k = k _w · k _{zw} · k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 154.263 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 154.263 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr} EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f b ² f _y /4) [verify]		
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T _t , τ _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.771 kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
V _{pl,T,Rd} = [√(1 - τ _f /(1.25f _y /√3)) - τ _w /(f _y /√3)] · V _{pl,Rd}	@ x = 0 m: T _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.099	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: θ _{max} ≤ 2.00°	0.2787 rad = 15.97° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	3.574	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

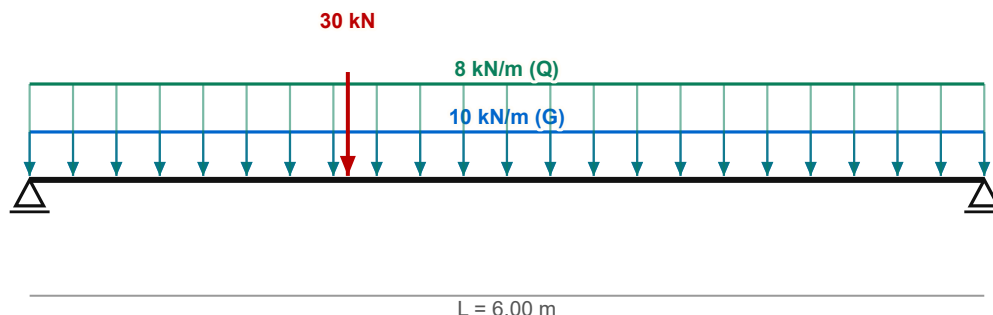
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

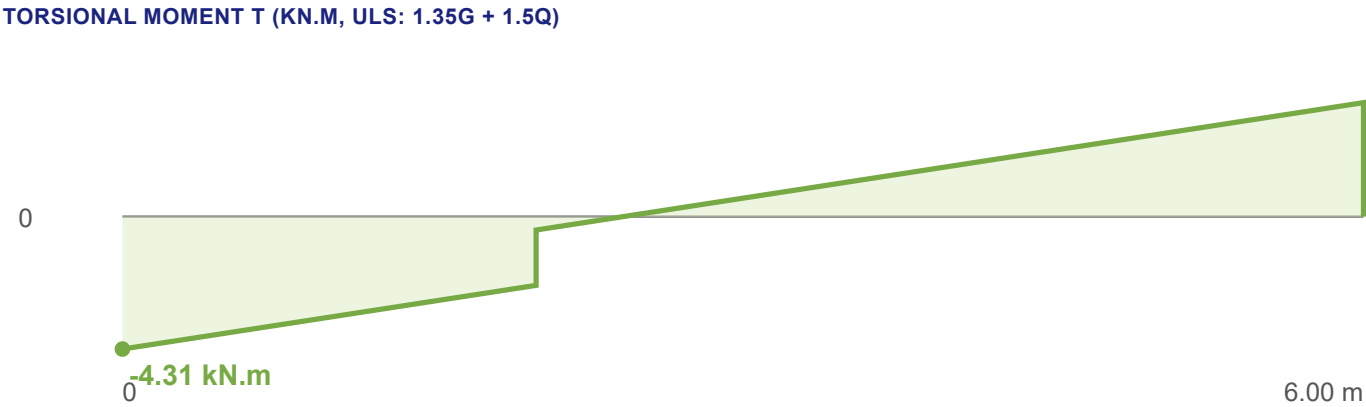
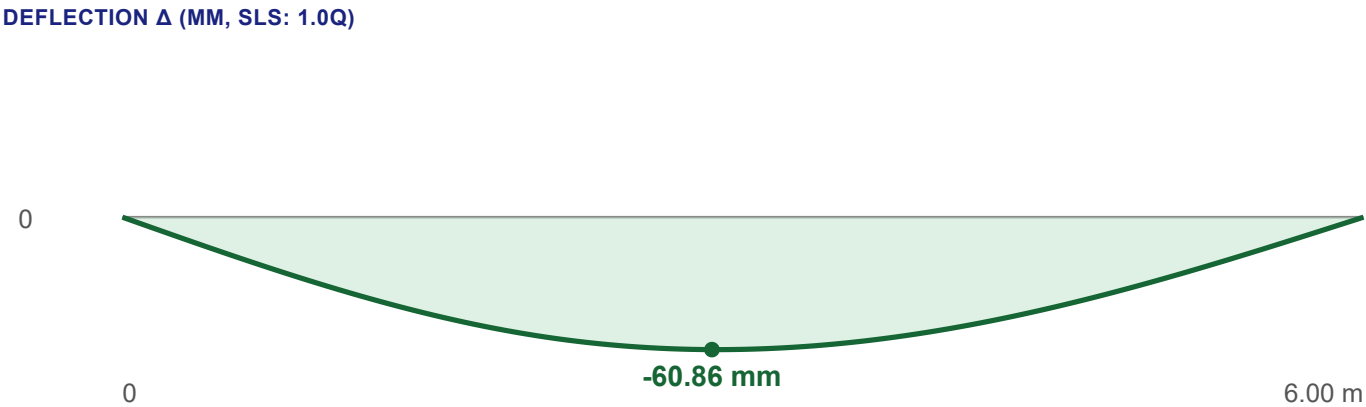
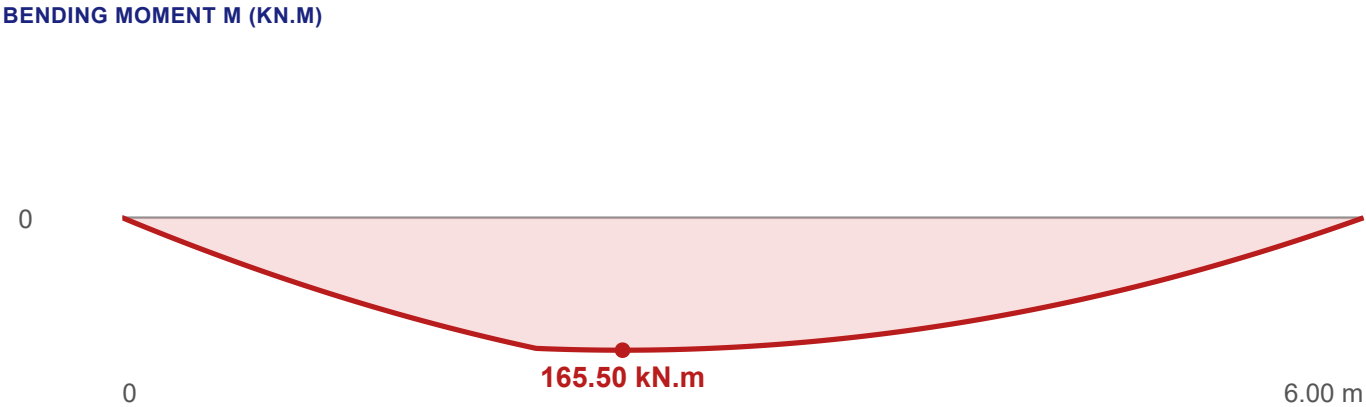
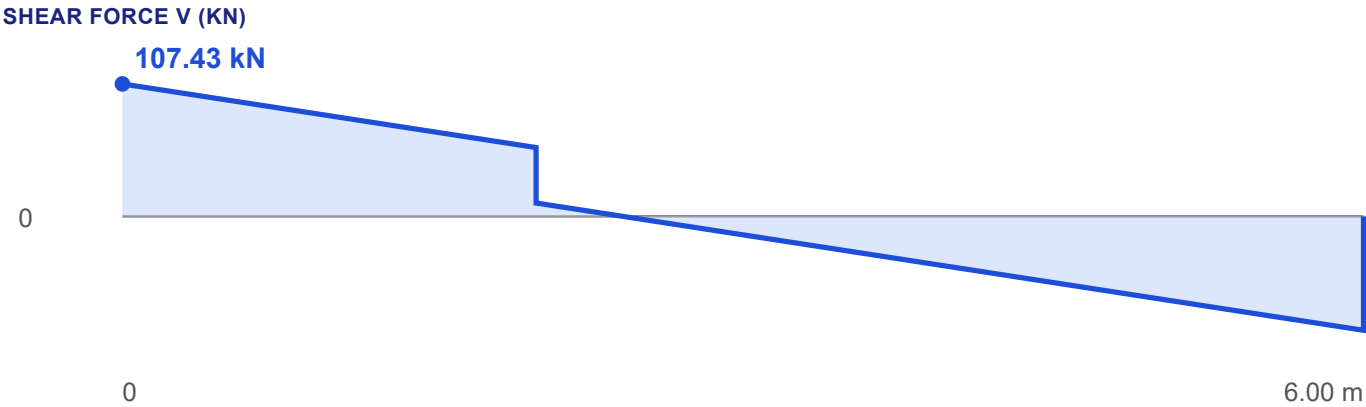
ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_{\alpha} = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 0.00, 165.50, 0.00	
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _y .f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	163.240	ULS: 1.35G + 1.5Q	816.199	20.405	restraint design force, advisory
4	lateral restraint 2	133.240	ULS: 1.35G + 1.5Q	666.199	16.655	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ'''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	109.717 kN.m	

$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y}$ = 62.425, $M_{pl,f}$ = 4.834, $M_{pl,z}$ = 16.665	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty (M_{y,Ed} \geq M_{cr} = 9.128 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y / 4$) [verify]		
End torques T_t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total T = $GI_T \phi'$ — $El_w \phi''' = 4.307 / 3.707 \text{ kN.m}$	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T_t, τ_w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.771 \text{ kN.m}$, V = 195.04 / 5.22 N/mm ² = 107.430 kN	P385
$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 20.965	5.124 Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm}$ @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

ADVISORY: Standard (closed-form) M_{cr} : the member is treated as ONE segment of length $L_E = 1.00 \times L$ with C_1 from the whole-member moment diagram; intermediate lateral restraints are not applied on this route (conservative on L_E). Use the FE eigenvalue method for the bay-by-bay M_{cr} , or verify each bay separately.

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R4 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: WARPING RESTRAINED
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R4 "WARPING RESTRAINED": warping (phi' = 0) held at both ends (ends warp flags; LTB eigen k_w = 0.5 type and warping-torsion FE)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, End Warping Fixed
Member PFC-R4 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x W restrained (pinned in plane; LTB fork, warping fixed); End 2: U_y U_z R_x W restrained (pinned in plane; LTB fork, warping fixed) — custom end conditions

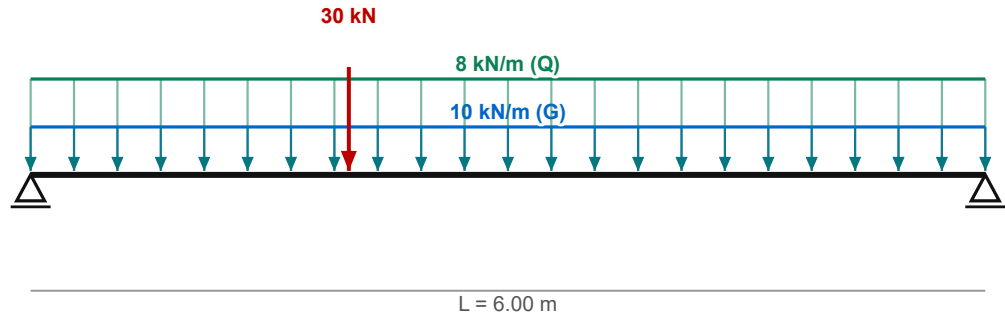
Member Loading and Member Forces

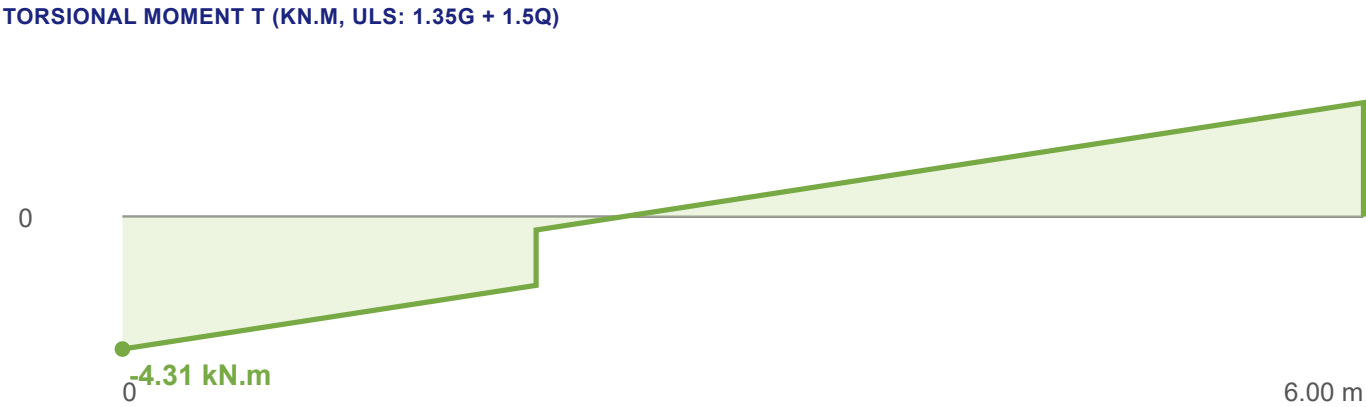
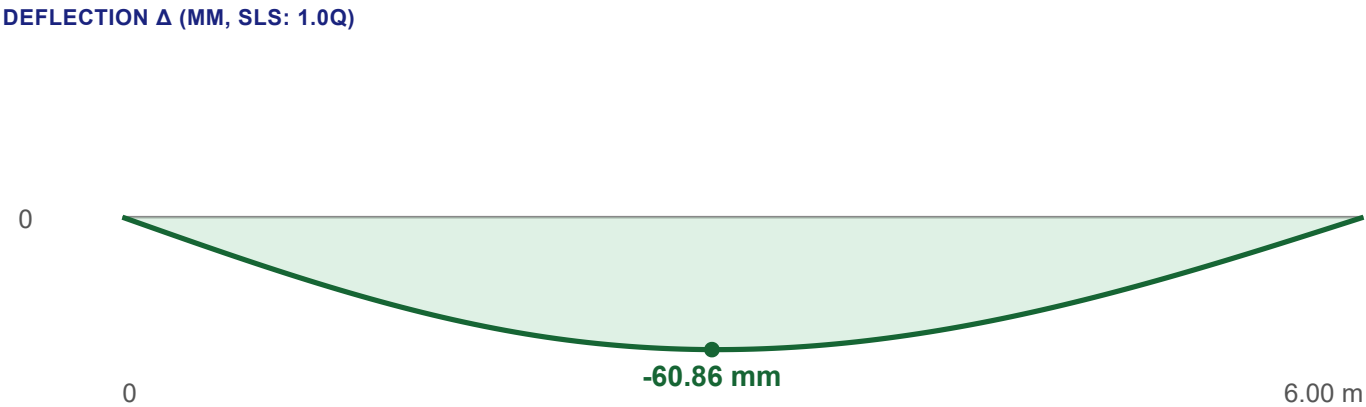
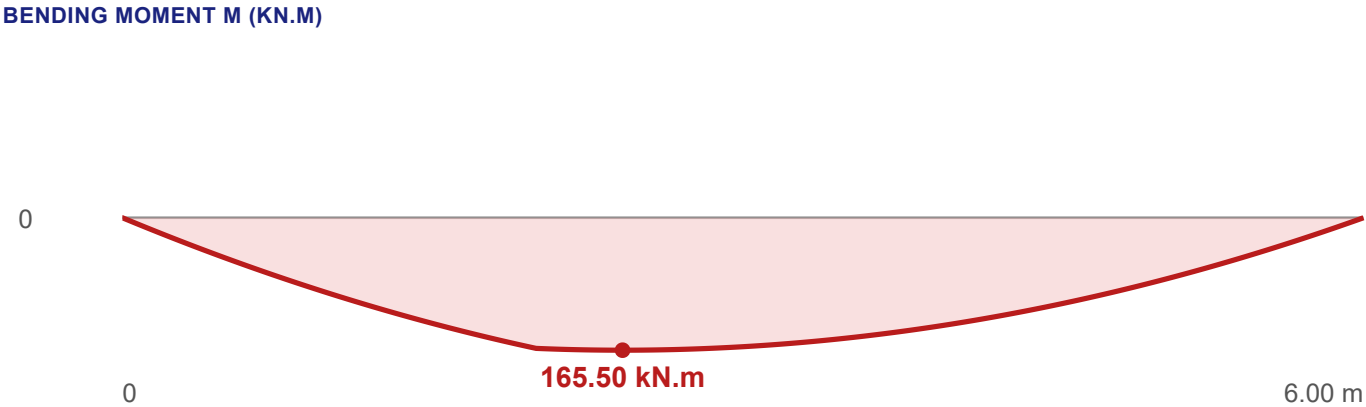
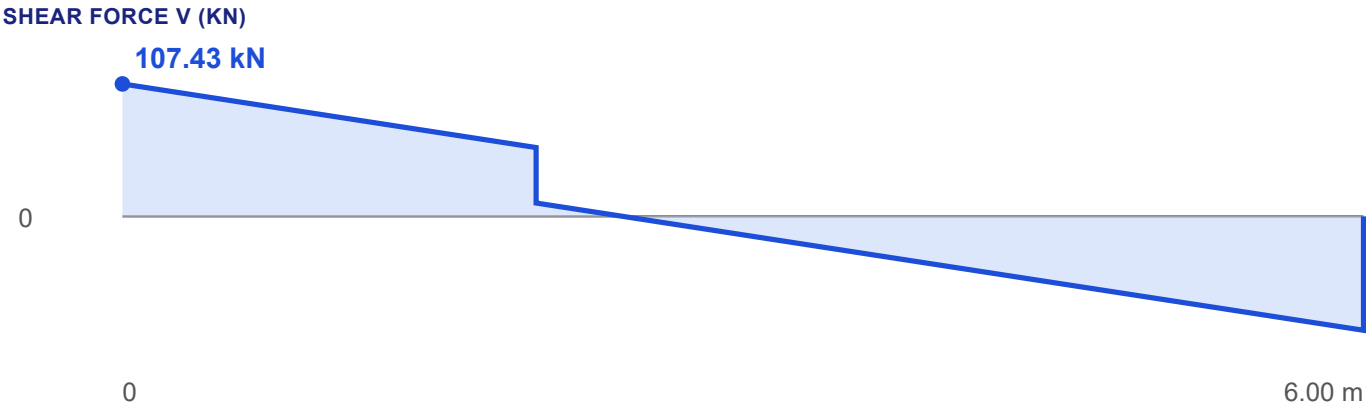
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.36	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.65	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
		(Axial: Non-Slender)
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.475 m, P1 flange tip	@ x = 2.475 m (ULS: 1.35G + 1.5Q): $M_y = 165.463$ kN.m, $V = 1.450$ kN, $\phi = 0.47893$ rad, $\phi' = 3.355e-5$ rad/mm, $\phi'' = 1.778e-7$ rad/mm ² , $\phi''' = 1.406e-10$ rad/mm ³ ; $M_z = \phi M_y = 79.246$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.20 + 2344.55 + 150.47 = 3339.22$ N/mm ² ; $\tau = G_t \phi' = 33.96$ N/mm ² ; $(3339.22/275)^2 + 3(33.96/275)^2 =$		
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.20 + 2344.55 + 150.47 = 3339.22$, $\tau = 33.96 \rightarrow 147.489$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.44 + 340.72 = 1132.15$, $\tau = 0.30 + 33.96 + 2.72 = 36.98 \rightarrow 17.003$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.44 + 1151.59 + 101.56 = 2044.58$, $\tau = 0.52 + 33.96 + 1.48 = 35.96 \rightarrow 55.328$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.67 + 1151.59 + 101.56 = 1991.82$, $\tau = 1.08 + 16.30 + 3.08 = 20.46 \rightarrow 52.477$; P4 web mid-depth: $\sigma_x = 0.00 + 1151.59 = 1151.59$, $\tau = 1.37 + 16.30 + 2.28 = 19.95 \rightarrow 17.552$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		147.489	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$
			6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$175.00, 6.00, 12.50, 75.00$ mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

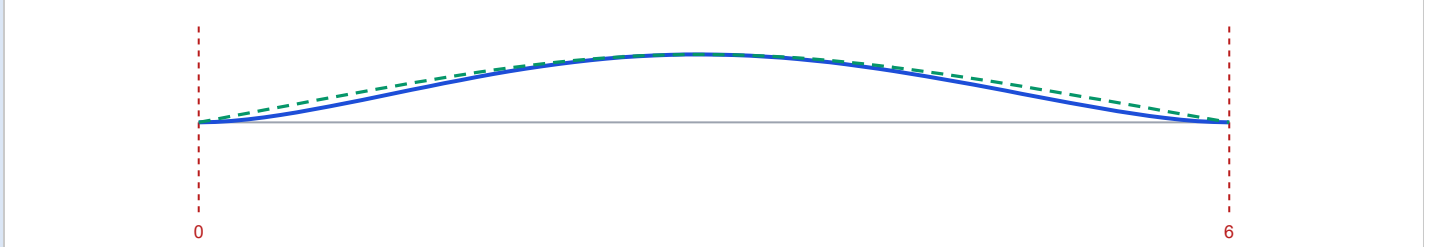
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	41.619 / 37.351 kN.m (shape only: $z_g = z_j = 0$)	1.114	k _c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m [$k = k_w = 1$: clamped / warping-fixed end(s) taken as forks]	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi, \phi' = 0$; End 2 v, $\phi, \phi' = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	41.619 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 41.619}$	1.225	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.225, 1.376, 0.76	0.444	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.444, 1.225, 0.947, 0.983	0.452	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.452 \times 227.0 \times 275 \leq 62.425 =$	28.191 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 28.191	5.871	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.425 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	El _w φ'''' - Gl _T φ'' = m _t (x): warping-torsion FE (240 elements); twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed); closed forms not applicable: warping-fixed end; 2 FE solve(s)	mesh error 0.008 %	FE (≤ 0.5 %)
φ _{max} (ULS)	warping-torsion FE, twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed)	0.4823 rad = 27.63°	
B _{Ed} = El _w φ'' (max)	@ x = 0 m	1.871 kN.m ²	
M _{w,Ed} = El _w φ''/(h-t _f)	max over span	9.979 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	79.532 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.425 m: 165.503, 2.219, 78.950 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	12.226	Warning P385 3.1.2

$k = k_w \cdot k_{zw} \cdot k_\alpha$ $M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} +$ $k_w k_{zw} k_\alpha M_w/M_{f,Rk}$ (A.1) scope	$x \times \infty (M_{y,Ed} \geq M_{cr} = 41.619 \text{ kN.m})$ k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 41.619 \text{ kN.m}$	— ≥ 99	EN 1993-6 A Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
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EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y / 4$)
[verify]
St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi'$
— $El_w \phi''' = 4.363 / 3.652 \text{ kN.m}$

End torques T_t

0.000 / 0.000 kN.m

Torsion Shear Design @ 0.625 m

τ_t, τ_w at the V-T station	@ $x = 0.625 \text{ m}$ (ULS: 1.35G + 1.5Q): $T_t = 2.450$ kN.m, $V = 91.299 \text{ kN}$	126.74 / 12.32 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0.625 \text{ m}$: $\tau_t = 126.74$ ($\tau_w = 12.32$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.524$	111.396 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	91.299 / 111.396	0.820	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.5 m: $\theta_{max} \leq 2.00^\circ$	$0.2047 \text{ rad} = 11.73^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	11.73°	> 2° advisory

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	5.871	3.652	0.505	0.452	1.666	12.226	147.489	0.820	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

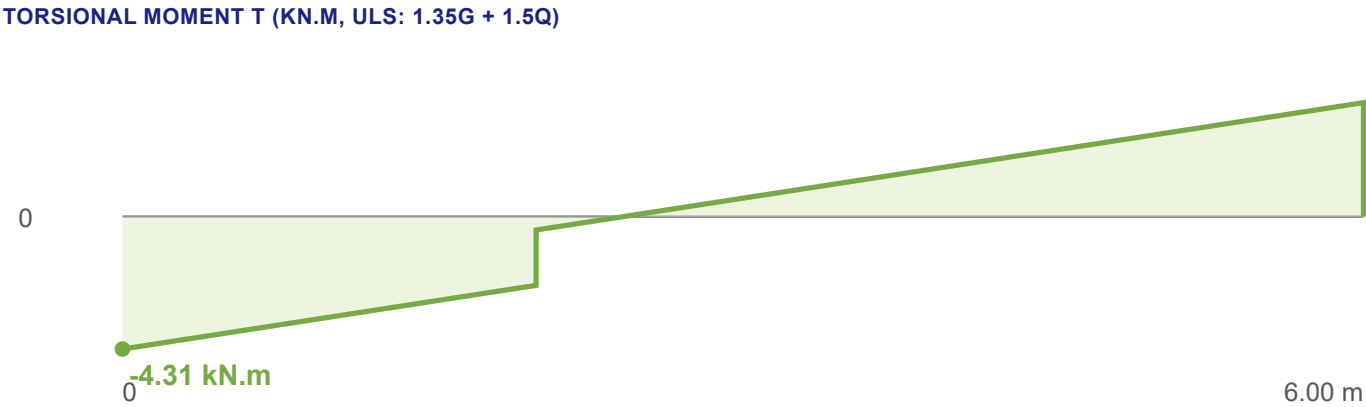
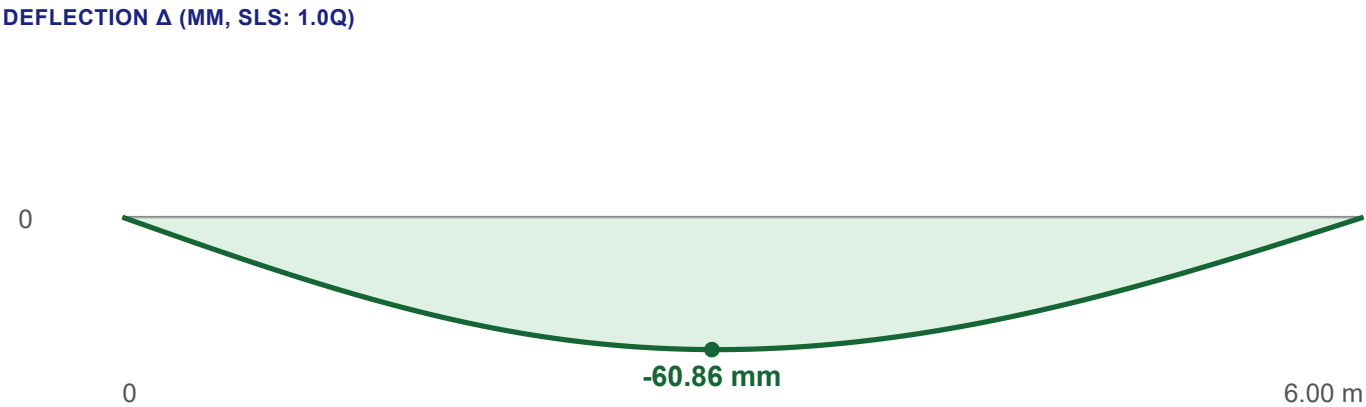
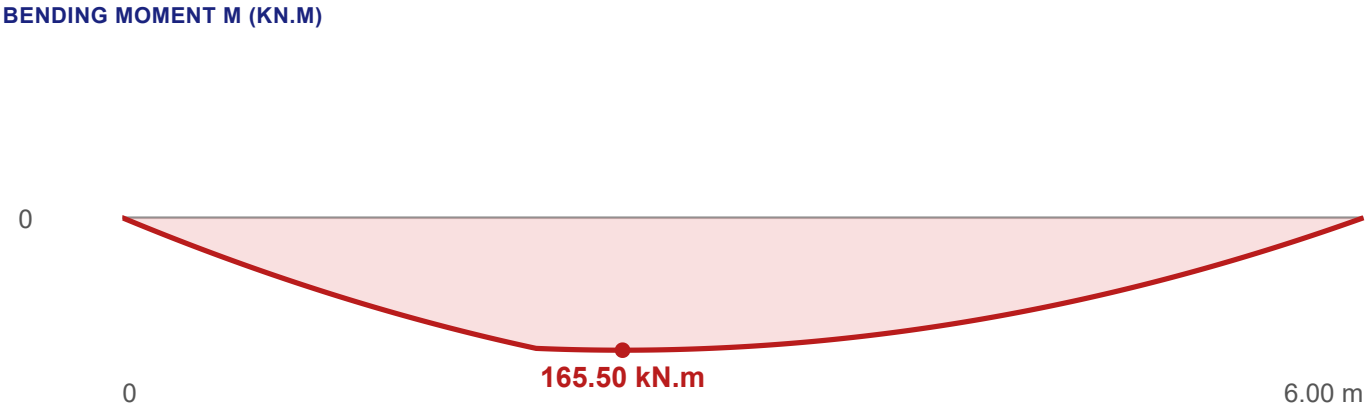
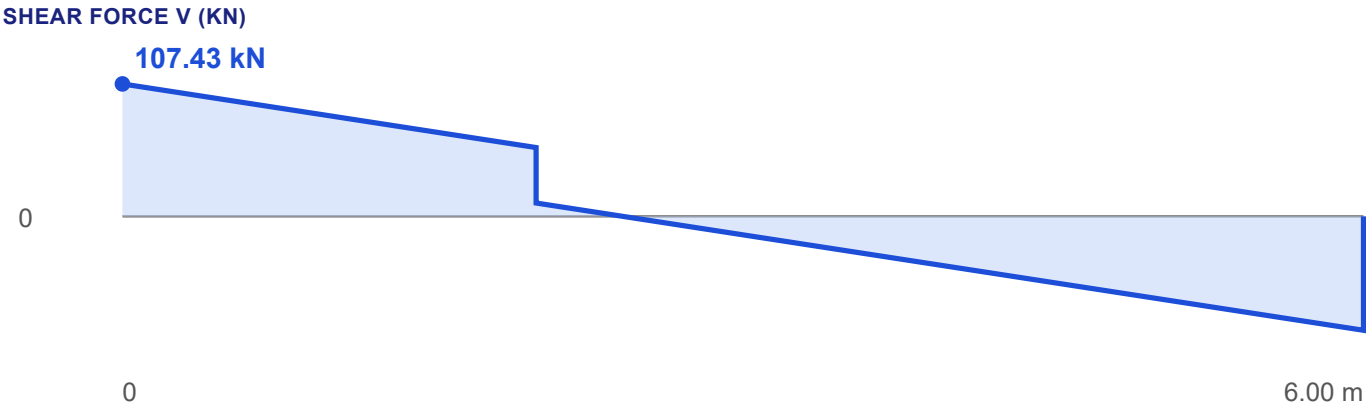
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 11.7^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385); check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 147.489 at $x = 2.48 \text{ m}$ (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).



Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$1340 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	$107.430 / 212.754 =$	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed}/V_{pl,y,Rd}$ (at max M)	$0.021 / 212.754 =$	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 227.0 / 1$	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.504 / 62.425 =$	2.651	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.475 m, P1 flange tip	@ x = 2.475 m (ULS: 1.35G + 1.5Q): $M_y = 165.463$ kN.m, $V = 1.450$ kN, $\phi = 0.47893$ rad, $\phi' = 3.355e-5$ rad/mm, $\phi'' = 1.778e-7$ rad/mm ² , $\phi''' = 1.406e-10$ rad/mm ³ ; $M_z = \phi M_y = 79.246$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.20 + 2344.55 + 150.47 = 3339.22$ N/mm ² ; $\tau = G_t \phi' = 33.96$ N/mm ² ; $(3339.22/275)^2 + 3(33.96/275)^2 =$		
σ_x , τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.20 + 2344.55 + 150.47 = 3339.22$, $\tau = 33.96 \rightarrow 147.489$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.44 + 340.72 = 1132.15$, $\tau = 0.30 + 33.96 + 2.72 = 36.98 \rightarrow 17.003$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.44 + 1151.59 + 101.56 = 2044.58$, $\tau = 0.52 + 33.96 + 1.48 = 35.96 \rightarrow 55.328$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.67 + 1151.59 + 101.56 = 1991.82$, $\tau = 1.08 + 16.30 + 3.08 = 20.46 \rightarrow 52.477$; P4 web mid-depth: $\sigma_x = 0.00 + 1151.59 = 1151.59$, $\tau = 1.37 + 16.30 + 2.28 = 19.95 \rightarrow 17.552$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; S_{w1} , S_{w2} , $S_{w3} =$ P385 Table A.2 points 1, 2, 3)		
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		
		147.489	> 1 information 6.2.7(5)
			$f_y/\gamma_{M0} = 275$

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	$175.00, 6.00, 12.50, 75.00 \text{ mm}$ ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00 / 12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00 \text{ mm}$ (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C_1	—	not used
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		P362

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e/i_z)/\kappa$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.425 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	$E I_w \phi''' - G I_T \phi'' = m_t(x)$: warping-torsion FE (240 elements); twist restrained at $x = 0$ m ($\phi = 0$, $\phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0$, $\phi' = 0$: warping fixed); closed forms not applicable: warping-fixed end; 0 FE solve(s) + 2 cached	mesh error 0.008 %	FE (≤ 0.5 %)
ϕ_{max} (ULS)	warping-torsion FE, twist restrained at $x = 0$ m ($\phi = 0$, $\phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0$, $\phi' = 0$: warping fixed)	0.4823 rad = 27.63°	
$B_{Ed} = E I_w \phi''$ (max)	@ $x = 0$ m	1.871 kN.m ²	
$M_{w,Ed} = E I_w \phi'' / (h - t_f)$	max over span	9.979 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	79.532 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.425$ m: 165.503, 2.219, 78.950 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	12.226	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 9.128 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]		EN 1993-6 A
End torques T _t	St Venant part $G I_T \phi'$ at $x = 0 / x = L$; total $T = G I_T \phi' - E I_w \phi''' = 4.363 / 3.652$ kN.m	0.000 / 0.000 kN.m	

Torsion Shear Design @ 0.625 m

T _t , T _w at the V-T station	@ $x = 0.625$ m (ULS: 1.35G + 1.5Q): T _t = 2.450 kN.m, V = 91.299 kN	126.74 / 12.32 N/mm ²	P385
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$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl.Rd}$	@ x = 0.625 m: $\tau_t = 126.74$ ($\tau_w = 12.32$) N/mm ² ;	111.396 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	$V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.524$ 91.299 / 111.396	0.820	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.5 m: $\theta_{max} \leq 2.00^\circ$	$0.2047 \text{ rad} = 11.73^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	11.73°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	12.226	147.489	0.820	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 11.7^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 147.489 at x = 2.48 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

ADVISORY: Standard (closed-form) M_{cr} : warping-fixed ends (End 1 and End 2) taken as fork ends, $k = k_w = 1$ (conservative: the SN003a $k = 0.5$ / $k_w = 0.5$ rows are not applied; the FE eigen route uses $v' = 0$ / $\phi' = 0$ and gives the higher M_{cr}).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R5 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: WARPING FREE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R5 "WARPING FREE": warping free at both ends (the default fork end; identical input to R2, kept under the owner's heading)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R5 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

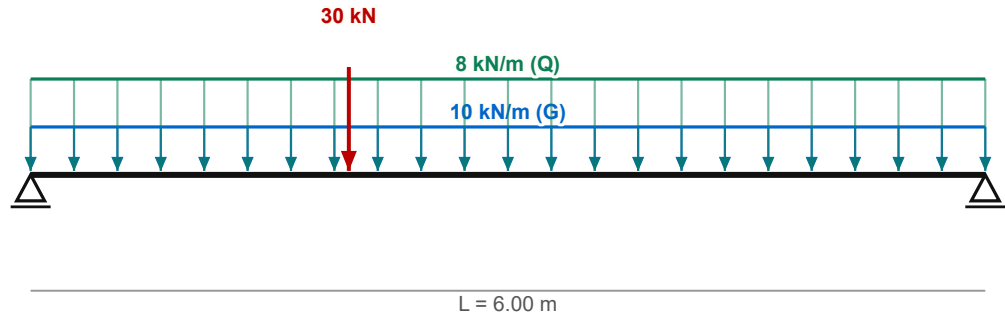
Member Loading and Member Forces

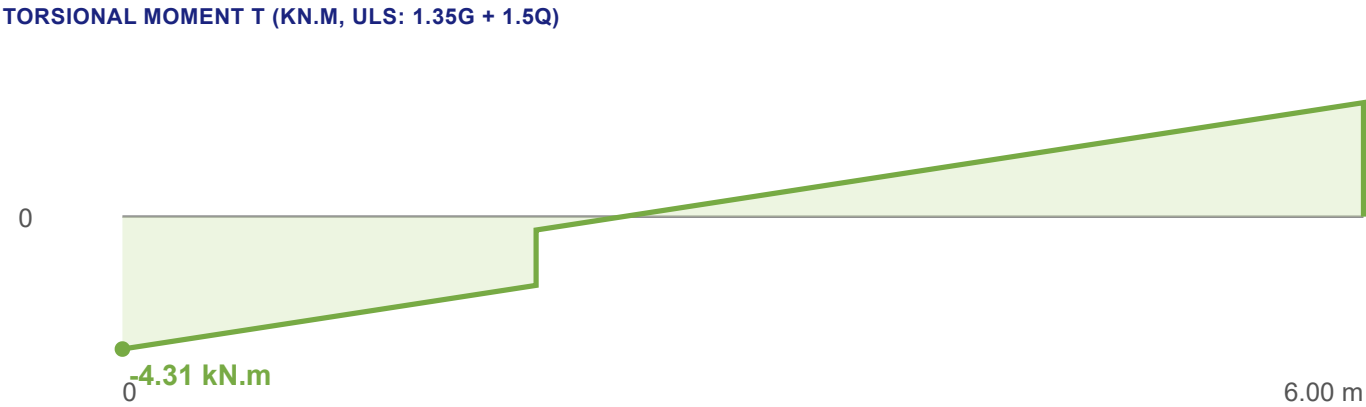
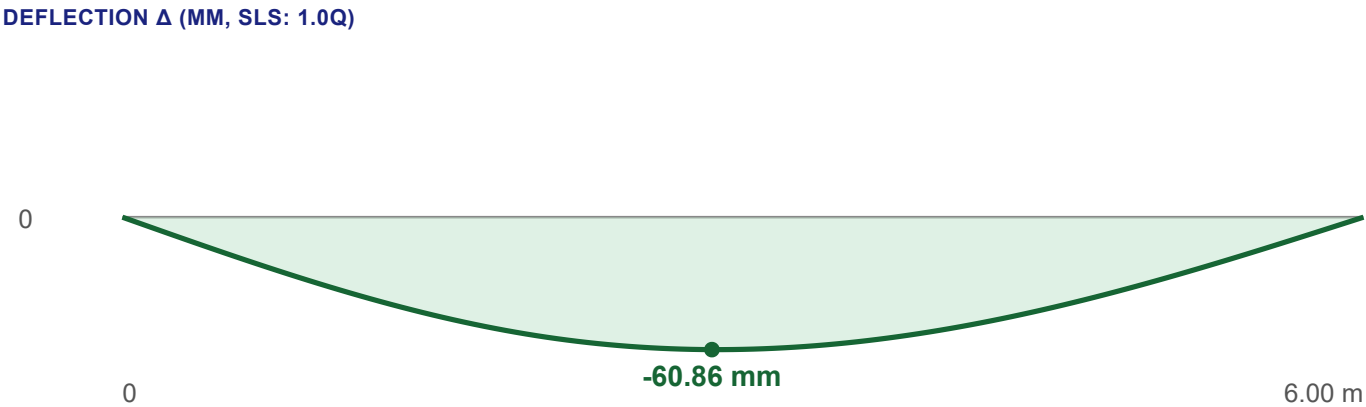
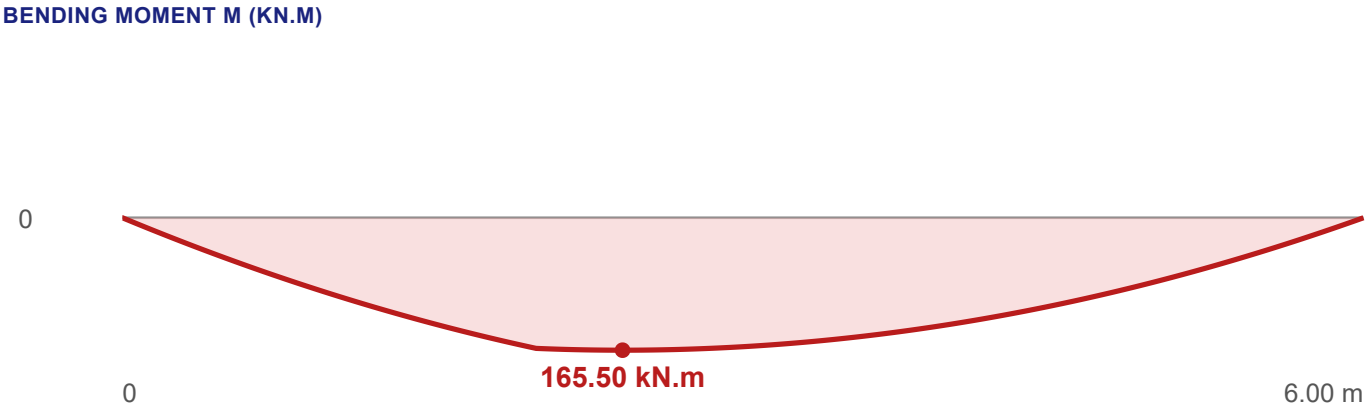
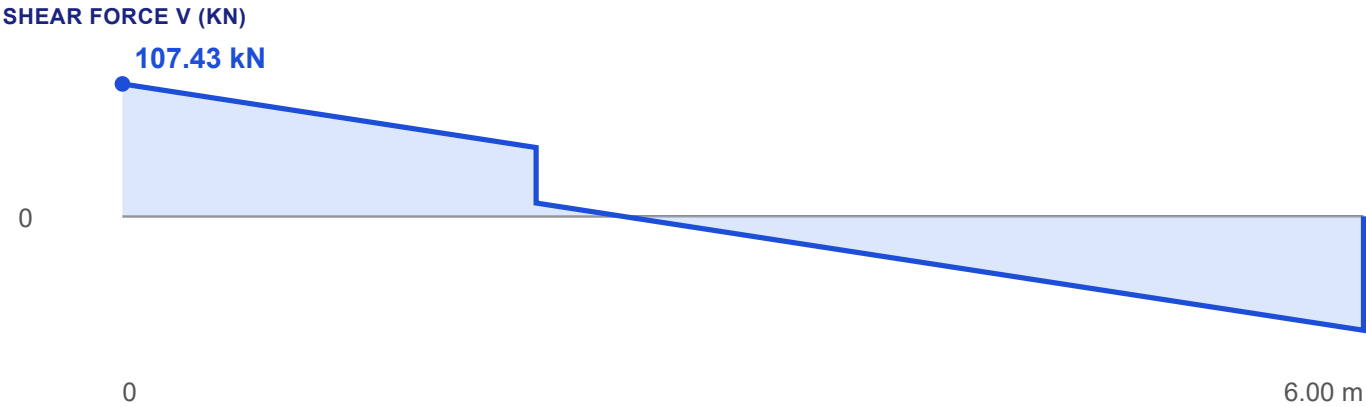
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)
 $Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]
 $Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

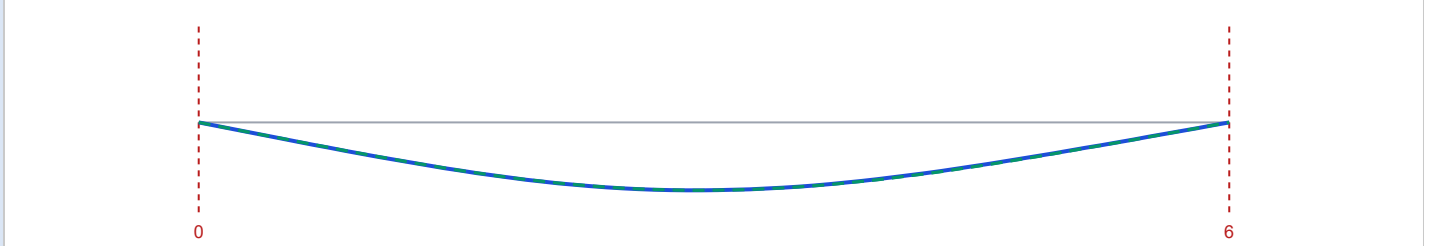
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 35.819 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_p b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	M _y (y.Ed)/M _b (b.Ed)	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R5 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: WARPING FREE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R5 "WARPING FREE": warping free at both ends (the default fork end; identical input to R2, kept under the owner's heading)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R5 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

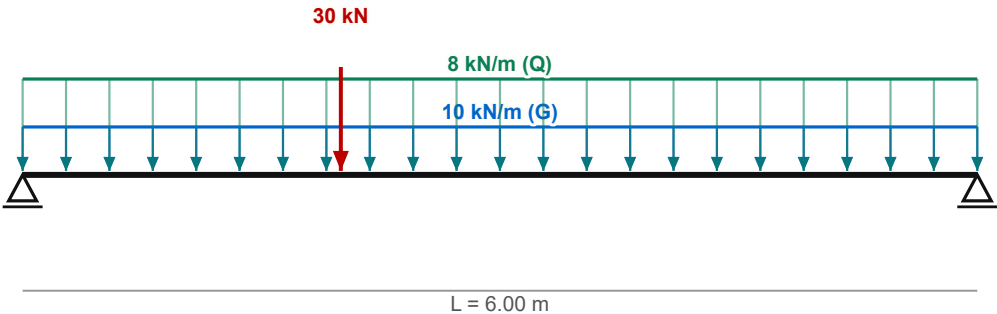
Member Loading and Member Forces

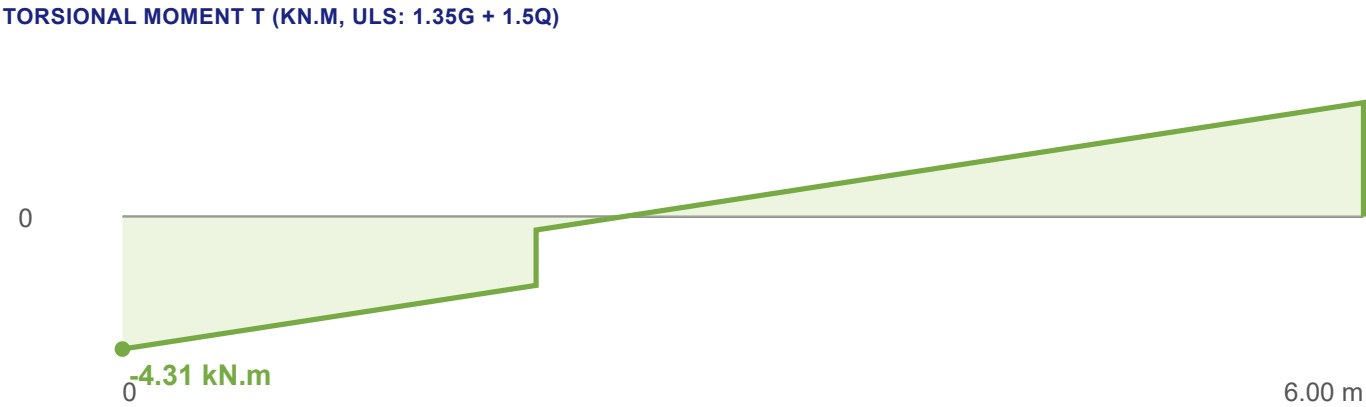
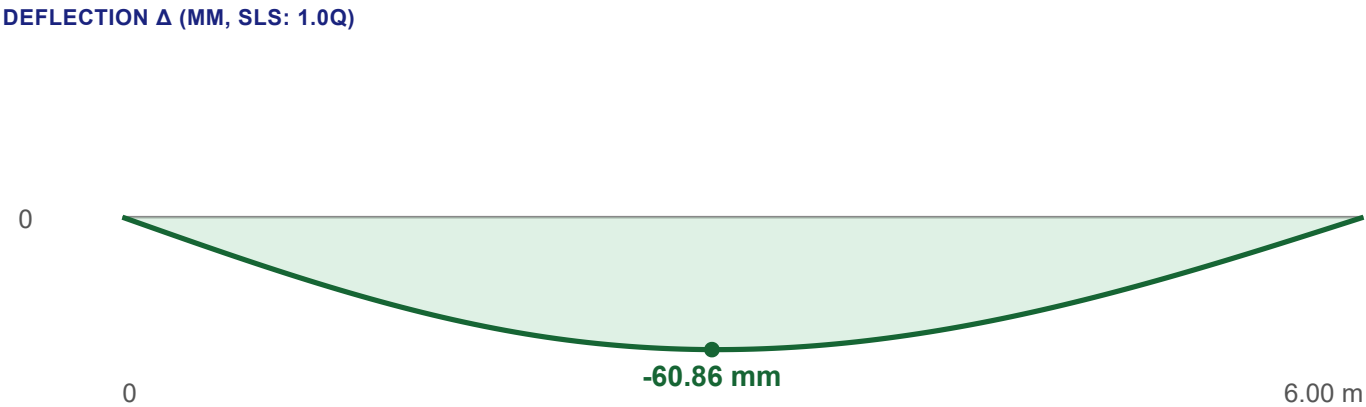
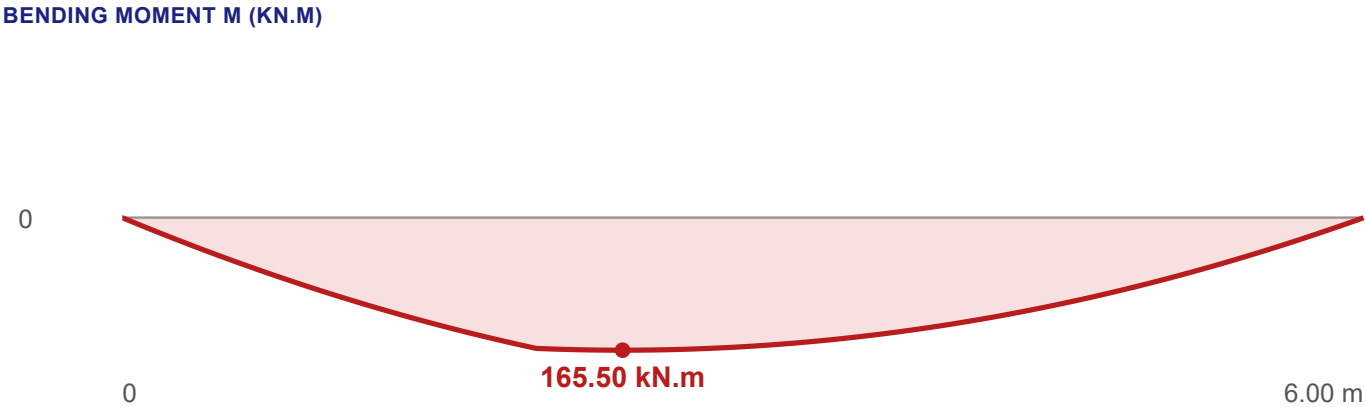
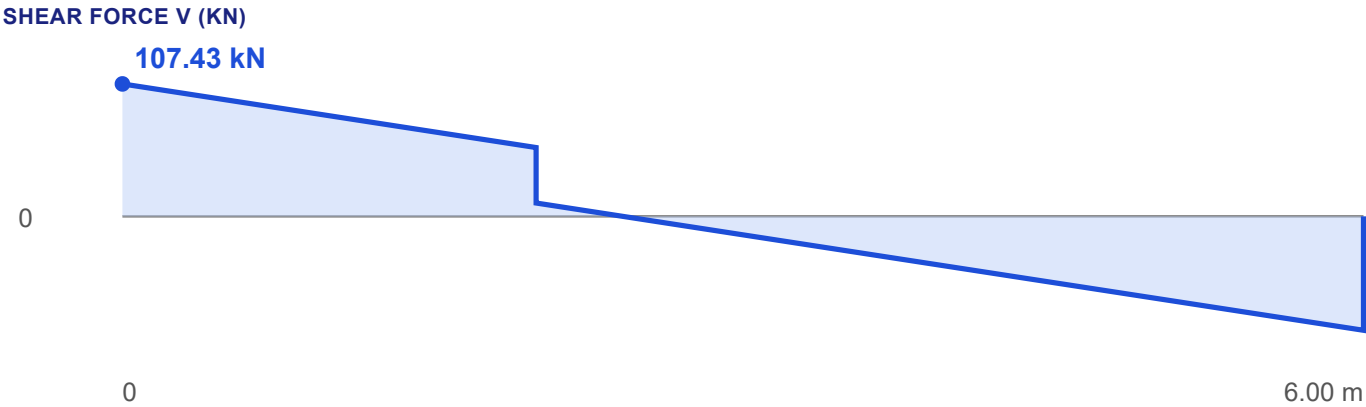
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]
 $Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the	—	not used
C_1 basis	P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C_1 channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		P362

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e/i_z)/\kappa$	(6000/23.9)/96 (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 12.27	P385 App C
φ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 9.128 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]		EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

τ_t , τ_w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{1 - \tau_t/(1.25f_y/\sqrt{3})} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	$107.430 / 20.965$	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M_(y,Ed)/M_(b,Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

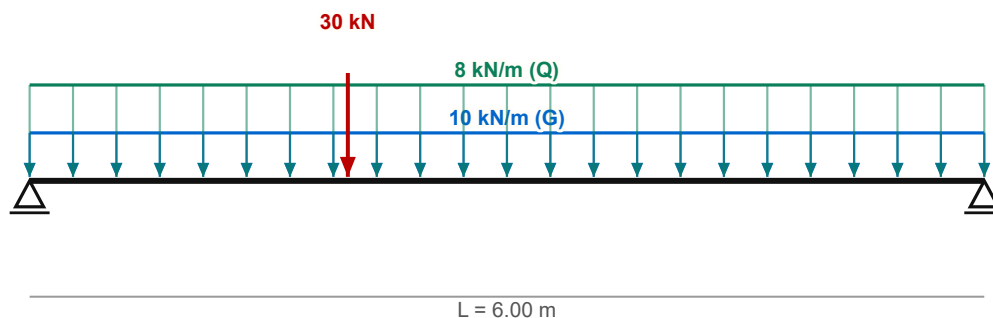
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

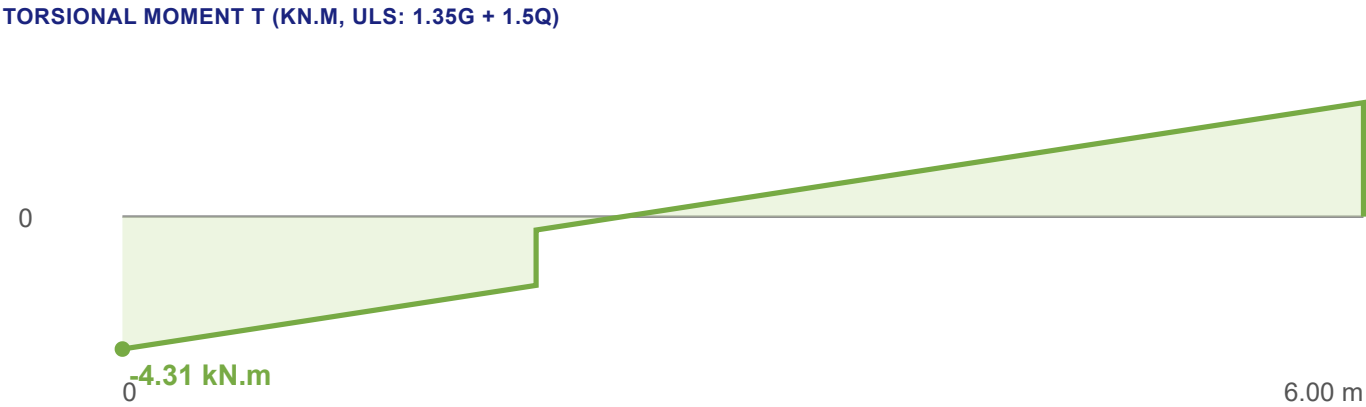
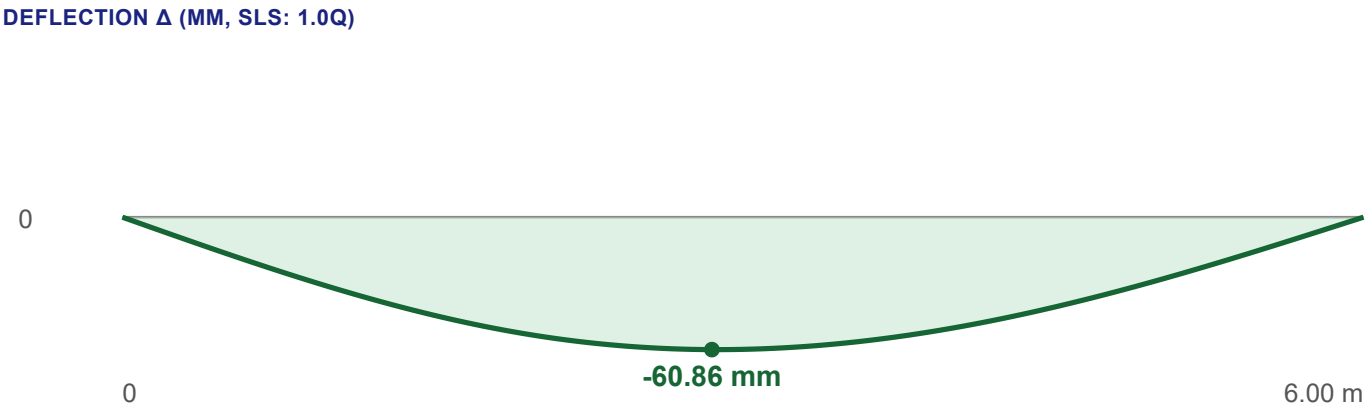
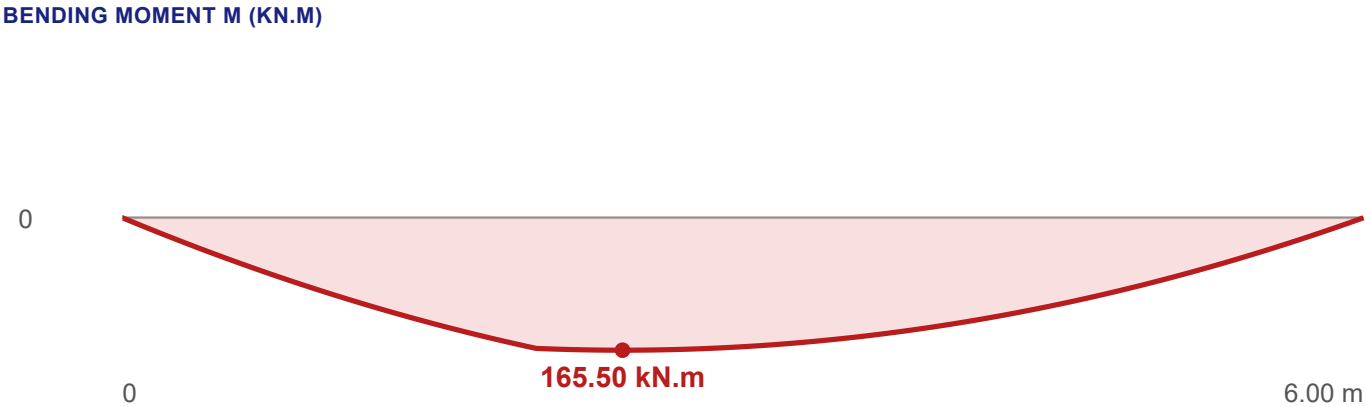
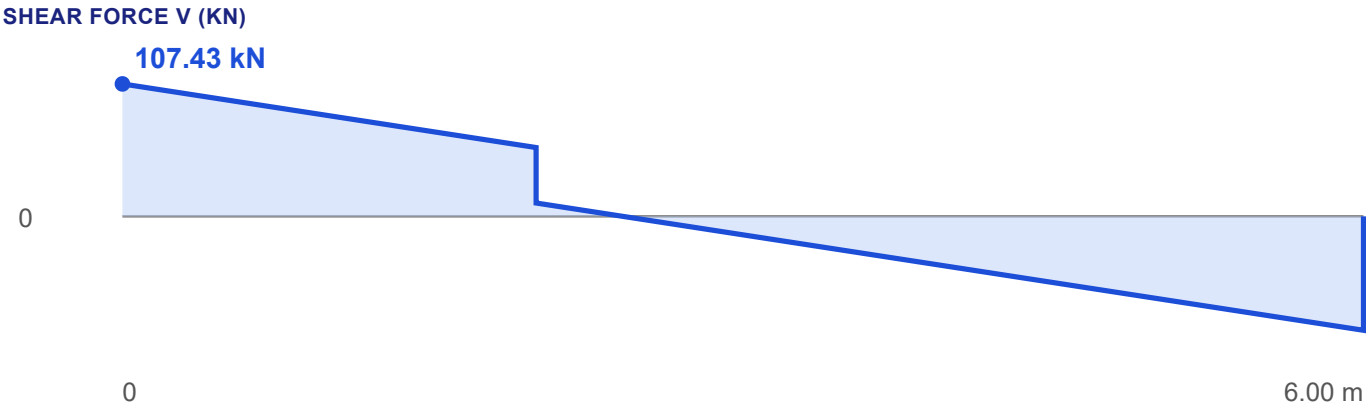
ADVISORY: Serviceability twist $\theta_{\text{max}} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3}$ x 1)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \epsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning

Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip

@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\phi = 0.66174$ rad, $\phi' = 3.186e-5$ rad/mm, $\phi'' = 1.851e-7$ rad/mm², $\phi''' = 1.535e-10$ rad/mm³; $M_z = \phi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \phi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm²; $\tau = G t \phi' = 32.26$ N/mm²; $(4240.69/275)^2 + 3(32.26/275)^2 =$

σ_x, τ , (6.1) at the 5 section points

P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; S_{w1}, S_{w2}, S_{w3} = P385 Table A.2 points 1, 2, 3)

$f_y / \gamma_{M0} = 275$

Elastic verification (6.1): information only

Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information

NOT VERIFIED	Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.	NOT VERIFIED
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Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)

$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) =$ $0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

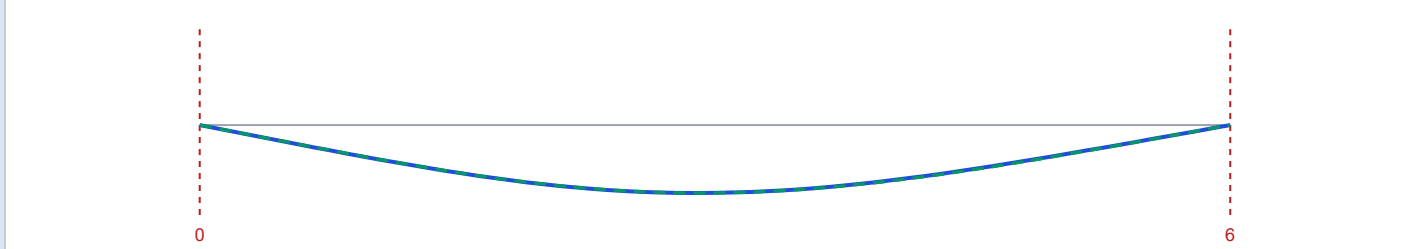
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0} , S_{w1}	40.3 cm ³ , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \phi'' / (h - t_f)$	max over span	3.645 kN.m	

$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y}$ = 62.425, $M_{pl,f}$ = 4.834, $M_{pl,z}$ = 16.665	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty$ ($M_{y,Ed} \geq M_{cr} = 35.819$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y / 4$) [verify]		

End torques T_t

St Venant part $GI_T \phi'$ at x = 0 / x = L; total T = $GI_T \phi'$ 3.771 / 3.200 kN.m
– $EI_w \phi''' = 4.307 / 3.707$ kN.m

Torsion Shear Design @ 0 m

T_t, T_w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, V = 195.04 / 5.22 N/mm²		P385
	= 107.430 kN		
$V_{pl,T,Rd} = [\sqrt{(1 - t_f/(1.25f_y/\sqrt{3}))} - T_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 195.04$ ($T_w = 5.22$) N/mm²; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

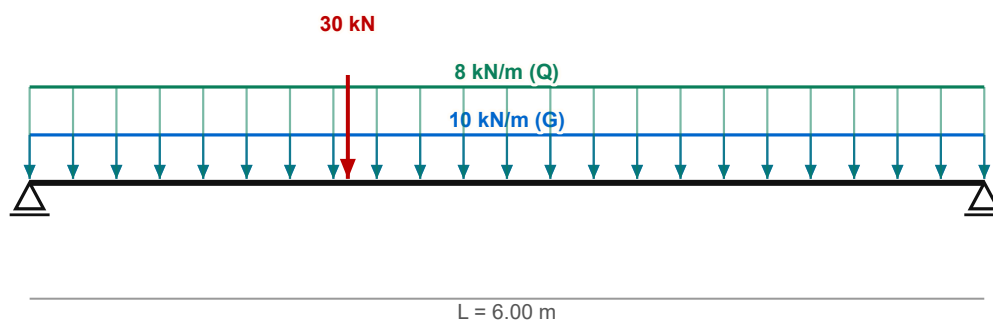
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

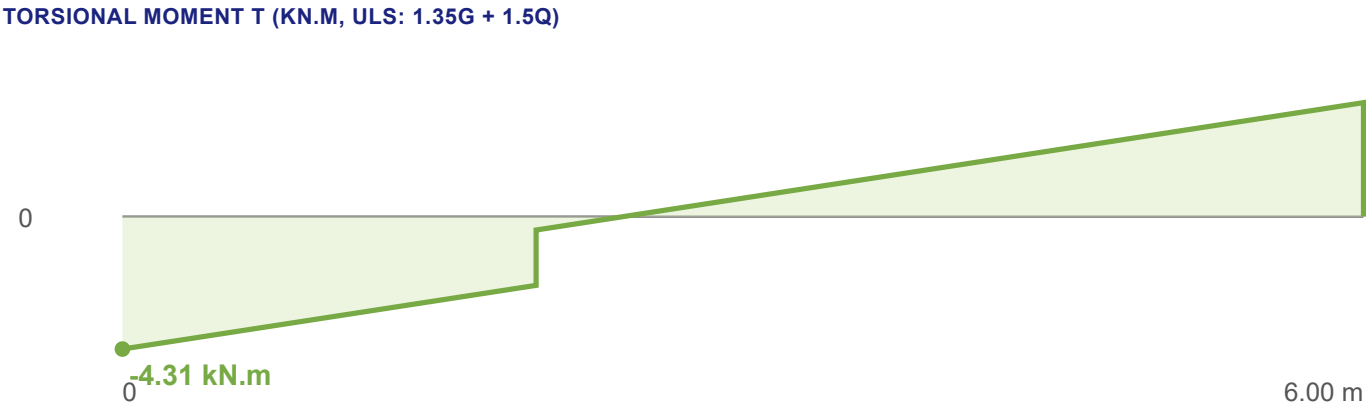
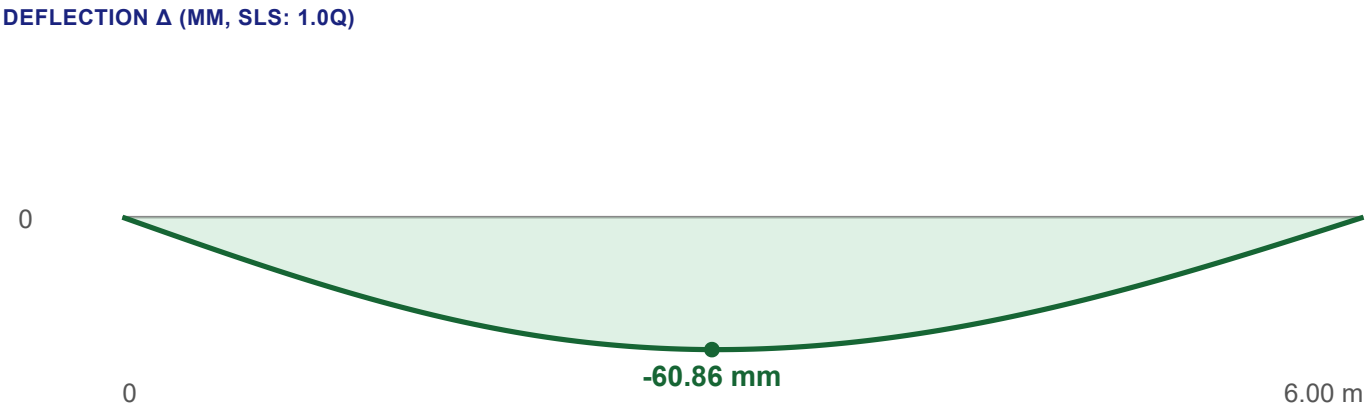
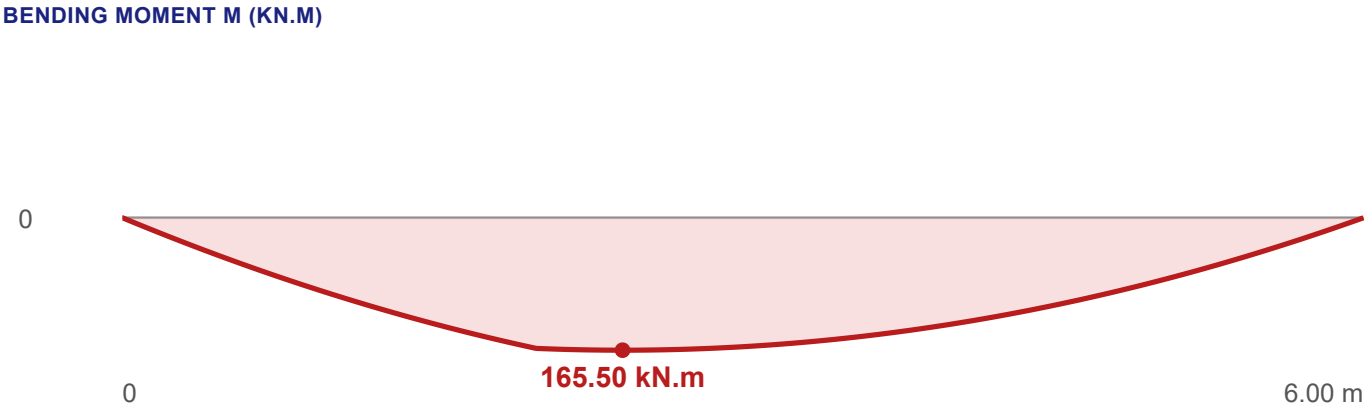
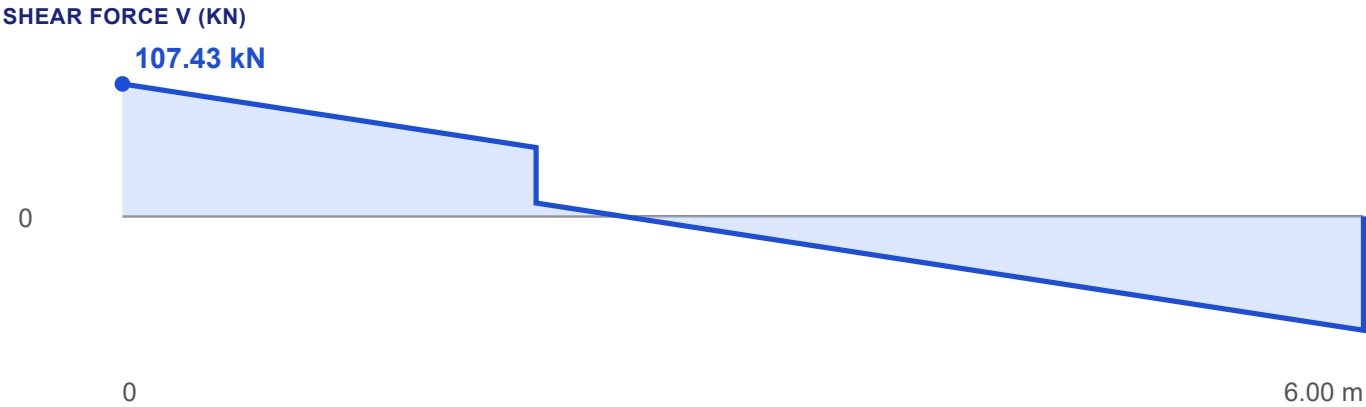
ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C_1	—	not used
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		P362

Lateral Buckling Check M.b.Rd

$L_e = 1.2 L$	1.2 x 6 =	7.2 m	
$\lambda_{LT} = (L_e/i_z)/\kappa$	(7200/23.9)/96 (S275)	3.138	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	3.138, 5.233, 0.76	0.102	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	0.102 x 227.0 x 275 =	6.339 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	227.0 x 275 / 3.138 ²	6.339 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	165.504 / 6.339	26.108	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \phi''$ (max)	@ x = 2 m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	x x ∞ ($M_{y,Ed} \geq M_{cr} = 6.339$ kN.m)	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 6.339$ kN.m	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]		
End torques T_t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total T = $GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

τ_t , τ_w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
$V_{pl,T,Rd} = [\sqrt{1 - \tau_t/(1.25f_y/\sqrt{3})} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M_(y,Ed)/M_(b,Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	26.108	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\text{max}} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R7 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)

Owner case: STABILIZING LOADS

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R7 "STABILIZING LOADS": all loads hung from the bottom flange $z_g = -100$ mm ($-D/2$) with $e = +40$ mm

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R7 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40$ mm $z_g = -100$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 40$ mm $z_g = -100$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = -100$ mm (kN)

G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)

$V_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$V_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$V_{G,inf} = 0.9$, EQU set A]

$V_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered V_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

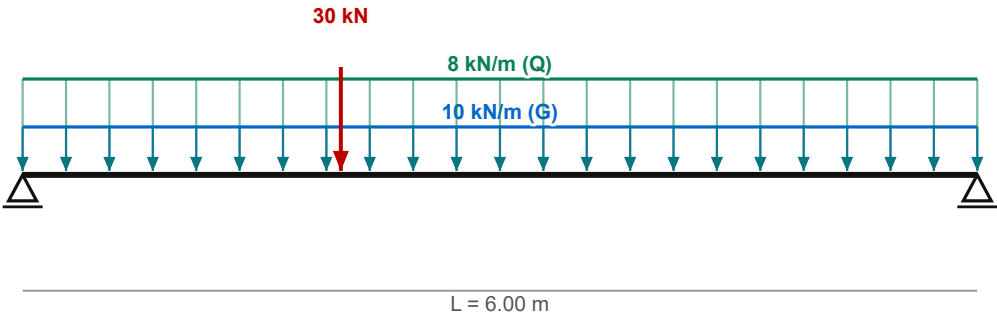
Uplift

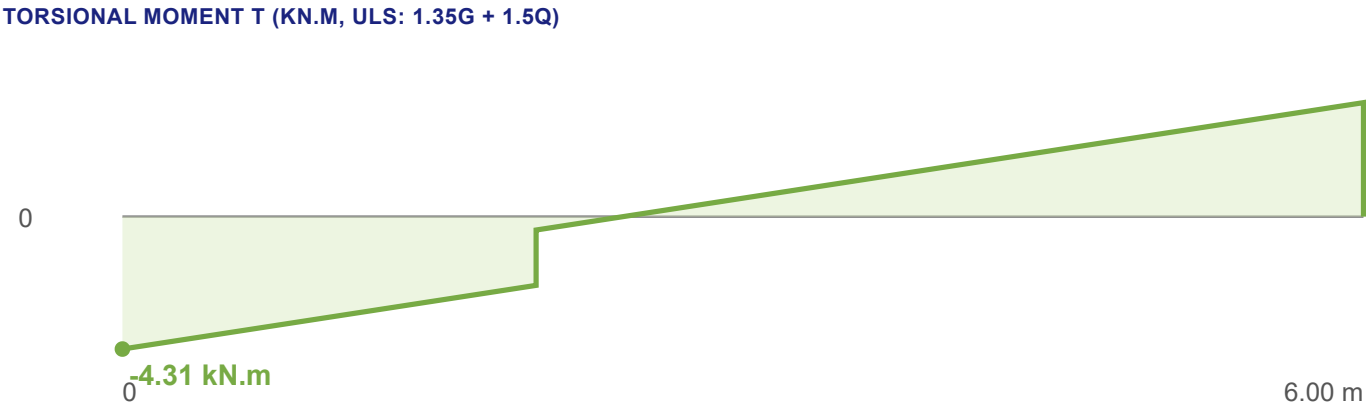
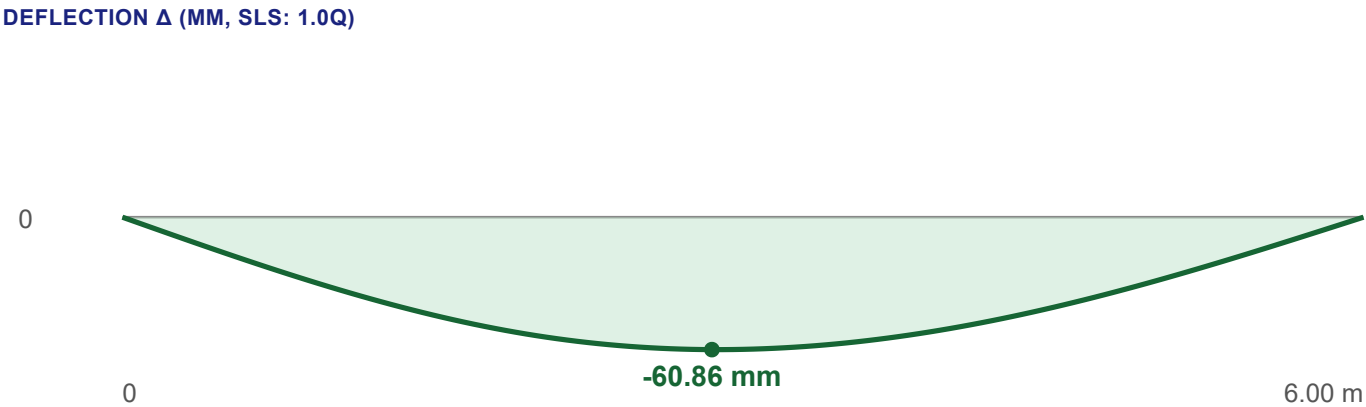
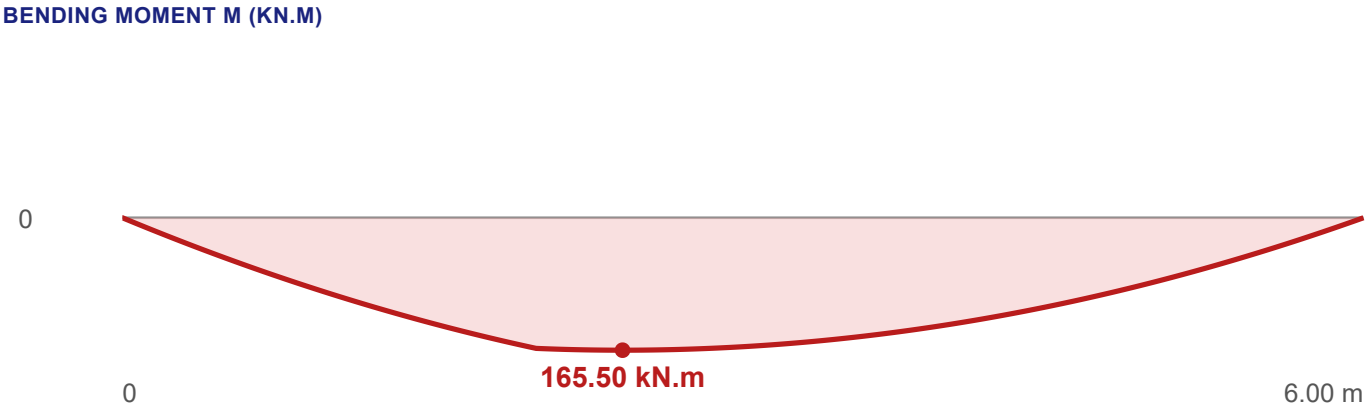
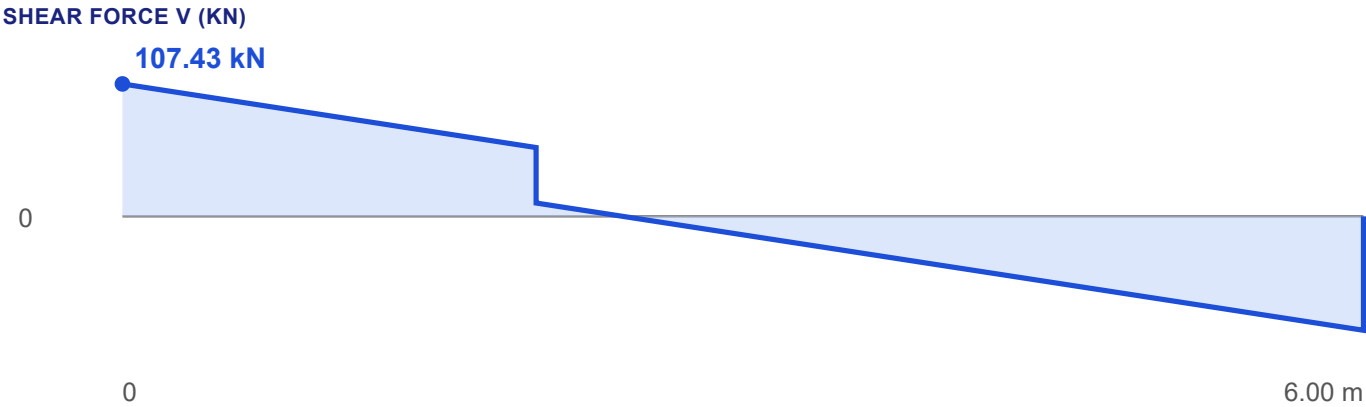
no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $V_{G,inf}$ companions with G at 1.0 and 0.9)

$R_{min} \geq 0$

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

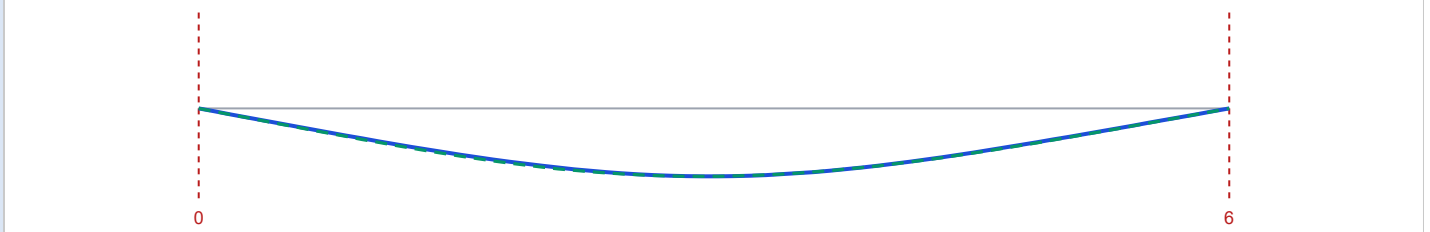
Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	41.391 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 41.391}$	1.228	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.228, 1.380, 0.76	0.442	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.442, 1.228, 0.935, 0.979	0.452	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.452 \times 227.0 \times 275 \leq 62.425 =$	28.204 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 28.204	5.868	Warning

Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \propto (M_{y,Ed} \geq M_{cr} = 41.391 \text{ kN.m})$	—	EN 1993-6 A
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 41.391 \text{ kN.m}$	≥ 99	Warning FAIL: $M_{y,Ed} \geq M_{cr}$

(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_f b^2 f_y / 4$) [verify]	EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m
Torsion Shear Design @ 0 m		
T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	20.965 kN 6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning
Deflection Check - Load Case 1 (SLS: 1.0Q)		
In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	M _y (y.Ed)/M _b (b.Rd)	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	5.868	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R7 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: STABILIZING LOADS
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R7 "STABILIZING LOADS": all loads hung from the bottom flange z_g = -100 mm (-D/2) with e = +40 mm

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R7 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = -100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = -100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = -100 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf}$ = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf}$ = 0.9, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

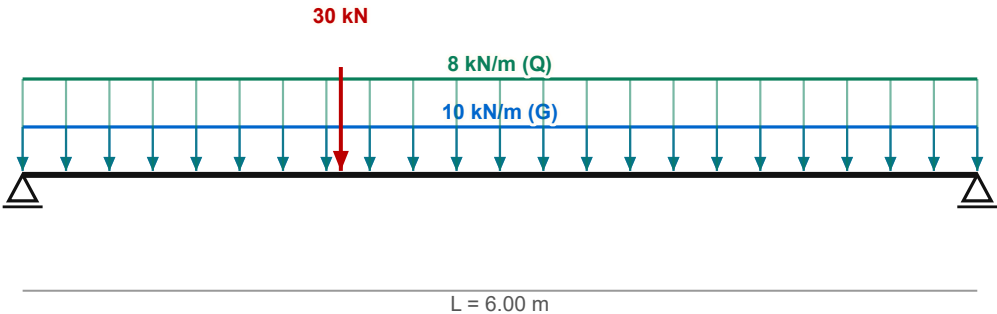
Uplift

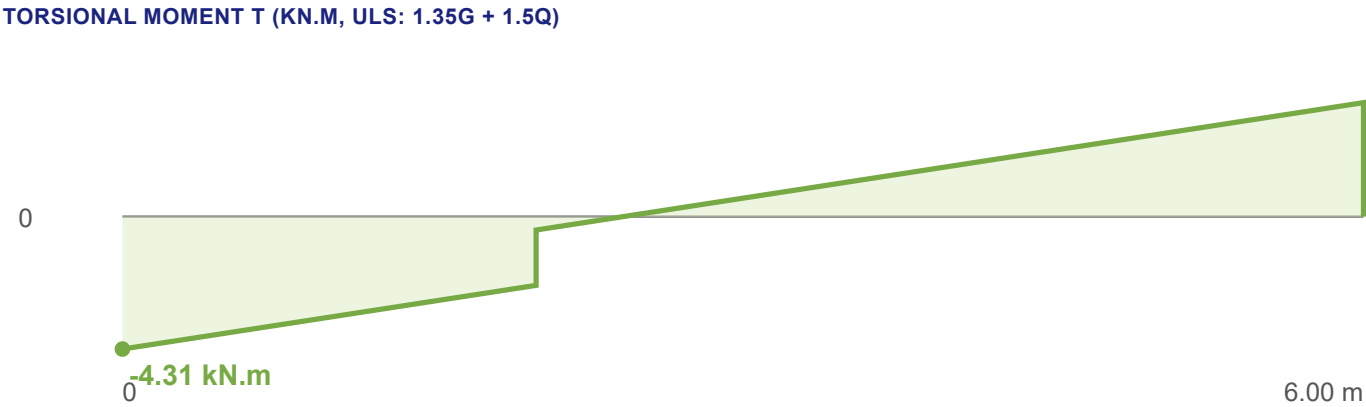
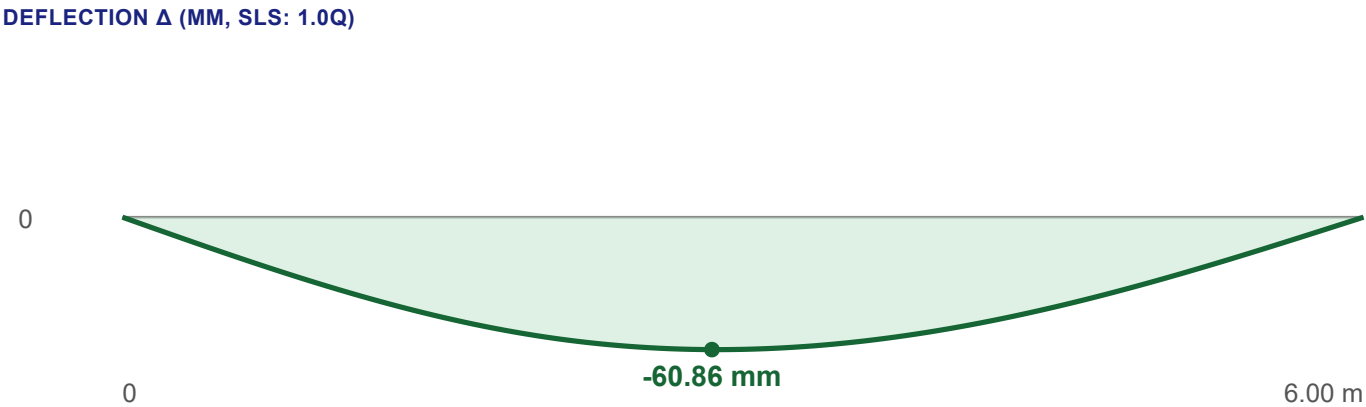
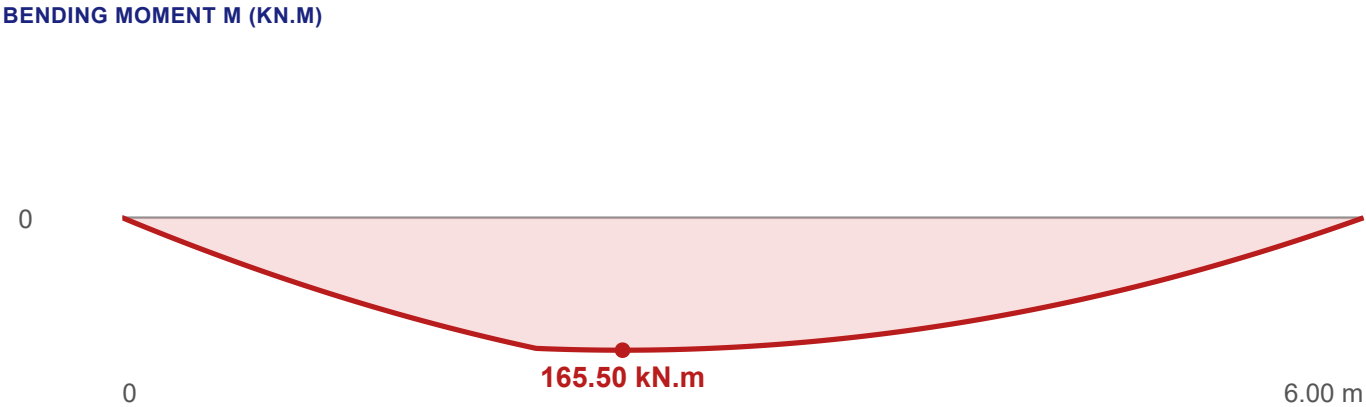
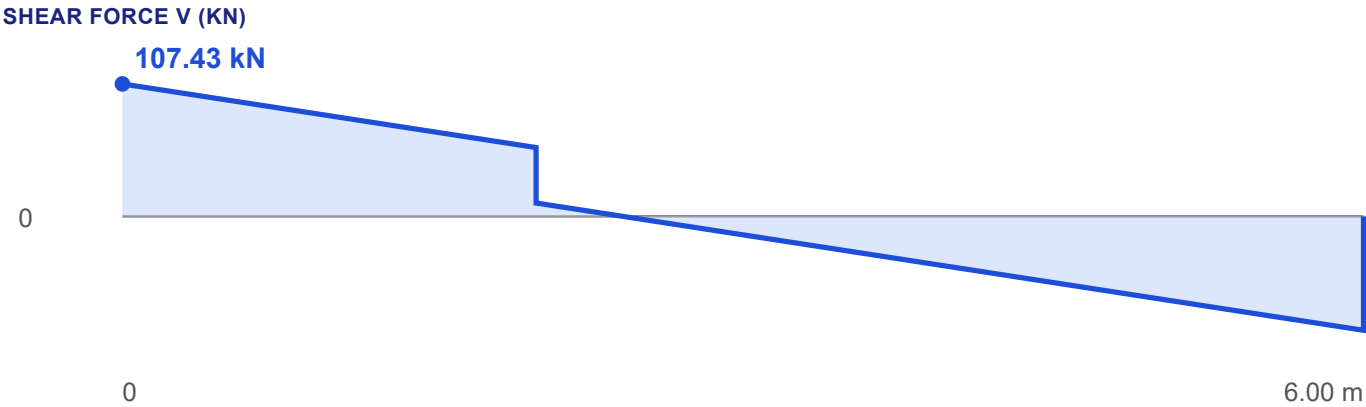
no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)

$R_{min} \geq 0$

OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = Fn(c/t, d/t, fy, N, My, Mz)	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
	(Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the P362 κ chain sets $\lambda_{LT} = (L_e/i_z)/\kappa$ without C ₁	—	not used
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
Z _g	Z _g = -100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L _E switch); the shear-centre M _{cr} route is not offered		load height
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
X _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W.f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 9.128 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f b ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

τ _t , τ _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.771 kN.m, V = 195.04 / 5.22 N/mm ²		P385
	= 107.430 kN		

$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl.Rd} = 212.754$; $S_{mod} = V_{pl.T.Rd}/V_{pl.Rd} = 0.099$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	0.2787 rad = $15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M _(y.Ed) /M _(b.Rd)	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed}: EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with M_{z,tot}); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: M_{y,Ed} reaches the elastic critical moment M_{cr} (M_{y,Ed} >= M_{cr}): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R8 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: EFFECTIVE LENGTHS OVERRIDE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R8 "EFFECTIVE LENGTHS OVERRIDE": $L_E = 0.85 L$ entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R8 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

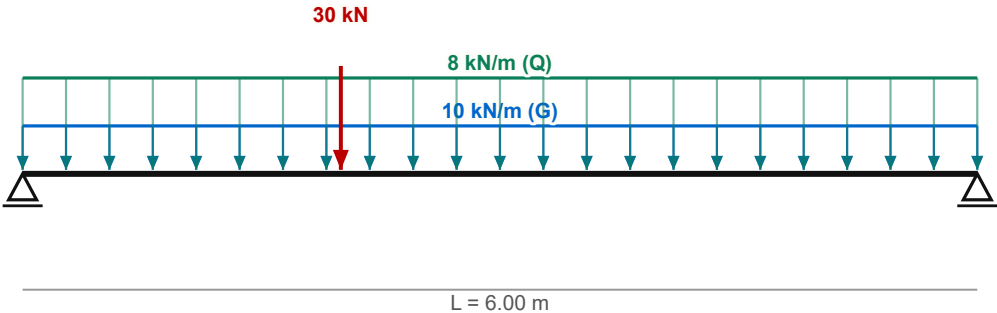
Member Loading and Member Forces

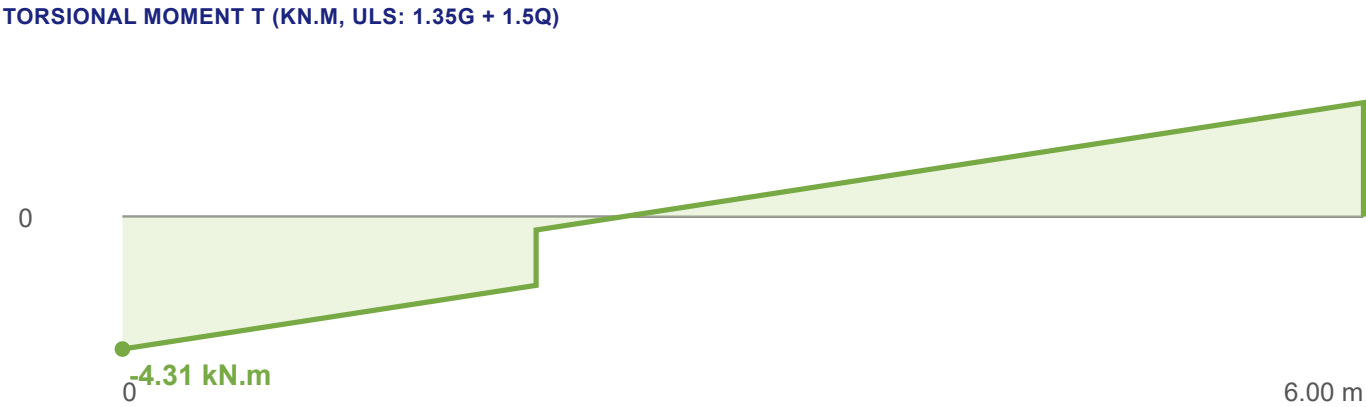
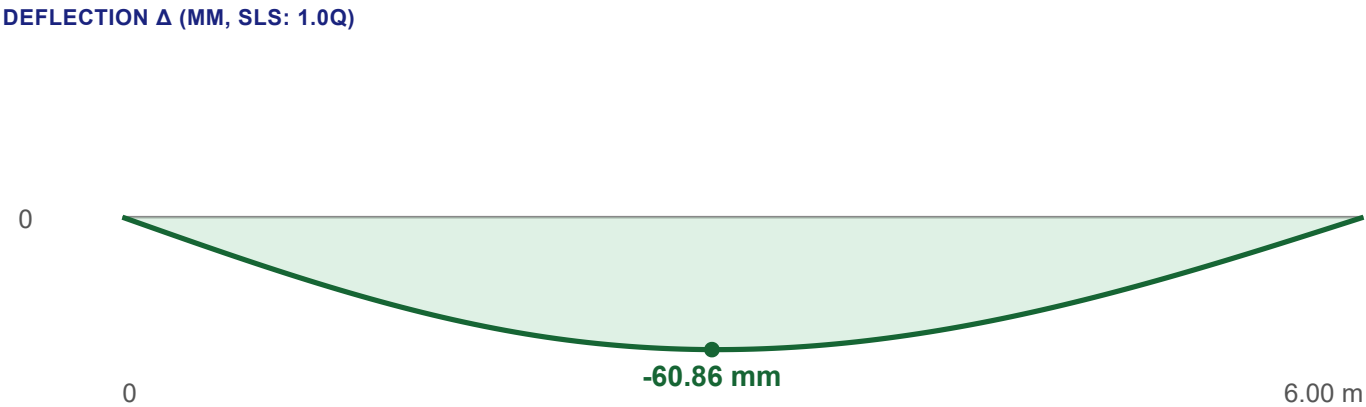
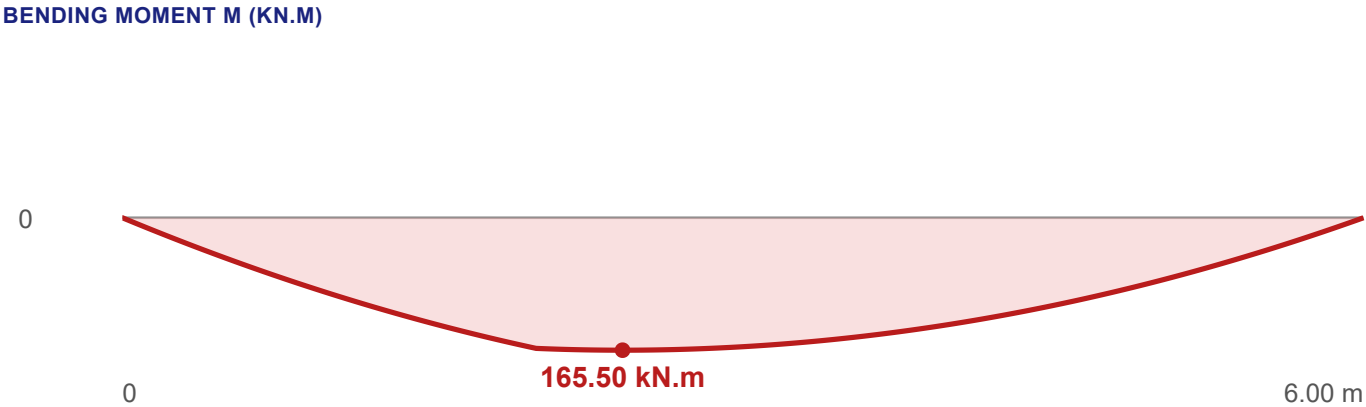
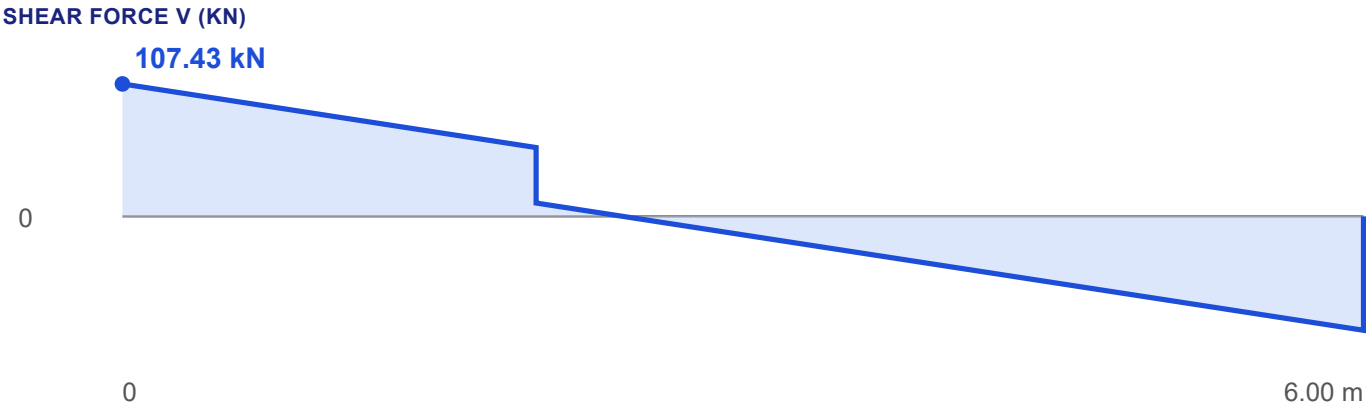
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0-6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0-6 m $e = 50.9$ mm (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + E W_{n0} \varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = G t \varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15 \varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

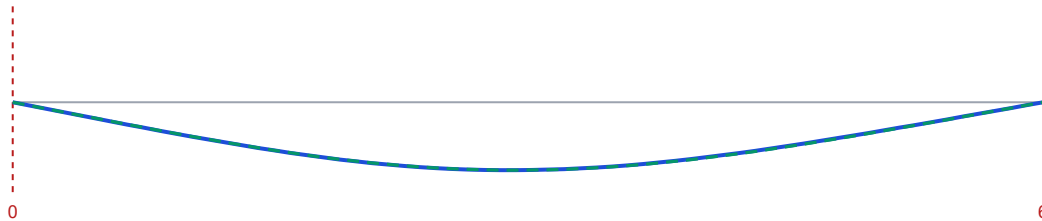
$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k _c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Advisory	The L_e factor does not affect EC3 LTB on the eigen route; the buckling length is set by the end restraints and the restraint positions. It overrides the strut lengths of both axes (the end-fixity defaults are printed with them).	advisory
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Buckled mode shape — critical eigenmode (normalised; twist ϕ solid, lateral v dashed; red: lateral restraint points; peak twist at $x = 2.93$ m)



Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ³ , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	L/a = 12.27	P385 App C
ϕ_{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = E I _w ϕ'' (max)	@ $x = 2$ m	0.683 kN.m ²	
M _{w,Ed} = E I _w $\phi''/(h-t_f)$	max over span	3.645 kN.m	
M _{z,Ed} = $\phi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
$k = k_w \cdot k_{zw} \cdot k_\alpha$	$x \times \infty$ (M _{y,Ed} ≥ M _{cr} = 35.819 kN.m)	—	EN 1993-6 A

$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$ (A.1) scope	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819$ kN.m EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with $M_{f,Rd} = t_b^2 f_y / 4$) [verify] St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi'$ – $EI_w \phi''' = 4.307 / 3.707$ kN.m	≥ 99 Warning FAIL: $M_{y,Ed} \geq M_{cr}$ EN 1993-6 A
End torques T_t	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T_t, T_w at the V-T station	@ $x = 0$ m (ULS: 1.35G + 1.5Q): $T_t = 3.771$ kN.m, $V = 195.04 / 5.22$ N/mm ² = 107.430 kN	P385
$V_{pl,T,Rd} = [\sqrt{(1 - T_t/(1.25f_y/\sqrt{3})) - T_w/(f_y/\sqrt{3})}] V_{pl,Rd}$	@ $x = 0$ m: $T_t = 195.04$ ($T_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 20.965$ kN 212.754; $S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099$	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124 Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm Warning
Torq in Case 1 @ 2.465 m: $\theta_{max} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97° > 2° advisory

MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} ($M_{y,Ed} \geq M_{cr}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,Ed}/M_{cr})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).

PFC-R8 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: EFFECTIVE LENGTHS OVERRIDE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R8 "EFFECTIVE LENGTHS OVERRIDE": L_E = 0.85 L entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

Beam & Beam-Portion (Member) (FAIL)
Includes Design for Torsion with Span Warping, Ends Free to Warp
Member PFC-R8 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

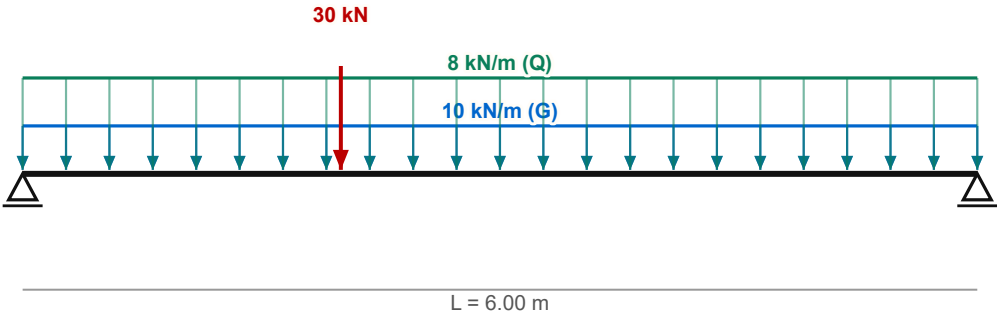
Member Loading and Member Forces

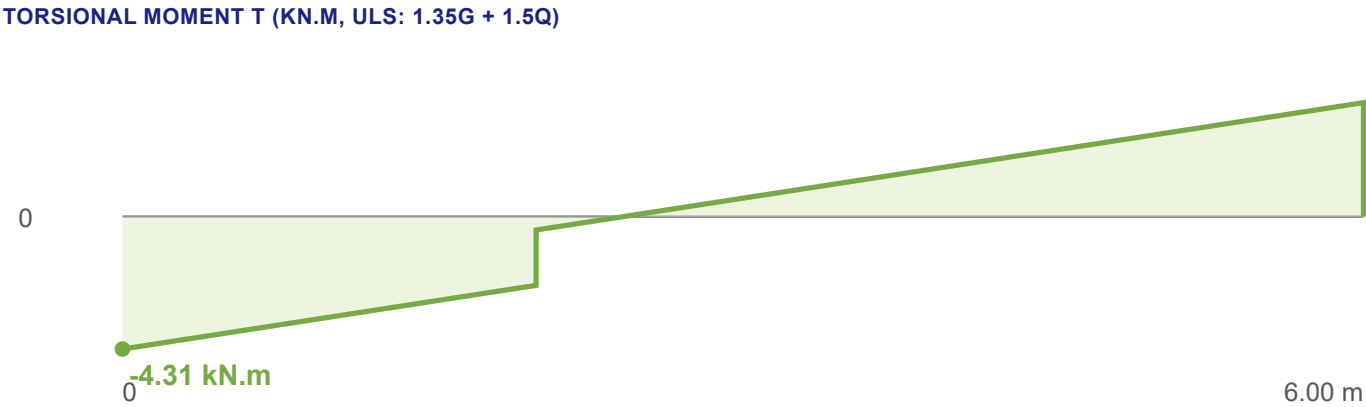
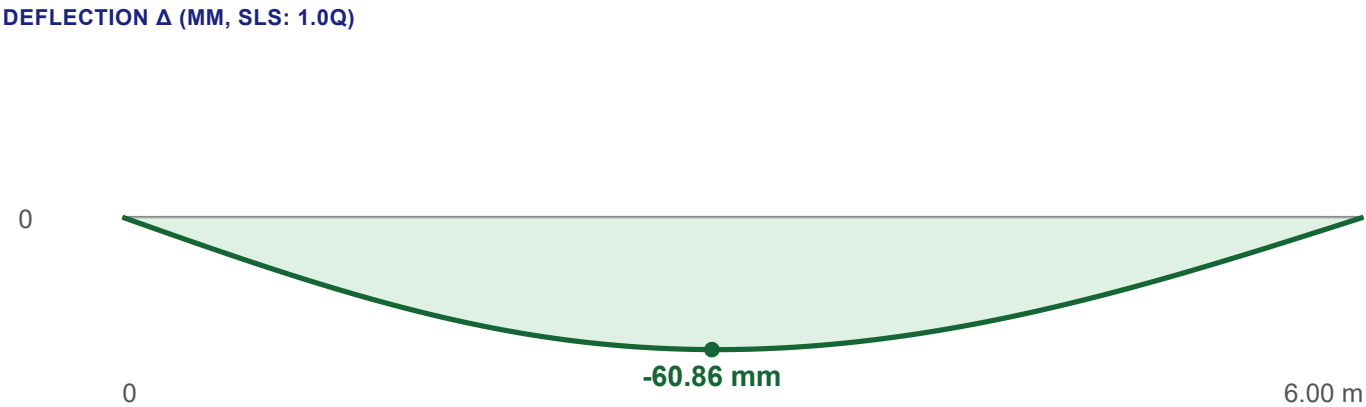
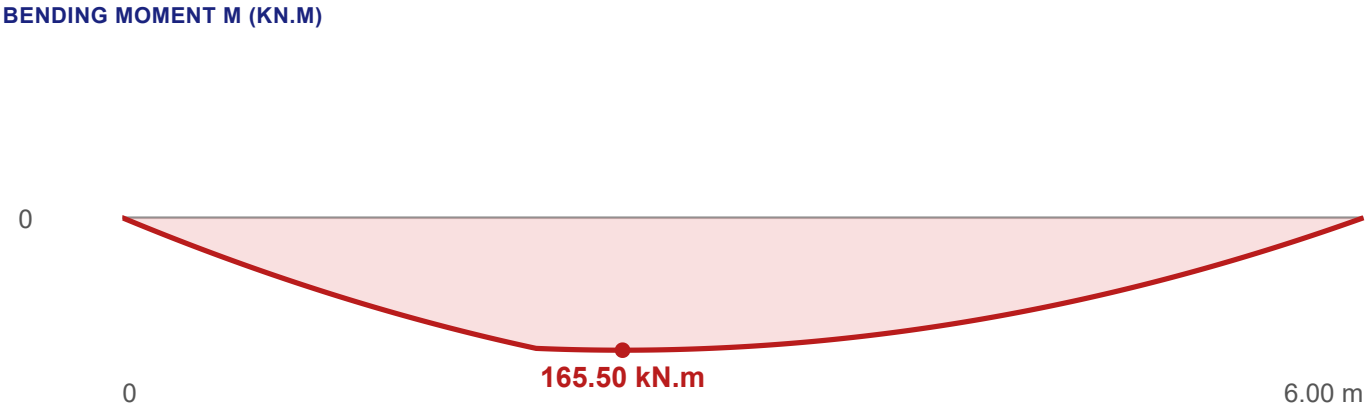
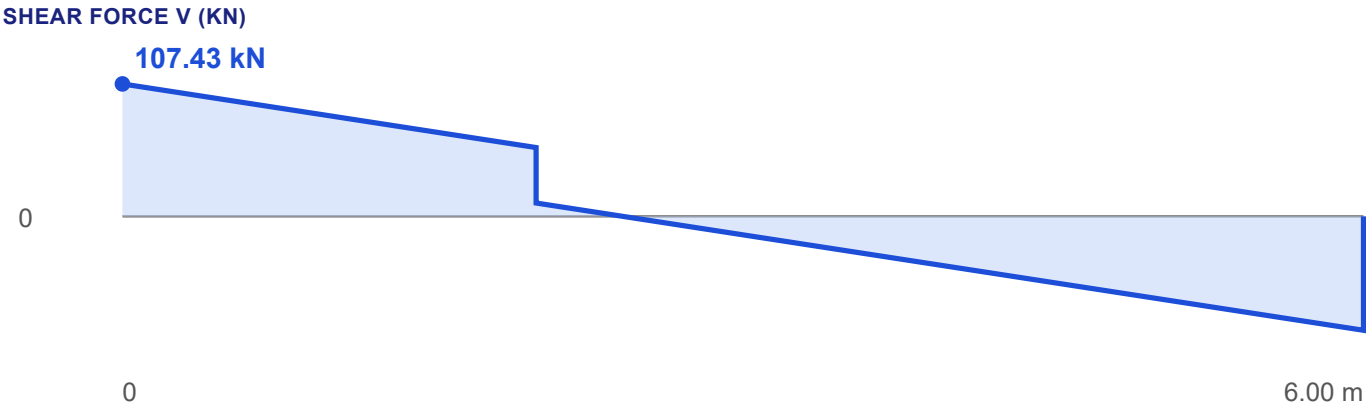
Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	4.31	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.71	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

LOADING





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	Class 1
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 0.00, 165.50, 0.00 (Axial: Non-Slender)	
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	$29.17 \leq 72 \times 0.924 / 1 = 66.56$	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000 — not used: the	—	not used
C ₁ basis	P362 κ chain sets λ _{LT} = (L _e /i _z)/κ without C ₁ channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		P362

Lateral Buckling Check M.b.Rd

L _e = 0.85 L	0.85 x 6 =	5.1 m	
λ _{LT} = (L _e /i _z)/κ	(5100/23.9)/96 (S275)	2.223	P362 channel
X _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.223, 3.045, 0.76	0.185	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.185 x 227.0 x 275 =	11.549 kN.m	
M _{cr} (back-calculated) = W _y .f _y /λ _{LT} ²	227.0 x 275 / 2.223 ²	12.634 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 11.549	14.331	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
k = k _w .k _{zw} .k _α	x x ∞ (M _{y,Ed} ≥ M _{cr} = 12.634 kN.m)	—	EN 1993-6 A
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 12.634 kN.m	≥ 99	Warning FAIL: M _{y,Ed} ≥ M _{cr}
(A.1) scope	EN 1993-6 Annex A (informative) states (A.1) for members with an I section; applied to the channel on the SCI P385 basis (P385 6.2/8.2 form with M _{f,Rd} = t _f ² f _y /4) [verify]		EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	

Torsion Shear Design @ 0 m

T _t , T _w at the V-T station	@ x = 0 m (ULS: 1.35G + 1.5Q): T _t = 3.771 kN.m, V = 107.430 kN	195.04 / 5.22 N/mm ²	P385
V _{pl,T,Rd} = [√(1 - T _t /(1.25f _y /√3)) - T _w /(f _y /√3)].V _{pl,Rd}	@ x = 0 m: T _t = 195.04 (T _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754; S _{mod} = V _{pl,T,Rd} /V _{pl,Rd} = 0.099	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	60.86 mm	Warning
Torq in Case 1 @ 2.465 m: $\theta_{\text{max}} \leq 2.00^\circ$	$0.2787 \text{ rad} = 15.97^\circ \leq 2.00^\circ$ (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict)	15.97°	> 2° advisory

MA/Mc	M_(y,Ed)/M_(b,Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	Yield 6.1 (info)	V+T	LTB+T	Max
2.651	14.331	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

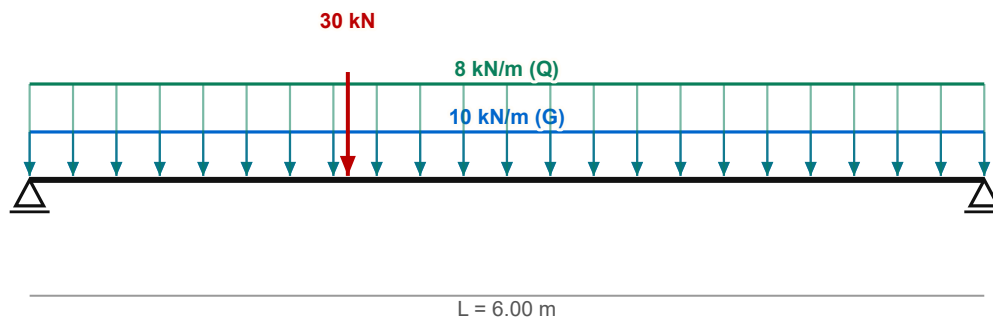
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

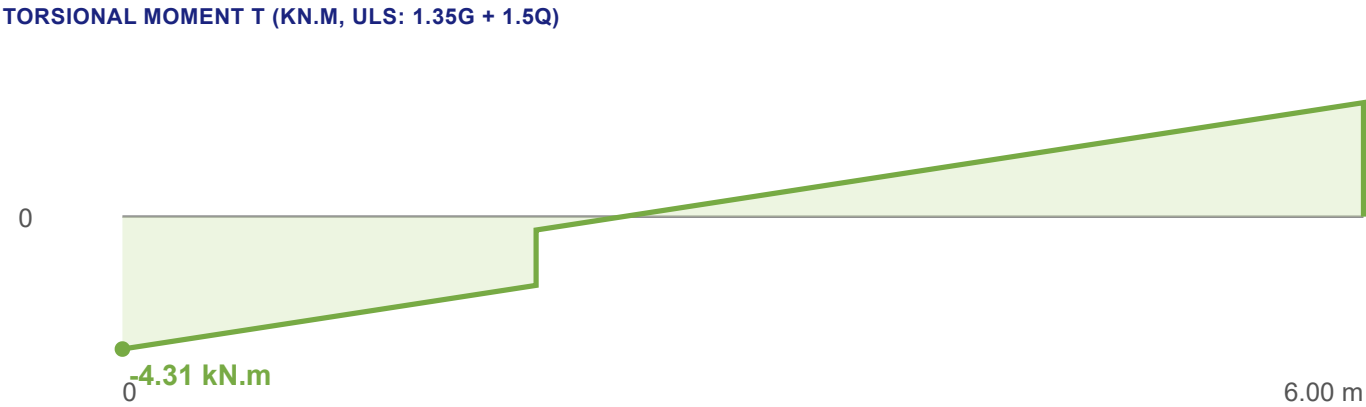
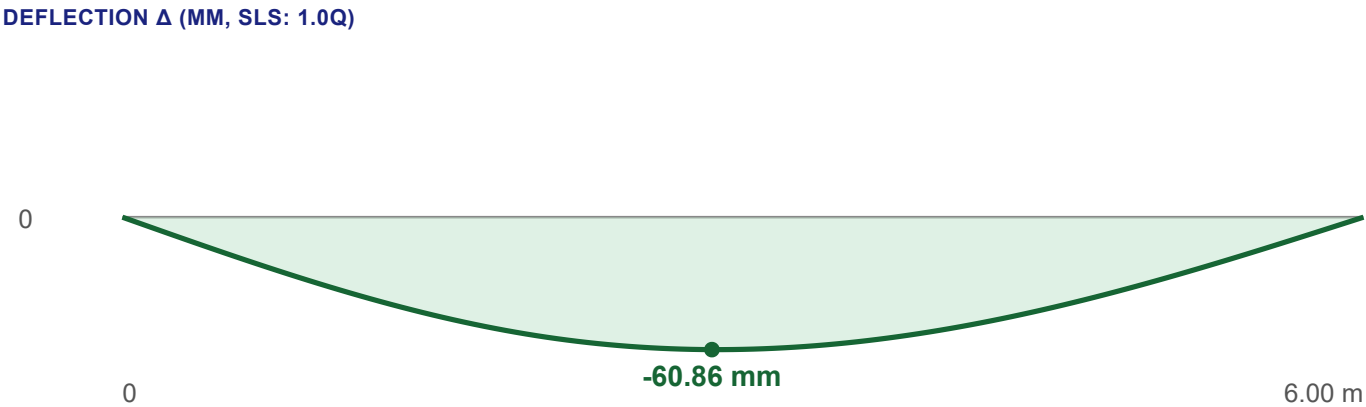
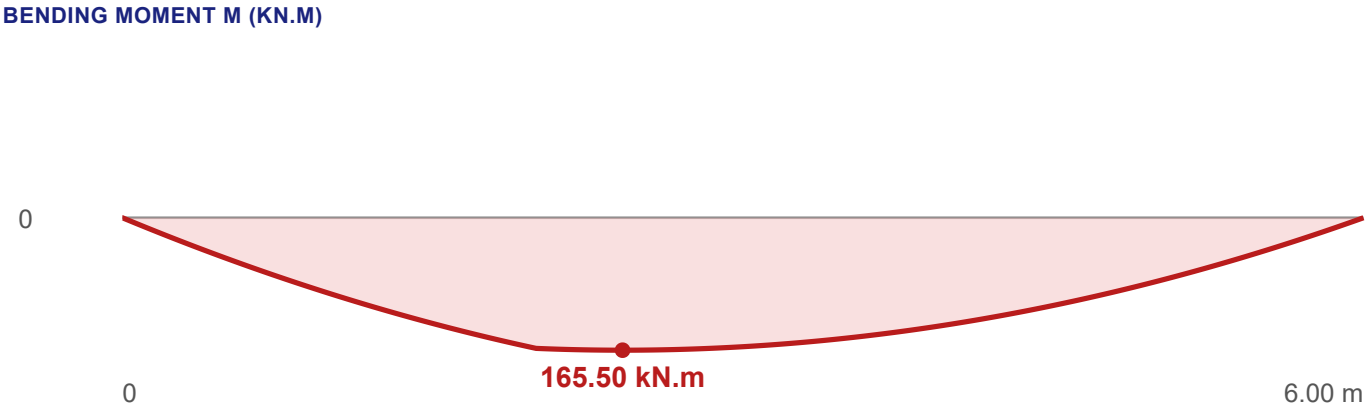
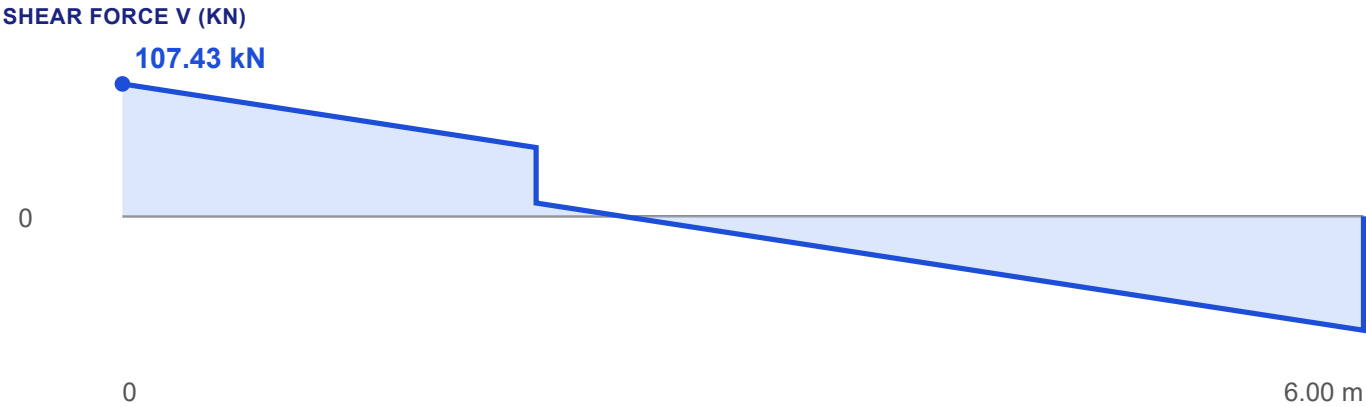
ADVISORY: Serviceability twist $\theta_{\text{max}} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at x = 2.46 m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,\text{tot}}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

FAIL: $M_{y,\text{Ed}}$ reaches the elastic critical moment M_{cr} ($M_{y,\text{Ed}} \geq M_{\text{cr}}$): lateral-torsional buckling governs before the torsion interaction can be evaluated - the EN 1993-6 Annex A amplifier $k_\alpha = 1/(1 - M_{y,\text{Ed}}/M_{\text{cr}})$ is unbounded and the utilisation is carried as 99; the member is inadequate as arranged (larger section, shorter unrestrained length or compression-flange restraint).





Bending-moment diagram drawn on the tension side (sagging down); shear and deflection to true sign (down = below the axis); hover over a diagram for the value at any point.

Classification and Effective Area (EN 1993: 2006)		
Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]	
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	4.56, 25.20, 275, 0.00, 165.50, 0.00	Class 1
(Axial: Non-Slender)		
Auto Design Load Cases	1; SLS 1	

Shear Capacity Check

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3} \times 1$)	212.754 kN	6.2.6
$V_{y,Ed} / V_{pl,y,Rd}$	107.430 / 212.754 =	0.505	OK
$h_w / t_w \leq 72 \varepsilon / \eta$	29.17 $\leq$ 72 x 0.924 / 1 = 66.56	no shear buckling	OK 6.2.6(6)

Moment Capacity Check M.c.y.Rd - Fully Restrained Beam

$V_{y,Ed} / V_{pl,y,Rd}$ (at max M)	0.021 / 212.754 =	0.000	Low Shear
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 227.0 / 1	62.425 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y,Ed} / M_{v,y,Rd}$ @ x	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl,Rd} = 20.965$: pure shear governs	2.662	Warning
Elastic yield criterion with torsion (cl 6.2.7(5), Eq 6.1) @ 2.465 m, P1 flange tip	@ x = 2.465 m (ULS: 1.35G + 1.5Q): $M_y = 165.476$ kN.m, $V = 1.185$ kN, $\varphi = 0.66174$ rad, $\varphi' = 3.186e-5$ rad/mm, $\varphi'' = 1.851e-7$ rad/mm ² , $\varphi''' = 1.535e-10$ rad/mm ³ ; $M_z = \varphi M_y = 109.503$ kN.m; P1 flange tip: $\sigma_x = N/A + M_y/W_{el,y} + M_z/W_{el,z} + EW_{n0}\varphi'' = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$ N/mm ² ; $\tau = Gt\varphi' = 32.26$ N/mm ² ; $(4240.69/275)^2 + 3(32.26/275)^2 =$	237.839	> 1 information 6.2.7(5)
σ_x, τ , (6.1) at the 5 section points	P1 flange tip: $\sigma_x = 0.00 + 844.27 + 3239.73 + 156.69 = 4240.69$, $\tau = 32.26 \rightarrow 237.839$ (governs); P1b flange at $W_n = 0$: $\sigma_x = 0.00 + 791.50 + 470.81 = 1262.31$, $\tau = 0.24 + 32.26 + 2.97 = 35.47 \rightarrow 21.120$; P2 web-flange junction, flange: $\sigma_x = 0.00 + 791.50 + 1591.28 + 105.76 = 2488.54$, $\tau = 0.42 + 32.26 + 1.61 = 34.30 \rightarrow 81.935$; P3 web-flange junction, web: $\sigma_x = 0.00 + 738.73 + 1591.28 + 105.76 = 2435.77$, $\tau = 0.89 + 15.48 + 3.36 = 19.73 \rightarrow 78.468$; P4 web mid-depth: $\sigma_x = 0.00 + 1591.28 = 1591.28$, $\tau = 1.12 + 15.48 + 2.49 = 19.09 \rightarrow 33.498$ (y_w = web back from the minor axis = 24.70 mm; P1b at $s_1 = 42.99$ mm from the toe, $y_1 = 7.31$ mm; $S_{w1}, S_{w2}, S_{w3} =$ P385 Table A.2 points 1, 2, 3)		$f_y / \gamma_{M0} = 275$
Elastic verification (6.1): information only	Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information		6.2.7(5)-(6)

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\varepsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^3 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$ Fully Restrained 62.425 kN.m

Torsion Design

J, H, a, Q_f, Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
W_{n0}, S_{w1} 40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 12.27 P385 App C
φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 0.683 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 3.645 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 109.717 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665 14.062 **Warning** P385 3.1.2
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.307 / 3.707 kN.m 3.771 / 3.200 kN.m

Torsion Shear Design @ 0 m

T_t, τ_w at the V-T station @ x = 0 m (ULS: 1.35G + 1.5Q): T_t = 3.771 kN.m, V = 195.04 / 5.22 N/mm² P385
= 107.430 kN
V_{pl,T,Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)] · V_{pl,Rd} @ x = 0 m: τ_t = 195.04 (τ_w = 5.22) N/mm²; V_{pl,Rd} = 20.965 kN 6.2.7(9)
= 212.754; S_{mod} = V_{pl,T,Rd}/V_{pl,Rd} = 0.099
V_{Ed}/V_{pl,T,Rd} 107.430 / 20.965 5.124 **Warning**

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 60.86 mm **Warning**
Torq in Case 1 @ 2.465 m: θ_{max} ≤ 2.00° 0.2787 rad = 15.97° ≤ 2.00° (SLS: 1.0Q; P385 guidance, advisory: does not enter the verdict) 15.97° **> 2° advisory**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	Yield 6.1 (info)	V+T	Max
2.651	2.651	3.652	0.505	2.662	0.452	1.666	14.062	237.839	5.124	14.062

FAIL — governing Bending+torsion cross-section (P385 3.1.2) = 14.062

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385); check that the rotation is acceptable for whatever the member supports.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: elastic yield criterion (6.1) with torsion, cl 6.2.7(5) = 237.839 at $x = 2.46$ m (P1 flange tip): first yield is reached under the design actions (information only, not a utilisation: Class 1/2 open section without N_{Ed} : EN 1993-1-1 6.2.7(6) permits the plastic moment resistance under bending + torsion with B_{Ed} from the elastic analysis (the SCI P385 3.1.2 interaction with $M_{z,tot}$); the elastic verification of 6.2.7(5) is permissive ("may be applied") and is printed for information).

PFC-R10 · PFC 200 x 75 x 23 · M_{cr} method: refused · verdict **REFUSED**

Owner case: BOTTOM FLANGE RESTREAINED

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R10 "BOTTOM FLANGE RESTREAINED" = tension-flange restraint only: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Tension-flange (bottom flange of a sagging beam) restraint is not modelled: the tool's restraints (end flags $U_y / R_x / R_z /$ warping and the intermediate lateral restraints v, v', ϕ, ϕ') act at the shear centre and have no height in the cross-section, so a restraint that holds the tension flange laterally while the compression flange is free cannot be described; it needs a restraint-height term in the LTB eigen model (lateral constraint at $z = -h/2$, which couples v and ϕ) or the tension-flange-restraint method of BS 5950-1 Annex G / SCI P093. Treating it as unrestrained (R2) is the conservative bound; treating it as fully restrained (R9) is unsafe.

Owner case: BOTTOM FLANGE RESTREAINED

PFC-R11 · PFC 200 x 75 x 23 · M_{cr} method: refused · verdict **REFUSED**

Owner case: PARTIALLY RESTRAINED

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R11 "PARTIALLY RESTRAINED" = elastic (spring) restraint: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Elastic (partial) restraint is not modelled: every restraint in the tool is rigid or absent (each of v , v' , ϕ , ϕ' is held or free), so a lateral or torsional spring of finite stiffness (a purlin, a sheeting rail, a slab with a stiffness) cannot be entered; it needs lateral / rotational spring stiffness terms added to the LTB eigen element matrices (and to the warping-torsion FE) with a stiffness input per restraint and per end, plus the EN 1993-1-1 BB.2 minimum-stiffness test to decide whether a spring counts as a full restraint. The bounds are R2 (no restraint) and R3 (rigid restraints at the same positions).

Owner case: PARTIALLY RESTRAINED